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THE ENVIRONMENTAL LAW OF THE SEA

edited by
Douglas M. Johnston

International Union
for Conservation of Nature and Natural Resources
Gland, Switzerland
1981

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Preface

At the time of the winding down of the Third United Nations Conference on the Law of the Sea, it is appropriate to reassess the emerging new framework for marine law and policy. This time also coincides with policy initiatives in the field of the environment which underline the relationships between conservation and development. These include the World Conservation Strategy, prepared by IUCN with the advice, co-operation and financial support of UNEP and the World Wildlife Fund and in collaboration with FAO and UNESCO, as well as the Charter for Nature, also prepared by IUCN upon the request of the Government of Zaire and introduced into the United Nations General Assembly by that Government.

IUCN's Commission on Environmental Policy, Law and Administration has thus undertaken this study to place the environmentally relevant aspects of UNCLOS III in global and historical perspective. A final chapter on conclusions and recommendations identifies legal and administrative follow-up measures that must be taken for sound management and development of ocean resources at the global, regional or national level. In the aftermath of UNCLOS III, a great deal of important work will have to be undertaken to implement and supplement the environmental provisions agreed upon. Even before the new Convention of the Law of the Sea comes into force, developments in international environmental law will require significant conservation initiatives to be taken within the framework of the Convention.

The study consists of five chapters. The first three provide a historical overview of marine conservation, the fourth is devoted to analyzing regional approaches to the subject, and the fifth provides (the aforementioned) conclusions and recommendations. Within this final chapter it is stressed that the new legal regime of the oceans involves a much higher degree of national responsibility for the management of marine resources and the protection of the marine environment. Considerable attention is thus devoted to the need for developing tools to meet these responsibilities and for co-ordinating national measures at the regional or global level.

Our study has been an ambitious undertaking. The subject is a complicated one, requiring a wide range of expertise on matters as diverse as fishery conservation, marine pollution from land-based sources, and sea-use planning techniques. As a result, the work has been a team effort, with several leading experts making contributions. It can be seen in the following acknowledgements just how extensive the range of participation and consultation has been.

It is our hope that this study will serve two functions: to provide useful background information about the present state of marine conservation measures and, more important, to generate future initiatives for marine conservation. The reader's responses will be welcomed as a stimulus for further refinement of our conclusions and recommendations.

Wolfgang E. Burhenne
Chairman
IUCN Commission on Environmental
Policy, Law and Administration

The Contributors

James N. Barnes, Centre for Law and Social Policy, Washington, D.C.

Françoise Burhenne-Guilmin, IUCN Environmental Law Centre, Bonn

Cyrille de Klemm, IUCN Consultant, Paris

Lawrence M.G. Enomoto, formerly of the University of Hawaii, now with the US Department of State

Douglas M. Johnston, Dalhousie University, Halifax, Nova Scotia

Norman G. Letalik, Dalhousie University, Halifax, Nova Scotia

Daniel B. Navid, Executive Officer, IUCN Commission on Environmental Policy, Law and Administration, Gland

Martine Rémond-Gouilloud, University of Paris I, Paris

Jan Schneider, Covington and Burling, Washington, D.C.

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This study would also not have been possible without the work of the other contributors and their readiness to put their talent and expertise at our disposal; we want to also thank them warmly for their co-operation.

Last but not least, many individual experts have been consulted during the last two years in the course of the elaboration of this study in organizational or substantive matters. We would like to particulary thank the following individuals for the advice they gave in their personal capacity, although at the same time pointing out that they bear no responsibility for the final content of this volume:

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Editorial Preface

This lengthy and complicated study has consumed masses of collective energy over a period of two years or more. Fortunately the total energy expended has been distributed over a fairly large number of willing and able colleagues — those named above — with whom we have consulted on a variety of organizational and substantive matters. As the most direct beneficiary of their generosity, I would like to add my own personal thanks to them for their valuable co-operation and advice.

Although the vehicle consists of many components, this Project has been propelled throughout by a team of skillful drivers and an unusually durable engine. Few editors have had the privilege of working with such a talented group of fellow-writers as the contributors to this collaborative work. Representing different areas or aspects of this large field, we have not always agreed on questions of emphasis or balance, but the involvement of the other contributors has always been valuable and ungrudging. I hope that the final edited product is not distortive of their joint and several inputs.

Above all, I am deeply indebted to the very special colleagues who have worked most closely with me. I am referring, of course, to Wolfgang Burhenne, Françoise Burhenne-Guilmin, Daniel Navid, and Cyrille de Klemm. No computer could quantify the time, effort, imagination and determination they have contributed. It is measured in friendship.

Douglas M. Johnston

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Chapter One

The Environmental Law of the Sea: Historical Development

Douglas M. Johnston

A. The Rise of Environmentalism

Environmentalism has become a value system for modern society. Like all other ideologies, it means different things to different people. Neither its supporters nor its detractors find it easy to agree among themselves on what to emphasize or repudiate. For the purposes of this essay, concerned only with the protection of the ocean environment, the relevant history is mostly that of conservationism.

Even the concept of conservation, however, is baffling to the historian who wishes to trace its roots. Long before scholars were advancing theories on resource scarcity, propounding the merits of wise use, or advocating techniques of efficient management, primitive cultures were developing different attitudes towards "nature", animals and plants[1]. If we incorporate the idea of "pollution", we are driven back to pre-historic attitudes towards purity or cleanliness[2]. Psychologically, our environmental consciousness, newly raised in the late 20th century, might be traced to ancient fears that a steady commitment to an excess of natural bounty, or a continued disregard for imbalance in the nature of things, will finally prove to be dangerous and even threaten the survival of the human species.

The first legal or institutional restraints with an environmental purpose may have been the municipal controls on the burning of soft coal, which were introduced in certain European cities in the Middle Ages, often with the sanction of surprisingly heavy penalties[3]. Conservation, as distinguished from pollution control, may have been first enforced through local regulations on hunting, perhaps as early as the late Middle Ages[4]. In England, the first enforceable conservation legislation was a wildfowl protection statute enacted in 1534, in the reign of Henry VIII. — apparently at the urging of falconers, who may have been the first successful lobbyists for conservation[5]! Outside England, the first conservation laws to be enacted within a legislative process may have been those passed by the Assembly of Bermuda for the protection of turtles in the first half of the 17th century[6]. In the American colonies, the idea of conservation inspired the forest protection policy of William Penn, when he founded his settlement in 1681[7], influenced no doubt by the tree-planting movement led by John Evelyn in England[8].

Right up to the Second World War, the main thrust for conservation after the 17th century continued to come from Britain and the United States. In England, John Evelyn's tree-planting campaign was supported by an influential number of the landowning gentry. Evelyn also pioneered the idea of banishing air polluting industries from the major cities and advocated the planting of trees and shrubs to create a more congenial urban environment. Although neither of these ideas was implemented legally in his lifetime, the second of them was supported by the royal family, leading to the cultivation of the royal parks of Inner London and, less direct-

ly, to the initiation of the English landscape movement[9]. This movement to enhance scenic values unfortunately lost most of its momentum during the Industrial Revolution[10], and the cause of conservation in Britain up to the late 19th century was left mainly to a few dedicated individuals and their admirers[11]. By the period of the First World War the cause had been taken up by a dozen or more organizations, such as the National Trust, the Fauna Preservation Society, and the Council for the Preservation of Rural England, and later they would be joined by a number of other organizations with biological interests, such as the Wildfowl Trust, the Royal Society for the Protection of Birds, and the Society for the Promotion of Nature Reserves. As a result of these pressures, it proved possible in the 1940s to persuade the British government to accept a national policy on the conservation of nature and nature reserves and to create a new agency, the Nature Conservancy[12]. Significantly, the crucial arguments prevailed during the period of hostilities — allegedly due to the wartime suspension of "the normal British mechanisms for ensuring inaction"[13]! A few years later, Julian Huxley, and a few other scientific naturalists like Peter Scott and Charles Elton, "helped to put Ecology on the biological map by founding an Ecological Society, which would lay the scientific basis for conservation in Britain"[14].

The American conservation story is quite distinct from the British, especially after the 1830s, when George Catlin, Ralph Waldo Emerson, and Henry Thoreau took up the cause in lectures and publications[15]. The most influential conservationist of all, however, was George Perkins Marsh, whose classic *Man and Nature* (1864)[16] is now regarded in the United States as "the fountainhead of the conservation movement"[17]. Although written as a diatribe against the follies of man, this famous work is essentially modern in its emphasis: man is a free moral agent working independently of nature; man can control the environment for good as well as ill; wisdom lies in seeking to preserve the "balance of nature"; the present generation has an obligation, above all, to secure the welfare of future generations[18].

American contributions to conservation[19] include the establishment of a national park system and the concept of wilderness preservation. The Yellowstone National Park, established in 1872, was the first national park in the world. Although not created primarily for conservation reasons[20], "it did crystallize thought and words in definite action, and so marks the first practical beginning of effective conservation practice" in the United States[21]. The father of American wilderness protection was a Scottish-born biologist, John Muir, who became an influential promoter of the aesthetical and ethical philosophy associated with conservation, and founded the Sierra Club to monitor governmental and industrial action (and inaction) that seemed to threaten these values[22]. His "nature for nature's sake" philosophy brought him eventually into collision with the other great champion of conservation in America at the turn of the century, Gifford Pinchot. Pinchot, a professional forrester who became chief of the U.S. Forestry Division in 1898, was essentially out of sympathy with Muir's philosophy of leaving nature alone. His advocacy lay in the emphasis on wise use under systematic management, based on the biological concept of "sustained yield"[23]. Along with his famous friend and admirer, President Theodore Roosevelt[24], Pinchot led the Progressive Conservation Movement of the 1890-1920 period[25]. The U.S. conservation movement had its origins in American thought and practice long before 1890, but it was in this 30-year period that most of

the contemporary ideas about conservation came together publicly for the first time, generating the need for policies and compromises at the governmental level, setting the framework for the discussion of modern environmental issues[26].

Ultimately, of course, the most important influences in the history of conservation were intellectual: philosophical, scientific, and economic. Many ideas about conservation, as a concept, are clearly derived from classical philosophy, where the roots of the metaphysical concept of "nature" may be discerned[27]. But it was not until the mid-20th century that anyone attempted to place "philosophical naturalism" — including conservation ethics — into ecological perspective. Perhaps the most famous and most diligent effort to articulate the conservation ethic at that time, prior to the advent of the "environmentalists", was that of Julian Huxley, whose "evolutionary humanism" provided a philosophical and ethical viewpoint that appealed to many conservationists[28]. Even more closely related to nature conservation was the popular sentiment cultivated by writers like Marsh and Muir in the United States, who shared a strongly held conviction that the wilful destruction of nature was a reprehensible, and ultimately stupid, form of waste[29]. In a traditionally "wasteful" society like the United States, this theme has assumed an obsessive character, that might be likened to guilt. In extreme cases the cause of conservation in the United States has become associated with that of human virtue, arming conservation ethic with an almost theological faith. On both sides of the Atlantic, most conservationists today would agree with Julian Huxley in viewing man both as a part of nature and yet truly unique[30], and emphasize the moral responsibility of man as the guardian of nature[31].

Yet the modern conservation movement could never have become a moral crusade without the support of science. Scientifically, conservation has its roots in the earliest phases of the history of biology, even as early as Aristotle. In more modern times there is almost no limit to the number of naturalists and biologists who could be viewed as making major contributions to the scientific foundation of modern ecology: Copernicus, Kepler, Galileo, Descartes, Newton, Harvey, Ray, Linnaeus, Buffon, Lamarck, Wilson, Cuvier, Audubon, von Humboldt, Lyell, Agassiz, Darwin, Wallace, Mendel, Thomas Huxley, and many others[32]. Moreover, it is necessary to add the names of those who made the most important early contributions to "applied ecology". Concern with problems of environmental health can be traced back to the 1st century A.D., when Pliny the Elder called for the use of protection masks by workers engaged in mining and grinding operations, which generated dust and other harmful particles[33]. The first to associate health with the working environment in a scientific manner was Bernardino Ramazzini, whose systematic treatise *De Morbis Artificium Diatriba* was published in 1700[34]. By the 19th century scientists were beginning to study different forms of "stress" on organisms. By the mid-1800s, for example, Jus Liebig had deduced that plant growth was limited less often by the supply of factors present in abundance, such as water and carbon dioxide, than by that of factors present in minute quantities, such as boron and phosphorus[35]. By the outbreak of the First World War applied ecologists such as V.E. Shelford had reached the point of identifying the basic principles of "toleration ecology"[36]. Yet the term "ecosystem", now basic to the discipline of ecology, was not coined until 1935, when it was introduced by Arthur H. Tansley[37]. Ecology today is still something of an infant science, but by no means a neglected area of inquiry. Grow-

ing agreement on the use of basic terminology may be taken as evidence that the discipline is settling down[38].

A third source of the intellectual growth of conservation is that of economic theory; or, more specifically, the doctrine of increasing natural resource scarcity[39]. Such a doctrine is essentially multi-disciplinary in its complexity, drawing upon the findings of philosophers, ecologists, demographers, and political scientists, as well as economists, but of these various contributions economic theory might be said to constitute the core. Much of this core is derived from the writings of Thomas Malthus and David Ricardo in the first quarter of the 19th century, and further developed by John Stuart Mill later in the same century[40]. Many economists today are still troubled by the concept known as the "Malthusian trap", a term applied to situations that are "trapped" in a vicious circle of subsistence: a small rise in income per capita above the subsistence level generates a rise in population which reduces income per capita to the former subsistence level[41]. Both the Malthusian and Ricardian versions of the early doctrine were based on the premise of a fixed supply of agricultural land and focussed primarily on the problem of scarcity applied to the production of food, but Ricardo also took mining into account, since he was chiefly interested in developing a general theory of rent[42]. Mill's approach was more "environmental", developing and extending the doctrine of natural resource scarcity to cover all sectors of living space, its quality as well as quantity[43]. Since then the specialized field of natural resource economics has become one of the most important in modern economics, and within this field it has become fashionable again to adopt an attitude of "guarded pessimism", somewhat in the manner of Malthus[44]. Within this field there has emerged a small but influential subset of specialists in "conservation economics"[45]. The impact of these writers on the theory of forestry and fishery conservation, in particular, has been considerable since the early 1960s[46]. One effect of the growing influence of resource economists in the field of fishery studies is the emergence of a clear distinction between the concept of "management" and that of "conservation". As these terms are now generally used by fishery specialists, conservation is regarded as one — but only one — of the functions of management[47].

B. Antecedents of International Environmental Law

1. Introduction

The environmental law of the sea is a relatively new development in the international legal system. It is represented by the area of convergence between the law of the sea and international environmental law. The law of the sea is, of course, of ancient heritage, traceable to rudiments in the Rhodian sea law of the 9th century B.C. The dating of the origins of international environmental law, on the other hand, is largely a matter of jurisprudential preference. The more one feels compelled by the strictures of the positivist school to look for international law in precise, clearly established, firmly sanctioned, obligatory rules, based on the universal consent of nation states, the more likely one is to emphasize the recent and still tentative character of international environmental law. The more warmly one responds to the imagery of natural law, projecting concepts of natural order, human reason, and

moral authority, the more likely one is to find the roots of international environmental law buried deep in the classical principles of the international legal system. Others, declining to make a choice between the inchoate and the immanent, may be content to note the "antecedents" of modern normative developments in international environmental law within a broadly conceived, multi-factoral framework of legal development[48].

2. The Traditional Framework of Customary International Law

In retrospect, we can see the potential environmental significance of several — arguably immanent, admittedly inchoate — maxims or principles in the classical literature of international law[49]. None of the classical norms of customary international law had been applied explicitly to the protection of the environment, as such, by the end of the 19th century, but this is scarcely remarkable in an age when environmental consiousness was virtually non-existent in public life. Yet the doctrine of *state responsibility* was sufficiently broadly framed by then to encompass the modern concept of state liability for environmental damage to another state. By the 1930s, in his formulation of the doctrine of an "international delinquency"[50], Lauterpacht had focussed on the scope of international legal duties falling upon sovereign states. The responsibility of a state, he maintained, "may become involved as the result of an abuse of a right enjoyed by virtue of international law"[51]. Significantly, he chose as his example the duty of a state not to interfere with the flow of a river to the detriment of other riparian states[52]. The prohibition against *abuse of rights,* said to be based on the old maxim *sic utere tuo ut alienum non laedas*, was admitted to be recent and uncertain, but available for judicial development[53]. Because of its application to international rivers and lakes[54], the abuse of rights principle has sometimes been called the principle of *neighbourliness*[55], and it received support, in the context of an air pollution problem, from the famous 1941 award arising out of the *Trail Smelter* arbitration between Canada and the United States[56]. The broader principle of *sic utere tuo* was upheld by the International Court of Justice in 1949 in the *Corfu Channel* case between the United Kingdom and Albania[57], when it referred to "every State's obligation not to allow knowingly its territory to be used for acts contrary to the rights of other States"[58]. The *sic utere tuo* principle was also discussed in the *Lake Lanoux* arbitration between France and Spain[59], and applied to the use of an international river, but it should be noted that this arbitration was not concerned with a pollution problem[60].

Several jurists argue that, as a result of the *Trail Smelter, Corfu Channel,* and *Lake Lanoux* decisions, there is now an evolving principle of *strict liability* for environmental injury in international law, but it may be premature to regard this as a firmly established norm[61]. Similarly, it is a matter of controversy whether to treat *sic utere tuo,* abuse of rights, neighbourliness, and other environmentally significant areas of doctrine (e.g. right of self-defence, right of self-help), as belonging to the category of "general principles of law" which the international Court of Justice is bound to apply[62].

3. Early Treaty Arrangements (1815-1945)

a) International Waterways and Lakes

Since the 17th century efforts have been made to negotiate international ar-

rangements to secure the right of free passage on certain, internationally important, navigable rivers, such as the Danube. In 1792 the Provisional Executive Council of the French Republic adopted the principle that "streams which flow through the territory of several states do not belong exclusively to these states with respect to the particular portion traversing their territory, but represent common, indivisible and inalienable property of all riparian states"[63]. The Treaty of Paris (1814) applied this principle to the river Rhine, but it was left to the Congress of Vienna in the following year to elaborate the principles of regulation to be applied to all users, both riparian and non-riparian, of all navigable rivers which separate or traverse the territories of two or more states. However rudimentary, this early regulatory system is perhaps the first applied to an international area or resource that could be viewed as an expression of common environmental concern.

Since 1815 numerous European treaty arrangements have been made for the allocation of fishing rights, navigation control, and other environmentally related purposes in international rivers and other boundary fresh water areas[64]. In more modern times regulatory systems for international rivers have been transformed into something like management systems, dealing with a variety of purposes ranging from resource extraction and conservation, energy generation, irrigation, and erosion control, to the prevention and control of various forms of pollution. By the time of World War Two there was a general trend toward the extension of such multipurpose management systems to cover entire river basins, and a more limited and gradual appreciation of the need to provide international protection for ground water as well as surface water supplies[65].

The history of international river law makes a particularly important contribution to the conceptual development of international environmental law, showing as it does a perceptible, albeit reluctant, movement away from the principle of *unrestricted territorial sovereignty* (viz. the Harmon doctrine) through that of *unrestricted territorial integrity* (or natural water flow doctrine) to the emerging principle of *limited territorial sovereignty* (as reflected in community of interest or equitable apportionment doctrine)[66].

In the case of international lakes, Canada and the United States developed a more ambitious system of international co-operation for environmental management purposes under the Boundary Water Treaty of 1909[67]. The International Joint Commission established under this treaty was given jurisdiction over "all cases involving the use or obstruction or diversion" of the boundary waters, and environmental management responsibility was conferred upon it by the assignment of the highest priority to "uses for domestic and sanitary purposes"[68]. The environmental significance of the Treaty was further emphasized in the provision that the "boundary waters and waters flowing across the boundary shall not be polluted on either side to the injury of health or property on the other"[69]. By 1945 the International Joint Commission had accumulated considerable administrative and quasi-judicial experience in the environmental management of the Great Lakes and adjacent waters, in dealing with air as well as water pollution problems. In favourable political circumstances the IJC was able to function in a more harmonious and creative manner than was possible for the Danube Commission and other international river management authorities[70].

b) Wildlife Protection

International measures for wildlife protection seem to have been first attempted in 1875, when Austria/Hungary and Italy signed a Declaration for the Protection of Birds Useful to Agriculture[71]. In the 1890s several national governments were asked to support a draft convention for the protection of migratory birds[72], and an effort was made to prohibit small-ivory export[73]. The first of these initiatives resulted in the 1902 Convention to Protect Birds Useful to Agriculture[74] which became effective in 1905 and contributed to other treaty arrangements such as the U.S.-Canadian Migratory Birds Convention of 1916[75] and the 1936 U.S.-Mexican for the Protection of Migratory Birds and Game Mammals[76]. The ivory trade in Africa was only one aspect of a wider effort to provide international protection for natural communities (e.g. wildlife parks). Although the attempt to create a permanent international commission for nature protection was unsuccessful[77], it was possible to place some limitation on the trophy trade through the 1933 London Convention for the Preservation of Fauna and Flora in their Natural State[78], and its counterpart concluded under the auspices of the Pan American Union[79].

After the First World War concern for the future of mammals in general[80] gave way to a special concern for the preservation of marine mammals, such as whales and seals, in particular[81]. In this context it became common to draw the distinction between the preservation and protection of wildlife, on the one hand, and the conservation of species of commercial value, on the other[82]. This distinction led in turn to two, rather different, thrusts in normative thinking: one emphasizing altruistic concepts of heritage, trust, guardianship, and other formulations of moral responsibility; the other focussing on long-range considerations of wise and efficient use of a scarce and economically valuable resource. The former movement owed much to the popularization of wilderness and nature preservation in the United States by pioneers such as Marsh and Muir in the 19th century; the latter owed more to the theories of forestry management pioneered by conservationists like Gifford Pinchot[83].

c) Plant and Animal Protection and Disease Control

In the 14th century the Black Death pandemic of plague transmitted by rat fleas killed one quarter of the population of Europe[84]. In modern times it has become more clearly recognized that insects and rodents not only constitute pests of man and animals but also act as vectors of disease. Despite major advances in medical research and public health, diseases like malaria are still traced chiefly to insects, and others such as dengue, dengue haemorrhage fever, and yellow fever are attributed to epidemics of arboviruses[85]. Out of modern research comes the realization that the vectors and reservoirs of disease are often multiplied by man's own actions, and can therefore be controlled, to some extent, by appropriate environmental measures[86].

The international treaty approach to these problems of environmental health seems to have originated in the second half of the 19th century[87], and an early system of monitoring was devised with the establishment of the International Office of Public Health in Paris in 1907. This Office, which was basically a rule-making body, was succeeded by the Health Organization of the League of Nations (1921-1946) and by an age of more direct action[88]. The same period witnessed international approaches

to agricultural problems such as plant and animal disease. The combined effect in the inter-war period was a mushrooming of conventions in specific areas such as locust control[89] and more general areas of plant[90] and animal protection[91]. Reacting to this record of tentative action before 1945, the World Health Organization was able to attract more substantial budgets after the Second World War and to adopt a more regulatory approach to these problems[92].

d) Conservation of the Living Resources of the Sea

Before the 1880s a few prominent biologists[93] had occasionally turned their attention to the problems of sea fisheries, but it was not until the 1890s that fishery science became a full-time professional pursuit[94]. Almost immediately the need for scientific collaboration was understood and a mechanism for this purpose was established in 1902. Indeed the International Council for the Exploration of the Sea (ICES)[95] not only laid the basis for the new science of fishery investigation, but provided a successful working model for all international research councils. Fears of depletion of fishery stocks had, of course, been expressed for hundreds of years[96], but the first efforts at international fishery conservation were not begun until 1839, when a Franco-British treaty was concluded to regulate fishing by the parties in the English Channel[97]. This produced detailed regulations in 1843[98], the first international set of regulations for a high seas fishery. Almost four decades later, in 1882, the North Sea fishery was brought under a multilateral policing arrangement[99]. In the Northeast Atlantic, where these early initiatives were proposed, the ICES was to play a major scientific role in international fishery conservation efforts in 1929, 1932, 1937, and 1943[100]. These efforts culminated in the 1946 Convention for the Regulation of the Meshes of Fishing Nets and the Size Limits of Fish (the London "Over-fishing Convention[101]"), which in turn would be replaced by the Northeast Atlantic Fisheries Convention of 1959[102].

The other region of major importance in the early history of international fishery conservation was the Northeast Pacific. The first species to be brought under a degree of international management was the Pacific fur seal[103]. A source of conservation concern as early as the 1880s[104], this species was brought under regulation as a direct result of the famous arbitration of 1892 between Great Britain and the United States[105]. In 1911 Russia and Japan joined these two nations in adopting the Convention for the Preservation and Protection of Fur Seals[106]. During the Second World War Canada and the United States continued to implement the 1911 provisions, with minor changes, under the 1942 Provisional Fur Seal Agreement[107], until all four of the states involved — Canada, Japan, the U.S.S.R., and the U.S.A. — were able to come together in 1957 to sign the Interim Convention on the Conservation of the North Pacific Fur Seals[108].

The sockeye salmon fishery of the Fraser River[109], exploited both by Canadians and Americans, was first studied internationally as early as 1892, when it was reported to be in a healthy condition. After several years of disappointing "runs" an effort was made to bring the fishery under a joint Canadian-U.S. fisheries commission, which had been established by the Bryce-Root Treaty of 1908, but this convention became inoperative. There followed a series of other unsuccessful efforts to negotiate and ratify a bilateral regulatory treaty for the Fraser River fishery, but finally in 1937 the fifth attempt prevailed[110] and the International Pacific Salmon

Fisheries Commission was brought into existence[111]. After the Hell's Gate Canyon disaster in 1941, it became necessary to grant the Commission "regulatory powers", and this was finally accomplished in 1946[112].

Of all these early fishery conservation arrangements, none was more important than the famous Convention for the Preservation of the Halibut Fishery of the Northern Pacific Ocean[113] concluded in 1923. As a result of the findings of a joint commission in 1918, this treaty created the International Fisheries Commission, the first effort at international fishery conservation on the high seas[114]. The regulatory powers of the Commission were increased by a series of new conventions signed in 1930[115], 1937[116], and 1953[117]. As the first international fishery agency with management responsibilities, this body (re-named the International Pacific Halibut Commission) deserves an honoured place in the history of international environmental law.

Finally, reference should be made to the early efforts to place whaling, especially Antarctic whaling, under international regulation[118]. Between 1924 and 1927 the League of Nations canvassed the feasibility of an international whaling control agency, and at the 1927 meeting of the ICES the suggestion was put forward that the whaling nations should prohibit any further expansion of whaling activities. No immediate agreement was reached on these proposals, but in 1930, at the request of the ICES, the Norwegian government established the International Bureau for Whaling Statistics at Oslo, and in the following year the International Whaling Convention[119] was concluded under the auspices of the League of Nations. Six years later, in 1936, this convention entered into force. In 1937, and on several other occasions before and after the Second World War, further effort was made to halt dwindling stocks of whales, until finally in 1946 most of the major whaling nations came together to sign the International Convention for the Regulation of Whaling[120]. Although the International Whaling Commission[121] is generally criticized today for its conservation failures, most of the blame rests with a minority of its membership and the organization itself deserves recognition as the first approximation to a global approach to a controversial issue in the conservation of the commercially valuable resource.

C. Environmental Hazard Control (1945-1972)

1. Introduction

Legal development is shaped by events; or, more exactly, by a mixture of both public and official attitudes, which have been conditioned by events perceived as critical and requiring a legal response. The early history of international environmental law was, in larger part, a gradual process of development arising from the kinds of treaty arrangements reviewed above. But this process has been accelerated since 1945 by the force of important events, which have altered man's perception of his environment and of the dangers that now threaten it. Particularly decisive are those events that have served to focus public attention on what appears to be a totally new order of *hazard* in human affairs[122]. Three events in particular seem to have had a special influence on the public perception of "environmental hazard"[123]: the advent of the nuclear age, the publication of Rachel Carson's *Silent Spring,* and the succession of catastrophic oil spills in coastal areas of

the ocean. All three are examples of events which have transformed the perception of routine hazard into that of potential crisis or catastrophe and created a world-wide awareness of the need for rapid and significant legal change.

It is generally agreed that the advent of nuclear capability at the end of the Second World War played a major role in the raising of environmental consciousness around the world. The emotional impact was most evident in the context of nuclear disarmament issues, such as that of the legality of nuclear testing[124], embittered by the evidence of genetic damage to Japanese fishermen after one such test. But as the danger of a nuclear holocaust seemed to become less imminent in the 1960s it gave way gradually to a rising apprehension about the effects of routine or accidental radiation that might occur in the process of energy production, and about the efficacy of existing safety standards in the industry[125]. By 1972, the year of the Stockholm Conference on the Human Environment, fears concerning the peaceful and military uses of nuclear energy had conditioned most governments to adopt a global view of the threat to the biosphere.

Rachel Carson's famous book was published in 1962[126]. Drawing upon the accumulation of scientific evidence, it convinced most readers that the environment was deteriorating rapidly as a result of the overuse and misuse of chemicals. The author set out to frighten her readers[127], and the spate of official and unofficial concerns with chemical abuse since then may be taken as a measure of her success[128]. With the gathering of increasingly reliable data, it has become more difficult to reject the view that the chemical danger to mankind may be at least as great as the nuclear[129]. Although *Silent Spring* aroused a storm of controversy in the 1960s[130], we can look back now, twenty years later, and recognize the legitimacy of her shocking assault[131].

Also, in the 1960s, the public became aware of the growing hazard of catastrophic oil spills, especially in congested coastal waters, aggravated by the rapid growth in the size of new tankers entering the shipping industry[132]. Certain, particularly dramatic, examples of collisions and groundings attracted worldwide publicity, providing graphic proof that, despite impressive accomplishments in modern construction and design, these vessels were as vulnerable as ever to the risk of human error[133]. With steadily increasing world consumption of cheap oil, the period up to 1972 contemplated an apparently indefinite trend towards larger and larger carriers of petroleum products, and larger and larger spillages. As much as the evidence of nuclear and chemical hazards, this prospect underlined that the world was paying an unexpectedly high environmental price for modern technology. To many, these spillages themselves were no more shocking than the discovery by explorers like Cousteau, Chichester and Heyerdahl that virtually no region of the ocean, however remote, was free from floating oil[134], and the realization that this diffuse damage to the marine environment was almost entirely caused by routine discharges[135], not by catastrophic spillages. Less well known were those estimates in the early 1970s that barely 20% of the pollutants in the ocean could be attributed to shipping accidents or practices[136]. Yet because the ocean was still conceived, in large part, as an international area, the problem of ship-generated pollution was seen as one that might be dealt with effectively through international co-operation and regulation[137].

2 The Regulatory Response to Unnatural Hazards

a) Control of Environmentally Dangerous Weapons

Most categories of weapons of mass destruction in the late 20th century are potentially "ecocidal". Consequently, most of the major arms control agreements concluded since 1945 must also be viewed as an important contribution to environmental preservation. Of these arrangements designed, at least in part, for the control of environmentally dangerous weapons, most are chiefly concerned with nuclear armaments[138].

The most famous, and perhaps most important, of the nuclear agreements is the 1968 Treaty on the Non-Proliferation of Nuclear Weapons (NPT)[139], which emerged after years of difficult negotiations. By March of 1970 enough countries had become parties to bring it into force. By the time of the first five-year review conference, held in 1975, 95 countries had become parties[140]. Under the NPT all non-nuclear-weapon parties pledge not to make nuclear weapons or to acquire such weapons or weapon-making materials from states possessing them[141]. Furthermore, the parties undertake to allow inspection of their peaceful nuclear activities to ensure that materials are not diverted from these activities to the manufacture of nuclear explosives. Such inspection is to be conducted by the International Atomic Energy Agency (IAEA), an international organization established in 1956 with its headquarters in Vienna. The NPT envisages that each non-nuclear-weapon party will accept a system of international safeguards under a bilateral agreement to be negotiated and concluded with the IAEA[142]. By 1975, 39 of the 95 parties had entered into such agreements, and an additional 18 safeguards agreements had been signed or approved by the IAEA board of governors[143].

Although the provisions of the NPT are formulated in terms of arms control objectives, it is clearly an agreement of major environmental significance. Accordingly, it is a matter of environmental, as well as political, concern that it is not yet universally accepted, especially by nuclear powers such as China and France, and by a number of "near-nuclear" countries[144]. Moreover, many of the parties, as we have seen, have not yet entered into safeguards agreements with the IAEA as required by the NPT, and the IAEA safeguards system[145] in any event falls short of the kinds of international controls sought in the early stages of negotiations before the NPT was concluded[146].

An equally famous arms control arrangement is the cluster of SALT I agreements concluded between the United States and the Soviet Union in 1969, and related agreements in the following years[147]. But, like the SALT II agreements concluded in 1979[148], SALT I is devoted to overall limitations on the strategic weapons systems of the two superpowers, and not simply to those that are environmentally dangerous.

The remaining arms control agreements of major environmental significance concluded up to 1972 may be divided into four categories[149]:

(i) *limited nuclear test ban:* viz. 1963 (Partial Test Ban) Treaty Banning Nuclear Weapons Tests in the Atmosphere, in Outer Space and Under Water[150];
(ii) *crisis communication arrangements:* viz. the 1963 ("Hot Line") Memorandum of Understanding between the United States and the Soviet Union for reducing the

possibility of a miscalculated nuclear attack[151], and the 1971 Agreement for improving this system of communication[152];

(iii) *prohibitions of biological and chemical weapons: viz.* the 1971 (BTW) Convention on the Prohibition of the Development, Production and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on their Destruction[153]; and

(iv) *limited, "regime-oriented" prohibitions of nuclear weapons:* viz. (a) the 1959 Antarctic Treaty[154]; (b) the 1967 (Outer Space) Treaty on Principles Governing Activities of States in Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies[155]; (c) the 1967 (Tlatelolco) Treaty for the Prohibition of Nuclear Weapons in Latin America[156]; and (d) the 1972 (Sea-bed Arms Control) Treaty on the Prohibition of the Emplacement of Nuclear Weapons and Other Weapons of Mass Destruction on the Sea-Bed and the Ocean Floor and in the Subsoil Thereof[157].

Of these agreements, five deserve particular comment. The Antarctic Treaty "System",now the subject of special attention possibly with a view to revision[158] and amplification[159], was the first of all these environmentally significant, arms control arrangements to be concluded. Its primary purposes are divided between arms control, scientific co-operation, environmental preservation, and non-prejudicial "freezing" of conflicting territorial claims. The environmental provisions take the form of prohibitions against "the testing of any weapons"[160] and "(a)ny nuclear explosions in Antarctica and the disposal there of radioactive waste material[161], but also of significance for the future development of international environmental law is the provision that each party has the right to designate observers to carry out inspections in all areas of Antarctica, "including all stations, installations, and equipment within those areas, and all ships-and aircraft at points of discharging or embarking cargoes or personnel in Antarctica"[162]. Since the purpose of the inspection system is "to promote the objectives and ensure the observance of the provisions" of the Treaty, it can be said to represent an early effort to enforce an international arrangement which is partly devoted to environmental management. The fact that Antarctica is, territorial claims notwithstanding, an "international area", analogous to outer space and the deep ocean floor, enhances the Treaty's significance in the history of international environmental law.

The Partial Test Ban Treaty is the first major treaty to be dedicated to the preservation of the human environment. It was preceded by a host of conservation and wildlife protection treaties, such as those mentioned above[163], but none had a declared purpose remotely comparable to that of this arms control agreement: "to put an end to the contamination of man's environment by radioactive substances"[164]. Though limited in its applicability to the atmosphere, outer space, and submarine areas, and subject to criticism on certain grounds[165], it is a major step towards the elimination of weapons which are unacceptable for environmental (as well as other) reasons. Unfortunately it is not yet adopted by all nuclear powers, and as an attempt to reduce to nuclear hazard from military weapons its efficacy must be evaluated along with that of the Non-Proliferation Treaty.

The Outer Space Treaty develops the legal regime of outer space (including the moon and celestial bodies) on the basis of the principle that outer space is an international area, and therefore "not subject to national appropriation by claim of

sovereignty, by means of use or occupation, or by any other means"[166]. This designation of an "international area" has been officially extended, in more recent years, to the deep ocean floor beyond limits of national jurisdiction[167], and might also be regarded by some as extending to the Antarctic landmass and the Southern Ocean[168]. The designation of certain sectors of the human environment as "commons" has an obvious importance for the future development of the concepts of common interest and universal responsibility in international environmental law[169]. Parties to the Outer Space Treaty undertake to "bear international responsibility for national activities in outer space, including the Moon and other celestial bodies, whether such activities are carried on by government agencies or by non-governmental entities"[170]. Specifically, the Treaty builds on the Partial Test Ban Treaty of 1963 and expressly forbids the parties to "place in orbit around the Earth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction, install such weapons on celestial bodies, or station such weapons in outer space in any other manner"[171].

The prohibition against the storage, testing and use of nuclear and other weapons of mass destruction is extended by the 1972 Sea-bed Arms Control Treaty to the sea-bed and ocean floor "beyond the outer limit of a sea-bed zone"[172]. This outer limit is coterminous with the 12-mile seaward limit of the contiguous zone, as recognized by Article 24(2) of the 1958 Geneva Convention on the Territorial Sea and the Contiguous Zone. The Treaty provides for verification, at the instance of any party, by means of observation and, if necessary after appropriate consultations, inspection of suspected "objects, structures, installations or other facilities" in the prohibited areas of the sea-bed and ocean floor[173]. It should be noted that this Treaty was negotiated outside the forum of the U.N. Sea-bed Committee, which in 1972 was deeply engaged in formal preparations for the Third U.N. Conference on the Law of the Sea (UNCLOS III). The successful negotiation of the Sea-bed Arms Control Treaty was a major reason for the decision of the Sea-bed Committee not to seek provisions on the military uses of the sea-bed at UNCLOS III[174]. Conversely, since the First Committee of UNCLOS III was assigned the task of elaborating the legal regime of the sea-bed and ocean floor beyond limits of national jurisdiction, it was considered inappropriate for the Sea-bed Arms Control Treaty to attempt to articulate constitutive provisions of that kind on the model of the Outer Space Treaty.

Finally, the 1971 BTW Convention represents a new departure in raising to the treaty level the prohibitions against chemical and biological weapons which had appeared previously in less formal documents[175]. The prohibition extends only to "(m)icrobial or other biological agents or toxins, whatever their origin or method of production"[176], but the Convention is seen as "a first possible step towards the achievement of agreement on effective measures also for the prohibition of the development, production and stocking of chemical weapons"[177]. In the case of an alleged breach, the Convention authorizes any party to lodge a complaint with the U.N. Security Council and obligates all parties to co-operate in carrying out any investigation which the Security Council may initiate[178].

b) Regulation of Peaceful Uses of Nuclear Energy

One of the most divisive environmental issues of the 1970s was how far and how fast to proceed in order to bring the nuclear energy industry under effective en-

vironmental safeguards. Even before the oil crisis of 1973, experts and non-experts were divided on the degree of risk involved in turning to the nuclear source of energy[179]. Characteristically, this controversy is conducted initially within the context of national, rather than international, policy, since any critical decisions have to be implemented, if not initiated, under the cover of territorial jurisdiction. Yet the central elements of the issue are universal in scope[180], and much of the regulatory impetus in the last 30 years has come from the international level[181].

The earliest international approach to regulation of the energy industry was regional, after the establishment of the European Atomic Energy Community in 1953, but, as we have seen, the International Atomic Energy Agency was constituted at the global level in 1958 and since then has become the principle vehicle for the international co-ordination and harmonization of national nuclear control policies[182]. In recent years the issue of regulation of peaceful uses of nuclear engergy has become conjoined with the larger question of international energy policy[183], and as a result the international aspect of nuclear safety seems more conspicuous than ever. Moreover, as the problem of nuclear waste disposal has become more acute, it has become necessary to impose stricter controls on the use of the ocean for this purpose, both under global and regional treaty arrangements[184]. The sensitivity of the disposal issue is, of course, aggravated by the overlapping of military and peaceful uses of nuclear energy.

c) Regulation of Industrial Discharge and Waste Disposal

In the years between the end of World War Two and the Stockholm Conference on the Human Environment national governments were slow to recognize the need to deal directly at the international level with the problems of industrial discharge and waste disposal. Before Rachel Carson's warning became famous, few governments exhibited any special concern with the chemical hazard, even in dealing with local pollution problems falling entirely with domestic jurisdiction. Apart from a few agreements concerning polluted international water courses[185], little effort was made to adopt an international regulatory approach to discharge and disposal problems until the early 1960s, when the littoral states in question concluded a succession of agreements for the environmental protection of Lake Constance[186], the river Saar[187], the river Mosel[188], Lake Leman (Geneva)[189], and the river Rhine[190]. In the years that followed, Europe continued to take the lead in such initiatives: a number of East European nations negotiated frontier water and "water economy" agreements[191] that were at least partly influenced by the awareness of chemical hazards; and the ECE[192], the Council of Europe[193] and the EEC[194] began to develop discharge and disposal control policies. The European "monopoly" on these kinds of treaty arrangements was finally broken in 1972 with the Canadian-U.S. Agreement on Great Lakes Water Quality[195]. This sophisticated and comprehensive Agreement identifies both general[196] and specific[197] water quality objectives; requires that the water quality standards and other regulatory requirements of the parties shall be consistent with these objectives[198]; and calls for programmes and other measures to be implemented in nine designated areas[199].

d) Prevention and Control of Ship-Generated Pollution

In the first two decades following the Second World War there was little concern about the deterioration of the ocean environment. Radiation was a threat to man,

not to the sea, and even Rachel Carson's warning of the chemical hazard in the early 1960s focussed on the land-based sources of emissions and effluents rather than on their ultimate destination in the sea. In the 1950s and early 1960s the run-off of industrial and municipal wastes was regarded at most as a potential problem in rivers, lakes, and estuaries. Applied to the immensity of the world's oceans, the pollution threat was generally regarded as diffuse and unalarming.

This attitude of complacency was shaken irreparably on March 18th, 1967, when the Liberian flag tanker *Torrey Canyon*, carrying a cargo of 117,000 tons of crude oil, ran aground on the Seven Stones Reef off the southwest coast of England[200]. The impact of this famous grounding on informed opinion was dramatic and profound. Among the direct consequences one must list: a concentration on the threat of shipping to the ocean environment; a fundamental re-appraisal of its role by the Intergovernmental Maritime Consultative Organization (IMCO); a succession of important treaty-making initiatives in the field of marine pollution control; a re-examination of national environmental policy by the British government and an assertion of special rights of self-protection by other vulnerable coastal states; and a marked upgrading of priority for the topic of ship-generated marine pollution control in a variety of global forums.

First, one must take note, albeit ironically, that although collisions and groundings of large tankers attract public attention and provoke official response, tanker accidents contribute only a small fraction of the total volume of identifiable pollutants in the ocean. Indeed ship-generated pollution in general may have provided no more than 20% of the marine pollution problem in 1972[201], when it was addressed at the Stockholm Conference on the Human Environment[202]. Moreover, accidental tanker discharges of the *Torrey Canyon* character account, according to the U.S. National Academy of Sciences[203], for only 33 percent of all vessel discharges[204]. Furthermore, oil is very far removed from being the most dangerous kind of pollutant in the ocean. It may seem strange indeed that the vast amount of diplomatic attention that has been devoted since 1967 to the preservation of the marine environment was provoked, more or less directly, by an incident of a kind that cannot be said to represent more than a small proportion of 20 percent of the general problems of marine pollution[205] — a general problem which is, moreover, only part of the even larger concern with the environmental protection of the ocean.

Yet it is generally conceded that the famous stranding, the biggest wreck up to that point in time, had the effect of transforming IMCO's self-image. "Before the loss of the *Torrey Canyon,* technical duties arising from the International Convention on the Safety of Life at Sea (SOLAS 1960) were the main concern of IMCO; after the sinking of the *Torrey Canyon* legal matters and legal conventions became a concern, and IMCO came to be a 'technical-legal' rather than just a 'technical' organization"[206]. On May 4th-5th, 1967, at the urging of the British government, an emergency meeting of the IMCO Council was convened and an 18-point programme adopted for the purpose of developing a major role for the Organization in the prevention and control of marine pollution[207], despite the fact that such a role was outside its original terms of reference[208]. With this additional responsibility, taken upon itself in what was perceived to be a response to a growing problem of crisis proportions, IMCO began to become less of a technical organization and more of a "forum"

organization[209]. As events have unfolded in more recent years, there is now a prospect that the IMCO forum may soon have to assume a more interventionist role as a regulator of shipping[210].

Certainly the most striking evidence of IMCO's new involvement in pollution control is its active promotion of a succession of major conventions. The first IMCO convention, the 1954 International Convention for the Prevention of Pollution of the Sea by Oil[211], did in fact confer some pollution control responsibility on IMCO in 1958, the year the 1954 Convention came into effect, and nine years before the sinking of the *Torrey Canyon.* But the preventive approach was limited essentially to a prohibition against the discharge within 50 miles of land, by sea-going ships of 500 gross tons and more which were registered in contracting states, of oil or any oil mixture which would foul the surface of the sea[212]. Some of the limitations and deficiencies of this early convention were removed by amendments adopted in 1962, 1969, and 1971, but it remained a cautious beginning[213].

The next stage in IMCO's treaty-making occurred at the Brussels Conference convened in 1969, where two new instruments, heavily influenced by memories of the *Torrey Canyon*, were adopted: the Convention on Civil Liability for Oil Pollution Damage[214], and the Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties[215]. The former establishes rules and procedures for determining questions of liability and providing compensation to persons who have suffered damage caused by pollution resulting from the escape or discharge of oil from ships, but it applies only to pollution damage caused on the territory or in the territorial sea of a contracting party. Under the Convention liability falls on the shipowner, not the charterer or cargo, but his liability is limited to specified amounts[216]. The Intervention Convention is of a more radical nature, juridically, for it permits any party to take "such measures on the high seas as may be necessary to prevent, mitigate or eliminate grave and imminent danger to their coastline or related interests from pollution or threat of pollution of the sea by oil, following upon a maritime casualty or acts related to such a casualty, which may reasonably be expected to result in major harmful consequences"[217]. The major purpose is to allow parties to resort to reasonable self-help in major oil spill situations that might attain emergency proportions. The principle of self-help on the high seas is not new, but this was the first international convention to apply it to the purposes of environmental protection[218].

In accordance with guiding principles[219] agreed upon at the International Legal Conference on Marine Pollution Damage held in 1969, a third convention was concluded, again at Brussels, in 1971: the International Convention on the Establishment of an International Fund for Compensation for Oil Pollution Damage[220]. Under this Convention the parties established the International Oil Pollution Compensation Fund to make supplementary awards to any person suffering pollution damage who has been unable to obtain full and adequate compensation under the Liability Convention of 1969. It should be noted that the International Fund may be wholly or partially exonerated from its obligations to provide relief, if it proves that, as a result of the actual fault or privity of the owner, the ship causing the oil pollution damage did not comply with the requirements laid down in specified conventions: the 1954 Oil Pollution Prevention Convention, as amended in 1962; the 1960 SOLAS Conven-

tion[221], the 1960 International Regulations for Preventing Collisions at Sea[222]; and the 1966 International Convention on Load Lines[223]. The Fund was designed, therefore, to operate as an external sanction of these earlier conventions[224].

Although IMCO's 18-point programme, initiated in 1968, was addressed to a wide range of hazardous and noxious substances, these treaty arrangements in 1969 and 1971 were concerned solely with oil. As awareness of the chemical hazard increased, it became necessary to assign a higher priority to the environmental dangers associated with the carriage of substances other than oil. This transfer of attention away from oil represented also an important change from the previous *remedial* approach to what would become an essentially *preventive* approach to this sector of environmental treaty-making. The process began outside IMCO with regional initiatives to bring the practice of dumping of wastes at sea under international regulation: first by a small group of Northeast Atlantic states at Oslo in 1971[225] and a larger number of states at Reykjavik in 1972[226]. This exercise culminated in the conclusion of a global convention, also outside IMCO, which was adopted in 1972 and kept open for signature by any state throughout 1973: the London Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter[227]. Designed along the lines of the Oslo and Reykjavik models, the London Convention established a "black" list of prohibited substances[228], which may not be dumped at sea by contracting states[229], and a "grey" list of somewhat less harmful substances and materials[230] "requiring special care", which may dumped, but only under a prior "special" permit[231]. The dumping of "all other wastes or matter" requires a prior "general" permit[232]. Although this Convention represents a sophisticated approach to the regulation of dumping at sea, its scope and effectiveness are reduced by the watering down of various provisions, which were discovered to be politically controversial[233]. Moreover, the applicability of the duty to implement measures under the Convention is still in question, pending the final outcome of UNCLOS III[234].

The final convention to be noted, the 1973 International Convention for the Prevention of Pollution from Ships[235], is certainly the most significant of all treaty arrangements in the prevention and control of ship-generated pollution, and the major contribution by IMCO to the cause of environmental management. Intended to supersede the 1954 International Convention for the Prevention of Pollution of the Sea by Oil, it provides a comprehensive framework for detailed regulations to be applied to the intentional discharge not only of oil but of any "harmful substance"[236]. The Convention applies not only to vessels of over 500 gross tons, but to vessels of any type, including hydrofoil and air-cushion vehicles, submersible and floating craft, and fixed or floating platforms[237]. Parties are required to enact and enforce measures to prohibit and punish violations by any vessel flying their flag, regardless of location; and, again without regard for the location of the violation, penalties are required to be sufficiently severe to have a deterrent effect[238].

Attached are five annexes, two mandatory and three optional. The first of the mandatory annexes[239] contains detailed rules for the prevention of pollution by oil[240]. The other mandatory annex[241] deals with "noxious liquid substances in bulk" (other than oil), and these are divided into four categories for purposes of differential treatment[242]. Also concluded was an important instrument, entitled Protocol Relating to Intervention on the High Seas in Cases of Marine Pollution by

Substances Other than Oil[243], to permit prompt "self-help" preventive action with respect to vessels carrying any of a wide variety of hazardous substances[244].

Like the other IMCO conventions the 1973 Convention is of limited applicability and subject to formidable problems of detection and enforcement[245]. Scientifically, there may be some difficulty in reconciling the *generic* approach to the annexes in the Dumping Convention and the *specific* approach in the IMCO Convention[246]. Moreover, it is still difficult to predict the acceptability of the 1973 Convention in the wake of UNCLOS III[247]. Yet, despite these flaws and uncertainties, it is likely to become a "classic" in the art of environmental treaty-making and a model to be studied in the years ahead, both inside and outside the context of marine pollution control.

Finally, three other effects of the *Torrey Canyon* grounding should be noted. The first of these is the important change in the British government's policy with respect to oil tanker accidents in the high seas. By amending legislation enacted in 1971[248], the Secretary of State for the Environment may, in sufficiently threatening circumstances, give directions for oil pollution prevention purposes and "require the person to whom they are given to take, or refrain from taking, *any action of any kind whatsoever"*[249] (emphasis added). The act also confers upon the Secretary of State the power "to undertake operations for the sinking or destruction of the ship, or any part of it, of a kind which is not within the means of any person to whom he can give directions" and "to undertake operations which involve the taking over of control of the ship[250]. This radical legislation, emanating from a normally conservative country in matters pertaining to the law of the sea, may be applied, in appropriate circumstances, to vessels of foreign as well as British registry[251].

The strength of the British reaction was significant, making it likely that other coastal states close to major shipping lanes, anticipating similar catastrophes, would react with similar special measures, even involving drastic unilateral action against foreign vessels outside the territorial waters of the coastal state. In the years following, the *Arrow, Argo Merchant,* and *Amoco Cadiz* disasters produced comparably dramatic changes of policy and law in Canada, the United States, and France, respectively[252]. In such a climate of expectations it became less difficult at UNCLOS III and other forums to present the case for a special right of self-protection on the part of vulnerable coastal states[253].

In conclusion, it should be added that the succession of tanker collisions and groundings after the *Torrey Canyon* together served to direct the world community's attention, and much of its diplomatic talent, toward the larger questions of marine pollution, not only in specific treaty-making contexts such as those reviewed, but also in more comprehensive forums such as the Stockholm Conference on the Human Environment and the U.N. Seabed Committee.

3. The Governability of Technocracy

Several general conclusions can be drawn from this review of responses to man-made hazards. First and foremost, mankind is facing an unprecedented volume and variety of environmental hazards, which together constitute a global threat of crisis proportions. Second, to the earlier scientific interest in the protection of nature, and the humanitarian concern for the welfare of animals, has been added a more gene-

ral and less altruistic anxiety about the future of the human species. Third, all these problems, in varying degrees of urgency, require effective action at the international level. Fourth, the effectiveness of preventive and remedial action depends, above all, on an understanding of the environmental impact of technology, old and new. Fifth, the impact of technology on the natural environment may be no less urgent than its impact on human society.

Interest in the impact of technology on human society is not new. Literature on this subject goes back at least as far as the writings of Karl Marx in Europe and of Thorstein Veblen in America. Since the Second World War — and for reasons associated with it — there has been a growing conviction that technology, generally defined, is the principal engine of social change, and that most of us now are ruled, in effect, by technocracy[254]. Some have taken the severe view that technology has become an autonomous force that is self-perpetuating, uncontrolled, and destructive of human values[255]. Others attack this attitude as much too negative, and ultimately self-defeating[256]. The ultra-pessimistic view ("technology is terrible") has been branded as "inadequate", along with the ultra-optimistic view ("technology is terrific") and the casual ("technology is unimportant")[257]. One prominent writer has argued that the world is moving through its "second great transition", from civilized to post-civilized (technological) society, and must learn to overcome its inability "to develop a genuinely stable high-level technology which is independent of exhaustible resources"[258]. Another, also stressing the importance of understanding the ecological consequences of policies, calls for environmentally sophisticated social planning, based on a new naturalism, holism, and immanentism[259]. Still another, in a famous essay[260], underlines "the tragedy of the commons" in selfish, densely populated society, urges the abandonment of individualism and *laissez faire,* and suggests the application of "mutual coercion mutually agreed upon" to natural resource planning, on the basis that "freedom is the recognition of necessity".

The impact of technology on human society has had the additional effect of raising doubts about the adequacy of orthodox modes of economic analysis[261]. A recent writer, calling for an "awakening from the technological trance"[262], goes so far as to condemn the discipline of economics — and, for that matter, the entire academic community — for creating a "reductionist" attitude to environmental and other modern problems of immense complexity, which are totally insoluble if viewed only within one intellectual compartment or another[263].

Indeed, without being unduly dramatic, it might be said that the new perception of environmental hazard, especially since the 1960s, has created an "intellectual crisis". Not only is this evident in the self-doubting sector of the literature of economics, but also in that of political science and international law. In recent years both of these disciplines have been increasingly concerned about the problem of "ungovernability" in human affairs. At one level, doubts have been expressed about the governability of modern industrial democracies. In the best known study[264], one author referred to "the upsetting of the intellectual world" as a "cultural crisis", as far as Western Europe is concerned[265]; and another referred to a decline in public trust, as far as the United States is concerned[266], and growing evidence that "(t)he public develops expectations which it is impossible for government to meet"[267]. All three authors, referring to "the rise of anomic democracy" and the modern phenomenon of

"consensus without purpose", admit that "no significant support has yet developed for any alternative image of how to organize the politics of a highly industrialized society"[268]. Reference is made to "the overloading of government"[269] and the extent of "parochialism in international affairs"[270] — matters of special concern in the difficult context of environmental problem-solving. It might be insisted that there is no relief in sight for governments committed to accept responsibility for the environmental impact of industrial technology. As one writer phrases it, while the technological revolution must "cease to be accidental", it must become "conscious, planned and democratic" through the political intervention of man[271].

Clearly, the problem of governability in western, indutrialized democracies is a special and advanced case of a more general disease afflicting all systems of government, at all levels of development, in the late 20th century. Inevitably, compliance with the rising demand for transfers of industrial technology will accentuate most of these difficulties in the recipient state, compounded by lower levels of education and technical skill. On the face of things, the most feasible approach to the problem of the overloading of national government is through various modes of international co-operation. Yet this may simply transfer the overload problem to a higher, and sometimes less effective, level of human organization, and create impossible expectations of the international legal system.

The record of international law and regulation in dealing with man-made hazards is not entirely dismal. In truth, it is a record of reasonable accomplishments, at least in those sectors where environmental concern can be presented credibly in terms of crisis and potential catastrophe. Fear is still the strongest emotion. Although it is possible to exaggerate the benefits in inducing nation states to subscribe to paper promises, the volume of environmental treaty arrangements on the record is not unimpressive, especially in the context of ship-generated pollution control, but even in the highly sensitive area of control over environmentally dangerous (nuclear) weapons.

However, some sobering reflections must be offered. First, the knowledge *of potential* environmental hazard, or the suspicion of *actual* hazard, may not be enough to instill the degree of fear needed to precipitate international action and co-operation. Second, conservation problems which cannot be perceived as endangering human life, or otherwise directly threatening to human welfare, may have difficulty in attracting the attention of national governments on the overloaded circuit of conference diplomacy. Third, the international legal system has a chronic problem of credibility in the eyes of self-styled "realists"[272]. Fourth, most of the more serious environmental problems have their origin in human activities that occur in areas under national jurisdiction and prerogative, and therefore enjoy a degree of license commensurate with the "pride of sovereignty" and the level of capability prevailing in the national political culture. Fifth, even in international areas left outside extended limits of coastal state jurisdiction[273], developing states are understandably reluctant to accept international standards and other kinds of obligations that will increase the cost, or retard the process, of economic development.

On balance, it is certainly possible, and probably wise, to take a pessimistic view of the prospect of governing technocracy through international law and regulation, but the recent record of the world community in responding to man-made hazards does provide a basis for realistic expectations of continued developments in international environmental law, at least in areas that are seen as threatening to human welfare. Since the oceans cover so much of the planet's surface, and constitute the most significant sector of its "international areas", we should expect these future developments to be profoundly affected by environmental concepts in the law of the sea.

D. Protection and Conservation of the Marine Environment (1945-1972)

1. Introduction: The Growth of Ocean Technology and Science

The history of ocean technology is one of dramatic endeavour by many cultures. The fisherman, the sailor, the architect, the engineer, the miner, the warrior: all have played their part[274]. Even in an age of engineering marvels, the sea still holds a romantic appeal for most of us on land. But for the users of the sea today, the ocean is mostly business, commanding the highest level of technical efficiency that one can muster[275].

Through the ancient skills of surface navigation and fishing, our ancestors contrived to make use of the sea for adventure, conquest, colonization, transport, and the harvesting of food. Boat fishing was practised in coastal as well as inland waters thousands of years before the introduction of exact navigation. Yet fishing remained wholly innocent of science until the late 19th century, when it seemed that new industrial techniques of capture might threaten the supply of fish[276]. Cargoes were carried by sea, beyond sight of land, long before the time of the Phoenicians. Yet the methods of navigation remained essentially the same until the 19th century breakthrough to steam[277]; and radical innovations in the handling of cargoes had to wait for the second half of the 20th century[278].

Overall, we might be excused today for judging the development of vessel technology as slow and uneventful for thousands of years, up to the Second World War, compared with the dramatic innovations of the last three decades[279]. However, none of these post-war improvements in vessel construction and design is more significant than the emergence of platform technology in the 1930s. The arrival of oil drillers in the offshore has probably done more than any other event to draw public attention to the ocean. The combination of these new capabilities through fixed as well as mobile technology has opened up a totally new vista of contemporary and prospective uses of the sea: in food production, mineral extraction, transport and communication, military security, weather forecasting and modification, scientific investigation, storage and disposal, power generation, recreation and therapy, and even residence[280]. As appetites for protein increase, we look to imaginative new methods of fishing[281], and in several imminent ocean projects, such as the use of ocean thermal energy conversion[282] and the mining of ferromanganese nodules[283], we see the promise of continued efforts to extend man's ingenuity in all dimensions of ocean space.

Unfortunately, these same exciting innovations give rise to profound anxieties. Can the ocean withstand this onslaught[284]? Are sufficient environmental safeguards being taken in the course of these new uses? Above all, do we know enough to act responsibly as the guardian of the ocean environment?

Environmental sophistication is normally presumed to rest upon an understanding of the relevant natural sciences and the adoption of an ecological perspective. Despite the importance of appreciating the nuances of environmental politics and policy-making — and, therefore, the techniques of the social sciences — it is difficult to deny the primacy of natural science and ecology in the process of environmental management. The role of science is no less crucial in the particular context of "ocean management".

It is commonly said that oceanography is new and international. Neither statement is entirely correct. Investigations in hydrography, bathymetry, and other oceanographic fields have been conducted for centuries[285], and most of the early scientific expeditions — such as the *Beagle* voyage (1831-36), the first Ross expedition to the Antarctic (1841), and the *Challenger* voyage (1870-72)[286] — were essentially national , not international, undertakings. Yet, oceanography, as an integrated science, is a relatively recent phenomenon: a large-scale and expensively mounted amalgamation of four distinct areas of investigation (physical oceanography, chemical oceanography, geology and geophysics, and marine biology)[287]. It is also true that the major centres of oceanographic research today tend to attract scientists of many different nationalities, cultivating the image of an international undertaking, albeit under the influence or control of the host government. This may point to a paradox: as many of the ocean areas of primary scientific interest are being divided into national compartments, under expanding regimes of coastal state jurisdiction, the teams of scientists engaged in such research seem to be becoming increasingly international in composition.

Of particular environmental importance is, of course, the rise of marine biology and chemical oceanography, and the eventual emergence of fishery science and marine ecology. The departure of *The Challenger* on its famous voyage in 1870 is often taken as the true beginning of oceanography, but in earlier decades it had been the practice of national governments to appoint naturalists to survey ships: such as Darwin to *The Beagle* and Hooker to the Ross expedition to Antarctica[288]. By the end of the 19th century the interests of marine biology and chemical oceanography were almost invariably represented on major scientific expeditions at sea.

By that time, moreover, a variant of the species of marine biologist, the fishery scientist, had evolved in response to a very practical concern about the need to check the depletion of commercial fishery stocks and to institute scientifically informed policies of fishery conservation and management. By the early 1950s the voice of biology had become dominant in the formation of fishery conservation objectives and methods[289]. When preparations were being made for the first United Nations Conference on the Law of the Sea (UNCLOS I), it was assumed to be natural and desirable that biological considerations should be predominant in the formulation of international fishery rights and obligations[290]. It should be remembered, however, that ecology was not quickly accepted as a legitimate member of the family of science, and that as recently as the 1950s ecological concepts were ridiculed

in the context of marine biology and fishery science[291]. But the ecological perspective proved to be compelling in the 1960s, in an age of mounting environmental concerns, which seemed to yield to solution only within spatially extensive, functionally integrated frameworks of analysis. In many cases, the ecological perspective on fishery management policy proved attractive to national governments anxious to secure a scientific rationalization for claims to extended fishery or multi-functional jurisdiction.

2 Changing Attitudes, Institutions, and Concepts (1962-1972)

These dramatic developments in ocean science and technology in the post-war period can be seen in retrospect as the initial impetus to the development of a modern enviornmental law of the sea, which was to emerge in the 1970s. Looking back, we can see that the UN Conference on the Human Environment, held at Stockholm in 1972, was a crucial preparation for these legal developments, and that the Stockholm Conference was itself preceded by an all-important 10-year period of "incubation".

New factors which served to underline the need for more effective protection and conservation of the marine environment took several forms: attitudinal, institutional, and conceptual. Special reference should be made to one example of each of these three types of changing perspectives: the advent of environmentalism, the rebirth of IMCO, and the Pardo vision of ocean space.

a) The Advent of Environmentalism

It was not until the 1960s that "environment", "pollution", "ecology", "quality of life" and related issues became matters of public and political concern. The emergence of environmentalism as a concept, as a mood, as a perspective[292] — but especially as a cause — owed much to the "moral entrepreneurial activity"[293] of a handful of individuals such as Rachel Carson. Almost immediately her *Silent Spring* succeeded in arousing public and official anxiety about the effects of pesticides, and by the end of 1962 over forty bills had been introduced, in U.S. state legislatures alone, to regulate their use[294]. Within a few years the environmental movement had been joined by social reformers such as Paul Ehrlich[295], who interpreted environmental problems as evidence of a deep-rooted social malady that could not be cured without fundamental structural change[296]. By the end of the 1960s the environmental movement had become a heterogeneous coalition of groups with diverse interests and values[297], but each capable of exerting some influence on the political system. Juridically, it was perceived that out of the original concept of resource conservation a much larger framework of environmental law must be developed, within which it might be necessary to re-examine the fundamental concepts of private property rights[298].

Most of these groups focussed entirely on local environmental issues, which they could reasonably hope to influence through political pressure. A few of the larger groups, however, were beginning to monitor governmental planning at the national and international levels, where preparations were being made for the United Nations Conference on the Human Environment, scheduled to be held in Stockholm in 1972. Yet even at this stage, the environmental groups, unlike a growing number of national governments, had relatively little interest in the problems specific to the

ocean environment. Rachel Carson, herself a marine biologist by training, gave surprisingly little attention to environmental protection problems in her earlier works on the sea: *The Sea Around Us* (1950) and *The Edge of the Sea* (1955). In the preface to the second edition of *The Sea Sround Us,* published in 1961, she drew attention to the dangers associated with the marine transportation of nuclear wastes[299], but the text remains essentially an introduction to marine biology.

b) The Re-Birth of IMCO

In the crisis atmosphere engendered by the *Torrey Canyon* catastrophe of 1967 the Intergovernmental Maritime Consultative Organization underwent a change of personality that was virtually metamorphic[300]. The special *ad hoc* Legal Committee established by the Council in May 1967 evolved quickly into a permanent feature of international organization and the most important legal mechanism for dealing with marine pollution within the U.N. family. The Legal Committee's responsibility in matters of "public law", such as the right of intervention on the high seas, was quickly conceded, and eventually, after acrimonious debate, the resistance of traditional commercial interests to its involvement in "private law" issues, such as liability and compensation, was also beaten down[301]. The Subcommittee on Oil Pollution was re-named the Subcommittee on Marine Pollution (SCMP) in 1969, so as to reflect its expanding concerns, but in 1973, with the pollution control priority still rising within the Organization, the SCMP was replaced with a new body, the Marine Environment Protection Committee (MEPC). Unlike its predecessors, which were answerable to the Marine Safety Committee (MSC), the MEPC was made a permanent subsidiary body of the Assembly itself, and under subsequent amendments to the IMCO Convention it was given formal equality with the MSC. Henceforward, IMCO has operated as a dual agency, equally concerned with its original mandate for the promotion of maritime safety and with its new responsibility for pollution prevention and control[302].

Thus "born again", with a split personality, IMCO now lends itself to some degree of comparison with its intergovernmental counterpart in aviation, the International Civil Aviation Organization (ICAO), and like that agency might be expected to develop a limited measure of regulatory responsibility[303]. Similarly, IMCO's symbiotic relationship with the Comite Maritime International, the private industrial-legal advisory body, might be analogized with ICAO's relationship with the International Air Transport Association.

The re-birth of IMCO, as a potential regulatory agency in the field of marine pollution control, has drawn attention to larger questions about the adequacy of existing international organizations for the purposes of environmental protection in general[304]. The fact that IMCO could undertake initiatives only in the area of ship-generated pollution, and that this area represented only a small part of the threat to the ocean environment, underlined the need to establish a new environmental agency with much broader terms of reference. In 1973, in the recommendation of the Stockholm Conference, this need was met with the establishment of the United Nations Environment Programme. Significantly much of UNEP's attention since 1973 has been directed at the special problems of preserving the ocean environment[305].

c) The Pardo Vision of Ocean Space

In his famous speech before the U.N. General Assembly, on November 1st, 1967, Ambassador Arvid Pardo of Malta spoke eloquently, from a universal perspective, on the need for the radical restructuring of the law of the sea[306]. What he envisaged was a legal revolution which would match the technological revolution taking place on the seabed and beginning in the superjacent waters. Pointing to the inadequacy of the old law of the sea, based on the arbitrary and clear-cut distinction between the "sovereign" areas of the territorial sea and the "free" areas of the high seas, he stressed the ecological unity of "ocean space"[307] and the real-life interactions between all areas and all uses of the marine environment. A new concept, the common heritage of mankind, was put forward as a replacement of the classical principle of the freedom of the high seas[308].

At first, practical considerations compelled Pardo to limit his conceptual and organizational proposals to the seabed, but on March 23rd, 1971, he delivered another major address proposing a new legal order for open space as a whole, one in which the overlying and overlapping boundaries of existing regimes would be abandoned in consonance with the fundamental unity ot the natural order of the ocean[309]. Throughout this speech Pardo repeatedly emphasized the vulnerability of the ocean environment under the impact of modern technology. Two years later, on March 14th, 1973, he dealt with the many pollution hazards confronting the oceans and urged that the forthcoming Law of the Sea Conference should adopt a comprehensive and ecologically sophisticated approach to the formulation of state responsibility in the ocean[310]. In the following month he advocated that scientific research in ocean space should be regarded as a world public interest entitled to special protection under the new law of the sea, albeit subject to a measure of non-discriminatory international regulation[311]. A few weeks later he expressed concern, on procedural grounds, with the way in which IMCO's pollution control responsibilities had grown[312], and continued to press for a broader conceptual and organizational approach to the preservation of the marine environment.

3. Pre-Stockholm Approaches to Marine Conservation

a) Introduction

By the end of the 1960s the world community was sufficiently aroused by mounting concerns to look for a new beginning to the preservation of the environment. At the very least, the Stockholm Conference on the Human Environment was seen as an opportunity to collate much of what was known about actual and potential threats to the biosphere[313]. In reviewing the range of measures that might be attempted to combat these threats, it was generally recognized that many should focus on the problem of preserving the marine environment, since ocean space represented over 70 percent of the surface of the planet and consisted largely of "international areas"[314].

Many of the participants preparing for Stockholm, moreover, were aware of the potential importance of legal development in this effort[315]. Other forums, it was known, were also contributing to the development of legal norms, institutions, and procedures for environmental protection and management. Special significance was attached to current legal developments for the prevention and control of marine pollution under IMCO and other auspices[316]. Some of the delegates were

even prepared to advocate that the United Nations should consciously adopt a synchronized strategy for developing a new global environmental law of the sea, through a variety of exercises in conference diplomacy dircted essentially at law reform. It was agreed, however, that the Stockholm Conference itself should not attempt a treaty-making initiative. Instead an attempt would be made to articulate the legal, institutional, and other requirements, at all levels of government, for the protection and conservation of the "biosphere", including, of course, the ocean environment. The Conference would, therefore, be essentially a stock-taking of what had been learned about conservation and related requirements.

In ocean affairs, the pre-Stockholm approaches to conservation might be divided into two categories: the conservation of marine species, and the prevention and control of marine pollution.

b) Conservation of Marine Species

This category of concerns encompasses what used to be called the conservation of nature, as applied to the ocean: that is, the protection of ocean fauna and flora. In more modern language, it includes the conservation of fisheries and the protection of marine mammals; the preservation of endangered marine species; the maintenance of ecological balance in the ocean through a variety of techniques, such as the protection of spawning grounds and other habitats of special importance; and the establishment of marine areas of special protection, such as nature reserves, sanctuaries, and parks.

As we have seen, in the period 1815 to 1945 the movement for nature conservation was international in scope, but the development of intergovernmental measures was very gradual and tentative. These measures dealt mostly with the protection of wildlife and, to a lesser extent, plant protection and the control of animal and plant diseases. After 1945 naturalists had continued the struggle for governmental recognition of their cause with new support from some of the U.N. specialized agencies, such as UNESCO and FAO. Whereas the latter's interest was chiefly in the practical problems of natural resource management, not least those of fishery conservation[317], UNESCO's scientific orientation embraced almost the entire range of traditional concerns for the conservation of nature[318].

With a view to promoting the cause of conservation at the international level, the first Director-General of UNESCO, Julian Huxley, was instrumental in establshing the International Union for Conservation of Nature and Natural Resources (IUCN)[319]. This organization was intended to monitor, co-ordinate, and promote intergovernmental action in the field of conservation, somewhat on the model of the Nature Conservancies, which Huxley and his scientific colleagues had recently persuaded the British government to establish: one for England and Wales, and another for Scotland[320]. From its inception, however, IUCN had to struggle with a seriously deficient budget, and for most of its scientific input it had to depend heavily on the voluntary services of biologists and other conservationists scattered around the world[321]. Finally, it was realized that the organization's efforts would have to be supported by a separate funding agency, and to this end the World Wildlife Fund (WWF) was established in 1961. As its name implies, however, the Fund was envisaged as being chiefly involved in the traditional, and somewhat restricted, area of nature

conservation[322], rather than equally in all sectors of the larger, and rapidly expanding, field of conservation.

Up to the 1960s most of UNESCO's interest in conservation continued to be channeled through non-governmental organizations like IUCN and WWF and the International Biological Programme (IBP)[323]. In 1968, however, UNESCO was provided with an intergovernmental infrastructure, the Man and the Biosphere Programme (MAB)[324]. Yet even in 1972, despite these improvements, underinvestment continued to plague the cause of nature conservation, even though ecologists now knew that the misuse of the natural environment was reaching critical proportions. As one writer observed, "(t)he wholesale destruction of natural habitats and of wild animal and plant species by man is a very recent phenomenon in human history"[325]. Until recently the "species that were brought to extinction were not numerous and mostly consisted of small isolated populations which were especially vulnerable to hunting .... With the industrial revolution of the 19th century and the colonization by European settlers of vast territories on all continents, the pace of the process of destruction and extinction began to increase. Recent trends of rapid worldwide population growth, uncontrolled economic development and technological advances have contributed, in the few decades elapsed since the end of the second World War, to a tremendous acceleration in the destruction process"[326].

Moreover, most of the evolving international measures for the protection of species and ecosystems were exclusively, or mostly, terrestrial in focus: the 1950 International (Paris) Convention for the Protection of Birds[327]; the 1971 (Ramsar) Convention on Wetlands of International Importance, Especially as Waterfowl Habitats[328]; the 1972 (UNESCO) Convention on the Protection of the World Natural and Cultural Heritage[329]; the 1973 (Washington) Convention on International Trade in Endangered Species of Wild Fauna and Flora[330]; the 1973 (Oslo) Convention on the Conservation of Polar Bears[331]; and other conservation treaties being drafted in the early 1970s[332]. In the context of ocean management, the cause of conservation, at the time the Stockholm Conference was convened, was almost entirely limited to that of fishery conservation and management.

As we have seen[333], the history of international co-operation in fishery conservation can be traced to the late 19th century. It was not until after 1945, however, that this trend in international affairs became a worldwide phenomenon, and not until the mid-1950s that fishery management developed into a permanent feature of international organization. Fishery issues had become recognized by then, in most nations, as an important, and sometimes exceedingly contentious, area of international diplomacy, and few governments were prepared to deny the need, at least in principle, for co-operation in the cause of fishery conservation on the high seas.

The primary reason for this growing interest in ocean fisheries varied, of course, with the national situation: the need for increased food production; the problem of unemployment or under-employment in coastal communities; the opportunity to earn foreign earnings through fishery experts; and the desire to instill a sense of autonomy or self-sufficiency in the political culture through local initiative[334]. In the post-war years of reconstruction and decolonialization an enormous diversity of techniques and technologies emerged within the world fishing industry. Of these the most destructively efficient were seen to be introducing an unprecedented level

of actual menace to the most actively exploited stocks. Indeed the general trend toward capital-efficient, industrial fishing was perceived by some to have potentially disastrous implications for the self-renewal of the ocean's living resources[335].

On the other hand, most of the newer, developing nations were, understandably, less immediately concerned with the threat of depletion than with the prospect of losing an opportunity to share, more equitably than in the past, in the available stock of wealth and protein off their shores[336]. As the developmental significance of fishery resources became more evident to the developing coastal states, the objectives of fishery regulation and management became even more complicated and more controversial than those of stock conservation.

Ironically, this growing confusion over the objectives of fishery regulation and management was scarcely evident in the 1950s, when the United Nations made its first systematic effort to codify and develop the international law of the sea. Relatively few of the developing nations had become independent by 1958, when UNCLOS I convened at Geneva, and of these only a handful had any appreciable influence on the course of the Conference. For one thing, preparations for UNCLOS I had begun as early as 1950, when the International Law Commission of the U.N. General Assembly was requested to begin the preparation of draft articles on the regimes of the high seas and territorial sea. The use of this body of independent jurists, mandated to "initiate studies and make recommendations for the purpose of .... encouraging the progressive development of international law and its codification"[337]; ensured that more attention would be given to the incremental and academic process of developing the science of international law than to the expanding, disparate, and often conflicting, objectives of ocean management. By the time the I.L.C.'s draft articles were ready for injection into the diplomatic forum, there was little chance that the delegations, confined to a single session of nine weeks[338], would undertake a major revision of the text.

The relative impotence of the LDC delegations at UNCLOS I was especially evident in the context of fishery management provisions. The I.L.C.'s fishery articles were heavily influenced by scientific advisers, most of whom approached fishery management from an exclusively biological perspective[339]. Fishery economics, it should be recalled, was still an infant science (or quasi-science) in the 1950s and the few sociologists with an interest in fishing communities were totally estranged from the process of conference diplomacy. It short, only a handful of the LDC delegates had the scientific credentials to challenge or correct the biological bias of the fishery articles prepared for their consideration.

In the Convention on Fishing and Conservation of the Living Resources of the High Seas[340], which was opened for signature at UNCLOS I[341], the term "conservation" was defined as "the aggregate of the measures rendering possible the optimum sustainable yield from those resources so as to secure a maximum supply of food and other marine products"[342]. This formulation — a biologically determined concept of maximum sustained yield recommended by a group of scientists — was attacked by many biologists, as well as all economists, as an unrealistic goal for regulation[343]. Yet it passed the scrutiny of the I.L.C. and UNCLOS I without serious criticism, apparently on the ground that it was a technical matter best left to experts.

The heart of the biological system of fishery conservation approved at UNCLOS I lay in the notion that all conservation measures should be based on scientific *(i.e.,* biological) findings[344]. For purposes of effective management and regulation, the Geneva system placed faith in the principle of negotiation[345], and therefore in the potential utility of regional and subregional co-operative arrangements[346]. A principled balance was struck between the prerogative of the coastal state, having "a special interest in the maintenance of the productivity of the living resources in any area of the high seas adjacent to its territorial sea"[347], and the obligations of all states to adopt "such measures for their respective nationals as may be necessary for the conservation of the living resources of the high seas"[348]. Provision was made for the submission of disputes arising out of the Convention to a special commission to be established for that purpose, unless the parties to the dispute agreed to an alternative method of settlement[349]. Disputes were to be adjudicated in light of the requirement that "scientific findings demonstrate the necessity of conservation measures"[350].

The Geneva system of international fishery conservation was intended, then, to be rational and equitable, but in the tradition of biologically inspired experiments which had been pursued since before the Second World War, and primarily within the framework of regional fishery conservation agreements which had evolved in the post-war years. After UNCLOS I there was no immediate prospect that the Geneva system would fail. Within a few years enough instruments of ratification had been deposited to bring the instrument into force[351], and in the years between 1958 and 1969 no less than nine new international fishery commissions came into existence[352], faithful in large part to the ideals and objectives of the Geneva system.

By 1972 the total number of intergovernmental fishery conservation commissions had risen to two dozen, covering virtually every area of the world's ocean and almost all species that could be said to have a significant commercial value[353]. A few, especially of the older ones, were bilateral or trilateral arrangements[354], but the majority were regional or subregional in scope. Of those established since 1945, half were either designed as part of a global system of regional arrangements under the auspecies of FAO[355], or at least the result of advocacy by FAO, albeit initiated outside the U.N. structure[356].

Perhaps the most useful method of classifying these two dozen fishery bodies is by reference to their *primary function,* as reflected in the language of the constitutive treaty or charter. Four could be said to be limited to *scientific* purposes: the collection of data, the exchange of information, the promotion and co-ordination of research, and similar activities[357]. Others are designed to serve a broader range of *advisory services,* both scientific and non-scientific, with a view to assisting the member governments with their policy-making and management responsibilities[358]. Two others are simply *consultative* mechanisms to facilitate the co-ordination of the fishery development and management policies of neighbouring countries[359]. The remaining 15 bodies might be described as having a *managerial* function of one kind or another. Most are equipped to collect scientific and statistical data for the express purpose of designing and implementing a joint conservation policy. Most of the commissions are limited to making management recommendations to the member governments; only a few have regulatory powers vested in them directly[360].

Some focus solely on the conservation of a single species[361]; others address themselves to the conservation of most or all the commercial species of a designated geographical area[362].

Whatever the differences of purpose and method among them, virtually all of these fishery bodies were conceived as necessary and desirable mechanisms for international co-operation in the conservation of the living resources of the high seas. All, without exception, were brought into existence at a time when an overwhelmingly large proportion of the ocean was conceded to belong to the classical regime of the high seas, under which the tasks of fishery regulation and management could simply not be discharged except through the processes of international co-operation. Without such co-operation, it was assumed, the major ocean fisheries were likely to remain unregulated, and the management of them would be fragmentary and unconcerted at best. The alternative, of national conservation and management beyond a narrow territorial sea, would have to await fundamental legal change at UNCLOS III and in state practices around the world[363].

c) Prevention and Control of Marine Pollution,

Reference has been made to important treaty developments initiated by the shock and alarm over a succession of dramatic oil spillages at sea[364]. Although the term "catastrophic" is often employed to characterize the consequences of tanker groundings and collisions, specialists know that oil is not by any means the most serious pollutant in the ocean, either in kind or in volume[365]. Indeed, except for wildlife in the immediate vicinity of a spillage, oil may be one of the least harmful of ship-generated wastes. Moreover, the volume of oil deposited in the ocean has been diminishing substantially in recent years as a result of a combination of factors: increasingly stringent regulations both of the preventive and remedial sorts; technical improvements in the construction and design of vessels carrying oil as cargo or fuel; the emergence of shipping control systems to reduce the problems of congestion and raise the standards of navigation; more scrupulous adherence to modern tank cleaning procedures at sea; increased efficiency in surveillance and monitoring and a resulting rise in the number of successful prosecutions; a significant reduction in the total tonnage of loaded tankers at sea; and, with the seven-fold increase in the price of oil over a six-year period, a marked reluctance by carriers to waste their precious cargo by deliberate discharges[366]. For similar reasons, substances other than oil — and often more harmful than oil — were about to be brought under international control through two major agreements, which were in the process of preparation or negotiation at the time of the Stockholm Conference: namely the London Dumping Convention (1972)[367] and the IMCO Marine Pollution (MARPOL) Convention (1973)[368].

4. The Stockholm Conference on the Human Environment (1972)

The U.N. Stockholm Conference on the Human Environment was held at Stockholm in June 1972, attended by representatives of 113 member states, of various U.N. bodies, most of the specialized agencies, the International Atomic Energy Agency, and the General Agreement on Tariffs and Trade (GATT), and by observers from intergovernmental and non-governmental organizations.

It was one of the largest United Nations conferences ever assembled, and the preparations for it, begun four years earlier, were on an unprecedented scale[369]. The Conference adopted a Declaration setting forth 26 principles for the preservation and enhancement of the human environment[370], and an Action Plan consisting of over 100 recommendations for environmental action at the international level[371]. In December 1972 the U.N. General Assembly adopted 11 resolutions concerning various environmental matters; established the United Nations Environment Programme (UNEP); and took decisions on a variety of environmental issues (International co-operation on environmental problems, measures to protect the environment, development and environment, human settlements, and other matters[372]).

Despite the years that have passed since 1972, it is still premature to make a firm assessment of the institutional and legal significance of the Stockholm Conference. Its institutional impact has to be measured by reference to UNEP's continuing efforts to implement parts of the Action Plan[373]. Its legal impact depends on a somewhat subjective interpretation of the Stockholm Declaration and of U.N. resolutions in general[374].

The drafting of the Declaration in committee proved to be an arduous process, and even at the Conference proper, and in the General Assembly debate that followed several delegations expressed their objections to some of the provisions[375]. The final version reflects the dual purpose of the document, containing on one hand, in its Preamble, the general, hortatory language of an inspirational address to "the peoples of the world", and on the other, in a set of Principles, the rather more specific, operational language of guidelines for official action. The constant tinkering with the text, it must be said, sadly diminished the quality of its inspiration and the clarity of its guidance. At least one basic principle appears only in the Preamble: the principle that "(t)he protection and improvement of the human environment is .... the duty of all Governments"[376]. Yet it is to the 26 Principles that one must look as a basis for the future development of international environmental law[377].

Juridically, the most important components of the Stockholm Declaration are Principles 21 and 22, and the most compelling of all is the second part of the former. *Principle 21: "States have, in accordance with the Charter of the United Nations and the principles of international law,* the sovereign right to exploit their own resources pursuant to their own environmental policies, and *the responsibility to ensure that activities within their jurisdiction or control do not cause damage to the environment of other States or of areas beyond the limits of national jurisdiction"* (emphasis added).

The underdevelopment of the principle of liability and the duty to compensate, and the need to correct these deficiencies in international environmental law, are clearly reflected in the following principle.

Principle 22: "States shall co-operate to develop further the international law regarding liability and compensation for the victims of pollution and other environmental damage caused by activities within the jurisdiction or control of such States to areas beyond their jurisdiction."

Assessments of how quickly and adequately these principles of international environmental law will evolve in the 1980s must be made in the light of new thinking about the concept and technique of "legal development" at the international level. Above all, it seems necessary to dispel the old idea that international law is created only through solemn, formal acts of consent couched in the form of hard promises of dispositive clarity, in the manner of a commercial contract. More normally, the principles of international law — not least in areas of wide-ranging environmental significance — will be developed in the tradition of constitutional, rather than contractual, drafting, inspired by the purposes of law reform as much as by those of legislation.

What must be expected is a continuation of difficult and protracted exercises in conference diplomacy, ranging from norm-setting forums such as the Stockholm Conference on the Human Environment to law-creating forums such as the Third U.N. Conference on the Law of the Sea. However much they vary in focus and purpose, these "constitutional" forums will be attended by lawyer-diplomats alert to the juridical uses of "soft law" pronouncements in the form of recommendations, declarations, guidelines, criteria, standards, and other quasi-obligatory norms. In favourable circumstances this new breed of lawyer-diplomats, living in a "romantic" not a "classical" age of legal history[378], can, and sometimes do, exceed the modest juridical limits set by the original idea that brings them together[379]. In this light it might be realistic to view the 26 principles of the Stockholm Declaration as "common convictions", which "collectively create a new atmosphere for international co-operation", so that "this first step toward the establishment of international environmental law on a firm foundation might prove to be more decisive than originally anticipated"[380].

E. Conservation Prospects at UNCLOS III (1973)

1. The U.N. Seabed Committee (1968-1973)

The Third United Nations Conference on the Law of the Sea (UNCLOS III) evolved out of six years of intensive preparatory discussions organized under the auspices of the U.N. Seabed Committee[381]. The environmental work of this Committee[382] was influenced, in one degree or another, by all of the events and considerations discussed above. Not least, however, it was affected, like the other work of the Conference, by the trend toward extended coastal state jurisdiction. These impacts can be discussed conveniently under four subheadings: the legacy of Geneva law; the perceptions of hazard and crisis; the need for conceptual development; and the rise of the coastal state.

a) Legacy of Geneva Law

At Geneva in 1958 the First U.N. Conference on the Law of the Sea (UNCLOS I) resulted in the conclusion of four conventions: the Convention on the High Seas[383]; the Convention of the Territorial Sea and the Contiguous Zone[384]; the Convention on the Continental Shelf[385]; and the Convention on Fishing and Conservation of the Living Resources of the High Seas[386]. Since the main thrust of its work in 1968 was at new issues related to the sea-bed beyond limits of national jurisdiction, the *Ad Hoc* Committee devoted little attention to the applicability of any of these four sectors of "Geneva law", other than emphasizing the inadequacy of the definition of the

shelf contained in Article 2 of the 1958 Convention on the Continental Shelf. As its scope of interest expanded, however, the Seabed Committee from 1969 onwards was forced to give increased consideration to all four Geneva conventions[387]. For environmental purposes, the main interest lay in these areas of "Geneva law": the freedom of navigation (and the licence to pollute?) on the high seas[388]; the freedom of fishing (and the licence to disregard conservation requirements?) on the high seas[389]; and the freedom to conduct scientific research on the high seas[390].

When the scope of the Seabed Committee's work was expanded, at the request of the U.N. General Assembly, in the fall of 1970, fishery conservation problems were assigned to Sub-Committee II, but most of the other environmental issues were concentrated in Sub-Committee III. At the beginning, in March 1971, the chairman of Sub-Committee III asked: "Should we think in terms of drafting articles for insertion in the four Geneva Conventions, which are open to revision — articles which would crystallize the substance of more specialized conventions prepared in other bodies; or should we think in terms of drafting a separate convention concerning the marine environment and scientific research[391]?" In the discussion that followed[392], most delegations took the view that a fresh start was needed, not least because of the fragmentary nature of the environmental provisions in the Geneva Conventions on the High Seas (Articles 24 and 25) and the Continental Shelf (Article 5). Some also pointed out that the Geneva Conventions, like the 1954 and 1969 IMCO conventions on oil pollution, were based on flag-state jurisdiction and did not sufficiently protect coastal states likely to suffer damage from marine pollution[393]. In the fishery discussions in Sub-Committee II in the summer of 1971 there was less explicit reference to the Geneva law of fishing, but the growing demand for increased coastal state control over resources, both living and non-living, seemed to guarantee that the fishery provisions negotiated at UNCLOS III would be far different from those contained in the 1958 Fishing Convention[394].

b) Perceptions of Hazard and Crisis

It is safe to assume — albeit difficult to document — that memories of the *Torrey Canyon* and the prospect of recurring catastrophies had a strong influence on the minds of delegates at the U.N. Seabed Committee and UNCLOS III. Nothing in the conference record is clearer than the determination of Sub-Committee III and the Third Committee to make a major effort to control ship-generated pollution, even at the risk of neglecting other important environmental issues and dangers. The concentration on vessel-source pollution was assisted by the fact that it presented less sensitive political difficulties, for those obsessed with the prerogative of national sovereignty, than other sources of marine pollution more directly linked with land-based activities. Developing countries, nervous about the short-term economic costs of rising environmental standards, were persuaded that most of the direct costs of controlling ship-generated pollution would fall, at least initially, on the shipping industry of the developed nations, and only indirectly and less immediately on themselves. The pre-occupation with shipping-related causes of pollution seemed appropriate at a law of the sea conference, especially since the foundation for an international approach had already been built in earlier treaties under IMCO and other auspices. Moreover, more than the other causes of the problem, shipping lent itself to a global approach at a forum such as UNCLOS III, and regional diplomacy seemed to have an auxiliary value at best. Hazard and crisis, then, con-

stituted a precipitating factor in the environmental work at the Seabed Committee — one which would constantly reassert itself, as news of other shipping casualties broke upon the conference[395] — but other influences were at least as persuasive.

Hazard and crisis also had an impact on delegates debating provisions on peaceful uses of the ocean. In 1970, for example, the Committee's apprehension about the urgency of pollution problems was sharpened considerably by reports of U.S. government practices in the dumping of nerve gas in the Atlantic Ocean[396]. Generally speaking, however, concern about environmental hazards associated with military activities at sea was gradually transferred to other forums, and alleviated in some degree by the kinds of treaty arrangements reviewed above[397].

c) Need for Conceptual Development

It is important to note than the Seabed Committee's decision to adopt a relatively comprehensive approach to marine pollution had formed in the summer of 1970, and therefore *before* the U.N. General Assembly's request that the Committee broaden its mandate in order to prepare for a full-fledged conference on the restructuring of the law of the sea. It needs to be emphasized that a holistic approach was not imposed upon the Conference; it came from within the Seabed Committee, despite its awareness of developments in other forums and of the need to avoid duplication of diplomatic effort[398]. Partly this holistic philosophy was the result of advocacy by a few enthusiastic delegations, which saw the forthcoming forum of UNCLOS III as an opportunity to promote the further development of international environmental law after the contributions of various IMCO treaty arrangements and of the Stockholm Conference[399]. Partly it was due also to the growing conviction of the United Nations Secretariat that UNCLOS III could play a major role in the reform of the existing international machinery for environmental as well as other reasons[400].

Despite the Committee's acceptance of the need for conceptual development of international environmental law through a comprehensive approach to the problems of marine pollution, it had become clear before the convening of UNCLOS III proper that the Conference would have to settle for something less than an overarching framework for effective international treatment of all environmental aspects of ocean management. Whether such an undertaking was ever feasible, given the constraints and demands of an increasingly ambitious law-making conference, may be debated, but by 1973 there was little doubt that the chief contribution of UNCLOS III to the environmental law of the sea would be limited to the area of ship-generated pollution, and that major contributions to the other facets of the problem would have to be left to other forums at global, regional, and national levels.

d) Rise of the Coastal State

Overshadowing all other considerations at work on the Seabed Committee in the early 1970s was the gathering momentum toward a new legal order, in which vastly extended regimes of coastal state jurisdiction would be permitted and developed[401]. From an environmental perspective, it would be difficult to exaggerate the potential significance of the proliferation of these regimes of national jurisdiction in the 1970s[402]. Under the new law of the sea, coastal states will be permitted to establish national zones of various kinds to cover immense areas of ocean space, including most of the more fertile waters and of the more accessible resources. Many

developing coastal states simply lack the capability to exercise environmental management responsibilities in an effective manner even in local waters, and few have the means to apply and enforce such authority throughout areas extending 200 miles from shore. Almost no problem in the preservation of the ocean environment can be regarded as totally unaffected by the way in which coastal states choose, or are able, to exercise their ocean management authority, under the neighbouring regimes of territorial sea[403], exclusive economic zone[404], and continental shelf[405] and in some regions also under the special regime established for mid-oceanic archipelagic states[406].

The future of effective environmental management in the ocean seems to depend very largely on the prospects for international co-operation: directly between states, through competent international agencies, or by means of generous transfers of appropriate skills and technologies. No one can foresee the future pattern of co-operative arrangements in ocean affairs, but it is generally assumed that much depends upon the stock of goodwill and understanding left among nations at the end of UNCLOS III. Even if most national governments are relatively satisfied with the final outcome of the Conference in the early 1980s, it would be naive to expect unprecedently massive programmes of co-operative global action for the preservation of the ocean environment. Some bold and constructive innovations might be expected at the regional level[407], but effective initiatives at the global level are likely to be spasmodic, influenced less by idealism than by a perception of common interest threatened by environmental hazards which seem capable of assuming crisis proportions.

2. The Conservation Agenda for UNCLOS III

In response to the U.N. General Assembly's request[408] in 1970, the U.N. Seabed Committee prepared a "list of subjects and issues relating to the law of the sea" in preparation for the forthcoming formal session of UNCLOS III. This list, adopted by the Committee on August 16th, 1972[409], formed the basis for the official agenda of the Conference.

Of the 25 items included on this preliminary list, only one dealt wholly and directly with conservation issues:

Item 12 Preservation of the Marine Environment

12.1. Sources of pollution and other hazards, and measures to combat them.
12.2. Measures to preserve the ecological balance of the marine environment.
12.3. Responsibility and liability for damage to the marine environment and to the coastal state.
12.4. Rights and duties of coastal states.
12.5. International co-operation.

Another dealt wholly with scientific research, a matter related to conservation matters:

Item 13 Scientific Research

13.1. Nature, characteristics and objectives of scientific research of the oceans.
13.2. Access to scientific information.
13.3. International co-operation.

In addition, a number of others dealt partly with conservation issues or matters related to them: e.g.

Item 4 Straits Used for International Navigation
4.1 Innocent passage

Item 5 Continental Shelf
5.6 Scientific research

Item 6 Exclusive Economic Zone Beyond the Territorial Sea
6.1 Nature and characteristics, including rights and jurisdiction of coastal states in relation to resources, pollution control and scientific research in the zone. Duties of states.
6.6 Fisheries
6.6.3 management and conservation
6.8 Prevention and control of pollution and other hazards to the marine environment.
6.9 Scientific research.

Item 7 Coastal State Preferential Rights or Other Non-Exclusive Jurisidiction Over Resources Beyond the Territorial Sea
7.4 Prevention and control of pollution and other hazards to the marine environment.
7.5 International co-operation in the study and rational exploitation of marine resources.

Item 8 High Seas
8.4 Management and conservation of living resources.

Item 20 Responsibility and Liability for Damage Resulting from the Use of the Marine Environment

The decision to "scatter" conservation issues under various headings was understandable within the context of a law-making conference, which was chiefly devoted to the task of elaborating treaty provisions for a widening variety of legal regimes. What is important is that this decision made it difficult, if not impossible, for the Conference to develop the environmental law of the sea in a systematic, holistic manner. How far the Conference would resort to an *ad hoc,* regime by regime, approach to conservation and protection of the ocean environment depended on two considerations:

(i) how widely the Conference would choose to interpret its environmental terms of reference under Item 12; and
(ii) how hard the Conference would try to effect linkages among the various committees assigned these items, with a view to reconciling divergencies of philosophy and reducing inconsistencies of language.

The Conference began its first session in New York on December 3rd, 1973. This session was devoted entirely to procedural matters[410]. Some procedural issues had to be held over to the second session, held at Caracas in the summer of 1974. Accordingly, it was not until its 15th meeting, on June 21st, 1974, that the Conference

made the official allocation of the substantive agenda items within the approved committee structure. The conservation-related items, identified above, were allocated as follows[411]:

Second Committee
Item 4 Straits Used for International Navigation
Item 5 Continental Shelf
Item 6 Coastal State Preferential Rights or Other
Item 7 Non-Exclusive Jurisdiction Over Resources Beyond the Territorial Sea
Item 8 High Seas

Third Committee
Item 12 Preservation of the Marine Environment
Item 13 Scientific Research

All Main Committees
Item 20 Responsibility and Liability for Damage Resulting from the Use of the Marine Environment

This, then, was the official framework for the debate on conservation issues at UNCLOS III. How these issues were treated is discussed at some length in the succeeding studies.

Notes

1 Max Nicholson, *The Environmental Revolution: A Guide for the New Masters of the World* (1970), pp. 132-140.
2 Mary Douglas, *Purity and Danger: An Analysis of Concepts of Pollution and Taboo* (1966).
3 Lord Kennett, "The Politics of Conservation" in A. Warren and F.B. Goldsmith, *eds., Conservation in Practice* (1974), pp. 465-475 at 466.
4 Nicholson, *supra* note 1, pp. 43, 193.
5 *Ibid.,* p. 141.
6 Kennett, *supra* note 3, p. 466.
7 Richard M. Highsmith, Jr., J. Granville Jensen, and Robert D. Rudd, *Conservation in the United States* (2nd ed., 1971), p. 9.
8 Nicholson, *op. cit.,* p. 162.
9 *Ibid.,* pp. 142-145.
10 "It was not so much a defeat as an abject surrender to the invasion of half-baked greedy materialism and to the ascendancy of barbarian values in aesthetics, in ecology, in economics, and in social responsibility. It initiated a new Dark Age for the natural environment, and placed the bulk of available human ingenuity and resources at the disposal of developments disastrous to the trustee interests of those generations towards their descendants". *Ibid.* at p. 148. Victorian Britain was not, however, totally unenlightened in environmental matters. The "revolutionary" year of 1848 witnessed the passing of the first-ever clean water legislation, which also gave rise to the public health inspector.
11 The torch for conservation was carried by miscellaneous and unrelated constituencies, such as pastoralists (with sheep farming interests), field sports enthusiasts (mostly of the hunting and fishing fraternity), professional botanists and ornithologists, patrons of zoological and botanic gardens, and scientific explorers like Cook and Banks and their middle-class admirers.
12 The idea seems to have been influenced by conservation efforts within the U.S. national parks system. The first initiatives were taken by a small group of scientists, such as Julian Huxley, Arthur Tansley, Max Nicholson, Clough Williams-Ellis, and others who served on the Committee on National Parks for England and Wales chaired by Arthur Hobhouse Julian Huxley, *Memories* (Vol. I, 1970), pp. 287-289.
13 Nicholson, *op. cit.,* at p. 158.
14 Julian Huxley, *Memories* (Vol. II, 1973), p. 79.
15 Hans Huth, *Nature and the American* (1957), *passim.* See also David Cushman Doyle, *Conservation: An American Story of Conflict and Accomplishment* (1957).
16 Republished in revised form under the title *The Earth as Modified by Human Action* (1885). Marsh, in his preface, describes the objective of his book in these words:
"to indicate the character and, approximately, the extent of the changes produced by human action in the physical conditions of the globe we inhabit; to point out the dangers of imprudence and the necessity of caution in all operations which on a large scale interfere with the spontaneous arrangements of the organic or the inorganic world; to suggest the possibilities and importance of the restoration of disturbed harmonies and the material improvement of waste and exhausted regions".
17 Lewis Mumford, *The Brown Decades: A Study of the Arts in America, 1865-1895* (1931), at p. 78.
18 "Appearing at the peak of American confidence in the inexhaustibility of resources, it was the first book to controvert the myth of superabundance and to spell out the need for reform. It showed how man differed from nature; it showed exactly what happened to nature when men cleared and farmed and built. Before Marsh wrote this book, few saw and fewer worried about how man affected his environment. Today March's insights are virtually taken for granted." Editorial introduction to *Man and Nature* (ed. David Lowenthal, 1965 reprint), p. vi.
19 For a general account, see Stewart L Udall, *The Quiet Crisis* (1963).
20 It was not until 1894 that wildlife protection measures were enacted for Yellowstone.
21 John D. Black, *The Management and Conservation of Biological Resources* (1968), at p. 16.
22 Traditionally, the Sierra Club has placed emphasis on "the study and protection of national (*i.e.* U.S.) scenic resources, particularly those of mountain regions". A decade ago it began to monitor UN and other international forums concerned with environmental issues. Gene Coan, "The International Role of the Sierra Club; A Proposal" (1972), *Sierra Club Bulletin,* vol. 57, no. 1, p. 12. In recent years Sierra Club representatives have attended UNCLOS III sessions as NGO observers.
23 See Gifford Pinchot, *The Fight tor Conservation* (1910) (reprinted with an introduction by Gerald D. Nash in 1967).
24 Black, *supra* note 21, pp. 17-18. Many commentators today credit Roosevelt with the initiation of conservation policy-making in the United States, when he convened the first Governor's Conference in 1908 with a view to inaugurating a political discussion of natural resource problems.
25 Samuel P. Hays, *Conservation and the Gospel of Efficiency: The Progressive Conservation Movement, 1890-1920* (1959).

26 *See*, for example, Edgar B. Nixon, *ed., Franklin D. Roosevelt and Conservation, 1911-1945* (in 2 vols., 1957).

27 For a brief discussion, see Harold J. Barnett and Chandler Morse, *Scarcity and Growth: The Economics of Natural Resource Availability* (1963), p. 22-29.

28 Huxley's philosophy was, of course, Darwinian in character. For a complete picture of his evolutionary humanism, *see Religion Without Revelation* (1927, rev. ed. 1957); *Evolutionary Ethics* (1943); (with Thomas H. Huxley), *Evolution and Ethics, 1893-1943* (1947); and *Essays of* a *Humanist* (1964).

29 "It seems incredible that with all our talents of rationality and skill, all our genius for creative thought and visible achievement, a shameless, reckless waste still rules much of human behavior." Sherman S. Hayden, *The International Protection of Wildlife* (1942), at p. 14.

30 See Huxley's famous essay, "The Uniqueness of Man" in *The Uniqueness of Man* (1941), pp. 1-33 (reprinted under the title *Man Stands Alone* in 1970), and in *Man in the Modern World* (1947), pp. 1-21.

31 "If only a sufficient number could bring themselves to look upon the natural world as a trust committed to our common care, with its inhabitants to be preserved alive where possible and its physical form to be radically changed and altered only as the general need may require, is it not thinkable that the whole destructive side of human nature might in time be softened as well? The cultivation of this spirit of moderation and care I believe to be the ultimate justification of all wild life preservation, individual or international, more truly than the economic, scientific or recreational value of the thing preserved, the grounds regularly advanced as prime reasons for taking action." Hayden, *supra* note 29, at pp. 14-15. The question of what, precisely, should be guarded in nature is not so easily answered; or, more properly, it can be answered in many different ways. For some the enemy is waste or misuse or induced imbalance; for others the basic purpose is to preserve diversity, a theme with social as well as scientific implications. One prominent conservationist has made "a plea for diversity — for the preservation of natural diversity — in the hope that the prevailing trend toward uniformity can be arrested and the world kept a fit place for the greatest possible human variety". Raymond F. Dasmann, *A Different Kind of Country* (1968), at p. vii. Max Nicholson has taken a similar stand, but on biological grounds. In replying to a question by Peter Scott about the objectives of nature conservation, Nicholson suggested "that nature had a value.... that we should not intervene at all unless we were certain that intervention was necessary, and that when we intervened we should do so in favour of diversity of varieties". E. M(ax) Nicholson, "Nature Conservation and the Management of Natural Areas" (1956) in International Union for Conservation of Nature and Natural Resources (IUCN), *Proceedings and Papers of Sixth Technical Meeting (held at Edinburgh in June 1956),* pp. 8-27 at 26.

32 *See* Alan C. Jenkins, *The Naturalists: Pioneers of Natural History* (1978), and Alexander B. Adams, *Eternal Quest: The Story of the Great Naturalists* (1969).

33 Robert S. DeSanto, *Concepts of Applied Ecology* (1978), p. 153.

34 *Ibid.*

35 *Ibid.,* p. 159.

36 *Ibid.,* p. 161.

37 A.G. Tansley, "The Use and Abuse of Vegetational Concepts and Terms" (1935), 16 *Ecology* 284. Tansley and his contemporary W.H. Pearsall were the leading synoptic ecologists of the 1930s. Their shift from the micro scale (autecology) to the macro scale (synecology) had a profound impact on the development of modern ecology. As early as 1887 the idea of an ecosystem was expressed by the term "microcosm". For a historical review of the concept, see Jack Major, "Historical Development of the Ecosystem Concept" in George M. Van Dyne, *ed., The Ecosystem Concept in Natural Resource Management* (1969), pp. 9-22. See also Stephen H. Spurr, "The Natural Resource Ecosystem", *ibid.* pp. 3-7, and Arnold M. Schultz, "The Ecosystem as a Conceptual Tool in the Management of Natural Resources" in S.V. Ciriacy — Wantrup and J.J. Parsons, *eds., Natural Resources, Quality and Quantity* (1967), pp. 139-161.

38 *See*, for example, Nicholson, *supra* note 1, pp. 239-242.

39 For a general discussion, see Barnett and Morse, *supra* note 27, pp. 17-97.

40 *Ibid.,* pp. 51-71.

41 *ibid.,* p. 47, n. 54.

42 *Ibid.,* pp. 63-64.

43 *Ibid.,* pp. 69-70.

44 Philip Appleman, "Malthus: The Continuing Controversy" (1976), *Sierra Club Bulletin* No. 2, p. 9. For a recent re-evaluation, see William Petersen, *Malthus* (1979).

45 *See*, for example, S.V. Ciriacy-Wantrup, *Resource Conservation: Economics and Policies* (1952), and Anthony Scott, *Natural Resources: The Economics of Conservation* (1955).

46 *See*, for example, Francis T. Christy, Jr., and Anthony Scott, *The Common Wealth in Ocean Fisheries: Some Problems of Growth and Economic Allocation* (1965).

47 For contemporary approaches, compare Lee G. Anderson, *The Economics of Fisheries Management* (1977) and Robert T. Lackey and Larry A. Nielson, *eds., Fisheries Management* (1980),

48 For a recent example of such a conceptual framework, see Jan Schneider, *World Public Order of the Environment: Towards an International Ecological Law and Organization* (1979). For an overview within a similar framework, see Myres S. McDougal and Jan Schneider, "The Protection of the Environment and

World Public Order: Some Recent Developments" (1974), 45 *Miss. L.J.* 1085. On the "nature" of international environmental law, see Alexandra C. Kiss, *Survey of Current Developments in International Environmental Law* (IUCN, Environmental Policy and Law paper, No. 10, 1976), pp. 14-15.

49 Kiss uses the term "fundamental principles of international environmental law" to include "not only generally recognized legal concepts like international responsibility for environmental harm, but also requirements such as prevention, or international co-operation for the protection of the environment, and even trends such as growing awareness In safeguarding the biosphere, or the need to draft specific rules to prevent pollution or to save endangered species". *Ibid.,* at p. 29. In short, he is prepared to err on the side of the "immanent". From this broad viewpoint, Kiss lists the right to information and the right to due process among these fundamental principles. *Ibid.,* pp. 29-40. For a more restrictive view, see Ian Brownlie, "A Survey of International Customary Rules of Environmental Protection" (1973), 13 *Nat. Res. J.* 179.

50 L. Oppenheim, *International Law: A. Treatise* (5th ed., Lauterpacht, 1937), Vol. 1 *(Peace),* pp. 271-298.

51 *Ibid.,* at p. 280. *See* also H. Lauterpacht, *The Function of Law in the International Community* (1933), pp. 290-294.

52 Oppenheim (Lauterpacht), *supra* note 50, p. 280. *See* also Ludwik A. Teclaff, "The Impact of Environmental Concern on the Development of International Law" (1973), 13 *Nat. Res. J.* 357.

53 "The maxim, *sic utere tuo ut alienum non laedas,* is applicable to the relations of states no less than of individuals; It underlies a substantial part of the law of (t)ort in English law and the corresponding branches of other systems of law; it is one of those general principles of law recognized by civilized states which the Permanent Court is bound to apply by virtue of Article 38 of its Statute. However, the extent of the application of the still controversial doctrine of the prohibition of abuse of rights is not at all certain. It is of recent origin in the literature and practice of international law, and it must be left to international tribunals to apply and develop it by reference to individual situations." Oppenheim (Lauterpacht), *supra* note 50, at pp. 280-281.

54 Kiss, *op. cit.* pp. 69-77.

55 UNEP, *Report of the Executive Director on Co-operation in the Field of Environment Concerning Natural Resources Shared by Two or More States* (1975), *passim.* The principle has been described as "little more than a policy slogan", which may, however, be an effective guide in the creation of more specific rules." L.F.E. Goldie, Liability for Damage and the Progressive Development of International Law" (1965), 14 *Int. & Comp. L..Q.* 1189 at p. 1227.

56 The Trail Smelter Arbitral Decision (United States v. Canada), 3 *U.N. Rep. Int'l Arb. Awards* 1905(1949). Kiss, *op. cit.,* pp. 43-47. For analysis, *see* John Read, "The Trail Smelter Dispute" (1963), 1 *Can. Y.B. Int'l L.* 213; and Alfred P. Rubin, "Pollution by Analogy: The Trail Smelter Arbitration" (1971), 50 *Oreg. L. Rev.* 259. For a recently suggested extension of the doctrine, *see* Edith Brown Weiss, "Who Pays for Weather Modification Damage?" (1978), 4 *Envir. Policy and Law 22.*

57 Corfu Channel Case (Merits), 1949 *I.C.J.* Rep. 4.

58 *Ibid.,* at p. 22.

59 Lake Lanoux Arbitral Decision (Spain v. France), 12 *U.N. Rep. Int'l Arb. Awards* 281 (1958). Kiss, *op. cit.*, p. 47.

60 Anthony Lester, "Pollution", in A.H. Garretson, R.D. Hayton, and C.J. Olmstead, *eds., The Law of International Drainage Basins* (1967), pp. 89-123 at p. 100. The decision is chiefly of environmental importance because of its rejection of the rule in the Declaration of Montevideo that "no state may, without the consent of the other riparian state, introduce into water courses of an international character, for the industrial or agricultural exploitation of their (sic) waters, any alteration which may prove injurious to the margin of the other interested state". Quoted in Teclaff, *supra* note 5, at p. 359. The Tribunal was not prepared to go further than to suggest that in the absence of specific obligations imposed by agreement, all that international law requires of states is to take into account the interests of the other riparian. *Ibid.,* p. 360. One commentator regards the decision as authority for the view that "the right of territorial sovereignty over an international river entitles a riparian state to use the river in any manner which will not result in a material alteration of the river or of its availability for use as it passes through the riparian state." William W. Van Alstyne, "The Justiciability of International River Disputes: A Study in the Case Method" (1964), *Duke L.J.* 307 at p. 314, quoted in Lester, *op. cit.,* at p. 101.

61 L.F.E. Goldie, "International Principles of Responsibility for Pollution" (1970), 9 *Colum. J. Transnat'l L.* 283. *See* Schneider, *supra* note 48, pp. 163-167.

62 Statute of the International Court of Justice, Article 38(1)(c). Berber, for example, concludes that abuse of rights doctrine does not qualify as a general principle of law. F.J. Berber, *Rivers in International Law* (1959), pp. 195-210. For a review of relevant doctrinal developments, see Robert E. Stein, *ed., International Responsibility tor Environmental Injury* (in press).

63 George E. Glos, *International Rivers: A Policy-Oriented Perspective* (1961), at p. 3.

64 France and Spain have dealt with problems concerning the Bidassoa river from as early as 1856. The Rhine and Lake Constance have been under some degree of international regulation since 1867, and in 1868 the littoral states even adopted regulations for the transportation on the Rhine of "inflammable, corrosive and

poisonous substances". Lake Geneva fishing has been regulated by France and Switzerland since 1880, and the Danube and Pruth since 1901. Bernd Raster and Bruno Simma, *eds., International Protection of the Environment: Treaties and Related Documents* (1975-1981) (hereafter Rüster-Simma), vol. IX, pp. 4319-4792, and vol. X, pp. 4793-5154.

65 Ludwik A. Teclaff and Eileen Teclaff, "Transboundary Ground Water Pollution: Survey and Trends in Treaty Law" (1979), 19 *Nat. Res. J.* 629. *See* also Dante A. Caponera and Dominique Alheritiere, "Principles for International Groundwater Law" (1978), 18 *ibid.* 589.

66 *Ibid.,* pp. 614-617. The first two of these principles have been condemned for being "grounded in an individualistic and anarchical conception of international law in which personal and egotistical interests are raised to the level of guiding principles". Berber, *supra* note 62, at p. 96. For a recent criticism of the equitable apportionment doctrine, see Albert E. Utton, "International Water Quality Law" (1973), 13 *Nat. Res. J.* 282 at pp. 297-299. For a more favourable view, *see* Jerome Lipper, "Equitable Utilization" in Garretson, Hayton and Olmstead, *supra* note 60, pp. 15-88.

67 *Treaty Series* No. 548 (1909); Malloy, *ed.,* Treaties, Conventions, International Acts, Protocols and Agreements between the U.S.A. and Other Powers, 1910-1923 (hereafter '"Malloy"), vol. III, p. 2607; Rüster-Smma, vol. X, p. 5158.

68 Article VIII.

69 Article IV.

70 On the pre-World War Two problems besetting the Danube Commission, see Stephen Gorove, *Law and Politics of the Danube: An Interdisciplinary Study* (1964), pp. 1-36. *See* also David G. LeMarquand, *International Rivers: The Politics of Co-operation* (1977).

71 Rüster-Simma, vol. IV, p. 1561.

72 Sherman S. Hayden, *The International Protection of Wildlife* (1942), p. 92.

73 *Ibid.,* p. 35.

74 Martens, 30 *Nouveau Recueil* (2nd), p. 686; Rüster-Simma, vol. IV, p. 1615. This convention was signed and ratified by 11 European states: Switzerland, Germany, Austria-Hungary, Belgium, Spain, France, Greece, Luxembourg, Monaco, Portugal, and Sweden. Great Britain, though active in conservation, chose not to join this convention. Apparently this was due to a preference for informal approaches, seeking the negotiation of specific agreements limited to parties with a particular interest, instead of the formal approaches through international bodies seeking wide-ranging conventions, said to be favoured by the rest of Europe. Hayden, *op. cit.,* p. 99.

75 Malloy, vol. III, p. 2645; Rüster-Simma, vol. IV, p. 1638.

76 178 *L.N.T.S.* 310; Rüster-Simma, vol. IV, p. 1723.

77 Hayden, *op. cit.,* pp. 16-17.

78 172 *L.N.T.S.* 242; Rüster-Simma, vol. IV, p. 1693. Signed by nine states (Egypt, Great Britain, Belgium, Sudan, South Africa, Italy, France, Portugal, and Spain), this convention was ratified by all except the last two, and later it was adhered to by India. It came into force in 1936. Hayden, *op. cit.,* pp. 44-55.

79 Convention on Nature Protection and Wildlife Preservation in the Western Hemisphere (1940). Bevans, *ed., Treaties and Other International Agreements of the United States of America, 1776-1949,* (hereafter Bevans), vol. 3, p. 630; Rüster-Simma, vol. IV, p. 1729. See also Hayden, *op. cit.,* pp. 63-66.

80 *See*, for example, H.F. Osborn and H.E. Anthony, "Close of the Age of Mammals" (1922), 3 *J. of Mammalogy* 219. These authors argued that today's mammals are "a sort of remnant class" left over after the domination of glacial ice. Mankind was "something like destiny's agent either for preservation or annihilation of the relatively few that remain". Quoted in Hayden, *op. cit.,* at p. 13.

81 For early international efforts to protect fur seals see Rüster-Simma, vol. VIII, pp. 3655 ff.

82 Hayden, *op. cit.,* p. 15.

83 *See supra,* pp. 17-19.

84 World Health Organization, *Health Hazards of the Human Environment* (1972), p. 108.

85 *Ibid.,* pp. 106-110.

86 Erik P. Eckholm, *The Picture of Health: Environmental Source of Disease* (WorldWatch Institute, 1977).

87 Peter Ruderman, "Public Health and the Human Environment" in Ronald St.J. Macdonald, Douglas M. Johnston, and Gerald L. Morris, *eds., The international Law and Policy of Human Welfare* (1978), pp. 624-625. For texts of early arrangements, see Convention of Measures to be taken against Phylloxera Vasatrix (1878), signed by Germany, Austria-Hungary, Spain, France, Italy, Portugal, and Switzerland (Rüster-Simma, vol. IV, p. 1565); and the Convention Designed to Remove the Danger of Epizootic Diseases (1887) signed by Austria-Hungary and Italy *(ibid.,* p. 1586).

88 This organization provided "epidemic intelligence and epidemiological studies, technical disease studies, the international standardization of vital statistics, efforts to control drug traffic, and a small amount of direct assistance to governments". Ruderman, op. *cit.,* at p. 626.

89 *E.g.* the Convention regarding the Organization of the Campaign against Locusts (1920), 164 *L.N.T.S.* 85, Rüster-Simma, vol. IV, p. 1642; the Agreement concerning the Creation of an International Office for Information regarding Locusts (1926), 109 *L.N.T.S.* 122, Rüster-Simma, vol. IV, p. 1657; and Soviet-Afghanistan

(164 *L.N.T.S.* 335; Rüster-Simma, vol. IV, p. 1715) and Soviet-Iran (176 *L.N.T.S.* 335; Rüster-Simma, vol. IV, p. 1718) agreements in 1935.

90 *E.g.* International Convention for the Protection of Plants (1929) (126 *L.N.T.S.* 305; Rüster-Simma, vol. IV, p. 1669); and the Soviet-Iran Convention concerning the Campaign against Plant Diseases and Parasites (1935). (171 *L.N.T.S.* 343; Rüster-Simma, vol. IV, p. 1721).

91 *E.g.* International Agreement for the Creation of the International Office for Dealing with Contagious Diseases of Animals (1924) (57 *L.N.T.S.* 136; Rüster-Simma, vol. IV, p. 1646); Convention on Prevention of Diseases in Livestock (1928) (Bevans, vol. IX, p. 959; Rüster-Simma, vol. IV, p. 1665); International Convention for the Campaign against Contagious Diseases of Animals (1935), (186 *L.N.T.S.* 175; Rüster-Simma, vol. IV, p. 1705); and the Danish-Swedish Agreement regarding Co-operation for the Campaign against Foot-and-Mouth Disease (1938) (188 *L.N.T.S.* 461; Rüster-Simma, vol. IV, p. 1727).

92 Stephen S. Goodspeed, *The Nature and Function of International Organization* (2nd ed., 1967), pp. 420-465).

93 For example, Buckland, Huxley, Mclintosh and Agassiz.

94 In the early decades of fishery investigation, between the 1890s and World War Two, the leading specialists were Petersen, D'Arcy Thompson, Johansen, Heincke, Baranov, Fulton, E.W.L. Holt, Hjort, H. Thompson, ES. Russell, Graham, W.F. Thompson, and Garstang.

95 Albert Koers, *International Regulation of Marine Fisheries: A Study of Regional Fisheries Organizations* (1973), pp. 77-79.

96 Douglas M. Johnston, *The International Law of Fisheries: A Framework for Policy-Oriented Inquiries* (1964), p. 74.

97 Convention between France and Great Britain for Defining the Limits of Exclusive Fishing Rights (1834). Parry, *ed., The Consolidated Treaty Series* (hereafter "Parry"), vol. 89, p. 221.

98 L. Larry Leonard, *International Regulation of Fisheries* (1944), pp. 35-42. For text, see Rüster-Simma, vol. VI, p. 2546.

99 Convention for Regulating the Police of the North Sea Fisheries (1882). Martens, *ed., Nouveau recueil general de traites,* 2 me serie, vol. IX, p. 556; Parry, vol. 160, p. 219; Leonard, *op. cit.,* pp. 42-48.

100 Johnston, *op. cit.,* pp. 359-360.

101 231 *U.N.T.S.* 200. Johnston, *op. cit.,* pp. 361-363.

102 486 *U.N.T.S.* 158. Koers, *supra* note 95, pp. 90-92.

103 J. Tomasevic, *International Agreements on the Conservation of Marine Resources, with Special Reference to the North Pacific* (1943), pp. 65-122. *See* also Hayden, *op. cit.,* pp. 114-137.

104 Conservation concerns first arose with the rapid development of pelagic sealing: that is, the taking of seals at sea. Tomasevic, *op. cit.,* pp. 75-80.

105 Johnston, *op. cit.,* pp.206-210, 264-266.

106 104 *Brit. For. State Papers* 175 (1911); Malloy, vol. Ill, p. 2629; Rüster-Simma, vol. VIII, p. 3678.

107 26 *U.N.T.S.* 363.

108 314 *U.N.T.S.* 105. This convention established the North Pacific Fur Seal Commission. Koers, *op. cit.,* pp. 85-87.

109 Tomasevic, *op. cit.,* pp. 219-265. See also Homer E. Gregory and Kathleen Barnes, *North Pacific Fisheries* (1939), pp. 70-76; and Johnston *op. cit.,* pp. 384-390.

110 Canadian-U.S. Convention for the Protection, Preservation and Extension of the Sockeye Salmon Fisheries in the Fraser River System, 184 *L.N.T.S.* 305.

111 Koers, *op. cit.,* pp. 82-85.

112 Johnston, *op. cit.,* pp. 387-388.

113 32 *L.N.T.S.* 93. See Tomasevic, *op. cit.,* pp. 125-215.

114 Koers, *op. cit.,* pp. 80-82.

115 121 *L.N.T.S.* 45.

116 181 *L.N.T.S.* 209.

117 222 *U.N.T.S.* 77.

118 Hayden, *op. cit.,* pp. 137-171; Johnston, *op. cit.,* pp. 396-411; and Leonard *op. cit.,* pp. 98-109.

119 155 *L.N.T.S.* 349.

120 161 *U.N.T.S.* 72.

121 Koers, *op. cit.,* pp. 86-90.

122 Ian Burton, Robert W. Kates, and Gilbert F. White, *The Environment* as *Hazard* (1978).

123 Environmental threats may be classified in a number of ways. One writer distinguishes four levels of threat: (in ascending order of seriousness), loss of amenity; danger to individual human health; genetic damage; and disturbance of ecological balance (threat to the life-supporting capacity of the planet). Sterling Brubaker, *To Live on Earth: Man and his Environment in Perspective* (1972), pp. 49-57. The same author identifies five major hazards in the present age: man's effect on the global climate (e.g. aerosols, water vapor in the stratosphere, carbon dioxide, and heat from man's energy conversion), radioactivity, pesticides, fertilizers, and erosion. *Ibid.,* pp. 59-103. Today, in the 1980s, a growing number of countries, especially in North America and Europe, regard the "acid rain" problem as the most serious environmental threat. Irene

Van Lier, *Acid Rain and International Law* (in press).

124 Emanuel Margolis, "The Hydrogen Bomb Experiments and International Law" (1955), 64 Yale L.J. 629; and Myres S. McDougal and Norbert Schiei, "The Hydrogen Bomb Tests in Perspective: Lawful Measures for Security", *ibid.* 648. For a commentary on the *Nuclear Tests* case (Australia v. France, New Zealand v. France), 1973 *I.C.J.* Rep. 320, *see* Günther Handl, "Territorial Sovereignty and the Problem of Transnational Pollution" (1975), 69 *A.J.I.L.* 50.

125 For an early warning, *see* John W. Gofman and Arthur R. Tamplin, *Poisoned Power: The Case against Nuclear Power Plants* (1971). For a recent assessment, *see* Elizabeth S. Rolph, *Nuclear Power and the Public Safety: A Study in Regulation* (1979).

126 Rachel Carson, *Silent Spring* (1962). This book was based on a number of articles which appeared originally in the *New Yorker* between 1960 and 1961.

127 *See*, for example, her warning of cancer hazard arising from polluted waterways, *ibid.,* p. 50.

128 Perhaps her most effective argument is the one which challenges the complacent assumption that man will inevitably learn to adjust in time to the chemical hazard. "The rapidity of change and speed with which new situations are created follow the impetuous and heedless pace of man rather than the deliberate pace of nature ... To adjust to these chemicals would require time on the scale that is nature's; it would require not merely the years of a man's life but the life of generations". *Ibid.,* at p. 7. These doubts about the limits of human adaptability were developed, several years later, by Alvin Toffler in his brilliant, and equally influential, book *Future Shock* (1970). The response to Carson seems to owe much to the fact that we live in a Tofflerian age of self-questioning.

129 "We are rightly appalled by the genetic effects of radiation; how, then, can we be indifferent to the same effect in chemicals that we disseminate widely in our environment". Carson, *supra* note 126 at p. 37.

130 Frank Graham, Jr., *Since Silent Spring* (1970), *passim.*

131 For an admiring assessment of her work, *see* Paul Brooks, *The House of Life: Rachel Carson at Work* (1972).

132 On the various factors contributing to the growing problem. *see* R. Michael M'Gonigle and Mark W. Zacher, *Pollution, Politics and International Law: Tankers at Sea* (1979), pp. 14-38.

133 *1979 IMCO News* (No. 1, pp. 12-15) has listed 55 tanker incidents between 1962 and 1978 known to have caused the spillage of at least 5,000 tons of oil each. Up to 1972 the largest and most famous of these spillages was caused by the grounding of the *Torrey Canyon* in 1967 (100,000 tons spilled), which caused a major change in British policy and legislation. But the impact on public opinion is not directly correlated with the volume of oil spilled. *The Arrow* (1970) spilled only 12,000 tons and the *Argo Merchant* only 32,000, yet both had an important effect on the raising of environmental consciousness, in Canada and the United States respectively. The *Amoco Cadiz* (1978), on the other hand, broke all records for a single tanker spill (230,00 tons) and led to a profound change in French policy and legislation.

134 M'Gonigle and Zacher, *supra* note 82, pp. 23-28. Thor Heyerdahl, "Ocean Pollution Seen from Rafts" in R. Gordon Pirie, *Oceanography: Contemporary Readings in Ocean Sciences* (2nd ed., 1977), pp. 315-321.

135 M'Gonigle and Zacher, *op. cit.,* pp. 15-23.

136 Douglas M. Johnston, "Marine Pollution Control: Law, Science, and Politics" (1972), 28 *Int'l J.* 69, at pp. 69-74.

137 On one level of conceptualization, the volume of attention devoted to ship-generated pollution in the high seas, prior to UNCLOS III, might be attributed to a growing sense of responsibility for environmental damage to the "commons". Kiss, *supra* note 48, pp. 48-52.

138 "Introduction: To Bell the Cats or Catch the Mice?" In Onkar Marwah and Ann Schulz, *eds., Nuclear Proliferation and the Near-Nuclear Countries* (1975), pp. 9-10.

139 21 *U.S.T.* 483; *T.I.A.S.* No. 6839; 729 *U.N.T.S.* 162; Rüster-Simma, vol. XV, p. 7848. *See* also 7 *Int'l Leg. Mats.* 811 (1968); and James Barros and Douglas M. Johnston, *eds., The International Law of Pollution* (1974) (hereafter "Barros-Johnston") 392-395.

140 Marwah and Schulz, *op. cit.,* pp. 323-336.

141 Article 11. Conversely, under Article 1, each nuclear weapon party accepts the obligation "not to transfer to any recipient whatsoever nuclear weapons or other nuclear explosive devices, directly or indirectly; and not in any way to assist, encourage, or induce any non-nuclear-weapon State to manufacture or otherwise acquire nuclear weapons or other nuclear explosive devices, or control over such weapons or explosive devices".

142 Article III.

143 Marwah and Schulz, *op. cit.,* pp. 335-336. See Henry D. Smyth, "The Need for International Safeguards", in Mason Willrich, *ed., International Safeguards and Nuclear Industry* (1973), pp. 3-20.

144 Marwah and Schulz, *op. cit., passim.*

145 Paul C. Szasz, "International Atomic Energy Agency Safeguards", in Willrich, *supra* note 143, pp. 73-144.

146 Bernhard G. Bechhoefer, "Historical Evolution of International Safeguards", *ibid.,* pp. 21-44.

147 These agreements were four in number. The most important was the Treaty on the Limitation of Anti-Ballistic Missile Systems (the ABM Treaty), which entered into force in October 1972. 23 *U.S.T.* 3435; *T.I.A.S.* No. 7503. See also 11 *Int'l Leg. Mats.* 784 (1972). Also entering into force at the same time was the Interim

Agreement on Certain Measures with respect to the Limitation of Strategic Offence Arms (and the Associated Protocol). 23 *U.S.T.* 3462; *T.I.A.S.* No. 7504. *See* also 11 *Int'l Leg. mats.* 791 (1972). The other two were executive agreements which entered into force on the day of signature, September 30, 1971: the Agreement on Measures to Improve the U.S.-U.S.S.R. Direct Communication Link (the Revised Hot Line Agreement) and the Agreement on Measures to Reduce the Risk of Outbreak of Nuclear War (the Accident Measures Agreement). Texts reproduced in 10 *Int'l Leg. Mats.* No. 6 (Nov. 1971), pp. 1173-1176. *See* also texts of ancillary instruments reproduced in John H. Barton and Lawrence D. Weiler, *eds., International Arms Control: Issues and Agreements* (The Stanford Arms Control Group, 1976), pp. 385-390.

148 18 *Int'l Leg. Mats.* 1112 (1979).

149 *Ibid.,* pp. 94-122.

150 14 *U.S.T.* 1313; *T.I.A.S.* No. 5433; 480 *U.N.T.S.* 43; Rüster-Simma, vol. 1, p. 422. *See* also 2 *Inf'l Leg. Mats.* 883 (1963).

151 14 *U.S.T.* 825; *T.I.A.S.* No. 5362. *See* also 2 *Int'l Leg. Mats.* 793 (1963).

152 *See* note 97, *supra.*

153 26 *U.S.T.* 583; *T.I.A.S.* No. 8062; Barros-Johnston, p. 381; Rüster-Simma, vol. XV, p. 7858.

154 12 *U.S.T.* 794; *T.I.A.S.* No. 4780; 402 *U.N.T.S.* 71; Rüster-Simma, vol. 1, p. 18.

155 18 *U.S.T.* 2410; *T.I.A.S.* No. 6347; 610 *U.N.T.S.* 205; Rüster-Simma, vol. 1, p. 30.

156 634 *U.N.T.S.* 281; Rüster-Simma, vol. XV, p. 7827. *See* also 6 *Int'l Leg. Mats.* 521 (1967).

157 23 *U.S.T.* 703; *T.I.A.S.* No 7337; Rüster-Simma, vol. II, p. 498.

158 The question of revision is raised as a result of the growing feasibility of extracting mineral resources from the Antarctic landmass. Frank Pallone, "Resource Exploitation: The Threat to the Legal Regime of Antarctica" (1978), 12 *Int'l Lawyer* 547; (Comment), "Antarctica: Gold Mine or Landmine?", 9 *IUCN Bulletin* 22 (No. 4, April 1978); and M.J. Peterson, "Antarctica: The Last Great Land Push on Earth" (1980), 34 *Int'l Organ,* 377

159 The parties to the Treaty have recently concluded an agreement for the establishment of a management regime for the living resources of the Southern Ocean. A draft version is reproduced in 5 *Environ. Policy and Law* 58 (1979). For comments, *see* "How Much Krill Can the Antarctic Spare?" (1979) *ibid.;* "Antarctica:- Inching toward a Krill Convention," 10 *IUCN Bulletin* 28 (No. 4, April 1979); Barbara Mitchell, 72 *Am. Soc. Int'l L Procs.* 321 (1978) at pp. 326-338, and Dayton L. Alverson, "Tug-of-war for the Antarctic Krill" (1980), 8 *Ocean Devel. & Int'l L.J.* 171. For a suggested new regime for Antarctica, see Edward Honnold, "Draft Provisions of a New International Convention on Antarctica" (1977), 4 *Yale Studies in World Public Order* 123.

160 Article 1(1).

161 Article V(1).

162 Article Vll(3). Note that the Treaty applies to "the area south of 60° South Latitude, including all ice shelves". Article VI.

163 For a fuller list, *see* Kiss, *supra* note 48, pp. 126-132.

164 Preamble to the Partial Test Ban Treaty.

165 For example, Article IV provides a discretionary right of withdrawal:
"Each Party shall in exercising its national sovereignty have the right to withdraw from the Treaty If it decides that extraordinary events, related to the subject matter of this Treaty, have jeopardized the supreme interests of its country".

166 Article 2.

167 In the latest text produced at the Third U.N. Conference on the Law of the Sea — the Draft Convention on the Law of the Sea (Informal Text) *Informal Composite Negotiating Text/Revision 1* (Doc. A/CONF. 62/W.P. 10/Rev. 3) — the "sea-bed and ocean floor and subsoil thereof beyond the limits of national jurisdiction" is designated merely as "the Area" (Article 1(1)). The dropping of the adjective "international" was presumably intended to meet the criticism that the high seas were also "international", albeit subject to a different kind of regime.

168 The absence of accepted territorial sovereignty over Antarctica seems to preclude the application of the territorial sea, continental shelf, and economic zone regimes to the waters and seabed adjacent to the Antarctic land-mass under the new law of the sea. Moreover, the common heritage of mankind principle applied to the deep ocean floor beyond limits of national jurisdiction might be in the process of being extended by analogy to other areas, such as Antarctica (and outer space) which cannot be appropriated by nation states.

169 John Kish, *The Law of International Space* (1973), pp. 52-94. On the prohibition of territorial sovereignty over international spaces, *see* colloquium on territorial sovereignty and Antarctic in 33 *Miami L. Rev.* 240 ff. (1978).

170 Article 6.

171 Article 4.

172 Article 1(10).

173 Article III.

174 Kenneth Booth, "The Military Implications of the Changing Law of the Sea", in John K. Gamble, Jr., *ed., Law of the Sea: Neglected Issues* (Proceedings of the 12th Annual Conference of the Law of the Sea Institute

held at The Hague, October 1978) (1979), pp. 328-397 at 378-380.

175 Barros-Johnston, pp. 379-381.

176 Article 1.

177 Preamble to the BTW Convention.

178 Article VI.

179 Compare, for example, Gofman and Tamplin, *supra* note 125, and Glenn T. Seaberg and William R. Corliss, *Man and Atom: Building a New World through Nuclear Technology* (1971).

180 Desaix Myers III, *The Nuclear Power Debate: Moral, Economic, Technical and Political Issues* (1977), *passim*.

181 Mason Willrich, *Global Politics of Nuclear Energy* (1971), especially pp. 43-66 and 85-95; and Seaberg and Corliss, *op. cit.,* pp.295-324.

182 I.A.E.A., *International Conventions on Civil Liability for Nuclear Damage* (Legal Series No. 4, rev. ed., 1976); Paul C. Szasz, *The Law and Practices of the International Atomic Energy Agency* (Legal Series No. 7, 1970). On I.A.E.A.'s oceanic concerns, see I.A.E.A., *Maritime Carriage of Nuclear Materials* (Proceedings of Symposium held at Stockholm in June 1972) (1973); and I.A.E.A., *Disposal of Radioactive Wastes into Seas, Oceans and Surface Waters* (Proceedings of Symposium held at Vienna in May 1966) (1966).

183 *See*, for example, John C. Sawmill, Keichi Oshima, and Hans W. Maull, *Energy: Managing the Transition* (The Trilateral Commission, The Triangle Papers, No. 17, 1978); Mason Willrich (with others), *Energy and World Politics* (1975), especially pp. 141-179; and Edward W. Erickson and Leonard Waverman, *eds., The Energy Question: An International Failure of Policy* (1974).

184 The global arrangement, the London Dumping Convention, was concluded in 1972, and came into force in August 1975. *See* page 32, *supra.* The regulation of dumping of nuclear wastes at sea under this Convention is supervised by IMCO, which acts upon technical recommendations by the I.A.E.A. There are also two regional arrangements. The Oslo Dumping Convention was concluded by the Baltic Sea countries in February 1972, several months before the London Dumping Convention. Nuclear Waste disposal at sea is also regulated regionally by the Convention for the Prevention of Marine Pollution from Land-Based Sources concluded at Paris in 1974 by a number of Northeast Atlantic states. Text reproduced in U.N. Doc. ST/LEG/Ser. B/18 at pp. 547-558. For a recent critique of these arrangements, see Lawrence M. Enomoto, "Disposal of Nuclear Wastes at Sea: The 'Hot Potato' in Marine Environmental Law" (publication forthcoming).

185 *See* note 64 *supra,* on early international watercourse agreements of a general environmental nature. For an early post-war example of a specific water pollution initiative, see the 1950 Protocol to Establish a Tripartite Committee on Polluted Waters (signed by Belgium, France, and Luxembourg), Rüster-Simma, vol. XI, p. 5553.

186 Agreement on the Protection of Lake Constance against Pollution (1960), *ibid.,* vol. X, p. 4820. Alphonse Droz, "The Protection of Lake Constance from Pollution", in O.E.C.D., *Environmental Protection in Frontier Regions* (1979) (hereafter "Transfrontier Pollution Studies"), pp. 345-392.

187 Protocol concerning the Constitution of an International Commission for the Protection of the Saar against Pollution, (1961), Rüster-Simma, vol. XI, p. 5613. Uta Hülshoff, '"Protection of the Environment in the Saar-Lorraine region", in *Transfrontier Pollution Studies, op. cit.,* pp. 266-281.

188 Protocol concerning the Constitution of an International Commission for the Protection of the Mosel against Pollution (1961), Rüster-Simma, vol. XI, p. 5618. Hülshoff, *op. cit.*

189 Convention concerning the protection of the Waters of Lake Leman against Pollution (1962), Rüster-Simma, vol. X, p. 4872. A. Droz, "Protection of Lake Leman Waters Against Pollution," in *Transfrontier Pollution Studies, op. cit.,* pp. 393-430.

190 Agreement concerning the International Commission for the Protection of the Rhine against Pollution (1963), Rüster-Slmma, vol. X, p. 4820. In 1976 two additional instruments were signed: the Convention on the Protection of the Rhine against Pollution by Chlorides, by the riparian states (France, the Federal Republic of Germany, Luxembourg, the Netherlands, and Switzerland) 16 *Int'l leg. Mats.* 265 (1977); and the Convention of the Protection of the Rhine against Chemical Pollution, by the riparian states and the E.E.C. (*ibid.* 242). For comments, see Günther Hartkopf "International Protection of the Rhine", (1975), 1 *Environ. Policy and Law* 166; and A.C. Kiss and M. Prieur, "Institutional and Administrative Practice Relating to the Management of the Environment and Land-Use Planning in the Frontier Regions of the Upper Rhine Basin", in *Transtrontier Pollution Studies, op. cit.,* pp.231-265.

191 *E.g.,* the Polish-Soviet Agreement concerning Use of Water Resources in Frontier Waters (1964), (552 *U.N.T.S.* 188; Rüster-Simma, vol. IX, p. 4641); and Austrian-Czechoslovakian Treaty concerning the Regulation of Water Management Questions relating to Frontier Waters (727 *U.N.T.S.* 352).

192 Rüster-Simma, vol. XI, pp. 5740, 5759, 5764, 5779.

193 *Ibid.,* pp. 5728, 5744, 5748, 5763, 5785. The most important of these initiatives were the 1968 European Water Charter, the Draft European Convention on the Protection of Fresh Water against Pollution (1969), and the Draft European Convention for the Protection of International Watercourses against Pollution (1974). In 1968 members of the Council of Europe concluded the European Agreement on the Restriction of the Use of

Certain Detergents in Washing and Cleaning Products. Rüster-Simma, vol. XI, p. 5657.

194 The most important of the E.E.C. Initiatives are the Council Directives on water quality. Rüster-Simma, vol. XI, pp. 5814, ff.

195 23 *U.S.T.* 301; *T.I.A.S.* No. 7312; Rüster-Simma, vol. X, p. 5292; Barros-Johnston, pp. 127-153.

196 Article II.

197 Article III and Annex I.

198 Article IV.

199 Article V. These areas are: (i) pollution from municipal sources; (ii) pollution from industrial sources; (iii) entrophication; (iv) pollution from agricultural, forestry and other land use activities; (v) pollution from shipping activities; (vi) pollution from dredging activities; (vii) pollution from onshore and offshore facilities; (viii) contingency planning; and (ix) the identification of hazardous polluting substances.

200 *The Torrey Canyon* (British Home Office report, cmnd. 3246, 1967); Crispin Gill, *et al.*, *The Wreck of the Torrey Canyon* (1967); M'Gonigle and Zacher *supra* note 132, pp. 143-154.

201 Johnston, *supra* note 136, at p. 70.

202 On the history of marine pollution diplomacy leading up to Stockholm, *see* Douglas M. Johnston, "International Environmental Law: Recent Developments and Canadian Contributions", In Ronald St. J. Macdonald, Gerald L. Morris, and Douglas M. Johnston, *eds., Canadian Perspectives on International Law and Organization* (1974), pp. 555-611.

203 National Academy of Sciences, *Petroleum in the Marine Environment* (1975).

204 For the distinction between accidental and operational discharges, and between tanker and non-tanker discharges, see M'Gonigle and Zacher, *supra* note 132, pp. 15-23.

205 "Almost all of (the) effluents and emissions from land are man-made: familiar waste products of our technological civilization. Accordingly, 80 per cent of the problem of controlling marine pollution is complicated by all the difficulties that stand in the way of a universal approach to pollution control problem on land ... Even the most ardent environmentalist states,... having assumed a leadership role in marine pollution diplomacy, have had to acknowledge the present intractability of 80 per cent of the problem and concentrate their diplomatic energy on the remaining 20 per cent: the problem of reaching agreement on adequate measures to curb the dumping and discharge of wastes at sea". Johnston, *supra* note 136, at pp. 70-71.

206 Robert I. McLaren, "Pollution Probe in the Global Village" (1974-75), 30 *Int'l J.* 127 at p. 130.

207 Lawrence Juda, "IMCO and the Regulation of Ocean Pollution from Ships" (1977), 26 *Int'l & Comp. L.Q.* 558, at pp. 562-564.

208 McLaren, *op. cit., passim;* M'Gonigle and Zacher, *op. cit.,* pp. 39-44; and Juda, *op. cit.,* pp. 560-562.

209 A "forum" organization is one which is useful above all for exchanging views, negotiating, providing collective legitimization, and serving as an outlet for propaganda. Robert W. Cox and Harold K. Jacobson, *The Anatomy of Influence: Decision-making in International Organization* (1973), pp. 5-6, quoted by Juda, *op. cit.,* at pp. 559-560, note 10. This transformation of IMCO is believed to be consistent with a change in the nature of international organizations generally. "We need to think of international organizations less as institutions than as clusters of intergovernmental and transgovernmental networks associated with the formal institutions." Robert O. Keohane and Joseph S. Nye, *Power and Interdependence: World Politics in Transition* (1977) at p. 240. In other words, "within these organizations there is such a continuous flow of officials dealing with such a variety of issues that the function of the agency may be as much a place to 'activate potential coalitions' as to engage in more formalized undertakings." M'Gonigle and Zacher, *op. cit.,* at p. 10.

210 Edgar Gold and Douglas M. Johnston, "Ship-generated Marine Pollution: The Creator of Regulated Navigation", (presented at the 13th Annual Conference of the Law of the Sea Institute at Mexico City, October 20th-24th, 1979). For a more critical view of IMCO, see Eldon V.C. Greenberg, "IMCO: An Environmentalist's Perspective" (1976), 8 *Case Western Res. J. Int'l L.* 131.

211 3 *U.S.T.* 2989; *T.I.A.S.* No. 4900; 327 *U.N.T.S.,* 3; Rüster-Simma, vol. I, p. 332.

212 Articles II and III.

213 For example, non-tankers are exempted from the prohibition when the discharge is made "as far as practicable from land" and the enforcement provisions are very weak. Johnston, *supra* note 136, pp. 75-76. For texts of amendments, see 17 *U.S.T.* 1523; *T.I.A.S.* No. 6109; 600 *U.N.T.S.* 332; Rüster-Simma, vol. I, p. 358; 28 *U.S.T.* 1205, *T.I.A.S.* No. 8505; Rüster-Simma, vol. I, p. 366; and 11 *Int'l Leg. Mats.* 267; Rüster-Simma, vol. I, p. 376.

214 Rüster-Simma, vol. 1, p. 470; 9 *Int'l Leg. Mats.* 45 (1920).

215 26 *U.S.T.* 765; *T.I.A.S.* No. 8068; Rüster-Simma, vol. I, p. 460,

216 Johnston, *supra* note 136, pp. 78-79; Juda, *op. cit.,* pp. 564-566. The 1969 Civil Liability Convention was amended by a 1976 Protocol. 16 *Int'l Leg. Mats.* 617 (1977).

217 Article 1(1).

218 Johnston, *op. cit.,* p. 78; Schneider, *supra* note 48, pp. 153-154.

219 9 *Int'l Leg. Mats.* 64 (1970).

220 Rüster-Simma, vol. 11, p. 504, 11 *Int'l Leg. Mats.* 284 (1972).
221 14 *Int'l Leg. Mats.* 959 (1975).
222 These Regulations are being replaced by the 1972 Convention on the International Regulations for Preventing Collisions at Sea. 28 *U.S.T.* 3461; *T.I.A.S.* No. 8587.
223 18 *U.S.T.* 1857; *T.I.A.S.* No. 6331.
224 Johnston, *supra,* note 136, pp. 79-80. The 1971 Convention was amended by a 1976 Protocol. 16 *Int'l Leg. Mats.* 621 (1977).
225 The Oslo Convention on the Control of Marine Pollution by Dumping from Ships and Aircraft (1972), Rüster-Simma, vol. II, p. 530, 11 *Int'l Leg. Mats.* 262 (1972).
226 In April 1972 draft articles for a new convention, similar to the Oslo Model, were approved and forwarded for further consideration and appropriate action to the forthcoming Stockholm Conference on the Human Environment. Johnston, *op. cit.,* pp. 88-89.
227 26 *U.S.T.* 2403; *T.I.A.S.* No. 8165; Rüster-Simma, vol. II, p. 537; Barros-Johnston, pp. 250-262.
228 Annex I. This list includes organohalogen compounds; mercury and mercury compounds; cadmium and cadmium compounds; persistent synthetic materials; various categories of oil; "high-level radioactive wastes or other high-level radioactive matter, defined on public health, biological or other grounds, by the competent international body in this field, at present the International Atomic Energy Agency, as unsuitable for dumping at sea"; and "materials in whatever form ... produced for biological and chemical warfare". *See* the 1978 devised definition and recommendations of the International Atomic Energy Agency concerning radioactive matter referred to in this Convention. 18 *Int'l Leg. Mats.* 826 (1979).
229 Article IV 1(a).
230 Annex II.
231 Article IV i(b).
232 Article IV 1(c).
233 For example, it was not possible to fix on the parties a legal responsibility for extraterritorial environmental damage caused by dumping (*cf.* Article X); nor to make the Convention applicable to "those vessels and aircraft entitled to sovereign immunity under international law" (Article VII (4)), although most vessels and aircraft engaged in dumping fell into the category; and broadly worded "escape clauses" had to be injected into Article V.
234 Article Vll(1) provides, *inter alia*, that each party to the Convention shall apply prohibitive or restrictive measures to "... vessels and aircraft and fixed or floating platforms under its jurisdiction believed to be engaged in dumping". Given the expansion of coastal state and port state jurisdiction for purposes of environmental preservation at UNCLOS III, it remains to be seen how this phrase "under its jurisdiction" is likely to be interpreted in the 1980s.
235 Rüster-Simma, vol. II, p. 552; 12 *Int'l Leg. Mats.* 1319 (1973).
236 "Harmful substance" Is defined as "any substance which, if introduced into the sea, is liable to create hazards to human health, to harm living resources and marine life, to damage amenities or to interfere with other legitimate uses of the sea, and includes any substance subject to control by the persent Convention". Article 2.
237 Article 3.
238 Article IV.
239 Annex I.
240 Juda, *supra* note 207, pp. 570-571.
241 Annex II.
242 *Category I:* substances which if released into the sea from tank cleaning or de-ballasting operations would present a *major hazard* to either marine resources or human health or cause *serious harm* to amenities or other legitimate uses of the sea and therefore justify the application of *special measures* to prevent their escape into the marine *environment*;
Category II: substances which when released into the sea from tank cleaning or de-ballasting operations may present a *hazard* to either marine resources or human health or cause *harm* to amenities or other legitimate uses of the sea and therefore justify the application of special antipollution measures;
Category III: substances which when released into the sea from tank cleaning and de-ballasting operations may present a *minor hazard* to either marine resources or human health or cause *minor harm* to amenities or other uses of the sea and therefore require *special operational conditions*; and
Category IV: substances, which when released into the sea from tank washing or de-ballasting operations are presently considered to present *negligible harm* to marine resources or human health or amenities or cither legitimate uses of the sea. Discharge of substances in the "black" list (Category I) is forbidden; discharge of those in the "white" list (Category IV) is unregulated; and the substances listed in the "dark grey" and "light grey" categories (II and III respectively) may not be discharged intentionally except when specified conditions have been satisfied.
243 Rüster-Simma, vol. II, p. 656; 17 *Int'l Leg. Mats.* 1103 (1978). *See* Juda, *supra* note 207, pp. 571-572.
244 For other parts of the Final Act, *see* Rüster-Simma, vol. II, pp. 660 ff.

245 Greenberg, *supra* note 210, at pp. 139-141. For a less critical view, *see* Thomas A. Mensah, "International Conventions Concerning Oil Pollution at Sea", *Ibid.* 110.
246 Johnston, *supra* note 136, pp. 98-101.
247 There may be some confusion over interpretation of Article 3(1), which provides that the 1973 Convention applies not only to flag states which are party to it, but also to ships "which operate under the authority of a Party". Yet important technical improvements were introduced into the 1973 Convention by the Protocol of 1978. 17 *Int'l Leg. Mats.* 546 (1978).
248 Prevention of Oil Pollution Act, 1971. (An Act to Consolidate the Oil in Navigable Waters Acts 1955 to 1971 and section 5 of the Continental Shelf Act, 1964), Public General Acts 1971, c. 60.
249 Section 12(3).
250 Section 12(4)(b).
251 Section 16(1).
252 *See*, for example, R. Michael M'Gonigle and Mark W. Zacher, "Canadian Foreign Policy and the Control of Marine Pollution", in Barbara Johnson and Mark W. Zacher, *eds., Canadian Foreign Policy and the Law of the Sea* (1977), pp. 100-157, especially 106-123.
253 Such a right, albeit a limited and conditional one, seems to be vested in the coastal state under Article 211 of the Caracas Convention on the Law of the Sea. For the special case of ice-covered waters, see also Article 234 of th same text. On the prerogative of coastal states with respect to vessel traffic control, *see* Gold and Johnston, *supra* note 210, *passim.*
254 On "the century of technocracy", *see* William G. Carleton, *Technology and Humanism: Some Exploratory Essays for Our Time* (1970), pp. 254-274. Carleton defines "technocracy" as the technicalized, cybernetic, computerized society run by scientists, engineers, and "technicians". This society, he asserts, is "drastically changing man's institutions and transforming his behaviour and basic values" (p. 225).
255 Jacques Ellul, *Technological Society* (paperback ed., 1967).
256 "By exaggerating the situation, Ellul creates a despair so profound as to render resistance hopeless, leaving many who accept what he has to say with the conviction that the only dignified thing left to do is to await the end, savouring one's knowledge of its inevitability, like a figure in a Greek tragedy". Victor C. Ferkiss, *Technological Man: the Myth and the Reality* (1969), at p. 87.
257 Emmanuel G.M. Mesthene, *Technological Change: Its Impact on Man and Society* (1970), pp. 15-20.
258 Kenneth E. Boulding, *The Meaning of the 20th Century: The Great Transition* (1964), at p. 26.
259 Ferkiss, *op. cit.*
260 Garrett Hardin, "The Tragedy of the Commons" (1968), 162 *Science* 1243. In feudal England, Hardin reminds us, all the farmers grazed their flocks in the commons, but when a few realized they could secure an advantage by grazing more animals than their neighbours, it was only a matter of time before others followed suit, with the result that the commons were eventually destroyed by overgrazing. If no individual (owner) or institution (regulatory agency) is responsible for the protection of a free, common property, resource, then it is likely to be destroyed completely. Hardin's "iron logic" leads him to call for population control as the only solution. Critics have condemned this as a "new barbarism" to be resisted. Barry Commoner, *The Closing Circle* (1972), pp. 296-297.
261 *E.g.* John Kenneth Galbraith, *The Affluent Society* (1958) and *The New Industrial State* (1967); and Kenneth E. Boulding, *Beyond Economics* (1968).
262 Hazel Henderson, *Creating Alternative Futures: The End of Economics* (1978), pp. 303-326.
263 *Ibid.*
264 Michel J. Crozier, Samuel P. Huntington, and Joji Watanuki, *The Crisis of Democracy* (Report on the Governability of Democracies to the Trilateral Commission, 1975).
265 Michel Crozier, "Western Europe", *ibid.,* pp. 30-33.
266 Samuel P. Huntington, "The United States", *Ibid.,* pp. 76-85.
267 *Ibid.,* at p. 104.
268 *Ibid.,* at p. 159.
269 *Ibid.,* pp. 163-164.
270 *Ibid.,* pp. 166-168.
271 Michael Harrington, *The Accidental Century* (Pelican paperback ed. 1965), pp. 13-42. Similarly, it might be argued, technology should not be "carelessly" applied to the developing nations. M. Taghi Farvar and John P. Milton, *eds., The Careless Technology: Ecology and International Development* (1972).
272 Events may conspire, sometimes quite unexpectedly, to feed suspicions that international law is an ineffective system of constraints. Recently the age-old institution of diplomatic immunity has been under strain. Whether this new "crisis" in international law will weaken or strengthen the system in the 1980s is difficult to forecast.
273 Douglas M. Johnston and Edgar Gold, "Extended Jurisdiction: The Impact of UNCLOS III on Coastal State Practices" (presented at the 13th Annual Conference of the Law of the Sea Institute at Mexico City, October 20th-24th, 1979).
274 Elisabeth Mann Borgese, *The Drama of the Oceans* (1975), pp. 83-188.

275 Brenda Horsfield and Peter Bennett Stone, *The Great Ocean Business* (1972).
276 Johnston, *supra* note 96, pp. 77-80, 321-326.
277 Courtlandt Canby, *A History of Ships and Seafaring* (1963).
278 R.B. Oram, *Cargo Handling and the Modern Port* (1965), pp. 64-80; A.A. Evans, *Technical and Social Changes in the World's Ports* (1969), pp. 5-40.
279 On the "technological revolution" in the shipping industry and its trade implications, see S.A. Lawrence, *International Sea Transport: The Years Ahead* (1972), pp. 153-195.
280 Borgese, *op. cit., passim;* and Douglas M. Johnston, "Law, Technology and the Sea" (1967), 55 *Calif. L Rev.* 449.
281 Jon L. Jacobson, "Future Fisheries Technology and the Third Law of the Sea Conference", in H. Gary Knight, *ed., The Future of International Fisheries Management* (1975), pp. 51-92.
282 H. Gary Knight, J.D. Nyhart, and Robert E. Stein, *eds., Ocean Thermal Energy Conversion: Legal, Political and Institutional Aspects* (1977).
283 Richard A. Frank, *Deepsea Mining and the Environment* (American Society of International Law, Report of the Working Group on Environmental Regulation of Deepsea Mining, 1976).
284 Colin Moorcraft, *Must the Seas Die?* (1973); Wesley Marx, *The Frail Ocean* (1969).
285 Margaret Deacon, *Scientists and the Sea, 1650-1900* (1971); and Richard Carrington, *A Biography of the Sea* (1960).
286 Henry Charnock, "H.M.S. *Challenger* and the Development of Marine Science", in Pirie, *supra* note 134, pp. 24-33.
287 William A. Anikouchine and Richard W. Sternberg, *The World Ocean: An Introduction to Oceanography* (1973).
288 Claiborne Pell (with Harold L. Goodwin), *Challenge of the Seven Seas* (1966), p. 166.
289 Johnston, *supra* note 96, pop. 49-55, 59-68.
290 Wit a view to placing the forthcoming fishery negotiations at UNCLOS I on a proper scientific footing, the Food and Agriculture Organization of the United Nations convened a meeting of experts at Rome in 1955. At this meeting, It was agreed that "the principal objective of conservation of the living resources of the sea is to obtain the optimum sustainable yield so as to secure a maximum supply of food and other marine products". *Report of the International Technical Conference on the Conservation of the Living Resources of the Sea* (FAO, 1955), para. 18. This biological objective of optimum sustainable yieid was accepted more or less uncritically by the International Law Commission, which was called upon by the General Assembly to prepare draft articles for the codification of the law of the sea; and the same objective found its way into Article 2 of the Convention on Fishing and Conservation of the Living Resources of the High Seas. *Ibid.*, pp. 411-413; Christy and Scott, *supra* note 46, pp. 6-16. On the alternative, economic, objective of maximization of rent, *see Ibid.,* pp. 215-230.
291 Johnston, *supra* note 96, pp. 335-338.
292 Johnston, *supra* note 202, pp. 555-559.
293 Neil Gunningham, *Pollution, Social Interest and the Law* (1974), at p. 30.
294 *Ibid.,* pp. 30-31.
295 Paul R. Ehrlich, *The Population Bomb* (1968).
296 Dennis Pirages and Paul R. Ehrlich, *Ark II: Social Response to Environmental Imperatives* (1974).
297 Gunningham, *op. cit.,* pp. 35-41. See also Stuart S. Nagel, *ed., Environmental Politics* (1974), and Lester W. Milbraith and Frederick R. Inscho, *eds., The Politics of Environmental Policy* (1975).
298 David Sive, "From Conservation to Environmental Law", in John A. Day, Frederic F. Fost, and Peter Rose, *eds., Dimensions of the Environmental Crisis* (1971), pp. 159-168.
299 Rachel L. Carson, *The Sea Around Us* (rev. ed., 1961), pp. xi-xiii.
300 McLaren, *supra,* note 206, *passim.*
301 M'Gonlgle and Zacher, *supra* note 132, pp. 154-155.
302 *Ibid.,* p. 42.
303 Gold and Johnston, *supra,* note 210.
304 (U.S.) National Academy of Sciences, *Institutional Arrangements for International Environmental Co-operation* (1972).
305 *See*, for example, the ocean-related activities of UNEP outlined in the 1979 Report of the Executive Director, Doc. UNEP/GC.7/7 (14 Feb. 1979), pp. 100-112.
306 For the text of Pardo's 1967 speech, "Ocean Space, Seabed, Common Heritage of Mankind", *see* Elisabeth M. Borgese, *ed., The Common Heritage: Selected Papers on Oceans and World Order, 1967-1974*, by Arvid Pardo (1975) pp. 1-41.
307 On his conception of "ocean space", Arvid Pardo, "Development of Ocean Space — An International Dilemma" (1970), 31 *La. L. Rev.* 45.
308 Pardo suggested three characteristics for the common heritage principle: the absence of any property (even common property); the management of all uses (not merely resources); and the sharing of benefits. Borgese, *supra* note 306, p. xi.

309 "All parts of the marine environment have, ofcourse, always been interconnected. Now, however, this interconnection extends to our uses of the marine environment. Instead of thinking of the marine environment in terms of distinct strata, we are increasingly being constrained to think of it in terms of ocean space as an indivisible whole". Arvid Pardo, *"From Seabed Regime to Ocean Space Regime", ibid.,* pp. 189-226, at. p. 190.

310 Arvid Pardo, "Pollution and Pollution Control", *ibid.,* pp. 273-281.

311 Arvid Pardo, "Scientific Research as International Public Interest, *Ibid.* pp. 283-291.

312 Arvid Pardo, "IMCO and Pollution Control", *ibid.,* pp. 293-298.

313 Early overviews include: Study of Critical Environmental Problems (SCEP), *Man's Impact on the Global Environment* (M.I.T., 1971); William H. Matthews and others, *Man's Impact on Terrestrial and Oceanic Ecosystems* (1971); and Barbara Ward Jackson and Rene Dubos, *Only One Earth: The Care and Maintenance of a Small Planet*, (1972). For a recent re-appraisal, conducted from a humanistic perspective, *see* John McHale, *Basic Human Needs: A Framework for Action* (A Report to the U.N. Environment Programme, 1977).

314 Of the 109 Recommendations approved at the Stockholm Conference, nine (86-94) dealt specifically with the problems of marine pollution, nine others (32, 33, 37, 46, 47, 48, 49, 50, 55) with fishery conservation and aspects of the preservation of the marine environment, and dozens of others with more general matters relevant to the ocean as well as the other sectors of the biosphere.

315 At least 19 of the Stockholm Recommendations seem to reflect an awareness of the importance of legal development: namely, Recommendations 32, 33, 37, 45, 46, 47, 48, 50, 51, 53, 55, 72, 82, 84, 85, 86, 92, 98, and 102.

316 *See*, for example, Recommendations 86 and 93.

317 Article I of FAO's Constitution puts maximum emphasis on "nutrition, food and agriculture". The text is reproduced in FAO, *Basic Texts*, Vol. 1, pp. 7-26 (Rome, 1964).

318 UNESCO's first major effort in natural resources research was in the field of arid zones and humid tropics research. In more recent years this has expanded to a wider range of ecological studies, as well as studies in hydrology, geology, and the soil sciences. For reports on UNESCO's conservation activities, *see* its *Nature and Resources* newsletter.

319 J.R. Baker, *Julian Huxley: Scientist and World Citizen, 1887 to 1975* (UNESCO, 1978), p. 39; Huxley, *supra* note 12, pp. 287-289; and Huxley, *supra* note 14, pp. 61-62. IUCN, founded in 1948, has been described (in *World Wildlife Yearbook 1969* at p. 6) as "an independent international body whose membership comprises states irrrespective of their political and social systems, government departments and private institutions as well as international organizations ... IUCN's main purpose is to promote or support action which will ensure the perpetuation of wild nature and natural resources on a worldwide basis not only for their intrinsic cultural or scientific values but also for the long-term economic and social welfare of mankind ... IUCN believes that its aims can be achieved most effectively by international effort in co-operation with other international agencies such as UNESCO and FAO". Since 1973 it has also co-operated with UNEP. For continuing reports on its activities, *see IUCN Bulletin.*

320 Nicholson, *supra* note 1, pp. 158-162.

321 *Ibid.,* pp. 226-227.

322 World Wildlife Foundation, established in 1961, is "an international charitable organization for saving the world's wildlife and wild places ... Its aim is to support the conservation of nature in all its forms (landscape, soil, water, flora and fauna) by raising funds and allocating them to projects, by publicity, and the education of the general public and young people in particular. For all these activities it takes scientific and technical advice from IUCN". *World Wildlife Yearbook 1969* at p. 6.

323 The IBP was originally conceived in 1959 as a biological counterpart to the International Geophysical Year, and with the support of the International Union of Biological Sciences and IUCN it gradually acquired operational viability in the 1960s. On the origins and early history, see E.B. Worthington, *ed., The Evolution of IBP* (IBP, 1975), pp. 3-16. On the constraints under which IBP has operated, see Nicholson, *supra* note 1, pp. 61, 251. On the conservation side of IBP, *see* Max Nicholson, *Handbook to the Conservation Section of the International Biological Programme* (IBP Handbook No. 5, 1968).

324 In the mid-1970s, the International Co-ordinating Council of MAB shifted from planning to operational activities. Among its ocean-related areas of research, MAB has developed projects that focus on island systems and emphasized biosphere reserve problems in coastal areas. For a general overview of MAB's current activities, *see* UNESCO Doc. 20C/78 (28 July 1978), "Report on the Man and the Biosphere Programme".

325 Cyrille de Klemm, "Species and Habitat Reservation: An International Task" (1975), 1 *Environ. Policy and Law 10* at p. 10.

326 *Ibid.,* at pp. 10-11.

327 638 *U.N.T.S.* 185; Rüster-Simma vol. 4, p. 1791.

328 Rüster-Simma, vol. 5, p. 2161.

329 Text in *U.N. Judicial Yearbook 1972*, pp. 89-99; and Rüster-Simma, vol. 14, p. 7238. This Convention was

adopted by the UNESCO General Conference at Paris in November 1972. It will enter into force when it has been ratified by 20 states. *IUCN* Bulletin, vol. 4, No. 2,February 1973) p. 6.

330 27 *U.S.T.* 1087; *T.I.A.S.* No. 8249; 12 *Int'l Leg. Mats.* 1085 (1973). After 10 years of promotional effort by IUCN, this Convention was opened for signature early in 1973. It entered into force on July 1, 1975, and now has a membership of over 60 states. The first meeting of the Parties was held in Bern in November 1976 (*see* Supplement to *IUCN Bulletin*, vol. 8, No. 4 (April 1977) and 2 *Environ. Policy and Law* 167), the second in San Jose, Costa Rica, in March 1979 (5 *Environ. Policy and Law* 82) and the third in New Dehli, February 1981. *See* also Michael W. Gordon, "The Third World and the Protection of National Patrimony: Oil, Art and Orchids" (1979), 2 *Hastings Int'l & Comp. L. Rev.* 281.

331 *T.I.A.S.* No. 8409, 13 *Int'l Leg. Mats.* 13 (1974). Text also reproduced in supplement to *IUCN Bulletin*, vol. 4, No. 12 (December 1973). Also initiated by IUCN, this instrument entered into force on 26th May, 1976, upon the ratifications of Norway, Canada and the Soviet Union.

332 For example, the Convention for the Conservation of Antarctic Seals, signed on 1st June, 1972, and entered into force on 11th March, 1978 (text in Rüster-Simma, vol. 8, p. 3753); and the Bonn Convention on the Conservation of Migratory Species of Wild Animals, signed in June 1979 (*see IUCN Bulletin,* vol. 10, No. 6 (June 1979), p. 1; 5 *Environ. Policy and Law* 138). Also in the early 1970s UNEP was beginning to work toward the formulation of Draft Principles of Conduct in the Field of the Environment for the Guidance of States of the Conservation and Harmonious Utilization of National Resources Shared by Two or More States (*see* 17 *Int'l Leg. Mats.* 1097 (1978)).

333 *See* pages 14-17, *supra*.

334 "Fishery objectives for underdeveloped and developing countries must be framed in broader terms than for developed countries, since they are more immediately related to general social and political objectives, such as the social welfare of the fisherman, the economic conditions of the fishing community, and the nutritional level of the national population". Johnston, *supra* note 96 at p. 48.

335 On the theory of stock depletion, *see* Christy and Scott, *supra* note 46, pp. 74-86.

336 Oscar Schachter, *Sharing the World's Resources* (1977), pp. 37-64.

337 U.N. Charter, Article 13(1)(a). By Article 15 of the Statute of the I.L.C., "the progressive development of international law" is defined as "the preparation of draft conventions on subjects which have not yet been regulated by international law or in regard to which the law has not yet been sufficiently developed in the practice of States"; and "the codification of international law" is defined as "the more precise formulation and systemization of rules of international law in fields where there already has been extensive state practice, precedent and doctrine".

338 The Conference met from 24th February to 27th April 1958.

339 Most of this advice was given in the *Report of the International Technical Conference on the Conservation of the Living Resources of the Sea* (Rome, 1955).

340 17 *U.S.T.* 138 (1966); *T.I.A.S.* No. 5969; 559 *U.N.T.S.* 285.

341 This Convention was concluded in Geneva at the same time as three other instruments: the Convention on the High Seas; the Convention on the Territorial Sea at the Contiguous Zone; and the Convention on the Continental Shelf.

342 Article 2.

343 Christy and Scott, *supra* note 46, pp. 215-230.

344 Articles 7(2), 8(1), and 10(1).

345 Articles 4(1), 5(2), 6(4), 7(1), 8(1), 9(1), and 12(1).

346 Article 1(1) provides that "(a)" States have the right for their nationals to engage in fishing on the high seas subject (a) to their treaty obligations....."

347 Article 6(1).

348 Article 1(2).

349 Article 9(1).

350 Article 10(1)(a)(i). For a similar characterization of the Geneva system, see H. Gary Knight, "International Fisheries Management: A Background Paper", in Knight, *ed., The Future of International Fisheries Management* (1975), pp. 1-49, at p. 11.

351 The Convention came into force on March 20th, 1966, almost eight years after it was opened for signature.

352 The Northeast Atlantic Fisheries Commission (1959); the Joint Commission for the Black Sea (1959); the Mixed Commission for Fishing (the German Democratic Republic, Poland, and the Soviet Union, 1962); the (FAO) Regional Fisheries Advisory Commission for the Southwest Atlantic (1962); the Joint Fisheries Commission (Japan and Republic of Korea, 1965); the International Commission for the Conservation of Atlantic Tunas (1966); the (FAO) Indian Ocean Fishery Commission (1967); the (FAO) Fishery Committee for the East Central Atlantic (1967); and the International Commission for the Southeast Atlantic Fisheries (1969).

353 For an evaluation, *see* Koers, *supra* note 95. *See* also Charles B. Heck, "Collective Arrangements for Managing Ocean Fisheries" (1925), 29 *Int'l Organ.* 711.

354 *E.g.*, the International Pacific Halibut Commission, the International Pacific Salmon Fisheries Commission, the International North Pacific Fisheries Commission, and the Japan-Soviet Fisheries Commission

for the Northwest Pacific.

355 FAO's Committee on Fisheries (COFI) is authorized, among other things, to recommend the establishment of new international fishery organizations. Upon COFI's recommendations, the Fishery Committee for the Eastern Central Atlantic and the Indian Ocean Fishery Commission were brought into existence to provide advisory services within the FAO structure, similar to the Regional Fisheries Advisory Commission for the Southwest Atlantic which was established, before the existence of COFI, under Article 6(1) of the FAO Constitution. Koers, *op. cit.,* pp. 106-113. The Indio-Pacific Fisheries Council and the General Fisheries Council for the Mediterranean also have an advisory function within the general framework of FAO, but they came into existence, under Article 14 of the FAO Constitution, on the basis of separate conventions and enjoy a certain degree of autonomy.

356 *E.g.,* the International Commission for the Conservation of Atlantic Tunas and the International Commission for the Southeast Atlantic Fisheries.

357 The International Council for the Exploration of the Sea, the International Council for the Scientific Exploration of the Mediterranean, the Commission for Fisheries Research in the Western Pacific, and the FAO's Advisory Committee on Marine Resources Research.

358 *E.g.,* the Japan-Republic of Korea Joint Fisheries Commission and those named in footnote 355, *supra.*

359 The Permanent Commission of the Conference on the Use and Conservation of the Marine Resource of the South Pacific and the Mixed Commission for Fishing.

360 *E.g.,* the International Pacific Salmon Fisheries Commission and the International North Pacific Fisheries Commission.

361 *Viz.,* the International Pacific Halibut Commission, the International Pacific Salmon Commission, the North Pacific Fur Seal Commission, the International Whaling Commission, the Inter-American Tropical Tuna Commission, and the International Commission for the Conservation of Atlantic Tunas.

362 The trend has been toward area-oriented, rather than species-oriented, commissions. Of the nine commissions listed in footnote 352 *supra*, all but one are area-oriented.

363 Johnston and Gold, *supra* note 273.

364 *See* pages 30-34, *supra.*

365 *Cf.* Max Blumer, "Scientific Aspects of the Oil Spill Problem" in Pirie, *supra* note 84, pp. 322-333.

366 For a recent review of several of these factors, *see* Daniel P. Finn, *et al, Oil Pollution from Tankers in the Straits of Malacca: A Policy and Legal Analysis* (The East-West Center, Open Grants Papers No. 6, 1979) pp. 44-63.

367 *See* page 29 *supra*

368 *See* pages 29-30, *supra*

369 Johnston, *supra* note 202, pp. 559-567.

370 For a detailed analysis, see Louis B. Sohn, "The Stockholm Declaration on the Human Environment" (1973), 14 *Harv. Int'l L.J.* 423.

371 The official texts are contained in the Report of the *U.N. Conference on the Human Environment*, U.N. Doc. A/CONF. 48/14.

372 (1973) 10 *U.N. Monthly Chronicle* 72-73. *See* also *Yearbook of the United Nations* 1972, pp. 323-329.

373 *See* U.N. Doc. RS/PAC (5 January 1980) ("Information on UNEP's Regional Seas Programme").

374 *See*, for example, Obed Y. Asamoah, *The Legal Significance of the Declaration of the General Assembly of the United Nations* (1966).

375 Sohn, *op. cit.,* pp. 424-434.

376 *Ibid.,* pp. 439-440.

377 The Canadian representative at Stockholm (Ambassador J. Alan Beesley) went so far as to describe the Declaration as "a first step toward the development of internation environmental law". U.N. Doc. A/CONF. 48/14 at p. 115.

378 Douglas M. Johnston, (commentary) in Gamble, *supra* note 174 at pp. 509-513.

379 Sohn, *op. cit.,* p. 514.

380 *Ibid.,* p. 515.

381 On 18th December 1967 the U.N. General Assembly responded to Pardo's famous speech six weeks earlier by establishing the *ad hoc*-Committee to Study the Peaceful Uses of the Seabed and the Ocean Floor beyond the Limits of National Jurisdiction. On 21st December 1968 this *ad hoc* body was replaced by the Committee on the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction. On 17th December 1970 the U.N. General Assembly decided to convene In 1973 a conference on the law of the sea which would deal not only with the establishment of an equitable international regime for the extra-national areas indicated in the titles of the two preparatory committees, but also with a broad range of law of the sea issues, including *inter alia* the preservatin of the marine environment, the conservation of living resources and scientific research. The Committee was requested to prepare a comprehensive list of subjects and issues related to the law of the sea and to present draft articles on these matters in preparation for the forthcoming conference.

382 Despite the limited terms of reference of the *ad hoc*-Committee, environmental issues arose at the earliest

stage. U.N. General Assembly, Official Records of the 23rd Session, *Report of the ad hoc-Committee to Study the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction* (1968), (hereafter "Report of *ad hoc*-Committee"), pp. 14, 48.

383 450 *U.N.T.S. 11;13 U.S.T.* 2312; *T.I.A.S.* No. 5200.

384 515 *U.N.T.S.* 205; 15 *U.S.T.* 1606; *T.I.A.S.* No. 5639.

385 499 *U.N.T.S.* 311; 15 *U.S.T.* 471; *T.I.A.S.* No. 5578.

386 559 *U.N.T.S.* 285; 17 *U.S.T.* 138; *T.I.A.S.* No. 5969.

387 Since many of the states represented on the Seabed Committee had chosen not to adopt one or more of these Geneva conventions, the delegates were faced with the contentious question, how much of the Geneva law existed, or had passed into existence, in *customary* international law, independently of treaty obligation.

388 This freedom is one of several enumerated in Article 2 of the 1958 Convention on the High Seas. It was referred to as early as 1969 at the Seabed Committee discussions of the pollution problem. U.N. General Assembly, Official Records of the 24th Session, *Report of the Committee on the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction* (1969) Hereafter "Seabed Committee Report 1969", pp. 24-26.

389 Fishery conservation questions received little attention until after the General Assembly's decision in December 1970 to expand the work of the Seabed Committee in preparation for UNCLOS III. The earliest detailed discussion took place in Sub-Committee II in the summer of 1971. U.N. General Assembly, Official Records of the 26th Session, *Report of the Committee on the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction* (1971) (hereafter "Sea-Bed Committee Report 1971"), p. 36.

390 This is *not* one of the enumerated freedoms contained in the 1958 Convention on the High Seas, and support for it depended upon the general assumption that it existed in customary international law and was not abridged by that Convention or any other. For an early discussion, *Seabed Committee Report, 1969*, pp. 22-24.

391 *Seabed Committee Report 1971,* p. 39.

392 *Ibid.,* pp. 41-45.

393 *Ibid.,* p. 41.

394 *Ibid.,* pp. 33-37.

395 The most dramatic example of this kind of influence was that of the *Amoco Cadiz* disaster in April 1978 which was referred to frequently at the following session of the conference and led to several textual amendments supported by France and other sympathetic delegations. *See* proposed amendments to Articles 221 and 222 in Third U.N. Conference on the Law of the Sea, *Official Records,* vol. X (1978), pp. 180-181.

396 U.N. General Assembly, Official Records of the 25th Session, *Report of the Committee on the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction* (1970) (hereafter "Seabed Committee Report 1970"), p. 8.

397 *See* pp. 27-29, *supra*.

398 *Seabed Committee Report* 1970, p. 11.

399 For example, the Canadian delegation advocated a "comprehensive approach", by UNCLOS III and other forums, to the preservation of the marine environment and the prevention and control of marine pollution. *See* Canadian Working Paper on Preservation of the Marine Environment, UN Doc. A/AC. 138/SC. III/L26.

400 *Sea-Bed Committee Report 1970*, Annex III (Report of the Secretary-General, "Study of International Machinery") pp. 61-123, at pp. 111-115.

401 Edgar Gold, "The Rise of the Coastal State", in Douglas M. Johnston, *ed., Marine Policy and the Coastal Community* (1976), pp. 13-33

402 IUCN, *Conservation and the New Law of the Sea*, reproduced in part in 4 *Environ. Policy and Law* 126 (1978), at pp. 127-129

403 Draft Convention on the Law of the Sea (Informal Text), Part II

404 *Ibid.,* Part V

405 *Ibid.,* Part VI

406 *Ibid.,* Part IV

407 *See*, generally, Douglas M. Johnston, *ed., Regionalization in the Law of the Sea (1978). The UNCLOS III texts have envisaged various kinds of "regional arrangements". See* L.D.M. Nelson, "The Funcitons of Regionalism in the Emerging Law of the Sea as Reflected in the Informal Composite Negotiating Text", *ibid.,* pp. 12-29

408 *See* J.D. Kingham and D.M. McRae, "Competent International Organizations and the Lw of the Sea" (1979), 3 *Marine Policy* 106. For some difficulties with the concept of "marine region" *see* Maria Edwarda Goncalves, "Concepts of Marine Region and the New Law of the Sea", *ibid.* 255.
See footnote 381, *supra*

409 The list is reproduced in S. Houston Lay, Robin Churchill, and Myron Nordquist, *eds., New Directions in the Law of the Sea* (Vol. III, 1973), pp. 745-749

410 For the agenda of the 1st session of UNCLOS III, see U.N. Doc.A/CONF.62/1, in Third United Nations Conference on the Law of the Sea, *Official Records,* Vol. III, p. 1

411 U.N. Doc.A/CONF.62/29, *Ibid.,* pp. 59-61

Chapter Two
Living Resources of the Ocean

Cyrille de Klemm

A. Introduction

The oceans cover 71 % of the surface of the earth and are the home of about 250.000 species of plants and animals. Each one of these species has its own unique anatomical, physiological and ecological characteristics, as well as its own particular geographic distribution, and is the result of an evolutionary process which has been going on for many thousands of years and is still going on today. In the course of this evolution, each species has become adapted to specific ecological conditions such as water temperature and salinity, the nature of the substrate on which bottom-dwelling species live, and the availibility of the particular plants or animals on which it feeds. It is the combination of these various ecological factors, in their infinite diversity, which is essential to the continuous existence of any species and constitutes its habitat. Conversely, the destruction of a species habitat, through the alteration, degradation or modification of any of its components will almost invariably result in the extinction of the species concerned.

Although only a small number of marine species are at present harvested by man, all need, however, to be preserved because of the part they play in the innumerable and intricate relationships that exist among all species in the marine environment as well as for their role in the various ecosystems of which commercially exploited species form a part. Last but not least, the protection of marine species, like that of any other species, is of paramount importance because of the enormous and practically untapped genetic resources they represent for mankind as potential new sources of food and pharmaceutical or industrial products.

The purpose of this paper is to review the state of the law emerging from the negotiations that have taken place at the Third United Nations Conference on the Law of the Sea(UNCLOS III) as it relates to the conservation and management of living resources, as well as existing institutional arrangements that have been set up to achieve these objectives.

After a short description of the living resources of the sea and of the threats to their continuing existence, consideration is given to the basic legal and institutional requirements which, it is felt, should be met if these are to be preserved. This is followed by a review of the historical development of the law of the sea, as it applied to living resources, until the convening of UNCLOS III. Finally, the last, and most important part of this paper consists in an analytic and critical review of those articles in the new convention that are of importance for the conservation of living resources as well as of other developments, including the conclusion of certain treaties, that have taken place during the time the negotiation of the convention was proceeding.

B. The Living Resources of the Ocean

1. The Nature of the Resources

Among the 250.000 species, at present known to science, which live in the sea, there are approximately 180.000 species of animals and 70.000 species of plants. These range in size from bacteria and microscopic unicellular algae and protozoa to the great whales and the giant kelp and include representatives of many different taxonomic groups, in particular molluscs, crustaceans and fishes. Out of a total number of about 20.000, 12.000 or 60% of all fish species are marine. The distribution of these species is, however, extremely unequal. Primary production in the sea, that is to say vegetable production, is the result of the synthesis by plants, mostly unicellular algae, of organic matter. Plants use as nutrients carbon dioxyde and mineral salts such as phosphates and nitrates that are dissolved in water, and sunlight as a source of energy. Solar radiation being quickly absorbed by the water, primary production can only take place near the surface. Furthermore, mineral nutrients are only really abundant near the coast as they are brought into the sea by run off waters and rivers or rise from the bottom in upwelling areas. Shallow coastal areas are for these reasons the most productive in vegetable matter and, as a result, of the planctonic animals that feed on microscopic algae, of the slightly larger animals which feed on the plancton, of the predators of these animals, and so on, up to the end of the food chain. A very large majority of all ocean species are, therefore, concentrated along the 280.000 miles of coastline that exist in the world or in relatively shallow waters on the continental shelf. As one goes further away from the coast and from the surface the abundance of animals and the diversity of species decreases. Mid-ocean areas are almost completely desertic. Recent discoveries, however, tend to suggest that the number of species living on the deep ocean floor may be far greater than it was thought earlier. Along the coast, species diversity increases with water temperature and reaches a peak in tropical seas. The coral reefs of the Indo-Pacific area or the Caribbean Sea are probably the marine habitats which are the richest in species and where, for instance, 500 to 1.000 different species of fish can occur in a very small area of shallow waters.

Most species have a fairly wide distribution. There are, however, several semi-enclosed seas which are rich in endemic species, that is to say in species that do not exist anywhere else in the world. Examples are the Red Sea, the Gulf of California, the Gulf of Mexico and the China Sea. Endemic species have also evolved along the coasts of oceanic islands. Others became adapted to the tremendous pressures that occur in deep ocean trenches. Many are restricted to very specific types of habitats or can live only if very particular ecological conditions are met. As a result, the maintenance of these habitats or conditions is essential if they are to survive. For most species, including commercial species, certain areas may play a critical part at a particular time of their life cycle. Thus, some species always use the same area, sometimes of very limited size, for spawning or as nurseries for their young. These are called critical habitats since their disappearance or destruction could often result in the severe depletion or extinction of a whole population or species. For many species, critical habitats are located very close to the shore or even in inshore areas such as shallow bays, lagoons or estuaries, mangroves, eel grass beds, coastal marshes or other biologically rich ecosystems.

Many marine species, especially fish, migrate[1], often over considerable distances, from spawning areas to feeding grounds. This is the case of many of the commercial species such as cod or tuna. A small number of species migrate back and forth between fresh water rivers and the sea. When they feed and grow in the ocean and return to fresh waters to spawn, these species are called anadromous. Examples are salmons, sturgeons and shads. When on the contrary, species spend most of their life in fresh waters and migrate to the sea to spawn, they are called catadromous. Examples are eels and the Indo-pacific chanos.

Conservation requirements may, therefore, vary considerably from species to species and must take into consideration ecological conditions, critical habitats and migration patterns.

2. The Exploitation of the Resources

Man has always caught fish in the sea. But until the present century the total catch of marine species had remained very small. In 1938 it reached 15 million tonnes and thereafter kept rising at a very fast rate, going up to 27 million tonnes in 1958 and 64 million tons in 1978. This tremendous increase was necessary because of the greater demand of fish resulting from economic development and from the very large increase in the world population, and it became possible because of the many technological advances that had taken place in the field of fisheries. These ranged from electronic gear to locate fish shoals, to the development of new and more efficient gear such as the purse seine and to the processing and deep-freezing of fish on board. Fishing flotillas composed of a mother-ship providing storage space and supplies and of a number of smaller catchers can now be found on all the richest fishing grounds of the ocean. As a result, whereas in 1955[2] almost all fish stocks, except in the North Atlantic and North Pacific, were unexploited or underexploited, nearly all commercial species fisheries of the world are now fully exploited and in some cases overexploited. This remarkable growth rate which had reached 6% to 7% per year during the 1960s dropped to about 1% to 2% during the 1970s. This decrease was essentially due to a considerable reduction of the number of commercially exploitable fish stocks in respect of which sustainable increase of the catch remained possible. According to the latest scientific estimates made by FAO it does not seem possible to increase the total world catch of conventional marine species by more than 20 or 30 million tonnes[3]. Furthermore, this increase in catches will only be obtained by a better management of the stocks that are already exploited and by considerable investments to develop exploitation of the less commercially attractive stocks. In other words, almost all readily accessible stocks have now become fully exploited and the world production of conventional marine resources is now reaching a ceiling of about 100 million tonnes[4]. It should, furthermore, be pointed out that commercial fish stocks are very unevenly distributed around the world. Indeed, although 99% of exploited resources are found within 200 miles from the coast, high production areas are relatively few and far apart. These areas include, in particular, upwelling regions in California, Peru, North-West Africa, South-West Africa and in certain parts of the Indian Ocean, temperate and sub-polar seas in the northern and southern hemispheres and the shallow waters over the continental shelf.

Non-conventional resources, at present little if at all exploited, include squid, lanternfish, pelagic crabs and krill. These are open ocean resources often found at

considerable distances from the coast. It has been calculated that the world catch of non-conventional species could reach several hundred million tonnes, including 50 million tonnes of krill, provided suitable and economic fishing methods and gear are developed and appropriate market outlets can be found.

Fish is essentially a source of animal proteins. A large part of the world fish catch is consumed directly by man either fresh, or salted, smoked, frozen, canned or otherwise processed. An increasing proportion of the catch, however, is now being converted into fish meal or processed for the extraction of oil. In 1950, 16% of the total world catch was reduced to fish meal and in 1975 more than 31%, that is about 18 million tonnes out of a total catch of nearly 60 million tonnes. The main raw material for the production of fish meal consists of small shoaling coastal pelagic fishes such as herrings, sardines and anchovies. These are found in enormous quantities in some upwelling areas. Unrestricted exploitation accompanied by unfavourable environmental conditions has brought about the depletion of several important stocks of these fishes. Most spectacular has seen, in that respect, the collapse of the anchoveta *(Engraulis ringens)* fishery off the coast of Peru. From a negligible quantity in the 1950s, catches of this species rose to more than 13 million tonnes in 1970, 22% of the world total fish catch. These quantities fell to 2 million tonnes in 1973 and, after a small increase, to an even lower figure in 1977, resulting at the same time in the collapse of the fish meal industry that had been created ashore.

Fish meal is mainly used as animal feed supplement for chickens, pigs and cultivated fresh water fishes such as trout. Marine oils, including baleen whale oil are consumed directly by man. Some oils that are inedible, such as sperm whale oil, have industrial applications.

Among other potential uses of marine species in the future are the harvesting or use of certain marine organisms for the production of pharmaceutical products[5] and the cultivation of sea weeds in large quantities as a source material for the production of food, chemicals and energy[6].

Finally, considerable developments are very likely to take place in the near future in respect of aquaculture or fish farming. This will, however, be dependent upon the maintenance of wild populations of the cultivated species if the genetic diversity which is essential to sound breeding programmes is to be kept.

3. The Threats to the Resources

So far only a very small number of marine species have become extinct or seriously endangered. Extinct species include the great auk *(Alca impennis)*[7], Steller's sea cow *(Hydrodamalis stelleri)* and most probably the Caribbean monk seal *(Monachus tropicalis)*. Endangered marine animals as listed in the IUCN Red Data Books[8] include most of the great whales[9], species of toothed whales[10], all sirenians[11], the chingungo, a South-American sea otter *(Lontra felina)*, several species of seals[12], about twenty five forms of sea birds[13], all marine turtles[14] and a very small number of fishes[15]. Almost all these species are endangered because of past or present over-exploitation by man. There is no information on the conservation status of marine invertebrates, but it is very likely that certain species, especially molluscs, are now threatened by over-collecting.

The threats to any species of plants or animals can roughly be divided into four categories, although these categories may in certain cases overlap.

a) Overexploitation

Except for the great whales which are particularly vulnerable to modern catching techniques and for seals and marine turtles which are generally taken on the shores where they come to breed, overfishing is seldom the cause of the extinction of a species. Most species of fishes and invertebrates have a very large reproductive potential and often also a wide distribution. As a result, depleted areas can be repopulated, and even when a species has been depleted throughout its range it will generally survive since fishing will no longer be economical and will therefore stop long before the species becomes extinct.

But if the extinction of species by overexploitation is, in most cases, unlikely, the collapse of a fishery, caused by overfishing, may have serious ecological and economic consequences. An overfished species may indeed never recover its former position in the ecosystem after fishing has ceased and may even be replaced in that position by another species. As an example, the California sardine *(Sardinops caerulea)* fishery collapsed presumably because of overfishing and that species was replaced by the Pacific anchovy *(Engraulis mordax).* Other consequences of the depletion of a fish stock can be the corresponding depletion of its predators or increase of its former preys. Overfishing will therefore, in most cases, result in a situation of imbalance, followed by a period of adjustments until a new balance is reached which can be very different from the initial one. There have been in the recent past many examples of fishery collapses due to overfishing. This has been, for instance, the case of the North Sea and Atlanto-Scandian stocks of herring and, as we saw, of the anchovetta fishery off the coast of Peru which collapsed in 1972.

Many other stocks have also been seriously depleted by overfishing in the recent years. This is particularly true of certain demersal fish stocks such as cod, haddock and halibut in the North Atlantic and North Pacific areas.

The depletion or collapse of a fishery will naturally have serious consequences on the livelihood of the fishermen as well as on the economy of the countries concerned. It is unnecessary, preventable and wasteful.

Studies of the effects of overfishing on marine species have so far been almost exclusively confined to effects on target species and have largely ignored the potential consequences of the incidental take of non-target species and of the resulting involuntary overexploitation of these species.

Incidental take is the accidental capture and killing of non-target species in the course of fishing for other species. It is one of the most serious and at the same time one of the most neglected problems of marine resource management. It has now become particularly acute in many areas and in many fisheries of the world and has reached in some cases the proportions of a major conservation issue. Two different aspects of this problem should, however, be considered.

i) When fishing for a certain target species, fishing boats often make incidental takes of other commercial species. When the harvesting of these other species

is regulated, such an incidental take, unless also regulated, may defeat the conservation programmes established for those species. This is particularly true if the incidental take involves the catching of undersized fishes, fishing in prohibited places or periods, or the taking of large numbers of fishes in excess of the total catch allowed for the main fishery. This problem can, however, be at least partly resolved by allowing for the incidental take when taking the conservation measures required for the main fishery.

ii) The problem is much more difficult to resolve and probably much more serious when the non-target species which is incidentally taken is not a commercial species. The trawling for bottom dwelling molluscs and shrimps, the harvesting of kelp and other fishing activities result each year in the killing of billions of fishes throughout the world. These fishes are used as fertilizer, converted into fish meal or merely thrown overboard. In the shrimp fisheries alone the quantities of fish that are taken incidentally and generally thrown back into the sea amount to about 5 million tonnes per year[16]. Very little seems to be known on the species composition of that catch or of its effects on the stocks of the many non-target species concerned.

Marine mammals, sea-birds and sea turtles can also be seriously affected. Dolphins and porpoises have been taken in large numbers in the Eastern Pacific tuna fishery and the North Pacific salmon fishery. The fully protected and endangered humpback whale has been caught in cod nets off the Canadian coast[17]. Very large numbers of sea birds, perhaps a million birds, mostly alcids, are killed incidentally each year in gill nets in the North Atlantic and the North Pacific. Finally marine turtles, which are all endangered species, are increasingly caught and drowned in shrimp trawls. The recent collapse of the populations of Atlantic ridley turtles *(Lepidochelys kempi),* from about 40.000 nesting females in 1947 to only a few hundred now, has been almost entirely caused by the incidental take of this species by shrimp trawlers.

b) Habitat Destruction

Habitat destruction should be understood here in a broad sense and includes not only the physical destruction of habitats but also any alteration of the ecological conditions that will make the continuous existence of a species impossible. This includes, *inter alia,* changes in temperature or salinity, the destruction of spawning areas, nurseries or feeding areas or a decrease in the supply of nutrients brought to certain areas by rivers or other paths. Most species being coastal, their critical habitats can be easily affected by human action. Dredging, trawling or, on the contrary, filling or silting can be extremely destructive when carried out in species-rich coastal habitats. Bottom fishes depend upon crabs, or little fishes that live in eel grass communities or on bivalve molluscs and worms buried in a sandy sea bed. It would seem obvious that the destruction of such habitats by trawling, dredging or filling may seriously affect the populations of fishes that live on the bottom. Yet, although the potential harmful effects of these practices have very often been emphasized, it does not seem that much research has, so far, been undertaken on the subject[18].

The exploitation of coral reefs for the production of dead coral or sand used in construction work similarly endangers large and varied communities of animals and plants and could in some cases lead to the extinction of rare or endemic species.

The very great importance of certain in-shore areas such as estuaries, lagoons, mangrove swamps, coastal marshes and shallow bays for many species of commercial fishes seems to have been only recently recognized by scientists. It now appears that these species are to a very great extent dependent upon these habitats for spawning, nurseries, growing or permanent residence. Yet these habitats, essential as they may be to the maintenance of many commercial fisheries, are being increasingly destroyed by man. The damming of estuaries[19], the filling or draining of coastal marshes, the dredging of estuaries and shallow waters, the development of construction along the shores, and pollution all combine to speed up the rate of destruction of many critical habitats upon which not only marine species[20], but also non-marine species, such as migratory water-fowl, depend for their existence.

c) Pollution

With the development of industrialization in many parts of the world in the past thirty years, the oceans have increasingly been used for the discharge or release of many kinds of waste. Furthermore, the increase in the size of ships and the more hazardous nature of their cargoes have resulted in a number of particularly spectacular accidents such as those of the Torrey Canyon and Amoco Cadiz. These accidents have undoubtedly contributed to a great extent to the awareness of governments and public opinion of the harmful effects of large-scale oil pollution. There are, however, other forms of pollution, albeit less spectacular and more insidious, that may have much more serious effects on the ecosystems and species in the sea.

A distinction should be made here between the economic and ecological damages caused by pollution, although the two may be in some cases closely interrelated. Economic damages are temporary losses of amenities, destruction of fish or mollusc farming operations or temporary losses in fish sales by fishermen. These can be compensated by the payment of sums of money as reparation for the damage caused, provided liability rules are clearly established and adequate insurance has been taken. Oil spills, even of the magnitude of those mentioned above, generally have limited effects on the natural environment both in space and time, even though the economic damage may be enormous. Economic damage, however, can, in principle at least, be repaired. Ecological damage, when reparable, requires sound management practices, hard work and time. It may, however, be irreversible and species, habitats or fundamental biological processes may then disappear forever.

For the purpose of this paper, only the ecological damage caused by pollution will be considered. Pollution can cause damage to the natural environment by reason of the nature of the pollutants released and also because of the place and time of the polluting activity or accident.

The most dangerous substances for the marine environment are certainly those that are released in large quantities, are long-lived, that is to say are not quickly degraded by chemical processes or bacteria, have the capacity to accumulate in certain parts of the system or in living organisms and finally are toxic either in the form in which they have been released or in any other form they may take afterwards.

These substances include halogenated hydrocarbons such as many pesticides, in particular DDT, polychlorinated biphenyls (PCBs) and heavy metals such as mercury[21], cadmium and lead. This type of pollution is usually land based. Pathways of these substances to the sea are run-off waters, rivers, and coastal effluents. DDT and similar products are easily vaporized and also reach the sea through the atmosphere and rain water. Other persistent and highly toxic substances are certain radioactive elements, especially transuranian elements, such as plutonium. The accumulation of there various substances in the sea will, if it is not reversed, result in a gradual poisoning of the marine ecosystem as a whole and in serious ecological imbalances[22]. In areas of greater concentration, irreversible effects on some species, ecosystems or biological processes will most probably occur[23].

Pollution by ships includes dumping and accidental releases. If the dumping of highly toxic substances can, in principle, be easily prohibited by strict legislation, accidents may not always easily be prevented. The accidental release of large quantities of highly toxic and persistent substances such as, for instance, DDT in a particularly rich and varied ecosystem like the Great Barrier Reef, could result in uncalculable ecological damage. An accident similar to that of the Amoco Cadiz occuring in an ecosystem containing endemic species could lead to the extinction of these species. Sea-birds are particularly vulnerable to oil pollution and have died in the past in large numbers as a result of major oil spills. If an accident to a large tanker or an oil well or pipeline were to occur in the vicinity of certain critical habitats, such as the breeding or moulting areas of certain species of sea-birds[24], seal rookeries, spawning areas of certain fishes or the nesting beaches of marine turtles[25], it could well result in the extinction or near extinction of whole populations or even of species. If would seem, therefore, that specific preventive measures are essential to protect particularly important ecosystems and species against accidental pollution.

d) Introduction of Alien Species

An important threat to the survival of many marine species, especially of isolated populations or endemic species with limited distribution, is the introduction of exotic species, particularly into isolated oceanic island environments. Introductions may be accidental, as for instance the introduction of the Chinese crab *(Eriocheir sinensis)* into Western Europe; they may be incidental to the establishment of a water connection between separate seas or oceans, as it has been the case after the opening of the Suez Canal when several Red Sea species invaded the Eastern Mediterranean; they may also be deliberate for the purpose of developing a new fishery.

Scientists have for a long time warned against the introduction of exotics. These species, which are generally introduced without their natural predators, often proliferate rapidly and cause considerable disturbance to the ecological balance of the environment. They compete with native species for the same ecological niches and may eliminate them completely. They may also be the carriers of parasites or diseases. In the marine environment of certain islands, such as Hawaii, for instance, where a high proportion of the native fauna is endemic (e.g. 30% of the fish species), the introduction of alien species could be a disaster[26]. It is therefore essential to institute strict control on these introductions and not allow them to be carried out without having proven scientifically that they will be harmless.

If a sea level canal across Central America was to be built parallel to the Panama Canal, direct communication would be established between the Pacific Ocean and the Caribbean Sea and a large number of incidental introductions would take place resulting probably in many extinctions.

4. The Conservation of the Resources

The objectives of the conservation of marine living resources, which are not substantively different from those that apply to terrestrial or fresh water resources, have been laid down by the World Conservation Strategy[27]. These are: the maintenance of ecological processes and life-support systems, the preservation of genetic diversity, and the sustainable utilization of species and ecosystems.

a) Ecological Processes and Life-Support Systems

This first objective constitues a necessary basis for the attainment of the other two. It implies that habitats that are critical for the preservation of genetic diversity or for sustainable fisheries be adequately managed and that the principal goal of such management be the maintenance of the processes on which the species depend. This means that these critical habitats, in particular estuaries, mangrove swamps, other coastal wetlands, shallows, and coral reefs should be managed in such a way that any factor, such as damming, dredging, draining, siltation, lowering of the water level or pollution which could adversely affect their capacity to sustain the wild animal and plant populations that depend on them for their existence should be adequately controlled. This also means that the overall pollution of the marine environment by toxic substances should be stopped.

b) Genetic Diversity

The preservation of genetic diversity, in other words of all living species of animals and plants, is essential to provide mankind with as large a number as "possible of future options and as the raw material for scientific and industrial innovation"[28].

This will often require specific measures to prevent the extinction of species. The taking of endangered species and the trade, whether domestic or international, in these species or their products should be prohibited, and protected areas should be established to preserve their habitat. Continuous monitoring and research are also necessary to ensure that all necessary steps are taken to maintain the numbers of the species at the highest possible level. With regard to species which are not considered endangered, the preservation of sufficient genetic diversity can be ensured by the establishment of protected areas which should include representative samples of all different types of marine ecosystems, and "only those uses that are compatible with the preservation of the ecosystem and its component communities of plants and animals should be permitted in areas protected for these purposes"[29]. Unique ecosystems should be protected as a matter of priority.

c) Sustainable Utilization

For some time now, sustainable utilization of marine living resources has been equated with the concept of maximum sustainable yield (MSY). This concept has now been widely accepted by lawyers and administrators and has been embodied in many treaties including the new draft treaty on the Law of the Sea, albeit subject to important qualifications. It is indeed paradoxical that the quasi-institutionalization

of MSY by international law should happen at a time when scientists are increasingly questioning the applicability of that concept to a large number of practical situations.

i) The maximum sustainable yield concept

"The maximum sustainable yield is the greatest harvest that can be taken from a self-regenerating stock of animals year after year while still maintaining a constant average size of the stock"[30].

This concept is based on the principle that the growth of an animal population is limited by environmental factors, such as the availability of food, and that it will always reach a saturation level which cannot be exceeded. This saturation level is called the carrying capacity of the habitat. A population which is not exploited by man will normally tend to remain at that level and natural mortality and recruitment will balance out. The population will then remain in a steady state provided environmental factors do not change. When exploitation occurs it reduces the population in question under the carrying capacity of its habitat. The population, to recover its losses, then starts growing at a rapid rate until the carrying capacity of its habitat is reached again. This rate of increase is the greatest when a population has been reduced to a particular level. This is where maximum sustainable yield occurs. Above this level, the rate of increase will be slower and the yield smaller. Under that level, mortality will be greater than recruitment and the population will eventually collapse. To sum up, MSY is obtained when both fishing mortality and recruitment are maximized at the same time.

Models have been developed to calculate MSY. A simple model establishes that MSY is obtained when an initial virgin stock is reduced to about one half of its original size. This means that, all factors influencing recruitment and natural mortality remaining equal, the net annual recruitment rate is maximized when a stock is reduced to that level.

Since, however, recruitment and natural mortality may and often are affected by environmental factors, such as the availability of food, it appears clearly that changes in these factors will affect the MSY. To resolve the problem, it became necessary to develop dynamic models where MSY was calculated for each age class of the fishes composing the stock on the basis of the recruitment rate of each one of these classes.

Such models are currently in use for the calculation of the MSY of many stocks of exploited fishes.

To be able, however, to calculate with a sufficient degree of accuracy the point at which MSY can be obtained it is necessary to have reasonable estimates of the total biomass of a stock and of its natural mortality. Research on the population dynamics of a stock is therefore essential.

ii) Recruitment and growth

Simple models of MSY were based on the assumption that fishing did not and could not significantly affect recruitment because the number of eggs produced by a spawning stock is so large that even if more than one half of the stock has been fished, there will still be enough eggs left for the stock to reach again the carrying

capacity of its habitat. Under this assumption overfishing does not really affect recruitment but affects the size of the fish that are caught making the fishery less and less profitable. In other words, the number of young fishes in the stock remains the same but they are caught before they are allowed to grow and reach a profitable size[31]. This has been called "growth overfishing". The problem was solved by regulating the size of the meshes of the nets that were used to catch these fishes. The larger the mesh size, the bigger were the fishes that could escape, and therefore grow to a larger size, and the more profitable became the fishery. This method of regulating the catch has been and is still applied in most of the major fisheries of the world.

The reproductive rate of animals and plants is not, however, the same for all species. Some species, which are called by scientists "r strategists", produce an enormous number of eggs. Most of these eggs or of the young larvae will be eaten by predators but enough will survive to produce a new generation. For those species, security is found in numbers. Examples of these "r strategists" in the sea are many of the benthic fishes such as the Gadoids and the flatfish.

At the other end of the scale are found the "K strategists". They lay a very small number of eggs, but these eggs and subsequently the larvae, are protected from predators by various means[32]. Between pure "r strategies" and "K strategies" many different intermediate types of strategies can be found; unlike most "r strategists" in the sea, "K strategists" often quickly reach their maximum size and stop growing. This is, for instance, the case of the great whales. Overfishing of such species will therefore not affect the size of the animals that are caught but the total number of breeding individuals in the stock and recruitment will decrease. This has been called "recruitment overfishing". When the mortality caused by fishing becomes great enough seriously to reduce recruitment, extinction, or at least a considerable depletion of the stock, becomes likely. This has been the cause of the recent collapse of certain fisheries such as the North Sea herring fishery and of the rapid decline of the great whales.

It is obvious that in the case of recruitment overfishing the establishment of a minimum mesh size will be largely inoperative, if the young fish that have been allowed to escape are caught as soon as they have become adults without having had the possibility to breed and thereby ensure the replacement of their generation.

It has furthermore become apparent that even in the case of the most fecund fishes which are able to grow for a long time after they have reached the adult stage, i.e. the fishes which, in principle, can only be affected by growth overfishing, it is the individuals that have the larger gonads that generate the greatest number of eggs and therefore of larvae. Growth, therefore, becomes a factor of fecundity and the systematic harvesting of the largest and most fecund fishes will probably result in this case also in recruitment overfishing. The consequence of this is that the difference between recruitment overfishing and growth overfishing will presumably tend to disappear. As a result, to quote S. Holt, measures taken to prevent the capture of young individuals such as the regulation of mesh sizes will have to be accompanied by measures limiting the fishing effort. If the two kinds of measures are not taken together, overfishing will not be prevented. ICNAF for two decades

regulated only mesh sizes of trawls, while any benefit was quickly removed by an increase in the intensity of fishing[33].

iii) Criticism of the maximum sustainable yield concept

The MSY concept, as we have seen, seems to provide a useful management tool, provided environmental factors affecting recruitment and natural mortality of the harvested species remain unchanged. In other words it would work satisfactorily in the case of a single isolated population living in a steady environment and with an unlimited supply of food. In the sea, however, the magnitude of recruitment is necessarily linked with the quantity of food available to larvae and with competition with other species. The problem becomes even more complicated in the case of multi-species fisheries since the decrease in the numbers of one species may very well affect, positively or negatively, the availability of food to others. It became therefore clear during the 1950s and 1960s that the study of single populations in isolation was no longer sufficient to solve fishery problems and that the concept of maximum sustainable yield needed to be considerably refined if sustainable production was to be obtained on a long-term basis for all species.

In addition, the effect of the exploitation of certain species must be considered in relation with the health of the ecosystem as a whole and not only with the increase or decrease in the numbers of the populations that are exploited.

Finally, allowance must be made for the unavoidable inaccuracy in the methods used for stock assessment, abundance, mortality and age determination, the collection of statistical data, or the evaluation of changes induced by exploitation[34].

For all those reasons, the IUCN General Assembly adopted at its twelvth session in Kinshasa, Zaire, in 1975 a resolution entitled Principles Replacing Maximum Sustainable Yield as Basis for Management of Wildlife Resources[35] which lays down four important principles:

1) "the ecosystem should be maintained in such a state that both consumptive and non-consumptive values can be realized on a continuing basis, ensuring present and future options, and minimizing the risk of irreversible change or long-term adverse effects;

2) management decisions should include a safety factor to allow for limitations of knowledge and imperfections of management;

3) measures to conserve one resource should not be wasteful of another;

4) survey or monitoring, analysis and assessment should precede planned use, and accompany actual use, of a resource and the results should be made available promptly for critical public review".

iv) The economics of fisheries

The rational management of fisheries cannot be limited to the consideration of purely biological factors. It must, necessarily, also take into account the economic aspects of the question. There would be little sense, indeed, in maximizing production if no attempts were made to maximize, at the same time, both the global profitability of a fishery and the profits of individual operators who take part in its exploitation.

Regulatory measures should, therefore, have as a goal not only the maintenance of a sustainable production and of a high catch level but also the preservation of economic efficiency. Faced, however, as they were, with a situation of unlimited access to finite resources, regulatory bodies could do no more than to try to conserve these resources by the adoption of regulations that inevitably resulted in a decrease in the economic efficiency of fishing operations.

Indeed, measures such as closed seasons, closed areas, restrictions on the types of vessels or gear that can be used or on the size of the fish which can be caught, all tend to contribute to the same result: an increase in the difficulty, for individual operators, to make a profit and hence a decrease in their economic efficiency. The purpose of these measures is, of course, to limit the catch to safeguard the productivity of the stocks. But, since these measures require from fishermen that they devote more time, money and efforts to take the same quantity of fish, they are economically undefendable.

Similarly, the imposition of quotas, which is the same thing as rationing, will lead, as new operators enter the fishery, to downward readjustments until the quotas fall so low that the fishery again becomes unprofitable.

The history of individual fisheries, therefore, almost always follows the same pattern. When a virgin stock is discovered, fishermen are few and catches and profits high. As the fishery develops, operators, often with the help of government subsidies, increase their investments and, in the expectation of even higher profits often over-invest. Parallelly, new operators begin to enter the fishery anticipating quick profits. After some time, the catch per unit of fishing effort begins to decline and signs of over-fishing appear. Regulatory bodies are then instituted and regulatory measures taken. Individual operators naturally oppose these measures on the understandable grounds that their have to amortize their investments and that they are entitled to a reasonable return on the money thus invested. In the face of this opposition, regulatory bodies have to compromise. The measures taken are, therefore, not sufficient to restore the situation. Quotas, for instance, are usually set too high. Thus the fishery continues to decline and eventually collapses. The sad story of the regulation of whaling is particularly illustrative of this process but is far from being unique in the history of living resources exploitation on land or in the sea.

Unlike sport hunting or fishing, where strict rules can be imposed and are even part of the game, the commercial exploitation of wildlife resources cannot be based on the promotion of economic inefficiency as the only means to try to maintain harvested populations at an exploitable level. Furthermore, as we saw, in practice the system results almost inevitably in the depletion or extinction of the stocks and, therefore, in the decline and demise of the fishery itself.

Thus, restriction in the amount of fishing effort seems to be the only solution to an optimization of both physical and economic yields. This is usually achieved by means of the licensing of fishing vessels allowed to fish for a particular stock or in a particular area. Inter-governmental fishery bodies seldom have, however, under their regulatory powers, the authority to impose restrictions on fishing effort and even when they have that authority, their member governments have shown con-

siderable reluctance to use It to adopt specific restrictive regulations. The matter is therefore mostly left to national authorities to decide whether, within the limits of their national quota, they will or will not limit the number of fishing vessels that can operate in a given fishery. It would be difficult, however, for a government to limit access to an international fishery if nationals of other countries continue to have free access to that fishery. Indeed, such measure would be resented as unfair and discriminatory, and would arouse considerable opposition. Thus, limitation of fishing effort can only be achieved by international agreement.

To what extent will the situation be much different in the newly established extended EEZs remains to be seen. Coastal states will, of course, be entitled to regulate fishing effort and license fishing vessels in a way such that the harvesting of their living resources be economically profitable. On the other hand, competition between individual fishermen or fishing companies enlisting the assistance of political pressure groups, and the economic incentive for many developing countries to conclude agreements with foreign nations for the harvesting of the "surplus" of their allowable catch may again tend to result in an uneconomical increase in the number and tonnage of vessels and, therefore, in economic inefficiency.

Assuming, however, that with appropriate measures, such as compensation for operators who have been reduced to inactivity by the action taken to limit fishing effort, it has become possible for fishermen to capture their allowable catch without increasing financial losses, the question arises as to the level of exploitation at which profitability can be maximized. When a stock is exploited at MSY level, its biological productivity is, subject to the many critical comments made on the MSY concept itself, in principle at its maximum. But this is not so of economic efficiency since, to produce its MSY a stock has first to be reduced to about 50% of its initial level. Therefore, although the annual net increment will be greater at that level, it stands to reason that more time and effort will be required to take that increment than it would have been the case for a smaller increment produced by a much more abundant stock. In other words, the gain in the quantities caught is not commensurate with the increased costs, and economic efficiency, and therefore profitability, are not maximized.

To quote S. Holt, "The catch per unit of fishing effort is at its greatest when the stock size is at its greatest — that is when fishing is very light"[36]. Thus, the optimum economic yield, maximizing profitability by unit of effort, is only possible at stock levels that are much higher than the one producing MSY Assuming that biological productivity of a stock is maintained and that no overfishing occurs, decision-makers will have to choose between greater amounts of fish on a sustainable basis at the cost of economic inefficiency, or smaller quantities resulting in a maximum profit. With the increased cost of inputs, particularly fuel, put into the fishing industry, and in view of the fact that cheaper sources of protein exist, it would seem that the MSY concept may tend to become as unpopular with economists as it is already now with biologists.

C. Basic Legal and Institutional Requirements for Resources Conservation and Management

If adequate conservation and management of the living resources of the sea are to be ensured, five basic requirements must be met: there should be clear jurisdictional rules establishing for any resource or area which is the State or authority which is legally competent to take regulatory measures; research and monitoring should provide the necessary information upon which regulations can be effectively founded; appropriate and efficacious regulations should be enacted; adequate enforcement mechanisms should be established and efficient institutional arrangements should be set up.

1. Jurisdiction

A clear determination of the authority which has jurisdiction over a particular area or resource is essential if binding regulations are to be enacted and enforced. Jurisdiction is normally exercised over certain areas such as for instance inland waters or the territorial sea, and, as a result, relates to all living resources present in those areas; but it may also be exercised over certain categories of species, such as sedentary or anadromous species, and would then apply to those species, whatever their location. In areas beyond national jurisdictional boundaries, unless special jurisdictional rules apply to certain species, states are only empowered to regulate the activities of their own nationals or ships and regulatory measures can only be adopted by international agreements. Such agreements are binding only upon their parties. Rules relating to the management of fish stocks must, however, and this is particularly true of migratory stocks, apply to the whole range of these stocks, irrespective of jurisdictional limits if effective management is to be achieved. For this purpose, the conclusion of agreements between all states, or other entities, having jursidiction over shared or migratory stocks is essential.

Jurisdictional problems are not limited to the international scene. At national level, areas which form part of the same ecological unit may very well come under the jurisdiction of different administrations. This is often the case, for instance, in federal states, where individual states are often competent to regulate fisheries in internal waters and in a three miles wide territorial sea[37], and the federal government in the remaining part of the territorial sea and in the exclusive economic zone. It is also the case when two different administrations under the same central government exercise jurisdiction over different parts of the habitat of the same species[38]. Thus, at national level as well, agreements between competent authorities are required if living resources are to be preserved and rationally managed.

2. Research

It is only towards the end of the nineteenth century that the need for fishery research became apparent as a result of the decline of certain fisheries, particularly in the North Sea and on the eastern coast of the United States, due to the introduction of new gear and steam boats[39].

During the past seventy years the sciences of fisheries biology and marine ecology developed slowly but steadily. Parallelly, the science of oceanography, which deals

with the scientific investigation of marine physical, chemical and biological phenomena, started also to develop about a hundred years ago. Considerable progress has now been achieved both in fisheries sciences and oceanography and our knowledge and understanding of natural processes and conditions in the ocean has been much improved. The adequacy of this knowledge is, however, still far from being perfect and considerable research efforts continue to be required before conservation and management regulations can, in all cases, be developed on a scientifically sound basis.

In the words of Christy and Scott, "The world has not yet sufficient knowledge to predict accurately the consequences of even small changes in the use of most parts of the oceans. Many questions have not yet been investigated thoroughly. The superficial exploration of the ocean bottom and the identification of the main currents have not been completed. Investigation of the dynamics of most fish populations has hardly been commenced. Their migrations during their life cycles are not documented. The interdependence of fish populations is merely guessed at. The ability of some fish to endure changes in temperature, food, salinity, or light is just now been recorded"[40]. In spite of all the work that has been carried out in the fifteen years that have elapsed since these words were written, they remain, to a large extent at least, still true in 1980.

Research is, therefore, essential to the conservation of species and the management of fisheries. The information required ranges from broad oceanographic data to the improvement of knowledge on the biology, ecology and population dynamics of individual stocks. It includes the results of research on such subjects as species critical habitats, species inter-relations, inter-specific competition, stock assessment as well as of the effects of fishing on individual stocks and on the ecosystemin general.

Statistics also play a vital role in the development of management measures. The calculation of fishing mortality, the determination of sustainable yields and the assessment of the size of a fish population are indeed impossible if fishery biologists cannot use as a basis accurate fishing statistics to evaluate the status of a harvested stock and recommend conservation measures if it is depleted. But statistics presently available are, however, often inaccurate, inadequate and incomplete.

One of the world's leading fishery experts could thus write in 1978[41]:
"A full review of the state of world fishery resources should give an account of the state of the individual fish stocks in terms of their biomass, their growth and mortality rates, and the general ecological conditions affecting the productivity of these stocks. Desirable as such information may be, it is available for only a small proportion of all stocks".

3. Regulations

Regulatory measures are essential to prevent the extinction or depletion of species, the destruction of habitats, pollution and the introduction of alien species. Such measures are also required if sustainable exploitation of harvested species is to be achieved. In many cases, these measures will have to be taken by international agreement. It will have to be so, in particular, whenever fish stocks are shared or are

migrating between two or several jurisdictional zones, or are harvested on the high seas by several countries, and when pollution or other harmful consequences of human activities are likely to affect areas under the jurisdiction of several States or the high seas.

With regard to fisheries, the type of regulations which are the most commonly found in national legislation and international agreements relate to the regulation of fishing gear, including the size of mesh of fishing nets, the regulation of the size limits of fish that may be retained on board or landed, the establishment of closed seasons and of closed areas, the establishment of total allowable catches, either globally, or by species or by area and the regulation of the amount of fishing effort, through the licensing of fishing vessels or by other means[42]. Some international conventions also provide for the allocation to each of their member States of a certain share of the total allowable catch or of the total amount of fishing effort authorized.

Regulations relating to the preservation of habitats and to pollution control include the prohibition or regulation of activities that may be damageable to the marine environment in general or to certain ecosystems, habitats or species in particular. Such regulations may concern the prohibition or control of dredging or filling, of the use of explosives or of the releasing or dumping of harmful or toxic substances, either in general or at certain places.

Regulations relating to the protection and management of species and ecosystems must come into force as early as possible after they have been adopted lest irreversible damage be caused to the environment if long delays are involved. The matter is relatively simple at national level where, once a general law has been adopted, specific regulations can be approved without delay by the competent administrative authorities and thereafter come into force as soon as they have been published in the official gazette. To take a recent example, under the Fishery Conservation and Management Act of the United States[43], which was adopted in 1976, Regional Fishery Management Councils have the obligation to prepare management plans in respect of the fisheries in their region. Management plans, once approved, are implemented by the Secretary of Commerce and regulations relating to any aspect of the plan, including catch quotas, are published in the Federal Register.

The problem is far more complex at international level when several independent states participate in the same fishery. Decisions cannot be imposed upon sovereign states against their will, in contrast with national legislation where laws are binding on all citizens. Conservation measures therefore require the unanimous approval of all parties to the treaty concerned. In addition, in normal circumstances, acceptance by states to be bound by an international convention is the subject of a long and protracted procedure, called the ratification procedure, which involves the submission of the convention to parliament for approval. It soon appeared that this system would be unworkable where fast management decisions must be taken to preserve living resources. This became clear as early as 1937 when new whaling regulations could not come into force until after the whaling compaign to which they applied had come to an end. As a result, a new procedure was established, which, whilst fully compatible with the right of sovereign states not to be bound

against their will, eliminated undue delays and allowed for fast action. This procedure can roughly be described as follows: a decision taken by a regulatory body set up under a fisheries or other conservation agreement becomes binding on all parties after a certain period of time, which usually ranges from three to six months, unless one or several parties to the agreement have lodged an objection; a decision is not binding on a party that has lodged such an objection; if more than a certain number of parties, for instance one half, have lodged an objection, the decision does not come into force at all.

The first fishery treaty to include this new procedure was the International Whaling Convention in 1946. It was introduced thereafter into most fishery agreements, into other conservation treaties relating, in particular, to the preservation of species, especially for the purpose of amending appendices listing species to be protected under these treaties, and even into conventions concerning pollution control to facilitate amendments to appendices listing dangerous pollutants or dealing with pollution prevention or control procedures.

4. Enforcement

To be effective, regulatory measures must be adequately enforced. This implies that coastal states should take steps to police fisheries and other activities under their national jurisdiction and that international agreements should provide for policing measures to control unlawful fishing practices or to prevent pollution in the high seas. It also implies that penalties inflicted by the courts to persons who have acted in violation of existing regulations are sufficiently severe to act as deterrents and thereby prevent, as much as possible, the repetition of the infractions.
In areas under national jurisdiction, coastal states have absolute and unshared authority to enforce their conservation laws and regulations. This did not present insuperable difficulties as long as the breadth of the sea area over which they exercised their jurisdiction was limited to the three or even twelve miles of the territorial sea. With the recent extensions of exclusive fishing or economic zones to a distance of 200 miles from the coast and the institutionalization of that practice by the new Convention on the Law of the Sea, coastal states, especially developing countries, will find it increasingly difficult to exercise effective control, and therefore to enforce their legislation. Adequate enforcement will, indeed, necessarily require the use of naval or coast-guard vessels, surveillance aircraft, coastal radars and other equipment as well as trained personnel and funds, which most countries either do not possess or cannot afford to use on a continuous basis. Enforcement is, therefore, bound to be haphazard and limited and only the imposition of heavy penalties by the courts of the coastal states concerned upon unlucky offenders will probably deter potential violators.

On the high seas, it becomes even more impossible for states to exercise effective control over their nationals and ships engaged in fishing operations in vast and often distant oceanic areas. It was, therefore, recognized long ago that some form of international action was required to ensure compliance with conservation regulations adopted by inter-governmental fishery bodies in respect of the high seas. Two possible methods could be envisaged in this regard. Enforcement of conservation measures could be entrusted to the fisheries commissions themselves which would, as a result, have their own patrol vessels and staff of inspectors. No fishery

body has, however, so far been granted the right to enforce its own regulations. Alternatively, each member state of a fishery commission could be given the right to exercise supervision and control over the activities not only of its own ships but of the ships of all other member states. This second method was adopted in a certain number of fishery agreements. It implies the right for authorized officials of a member state to board, search and seize vessels of other member states which have acted in violation of the convention or of regulations applicable thereunder. As to penal sanctions, however, only those states that have jurisdiction over the ships concerned are entitled to institute proceedings and apply penalties for violations[44].

With regard to violations of international rules and standards established for the prevention, reduction and control of pollution that have been committed on the high seas, enforcement would normally be entrusted to flag states. A growing tendency, however, which has been taken up in the new convention on the law of the sea[45] consists in allowing a state, in whose ports a ship having committed a violation of international rules or standards on the high seas has called voluntarily, to institute proceedings in respect of that violation.

5. Institutional Arrangements

Fisheries agreements, like many conservation treaties, are different from most international conventions in that if a party fails to comply with its obligations it is not deterred by the risk of seeing the other parties decide not to stand up to their own obligations vis a vis the defaulting state[46]. In other words, there is no quid pro quo. As an example, in a trade agreement, if one of the parties does not deliver the goods which are the subject of the agreement the other party has the possibility of suspending payments. In conservation treaties this possibility of exercising retaliation generally does not exist. Indeed, if the other parties ceased to apply the conservation measures established under such a treaty they would in most cases do harm to the resources and to themselves much more than to the defaulter[47].

Institutions are therefore required to implement these treaties, provide a forum for discussion, coordinate activities of member countries and establish procedures, thus making it much more difficult for parties not to fulfill their obligations. Institutions should normally comprise at least a commission composed of delegates of member countries and a secretariat to service commission meetings, maintain liaison between the parties during the intersessions and prepare the necessary surveys and reports. Whenever possible, a scientific committee should also be established.

The idea of a world organization to regulate fisheries on all the oceans was put forward on several occasions. Leonard, as early as 1944[48], proposed the creation of an International Fisheries Office to conserve the products of the sea and to establish a system of rational exploitation[49]. Later, the International Law Commission when preparing draft articles for the 1958 Geneva Convention on Fishing and Conservation of the Living Resources of the High Seas, had proposed the establishment of an international fisheries authority. This proposal was, however, subsequently dropped. During the negotiations of the new Convention on the Law of the Sea several proposals were made to the effect that, just as the sea-bed and ocean floor were to be proclaimed the common heritage of mankind, their superjacent waters

and the resources they contain should be recognized the same status and be managed by an international authority as well. These proposals, however, met with little support[50]. The new treaty, as a result, does not provide for a world-wide decision-making body which would be competent to regulate fisheries beyond the limits of national jurisdiction.

Admittedly, some of the United Nations agencies, such as FAO or UNESCO exercise competences in the fields of marine sciences and fisheries, but these organizations are essentially of an advisory nature and are not empowered to take binding decisions.

At regional level, however, or for the purpose of conserving and managing particular species or groups of species such as whales or tunas, it appeared from the very beginning of the development of fisheries agreements that some forms of institutional arrangements were essential if the provisions contained in these agreements were to be effectively implemented. Most fisheries treaties, therefore, as well as several conventions relating to the control of pollution, provide for the establishment of a commission whose task is generally to follow the implementation of the convention, promote and co-ordinate research, make recommendations to member states and, in a few cases, to take binding decisions provided they are approved by all member states.

D. The Development of Marine Conservation Law

1. The Traditional Law of Fisheries

a) Introduction

Subsistence fishing is probably almost as old as mankind. Commercial fisheries started to develop in Europe at the end of the Middle Ages when Basque sailors began to capture whales first in the Bay of Biscaye and later in the North Atlantic. In the seventeenth century whaling was taken over by the Dutch who extended their catching activities well into the Arctic Ocean. Subsequently cod fisheries were developed in the North Sea and near Newfoundland. The advent of the trawl in the late nineteenth century and the invention of the steamer trawler mark the beginning of the era of industrial fisheries; many new fishing grounds were discovered and world production of fish started to increase steadily. In 1938 the total world catch of marine fish was approximately 15 million tonnes. In 1957 it exceeded 27 million tonnes and in 1968 56 million tonnes.

b) The Legal Regime of the Sea

i) Internal waters and territorial sea

For many centuries the use of the sea and of its resources remained unregulated. States did not have sovereign rights over the adjacent sea and did not claim such rights. This situation changed in the sixteenth century when several important seafaring nations claimed sovereignty over large areas. These claims were opposed by certain jurists, the most famous of whom was the Dutchman Grotius, whose views eventually prevailed. Thus, by the middle of the eighteenth century, the freedom of the seas, which included the freedom to fish, became an established rule of international law. At the same time, mostly for reasons of national security, bays, gulfs and other inshore waters, as well as a narrow belt of water along the coast began to be

considered as susceptible of national occupation[51]. This is the origin of the concepts of internal and territorial waters. These waters were considered as forming part of the national territory. The breadth of the territorial waters was first established on the basis of the cannon-shot rule, that is to say on the basis of the distance within reach of a cannon shot from the coast. This rule was subsequently gradually abandoned in favour of a uniform breadth of three miles[52].

ii) The continental shelf and sedentary fisheries

The continental shelf is the sub-marine continuation of the continental land masses, extending under the sea to where the sea bed begins to fall off steeply into oceanic depths[53] forming what has been called the slope. The outer limit of the continental shelf, and therefore the beginning of the slope, often occur at a water depth of approximately 200 meters, but there are many exceptions. The width of the shelf is very uneven, varying from zero to more than 800 miles. The areas of greatest extension of the continental shelf are in the Arctic Ocean, the Gulf of Mexico and Caribbean Sea, the South China Sea, the East China Sea, the Java Sea, the Arafura and Timor Seas, off the northern coasts of Australia and in the South West Atlantic, off the coast of Argentina.

The continental shelf is often rich in mineral resources, in particular petroleum, and in living resources such as bottom living fishes, crustaceans and molluscs. Certain of these species, especially sponges and many molluscs such as pearl oysters or chanks, are, at least at their adult stage, either permanently fixed to the sea bed or buried in the sub-soil and are therefore completely sedentary. The exploitation of these sedentary resources by certain coastal states has been going on in some cases for many centuries and jurists generally recognized that coastal states had exclusive rights and therefore jurisdiction over these resources even beyond the outer limits of their territorial waters[54].

iii) Anadromous species

Anadromous fish are species which spend part of their life in the sea and migrate periodically to fresh water, generally to the very river where they were hatched, to spawn. Although the term is only used by biologists in the case of fish species, from the legal point of view it would seem that all species that perform such migrations and breed only on land or in waters under national jurisdiction should fall in the same category. These species include not only many forms of fishes but also certain whales, such as the grey whale, seals, sea turtles and sea birds.

Controversies arose as to the ownership of North Pacific fur seals and salmon. The seals breed almost exclusively on the Pribiloff islands which belong to the United States. Sealing on the high seas by British boats had developed to such an extent by the end of the past century that the seal stock had become severely depleted. The United States claimed that the seals, which were breeding on American land, were, for that reason, their property and that sealing on the high seas should come to an end. British boats were seized and their captains fined. The dispute was eventually submitted to arbitration. The British replied to the American claim that seals were "ferae naturae" universally regarded as "res nullius" until they were caught. The Court of Arbitration ruled in 1893 that "The United States had not any right of protection or property in the fur seals frequenting the islands of the United States in the Bering Sea, where such seals are found"[55] outside the ordinary three mile limit.

The 1893 arbitral award seems to have settled the legal matter In a final way. In 1937 Japanese fishing boats started to catch salmon of American origin off the Alaskan coast. Public opinion in Alaska demanded that the salmon be declared property of the United States, but the American government did not take up that claim.

iv) Conclusion

Thus the situation was clear. Within their internal waters and territorial sea coastal states exercised full sovereign rights including fishing rights. Outside these areas, the high seas were open to all nations to use as they pleased and fishing, on the basis of the postulate of an inexhaustible sea, was unrestricted. Even anadromous species could be taken on the high seas. The only exception concerned sedentary species on the continental shelf which were recognized as falling under the jurisdiction of the states that exploited them.

c) Early Fisheries Agreements

It appeared relatively early in the recent history of fisheries that some species at least were not inexhaustible. Extinctions or serious depletions first affected marine mammals which are K-strategists with a low reproduction rate and which are in addition, particularly vulnerable because they must come to the surface to breathe. Steller's sea cow became extinct in 1768[56]. The sea otter was thought to be extinct by the turn of the century. Fortunately small populations survived[57]. The North Pacific fur seal was, as we saw earlier, almost brought to extinction by uncontrolled pelagic sealing[58]. Several species of whales, in particular the right whale, the bowhead and the grey whale had almost completely vanished by the end of the nineteenth century[59].

Signs of depletion began also to appear in the early twentieth century in some heavily exploited stocks of fish in the North Atlantic and North Pacific areas. The stocks of North Pacific halibut and salmon were particularly affected. Furthermore, with the beginning of pelagic whaling in the Antarctic in 1928, it appeared that some restrictions should be put on the catches to prevent progressive reduction of the stocks.

It became clear, therefore, that the freedom of fishing on the high seas was no longer compatible with the fact that some of its living resources, at least, were not inexhaustible. The world community was, however, not prepared to forego that freedom and the first attempts at codification of the law of fisheries by the League of Nations failed. Except for a small number of areas or species, there did not seem, however, that there was, at the time, any risk of serious depletion and it was felt that the best approach to specific fisheries problems consisted in the conclusion of treaties under which States fishing for a particular species or in a particular area would commit themselves, by mutual consent, to adopt and apply certain conservation measures. But treaties are only binding upon parties and cannot therefore be enforced upon non-parties. The effectiveness of a fishing agreement relating to a certain area or to certain stocks was, as a result, dependent upon the number of countries fishing in that area, or exploiting these stocks, that had accepted to become parties to that agreement. For all other countries the principle of the freedom of fishing on the high seas continued to apply.

Among the most important of the earlier fisheries treaties are the Convention for the Preservation and Protection of Fur Seals concluded in 1911 for the conservation of North Pacific Fur Seals and regularly renewed thereafter which outlaws pelagic sealing; the Convention for the Preservation of the Halibut fishery of the Northern Pacific Ocean of 1923, the Convention for the Protection, Preservation, and Extension of the Sockeye Salmon Fisheries in the Fraser River System of 1937 and the International Whaling Conventions of 1931 and 1946. These treaties are all concerned with the conservation of particular species or groups of species. Other conventions were, however, also concluded for the purpose of managing fisheries in specific areas. As early as 1929 a Baltic Convention was concluded between some of the Baltic coastal states to regulate fishing in that semi-enclosed sea. In 1946 the Convention for the Regulation of the Meshes of Fishing Nets and the Size Limits of Fish was signed in London. This Convention applied to all the waters of the North-east Atlantic. In 1949, the International Convention for the Northwest Atlantic Fisheries was concluded in Washington by the representatives of eleven countries "sharing a substantial interest in the conservation of the fishery resources of the Northwest Atlantic".

Thus gradually a system of fishery regulatory measures was getting established. The problem of the new entrants into an already regulated fishery was, however, by no means resolved. Specific solutions were nonetheless found in a few important cases. In respect of the North Pacific Fur Seals all four coastal states in that area, namely Canada, Japan, the United States and the USSR agreed under the North Pacific Fur Seal Treaty to abstain from pelagic sealing. A limited number of seals, compatible with the maintenance of the herd at an optimum level, are slaughtered annually on land, mostly on the Pribiloff islands which are part of the United States. The other three parties receive in compensation for their abstention, a share of the proceeds of the sale of the pelts. A commission where the four signatory governments are represented decides on the necessary research work and makes recommendations to the Parties on all matters relating to the implementation of the treaty. The system worked remarkably well and the fur seal herds soon fully recovered. It must, however, be pointed out that no non-party state had expressed an interest in carrying out pelagic sealing in the area. Had this occurred, and had it been at the same time impossible to convince such a State to accede to the treaty, the North Pacific fur seal could well have been threatened with extinction again.

The principle according to which a potential entrant into an existing regulated fishery agrees to abstain from fishing certain species on the high seas has been called the "abstention principle". It was embodied in the 1952 Convention for the High Seas Fisheries of the North Pacific Ocean. Under this Convention Japan agreed to abstain from fishing halibut, herring and salmon in certain areas. Canada also agreed to abstain from fishing salmon in certain waters. Conservation measures were to be taken by the United States and by Canada in the areas where the abstention principle applied. The commission established under the convention was empowered to decide whether the stocks of the species mentioned above continued to qualify for abstention. No compensation was granted to the countries which were abstaining from fishing and in that respect the system differed substantially from the one established under the fur seals convention. It seems to have been chiefly designed to prevent Japan from entering into certain particularly pro-

fitable fisheries and was probably only possible because the convention was concluded almost immediately after the Japanese peace treaty a few years after the end of the last war. The "abstention principle" was not taken up in any subsequent fisheries agreements, nor was it accepted by the First United Nations Conference on the Law of the Sea in 1958. It has not therefore become a principle of international law[60].

2. The Codification of the Law of Fisheries — UNCLOS I

The Assembly of the League of Nations adopted in 1924 a resolution requesting the Council of the League to convene a committee of experts to report on questions of international law that were sufficiently ripe for codification. Among the subjects selected by the committee were the territorial sea and the exploitation of the products of the sea[61]. The Hague Conference on codification, in 1930, however, failed to agree on the question of the territorial sea and the subject of exploitation was not even put on the agenda of that conference. After World War II, the International Law Commission (ILC) within the United Nations, chose again a number of topics which could be the subject of codification, among which the regimes of the high seas and of the continental shelf, and prepared from 1949 to 1954 several draft articles on these questions[62].

Subsequently, a technical conference was convened in Rome in 1955 by a resolution of the UN General Assembly[63]. The terms of reference of this conference were to study the problem of the international conservation of the living resources of the sea and to make appropriate and technical recommendations.

The Rome Conference, as a technical conference, was of the opinion that it was not competent to express views regarding jurisdictional matters[64]. It laid down as the primary objective of conservation the obtainment of an optimum sustainable yield with a view to secure a maximum supply of food and other marine products[65], stressed the importance of scientific research and statistical data as a basis for adequate conservation measures[66], listed the types of conservation measures that should be applied to exploited fish stocks[67], enumerated the essential provisions that fisheries agreements should contain[68] and underlined the importance of adequate enforcement[69].

The Conference placed particular emphasis on the problem of the new entrants into a regulated fishery, and noted, in particular, that for a stock to be exploited in a manner compatible with its conservation, it was necessary that all States actively exploiting this stock should accede to the agreements made for its rational exploitation[70].

The Conference also drew attention to problems arising from disagreements among parties to fisheries treaties, on the need for conservation measures as well as on the nature of the measures to be taken, and suggested that such disagreements should be referred to suitably qualified and impartial experts[71]. Finally, the Conference recognized that a particular problem arose when intensive exploitation took place in an area of the high seas adjacent to already exploited coastal waters and could thus seriously affect the abundance of fish in these waters. It noted that the problem could be resolved by bringing the whole of the exploited area under a single conservation agreement with a view to maintaining an optimum sustainable yield of

the stocks concerned. The Conference, however, was divided as to what should be done in case no such agreement existed or could be concluded on time to avoid the depletion of these stocks and as to whether coastal States, in such an event, were to be entitled to take conservation measures in areas of the high seas adjacent to their coast[72].

The Rome Conference has been, so far, the only world conference organized by the United Nations on the subject of fisheries. It provided, for the first time, the opportunity to carry out a comprehensive review of all problems relating to the exploitation of the living resources of the sea, to take stock of the state of the art in the field of fisheries biology and to list guiding principles to be followed in formulating international fisheries conventions in the future. It certainly has had considerable influence on the drafting of many subsequent treaties. Unfortunately, at the time of the Conference, unqualified belief in the validity of the maximum sustainable yield concept was still in the minds of many biologists. The Conference therefore laid down, as the main objective of living resources conservation, the necessity of attaining such an objective and this concept has found its place in almost all fisheries conventions concluded ever since.

Immediately after the Rome Conference the ILC adopted in 1955 and 1956 a series of draft articles relating to the conservation of the living resources of the sea, thus revising its initial draft in the light of the report of the Rome Conference. Finally, at its eleventh session the UN General Assembly decided to convene a conference with a view to negotiating a convention.

The First United Nations Conference on the Law of the Sea (UNCLOS I) met in 1958 in Geneva. It adopted four separate conventions: the Convention on the High Seas[73], the Convention on the Territorial Sea and Contiguous Zone[74], the Convention on the Continental Shelf[75] and the Convention on Fishing and Conservation of the Living Resources of the High Seas[76].

The main features of these conventions as they pertain to the conservation and exploitation of living resources are as follows:

a) Jurisdiction

The Conference did not succeed in reaching agreement on the breadth of the territorial sea. All proposals made in that respect to the Conference were rejected. As a result the Convention on the Territorial Sea and the Contiguous Zone contained no provision on this matter. To get out of this deadlock another conference was convened in 1960 in Geneva. This was the Second United Nations Conference on the Law of the Sea (UNCLOS II). But this second conference also failed.

UNCLOS I succeeded, however, in establishing an elaborate system for the purpose of establishing the baselines for the measuring of the breadth of the territorial sea. In doing so it legitimized by treaty the expansion of internal waters that had already started in several countries some years before, after the International Court of Justice had judged, in the Anglo-Norwegian Fisheries case in 1951, that straight baselines could be used to delineate the landward limit of the territorial sea.

In respect of the continental shelf, the Conference agreed that the coastal state exercised sovereign, and therefore exclusive, rights for the purpose of exploring it and

exploiting its natural resources[77]. It was not able, however, to come to an agreement on the outer limit of the shelf. Article I of the Convention on the Continental Shelf defines that term as "referring to the seabed and subsoil of the submarine areas adjacent to the coast, but outside the area of the territorial sea, to a depth of 200 metres or, beyond that limit, to where the depth of the superjacent waters admits of the exploitation of the natural resources of the said areas, ...". Thus to a clear physical criterion was added a vague criterion of exploitability.

With regard to the living resources of the continental shelf, the Conference endeavoured to legitimize the earlier claims of coastal states over sedentary fisheries[78]. It gave jurisdiction to the coastal state over the regulation of fisheries conducted by means of equipment embedded in the floor of the sea in areas of the high seas adjacent to the territorial sea[79]. It also gave the right to the coastal state to exploit, on the continental shelf, all living organisms belonging to sedentary species and defined these species as including "all organisms, which at the harvestable stage, either are immobile on or under the seabed or are unable to move except in constant physical contact with the seabed or the subsoil"[80]. This was not, however, a scientifically based definition, and although there was no doubt that such organisms as sponges, coral or most molluscs were covered by the new text, uncertainties clearly remained with regard to the inclusion of other species, particularly crustacea. As a result, disputes were soon to arise between states fishing for certain species of Crustacea on the high seas[81].

The Conference also examined the question of the extension of jurisdictional fishery limits beyond the territorial sea but was unable to reach agreement on this point[82]. It recognized, however, that some states had a special interest in the conservation of certain resources of the high seas. Most prominent among these states are coastal states because they have "a special interest in the maintenance of the productivity of the living resources in any area of the high seas adjacent to their territorial sea"[83]. For that reason "a coastal state is entitled to take part on an equal footing in any system of research and regulation for purposes of conservation of the living resources of the high seas in that area even though its nationals do not fish there"[84]. Other states, however, even if their nationals are not fishing in a particular area of the high seas, may, if they have a special interest in the exploitation of the resources of that area, request the states that are fishing in that area to take the necessary conservation measures[85].

With regard to the high seas, the conventions reaffirmed the principle of the freedom of fishing for all states subject only to their treaty obligations and to the rights and obligations of coastal states as provided for in the Convention on Fishing and Conservation of the Living Resources of the High Seas[86].

Finally, nothing was said on the subject of the legal status of anadromous species or of any species breeding exclusively in areas under national jurisdiction but which subsequently migrate to the high seas. The earlier regime remained therefore applicable and species migrating from one jurisdictional area to another continued to be governed by the rules that applied to the particular area in which they happened to be at a certain time in their migration cycle.

This situation resulted some years later in a serious controversy on the subject of the Atlantic salmon.

From time immemorial salmons were caught in the rivers, estuaries and occasionally coastal waters of the countries where they spawned. But it is only very recently, in the late 1950s, that one of the feeding grounds of the Atlantic salmon was discovered off the west coast of Greenland. Salmons from the east coast of North America and from Western Europe were soon found to congregate and intermingle there in large numbers[87], and a commercial fishery, mostly by Danish vessels, started to develop fast. Some of the countries of origin of these fishes, such as Canada, the United States and the United Kingdom objected to the taking of salmon on the high seas on the grounds that their own highly valuable sports fisheries would be damaged and that high seas fishermen were unduly benefitting from conservation measures, such as the erection of fish ladders or restocking operations to which they did not contribute.

The states whose fishermen were taking the resource in international waters rightly replied that under international law they were perfectly free to do so[88]. After several unsuccessful attempts the dispute was eventually settled through ICNAF by a decision to phase out the high seas fishery and to limit the quantities of salmon that could be taken by local Greenland fishermen[89].

b) Conservation, Exploitation and the Problem of the New Entrants

The Convention defines conservation of the living resources of the high seas as "the aggregate of the measures rendering possible the optimum sustainable yield from those resources so as to secure a maximum supply of food and other marine products" thus taking up the formula which had already appeared in the report of the Rome Technical Conference and making of this concept a principle of international law but without providing any definition[90].

The Rome Conference, however, had established that the optimum sustainable yield was only the main objective of living resource conservation, thus implying that conservation could have other goals than the mere production of food or other products[91]. The Geneva Convention, on the other hand, merely equates conservation with optimum sustainable production[92]. With regard to fishing, the Convention on Fishing and Conservation of the Living Resources of the High Seas merely lays down the general principle that "all States have the duty to adopt or to co-operate with other States in adopting such measures for their respective nationals as may be necessary for the conservation of the living resources of the high seas"[93], thus simply providing a very general framework for the conclusion of subsequent more limited, but also more detailed, fisheries agreement. In establishing this principle, however, the Conference did indeed try to do more than merely legitimize in international law an idea which by that time had become generally accepted by the community of nations. It also tried to solve the irksome problem of the new entrants into an existing fishery by making the access to such a fishery contingent upon the acceptance by the entrants of the conservation measures taken in the area of that fishery by the state or states that were engaged in fishing in that area previously[94]. For the first time in the history of international law the freedom of fishing on the high seas was no longer absolute: states were bound to cooperate in the conservation of living resources and new entrants into a fishery could no longer, by wreckless exploitation, nullify the conservation efforts of those who had started to exploit the fishery first.

c) Institutions

The Geneva Conventions did not provide for the establishment of any commissions or other agencies to assist in their implementation. The ILC, however, at its third session in 1951 had adopted the principle of the establishment of an international authority, to be created under the United Nations, empowered to regulate fisheries on the high seas and whose decisions would be binding upon all parties[95].

The matter was little discussed at the Rome Conference. Several delegations felt, however, that the establishment of such an international authority would be difficult and that the problem lent itself better anyhow to a regional approach through the conclusion of regional agreements the importance of which was clearly emphasized.

The idea of an international, authority was subsequently, therefore, abandoned by the ILC in favour of a system of compulsory arbitration[96]. No permanent bodies were established, and no secretariat designated, and the system of compulsory arbitration which was designed, *Inter alia,* to compel states, including new entrants to an existing fishery, to take necessary conservation measures has not been used a single time since the Conventions came into force.

3. From UNCLOS I to UNCLOS III (1958-1973)

a) The Development of Ocean Exploitation

During the years that followed the Geneva Conference the world went through a period of considerable economic development. This development affected the oceans in several ways. Coastal habitats were increasingly being destroyed by the expansion of urban areas, industrial development, the construction of tourist facilities and pollution due to the discharge of industrial and domestic sewage.

Global pollution of the oceans increased manifold as a result of considerable developments in the use of certain toxic chemicals such as pesticides, PCBs and heavy metals in agriculture and industry. Pollution by oil, a by-product of the considerable increase in the demand of energy in the world, caused by wilful discharges from ships clearing their ballasts or by accidents such as that of the Torrey Canyon, became a serious environmental problem in many parts of the world. Dumping of wastes of all sorts into the sea, especially toxic and radioactive waste became also more and more alarming. All possible pathways of pollutants into the seas: ships, rivers, coastal discharges and the atmosphere contributed to raise the pollution of the seas to a disturbing level.

Parallelly, oceanic fisheries continued to develop at a rapid rate. From about 34 million tonnes, on the average, between 1958 and 1962, annual fish landings of marine species went up to more than 46.5 million tonnes between 1963 and 1967 and to 57.5 million tonnes between 1968 and 1972. In 1974 world landings reached 60.2 million tonnes and in 1978, 64 million tonnes[97]. It must be noted, however, that the rate of growth of the world fishing industry is now slowing down, a sign that many of the traditional fisheries are now fully exploited or overexploited and that fewer new fisheries are being developed[98]. Furthermore, it also seems that the marginal costs of securing these catches is increasing as abundance of stocks is reduced as a consequence of fishing. Yield per unit of fishery effort is declining.

This is evidenced by the fact that total registred tonnage of fishing vessels has everywhere been rising faster than fish catches[99].

Most major technical innovations in the field of fisheries appeared after the 1958 Geneva Conference. These include the use of sonar to locate fish which is especially important for purse-seiners, the development of new gear such as the mid-water trawl and fish pumps and considerable improvements in on-board processing and freezing techniques. As a result, considerable investments were made by a certain number of nations in the expansion and modernization of their fishing fleets. Many large and modern trawlers were built that were able to engage in fishing in all parts of the world. In addition, flottillas composed of a large mother-ship, with all necessary processing and freezing facilities, accompanied by a number of smaller catcher-boats have now almost become standard practice for the distant waters campaigns of a number of important fishing nations. Parallelly to this development, a world network of port facilities where fishing vessels can be unloaded, supplied and repaired and their crews relieved was gradually established[100]. There are at present in the world about 20.000 boats over one hundred tonnes (GRT) some of which reach 4.000 tonnes (GRT)[101]. The aggregate tonnage of these boats is of the order of 12 million tonnes (GRT). Japan and the USSR account for a large fraction of this tonnage and have now for many years been engaged in long range fishing with mothership and catcher boats. Other important distant water fishing fleets are those of Poland, Cuba, Spain, the Republic of Korea, the Federal Republic of Germany and the German Democratic Republic.

b) The Conclusion of New Fisheries Agreements

After the conclusion of the Geneva Convention on Fishing and Conservation of Living Resources of the High Seas, a number of new fisheries agreements were negotiated and came into force. Most of these agreements applied to specific areas of the ocean where risks of overfishing occured in consequence of the exploitation of fish stocks by fishing fleets of different nations. The most important of these agreements are: the North-East Atlantic Fisheries Convention[102], signed in London on 24 January 1959, replacing the Convention for the Regulation of the Meshes of Fishing Nets and the Size Limits of Fish (London, 5 April 1946); the Convention concerning Fishing in the Black Sea[103], signed at Varna on 7 July 1959; the Convention on the Conservation of the Living Resources of the Southeast Atlantic[104], signed in Rome on 23 October 1969, and the Convention on Fishing and Conservation of the Living Resources in the Baltic Sea and the Belts[105], signed at Gdansk on 13 September 1973.

A few agreements were also concluded on the subject of the conservation of particular species in certain marine areas. These were: the International Convention for Conservation of Atlantic Tunas signed at Rio de Janeiro on 14 May 1966[106] and, with respect to marine mammals, the protocol to the International Convention for the Northwest Atlantic Fisheries, extending the provisions of the Convention to Harp and Hooded Seals[107], signed at Washington on 15 July 1963; the Agreement on Conservation of Polar Bears[108], signed at Oslo on 15 November 1973 and the Convention for the Conservation of Antarctic Seals[109], signed at London on 1 June 1972.

Between 1973 and the date of the signature of the Convention on the Conservation of Antarctic Marine Resources on 21 May 1980, no fisheries treaties were conclud-

ed, pending presumably the outcome of the Third United Nations Conference on the Law of the Sea (UNCLOS III).

Most of the agreements mentioned above refer to the objective of an optimum yield, maximum sustainable catch or maximum productivity, following in that the Geneva Convention. All contain provisions on research, provide for the establishment of a Commission with regulatory powers to make recommendations or decisions subject to objections by parties and a few create, in addition, an enforcement mechanism. Thus, in 1973 many of the important fishing areas and commercial stocks came under the regulatory authority of an international fisheries commission. On the other hand, many others did not. In the absence of a world fisheries conservation authority empowered to take action and to establish regulatory mechanisms wherever required, this situation was, however, inevitable.

c) Research

Marine research is almost always undertaken by government agencies or national scientific institutions and very seldom by international organizations with their own equipment and staff. At world level, however, certain international agencies have, nonetheless, an important role in co-ordinating national research activities, opening up new avenues for research and helping their member countries to develop their own research potential. The International Oceanographic Commission (IOC) an organization closely associated with UNESCO, plays in that respect, a particularly significant part.

The leading non-governmental organization in the field of oceanography is the Scientific Committee on Ocean Research (SCOR) which was established in 1957 by the International Council of Scientific Unions(ICSU). SCOR, which is composed of all the leading marine scientists of the world provides scientific advice to IOC and other agencies in the United Nations system[110].

Regarding research in the area of fisheries, as distinct from general marine biological research, it was felt from the very outset of fisheries research and development that it was best dealt with on a regional basis, as species and conditions varied so much from one part of the ocean to another that the result of global research would be inapplicable to specific situations[111]. Fisheries research has, therefore, been associated from the very beginning to regional fisheries organizations or to organizations dealing with particular species or groups of species.

An important and noteworthy exception, however, is the International Council for the Exploration of the Sea (ICES) which is an intergovernmental organization established by treaty[112] and whose only tasks are to promote research, particularly in respect of living resources, to draw up research programmes, to organize research in agreement with its contracting parties and to dissimenate the result of the research thus undertaken[113]. ICES does not carry out any research itself and relies, for that purpose, entirely on its member countries. The council is the oldest fishery body in existence. It was established in Copenhagen in 1902. The treaty under which it was set up was revised in 1964 and the geographic competence of ICES extended to cover the whole Atlantic Ocean and its adjacent seas[114]. The Council is, under the relevant international agreements, the scientific advisory body to the Northeast Atlantic Fisheries Commission (NEAFC) and the International

Baltic Sea Fisheries Commission (IBSFC). It had working relations with the International Commission for the Northwest Atlantic Fisheries (ICNAF)[115]. One of the main tasks of ICES has been for many years to study and monitor stocks of fish in the North Atlantic, the North Sea and the Baltic. Although it has been and remains a purely scientific body with no regulatory authority it has, nonetheless, through its advice to governments, been instrumental in the conclusion of several international fisheries agreements, and in particular of the first Convention for the Regulation of Whaling of 1931[116] and the fisheries agreements for the Northeast Atlantic and the Baltic. It is considered as an organization which has particularly well succeeded in its tasks.

A similar body, the International Commission for the Scientific Exploration of the Mediterranean Sea, was established in Madrid in 1919 to cover the Mediterranean and its adjacent seas, but it never assumed the importance and international recognition of ICES.

Few fishery bodies are empowered by the treaties under which they have been established to carry out research themselves, that is to say with their own staff and budget. Noteworthy exceptions to this rule are the International Pacific Salmon Fisheries Commission[117] and the Inter-American Tropical Tuna Commission[118]. Fishery Commissions, therefore, have to rely to advise them on scientific matters on their party states, or, in a few cases, on organizations like ICES. Many fishery conventions provide for the establishment of a scientific council or committee. But these organs are composed of scientists who have been appointed by member governments. Thus, most fishery commissions do not benefit from the result of independent or neutral research and scientific arguments are, on the contrary, often used as a means to support a political position, controversial as it may sometimes be. As just one example, one should mention the International Whaling Commission where repeated excessive quotas, as evidenced by the subsequent decline in the stocks, were always supported by seemingly sound scientific arguments on the part of those countries who opposed stricter conservation measures.

Research, however, should not be limited to the gathering of information to be used by fisheries commissions for the purpose of regulating the harvesting of exploited stocks. Data on species habitat, on the effects of pollution on marine animals and plants in general and on the consequences of the introduction of alien species are also of great importance for the preservation of the living resources of the sea. Likewise, research on marine ecosystems, species interrelationships, and other ecological factors, including basic life-supporting processes, is essential if adequate conservation measures are to be taken in the sea. So far, research in those fields has been mostly concentrated on pollution, the effects of which on marine fauna are being studied by an increasing number of organizations, both national and international. On the other hand, the problem of habitats and their destruction or degradation has been barely touched upon. It is, however, a comforting sign that ICES is now expressing considerable concern on that matter[119]. As to general ecological and biological research on the marine environment, it does not seem that IOC which is the only inter-governmental organization that could play the role of coordinator, has either the means or, at least for the time being, the willingness to shift to living resources part of the emphasis which has been so far heavily placed on the side of physical oceanography.

d) The Control of the Threats Affecting Living Resources

i) Overexploitation

Fisheries commissions empowered to take conservation measures cannot be more conservation-minded than the sum of their members. The decisions they take are, therefore, often influenced by economic or political considerations and are the result of compromises between conservation requirements, as determined by scientists, and the search for immediate short term economic benefits, as demanded by some of their members which have to try to amortize over-investments or placate some important economic or political pressure-groups. This explains why many of the existing commissions, although provided with the best possible legal mechanisms to establish sound conservation measures, have nonetheless often failed to achieve a satisfactory conservation record.

These mechanisms have, for that reason, been criticized by conservationists who considered them inadequate. But as international law stood, and still stands, even better mechanisms would have failed since the real problem lies more in the willingness of states to accept, apply and enforce necessary conservation measures than in the perfectiveness of the legal mechanisms concerned. Short of a system of fisheries management such as the one proposed by Leonard in 1944[120], where decisions would be taken and enforced by majority votes, and that states are certainly not prepared to accept, there does not seem to be much hope, therefore, to modify, to any great extent at least, the present mechanisms. We have seen[121] that, in the opinion of scientists, overexploitation of marine species could be prevented only if regulations combined a limitation of the size of captured individuals with an overall catch quota set at a level somewhat lower than the maximum sustainable yield. It followed that, if fishing was to remain profitable for individual operators, some form of limitation of the fishing effort was also required.

Excessive effort on traditional species, sudden application of massive effort to little-studied species, conflicts between local and long-distance fleets have now become common problems throughout the world[122]. Faced with these problems, fishery bodies were often, however, unable to act with sufficient force or timeliness to prevent damage to the fisheries and, in consequence, to the economic interests of the fishermen themselves. Examples are numerous, but one which seems to be of particular interest is that of the International Commission for the Northwest Atlantic Fisheries (ICNAF) because of the very broad regulatory authority with which it is empowered and also because it "has often been a pioneer among international fishery bodies by its initiatives in the formulation and introduction of management measures, in particular in the field of quotas"[123]. Fishing effort increased rapidly in the Northwest Atlantic area in the 1960s, but despite the exploitation of some previously unfished stocks the total catch no longer increased after 1966 and several of the important stocks that supported the traditional fisheries were depleted. "Recognizing the damage, the ICNAF countries in a series of meetings from 1972 to 1974 set quotas on catches that could be taken from all the major stocks, and further negotiated a division of each quota among countries. While in principle the Commission had decided to limit the catch from each stock to the level recommended by its scientific committee, in practice the negotiators tended to agree on the most optimistic of the range of uncertain estimates produced by the scientists, and in some cases set the quotas even higher. As a result, the quotas

were, in many cases, too high to prevent the continuing decline of the stocks, and it was necessary to reduce the quotas each year. For the same reason that the quotas were not low enough when they were first imposed neither were the reduced quotas usually low enough to conserve the stocks in their worsened condition". Moreover, the principle of the limitation of the fishing effort was not accepted by the commission. Finally "ICNAF did not really succeed in enforcing its regulations, since each state retained authority over vessels under its flag, and, for various reasons, governments were not always able to make their nationals at sea comply with the regulations"[124].

A very similar situation prevailed in the Northeast Atlantic where all the larger demersal fishes are fully exploited and the herring stocks seriously depleted if not almost extinct. Here again, the Northeast Atlantic Fisheries Commission (NEAFC) started to set catch quotas, at least in certain areas, but "the quotas for herring have generally remained well above the potential of the stocks, with the result of continued serious depletion"[125].

Another, and most notorious, example of overexplotation in a regulated fishery is that of the great whales. The right whales and the grey whale were already close to extinction in the early part of this century and were given full protection under the Convention for the Regulation of Whaling of 1931[126] and the International Agreement for the Regulation of Whaling of 1937[127].

The International Convention for the Regulation of Whaling, signed at Washington in 1946[128], which established the International Whaling Commission (IWC) empowered that commission to regulate whaling anywhere in the world. Right whales and grey whales were again placed under full protection and the taking of the other species regulated. This did not prevent, however, a considerable decrease in the numbers of the regulated species. Blue whales, from an initial population of over 200.000 animals were reduced in 1965 to only a few thousands[129], the stock of humpback whales in the Southern Hemisphere, certainly the largest stock of this species was also the most heavily exploited. From an original size of about 100.000 it has now been brought down to 2.500 to 3.000[130]. The initial stocks of fin whales in all oceans probably ranged between 420.000 and 500.000 animals. They have now been reduced to a little more than 100.000 individuals. In the Antarctic alone,"the gross exploitation between 1955 and 1965 reduced the species from around 350.000 —400.000 to 70.000"[131].

In 1960 the Scientific Committee of the IWC urged the Commission to give also full protection to blue and humpback whales. But it took five or six years, during which these species were brought even closer to extinction, before this protection was eventually decided upon by the IWC. In respect of the other species under its regulatory authority, the Commission set annual quotas "intended to sustain catches but not necessarily to maximize them sustainably. It took more than twenty years of effort to reach the point at which quotas were agreed for all areas, and not only for the Antarctic, for all species and not only for the ones which were under heaviest pressure, and for all commercial whaling operations, not just for pelagic expeditions"[132].

Thus many stocks of marine animals were heavily exploited, quite a few already overexploited and several important world fisheries had completely collapsed. As to the great whales, they were probably saved from near extinction by a considerable arousal of interest on the part of the general public and by many groups of conservationists, especially dedicated to the cause of whales, which had become powerful enough pressure-groups themselves to countervail the opposite interests of a gradually vanishing industry.

Parallelly to the decline of many stocks, the regulatory measures taken by fishery commissions tended to become increasingly comprehensive. Earlier measures, which were often limited to the regulation of mesh sizes, were supplemented by the imposition of overall quotas and even, sometimes, by the allocation of national quotas. It appeared, however, that without limiting at the same time the amount of fishing effort, there was noway to improve the situation substantially.

The problem posed by the incidental catches of non-target species continued generally to be ignored almost everywhere with, however, two important exceptions. The convention establishing the Inter-American Tropical Tuna Convention Commission includes in the definition of the species it covers "other kinds of fish taken by tuna fishing vessels"[133]. This provision constituted probably the legal basis upon which the Commission decided at its 34th meeting in 1977 to launch a comprehensive programme of research into the incidental taking of porpoises by tuna purse-seiners in the Convention area and to take measures to reduce this incidental catch. The International Convention for the Conservation of Atlantic Tunas[134] provides in Article IV.1 that, in addition to tuna and tuna-like fishes, the Convention also covers "such other species of fish exploited in tuna fishing in the Convention areas as are not under investigation by another international fishery organization". Discussions at the Conference of plenipotentiaries where the Convention was concluded clearly showed that there was a consensus to cover incidental catches. The Conference, however, decided that in respect of catches of species other than tuna and tuna-like fishes, the Commission should be restricted to research activities and should be expressly precluded from recommending conservation measures[135].

Another aspect of marine resources conservation is the protection of endangered species. Mention should be made here, in this respect, of the protection accorded to certain marine species by international conventions dealing with the subject of wildlife conservation. A recent example of this type of conventions is the treaty concluded in Algiers in 1968 on the conservation of nature and natural resources in Africa[136]. This treaty gives full protection to the dugong, the African manatee, the Mediterranean monk seal and to all species of marine turtles. An even more recent example of a conservation treaty according protection to certain marine species is the Convention on the Conservation of European Wildlife and Natural Habitats which was concluded in Bern on 19 September 1979 under the auspices of the Council of Europe[137]. Reference should also be made here to the Antarctic Treaty[138] which, in Article IX, provides for meetings of the Parties to recommend, *inter alia,* measures regarding the preservation and conservation of living resources in the Treaty area. These measures were approved in 1964 and bear the name of "Agreed Measures for the Conservation of Antarctic Fauna and Flora"[139]. They provide, in particular, for the full protection of certain species of seals and seabirds. The suc-

cess, however, of most of these international conservation conventions in ensuring the protection of endangered species has, so far, been somewhat limited. This is probably, partly at least, due to the fact that there are generally no provisions establishing permanent institutions and that enforcement, except in the case of the Antarctic Treaty where a system of observers is provided for[140], is invariably left to national authorities.

Finally, it also appears appropriate to refer here to the limitation of exploitation that may be achieved by prohibiting or controlling the international trade in endangered species of wildlife. International conservation agreements generally contain provisions on the control of such trade but their enforcement is left to the authorities of exporting countries. Concerned by the serious decline in the numbers of many species of wildlife, including marine species, the states that met in Washington in 1973 to conclude the Convention of International Trade in Endangered Species of Wild Fauna and Flora (CITES) decided to go one step further and instituted a strict system of export and import controls[141]. Among the marine species that are listed in the appendices to the Convention, the trade of which, including the trade in products and derivatives, is subject to these controls, are all the cetaceans, several species of seals and all species of marine turtles. By the middle of 1981, 70 states had become parties to the Convention. There is an institution, the Conference of the Parties, which meets regularly, and a permanent Secretariat, and party states are gradually setting up the necessary import and export control mechanisms required to implement effectively the Convention.

ii) Habitat destruction

Few if any fisheries agreements contain provisions on the preservation of the habitats of the species they purport to preserve. This is probably due to the belief that the main threat to marine resources lies in overexploitation as well as to insufficient knowledge of the location and ecological characteristics of species critical habitats. Admittedly, some measures have been taken to close to fishing, either permanently or at certain times, certain areas such as spawning grounds or nurseries. But that kind of action was generally designed to protect the spawners or the young fish more than to preserve the ecological conditions that are essential to their reproduction or growth. An exception to this rule can be found, however, in the treaty placing under joint regulation by Canada and the United States an anadromous stock, the sockeye salmon of the Fraser river system[142] which, in its Article III, instructs the Commission established under the agreement, to make a thorough investigation into the conditions of the salmon's spawning grounds and empowers it to recommend the removal or the overcoming of obstacles to the ascent of the fishes in the rivers.

In areas under national jurisdiction, such as internal waters or the territorial sea, where protective measures can be taken through national legislation, very few laws were, in the recent past, really concerned with the preservation of important marine habitats. A noteworthy exception in the Fisheries Habitat Reserves Regulations enacted by the State of Queensland in Australia in 1968 which provide for the total protection of mangroves in the internal waters of that state[143].

The establishment of protected areas such as parks or reserves for the purpose of protecting particularly important ecosystems or the habitat of endangered species

has, for a long time, almost completely been confined to land areas. Marine parks and reserves were few and the area they covered was generally small. The recognition of the necessity to create marine protected areas as an important conservation priority is thus relatively recent, as awareness of the considerable value of many coastal habitats, together with their accelerated destruction, increased. Several international conservation conventions provide for the obligation to establish protected areas, among which the Western Hemisphere Convention[144], the African Convention[145] and the World Heritage Convention[146] stand most prominently. So far, marine parks or reserves created under these treaties have not been many. The Convention on Wetlands of International Importance, Especially as Waterfowl Habitat, otherwise known as the Ramsar Convention[147], from its place of signature, was concluded in 1971. It is the only worldwide treaty, so far, that deals with the preservation of a certain type of ecosystem. Its only aim is to ensure the conservation of wetlands, whether inland or coastal. Parties are under the obligation to formulate and implement their planning to promote, as far as possible the wise use of wetlands on their territory and to establish nature reserves. They must also designate at least one wetland for inclusion in a list of wetlands of international importance. In spite of its many imperfections[148], including its lack of force and the absence of a secretariat, the Ramsar Convention has met with some success in promoting the preservation of many wetlands of international importance, including a number of coastal wetlands[149].

As regards national legislation on marine parks and other protected areas, a corpus of legislation is now gradually developing since about 1975. It will be examined in more detail in the latter part of this paper. Before that date specific legislation on the subject was almost non-existent.

Finally on the high seas, where the establishment of protected areas can only be achieved through treaties, it does not seem that the very idea of creating marine parks was ever explored.

In 1975, the twelfth General Assembly of the International Union for Conservation of Nature and Natural Resources (IUCN), meeting at Kinshasa, Zaire, adopted three resolutions urging governments to establish marine parks, to conserve critical coastal marine habitats and to conserve mangroves, salt marshes and related habitats[150]. These resolutions exemplify the new awareness of the problem, the magnitude of the task, and the urgency there is to accomplish it.

iii) Pollution

This paper does not deal with the subject of pollution as a general threat to the health of the oceans. Certain pollutants, or certain pollutions at certain places or at certain times may, however, have far-reaching consequences on the survival of certain species or the continued maintenance of certain ecosystems. A short analysis of measures taken to control pollution as it may affect important living resources seems therefore in order.

Chlorinated hydrocarbons, in particular DDT and PCBs and heavy metals such as mercury or cadmium have now been recognized for some years as particularly hazardous because of their toxicity, persistence and concentration by marine organisms[151]. It is now increasingly recognized that their release into the environ-

ment, and therefore eventually into the sea through run-off waters and rivers or through the atmosphere, should be controlled and if possible banned. In the case of these substances ships do not constitute a source of pollution, except in the event of an accident. Chemicals are dumped into the sea as waste or used or disposed of in the terrestrial environment. In other words, pollution by these toxic substances is almost entirely land-based.

Solutions, therefore, lie in the prohibition of dumping, and in the taking of drastic measures to reduce land-based pollution to a sufficiently low level. Accordingly, a World Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matters was concluded on 29 December 1972[152]. Article IV of that Convention prohibits the dumping of substances listed in Annex I to the Convention. This list includes, *inter alia,* organohalogen compounds, mercury and cadmium and their compounds, and highly radioactive wastes. A few months earlier, on 15 February 1972, a regional Convention had been concluded in Oslo on the same subject for the Northeast Atlantic area[153]. Here again the dumping of the substances listed in Annex I to the Convention is prohibited, and here again organohalogen compounds, which include DDT and PCBs are listed together with mercury, cadmium and their compounds. The Convention on the Protection of the Marine Environment of the Baltic Sea Area, signed at Helsinki on 22 March 1974[154], covers both dumping and land-based pollution. Under Article 5 of that Convention, "the Parties undertake to counteract the introduction, whether airborne, waterborne or otherwise, into the Baltic Sea Area of hazardous substances as specified in Annex I of the present Convention". This time the only substances listed in that Annex are DDT and its derivatives, and PCBs[155]. Mercury, cadmium and their compounds appear only in Annex II which does not call for an absolute prohibition[156].

As a counterpart to the 1972 Oslo Dumping Convention, a Convention for the Prevention of Marine Pollution from Land-Based Sources was signed in Paris on 4 April 1974[157]. It covers the Northeast Atlantic and Arctic Oceans and the dependent seas excluding the Baltic and the Mediterranean. Under Article 4 of the Convention, the Parties untertake to eliminate, if necessary by stages, pollution of the maritime area from land-based sources by substances listed in Part I of Annex A to the Convention. Here again, organohalogen compounds, and mercury and cadmium and their compounds appear on the list.

New Conventions have since been signed or are in the process of being concluded within the framework of the United Nations Environment Programme's (UNEP) Regional Seas Programme[158]. By the end of 1980, two Protocols to the Convention for the Protection of the Mediterranean Sea Against Pollution[159] relating to dumpIng[160] and land-based pollution[161] had already been agreed upon by Mediterranean coastal countries. Both provide, *inter alia,* for the gradual elimination of the release of the highly toxic substances mentioned above.

Mention should finally be made of the measures taken by OECD in respect of these substances. A decision taken by the OECD Council on 13 February 1973 provides that PCBs stall not be used on the territory of member countries for industrial or commercial purposes except in a small number of uses in closed systems[162]. This is, so far, the only decision[163] of the OECD Council in this field. There have been, however, a number of recommendations to member countries that have also been

adopted by the Council in the field of toxic chemicals which, if they are not binding, have nonetheless a considerable hortatory value. Such is, for instance, the case of the recommendation adopted on 18 September 1973 on measures to reduce all man-made emissions of mercury to the environment[164].

With regard to ship-generated pollution, a distinction should be made between intentional or, as they are called, operational discharges by tankers or other ships cleaning their tanks, and which consists essentially of oil, and discharges that are the consequence of an accident and which may result in the release of large quantities of oil or of any other dangerous cargo.

To reduce operational pollution, strict regulations, appropriate enforcement procedures and the provision of adequate reception facilities in ports for the disposal of residues carried on board are required.

To limit as much as possible accidental pollution, preventive measures are essential. These include traffic separation schemes in areas of heavy traffic and the regulation of the design, construction, equipment, operation and manning of ships.

The intentional discharge of oil into the sea has been regulated since 1954 by the International Convention for the prevention of pollution of the sea by oil[165], as amended in 1969[166]. This Convention provides that the discharge of oil by tankers shall be prohibited in all sea areas within 50 miles from the nearest land[167]. For ships other than tankers discharges must only be made as far as practicable from land.

A new Convention was, however, concluded in 1973 as the 1954 treaty was increasingly considered inadequate to deal with ship-generated pollution. It is the International Convention for the Prevention of Pollution from Ships[168] which, by the end of 1980, had not yet come into force. Discharges of oil by tankers remain prohibited within 50 miles of the nearest land and are now also prohibited for other ships within 12 miles of the nearest land. But in addition, the new Convention establishes a certain number of special areas where discharges are totally prohibited. These special areas are the Mediterranean, the Baltic, the Black Sea, the Red Sea and the "Gulfs", which being almost completey enclosed bodies of water, are from the ecological point of view particularly vulnerable to pollution. In addition, the new Convention lays down an obligation for the provision of residues reception facilities, establishes certain standards for the construction of new tankers, regulates the control of pollution by toxic liquid substances in bulk, and contains provisions on the prevention of pollution by harmful substances carried in packaged form and the dumping of garbage from ships.

The London Conventions do not contain any specific provisions for the protection of particularly important ecosystems, critical habitats or vulnerable species. They rely almost entirely, for that purpose, on the 50 miles rule and on the prohibitions applicable to special areas. There is, however, one exception. A special baseline was established off the coasts of Australia to protect the Great Barrier Reef[169]. There are, however, certainly other areas which could benefit from this type of protective measure.

With regard to the prevention of accidents, improved regulations concerning safety at sea may go a long way towards the preservation of important ecosystems,

habitats or species. In that respect, treaties such as the Convention on the International Regulations for Preventing Collisions at Sea[170] and the traffic separation schemes adopted by IMCO under that Convention have certainly had and will continue to have a considerable importance. It remains, however, that accidental discharges of certain very toxic substances could have catastrophical consequences on unique ecosystems or endangered species and that stricter preventive rules, including restriction of navigation for ships carrying certain cargoes in certain areas could be a means better to prevent the occurrence of such accidents.

iv) Introduction of alien species

The introduction of alien species has not until very recently been considered as a possible threat to the marine environment. Furthermore, provisions regulating such introductions would hardly have been relevant in treaties dealing with fisheries or pollution. Admittedly, the matter could have been covered by the Geneva Convention on the High Seas, but was probably not in the minds of the negotiators at the time.

The only international instrument of some relevance to the conservation of marine resources where this question appears is the Agreed Measures for the Conservation of Antarctic Fauna and Flora[171], Article IX of this instrument provides that "each Participating Government shall prohibit the bringing into the Treaty Area of any species of animal or plant not indigenous to that Area ..."[172].

e) The United Nations Agencies and the Conservation of Marine Living Resources

United Nations specialized agencies and other bodies were initially set up on a functional basis just like ministerial departments within a government. Agencies were created to deal with such matters as food and agriculture (FAO), science, education and culture (UNESCO), public health (WHO) or maritime transport (IMCO). At the time when most of these agencies were established, a few years after the end of the war, the problems of the sea, particularly resource conservation and pollution, probably did not appear sufficiently important to warrant the setting up of a separate agency. Thus, marine matters have been dealt with by a wide range of different agencies: oceanic meteorology by WMO, the discharge into the ocean of nuclear substances by IAEA, the effects of pollution on public health by WHO, navigation and ship-generated pollution by IMCO, oceanography, marine sciences and ecology by UNESCO and the International Oceanography Commission (IOC), and fisheries by FAO.

Short of a reorganization of the United Nations System[173], which does not seem likely because of the natural resistance agencies are certain to display to any change that would affect their structure and deprive them of an important aspect of their activities, it is highly probable that this atomization and scattering of marine responsibilities will, with some adjustments, continue to exist in the future.

The creation of UNEP in 1972[174], immediately after the United Nations Conference on the Human Environment (Stockholm, June 1972), had as a purpose, however, to achieve better inter-agency coordination of all matters relating to the environment, including marine questions, and to catalyze the taking of new initiatives and the development of new environmental policies.

United Nations agencies, although not empowered to adopt regulatory measures, have had and continue to have, because of their worldwide membership and moral authority, considerable influence on the determination of policies, standards and practices relating to the preservation of the environment in general, and, in particular, of the marine environment.

Activities of the UN agencies in that respect have encompassed such subjects as research, data-collecting, policy formulation, and technical and financial assistance to developing countries. UN agencies have also been active in promoting the conclusion of several important international conventions concerning the conservation of natural resources and in providing the necessary secretariat services to the managing bodies established under these conventions[175].

The agencies which have been the most active in the field of marine resources proper, excluding therefore general measures to combat marine pollution, are UNESCO, and FAO. For the purpose of this paper, consideration of the activities of UN agencies will therefore be limited to those two organizations. Co-ordination mechanisms set up within the United Nations System will also be examined.

i) UNESCO

UNESCO, one of the specialized agencies of the United Nations, was created in 1946. Its terms of reference cover the whole field of education, science and culture. The activities of UNESCO in the field of marine sciences are exercised through the Intergovernmental Oceanographic Commission (IOC), the Division of Marine Sciences and, in respect of research and conservation in coastal areas, the Division of Ecological Sciences.

(a) Intergovernmental Oceanographic Commission (IOC)

The IOC is an autonomous body that was established within UNESCO in 1960. Its purpose is "to promote scientific investigation with a view to learning more about the nature and resources of the oceans through the concerted action of its members"[176]. It "functions as the co-ordinating body within the United Nations System for marine sciences and related activities. Co-operative scientific investigations, combined with a programme of training, education and mutual assistance, form the chief components of IOC's work"[177].

Among the main research programmes that have been carried out through the concerted efforts of IOC members, one should mention the International Indian Ocean Expedition (1959-1965), the International Co-operative Investigations of the Tropical Atlantic (1963-1964), the Co-operative Study of the Kuroshio and Adjacent Regions (1965-1977), and the Co-operative Investigations of the Caribbean and Adjacent Regions (1967-1976)[178]. In 1969 the IOC, upon request of the U.N. General Assembly, adopted a Long-term and Expanded Programme of Oceanic Exploration and Research (LEPOR). This programme has as a purpose "to increase knowledge of the ocean, its contents and the contents of its subsoil, and its interfaces with the land, the atmosphere and the ocean floor and to improve understanding of processes operating in or affecting the marine environment, with the goal of enhanced utilization of the oceans and its resources for the benefit of mankind"[179]. This very comprehensive programme includes the study of living recources and their relations with the marine environment, research on marine pollution and international in-

vestigations in specific regions[180]. During the first ten years of implementation of LEPOR, which were characterized by an accelerated phase of its development, the International Decade of Ocean Exploration, only one programme on living resources was launched whereas several dealt with pollution. In that respect it was decided to undertake, within the framework of LEPOR, a Global Investigation of Pollution in the Marine Environment (GIPME), whose ultimate objective is "the establishment of a sound scientific basis for the assessment and regulation of the marine pollution problem[181].

With regard to regional arrangements, IOC, after having for many years promoted regional programmes for the study of specific areas of the ocean, has, for the first time, established in 1975 a "Regional Association" of States for the Caribbean area, which was given the name of IOCARIBE[182]. This organization has its own governing body and secretariat. Its task is to oversee IOC projects in the region and to co-ordinate its activities with those of other U.N. agencies in the Caribbean region. Other "regional associations" may be expected to be set up in other regions in the future[183].

(b) Division of Marine Sciences

In 1972 the UNESCO Office of Oceanography was separated from the IOC Secretariat and became the Division of Marine Sciences[184]. This Division, which belongs to the Science Sector of the Organization specializes in the development of research programmes, training and assistance to developing countries, particularly on the subject of coastal ecosystems and coastal zone management, thus filling a considerable gap in present research efforts.

(c) Man and Biosphere Programme

In 1968 UNESCO organized, for the first time in history, an important, worldwide Conference on the subject of the rational use and conservation of the resources of the biosphere. This Conference reviewed the various threats to which the natural systems and wild species of the earth were exposed and recommended that an international research programme on Man and the Biosphere be launched on an intergovernmental basis as a follow-up and strengthening of the non-governmental International Biological Programme (IBP) which had been going on for some years.

The Man and Biosphere Programme (MAB) was officially launched by Resolution 2313 of the UNESCO General Conference which was adopted by the sixteenth session of the Conference in November 1970. The MAB programme is "a long-term inter-governmental and interdisciplinary programme on Man and the Biosphere focusing on the general study of the structure and functioning of the biosphere and its ecological regions, on the systematic observation of the changes brought about by man in the biosphere and its resources, on the study of the overal effects of these changes upon the human species itself and on the education and information to be provided on these subjects"[185]. The broad outlines of the programme were established at the first meeting of the MAB Co-ordinating Council in November 1971. Fourteen project areas have been proposed as subjects for research. Of particular significance for the preservation of marine living resources are project 5 which deals, *inter alia,* with the ecological effects of human activities on the value and resources of marshes, deltas, estuaries and coastal zones, and project 8 which is concerned with the conservation of natural areas and of the genetic material they

contain. Under the latter project a worldwide network of protected areas, which bear the name of biosphere reserves, is being established for the purpose of conserving species and genetic diversity and for use in a programme of monitoring, research and training. Among the many biosphere reserves which have been created so far, a certain number are designed to protect coastal ecosystems.

(d) World Heritage Convention

Mention should finally be made of the World Heritage Convention which was adopted by the General Conference of UNESCO on 23 November 1972[186], and which, in respect of the natural heritage is serviced, just like the MAB programme, by the Division of Ecological Sciences. Under this Convention, Parties are under the obligation to identify, protect and conserve natural sites, or the habitats of endangered species of outstanding universal value, included in the world Heritage List. Outstanding marine coastal areas may well, therefore, be so listed and benefit from the protection measures provided for by the Convention.

ii) FAO

FAO was established as a specialized Agency of the United Nations as early as 1945. For twenty years FAO's activities in the field of fisheries remained somewhat limited. In 1965, the FAO conference decided to establish a Committee on Fisheries (COFI) and to upgrade the FAO Fisheries Division to the status of Department.

Major activities of the Department of Fisheries include: assessment of fishery resources, review of management measures, collection and publication of fishery statistics on a world scale, studies of the effects of pollution on living resources, assistance to governments, particularly of developing countries, on specific fisheries management problems, fisheries planning, operations, training and administration and preparation of national, regional or inter-regional fisheries development projects which are funded by UNDP or by other sources[187].

Standing bodies concerned with the conservation of marine resources within the FAO structure are the Advisory Committee of Experts on Marine Resources Research (ACMRR) the Committee on Fisheries (COFI) and Fishery bodies established within the framework of FAO.

(a)ACMRR

This Committee was established as an advisory body to the Director General of FAO. Subsequently it also became advisory to IOC. It is composed of experts appointed by the Director General of FAO in their personal capacity[188].

(b) Committee on Fisheries

The FAO Committee on Fisheries is at present the only inter-governmental body, with worldwide membership, where all fishery problems can be the subject of periodic examination. It was set up under Article VI of the FAO Constitution, thereby making it merely a standing committee of the Organization reporting to the Council. At the time the Committee was established, however, some delegations felt that a body set up by means of an International Convention, as provided for under Article XIV of the Constitution would benefit from more flexibility in its operations, thus perhaps creating the nucleus of an International Fishery Organization which could thereafter acquire some regulatory powers. This proposal was not accepted but has come under review, however, several times since.

The two main functions of the Committee on Fisheries are to review the programmes of work of FAO in the field fo fisheries and to undertake periodic general reviews of fishery problems of an international character. Thus, the functions of the Committee are purely advisory[189].

(c) FAO Regional Fisheries Organizations

FAO has established nine regional fishery bodies. Three of these bodies deal only with inland fisheries. The remaining six commissions can be divided into two categories: bodies that have been set up under Article VI of the FAO constitution, by the Conference or by the Council and which, as a result, are merely commissions of the Organization, and bodies established, under Article XIV of the constitution, by means of an international agreement and which, although related to FAO, enjoy nonetheless a certain degree of autonomy.

Commissions belonging to the first category have mainly advisory functions. They include the Regional Fisheries Advisory Commission for the Southwest Atlantic (CARPAS), established in 1961, the Indian Ocean Fishery Commission (IOFC) and the Fishery Committee for the Eastern Central Atlantic (CECAF), both set up in 1967, and the Western Central Atlantic Fishery Commission (WECAFC), created in 1973.

Organizations established by International agreement under Article XIV are the Indo Pacific Fisheries Commission (IPFC)[190], set up in 1948, and the General Fisheries Council for the Mediterranean[191] (GFCM), established in 1949. Although not initially empowered to take regulatory measures, these two bodies had wider responsibilities than the commissions in the fields of resource management and development as well as for the protection of the marine environment against pollution. They were also entitled to make recommendations. The agreements establishing these two organizations were amended in 1976 with a view to strengthening their powers. The GFCM may now even take management decisions subject to an objections procedure.

(d) Conclusion

It seems important to note that although FAO is the only worldwide organization dealing with fisheries it seems to have for a long time deliberately avoided to enter the field of regulatory measures. The Committee of Fisheries, had it been set up under Article XIV of the FAO constitution and empowered with regulatory authority, could well have become an effective world fisheries organization either directly or through regional subsidiary bodies with the power to adopt regulatory measures. The member states of FAO, however, decided against it. The reasons for that decision, however, are not clear. Could states have been afraid to give too much authority to an international bureaucracy? Did they fear that regulatory measures would harm their immediate economic interests? Or were they simply anxious to avoid increase in the Organization's budget? Whatever the reason, the facts are that not only has the Committee on Fisheries remained an advisory body, but also that the various fishery commissions set up by the Conference or the Council of FAO to deal with fishery matters in areas of the world where no other fishery commission existed, and where management and regulatory requirements were certainly important, were nonetheless deprived of any authority. As to IPFC and GFCM, which have received, through the years, increased responsibilities, it is only in 1976 that they

were empowered to adopt management decisions. Furthermore, attention should be drawn to the fact that these two bodies, the only ones to be established under Article XIV, were set up in 1948 and 1949 and that the other fishery commissions created by FAO since that time are all advisory bodies established under Article VI.

iii) Inter-Agency co-ordination

The activities of the many UN Agencies dealing with marine matters have greatly expanded during the past twenty years. They include, at present, several large-scale programmes of a multidisciplinary character, such as for instance LEPOR or GIPME and the implementation of broad fishery research and development programmes involving several different agencies. Jurisdictional conflicts and unnecessary overlappings between Agencies are therefore bound to occur[192]. Co-ordination mechanisms were therefore essential.

The Advisory Committee of Experts on Marine Resources Research (ACMRR)[193] can hardly be considered to be a co-ordinating mechanism. As an advisory body, however, to two different agencies, FAO and IOC, it is rather unique in the United Nations system and, in that respect, necessarily ensures some degree of co-ordination between these organizations.

The Inter-Secretarial Committee on Scientific Programmes Relating to Oceanography (ICSPRO) consists of representatives of UNESCO, FAO, WMO, IMCO and the United Nations[194]. It was established in 1969. Its functions are to co-ordinate scientific programmes and to develop agreed proposals for joint action taking into consideration both the needs of IOC and those of the other organizations.

The Administrative Committee on co-ordination (ACC) of the United Nations established in 1960 a sub-committee on oceanography which was renamed in 1977 Sub-committee on Marine Affairs[195]. Its terms of reference are not limited to oceanography programmes but encompasses the whole range of marine activities undertaken by United Nations Agencies. Its functions are to ensure regular consultation between organizations of the United Nations System, to identify the means to respond to decisions of inter-governmental bodies within the UN system requiring collective action, to promote systematic prior consultations and joint planning, to identify programme areas and projects for concerted action and to recommend, where appropriate, the designation of "lead agencies".

The Joint Group of Experts on the Scientific Aspects of Marine Pollution (GESAMP)[196] is composed of independent experts appointed by the various UN Agencies concerned with marine pollution. Its functions are to advise these organizations on the scientific aspects of marine pollution, to assist in the development of LEPOR, to advise on the scientific aspects of the needs and criteria for inter-governmental instruments for the control of marine pollution and to provide advice, on request, to member States.

Mention should also be made of the role of the United Nations Environment Programme (UNEP). UNEP has, within the UN system, a global responsibility for all problems pertaining to the environment. It must, therefore, in that capacity review the state of the environment, promote monitoring programmes such as the Global Environmental Monitoring System (GEMS) and co-ordinate environment related pro-

grammes among the various UN Agencies. It would seem that "some internal conflicts of competence emerged among the organizations within the UN system, as a result of UNEP interaction with these bodies particularly in connection with direct involvement in operational matters"[197]. Co-ordination among the UN Agencies concerned is carried out by the Environment Co-ordination Board (ECB).

Mention should finally be made of another co-ordinating mechanism, the Ecosystem Conservation Group which is composed of representatives of UNEP, UNESCO, FAO and IUCN. It deals with the co-ordination of the activities of its member agencies in the field of ecosystems and species conservation, including marine habitats and species.

It is not very easy, from the outside, to pass a judgment on these various arrangements. It would seem, however, that their effectiveness could be improved and that co-ordination, for some matters at least, could be better achieved through an inter-governmental body rather than by meetings between representatives of Agency secretariats.[198]

iv) The Stockholm Conference

The United Nations Conference on the Human Environment which met in Stockholm in June 1972 can be considered as marking the end of the period of growing awareness of environmental problems and of haphazard action to resolve them, and the beginning of a period of legal recognition of these problems and of co-ordinated action to give them a solution.

The Conference laid down 26 principles and made 109 Recommendations constituting an Action Plan[199]. The principles are embodied in a Declaration which was adopted by the Conference. Although they have no binding force in law, these principles have nonetheless a considerable moral value, having been solemnly proclaimed by the Conference as the expression of the common conviction of all the States represented thereat. For that reason they must be considered as the basis for the future development of environmental law.

The Recommendations contained in the Action Plan, on the other hand, are generally of a more technical nature and are designed primarily to bring about the initiation or continuation of certain specific actions on the part of the Governments or of the International Organizations to which they are addressed.

Several of these principles and recommendations are relevant to the general problem of living resources conservation and in particular to the conservation of marine resources.

Most important, in this respect, are: Principle 2: "The natural resources of the earth including the air, water, land, flora and fauna and especially representative samples of natural ecosystems must be safeguarded for the benefit of present and future generations through careful planning or management, as appropriate."

Principle 5: "The non-renewable resources of the earth must be employed in such a way as to guard against the danger of their future exhaustion and to ensure that benefits from such employment are shared by all mankind".

Principle 7: "States shall take all possible steps to prevent pollution of the seas by substances that are liable to create hazards to human health, to harm living resources, and marine life, to damage amenities or to interfere with other legitimate uses of the sea".

Principle 21: "States have .... the responsibility to ensure that activities within their jurisdiction or control do not cause damage to the environment of other States or of areas beyond the limits of national jurisdiction[200].

Among the recommendations, particularly relevant are:
Recommendation 46 which stresses the importance of programmes such as LEPOR, GIPME and the International Biological Programme (soon to become MAB under UNESCO), presses for continued exchange of data on living aquatic resources *and their environment*[201] and fisheries activities, emphasizes the necessity to evaluate and monitor world fishery resources, *environmental conditions*[202], stock assessment, including statistics on catch and effort, and the economics of fisheries and support programmes and recommendations for management of stocks of fish and other aquatic animals proposed by the existing fishery bodies. This recommendation also draws attention to the fact that damage to stocks has often occurred because regulatory action is taken too slowly and that the need for management action to be nearly unanimous has reduced action to the minimum acceptable level.

Recommendation 47 recommends close participation of fishery agencies and interests in the preparations for UNCLOS III.

Recommendation 48 emphasizes the importance of international co-operation in the research, control and regulation of the side effects of national activities in resource utilization where these affect the aquatic resources of other nations. It underlines the importance of estuaries, intertidal marshes and other near-shore and in-shore environments in the maintenance of several marine fish stocks, warns against the effects of the discharge of toxic chemicals, heavy metals and other wastes on marine living resources even in the high seas, and guards against the deleterious effects of unregulated unilateral introduction of exotic species.

Recommendation 49 recommends the further development and strengthening of facilities for collecting, analysing and disseminating data on living aquatic resources *and the environment in which they live*[203].

Recommendation 50 urges governments to co-operate by strengthening the existing international and regional machinery for development and management of fisheries *and their related environmental aspects*[204], and, in those regions where these do not exist, to encourage the establishment of fishery councils and commissions as appropriate.

Other recommendations, although they do not specifically refer to the living resources of the sea, are nonetheless of direct relevance to the subject since they address general conservation matters and are therefore applicable to both land and sea. Of particular importance in this respect are:

Recommendation 37 which recommends to governments to co-ordinate and co-operate in the management of neighbouring or contiguous protected areas.

Recommendation 38 which recommends that governments set aside areas representing ecosystems of international significance for protection under international agreement.

Recommendation 39 which stresses that the conservation of genetic resources represents the crucial element to which all other programmes concerning these resources relate and recommends the establishment of an international programme to preserve them.

Recommendation 40 which presses for the making of inventories of the genetic resources most endangered by depletion or extinction, emphasizes that all species threatened by man's development should be included in such inventories, calls for special attention to be given to locating those areas of genetic diversity that are disappearing and stresses that these inventories should be reviewed periodically and brought up-to-date by appropriate monitoring.

Finally, Recommendation 32 recommends "that Governments give attention to the need to enact international conventions and treaties to protect species inhabiting international waters or those which migrate from one country to another". It is as a follow-up to this recommendation that a Convention on the Conservation of Migratory Species of Wild Animals was concluded in Bonn on 23 June 1979[205].

To sum up, the Stockholm Conference has laid down as a principle that water, flora and fauna, and, in particular, representative samples of natural ecosystems must be safeguarded. The recommendations made by the Conference are a consequence of this principle as well as of the conclusions it drew from the state of the environment as it was becoming apparent at that time. Of particular significance is the fact that overexploitation and pollution are no longer recognized as the only factors affecting the status of species and ecosystems. Indeed, emphasis is repeatedly placed on the necessity of preserving habitats, such as coastal habitats, and on the importance, not only of the living resources themselves but also of the environment in which they live. Thus, rational exploitation, habitat protection, pollution control and now even the regulation of the introduction of exotic species have all, in the eyes of the world community, become, at last, important objectives that must all be reached if living resources are to be preserved.

f) Conclusion - The Extension of Coastal State Jurisdiction

As the Third United Nations Conference on the Law of the Sea was about to be convened, the state of the living marine resources of the world and the effectiveness of the conservation measures taken to preserve them could be summarized as follows:

Fisheries for traditional species were progressively reaching a state of full exploitation, and unexploited or underexploited stocks were becoming fewer and fewer. Many stocks were already overexploited resulting in the collapse of several valuable fisheries. The unregulated taking of non-target species through incidental catches was disturbingly increasing and the problem was still barely recognized as requiring serious study and action. The destruction of habitats, especially of coastal

habitats that are critical to many commercial species was accelerating, and generalized pollution of the marine environment from all sources was increasing.

From the *scientific* point of view the very basis upon which most resources conservation agreements had been founded, the concept of Maximum Sustainable Yield had been, to say the least, questioned, but the development of new models applicable to multispecies fisheries was still in its infancy. With the improvement of fishery statistics it had become possible to carry out better stock assessments but in many cases further improvement in statistics continued to be required. There had been a considerable development of research activities resulting in new and important advances in our knowledge of the marine environment and its living resources, but further research was still needed in such fields as the ecology of many species, the determination of critical habitats and species interrelations.

From the *economic* point of view overinvestments by many fishing fleets in vessels and gear had often resulted in a gradual decrease of the catch per unit of effort making fishing operations, even where stocks were not overexploited, more and more unprofitable for individual operators. The increase in the cost of energy was soon to bring operating costs to an even higher level.

From the *legal* point of view, there was still no clear determination of the extent of coastal state jurisdiction, since both UNCLOS I and UNCLOS II had failed to arrive at an agreement on the breadth of the territorial sea. The problem of the new entrants into a fishery was, in principle, resolved and did not seem to give rise to major difficulties in practice. Regulatory measures taken by fishery bodies often proved insufficient to stop stock depletion or were taken too late. Regulation of mesh sizes had to be supplemented first by the determination of overall quotas, then by the allocation of individual national quotas. Fishing effort remained, however, unregulated.

The most important progress, however, from the legal side, has probably been that the conservation of the living resources of the sea was now recognized as an obligation under international law, embodied as such in the 1958 Geneva Convention on Fishing and Conservation of the Living Resources of the High Seas[206]. This can be exemplified by a quotation from the judgment rendered by the International Court of Justice in 1974 in the Fisheries Jurisdiction Case[207]: "It is one of the advances in Maritime International Law, resulting from the intensification of fishing, that the former *laissez faire* treatment of the living resources of the sea in the high seas has been replaced by a recognition of the duty to have due regard to the rights of other States and the needs of conservation for the benefit of all".

It seems obvious, however, that, in order to arrive at this result, the extent of state's rights and obligations over the living resources of the seas must be clearly established by international law on the basis of well-determined and fixed jurisdictional boundaries. It appears equally obvious that the powers of coastal states or international fishery bodies, as the case may be, to take conservation measures must derive from undisputable rules of international law.

There are only three methods which can be used to achieve this.

The first would be the establishment of a worldwide fisheries authority with sufficient regulatory and enforcement powers to impose effective conservation

measures. This method entails the transfer by the states of some of their sovereign rights to a supranational authority. As we saw, it was never accepted.

The second method consists in the conclusion of a large number of bilateral or multilateral agreements and the establishment under these agreements of fishery bodies with regulatory powers covering all major fishing grounds and migratory stocks. This is the system which developed from the early days of fishery regulation and further developed after the Geneva Conference in 1958. It was indeed the method which was apparently the best adapted to resolve the problem resulting from the complexity of fish stocks distribution and fishing patterns in the world. It was advocated as early as the 1930s by ICES and again in 1955 by the International Technical Conference in Rome, and in 1958 by the Geneva Conference itself[208].

It cannot be said, however, that, in view of the results, this method has been entirely successful in safeguarding the living resources of the sea.

The third method consists in extending the jurisdiction of coastal states over the resources living off their shores thereby vesting in these States sole and unshared authority to exploit and conserve these resources.

The question first arose in 1945 when President Truman declared by proclamation[209] that the United States considered the natural resources of the continental shelf adjacent to its coasts as subject to its jurisdiction and control. The resources referred to in this proclamation were, however, understood as including only mineral resources. Several other countries, mainly in Latin-America, followed suit and proclaimed their sovereignty or asserted their jurisdiction over large areas of the high seas including this time all living resources. Such was the case, for instance, of Panama and Argentina in 1946, Chile and Peru in 1947, Costa Rica in 1949, El Salvador and Brazil in 1950[210]. Some of these areas extended as far as 200 miles from the coast. On 18 August 1952, Chile, Peru and Ecuador signed the Santiago declaration by which they established a maritime zone 200 miles which they declared subject to their exclusive jurisdiction. In 1956, a Conference organized by the Organization of American States in Ciudad Trujillo adopted a series of principles affirming the rights of coastal states to extend the breadth of their territorial sea, to protect living resources in waters adjacent to their territorial sea and to exercise exclusive exploitation of these resources.

In other parts of the world unilateral extensions of jurisdiction were proclaimed by Iceland in 1950 and the Republic of Korea in 1952.

These claims were naturally opposed by other nations on the basis of the principle of the freedom of the seas, fishing vessels were seized and disputes started to arise.

As we saw, the Geneva Conferences in 1958 and 1960 were unable to come to an agreement on fishery jurisdiction, divided as they were between groups of nations with opposing interests. The Convention on Fishing and Conservation of the Living Resources of the High Seas recognized, however, the special interest of the coastal State in the maintenance of the productivity of the living resources in any area of the high seas adjacent to its territorial sea.

Unilateral extensions of jurisdiction continued thereafter at an increasing rate. These extensions concerned both the territorial sea, which was generally extended to a breadth of 12 miles, but in some cases to 100 or even 200 miles, and the exclusive fishing zone the breadth of which was almost always fixed at 200 miles from the base-line.

These developments, however, did not come as a surprise. Coastal states, especially developing countries, faced as they were with what they called, and in most cases rightly so, the plundering of marine living resources near their shores by large industrial fishing fleets of developed nations felt that they had no other alternative if they wanted to protect their legitimate interests. In spite of the resistance opposed by some fishing nations, as exemplified by a number of disputes such as the one which opposed the United Kingdom and the German Federal Republic of Iceland in the Fisheries Jurisdiction Case before the International Court of Justice[211], the trend towards extended fisheries jurisdiction continued to develop[212]. Thus, in the absence of a clear rule established by treaty, a customary rule of international law was slowly coming to life. The legitimization by treaty of this still emperfect custom was to become one of the main objectives and achievements of the Third United Nation Conference on the Law of the Sea.

E The New Regime of the Sea

1. The New Convention on the Law of the Sea

a) Introduction

i) Historical background

The General Assembly of the United Nations decided in 1968 to establish a Committee on the Peaceful Uses of the Sea-bed and the Ocean Floor beyond the Limits of National Jurisdiction[213]. This Committee agreed in 1970 to a draft declaration of principles which was adopted on 17 December 1970 by the General Assembly[214]. At the same time, the General Assembly decided to convene a Conference in 1973 to establish an equitable regime for the exploitation of sea-bed resources and to consider a large range of other questions such as the regimes of the high seas, the continental shelf, and the territorial sea, fisheries, conservation of living resources, protection of the marine environment and scientific research. It also invited the Sea-bed Committee to draw up a comprehensive list of subjects and issues related to the law of the sea and to prepare draft articles on these questions for submission to the Conference[215].

By the end of 1973, the Sea-bed Committee had considered a considerable number of substantive proposals submitted by delegations but had not succeeded to arrive at an agreement on draft articles. It was, therefore, only able to present a list of subjects and issues[216].

The Third United Nations Conference on the Law of the Sea was officially opened in New York on 3 December 1973 for a formal session. The second session was held in Caracas from June to August 1974. A first set of draft articles was produced at the end of the third session of the Conference in Geneva in the spring of 1975. This draft was entitled the Revised Single Negotiating Text (RSNT)[217]. A revised version of this draft followed the discussions which took place at the fourth session of the Con-

ference in New York in the spring of 1976[218]. At the end of the sixth session of UNCLOS III, in New York, in May-July 1977, a consolidated version of the draft was issued, under the name of Informal Composite Negotiating Text (ICNT)[219]. This draft was subsequently revised at the 8th session of the Conference in Geneva in the spring of 1979[220] and at its 9th session in New York in April 1980[221]. The 9th session, which was resumed in Geneva in July and August 1980 subsequently produced on the 27 August of that year a new draft entitled this time Draft Convention on the Law of the Sea (informal text)[222].

ii) Procedural background

Mention should be made here of a number of important differences in the procedures which were followed both at the preparatory stage and at the Conference itself between the Geneva Conference and UNCLOS III.

UNCLOS I, in 1958, was preceded by the International Technical Conference that was held in Rome in 1955. The International Law Commission and the Geneva Conference itself had, therefore, available to them the latest scientific information and scientists' recommendations on the basis of which they could formulate the relevant articles of their drafts. In 1970, when the General Assembly decided to Convene UNCLOS III, it did not at the same time decide to provide the Conference with up-to-date scientific information either by means of a Conference, as the Rome Conference, or by other means.

The reasons for this lack of action are unclear. It was perhaps felt that the scientific data available in 1955 were still, to a large extent, applicable fifteen years later, or that the new law of the sea would be dealing with general matters upon which scientific considerations would be of little consequence. Be it as it may, the Sea-bed Committee and later the Conference itself, delegates to which were mostly fisheries administrators and diplomats, were unable or perhaps unwilling to benefit from latest scientific developments and most especially of the increasing doubts in the minds of scientists as to the validity of the Maximum Sustainable Yield concept.

The second important procedural difference between the two Conferences consists in the fact that the Sea-bed Committee and UNCLOS III decided to work on the basis of consensus whereas the ILC and UNCLOS I had taken decisions by majority votes. The lack of consensus at Geneva resulted, however, in that less than half of the world states eventually ratified the Geneva Conventions.

The consensus rule which was first adopted by the Sea-bed Committee prevented that committee from adopting draft articles to be presented to the Conference. On the other hand it allowed for full discussion of all proposals submitted by delegations and certainly made easier the reaching of a consensus in the Conference later. The decision to take decisions by consensus in the Conference itself was taken by the General Assembly and subsequently embodied in the rules of procedure of the Conference. This new procedure in international negotiations admittedly requires much time and effort, but by avoiding the formation of majority and minority groups, in allowing, in the event of disagreements, sufficient time to reach compromises and in eliminating the risk of freezing states positions through formal votes, it seems to have considerably facilitated the drafting of a convention susceptible of gaining the largest possible support from individual states and, as a consequence, the greatest number of ratifications.

iii) The new Convention

The main subjects that are covered by the new Convention are the territorial sea, straits, archipelagic States, exclusive economic zone, continental shelf, high seas, islands, enclosed or semi-enclosed seas and land-locked States; the creation of a new regime applicable to the zone which consists of the sea-bed and the ocean floor beyond the limits of national jurisdiction and which is called the "Area"; the protection and preservation of the marine environment, scientific research, the transfer of technology and settlement of disputes. Of particular relevance to the exploitation and conservation of marine living resources are the provisions of the new Convention which relate to jurisdiction over living resources, fisheries, protection of the marine environments and scientific research. In the following pages, consideration will be given to matters of jurisdiction, regulation of fisheries, conservation of living resources, and other aspects such as scientific research, transfer of technology, enforcement, institutional arrangements and settlement of disputes.

b) Jurisdiction

The new Convention reaffirms the sovereignty of coastal states over their internal waters, territorial sea and continental shelf and as a consequence, on all the living resources in those areas. It establishes, in addition, an exclusive economic zone (EEZ) extending seaward beyond the limits of the territorial sea. It also institutes a special regime for anadromous and catadromous species and recognizes the special interest of coastal states, in particular of the states of origin or destination over these species.

Finally, it settles, once and for all, the difficult problems of the breadth of the various jurisdictional zones.

i) Internal waters

The straight baseline system followed by the Convention to determine the landward limit of the territorial sea is, with minor adjustments, the same as in the Geneva Convention on the Territorial Sea and Contiguous Zone. The waters on the landward side of the baseline form part of the internal waters of coastal states and thereby of their national territory[223]. All resources in the internal waters fall under the sovereignty of the coastal state.

ii) Territorial sea

States have the right to establish the breadth of their territorial sea up to a limit not exceeding 12 nautical miles, measured from the baselines as determined under the Convention[224]. This provision should therefore have as an effect to bring to an end unilateral extensions of the territorial sea beyond the distance of 12 miles and to render illegitimate earlier extensions beyond that limit. It provides a final settlement to this difficult matter. The sovereignty of a coastal state, and therefore its jurisdiction over living resources extends to its territorial sea[225]. This sovereignty is only limited by the right of innocent passage of foreign ships[226].

iii) Archipelagic waters

An archipelagic State may draw straight archipelagic baselines joining the outermost points of the outermost islands of the archipelago[227] from which shall be measured the breadth of the territorial sea, the exclusive economic zone and the continental shelf[228]. Since states enjoy the right of innocent passage in archipelagic

waters[229] there does not seem to be any real difference from the point of view of jurisdiction, between archipelagic waters and the territorial sea.

iv) Exclusive economic zone

The exclusive economic zone (EEZ) extends to a maximum of 200 nautical miles from the baselines from which the breadth of the territorial sea is measured[230]. In the EEZ, the coastal state has sovereign rights for the purpose of exploring and exploiting, conserving and managing the natural resources and jurisdiction with regard to marine scientific research and the protection and preservation of the marine environment[231].

Thus , under the new convention, most of the fish resources of the world come now under the jurisdiction and control of coastal states. There is, however, one limitation to the absolute powers of these states over the resources of their EEZ: they are required to determine the entire allowable catch for the living resources of that zone and if they do not have the capacity to harvest that catch by themselves they must give access to the resulting surplus to other states[232].

v) Continental shelf

The problem of the outer limit of the continental shelf which had remained unresolved at Geneva in 1958 was eventually settled after protracted negotiations by UNCLOS III. The continental shelf of a coastal state comprises the sea-bed and subsoil of the submarine areas that extent beyond its territorial sea throughout the natural prolongation of its land territory to the outer edge of the continental margin or to a distance of 200 miles from the baseline[233].

In other words, even when the continental shelf does not physically extend to a distance of 200 miles, and this is often the case, coastal states will exercise their sovereign rights over the whole of the sea-bed up to that distance.

Whenever the continental shelf extends physically beyond the 200 miles limit, as it happens in a few cases, the sovereignty of the coastal State extends to the outer limit of the shelf. To determine the limit of the shelf beyond the 200 miles boundary, the Convention provides that the shelf extends to the outer edge of the continental margin which is defined as including the slope and the rise and further provides for a formula to delineate the line corresponding to that outer edge.

Coastal states exercise over the continental shelf sovereign rights for the purpose of exploring it, and exploiting its natural resources. The living resources of the continental shelf are the living organisms belonging to sedentary species, that is to say, organisms which at the harvestable stage either are immobile on or under the sea-bed or are unable to move except in constant physical contact with the sea-bed or the subsoil[234].

The rights of coastal states over the sedentary species of the continental shelf are of the same nature as those exercised by these states in their internal waters or territorial sea: they are absolute, whatever the jurisdictional status of the superjacent waters, and whatever the rights and obligations of states over the non-sedentary species in those waters. Thus, sedentary species of the continental shelf underlying the EEZ do not form part of the resources of that zone and such rules as the obliga-

tion for the coastal state to determine the allowable catch and to give other states access to the surplus are not applicable to those species.

It will be recalled that the same definition of sedentary species had already been used in the Geneva Convention on the Continental Shelf and that, since it is not always clear which species actually remain in constant physical contact with the sea-bed, a number of disputes had arisen concerning, in particular, crustaceans. The problem thus remains unresolved and similar disputes could, thus, well occur in the future.

vi) High seas

The high seas are defined as all parts of the sea that are not included in the EEZ, in the territorial sea or in the internal waters of a state, or in the archipelagic waters of an archipelagic state[235]. The high seas are open to all states. Freedom of the high seas comprises freedom of fishing and freedom of scientific research subject to conditions laid down in other parts of the Convention.[236].

vii) Highly migratory species

No definition of these species is given by the Convention but a list of species names covered by this term appears in Annex I. This list includes the various species of tuna, other far-ranging migratory fish species such as billfishes, marlins and large sharks, and all marine cetaceans.

No simplified procedure is provided, however, to amend this annex if it appears that new species should be added to the list or species currently listed removed. Amending the list will, therefore, necessitate the use of the procedures required to amend the Convention itself and therefore a consensus.

Highly migratory species are not governed by specific jurisdictional rules but, under Article 64, are to be the subject of global management measures both within and beyond the limits of the EEZ.

viii) Anadromous and catadromous species

In view of the many disputes that had arisen in the past in respect of anadromous species, in particular salmon, UNCLOS III has attempted to resolve the problem in a definitive way. Article 66 of the Convention provides that states in whose waters anadromous stocks originate shall have the primary interest in and responsibility for such stocks. Fisheries for anadromous stocks shall, however, be conducted only in waters landwards of the outer limit at the EEZs. Fishing of these species on the high seas is therefore prohibited to all states, including the states of origin.

There is, however, one exception to this rule which concerns those cases where this prohibition would result in "economic dislocation" for a state other than the state of origin. In such cases the country of origin is entitled to establish, in consultation with the other states concerned the total allowable catch for stocks originating in its rivers and to conclude agreements with these states in respect of enforcement measures.

These provisions give almost complete control to the state of origin over its anadromous stocks but not jurisdiction since that state is not entitled to exploit the resource on the high seas or in the EEZ of another state. On the other hand, the

freedom of fishing on the high seas does not apply to these stocks and ail states, except in the case of "economic dislocation" have to abstain from fishing such stocks in that zone. This is a revival of the "abstention principle" that had been rejected in 1958. It was probably the only way to arrive at a satisfactory solution for the management of these stocks.

Article 67 deals, in a similar fashion, with the problem of catadromous species. Coastal states in whose waters catadromous species spend the greatest part of their life cycle shall have the responsibility for the management of these species and shall ensure the ingress and egress of migrating fish. Harvesting of catadromous species shall be conducted only in waters landwards of the outer limits of EEZs. Since there does not seem to be any fishing for catadromous species on the high seas, there is no provision to cover the cases of "economic dislocation". Control, but here again not jurisdiction, over these species is in this case given to the state of destination of the migration.

A basis weakness in these two articles seems to be the absence of clear definition of either anadromous or catadromous species. The intention of the drafters may well have been to cover only salmon and perhaps a few other species such as sturgeons in the first case, and eels in the second. Indeed, the reference to the rivers where anadromous stocks originate would tend to substantiate that view with regard to these stocks. This would then mean that the very numerous species which breed or spawn in coastal lagoons, estuaries or marshes, or even on land like seals or sea turtles, are not covered by this provision and can therefore be exploited on the high seas even when the state of origin has taken specific measures and incurred expenditures to protect or renew these stocks in areas under its jurisdiction.

With regard to catadromous species, on the other hand, the reference which is made is only to those waters where these species spend the greatest part of their life cycle. These waters being undefined could be understood as including not only internal waters, which are the destination of eels, mullets and of many species of shrimps, but also the territorial sea.

Thus in the case of anadromous stocks the coverage seems to be too narrow, and too broad in the case of catadromous species. In the absence of an adequate definition, or alternatively of an appendix listing the species which are actually covered by these two articles, the occurrence of disputes seems inevitable.

ix) Summary of the jurisdictional situation

The coastal state enjoys sovereign rights over the living resources of the sea in its internal waters and territoral sea, on its continental shelf and in the EEZ. In the latter, however, it must give other states access to the unharvested surplus.

On the high seas, that is to say beyond the outer limit of the EEZ, the principle of the freedom of fishing remains applicable. There is, however, an important restriction to the general applicability of this principle since both anadromous and catadromous species can no longer be exploited in that zone, except, but only in the case of anadromous stocks, by agreement with the state of origin of the stocks concerned.

c) Regulation of Fisheries

The new Law of the Sea Treaty does not provide for specific regulations for the management of fisheries as it was certainly felt that the taking of detailed regulatory measures was best left to the competent authority, whether it be the coastal state or international fishery organizations. It provides, however, for a general duty to co-operate in the exploitation of shared stocks or of the living resources of the high seas.

i) Regulation by unilateral action

In their internal waters, territorial sea and EEZ as well as on their continental shelf, coastal states have exclusive jurisdiction to regulate fishing and adopt and enforce conservation measures in respect not only of their nationals but also of nationals of any state they may have authorized to exploit living resources. With regard to the EEZ, Article 62 of the Convention gives a long and non-exhaustive list of the types of regulatory measures coastal states are entitled to apply to foreign nationals fishing for the unharvested surplus. There is no doubt that these measures can also be applied to other areas or resources under coastal state jurisdiction, if foreign fishing vessels, by agreement with the coastal state concerned, were authorized to fish in these areas or to take part in the exploitation of these resources.

Finally, in respect of the high seas, Article 117 provides that all states have the duty to take, or to co-operate with other states in taking, such measures for their respective nationals as may be necessary for the conservation of the living resources of the high seas.

ii) The duty to co-operate

Perhaps one of the most significant features of the new Convention in that it establishes a clear duty to co-operate whenever several states are engaged in the exploitation of the same stock, thus not only forcibly re-stating the principle that had already been adopted in Geneva in 1958 to resolve the problem of the new entrants into a fishery on the high seas, but also to give a solution to the new problem arising from the existence of stocks shared between two contiguous EEZs or between the EEZ and the high seas or of stocks migrating between different jurisdictional zones.

(a) Stocks occurring within the EEZs of two or more coastal states.

Article 63.1 provides that the states concerned shall seek to agree upon the measures necessary to co-ordinate and ensure the conservation and development of such stocks.

(b) Stocks occurring both within the EEZ and in an area beyond and adjacent to it.

Article 63.2 provides that the coastal state and the states fishing for such stocks in the adjacent area shall seek to agree upon the measures necessary for the conservation of these stocks in the adjacent area.

It should be noted, however, that under this article, the interest of the coastal state seems to be given much more importance than the interests of the states fishing for the resource on the high seas. Fishing in the high seas may indeed affect the conservation status of the resource in the EEZ and, if this is to be prevented, an agreement on conservation measures to be taken must be arrived at between all the

states concerned. Conversely, depletion of the resource in the high seas may be the result of inadequate conservation measures in the EEZ, but in this case, no co-operation agreement is required by the Convention. This provision, therefore, seems to have been drafted in the only interest of the coastal state whose "special interest" is now extended to cover the whole of the stocks straddling the outer limit of the EEZ whereas states fishing for the same stocks on the high seas cannot require the coastal state to take conservation measures in its EEZ.

(c) Highly migratory species

An exception to the provision of Article 63.2 has, however, been made in respect of highly migratory species a list of which is appended to the Convention of Annex I. Indeed the application of such a rule to these species, which often migrate over very long distances, as tunas do in particular, would have been especially unreasonable since it would have prevented any real possibility of rational international management of these stocks.

Article 64 provides, therefore, that the coastal state and other states whose nationals fish in the region for the highly migratory species listed in Annex I shall co-operate with a view to ensuring conservation and promoting the objective of optimum utilization of such species throughout the region both within and beyond the EEZ.

(d) Marine mammals

Articles 65 and 120 provide that states shall co-operate with a view to the conservation of marine mammals. These articles apply to the EEZ and to the high sea.

(e) Anadromous species

As we saw, the taking of these species on the high seas is prohibited by the Convention except when this would result in "economic dislocation" for a state other than the state of origin of these stocks. In such cases, Article 66.3 provides that the country of origin and the states entitled to fish for the resource on the high seas shall maintain consultations with a view to achieving agreement on terms and conditions of such fishing giving due regard to the conservation requirements and needs of the state of origin.

Co-operation in respect of anadromous stocks must also take place in another case. There are several instances where anadromous fishes during their migration cycle pass through or stay for some time in waters under the jurisdiction of another state than the state of origin. This is the case, for example, of the Sockeye Salmon of the Fraser river system in Southern British Columbia in Canada which crosses waters under the jurisdiction of the United States during its spawning run and also of a considerable number of Atlantic salmon which come to feed in the waters of Greenland under the jurisdiction of Denmark. Article 66.4 of the Convention provides that in these cases the states concerned which are naturally entitled to exploit the resource in their waters, shall co-operate with the state of origin with regard to the conservation and management of these stocks.

(f) Catadromous species

Article 67.3 provides that where catadromous fish migrate through the EEZ of a state or states other than the one in whose waters these fish spend the greatest

part of their life cycle, the management, including harvesting, of such fish shall be regulated by agreement between all the states concerned.

(g) High Seas

The matter of co-operation in the management and conservation of the living resources of the high seas is dealt with in Article 118 of the Convention. Under that article, states are under the obligation to co-operate in the management and conservation of these resources and, more specifically, states whose nationals exploit identical resources, or different resources in the same area, must enter into negotiations with a view to adopting the means necessary for the conservation of the living resources concerned.

(h) Enclosed or semi-enclosed seas

This expression is defined in Article 122 of the Convention as meaning gulfs, basins or seas surrounded by two or more states and connected to the open seas by a narrow outlet. Examples of such seas are the Mediterranean, the Black Sea, the Baltic Sea and the Red Sea. It is, however, further defined as meaning also gulfs, basins or seas consisting entirely or primarily of the territorial seas and EEZs of two or more coastal states. Whereas it would, therefore, seem that completely enclosed seas, such as the Caspian, are covered by this definition, doubts could be raised as to its applicability to certain other marine areas, which, although completely enclosed by the EEZs of their riparian states, might nonetheless encompass at the same time, in their central part, a fairly large area of waters that fall under the regime of the high seas.

State practice will show in the future whether a broad or restricted interpretation is to be given to this provision. The matter is of some importance because it is obvious that only close co-operation between all the coastal states of enclosed or semi-enclosed seas can ensure adequate conservation and management of the living resources they contain. In that respect, Article 123 provides that states bordering these seas should co-operate with each other in the exercise of their rights and duties under the Convention and shall, directly or through an appropriate regional organization, endeavour to co-ordinate the management, conservation, exploration and exploitation of the living resources of the sea, scientific research policies and the implementations of their rights and duties with respect to the protection and preservation of the marine environment. Although not a binding provision, this article is, no doubt, of very great importance for the development of future conservation policies and programmes in many areas of the world oceans, particularly in areas that are rich in endemic species. As it stands it already constitutes one of the bases for the Regional Seas Programme of the United Nations Environment Programme.

(i) Summary and appraisal

The new Law of the Sea Treaty calls for co-operation and the taking of management measures on the part of all States that are exploiting the same stock or stocks on the high seas, in the EEZ and the adjacent part of the high seas, or in adjacent EEZs; it also calls for co-operation for the conservation of marine mammals and for the management of anadromous stocks and catadromous species. But these provisions only apply to the EEZ and the high seas. Internal waters and the territorial sea are not referred to even though many stocks which are subject to regulatory

measures in other zones often extend landward to areas very close to the coast if not to in-shore areas where they spawn or feed.

Thus, the text of the treaty requires no co-operation between states in the following cases: stocks, whether migratory or not, occurring within the internal waters or territorial sea of two or more adjacent coastal states; anadromous stocks crossing or staying in the internal waters or territorial sea of states other than the state of origin, marine mammals on land, in internal waters or in the territorial sea. As a result, when stocks of any species, including marine mammals, highly migratory and anadromous species, occur, and this will necessarily be the most frequent situation, both in the territorial sea and the EEZ of two or several adjacent states, these states will be under the obligation to co-operate only in regard to that part of these stocks which are found in the EEZ. Since many of these stocks are likely to migrate between areas close to shore and the EEZ, this co-operation, if it is not extended to the whole of the stocks concerned, is bound to be illusory. A very similar situation will also arise, as we saw, in the case of non-highly migratory stocks that occur partly in the EEZ and partly in the high seas.

The absence of an obligation to include internal waters, the territorial sea, and even the EEZ in the case mentioned above, among the areas where agreed conservation measures must be taken by all the states concerned in respect of common stocks, can certainly be considered a major gap in the new Convention. Indeed the same stock, which will often be, in addition, a migratory stock, cannot be subject to the unilateral discretion of the coastal state in certain zones and to joint management measures in others if a reasonable degree of effectiveness in the conservation measures taken is to be expected. Stocks should be managed as units, irrespective of the jurisdictional zone where their individual members happen to be at a given point in time. The new Convention, however, does not seem to preclude the conclusion of fishery agreements covering the whole range of any stock, species or groups of species and there is no reason to believe that such agreements would be considered incompatible with the principle of full sovereignty of the coastal state over the living resources of its in-shore waters.

This principle has now been enshrined in international law for many decades and has not been modified or qualified by the new treaty in any way. It was never interpreted as meaning that sovereign states could not enter into agreements concerning the management of living resources in their internal waters and territorial sea as evidenced by several fishery agreements, such as, for instance the Fraser River Sockeye Salmon Convention. On the other hand, by not making it an obligation for parties to include their in-shore waters in the areas covered by fishery agreements, the negotiators of the Convention presumably felt that it was a matter directly related to state sovereignty which was better left to the initiative of individual states and in respect of which binding rules could not be formulated in a general treaty. Time will tell whether, in the absence of a binding obligation, states will nevertheless come to the necessary agreements to conserve and manage shared stocks as inseparable units.

d) Conservation of Living Resources

As we saw in the first part of this paper, the main threat to the survival of species and to the maintenance of sustainable populations of harvestable animals or plants

are overexploitation, destruction or alteration of habitats, pollution, and the introduction of alien species. A review of the new Convention to ascertain the extent to which these four threats are met by its provisions seem therefore to be of importance.

i) Exploitation

Three main requirements have been laid down by the new treaty in respect ot the exploitation of the living resources of the sea: optimum utilization, prevention of overexploitation and the necessity to take into consideration the effects of harvesting on associated or dependent species. These requirements apply, however, only to the exploitation of resources in the EEZ and, with the exception of optimum utilization, on the high seas. The treaty lays down no objectives or requirements related to living resources exploitation by coastal States in their internal waters or territorial sea or on their continental shelf.

(a) Optimum utilization

Article 62.1 provides that coastal states shall promote the objective of optimum utilization of the living resources in the EEZ. There is no corresponding provision in the part of the treaty that deals with fishing in the high seas.

Article 64, however, which deals with the question of fishing for highly migratory species both within and beyond the EEZ also provides for the promotion of maximum utilization of these species. Since a very large part of harvestable stocks are either confined to coastal areas or highly migratory, the objective of optimum utilization can be said, therefore, to be applicable to most of the species which are at present exploited. Exceptions would be stocks of non-highly migratory species that are only found in the high seas, but these are probably very few[237], and the high sea part of stocks shared between one or several EEZs and the high seas. In the latter case, in view of the fact that most stocks would generally be migrating to and fro between the two different jurisdictional zones, the fact that optimum utilization must be promoted in one part of the range of these stocks and not in the other does not really make much sense. Another important exception to the obligation of promoting the objective of optimum utilization is provided by Articles 65 and 120 which deal with the conservation of marine mammals. Under these articles, coastal states or competent international organizations, such as for instance, the International Whaling Commission, may prohibit, limit or regulate the exploitation of marine mammals more strictly than provided for in the Convention for other species of marine life. The provisions of Article 62.1 and Article 64 which relate to optimum utilization do not apply, therefore, to these species.

The principle of optimum utilization must not, and has never been, considered as a conservation principle. It was incorporated into the treaty as a basis for a compromise between those states, mostly developing nations, who favoured complete control over the living resources of their EEZ and other states, mostly nations with long-distance fishing fleets, or with traditional interests in certain coastal fisheries abroad, who feared they would be losing some economic benefits to no corresponding advantage for those states who do not have the capacity to harvest themselves the resources in their newly established EEZs. In addition, land-locked states and other geographically disadvantaged states claimed that they should also be entitled to get a share of the living resources of the sea.

Finally, the Law of the Sea Conference probably also felt that in a hungry world, underexploitation of food resources was as much to be avoided as overexploitation and, therefore, that the harvestable fraction of these resources in the EEZ could not be allowed to be wasted. Admittedly, this latter argument has now been seriously criticized by some scientists on the ground that there is no real shortage of protein in the world and that furthermore there is no real waste in natural systems since the decay of dead matter provides the nutrients essential to life in an almost completely closed cycle[238].

But, be it as it may, the fear of underexploitation was certainly also very much present in the minds of the law of the sea negotiators when they adopted these provisions.

It is, therefore, on the basis of this objective of optimum utilization that coastal states or, in the case of highly migratory species, the fishery bodies dealing with these species, have to make decisions regarding the harvesting of the resources. For that purpose, coastal states must first determine the allowable catch, as provided for in Article 61.1 and then determine their own capacity to harvest that allowable catch as laid down in Article 62.2. When it appears that a coastal state does not have the capacity to harvest the entire allowable catch, Article 62.2 further provides that it shall, through agreements or other arrangements, give other states access to the surplus of that allowable catch.

It should be noted here that the new treaty does not provide for any definition of optimum utilization which, admittedly would have been difficult to establish. Neither does it provide any standards to assist coastal states in determining the allowable catch or their harvesting capacity. The problem is further aggravated by the fact that most stocks being either common to the EEZs of two or several countries or migratory stocks, coastal states will have, to be able to determine the allowable catch, to rely heavily on statistical data and biological information from other states. Since such information is generally best channelled and disseminated through regional fisheries organizations it is to be expected that such organizations will be called to play an increasingly important role in the future and will assist coastal states in determining the allowable catch. The requirement for the contribution and exchange of scientific information, through the competent subregional, regional or global organizations with the participation of all states concerned was felt to be, in that respect, of such importance that it was specifically embodied by UNCLOS III in paragraph 5 of Article 61 of the treaty.

In respect of highly migratory species, Article 64 does not refer to the determination of an allowable catch by the competent international organizations regulating fishing for these species. On the other hand, it appears clearly that coastal states are under the obligation, according to Article 61.1[239], to determine their allowable catch for these species in their EEZ. There will be, therefore, as a result of this provision, an allowable catch for each of the EEZs, that are included in the range of a particular stock and, at the same time, no allowable catch determined for the high seas part of the same fishery, and therefore no allowable catch for the fishery as a whole.

There is, therefore, a serious risk that if allowable catches are determined individually by the coastal states concerned, their sum could equate the total sus-

tainable yield of the stock, thus either making high seas fisheries impossible without overexploitation, or even exceed the sustainable yield, thus making overexploitation inevitable. Yet fishery commissions are not empowered by the new treaty to determine allowable catch in areas under national jurisdiction, since this is a matter which is entirely left to the discretion of the coastal state.

Similar situations could arise where the same stock is shared between two or several adjacent EEZs if the sum of allowable catches exceeds the sustainable yield of the stock.

(b) Prevention of overexploitation

Article 61.2 provides that the coastal state, taking into account the best scientific evidence available to it, shall ensure through proper conservation and management measures that the maintenance of the living resources in the EEZ is not endangered by overexploitation. This provision, which does not appear in the articles dealing with fishing on the high seas, is the logical counterpart of the requirement of optimum utilization. These two provisions must therefore really be considered together since they are actually nothing else than the two sides of the same coin. In other words, utilization ceases to be optimum where overexploitation begins, and overexploitation must stop if optimum utilization is to be achieved.

How is then this delicate point of balance, which should at the same time be the allowable catch, to be reached? As it was stated earlier, the new Convention does not provide for any standards relying for this on fishery treaties and commissions[240]. It does, however, refer to the old concept of maximum sustainable yield. This concept appears both in Article 61.3, relating to the EEZ, and in Article 119.1 on the high seas. The text is the same in both articles. Under these two provisions, coastal states in their EEZ and states fishing on the high seas shall take measures designed to maintain or restore populations of harvested species at levels which can produce the maximum sustainable yield. In spite, therefore, of the criticism which had been voiced in recent years against this concept, the negotiators of the new Law of the Sea Treaty decided, notwithstanding, to establish it again as the basis for future conservation and management action. But this time, however, and perhaps for the first time, this concept is now embodied in an international instrument with a long list of qualifiying factors. These factors, which are also listed both in Article 61.3 and 119.1, are environmental and economic. But to a certain extent they counterbalance each other. Environmental factors include the interdependence of stocks and minimum standards as may be recommended by subregional, regional or global organizations. Such factors would probably tend to bring the level of the allowable catch to a value below MSY On the other hand, economic factors, such as the needs of coastal fishing communities, the special requirements of developing states and fishing patterns will generally tend to raise the allowable catch above the MSY level. In other words, in all cases, and even when all relevant scientific and statistical data are available, which will seldom be the case, decisions on the allowable catch will have to be political decisions and overexploitation will often ensue.

(c) Associated and dependent species

An important innovation in the new treaty is the recognition of the fact that harvesting may have as a consequence harmful effects on the population level of

other species. These effects can be caused by the incidental taking of non-target species which, when associating with target species are caught with the same nets or other gear. This would be the case for instance of porpoises which are caught by tuna fishing vessels. This can also be due to the fact that harvesting of a species which is the principal food of another species will necessarily affect the numbers of that other species or prevent overexploited stocks to recover. That would be the case, for instance, of excessive harvesting of krill in the Antarctic, resulting in the decrease in the numbers of seals, penguins, squids and certain fishes and preventing the recovery of the overexploited great whales.

Article 61.2, as we saw, provides that the coastal state shall ensure, through proper conservation and management measures that the maintenance of the living resources in the EEZ is not endangered by overexploitation. This provision can and should be interpreted as meaning that the maintenance of any living resource must not be endangered by the overexploitation of any other of these resources. This interpretation is substantiated by the text of Article 61.4 which specifies that the coastal state shall take into consideration the effects on species associated with or dependent upon harvested species with a view to maintaining or restoring populations of such associated or dependent species above levels at which their reproduction may become seriously threatened. An identical provision appears in respect of the high seas in Article 119.1(b).

What is of particular importance in these provisions is that they apply to all species, whether they are the subject of a commercial fishery or of no economic importance, and that they contain the obligation to regulate direct or indirect effects of fishing on these species so that their populations be maintained at sufficiently high a level to make their extinction impossible. It is indeed the first time in the history of fishery agreements that states will become bound to preserve species not for their immediate economic value but because of the role they play in the ecosystem.

As it stands, however, this provision is limited in two important ways. It is applicable only to the EEZ and the high seas and not therefore to internal waters or the territorial sea or to the sedentary species of the continental shelf. Thus, for instance, trawlers operating on the continental shelf will have to abide by this provision in respect of incidentally caught non-target fin fishes but not with regard to echinoderma or molluscs which are sedentary species. In view of the deliberate vagueness of the definition of these species in Article 77.4, the fate of crustaceans will remain uncertain.

Another limitation in the scope of the provisions on associated or dependent species consists in the fact that there are species which cannot be actually qualified as such but which may nonetheless suffer severe depletion as the result of commercial fishing for other species. Such would be the case, for instance, of the Atlantic ridley turtle in the Gulf of Mexico which has been caught in large numbers in trawls by boats fishing for shrimps. There is certainly no ecological dependency or relationship between the two species, nor is there any close association as it is the case between porpoises and tunas. It just so happens that the two species are found at the same place and at the same time. A broad interpretation of the term associated species would, however, go a long way towards solving this particular problem.

(d) Marine mammals

Marine mammals benefit from a special status in the new Convention on the Law of the Sea. This is most certainly due to increased recognition on the part of many States and of a large fraction of the public in general that marine mammals deserve to be singled out among the many marine species because of their intelligence, their considerable potential for research in animal behaviour, their vulnerability to man's depredations and the present depletion of many of their populations, particularly those of the great whales and of other species of Cetaceans.

As we saw, marine mammals are treated differently from other species in that the obligation to seek an optimum utilization of marine living resources in the EEZ and on the high seas does not apply to them. This is clearly and unmistakenly the result of Articles 65 and 120 of the Convention. As a consequence, coastal states are under no obligation to determine the allowable catch for marine mammals in their EEZ, and to allow other states access to a possible surplus. They may, therefore, prohibit, limit or regulate the exploitation of marine mammals, either directly by national legislation, or through international organizations such as the International Whaling Commission, as they please.

The other provisions of the Convention relating to the prevention of overexploitation and the maintenance or restoration of populations of harvested species at levels that can produce the maximum sustainable yield naturally remain applicable to marine mammals. Therefore the discretion given to states with regard to these species only concern the possibility of taking stricter measures than those provided for by the treaty, and should not be regarded as a licence to overexploit marine mammals should certain states decide that they wish to harvest them.

All these provisions apply to marine mammals in general. The Convention, however, further provides that among these species Cetacea must benefit from an even more special status.

Articles 65 and 120 require states to co-operate with a view to the conservation of marine mammals and in the case of Cetacea, in particular, to work through the appropriate international organizations for their conservation, management and study, thus making it an obligation for states to become parties to international agreements dealing specifically with the conservation and management of Cetacea, such as for instance, the International Whaling Convention. It must be noted, in that respect, that the Convention does not limit this obligation to those States that have a direct interest in the exploitation or conservation of Cetacea. It would seem, therefore, that, underlying the provisions of Articles 65 and 120, there has been, on the part of the Convention negotiators, a recognition of the validity of a new trend in conservation circles and public opinion towards the acceptance of the principle that certain species, Cetacea in particular, should be preserved not only for their commercial importance, and therefore for the only benefit of those who exploit them or could exploit them, but also for their intrinsic value, their place in natural ecosystems and other non-monetary values, for the benefit and enjoyment of mankind as a whole, and therefore, that states who do not exploit these resources are entitled, as much as those who exploit them, to take part in conservation and management decisions. As early manifestations of this trend at international level there should be recalled the moratorium on whaling unanimously

adopted by the Stockholm Conference in 1972, although not subsequently implemented by the International Whaling Commission, and the accession to the International Whaling Convention of an increasing number of non-whaling states[241].

Another part of the Convention where Cetacea are specifically mentioned is Annex I which contains the list of highly migratoy species to which Article 64 applies. All families of Cetacea are listed in that Annex with the exception of the Platanistidae, a small family with only four species of fresh water dolphins. The classification of Cetacea as highly migratory species provides for a fall-back position in the event where Article 65 would not apply either because a coastal State will have chosen not to exercise its right under that article to adopt in respect of certain species stricter measures than those provided for by the Convention or because there is no existing international organization competent for the conservation and management of some particular species of Cetacea.

In such an event, the coastal state concerned would not be entitled to proceed to the exploitation of such species by itself. It would have, under Article 64, to cooperate for the conservation and management of these species with any other state harvesting these resources in the region, within or beyond the EEZ boundaries. Furthermore, if no appropriate international organization exists, the states concerned would be bound to establish one and to take part in its work. Once established, such an organization, if the necessity arises, could then, pursuant to Article 65, decide to abandon the objective of optimum utilization and adopt stricter protection measures for the species of Cetacea in respect of which it is competent.

ii) Habitat preservation

The preservation of habitats, in particular critical habitats is essential to the conservation of all species, especially endangered or rare species, and to the maintenance of the numbers of exploited species at levels allowing for a sustainable yield. Yet, the new treaty, like the earlier Geneva conventions and most fisheries agreements is astonishingly silent on the question. The reasons for this silence are unclear. It could be argued, admittedly, that this situation is, at least partly, due to the absence of marine ecologists at fishery negotiations, or to the fact that the importance of certain critical habitats has only been discovered very recently; it could even further be argued that the protection of habitats in areas under national jurisdiction is best left to national legislation as any attempt to regulate that matter internationally would be infringing upon state sovereignty. In view, however, of the very great importance of habitat preservation for the conservation and management of exploited species, it seems difficult to imagine that this latter argument could have completely prevailed at the negotiations.

Be it as it may, the fact remains that habitat preservation is never referred to by the new treaty as an essential tool of fishery management. Thus, coastal states may, without apparently violating the letter or even the spirit of the Convention, in destroying critical habitats cause a decrease in the levels of fish populations off their coast, and as a result of the allowable catch, or in the case of migratory species a decrease in the quantities of fish that can be taken by neighbouring contries in their EEZs or even that can be caught on the high seas. Since the objectives of optimum utilization and prevention of overexploitation seem to be entirely linked to fishing operations proper nothing would prevent a state from draining, polluting,

filling, dredging or otherwise destroying coastal marshes, lagoons or estuaries which produce the fish upon which depend the fishermen and fisheries of a neighbouring country. As the law stands, it is doubtful that a state could be held liable for such an action.

Provisions relating to habitat preservation therefore appear in none of the parts of the new law of the sea treaty which are dealing with fisheries or the conservation of living resources whether in the territorial sea, the EEZ, on the continental shelf or in the high seas. Furthermore, there are no provisions for the establishment of marine protected areas or marine parks and no provisions on the preservation of representative types of habitats or ecosystems.

There are, however, some general provisions concerning habitats in Part XII on the treaty which deals with the preservation of the marine environment mainly against pollution, and a few more specific provisions in some of the articles which relate to the control of deep-sea mining activities in the "Area".

(a) General provisions

Part XII on the Convention is entitled "Protection and Preservation of the Marine Environment". Most of the articles included in this part deal only with the more limited subject of pollution, but several of them refer expressly to protection and preservation of the marine environment and it is quite clear from the context that this latter expression has a broader meaning than the mere prevention, reduction and control of pollution. Unfortunately, although a definition of "pollution of the marine environment" is given in Article 1.4 of the Convention, no definition is provided for the marine environment itself nor is one provided for the terms "Protection and Preservation".

A precise definition of the "marine environment" would have served two purposes: firstly to make clear that the term applied to all waters, irrespective of jurisdictional zones, and secondly to show that all the different elements that form the marine environment, including flora and fauna, were included. A definition of "protection and preservation" would have helped in ascertaining what types of harmful actions or practices, in addition to pollution, should be monitored, regulated or controlled.

Provisions included in Part XII apply to all jurisdictional zones and it should therefore be presumed that they also apply to internal waters. This can be substantiated by Article 1.4 which provides for a definition of "pollution of the marine environment" from which it appears that, in respect of pollution at least, the marine environment is deemed to extend to coastal estuaries. In addition, Article 145 which deals with the protection of the marine environment from activities in the "Area", includes the coastline as an element of the marine environment which should be preserved from the effects of offshore activities. The same article also refers to "the prevention of damage to the flora and fauna of the marine environment". Animals and plants are, therefore, considered, in this article at least, to form a part of the marine environment itself, and there are no reasons to believe that they would be excluded from the marine environment in other articles, especially in Part XII.

With regard to the meaning of the terms "protection and preservation" again the only article in the Convention which provides some information about harmful ac-

tivities other than pollution is Article 145. This article refers, in addition to pollution, to interference with the ecological balance of the marine environment, particular attention being paid to the need for protection from the consequence of such activities as drilling, dredging, excavation, disposal of waste and construction of installations. Admittedly, here again, the intent of this article is limited to controlling the effects of deep-sea mining in the "Area". On the other hand, there is no reason to believe that references in Part XII to the "preservation and protection of the marine environment" are not meant to cover the type of activities or threats mentioned above. Otherwise, it could prove difficult to assign a meaning to these terms.

Article 192 lays down the very general rule that states are under the obligation to protect and preserve the marine environment. This is an important provision because it could well form the basis of many future developments in the international law of the sea.

Article 197 requires states to co-operate in formulating international rules, standards and recommended practices and procedures for the protection and preservation of the marine environment. This provision seems to be of particular importance in view of the fact that so far almost all the work that has been done at both national and international level for this purpose has been limited to the control of pollution. This article, it would seem, could open the way to research, co-operation and legislation on methods to control other equally destructive activities. Article 202, in addition, deals with scientific and technical assistance to developing States for the protection and preservation of the marine environment, as distinct from pollution control. Taken together, these articles seem to open a new field in marine conservation activities which should necessarily cover the protection of marine ecosystems against all forms of degradation as well as the preservation of life-supporting processes in the sea.

Another important provision of Part XII is Article 194.5 which provides that the measures taken in accordance with that Part shall include those necessary to protect and preserve rare or fragile ecosystems as well as the habitat of depleted, threatened or endangered species and other forms of marine life. As all other provisions in Part XII, this article applies to all waters, including presumably internal waters. It calls for the taking of protection control measures and other measures for the protection of the marine environment to protect important ecosystems and species habitat.

Article 194.5 therefore, although it does not go as far as making it an obligation, under the Convention, to set aside representative samples of all types of ecosystems or critical habitats of commercial species by the creation of marine protected areas and to preserve these ecosystems and habitats against all forms of degradation, is nonetheless of very great importance as it provides for a legal basis upon which coastal zone management, the preservation of unique ecosystems and the protection of the habitats of endangered species can be developed in the future, and protected areas established in all jurisdictional zones including the high seas. For these reasons it must be considered as one of the major achievements of UNCLOS III from the point of view of conservation.

Finally, Article 206 of the Convention, which Is also contained in Part XII, deals with the assessment of the potential effects of human activities on the marine environment. This article is not limited to the effects of pollution but addresses itself to all activities that may cause significant damage to the marine environment, to the marine environment in general and to the ecosystems and habitats referred to in Article 194.5 in particular.

Broadly interpreted, this article could mean that all planned activities which could result in habitat alteration and destruction, affect the sustainable yield of fisheries or endanger the survival of species, should be the subject of an environmental assessment. Such activities could, for instance, include, in addtion to pollution, dredging, filling, silting, damming of estuaries, draining of coastal marches, introduction of alien species or even large scale fishing operations.

This provision, however, is only hortatory, and would be, moreover, difficult to enforce in the absence of institutional mechanisms which would assist states in identifying cases where assessments are necessary, developing criteria and procedures, carrying out the assessment themselves and integrating the results of these assessments into their planning. As Article 206 stands, states must, however, communicate the result of the environment assessments they make to the competent international organizations which, in turn, should make these reports available to all states. But among the many United Nations agencies that deal with the sea there does not seem to be any of them which could fulfill this role at present. Considerable developments, both in the legal and institutional fields are, thus, required if this important provision is to be of any effectiveness in the future. In fact, the same holds true for all the provisions in Part XII that deal with the protection and preservation of the marine environment in general as distinct from pollution prevention and control.

(b) The Area

The "Area" as defined by Article 1 of the treaty includes the sea-bed, ocean floor and subsoil beyond the limits of national jurisdiction. The mineral resources of the Area, and therefore not the living resources which remain governed by the relevant provisions relating to the high seas, are to be managed by a new body, the International Sea-Bed Authority.

The principal organs of the Authority, as laid down by the Treaty, are as follows:

a) The Assembly which is the policy making organ of the Authority and which is composed of all its members;
b) The Council which is the executive body of the Authority and which consists of 36 members and subsidiary organs of the Council which are the Economic Planning Commission, and the Legal and Technical Commission;
c) the Secretariat;
d) the Enterprise which is the organ through which the Authority will carry out its mining activities in the Area;
e) a Special Sea-Bed Chamber within the Law of the Sea Tribunal to settle disputes relating to the sea-bed.

Since it is clear that both exploration and exploitation of mineral resources in the Area may result in harmful effects on sea-bed species and ecosystems, as well as

on the marine environment in general, the treaty contains a number of provisions the purpose of which is to prevent or minimize these effects. It was clear, indeed, to the negotiators that the "Area" being outside the limits of national jurisdiction and having been proclaimed the "common heritage of mankind" by Article 136 of the treaty, required specific environmental protection rules and that these rules could only be applied by an international authority.

The threats to marine life resulting from deep-sea mining operations are habitat destruction, or degradation, and pollution. The Treaty deals, therefore, with these two aspects although they may not always be easily separated in specific provisions. Regarding habitat preservation[242], in particular, Article 145 requires the Authority to adopt appropriate rules, regulations and procedures for the protection and conservation of the natural resources of the Area and the prevention of damage to the flora and fauna of the marine environment.

It is obvious, however, that no deep-sea mining can go on without same degree of damage to animals, plants and their habitats. This article must, therefore, be considered only as a basis for more specific rules to be adopted by the Authority to prevent that form of damage, which because of its geographic extent or by reason of the importance of the species or ecosystems that would be affected, would clearly have unacceptable results.

The Legal and Technical Commission which is a subsidiary organ of the Council of the Authority, has the duty, under Article 165 of the Convention to prepare assessments of environmental implications of activities in the Area[243], supervise activities in the Area[244], make recommendations to the Council on the protection of the marine environment[245] and formulate and submit to the Council, which in turn will present them to the Assembly, rules, regulations and procedures which must take into account assessments of the environmental implication of activities in the Area[246].

The functions of the Council, with regard to the prevention of damage to the marine environment, appear in Article 162. They include in addition to the adoption and provisional application, pending final adoption by the Assembly, of the rules, regulations and procedures presented by the Legal and Technical Commission, the issuance of emergency orders, including orders for the suspension or adjustment of operations to prevent serious harm to the marine environment arising out of any activity in the Area, the disapproval of areas for exploitation in cases where substantial evidence indicates the risk of serious harm to a unique environment, and the initiation of proceedings before the Sea-Bed Authority in case of non-compliance.

In addition, Annex III to the Treaty establishes the "basic conditions of prospecting, exploration and exploitation" in the "Area". Under these conditions, contractors are under the obligation to comply with the Convention and the relevant rules and regulations of the Authority concerning protection of the marine environment[247], to carry out exploration and exploitation only in areas specified in plans of work approved by the Authority[248], to accept as enforceable and comply with the provisions of the Convention that are applicable to the "Area" and with the rules and regulations adopted by the Authority or by its organs[249], and to accept control by the Authority of activities in the "Area"[250]. Finally, Article 17 of Annex III contains a long

list of matters in respect of which the Authority is under the obligation to adopt and uniformly apply rules, regulations and procedures for the implementation of its functions.

Regulations made under this provision, where they relate to the protection of the marine environment, must be drawn up in order to secure effective protection of the marine environment from harmful effects resulting directly from activities in the "Area" or from shipboard processing immediately above a mine site[251].

The system provided for under the Convention to preserve natural habitats from the harmful effects of deep-sea mining seems to be both comprehensive and effective in regard of the institutions and procedures that have been established. Admittedly, no provision is made for the establishment of protected areas to preserve unique ecosystems and species, or representative samples of other ecosystems, but it could be argued that the creation of such areas is well within the environmental protection powers of the Authority and that nothing would, therefore, prevent that Organization from setting up protected areas in the future when the necessity arises.

As to the effective implementation in practice of conservation provisions, the Authority may use a variety of procedures including orders for the suspension of operations and the initiation of judicial proceedings in the case of non-compliance. In addition, particular attention should be given to the preparation of environmental impact assessments which the Legal and Technical Commission is under the obligation to carry out under Article 165.2(d) of the Convention. The extent to which these assessments are to be taken into consideration by the Authority in its rules, regulations and procedures remain, however, unclear. Admittedly, the Legal and Technical Commission when formulating and submitting to the Council draft rules, regulations and procedures has the duty, under Article 165.2(f) to take into account all relevant factors, including assessments of environmental implications of activities in the Area. But, on the other hand, the Council and the Assembly in adopting these rules, regulations and procedures are under no similar obligations. Decisions will, therefore, necessarily be taken on the basis of many factors, environmental considerations being only one of them. If, however, disputes arise between a state party to the Convention and the Authority concerning acts or omissions of the Authority that are alleged to be in violation of the Convention, its Annexes, or regulations made thereunder, including, therefore, disputes relating to environmental problems, they can be submitted to the Sea-Bed Disputes Chamber of the International Tribunal for the Law of the Sea, the establishment of which is provided for under Articles 186 and 187 of the Convention.

iii) Pollution

Pollution as it may affect living resources and ecosystems is hardly dealt with at all in the Convention. Most of the provisions relating to pollution in Part XII, which deals with all sources of pollution are of a general nature and concern more the marine environment as a whole than the preservation of particular areas or resources. Admittedly, most if not all of these provisions are important in that their implementation will assist in reducing the overall level of ocean pollution. This is particularly true of articles dealing with land-based pollution such as Article 207 which specifically provides that laws, regulations, measures, rules, standards and

recommended practices and procedures established thereunder shall be designed to minimize, to the fullest extent possible, the release of toxic, harmful or noxious substances, especially persistent substances, into the marine environment. But provisions to control pollution in particularly sensitive areas are conspicuously few.

(a) The EEZ

i) Ice-covered areas

Article 234 gives the right to coastal states to take measures in their EEZ, and doubtless also in their territorial sea and internal waters, for the prevention, reduction and control of marine pollution from vessels in ice-covered areas where the presence of ice for most of the year creates exceptional hazards to navigation, and where pollution of the marine environment could cause major harm to or irreversible disturbance of the ecological balance. There are no restrictions laid down as to the nature of these measures except that they must be non-discriminatory and give due regard to navigation and the protection and preservation of the marine environment. The meaning of this provision, therefore, is that contrarily to the rules established in other articles dealing with vessel source pollution, coastal states may in ice-covered areas adopt laws and regulations governing the design, construction, manning and equipment of vessels or the establishment of areas closed to navigation provided such areas are not so large as to make navigation impossible.

This important provision legitimizes unilateral measures that had been taken by Canada in 1970[252] for the preservation of the Atlantic Arctic from oil spills and other accidents which had been strongly opposed at the time in the name of the freedom of the seas. It could constitute a useful precedent for other ecologically vulnerable areas of the world.

ii) Special Areas

Article 211.6 of the Convention allows coastal states to adopt, in certain parts of their EEZ, where oceanographical and ecological conditions, the protection of living resources, and the special character of ship traffic justify it, mandatory rules for the prevention of pollution from vessels which may be more stringent than those generally accepted for the rest of the ocean. This system of "special areas" where stricter rules are in force is obviously draw from the 1973 London Convention for the Prevention of Pollution by Ships. But "special areas", under the London Convention are, as we saw[253], to cover the whole surface of semi-enclosed seas such as the Baltic or the Mediterranean, whereas Article 211.6 could presumably apply to vulnerable areas in an EEZ anywhere in the world. In order to establish such a "special area" a coastal state has, however, to start a long and protracted procedure and request the approval of the competent international organization. If states are not deterred by these difficulties, "special areas" could prove to be anim-portant tool to protect from pollution particularly important ecosystems.

iii) Routing systems

Article 211.1 requires states to promote the adoption, "through the competent international organization or general diplomatic conference," of routing systems designed to minimize the threat of accidents which might cause pollution of the marine environment. Here again, the establishment of such systems, including traffic separation schemes, could help considerably to preserve ecologically important and vulnerable areas.

(b) The "Area"

Most of what has been stated above regarding the preservation of habitats from destruction or degradation in the "Area" also applies, mutatis mutandis, to the protection of these habitats from pollution. Of particular relevance in this respect is Article 145 which requires from the Authority to adopt appropriate rules, regulations and procedures for the prevention of pollution and contamination and other hazards to the marine environment, including the coastline and of interference with the ecological balance of the marine environment.

iv) Introduction of alien species

The inclusion of a provision dealing with the prevention or control of the introduction of exotic species is a new feature of international conservation agreements[254]. Since most of these agreements apply to coastal zones as well as to land areas, the introduction of alien species into the marine environment, and in particular into unique island ecosystems seems to be slowly but gradually brought under stricter controls.

The new Law of the Sea Convention goes, however, a step further in that Article 196 provides that "States shall take all necessary measures to prevent, reduce and control. . . . .the international or accidental introduction of species, alien or new, to a particular part of the marine environment, which may cause significant and harmful damage thereto". This article is included in Part XII of the Convention which deals with the protection and preservation of the marine environment in general, irrespective of jurisdictional zones. It is therefore applicable to introductions effected into any part of the ocean. Non-compliance with that provision, resulting in significant damage to species and ecosystems in waters under the jurisdiction of another state or in the high seas should then necessarily entail recognition of the liability of the state who has allowed the introduction to take place, and Article 235 on responsibility and liability should apply.

Introductions, however, when they succeed, are generally irreversible and curative methods ineffective. Prevention is therefore the best and the cheapest method. Since states are to be the only judges of the advisability of the introductions they may wish to carry out, some form of procedure to safeguard the marine environment and the interest of other states from harmful introductions would seem to be essential. This would probably be best achieved under the environmental assessment provision of Article 206 provided adequate procedural and institutional mechanisms are set up to allow for full discussion between all the parties concerned of the environmental implications of any proposed introduction.

e) Other Aspects of the New Convention which are Relevant to the Conservation of Living Resources

i) Environmental assessments

The new Convention on the Law of the Sea provides, for the first time perhaps in an international agreement, for a requirement for environmental assessments. This requirement appears in Article 165, which relates to the "Area" and in Article 206, which, being included in Part XII of the Treaty, is, therefore, applicable to all jurisdictional zones.

Problems relating to environmental assessments in the "Area" have already been discussed in the section of this paper which deals with that particular zone. (See page 139 above.) Environmental assessments are, in this case, limited to deep sea mining beyond the limits of national jurisdiction and must be prepared by the international organization which is competent over activities in the "Area", namely the Authority. If a dispute arises in respect of the performance or contents of an environmental assessment, or of the procedures followed in the preparation of such an assessment, it could, presumably, be submitted for judicial adjudication to the Sea-Bed Disputes Chamber of the International Tribunal for the Law of the Sea.

Article 206, on the other hand, provides that "when States have reasonable grounds for expecting that planned activities under their jurisdiction or control may cause substantial pollution of, or significant and harmful changes to, the marine environment, they shall, as far as practicable, assess the potential effects of such activities on the marine environment ...".

Once they have carried out an environmental assessment, states have the obligation, under Articles 205 and 206 to communicate reports of such assessment "to the competent international organization which should make them available to all States".

In this case environmental assessments are to be prepared by states and may concern any jurisdictional zone. Admittedly, the provision of Article 206 is not really binding. It could nonetheless constitute a powerful weapon to prevent damaging changes to the marine environment since it applies to all activities which may cause significant and harmful damages including, therefore, not only polluting activities but also dredging, habitat destruction, trawling, the introduction of alien species and even, presumably, certain fishing operations. As an example, it would seem that large scale fishing for a species such as krill should be preceded by an environmental assessment so that the effects of fishing at different catch levels be assessed in respect of other species in the same ecosystem and of the ecosystem as a whole.

It may, however, in practice, prove difficult to enforce this provision. States remain free to decide whether an environmental assessment must be carried out. Furthermore, even if an assessment is actually made, there is no way to ensure that all relevant factors, including the effects of the proposed activity on marine fauna and flora, are really taken into consideration. Finally, when the results of an environmental assessment show that an activity is likely to cause serious damage to the marine environment, the state concerned is under no obligation to refrain from undertaking that activity.

It seems clear, therefore, that if environmental assessments are to become effective instruments for the prevention of significant and harmful changes to the marine environment, appropriate guidelines, criteria, standards and procedures will have to be developed, the content and scope of such assessments defined, and all possible steps taken to avoid environmental damage once it has been shown that a proposed activity may result in such damage. Furthermore, technical and financial assistance will, in many cases, have to be provided when environmental assessments are to be carried out by developing countries.

Article 206 should, therefore, only be considered as a basis upon which international organizations, aid agencies and the world community in general could develop review mechanisms for the purpose of assessing major development projects for their environmental impact and establish procedures reading to the taking of decisions on the advisability of financing and implementing such projects when environmental assessments have shown that significant and harmful damages to the marine environment may be caused.

ii) Scientific research

The importance of scientific research, including the collection of adequate statistical information, for the conservation and exploitation of marine living resources is emphasized in many articles of the Convention[255]. The subject of scientific research is, however, specifically dealt with in Part XIII. Coastal states, under Article 245 have the exclusive right to regulate, authorize and conduct marine scientific research in their territorial sea and, under Article 246, the right to regulate, authorize and conduct such research on their continental shelf and in their EEZ. In these two zones, research by other states can only be conducted with the consent of the coastal state[256]. In normal circumstances coastal States are under the obligation to grant this consent[257].

They may, however, withhold their consent if the proposed research project is, *inter alia,* "of direct significance for the exploration and exploitation of natural resources, whether living or non-living"[258]. Furthermore, coastal states have also the right, under Article 253 to require the suspension of any marine scientific research in progress. Finally, the discretion of coastal States to withhold their consent or require the suspension of research is absolute, and disputes arising from the exercise of that discretion cannot, under Article 297.2, be submitted to the adjudication procedure laid down in Articles 286 to 296 without the consent of the coastal state concerned.

As a result, there is no way, unless the coastal state gives its consent, to determine whether the obligation of that state in respect of optimum utilization of the resources, prevention or overexploitation and the establishment of an allowable catch, with a possible surplus accessible to other states, are actually implemented.

On the high seas, all states and competent international organizations have the right to conduct marine scientific research in the water column beyond the limits of the EEZ[259], as well as on the sea-bed, the ocean flora and the sub-soil, that is to say in the "Area"[260]. Research activities in the "Area", however, must be carried out for the benefit of mankind as a whole and the result of such research must be effectively disseminated[261].

The Convention emphasizes the importance of international co-operation and treaties in promoting research in all areas[262] and stresses in particular the role that states and competent international organizations, in particular fishery bodies, should play in the collecting, dissemination and exchange of information resulting from research[263]. It also lays down an obligation on the part of States or international organizations to provide to coastal states in whose EEZ they are conducting research comprehensive information on the results of that research[264]. Finally, marine scientific research must, in all cases, comply with any regulations relating to the protection and preservation of the marine environment[265].

iii) Transfer of technology

In view of the considerable extension of coastal state jurisdiction over living resources that will be taking place under the new Convention, many states are or will be soon confronted with the necessity of managing these resources themselves. However, the scientific information, infrastructure, equipment and trained staff that are essential for that purpose will often not be available to them. The Convention, therefore requires states to co-operate to promote the development and transfer of marine science and technology[266] and to promote the development of the marine scientific and technological capacity of states "which regard to the exploration, exploitation, conservation and management of marine resources, the protection and preservation of the marine environment, marine scientific research and other activities ..."[267]. It order to achieve these objectives, the Convention further provides that states shall endeavour to establish programmes of technical co-operation[268] and promote the establishment of national[269] and regional[270] marine scientific research centres. The potential role of the competent international organizations is, in this regard, strongly emphasized.

The functions of these regional centres in the fields of training, management studies, research in the preservation of the marine environment, acquisition and dissemination of data, and technical co-operation with other states of the region[271] should enable them, if adequately staffed and financed, to play an important role in the conservation of marine living resources in the areas where they are located.

iv) Enforcement

The new Convention does not provide for new specific rules regarding enforcement of conservation regulations relating to living resources. In areas under national jurisdiction, including the EEZ, enforcement powers belong to the coastal state. Nationals of other states fishing in the EEZ, when authorized to do so, must therefore comply with the enforcement procedures laid down by this state[272]. On the high seas, under Article 117, all states have the duty to take such measures for their respective nationals as may be necessary for the conservation of the living resources of the high seas. This clearly includes enforcement measures. Article 117, however, further provides that the measures in question may be taken in co-operation with other states. Enforcement measures may, therefore, as they always have been, be taken by international fishery bodies in the high seas. The Convention is silent as to how this clear division of enforcement powers between coastal States, in areas under national jurisdiction, and fishery commissions in the high seas would work in practice in the case of shared stocks. The problem is a relatively new one in that most of the extensions of coastal state jurisdiction have taken place only recently. But the matter, being complex and fraught with political difficulties, should preferably be left to bilateral or multilateral negotiations within the framework of fisheries commission.

Regarding habitat preservation, introduction of alien species, and protection of particular ecosystems from pollution, problems will generally arise in areas under national jurisdiction and enforcement will, therefore, depend on the coastal state. In the high seas, enforcement, if the need arises, will require international agreements. In the "Area" enforcement is to be carried out by the Authority.

v) Institutional arrangements
The Convention does not specifically provide for the establishment of new international organizations other than the Authority whose task is exclusively confined to the exploitation of the mineral resources of the "Area".

It relies, therefore, and to quite a considerable extent, either on existing organizations or on organizations to be created in the future, to deal with all those aspects of living resources conservation where international co-operation is essential.

No detailed indications, however, are given on such matters as the competence, functions or financing of these organizations, decisions on these matters being best left to the states concerned at the time individual agreements are negotiated.

With regard to the membership of fishery organizations, the Convention lays down, however, a clear obligation on the part of all states who share the same resources or who fish for identical resources or in the same area of the high seas to co-operate in the management of these resources and, in the absence of existing organizations, to establish sub-regional or regional bodies to this end[273].

Another type of regional organizations are the regional marine scientific and technological centres which have already been mentioned earlier[274]. The Convention merely provides a non-exhaustive list of the functions of these centres[275] and lays down the obligation for all states of a region to co-operate with these centres "in order to ensure the more effective achievement of their objectives"[276]. It does not, however, give any indications as to whether these centres should be independent bodies or be connected to existing global or regional organizations, nor does it provide any guidance as to the establishment, structure or financing of these organizations. Here again, these matters are certainly best left to the decision of the states concerned in each particular case.

Finally, Article 278 requires that all competent international organzations dealing with research and the transfer of technology should co-operate among themselves to discharge effectively the functions and responsibilities assigned to them under the relevant Parts of the Convention. The institutions and procedures through which such co-ordination would be achieved are, however, not specified.

vi) Settlement of disputes
The Convention provides for an elaborate and complex system of dispute management and settlement which cannot be dealt with in this paper in any detail. The main lines of the system, however, as it applies to the conservation of living resources can be briefly described as follows:

Any dispute relating to the interpretation or application of the Convention, where no settlement has been reached by conciliation or by any other means agreed upon by the parties to the dispute themselves, must be submitted to an international court or tribunal whose decisions are binding. Disputes relating to fishing in the high seas or to the preservation of the marine environment, irrespective of jurisdictional zones, can thus be settled in that way. An important exception to this principle has, however, been accepted in respect of the rights of coastal states. This exception relates to the discretionary powers of the coastal state over the exploitation of living resources in the EEZ and the right to conduct scientific research. Coastal states

are not obliged to accept the submission to the dispute settlement procedures provided for by the Convention of any dispute relating to their sovereign rights over the living resources in their EEZ or to the exercise of these rights. This means in practice that the discretionary powers of coastal states with regard to the determination of the allowable catch, their harvesting capacity, the allocation of surpluses to other states and the terms and conditions established in their conservation and management regulations, are absolute[277].

Exceptions to this rule are, however, allowed in three cases: if a coastal state has arbitrarily refused to determine the allowable catch and its harvesting capacity with regard to stocks that another state is interested in fishing[278], if a coastal state has arbitrarily refused to allocate a declared surplus[279], if a coastal State has manifestly failed to comply with its obligations to ensure through proper conservation and management measures that the maintenance of the living resources in its EEZ is not seriously endangered[280]. This latter provision is of particular importance because of the possibility it implies to exercise some pressure an coastal states guilty of mismanagement of the resources of their EEZ. Broadly interpreted, it could also apply to the cases where depletion of the resource may be attributed to other causes than overfishing such as for instance, the destruction of critical habitats.

Cases failing under these three exceptions cannot, however, be submitted to the compulsory procedures for the settlement of disputes which are provided under Articles 286 to 296 of the Convention. They can only be submitted to the conciliation procedure established under Article 284 and under Annex V to the Convention. This procedure is, however, obligatory and cannot, therefore, be refused by any party. But the conciliation commission is not entitled to substitute its discretion for that of the coastal state[281]. As a consequence, in the absence of an agreement on the subject matter of the dispute the only obligation of the conciliation commission is to communicate its report to the appropriate international organizations[282]. Thus, although they can to a certain extent at least be challenged, the powers of the coastal state in its EEZ remain, nonetheless, almost absolute.

The same rule applies with regard to scientific research conducted in the EEZ or on the continental shelf. Coastal states are not obliged to accept to submit to a compulsory adjudication procedure disputes arising out of the exercise of their rights or discretion to withhold their consent to the carrying out by other states of scientific research in zones under their jurisdiction. Neither are they obliged to submit to such settlement decisions on their part to suspend or cease research projects in these zones[283].

f) Conclusion

The new Convention on the Law of the Sea will result in a revolutionary change in the law of fisheries and in jurisdiction over living resources.

Whereas areas under national jurisdiction were for a long time only constituted by the narrow strip of waters along the coast that formed the territorial sea, the new regime will bring under the jurisdiction of coastal states more than half of the total area of the world oceans and seas, an area which is just a little smaller than the sum of all land areas in the world[284]. At the same time, about 99% of the world's presently exploited marine living resources will come under the jurisdiction of coastal states as against a negligible percentage previously.

As a consequence, coastal states may now well be considered as enjoying a monopoly on the exploitation of living resources under their jurisdiction, as well as on scientific research concerning these resources. They can, in a discretionary way, establish the allowable catch, their own harvesting capacity and allocate any surplus to other states. If they allow other states to fish in their EEZ, they are entitled to regulate the fishery and enforce these regulations as they please. Finally, disputes relating to fisheries or to marine scientific research as it relates to living resources cannot be submitted to compulsory procedures for the settlement of dispute without the consent of the coastal state concerned. Even in the case where a state will have manifestly failed to comply with its obligations to conserve and manage the resources of its EEZ, a dispute can only be submitted to a conciliation commission which is not entitled to "substitute its discretion" to that of the coastal state.

As a counterpart to these very extensive rights, coastal states have the obligation to conserve and manage resources under their jurisdiction and to co-operate with other states in the conservation and management of shared stocks.

With regard to the exploitation of the resources, the Convention endeavours to find a middle road between two seemingly equally detestable evils: underexploitation and overexploitation. The continuing use of the maximum sustainable yield concept, and, in addition, of several qualifying factors that more or less countervail each other, will probably be of little help to decide on a sound level of exploitation from both the biological and economic points of view. On the other hand, the requirement to take into consideration the effect of fishing on associated and dependent species must be considered as an important step forward towards the recognition of a solution to the difficult problem of the incidental taking of non-target species.

The necessity of preserving habitats, especially critial habitats, is rarely touched upon by the Convention. On the other hand, there is now an obligation on the part of all States, and in all jurisdictional areas, to protect rare or fragile ecosystems and the habitat of endangered species. This is the first time that such a provision appears in a convention relating to the living resources of the sea. Likewise, the article dealing with the introduction of alien species also constitutes an interesting new feature in that type of convention.

Except in respect of the "Area" where specific rules apply to the protection of living resources from pollution and other environmental causes of damage, provisions governing the prevention or abatement of pollution are mostly of a general nature and are not addressed to the protection of particularly vulnerable ecosystems. Noteworthy exceptions, however, are the articles on ice-covered areas and on the establishment of special areas in the EEZ.

Coastal states, from now on, will have to bear the full burden of conserving the living resources under their extended jurisdiction and of preserving the marine environment off their coasts. To perform these enormous tasks with any degree of effectiveness, they will have to proceed to a thorough re-examination of their fishery and environmental policies, revise their legislation, develop research, establish the necessary administrative mechanisms, adopt conservation-oriented regulations

and provide for the necessary enforcement machinery, including policing of their EEZ to a distance of 200 nautical miles from their coasts. Many will not succeed in accomplishing this formidable task without assistance from the outside. But if this cannot be done, many resources may well soon disappear forever.

2. The Times of UNCLOS III

It is proposed to consider in this section the new developments in the evolution of the law of the sea and in the relevant activities of international organizations that have taken place after the negotiation of the new Convention began.

Some of these developments are direct consequences of the elaboration of the new treaty and take into consideration, therefore, the probable outcome of the negotiations, thus providing a unique example in international law of a gradual implementation of a treaty well before its text was actually finalized, signed, and came into force.

Other developments, although necessarily influenced by the negotiations in UNCLOS III, were the result of earlier initiatives, such as for instance the recommendations of the Stockholm Conference, or decisions taken by the Governing bodies of international organizations.

a) Fisheries

i) Enactment of new national fishery legislation

As soon as it became clear that the new Convention would establish a 200 miles EEZ, unilateral extensions of jurisdiction by coastal states, which had already started before UNCLOS III was convened, began to develop at an increasing rate. By the end of 1980, it could be said that most of the countries in the world with an access to the sea had, by legislative action, acquired an EEZ. Exceptions were few and concerned primarily semi-enclosed seas, such as the Mediterranean, where the existence of narrow areas and of many island, some of them being independent states, would have made the delimitation of opposite and adjacent EEZs technically and politically difficult[285].

Most of these new laws, however, merely assert the new jurisdictional rights or sovereignty of the coastal states concerned over their EEZ and the living resources it contains, often reproducing verbatim clauses such as Article 56.1 of the new Convention. This article, one will remember, provides that the coastal state has, in the EEZ, sovereign rights for the purpose of exploring and exploiting, conserving and managing the natural resources of the sea-bed and sub-soil and the superjacent waters. Very few laws, for the time being at least, have taken up the provisions of the new Convention that lay down the rights and obligations of coastal states in respect of the conservation and utilization, of living resources, the principle of optimum utilization and the requirement to determine the allowable catch, the coastal state's harvesting capacity and the possible allocation of surpluses[286].

For most states, therefore, it would seem that the establishment of an EEZ is not very different, in fact, from a new extension of their territorial sea, and that conservation regulations that are provided for in their new fishery legislation are very much the same as those that were applicable earlier in their coastal waters only[287]. Admittedly, many of these new laws are very recent and states certainly felt that

they should start with what they thought to be the most urgent, namely to enact general legislation to enable them to bring under their control the activities or large foreign fishing fleets in the vicinity of their coasts. Further legislation will, therefore, follow and in some cases has already followed the enactment of general legislation on the extension of national jurisdiction zones[288]. Conservation measures will, however, remain entirely at the discretion of coastal States and enforcement of the obligations provided for under Article 61 of the Convention may not be easy.

An interesting development in national legislation has been the enactment in the United States of the Fisheries Conservation and Management Act of 1976[289]. One of its most important features is the establishment of an optimum yield concept for each fishery. This concept which was further developed by government regulation takes into consideration not only the stocks of fish but also the economic well-being of the commercial fishermen, the interests of recreational fishermen, the habitat quality, the national interest in conservation and management of the fisheries and the National Standards for fishery conservation and management that are laid down in the Act itself.

The provisions are obviously drawn from Article 61.3 of the new Law of the Sea Convention, but their interest lies in the fact that they constitute a first attempt to clarify the ambiguities of that article. However, the reconciliation of a host of divergent if not opposed interests will certainly not be an easy task.

Another Act, the American Fisheries Promotion Act (Public Law, 96-561 of 22 December 1980), provides for the phasing-down of foreign fishing in the exclusive fishery zone of the United States as American fishermen, aided by the incentives provided by the Act, gradually increase their own catch. The Act lays down criteria for the calculation of the Total Allowable Level of Foreign Fishing which include the degree to which foreign states co-operate with the United States in enforcing U.S. fishing regulations and the extent to which the states concerned require the fish taken in the United States fishing zone for their own domestic consumption. Official observers from the United States will be placed on almost all foreign fishing vessels operating in the 200 miles zone.

ii) Conclusion of bilateral agreements

A large number of bilateral agreements, perhaps over two hundred, have been concluded between 1976 and 1980 between coastal states that had extended their jurisdiction over fisheries and other states who wished to continue to carry out fishing operations in those areas. Most of these agreements are designed to define the terms and conditions under which vessels of one country may continue to be authorized to fish in the EEZ of another country.

Many of these agreements merely take up the relevant provisions of the new Convention with regard to the allowable catch of each fishery, the methods of determining that catch, the harvesting capacity of the coastal state and the part of the surplus which can be allocated to the other state party to a particular bilateral agreement[290].

iii) Renegotiation of fisheries conventions

In view of the extension of areas under national jurisdiction in anticipation of the conclusion of the new Convention on the law of the sea, many governments felt that some of the fishery agreements in force were no longer consistent with the new regime of the seas and that they should, therefore, be renegotiated.

By the end of 1980, renegotiation had already taken place in respect of several of the most important agreements, in particular ICNAF[291], NEAFC[292] and the North Pacific Fisheries Convention[293]. Under some of these renegotiated agreements fishery commissions are no longer entitled to regulate fishing in those parts of the Convention areas that are now included in the EEZ of their members[294].

As a result, coastal states may unilaterally determine the allowable catch in their EEZs and decide conservation and regulatory measures. This would not normally pose any problem when stocks are confined to the EEZ. This, however, will seldom be the case, especially in the North Atlantic and North Pacific where the continental shelf extends to great distances from the coastline. The coastal state finds itself in such situations in a particularly favourable position in that it may take part in any decision to regulate fishing for a particular stock on the high seas, but is the only one to be entitled to adopt regulatory measures in respect of the same stock, which could well be a migrating stock moving in and out of the area, in its EEZ. Thus, the coastal state is guaranteed against the taking of inappropriate conservation measures outside its EEZ, but states fishing for the same resource on the high seas are not protected against the mismanagement of that resource by the coastal state in waters under its jurisdiction.

To reconcile, however, global management requirements with the sovereignty of the coastal state over the resources of its EEZ the renegotiated Conventions contain non-mandatory provisions to the effect that the commissions set up under the agreements must seek to ensure "consistency" between proposals to regulate shared stocks in the high seas and measures taken by the coastal state in the area under its jurisdiction. In addition, commissions must in that respect also take into consideration effects produced on other stocks through "species interrelationships". Finally, the Commission and the coastal states concerned must promote the co-ordination of these measures or decisions[295]. In the case of the North-East Atlantic Fisheries Convention, the Commission is, however, also entitled to make recommendations concerning fisheries conducted within an area under fisheries jurisdiction of a Contracting Party, provided that the Contracting Party in question so requests and the recommendation receives its affirmative vote"[296].

To what extent will these provisions help in the conservation and management of stocks which are shared between the high seas and adjacent EEZs? Only time and practice will tell. But it appears clearly that one of the perverse effects of the new Convention is that, in some cases, it becomes no longer possible to manage individual stocks as single units.

The same problem necessarily also arises in respect of stocks shared between adjacent EEZs. In the ICNAF Convention area, the problem seems to have been resolved by the conclusion of an agreement between Canada and the United States[297] for the purpose of establishing a bilateral commission with the task to study and decide

management measures applicable to certain stocks shared between the two countries.

Another example relates to the Baltic Sea[298]. In 1978 all the coastal states of that sea decided to establish exclusive fishing zones beyond their territorial sea. As a result, all the waters of the Baltic came under national jurisdiction and it became necessary to amend the Convention on Fishing and Conservation of the Living Resources in the Baltic and the Belts to take the new situation into consideration. In many ways this situation was a simpler one in that, the problem of high seas fisheries having being eliminated, all that was required was a system for the management of stocks shared by neighbouring countries. Parties had to choose between a system of national management, where the International Baltic Sea Fishery Commission would only have a technical and advisory role, and a joint management mechanism with some degree of regulatory authority vested in the commission. They wisely chose the latter course, it being understood that commission recommendations which relate specifically to the fishing zone of a Contracting State must first be approved by that state before they come into force.

Finally, another aspect of the renegotiation of fishery agreements is that renegotiated agreements no longer apply to species in respect of which special regimes have been established by the new Convention on the Law of the Sea. Thus, sedentary species, which are defined both in the Northwest and North-East Atlantic Fisheries Convention by the same unprecise and dispute-prone terms as in the Law of the Sea Convention itself, and also sea mammals[299] or Cetacea[300], highly migratory species and salmon or other anadromous stocks are all specifically excluded from the species covered by the new agreements. In respect of the renegotiated North Pacific Convention, the amendment protocol no longer mentions the "principle of abstention" which had been so controversial in the past. The protocol, however, further restricts the taking of salmon by Japanese vessels both in the high seas and in the EEZ of the United States.

iv) The Convention on the Conservation of Antarctic Marine Living Resources
This Convention was adopted in Canberra on 12 May 1980. It was not yet in force in 1981. Its main purpose is the conservation and rational use of Antarctic marine living resources[301]. It applies to all such resources south of the Antarctic convergence[302]. Whales and seals, however, although nominally covered by the Convention are in practice excluded and remain, therefore, under the management authority of the International Whaling Commission and of the Commission established under the Convention for the Conservation of Antarctic Seals.

Many of the provisions of this new agreement are not much different from those contained in earlier fishery treaties in other parts of the world. There are, however, a number of important differences, resulting apparently from the new Law of the Sea Convention as well as from an increasing awareness of conservation requirements, between this new agreement and other fishery treaties.

The Convention indeed, and this is perhaps the first time in the history of fishery conventions, adopts an ecosystem approach to the conservation and management of living resources. For that purpose, it requires, for instance, that ecological relationships between harvested, dependent and related populations be maintained[303]

and that changes in the marine ecosystem which are not potentially reversible over two or three decades be prevented or at least that the risk of such changes be minimized[304]. To achieve these objectives, the Commission established under the Convention is empowered to take conservation measures concerning the effects of harvesting and associated activities on components of the marine ecosystem other than the harvested populations[305]. The Commission may also designate special areas for protection and scientific study[306] and thus establish true marine reserves to protect ecosystems, species or habitats. Finally, the Commission is required, when adopting conservation measures, "to take full account of any relevant measures established or recommended by the consultative Meetings of the Antarctic Treaty Powers or by existing fisheries commissions responsible for species which may enter the area to which the Convention applies"[307]. It must also "seek to co-operate with contracting parties which may exercise jurisdiction in areas adjacent to the Convention area in respect of the conservation of any stock or stocks of associated species which occur both within those areas and the area to which the Convention applies with a view to harmonizing the conservation measures adopted in respect of such stocks"[308].

On the other hand, the MSY concept, although qualified by the requirement to take Into consideration the effects of harvesting on associated or related species, remains nonetheless solidly entrenched in the text of this Convention, albeit under a different name, "the greatest net annual increment". The problem is of particular importance in the Antarctic area because the main harvested species will be Krill, a pelagic shrimp which congregates in enormous numbers and constitutes the principle food of a large number of animals such as many species of fish, squids, seabirds, seals and the great whales. Fishing for krill, if it involves the taking of large quantities, will, therefore, necessarily affect the food supply of other species and as a result their numbers and reproduction rate. If other species than krill are harvested, it will be, as a consequence, impossible to maximize the catch of both krill and species feeding on krill. If some of these species are alreday depleted, as it is the case for the great whales, their recovery will take a much longer time or will be impossible[309].

Article 113 of the Convention, therefore, contains a contradiction in terms since it requires parties not to allow the size of any harvested population to fall below the level which ensures the "greatest net annual increment" and at the same time to restore depleted populations to that level, an Impossible task, as we saw, when both predators and their prey are harvested in the same ecosystem[310].

Thus, in the Southern Ocean, if depleted whale stocks are to recover, Krill populations should most probably be maintained at a level much higher than the one at which MSY can be obtained.

v) The European Economic Community

The European Economic Community as an economic integration organization with ten member countries in 1981 has as a purpose the achievement of a European common market. To this end, in the particular field of fisheries, the Council of the Community decided to establish a common organization of the market in fishery products and a common structural policy for the fishing industry. Under this policy

Community fishermen must have equal access to the fishing grounds in maritime waters coming under the sovereignty or within the jurisdiction of member States.

A Regulation adopted on 19 January 1976 by the Council of the European Communities[311] provides that common rules shall be laid down for fishing in maritime waters and empowers the Council, where there is a risk of overfishing of certain stocks, to adopt the necessary conservation measures[312].

A special Standing Committee for the Fishing Industry was established under that same regulation to promote the co-ordination of structural policies and to ensure close and constant co-operation between member States and the EEC Commission[313]. Pursuant to this Regulation a new Regulation was adopted on 30 September 1980[314] providing for detailed technical measures for the conservation of fishery resources.

These rules relate to mesh size, fish sizes, limitation of the use of certain gear and vessels, prohibition of fishing for certain species in certain areas and periods and contains important provisions dealing with the by-catches, *i.e.* the incidental taking, of non-target species[315]. These provisions, however, concern mainly certain commercial species taken incidentally to the catching of other species. The Regulation also contains rules prohibiting fishing for certain species at certain places for the protection of spawning grounds and nursery areas[316], an interesting innovation in an international instrument. Another important Regulation, adopted by the Council on 26 March 1980, deals with the establishment of total allowable catches (TAC) per stock or group of stock and determines the share available for the Community[317] and, as a result, the "surplus" available to third countries. Negotiations were still proceeding in early 1981 as to the allocation of the TAC between Member States.

In the face of many difficulties, due to the exclusion of EEC fleets from some of their traditional fishing grounds after the establishment of 200 miles EEZs by many coastal States, over-investment and resulting overfishing, disagreements on national quotas and demands for certain exclusive fishing zones within the Community waters, a common fisheries policy is nonetheless gradually developing. It provides a unique example of common management decisions applicable to shared stocks.

vi) The new role of FAO

With the extension of areas under national fishery jurisdiction developing, at an increasing rate, everywhere in the world and in anticipation of the conclusion of the new Law of the Sea Convention, FAO is now confronted with the gigantic task to advise and provide assistance to its member States, particularly developing countries, in the conservation, management and development of the stocks of fish and other living resources over which they now exercise their sovereign rights.

As a consequence, FAO has now established a comprehensive programme of assistance to help coastal developing states in the development and management of fisheries in their EEZ. This programme was approved by the FAO Conference at its twentieth session in Rome in November 1979. Its three main objectives are to promote rational management and full use of fishery resources in the EEZs to strengthen the capabilities of coastal countries and groups of countries to manage

and develop their fisheries, and to enable developing countries to secure a greater share of living marine resources. It includes a medium-term action plan to meet immediate needs, including identification of national fisheries opportunities, assistance in the preparation of fisheries policies, resource assessment, the drafting of national legislation, the establishment of enforcement procedures, the training of administrators and specialists, and funding for development projects and technical assistance and basic long-term studies relating to fisheries management at national, regional and subregional levels and to the preparation of fishery development options based on socio-economic and technical studies of all aspect of fisheries[318]. It does not seem that the identification and preservation of critical habitats, the protection of endangered species and problems arising from the interrelation of stocks and from incidental take have been given high priority in that programme. On the other hand, other important conservation aspects such as the identification of stocks shared by two or more countries, the development of principles and practical methods of fisheries management, problems of enforcement and legal and institutional matters with particular emphasis, in all cases, on implications or shared stocks, have been given due consideration.

The implementation of FAO's EEZ Programme is to be carried out by a network of technical support units associated with regional fisheries bodies and based on natural management areas, that is to say areas when there are shared stocks or fisheries and where countries have common problems and opportunities. By 1980 seven of these units had already been established.

Another consequence of the new Convention on the Law of the Sea, is that, in respect of regional arrangements, FAO felt that its own regional fishery bodies should become better adapted to the new situation resulting from the extension of coastal state jurisdiction in many parts of the world. The Fishery Committee for the Eastern Central Atlantic (CECAF) decided accordingly to set up a sub-committee on resource management within the limits of national jurisdiction. This sub-committee which is open only to coastal states may recommend conservation measures and advise on the co-ordination of these measures with those taken on the high seas[319].

Other FAO regional bodies, in particular those whose area of competence extends over very vast areas of the ocean such as the Indian Ocean Fishery Commission and the Indo-Pacific Fishery Commission, are now establishing sub-regional committees to deal with the specific problems of particular geographic areas such as for instance gulfs and semi-enclosed seas which constitute natural management areas or ecological units. Membership of these sub-committees will be limited to coastal states[320].

Finally, with most of the world fisheries now under national jurisdiction, it was felt that the Committee on Fisheries of FAO should remain a purely advisory body and earlier ideas to give to that Committee a more independent status by means of a treaty concluded pursuant to Article XIV of the Organization's constitution were dropped.

On the other hand, at the Tenth Session of the Committee on Fisheries in June 1975, there was considerable discussion of the role of the Committee vis-a-vis regional fishery bodies both outside and inside FAO. It was felt that the proper function of

the Committee was to review critically the overall performance of these bodies and to examine the degree to which they were maintaining the stocks and the fisheries in a satisfactory condition. At the same session the Committee requested its Subcommittee on the Development of Co-operation with International Organizations concerned with Fisheries to propose to the Committee measures to improve existing mechanisms and arrangements and particularly the role to be played by regional fishery bodies (COFI/77/10, p. 1). The Committee on Fisheries has thus become a global reviewing and co-ordinating body covering the activities of all fishery organizations in the world. Reports on the activities of all fishery commissions are now, as a result, submitted to the Committee at each of its sessions.

b) The Regional Seas Programme

Activities of UNEP relating to the marine environment have mainly been concentrated on the development of the Regional Seas Programme. Although the environmental problems of the oceans are worldwide it was, indeed, felt that a regional approach could, in a first stage at least, provide more concrete results and allow for the solution of particularly acute problems especially in certain semi-enclosed seas. The UNEP programme is based on co-operation with the governments of each region concerned and implies full co-ordination of conservation activities with all the other UN agencies. This was the basis for the development of "Action Plans" which consist of three basic components: environmental assessment, environmental management and supporting measures, including, at least initially, financial support.

By the end of 1980, ten regional seas areas had been selected and Action Plans for these areas were either already in operation or under development. These regional seas were the Mediterranean, the Red Sea, the Kuwait Region, the Caribbean, the waters off the western coast of Africa, the East Asian Seas, the South-East Pacific; the South-West Pacific, East Africa and the South-West Atlantic[321].

The procedure which, so far, seems to have been generally followed in the development of a Regional Seas Programme consists, after some preparatory work to assess the nature and extent of the threats to the marine environment in the area, in the adoption by the coastal states of that region of an Action Plan, and finally in the signing of a general framework convention which provides for the conclusion of protocols to deal in more detail with certain specific aspects of pollution control and environmental protection and enhancement.

The general convention lays down general obligations to prevent, abate and combat pollution from all sources, establish a pollution monitoring system and co-operate in the development of procedures to control its application. It also provides for institutional arrangements. Protocols deal with such matters as regional co-operation in combatting pollution in cases of emergency, prevention of pollution by dumping and protection from pollution from land-based sources. By the middle of 1981 general Conventions had already been concluded for the Mediterranean[322], the Kuwait area, and the West and Central African Region[323]; in addition, three protocols to the Mediterranean Convention[324], one to the Kuwait Convention and one the convention on the West and Central African region[325] had been adopted. Similar conventions and protocols are expected to be concluded in respect of other regional seas in the realtively near future after completion of the preparatory work[326].

Two important aspects must be underlined in respect of the UNEP Regional Seas Programme. There seems to be, firstly, an increased awareness on the part of coastal states of the need to stop the gradual degradation of the marine environment. This is evidenced by the fact that since its inception in 1975, the development of the programme has been moving very fast. The Mediterranean Action Plan and Convention were adopted in 1975 and 1976 respectively, the Convention and its first protocols came into force as early as 1978 and new protocols have now been concluded or are being negotiated. Action Plans have also already been adopted for some of the other regional seas that form part of the UNEP programme, and all Action Plans, together probably with several conventions and protocols are expected to be approved by the end of 1981[327]. This tends to show that a regional approach to environmental marine problems seems to have better chances to succeed as all coastal states in a given region probably feel that they are directly concerned, and should therefore be also directly involved in the taking of appropriate preservation measures.

Another important feature of the UNEP Regional Seas Programme is that it is not, in principle, confined to the prevention or control of pollution and that it endeavours to take other factors into consideration, in particular the preservation of important habitats, the rational management of coastal ecosystems, the development of mariculture, the preservation of freshwater resources and the protection of soils[328]. Of particular importance to resource conservation is the proposed protocol to the Mediterranean Convention on the subject of the selection, establishment and management of specially protected areas.

The significance of this protocol will be twofold: it will be the first example in the world of a regional treaty concerned exclusively with the protection of particular types of ecosystems and at the same time the first treaty ever to develop the concept that protection of the marine environment in general must be supplemented by the protection, in coastal areas or in the high seas, of unique ecosystems, critical habitats, habitats of endangered species and other areas of great ecological, biological or even aesthetic value, thus implementing Article 194.5 of the Law of the Sea Convention well before it has come into force. A preliminary meeting to discuss the feasibility of a protocol has taken place in Athens in October 1980 and it is expected that a diplomatic conference for the conclusion of that protocol will be convened in 1981. The 1981 Abidjan Convention relating to the coastal areas of Central and West Africa contains a provision based on Article 194.5 of the Draft Convention on the Law of the Sea which specifically calls for the establishment of marine protected areas. It is therefore highly probable that a protocol to that convention relating to that particular subject will be adopted at a later date. Similar protocols are also likely to be concluded in respect of other regional seas in the future.

Another significant conservation development which has emerged from the UNEP Regional Seas Programme is the new emphasis on coastal zone protection and management. This subject has now become an integral part of several of the proposed or already adopted regional seas Action Plans. The 1981 Abidjan Convention, in particular, contains an article requiring parties to prevent, reduce or control coastal erosion resulting from man's activities.

Finally, a provision on environmental assessments has, for the first time in a Regional Seas Convention, been included in the Abidjan Convention. This provision closely follows Article 206 of the Draft Law of the Sea Convention thus providing another example of a regional agreement implementing this treaty before it has even been signed.

Thus, as new Regional Seas Conventions are adopted, new important conservation provisions tend to be added. As a result, these conventions, as they gradually evolve, may well eventually consistute an ideal basis upon which many of the articles of the Law of the Sea Convention will be enforced.

c) The Advent of Coastal Zone Management and the Development of Marine Protected Areas

Coastal habitats are being increasingly destroyed, mostly in developed countries, but now also more and more in developing countries, by reclamation, industrial and urban development, tourism, the construction of ports, dredging or silting, pollution and other human activities. Awareness of the importance of these habitats as the locus of important life-supporting processes, spawning grounds or nurseries for many species of fish or other forms of marine life and permanent residence for many other species was, however, slow to develop. Scientific studies on coastal areas have been started, for the most part, fairly recently and the enactment of national legislation for the purpose of preserving from development at least certain fractions of the coastline and the taking into consideration of coastal habitat conservation in physical planning schemes only began a few years ago.

Likewise, the establishment of marine parks or other forms of protected areas to preserve important marine ecosystems and habitats was very late to develop as compared with the creation of parks and reserves on land.

New developments in this field are the result of international agreements, national legislation, and the action of international organizations.

i) International agreements

A recent bilateral treaty provides a unique example of the establishment of a marine protected area by agreement between two states. This is the treaty concluded in 1978 between Australia and Papua New Guinea on the subject of sovereignty and maritime boundaries. (18 International Legal Materials (1979) 291.) This agreement provides, *inter alia,* for the creation of a large "Protected Zone" in the Torres Strait which extends to the seabed and subsoil. The purpose of that "Protected Zone" is to protect the traditional way of life and livelihood of the traditional inhabitants of the area and to protect and preserve the marine environment and indigenous fauna and flora in and in the vicinity of the zone (Article 10).

ii) National legislation

(a) Coastal zone management

An increasing number of countries are now including fairly strict rules into their physical planning legislation for the purpose of limiting or controlling destructive human activities in coastal areas. These may include, for instance, the prohibition in certain areas of any construction within a certain distance from the sea, or of any other activity which may be harmful to the coastal environment. The Danish Conser-

vation of Nature Act is a good example of that type of legislation[329]. Other laws provide for possible acquisition by the Government of certain coastal areas to preserve them from harmful developments. In France, for instance, a government body, which is called the "Conservatoire du littoral"[330] is empowered to acquire, expropriate, or receive as donations or legacies, areas of coastline which thus become inalienable and can therefore be preserved in perpetuity.

In a small number of countries, special legislation has been enacted for the specific purpose of applying special management measures to the coastal area, thus recognizing the specific nature of coastal habitats and of the threats thereto. Of particular relevance in this respect is the legislation of the United States.

The Coastal Zone Management Act of the United States was adopted in 1972[331]. Its primary objective is to preserve, protect, develop and where possible to enhance, the resources of the Nation's coastal zone. It operates through a fairly intricate system of Federal Government-State interrelationship. States that remain competent over their coastline may receive federal grants if they develop coastal zone management plans consistent with the Act[332]. One of the main aspects of the Act is that it is not limited to mere planning but emphasizes control of land and water use by individual states. Another important feature of the Act is that it provides for the setting aside of estuaries areas for protection and research purposes. These areas have been given the name of estuarine sanctuaries. Although, in principle, the Act is also applicable to internal waters and to the territorial sea under state jurisdiction, it does not seem that individual states have yet, in general, availed themselves of this opportunity to manage coastal waters and, in particular, to preserve the critical habitats of coastal marine species.

(b) Marine protected areas

The importance of creating marine protected areas has only been recently recognized by conservation organizations[333]. Just like fish seemed to be inexhaustible, marine habitats were probably thought to be unalterable or at least capable of quick recovery. In view of the increasing rate of destruction of coastal habitats, initiatives were eventually taken to stress the seriousness of the problem and propose remedial action[334]. As a result of this new awareness the number of marine protected areas has now increased although it is still far from satisfactory both in numbers and types of ecosystems covered.

Marine parks or reserves are often governed, with regard to their establishment and management, by the same laws that are applicable to protected areas in general. A few states, however, perhaps for jurisdictional reasons, have adopted special legislation relating only to these parks or reserves. Important examples of this special legislation, which is everywhere very recent, are the Marine Reserves Act of New Zealand, of 1971[335], the United States Marine Protection Research and Sanctuaries Act of 1972[336] and for Australia, the Great Barrier Reef Marine Park Act of 1975[337] which is designed to preserve one of the richest and most spectacular ecosystems in the world.

The conclusion of regional agreements, under the UNEP Regional Seas Programme, for the establishment of marine protected areas will certainly give an added impetus to the enactment of special marine parks legislation in many countries.

iii) International organizations

When it began to appear in the 1970s that coastal habitats required much more attention, both from the research and management points of view, than it was thought in the past, it became clear at the same time that, in view of the structure of the United Nations system, there was indeed no single organization which could deal with the problem in its entirety. Furthermore, since it was not a matter which could be qualified as belonging exclusively to the fields of oceanography or fisheries, neither the IOC nor FAO, although necessarily interested by certain of its aspects, was either competent or probably willing to deal with it. On the other hand, UNEP, restricted as it is by its almost purely catalytic functions and its limited funds, could not undertake to sponsor major scientific research projects. In addition to its scientific aspects, the subject of coastal zone management has also important economic and land use planning implications that must be taken into consideration if comprehensive and effective programmes for the preservation and rational use of coastal habitats are to be successfully established and implemented.

(a) Scientific and management aspects

The only organization within the UN System which could deal with the problem of developing research in the subject of coastal ecosystems was UNESCO through its Man and Biosphere Programme[338], and indeed a number of activities were undertaken within the framework of MAB projects 5 and 8 with a view to the study and conservation of certain coastal areas[339]. In view of the magnitude of the task, however, and presumably also because of the specifically marine aspects of the subject, it was thought preferable to bring this new activity under the purview of the Division of Marine Sciences[340] which, as a result, benefited from substantial increases in its approved budget.

The new emphasis on coastal ecosystems research appears clearly from the past and proposed future activities of the Division. Studies and surveys have been conducted and meetings and seminars organized on such subjects as coastal lagoons and their productivity, human impact on mangroves, coral reef research and the study of particular ecosystems as, for instance, southern mediterranean lagoons, deltas and coastal marshes, and benthic ecology and sedimentation on the continental shelf in the Southern Atlantic[341]. The Division will also launch, in the near future, a major regional project on the subject of research and training for the integrated management of coastal ecosystems (COMAR). This project will consist in a series of pilot projects in tropical and sub-tropical areas. Pilot projects will be centered on research on coastal ecosystems for the purpose of evaluating their resources and managing them for development[342].

State of the art studies on particular types of ecosystems, offshore-inshore interactions, classification of fishes and other subjects will be published as a result of the research work undertaken[343]. No mention has been made for the time being, of the identification of critical habitats of commercial species, of ecosystems of unique ecological importance or of critical life-supporting phenomena on which the survival of marine life might depend. This information may, however, emerge subsequently from the results of research. In view of the magnitude of the task ahead, this research work will necessarily have to extend over a long period of time. When such information becomes available, co-ordination with the other competent divisions of

UNESCO and with the governments of the countries concerned should ensure that unique areas and important critical habitats be granted protected area status, possibly as biosphere reserves or world heritage sites.

(b) Economic and land-use aspects

The Council of Europe, by a resolution adopted by the Committee of Ministers in 1973[344], and OECD in a recommendation approved by its Council on 12th October 1976 have laid down the principles upon which the coastal protection, management and development policies of their member States should be based. These principles, the formulation of which was felt to be essential in view of the increased industrialization and urbanization of many coastal areas, emphasize the necessity for the governments concerned to control further developments in coastal areas through the use of zoning techniques and to protect coastal ecosystems, wildlife and natural processes.

These texts are of course not binding upon the member States of the two organizations in question. They seem nonetheless to have played an important role in the development of new national legislation relating to coastal areas[345]. They are addressed, however, to developed countries where coastal zones have been going through a process of urbanization for a long period of time and where further degradation of the last remnants of coastal natural areas had to be brought to an end. In developing countries with still large stretches of undeveloped coastline, it would seem that it is still time to combine preventive measures to ensure the preservation of essential ecosystems with rational economic development.

This matter was taken up the United Nations Oceans Economics and Technology Branch (OETB)[346], a branch of the United Nations Secretariat in New York whose overall goal is to provide assistance at the international level to introduce the marine dimension into development planning. The programme of OETB includes, therefore, research in the field of integrated coastal area development and assistance for coastal area management and development. Research was particularly devoted to the subject of the economic issues of coastal management as related to the general development planning process and to the legislative and institutional aspects of coastal area management[347].

At regional level, OETB has been co-operating with UNEP and the other UN agencies concerned in the development of the Regional Seas Programmes and in promoting coastal zone management, particularly in the West African[348] and Caribbean[349] areas.

In the Mediterranean, FAO's General Fisheries Council for the Mediterranean discussed the question of a coastal zone management policy at its Fifteenth Session in 1980. It recognized that the management of coastal areas implies a global process of reflexion, discussion and decision-making and recommended to its members to define national coastal strategies and the "drawing up of coastal occupation plans according to the types of utilization".

d) The Convention on the Conservation of Migratory Species of Wild Animals[350]

This Convention has its origin in Recommendation 32 of the Action Plan adopted by the Stockholm Conference in 1972 which invited Governments to give attention to

the need to conclude international conventions to protect species inhabiting international waters or those which migrate from one country to another.

The Convention was concluded in Bonn on 23 June 1979. By the end of 1980 it was not yet in force. The Migratory Species Convention is based on the principle that all the states whose territory or waters under their jurisdiction form a part of the range of a migratory species, at any time during the migration cycle of that species, as well as all the states that take part, when this occurs, in the harvesting of that species on the high seas should co-operate in the conservation and management of that species[351].

The migratory species to which the Convention applies are listed in an Annex (Appendix II). All parties that are range states[352] of any one of these species are encouraged to conclude an Agreement in respect of that species.

Guidelines are provided on the contents of such Agreements[353]. These guidelines place particular emphasis on the preservation of suitable habitats, the elimination or, if it is not possible, compensation for activities and obstacles which impede or hinder migration, the prevention, reduction or control of the release into the habitat of the migratory species covered by a particular Agreement of substances harmful to those species, co-ordinated conservation and management plans and, in respect of exploited species, whether for commercial or sport purposes, measures based on sound ecological principles to control and manage the taking of these species.

A certain number of marine species are listed in Appendix II and should therefore become the subject of management agreements between the range states concerned. Those species include all sea turtles, the beluga or white whale, the dugong and the Mediterranean monk seal. There are no marine fishes. Additional species may be added in the future when the Convention has come into force. The Convention deals in no way with jurisdictional matters and simply provides for a requirement, on the part of the range states of listed species to co-operate in the joint management of these species. This is in no way different from the provisions of the new Law of the Sea Convention which also lays down an obligation to co-operate among the States concerned in the case of shared stocks or when fishing for the same species on the high seas.

But whereas the Convention on the law of the sea is silent on the type of conservation and management measures to be taken, the Migratory Species Convention goes a step further and provides guidelines relating to the content of the specific agreements that will be concluded thereunder. In other words, the Bonn Convention takes over where the Law of the Sea Convention stops and provides for a content to some of the co-operation agreements required under that Convention. The Migratory Species Convention could, therefore, well be considered as an instrument giving effect, in respect of a certain number of migratory species, to the obligation to co-operate provided for in the articles of the Law of the Sea Convention that relate to fisheries in the EEZ and in the high seas. It could be of particular importance as a means to implement on a sound ecological basis the provision of Articles 65 and 120 on marine mammals, of Article 64 on highly migratory species and of Article 66 on anadromous species.

F. General Conclusions

Many of the provisions contained in the new Convention on the Law of the Sea are self-executing and therefore require no future action. Such is the case, for instance, of the clauses dealing with jurisdictional matters, the establishment, functions and procedural rules of the Sea-bed Authority, and dispute settlement procedures. On the other hand, many other articles require, if their purpose is to be achieved, that active steps be taken to implement them. If this were not possible, or suffered too long delays, some of the most important environmental provisions in the convention could well remain empty shells. Laws and regulations must, therefore, be adopted, regional treaties concluded, institutional arrangements made and international co-operation promoted.

Important as they may be, however, the provisions of the new Convention constitute but a part, albeit a large part, of the recent developments that have taken place in the field of the environmental Law of the Sea. Other recent treaties such as the Migratory Species Convention and the Convention on the Conservation of Antarctic Marine Living Resources, earlier fishery treaties that have been renegotiated in the light of the new Law of the Sea, some of the recommendations of the Stockholm Conference as well as new or expanding activities of certain international organizations, especially UN agencies, are all contributing to the development of a new political and economic order of the oceans and therefore form also part of the basis for future conservation action.

Last but not least, full account must also be taken of the objectives set out by the World Conservation Strategy and, in view of the immensity of the conservation task ahead, priorities assigned to meet the most significant and most urgent requirements to prevent further irreversible damage to living resources and their habitat[354].

The major objectives of the World Conservation Strategy, it will be remembered, are the maintenance of essential ecological processes and life-support systems, the preservation of genetic diversity and sustainable utilization of species and ecosystems. To achieve these objectives an integrated approach is indispensable. Measures required include further research in oceanic processes, ecosystems, habitats and species, increased knowledge of population dynamics and biology of harvested species, better understanding of present and potential threats to natural systems and species and improved statistics on harvested species including those that are incidentally taken. Among other essential conservation requirements are new or improved national laws and regulations, better coverage of the world ocean and exploited stocks by international fishing agreements and commissions established thereunder, adequate regulatory authority vested in fishery commissions and proper enforcement of regulations by commissions or by coastal States. Last but not least, there is a need to protect marine habitats against all forms of degradation, to manage coastal areas in a rational way, to set up protected areas to preserve valuable ecosystems and species both in the high seas and in waters under national jurisdiction, to control the introduction of alien species, to prohibit or control strictly the release into the environment of certain substances which are particularly harmful to marine life and to protect particularly sensitive areas, ecosystems or species from accidental pollution.

This list is certainly far from being exhaustive. It is sufficient, however, to give an idea of the task that lies ahead if the general obligations to protect and preserve the marine environment which appear in the new Convention on the Law of the Sea are to be implemented. The difficulty of this task must, however, not be underestimated. Action limited to certain areas or to certain species will often prove to be ineffective if not accompanied by other measures to preserve biological processes or related species, the maintenance of which is indispensable if the conservation action proposed is to succeed. To quote Dr. Carleton Ray — ... "unless conservation is ecosystem-wide, it is doomed, eventually, to failure" — marine eco-systems are immensely larger, more complex, and less well known than terrestrial ones. "In the sea, systematic conservation is fraught with difficulties, not the least of which concerns a matter of scale"[355]. Inter-relations and inter-actions between phenomena, areas, species and human activities, sometimes over considerable distances, also occur frequently through ocean currents or the movements of animals. Thus, pollution in the open sea can contaminate distant shores, toxic substances ingested by marine animals become lethal to other species higher in the food chain, and the destruction of coastal marshes or mangroves adversely affect high seas fisheries.

It would seem therefore that there is an urgent need to develop, within the framework of the World Conservation Strategy, a global marine strategy providing general guidelines on the whole range of conservation measures that have to be taken to preserve the ocean ecosystem in its entirety, and would serve as a basis for the development of regional and national strategies. The following conclusions may be considered as a contribution to this undertaking.

1. Institutional Arrangements

Present institutional arrangements relating to the study, exploitation and preservation of the marine environment and its living resources are clearly unsatisfactory.

At world level, within the United Nations system, marine matters, often closely related to each other, are dealt with by several different agencies. There is little co-ordination and no real possibility of identifying gaps in the subjects covered or assigning new tasks to a particular organization. At regional level, a large number of fishery commissions which have been established outside the UN system have little means of benefiting from scientific and technical inputs and assistance from UN agencies, and have to rely almost entirely on the results of their own research or of research carried out by their party States. FAO regional fishery commissions or councils, IOC regional programmes and UNEP regional seas programmes are not co-ordinated by regional institutions. Finally, at national level, marine matters are often, here again, distributed among a number of different government departments and co-ordination is probably again difficult to achieve. To make things worse, the high tide mark is almost everywhere a line of demarcation of jurisdiction between departments competent over marine matters and those which deal with the land, thus making comprehensive coastal zone management plans difficult to prepare, adopt and implement.

It would seem therefore that there is a pressing need to create an intergovernmental body whose primary tasks would be to develop and implement a world marine conservation strategy, identify priorities, assign new tasks to existing agencies and in a general way cover all general aspects of oceanic research and liv-

ing resources study, exploitation, management and conservation, and thus implement the relevant recommendations of the Stockholm Conference and the conservation provisions of the new Convention on the Law of the Sea.

In view of the fact that there does not seem to be any possibility of establishing a new agency that would be responsible for all marine questions, this intergovernmental body need not be any more than a co-ordination structure. But it should, at the same time, have enough authority to orientate the action of the various agencies concerned towards achievement of the strategic objectives that it would have set up.

At regional level, particular attention could be paid to the institution of regional centres as provided for by Article 276 of the new convention. It would seem that these centres should neither be completely independent, lest they might lose the benefit of the assistance that could be provided to them by the UN agencies concerned, nor should they come under any one of the individual agencies lest they become focused on certain aspects only of the marine environment and might, therefore, tend to overlook the others. The functions of these centres should not be limited, as provided for in Article 277 of the Convention, to the acquisition, processing and dissemination of scientific and technological information, the carrying out of management studies, the study of programmes related to the protection and preservation of the marine environment and the organization of training and educational programmes, but their terms of reference should be broad enough to enable them to develop regional marine conservation strategies, to assist and advise governments and national marine scientific and technological centres in the development and implementation of national marine conservation strategies, to advise, and, even possibly co-ordinate, the activities of regional fishery bodies and other organizations concerned with marine matters in the region and to ensure liaison with other regional centres and with the inter-governmental body referred to above.

At national level, the national centres provided for in Article 275 of the Convention should be able to develop national marine conservation strategies as a part of the general national conservation strategy of each country.

The methods which could be used to establish these bodies and their relationships with other organizations need, however, to be elaborated a little further.

a) Inter-Governmental Co-ordinating Body

Since it does not seem possible to establish a new specialized agency, there would be no need to formalize the creation of this new body by way of a treaty. A resolution of the General Assembly would, therefore, probably be sufficient. The intergovernmental co-ordinating body should, above all, be considered as the body entrusted with the task of bringing into effect all the provisions of the new Convention that concern the conservation and management of living resources and the protection and preservation of the marine environment. Each individual agency within the UN system would naturally continue to deal with the subjects it was concerned with previously. The only difference, but, it is felt, an important difference with the present situation, would be that this new body, on the basis of the strategy that it will develop, will be able to determine priorities, identify gaps, allocate new tasks and in a general way, co-ordinate all relevant activities of the agencies concerned. It could

usefully benefit from the advice of ACMRR which could become its standing scientific advisory body as well as of SCOR and other competent organizations.

b) Regional Arrangements

Arrangements at regional level should relate as much as possible to areas constituting ecological units as it has been done, for instance, in the UNEP regional seas programme. Regional centres could, in addition to the development of regional conservation strategies, co-ordinate the activities of all UN bodies concerned with marine resources and the protection and preservation of the marine environment in their area of competence and, furthermore, entertain close relationships with bodies outside the UN system, such as fishery commissions. It would thus become easier to co-ordinate within the same ecological unit, such matters as the protection of coastal habitats, the regulation of fisheries, the preservation of sensitive areas from pollution and the development of national marine conservation strategies.

Priority areas for the establishment of regional centres could be certain semi-enclosed seas such as the Mediterranean or the Caribbean, or coastal areas of high biological productivity as, for instance, upwelling areas; zones with a high degree of species diversity, especially coral reefs should receive particular attention.

Article 276 of the Convention does not provide for any specific method for the establishment of regional centres. These centres are merely to be established by states in co-operation with the competent international organizations. They should, as a result, be small international organizations themselves. They could, it would seem, be established by means of a treaty concluded between all the states of the region concerned. When there exists already a treaty dealing with the conservation of a particular sea, and such would be the case of the treaties concluded under the UNEP regional seas programme, regional centres could be set up by means of protocols to these treaties. Alternatively, although this would not necessarily preclude the conclusion of a treaty, regional centres could be set up as joint regional organizations by the main UN agencies concerned, i.e. UNESCO, IOC, FAO, UNEP and possibly IMCO. The inter-governmental co-ordination body would provide general policy guidance to these centres, either directly, which would appear to be preferable, or through the governing bodies of its parent organizations. Functional links would probably have to be established, under the treaty, both vertically with the inter-governmental co-ordinating body, and horizontally with regional organizations or subsidiary bodies of UN agencies dealing with the marine environment or marine living resources in the area in question. Appropriate co-ordinating mechanisms would also have to be set up. Regional centres should naturally help in the creation of national centres, co-ordinate the activities of these centres and provide them with the assistance they require.

c) National centres

As much as possible, national centres should be established as inter-departmental bodies with powers to review the entire range of marine-linked activities and to co-ordinate the action of the various departments concerned. Particular attention should be paid to the necessity of giving them jurisdiction over coastal areas, including land areas which interact with the sea, estuaries, coastal marshes, mangroves, lagoons and other in-shore waters. They should also have the possibili-

ty to advise on problems of land-based pollution even when the source of this pollution is distant from the coast. National centres shall, naturally, be established by the states themselves. Article 275 of the new convention provides, however, that they can be set up through competent international organizations. Regional centres could, in that respect, play an active part in providing guidelines for the creation of national centres, the drawing up of their terms of reference and the methods they could use to co-ordinate the action of the various departments concerned. They could, once these centres are established, also provide advice as to the type of research they should undertake as a matter of priority, as well as on the methods to be used to develop and monitor the implementation of national marine conservation strategies.

d) Conclusion

In several parts of the world, regional organizations or programmes established or developed by different UN agencies are operating concurrently. Examples are the Caribbean and the Mediterranean seas. In the Caribbean, IOC's IOCARIBE and FAO's Western Central Atlantic Fishery Commission (WECAFC) are already in existence, and a Caribbean UNEP Action Plan is now under development. In addition, the United Nations Development Programme (UNDP) has an important programme for the development of fisheries in the area. In the Mediterranean, UNEP's Mediterranean Action Plan, FAO's General Fisheries Council for the Mediterranean (GFCM) and IOC's co-operative investigations in the Mediterranean (CIM) form already, if taken together, the basis for a comprehensive conservation and management programme. Similar situations exist or will develop in other parts of the world. Marine regional centres where all such activities could be regrouped and co-ordinated could become powerful tools for the national development of fisheries and the preservation of the marine environment at regional level. Co-ordination with organizations which do not belong to the UN system is bound, however, to be more difficult. It can indeed be stated that, so far, such a co-ordination has been almost non-existent. It would, however, seem important that efforts be made, within the UN system, to develop such co-ordination both at regional level and at world level. In the latter case, co-ordination should take place through the inter-governmental body described above and extend not only to general matters but also to more specific, albeit important questions such as high seas fisheries, habitat preservation, protection of unique ecosystems and endangered species, pollution control and the introduction of alien species.

2. Fisheries

a) Obligation of Coastal States in Areas under their Jurisdiction

Under Article 61 of the new Law of the Sea Convention coastal states must ensure that the maintenance of living resources in their EEZ is not endangered by over-exploitation, that populations of harvested species are maintained or restored at levels that can produce MSY and that populations of harvested and associated species are maintained above levels at which their reproduction may become seriously threatened.

These obligations apply, in principle, only to the EEZ. It must be noted, however, that over-exploitation of species or the absence of measures to protect associated or dependent species in other areas under national jurisdiction, such as the ter-

ritorial sea, will in most cases, because stocks are seldom confined to the territorial sea, affect population levels in the EEZ as well. As a consequence, if coastal states are to comply with their obligations under the treaty, they will have to control exploitation in all waters under their jurisdiction, and consider any stock in these waters as a unit, irrespective of jurisdictional areas, and manage any such stock accordingly.

To do that, coastal states will require scientific and statistical data and, in the case of many developing countries, technical assistance. This can probably best be provided through regional research centres and fisheries organizations. States will also require adequate legislation and enforcement measures. National fisheries legislation guidelines, founded on the best scientific knowledge, ecological principles and sound economic considerations could, to that effect, be drawn up and made available to coastal states and regional centres for adaptation to local situations.

There should also be a system to warn coastal states that they are running into a situation of over-exploitation in respect of one or several stocks. Article 61.4 requires that coastal states, and states authorized to fish in the EEZs, submit catch and fishing efforts statistics through competent international organizations. Mechanisms should, as a consequence, be instituted to warn all States concerned that over-exploitation is taking place in violation of the Convention and that remedial action should be taken.

Another important factor to be determined by coastal states is the MSY for stocks under their jurisdiction. However, MSY, as qualified by Article 61.3, is no longer a purely scientific factor. Since sub-regional, regional or global minimum standards are to be taken into consideration for its determination, it will be necessary to develop these standards which, for the time being, do not exist. Overall general standards may presumably be established by a world organization, i.e. FAO, and regional standards by regional centres in close co-operation with regional or sub-regional fishery bodies. Socio-economic aspects should likewise at least be the subject of guidelines developed at world level and refined, to adapt them to local fishing patterns and needs of coastal communities, at regional level. This could also be performed by regional centres and regional fishery organizations. The same is true of technical guidelines such as those related to the interdependence of stocks.

There is, however, an important aspect of the problem which should not be overlooked, namely that, although the determination of MSY is governed by a number of considerations, many of which are not related to fish biology and population dynamics, whatever adjustment to the scientific MSY is made should result in a decrease of the allowable catch and not an increase, since catches exceeding MSY would inevitably result in over-exploitation, thus violating the clear obligation of Article 61.2.

Under Article 62, coastal states must determine their capacity to harvest the living resources of their EEZ and allow other stocks to catch the surplus. Admittedly, paragraph 4 of this Article makes it an obligation on the part of these other states to comply fully with the conservation regulations adopted by the coastal state. This,

however, gives rise to a problem of enforcement. Enforcement will be particularly difficult in the case of large foreign fishing fleets authorized to fish in the EEZ under bilateral agreements or joint venture arrangements. Such fleets will naturally tend to reap maximum profits from their operations and may tend to disregard conservation regulations. Agreements or contracts with foreign states authorizing them to fish in the EEZ should therefore provide for procedures to control their operations and penalties to be applied in cases of violation. Model contracts, rules and procedures should, therefore, perhaps be developed and proposed as guidelines to the coastal states concerned.

b) Stocks Shared Between Neighbouring EEZs

The problem is dealt with by Article 63.1. In view of the large number of stocks belonging to that category, it is inevitable that a considerable number of bilateral or multilateral agreements will have to be concluded. Such agreements will have to provide for joint management measures and include provisions on associated and dependent species pursuant to Article 61.4. As much as possible, it would seem that such agreements should be concluded and administered under the umbrella of appropriate regional or sub-regional organizations as provided for by Article 63.1. Guidelines for the conclusion of these agreements would certainly be useful. The Baltic Convention could, in that respect, provide a valuable model.

c) Stocks Shared Between the EEZ and an Adjacent Area of the High Sea

This situation is governed by Article 63.2. Existing regional agreements have already been, for the most part at least, amended to conform with the new convention. In any event, the matter of associated and dependent species would have to be included in any revised or new agreement. Article 63.1 lays emphasis on the fact that conservation and management measures taken in respect of stocks shared between one or several EEZs and the adjacent high seas should only apply to that part of the range of these stocks that extend to the high seas. From the point of view of resource management, however, this cannot work and stocks should be managed as units, irrespective of jurisdictional zones. The revised ICNAF and NEAFC treaties have tried to resolve the problem by requiring that their respective fisheries commissions seek to "ensure consistency" between measures they may take concerning the high seas and measures taken by coastal states in areas under their jurisdiction and that coastal states and commissions co-ordinate their action in this respect. The new NEAFC convention provides, in addition, that its commission may make recommendations concerning fishing conducted in an area under the jurisdiction of a contracting party, provided the contracting party so requests and the recommendation receives its affirmative vote. It provides furthermore that the commission may give advice concerning such fisheries if the contracting party so requests. It is perfectly understandable that recommendations relating to conservation in areas under the jurisdiction of a state should be accepted by that state before they come into force. Indeed, if they were not, there would be no way to enforce them! But why should the commission have to wait until a coastal state has made a request before submitting a recommendation or even giving advice, is difficult to comprehend. Surely a mere advice on conservation measures required cannot be taken in a technical body as infringing upon state sovereignty. Moreover, conservation measures taken in waters under national jurisdiction may be essential, in the case of shared stocks, to the sustained productivity of these stocks on

the high seas. Mismanagement by the coastal state would then hurt the high seas fishery and, as a result, the interests of other states. Admittedly, these states could, if this happened, submit the matter to the conciliation procedure specified in Annex V of the Convention, but preventive measures would, nonetheless, be much preferable. For all these reasons, procedures for joint management of shared stocks which, whilst respecting the sovereign rights of states over the resources under their jurisdiction, would at the same time ensure adequate conservation of these resources and compliance with the obligations of the same states under the Convention, should be the subject of active legal research and be developed as early as possible. The Migratory Species Convention could, perhaps in this respect, be taken as a model. Finally, it is obvious that the determination of the total allowable catch for a shared stock and the allocation of this catch between the various states concerned, can only be done by agreement between these states. Although, in principle, coastal states are free to establish at their discretion the allowable catch of that part of the shared stock which is found in waters under their jurisdiction, in practice this will not be possible if serious disputes are to be avoided. Here again, practical procedures will have to be developed by fishery bodies to allow for a harmonious exploitation of these resources.

d) Highly Migratory Species

Much of what has been said in the previous paragraphs applies, mutatis mutandis, to highly migratory species. Article 64, however, provides, in respect of those species, for conservation measures to be taken by all states concerned throughout the region where a species occurs and furthermore lays down the obligation to establish international organizations for the management of these species where no such organization exists. Recommendations by the competent organization would therefore apply, in this case, to the EEZs of the states concerned, and the difficulty referred to in the preceding paragraph would disappear in that respect. Recommendation would not, however, be applicable to the territorial sea and internal waters. It is well known, however, that certain highly migratory species, such as the red tuna in the Mediterranean, for instance, pass quite close to the coast during their migration. Here again, therefore, provisions should be made to manage these stocks as single units and procedures will have to be developed by the international organizations concerned to recommend appropriate management measures and to determine the allowable catch in a way acceptable to the coastal states concerned. Regarding international organizations dealing with highly migratory species, it is, here again, essential that their area of competence be broad enough to encompass the whole range of the species or stocks they purport to cover. This would not preclude the establishment of sub-regional management groups within these organizations if it is found more practical. But for far-ranging species, as for instance tuna, global management policies should certainly be established for very large areas such as for instance the Indo-Pacific region as a whole. There even could be situations where a world-wide agreement would be preferable, as it has been the case for whales.

Finally, some form of a procedure will be needed to amend, as required, the list of highly migratory species in Annex I to the Convention. For that purpose the simplified revision procedure provided for by Article 313 would seem adequate. It would have, however, to be preceded by discussions and a preliminary agreement

on the advisability of adding new species to the list. This could perhaps best be performed by FAO, by means of a recommendation of COFI and with the advice of AC-MRR.

e) Marine Mammals

Here again, management measures should apply to the whole range of the species, irrespective of jurisdictional zones. At the present time, the International Whaling Convention is applicable to all waters where whaling is carried out. It is essential that this provision be retained in any possible renegotiation of that convention since it is in no way incompatible with the new Law of the Sea Convention which does not prohibit states from entering into specific agreements relating to the protection of living resources in waters under their jurisdiction, whether internal waters, the territoral sea or the EEZ, any more than general international law prevents states from concluding agreements for the protection of animal or plant species on land.

Regarding marine mammals which breed on land, like most species of seals, their conservation and management would be probably facilitated if they could enjoy the same status as anadromous fishes, to which, from the purely legal point of view, they are very similar.

f) Anadromous and Catadromous Species

The definition of these species certainly need clarification. An extended definition could cover all marine species breeding on land or exclusively in waters under the jurisdiction of coastal states[356]. The Convention, however, in Article 66 gives a restricted definition using only rivers as a criterion of origin for anadromous species. Thus, estuaries would probably be covered but not coastal lagoons. If it appears impossible to add a new appendix, the matter will have to be dealt with on a regional basis by inter-governmental fishing bodies.

The provisions of the Convention that concern these species underline the special interest of the states of origin or of destination and provide for the conclusion of agreements between states of origin, states authorized to fish for these resources in the high seas and states into or through the waters of which these fish migrate during their migratory cycle. The Convention, therefore, provides for management of individual stocks as a whole by all the states concerned. To implement these provisions, the Migratory Species Convention would seem to be an appropriate instrument.

g) High Seas

The cases where living resources exploited on the high seas are not at the same time either highly migratory species or stocks shared with an adjacent EEZ are bound to be very few, at least until new and unconventional resources begin to be harvested. Comments and recommendations made above regarding the determinations of the MSY and hence of the allowable catch are also applicable, mutatis mutandis, to the high seas. Prevention of over-exploitation in the high seas is not an obligation under the new Law of the Sea Convention. Since, however, the conservation of the living resources concerned is mentioned as a duty in Articles 117, 118, 119 and 120, it follows almost automatically that over-exploitation is deemed to be, at least, reprehensible.

As in other cases, agreements which might be concluded in the future should cover the whole range of the harvested stocks or, if this is not possible, at least provide for working agreements and co-ordination between commissions established under conventions covering adjacent areas.

h) Incidental Take

In view of the seriousness of the problem, general policy rules should be laid down at world level, preferably by the inter-governmental co-ordinating body mentioned above, with the advice of FAO to provide the basis of a solution to the vast variety of specific problems that, no doubt, occur in various regions of the world. These should include the setting up of monitoring mechanisms to provide basic data on what is actually happening and of research programmes to develop gear allowing for the escape of non-target species. Regional centres would seem to be well suited to assist countries in a particular region to identify problems and find appropriate solutions in areas under national jurisdiction. Fisheries agreements should be modified by the inclusion of a clause on incidental fishing requiring Contracting States to collect and provide statistics on species incidentally caught and to take remedial action as recommended by the commissions set up under these agreements. Although perhaps not comprehensive enough to cover all types of incidental take, the provisions in Article 61, 63 and 119 relating to associated and dependent species will in many cases provide a sufficiently sound legal basis for such action.

i) Fisheries Research

In the field of fisheries, research is almost always carried out by national authorities and institutions and seldom by the staff of the fishery commissions themselves. Many commissions, however, have established a scientific council or committee where the results of national research are analysed. Scientific committees have been of considerable use to fishery bodies in assisting them in taking management decisions. There is, it would seem, however, a requirement for independent research to provide, in certain cases at least, more neutral information to the commissions. This can be done by setting up a research department within the commission secretariat or by calling upon an external organization. If this latter solution, desirable as it may be in certain cases, appears nonetheless to be impractical because, for instance, of its cost, research could perhaps be carried out for a whole region, covering thereby several fisheries organizations, by the same specialized research agency, as it is the case in the North Atlantic with ICES.

Although it depends entirely on its member governments for the actual research work carried out, ICES, as an international organization which analyses the information it receives and prepares scientific reports, is considered more as a forum where scientific evidence is being discussed and established than as an expression of the political positions of its member governments.

The unchallenged success of ICES should, hopefully, lead to the establishment of similar organizations in other parts of the world, preferably in combination with some of the Regional Centres provided for by the Convention.

3. Protection of Species

The new Law of the Sea Convention is silent on the subject of the preservation of species other than those which are affected by fishing. Protection of species,

however, forms an integral part of the protection of the marine environment in general and the safeguarding of species is one of the basic requirements listed in Principle 2 of the Stockholm Declaration. Furthermore, recommendations 39 and 40 of the Stockholm Conference recommend that an international programme to preserve the world's genetic resources be launched and that inventories be made of endangered species, special attention being given to locating areas of natural genetic diversity that are disappearing. It seems, however, that since the Stockholm Conference little has been done to apply this principle and these recommendations to marine species.

Inventories of species with limited distribution and of centres of genetic diversity could perhaps first be carried out within the framework of the UNEP Regional Seas Programme and wherever special measures are required to protect particular species, appropriate protocols to the regional seas conventions could be concluded for the purpose of their protection. Close collaboration between UNEP and IUCN, especially its Species Survival Commission, will naturally be required. Alternatively, or where regional seas programmes have not been established, species conservation measures could be developed and implemented through Regional Centres. Among the species that could benefit from such measures as a matter of priority, are many coastal or coral reef species, in particular species living around oceanic islands or confined to semi-enclosed seas. Migratory species such as for instance sea-turtles or marine mammals could be the subject of agreements under the Migratory Species Convention. Finally, there should be some form of monitoring of the trade in certain species such as, for instance, aquarium species or certain sea-shells for which there seems to be an increasing demand, and wherever it appears that international trade is an important factor in the unsatisfactory conservation status of a species, that species should be listed on one of the CITES appendices.

Among the conservation measures that could be taken for the preservation of endangered species is the creation of sanctuaries where not only would the taking of such species be prohibited but where other activities which could harm these species could be prohibited or controlled. A recent example of a sanctuary for the preservation of certain species is the one recently established in the Indian Ocean by the International Whaling Commission. Although, for the time being, only the killing of whales is forbidden in that area, other international conservation measures could be adopted in the future, possibly by other organizations competent in the area or under other agreements relating to the area. In certain cases, the conclusion of treaties, either independently, or in the form of protocols to existing treaties such as the Regional Seas Conventions, could provide for the necessary machinery to set up a management authority for the sanctuary area and would have, furthermore, a considerable and very positive impact on public opinion[357].

4. Protection of Habitats

A distinction must first be made here between the habitats of species which are the subject of commercial exploitation, and these of other marine species. Preservation of the habitats of commercial species is essential to the sustained exploitation of these species. Maximum production will, therefore, depend, for commercially exploited fishes and other marine organisms, on the maintenance of suitable habitats. For non-commercial species, where maximum production is not the objec-

tive sought for, priority should be given to the preservation of endangered, rare or endemic species, unique or scientifically important communities and representative samples of more common types of ecosystems. In most cases, specially protected areas will have to be established.

a) Commercial Species

The connection between fishery productivity and the safeguarding of critical habitats such as spawning grounds or nursery areas is obvious. Yet it seems to start barely to be perceived. For many species, critical habitats are still to be identified and few protective measures have ever been taken. No population dynamic model seems to have taken into consideration the eventuality of reduced recruitment due to critical habitat degradation or alteration and it does not seem to be known whether habitat destruction has contributed to the collapse of sea fisheries as has occurred in many fresh water fisheries, such as the salmon fishery in many European rivers[358].

Further research, identification of critical habitats and assessment of the effects habitat destruction on the recruitment of commercial species is certainly required. For those species which fall under the regulatory authority of an inter-governmental fishery body, it would seem that this type of research work and the habitat preservation measures that would possibly ensue should be linked with the regulatory measures taken in respect of harvesting within a global management programme. In other words, the habitat preservation factor should be added to the subjects under review by fishery commissions and these commissions empowered to make recommendations to their member countries in respect of the protection of the habitats of the species over which they have regulatory authority.

When there is no competent commission, or in respect of purely coastal species that never leave waters under national jurisdicton, appropriate legislation will have to be developed by national authorities with guidelines from Regional Centres.

Coastal habitats which are important for the production of fish and other commercial species will require special consideration as already laid down by Recommendation 48 of the Stockholm Conference. Their preservation requires the adoption and adequate enforcement of strict regulations under existing physical planning laws, or under new legislation to be developed for the specific purpose of coastal zone management.

Wherever Regional Seas Programmes have been launched by UNEP, the preservation of habitats essential to the maintenance of sustainable fisheries should become an integral part of such programmes. Protocols to regional seas conventions dealing with the subject of protected areas could provide a useful tool for that purpose. Special provisions relating to critical habitats of commercial species would, if inserted in these protocols, which would normally cover all types of marine protected areas, point out the particular importance of certain habitats for commercial fisheries.

Finally, at world as well as at regional level, and especially in areas covered by Regional Seas Programmes, the UN agencies concerned, i.e. UNESCO, FAO and UNEP, should co-operate closely in the development of integrated conservation pro-

grammes, provide technical assistance to countries participating in these programmes and work in close association with Regional Centres.

b) Other Species

Marine fauna and flora should be considered as one of the components of the marine environment. As a result, protection of the marine environment necessarily means also the preservation of the animals and plants of which it is the habitat. Article 192 of the Law of the Sea Convention lays down the obligation for states to protect and preserve the marine environment. Article 194.5 more specifically provides for the preservation of rare or fragile ecosystems as well as the habitat of depleted, threatened or endangered species and other forms of marine life. In addition, the conservation of representative samples of natural ecosystems is particularly called for by Principle 2 of the Stockholm Declaration, and Recommendation 38 of the Stockholm Action Plan recommends to governments to set aside ecosystems of international significance for protection under international agreement. It goes without saying that this can only be achieved progressively and that, therefore, plans of action will have to be established and priorities determined. This task should be started as soon as possible with the assistance of conservation organizations, primarily IUCN, and inventories made and threats identified so that remedial measures may be proposed. The establishment of priorities is here absolutely essential as it will not be possible in the first stages of the development of a programme to protect all important sites. — A mechanism should be established for this purpose with IUCN participation and governmental support, to review inventories, determine priorities, make specific recommendation to Governments with regard to protection measures required, find sources of technical and financial assistance, assist in the conclusion of any treaties that would seem to be necessary and allocate the task of actual assistance in the field to competent bodies such as the Regional Centres or institutions set up under regional conservation treaties. Protocols under Regional Seas conventions will also constitute a useful tool for that purpose.

c) Marine Protected Areas

The regime and mode of establishment of marine protected areas will vary according to jurisdictional zones.

On the high seas, protected areas can only be established by treaty. If and when important areas in the high seas are identified as requiring protected areas status, it will be essential at the same time to determine possible threats and therefore activities which should be prohibited. These may include all forms of exploitation, pollution, dumping, filling, dredging and even navigation if it involves disturbance of certain important species or the risk of accidents resulting in severe pollution by harmful substances.

On the seabed and ocean floor beyond the limits of national jurisdicton, the Authority would undoubtedly be competent to set up protected areas and could therefore adopt special rules for their establishment, preservation and management. Such areas should comprise unique ecosystems, such as deep ocean trenches and representative samples of benthic fauna habitat some of which could be used as controls when monitoring the effects of deep sea mining in other areas.

In the areas under national jurisdiction, protected areas can be established by decision of the coastal states concerned. However, if navigation is to be prohibited or controlled in certain areas, the "Special Areas" procedure provided for under Article 211.6 at the Law of the Sea Convention will have to be followed. It might be advisable, in this respect, to develop within IMCO a set of specific guidelines relating to special areas established to protect particular ecosystems or habitats. These could include the possiblity of prohibiting or regulating navigation at certain times only, for instance to protect a migratory species which is present in the area only at a certain period of its migratory cycle, or to prohibit at all times, or at certain times only, the passage of ships carrying particularly dangerous cargoes. This procedure could include a system to notify ships that an area will be closed as from a certain date and for a certain period[359].

There exists a variety of conservation treaties which could be used for the purpose of establishing protected areas in zones under national jurisdiction. Protocols to the UNEP Regional Seas Convention seem to be particularly well adapted to this end and could cover, therefore, both critical habitats of commercial species and other ecosystems and species habitats. The possibility of using other existing treaties would have to be investigated. The Ramsar Convention would probably be well suited for certain coastal habitats as would also be the network of biogenetic reserves of the Council of Europe. The African and Western Hemisphere conventions may, however, need to be strengthened. The World Heritage Convention would be especially well adapted to the conservation of unique ecosystems. But new arrangements could also the concluded, preferably under the auspices of Regional Centres where there is no perspective of a Regional Seas Convention in the foreseeable future. By and large it would seem that although the idea of using existing conventions and networks is appealing, the specific nature of marine ecosystems would normally require special protective rules, including in some cases, perhaps, restrictions to navigation and would, therefore, lend itself better to the conclusion of separate and more specialized treaties.

5. Pollution

Pollution will be considered here only to the extent to which it may affect species and ecosystems.

There seems to be no doubt that certain chemical substances such as DDT and its metabolites and PCBs are gradually poisoning the marine environment because of their toxicity, persistance and concentration at higher trophic levels. Although the situation has somewhat improved during the past decade because of prohibitions or restrictions in the use of these products in several countries these measures were not sufficient to reverse the situation.

These products mostly penetrate the marine environment by the rivers or the atmosphere. All treaties concluded so far on land-based pollution have banned them. But as long as a world ban has not been agreed upon and been effectively enforced, the poisoning of the ocean will go on, as producing countries continue to export these products to countries where no restricting law is in force and will not agree to bring this trade to an end lest some of the consumer countries establish their own manufacturing units. Thus only a world treaty could succeed in achieving a ban.

Since these substances can be considered as one of the major threats to the marine environment as a whole, a world treaty banning their manufacturing, trade and consumption and requiring the destruction of existing stocks would go a long way in implementing Article 192 of the Convention and several other articles in Part XII. Some exceptions will, naturally, have to be accepted to provide for the very limited uses, involving small quantities of these substances, or their use in closed environments, for which no substitutes have yet been found. But these exceptions should be temporary and disappear as soon as replacement products have been developed and marketed at acceptable costs.

Another very serious danger to the marine environment are accidents at sea involving ships carrying particularly dangerous cargoes such as certain chemical substances, nuclear material or highly radioactive waste. Highly toxic substances dispersed into the sea after such an accident could indeed have far more catastrophic effects than the wreckages of the Torrey Canyon or the Amoco Cadiz. It is reported in the French daily, "Le Figaro", dated 11 February 1981, that, according to the Lloyd's register, 279 merchant ships were sunk in 1980. So far it would seem that we have been fortunate and that no particularly toxic substances were released into the sea. That could well, however, happen any day. Barrels containing a very dangerous chemical were salvaged from a ship sunk off the south-eastern coast of Italy a few years ago before they could leak into the sea. In early 1981, according again to "Le Figaro", six containers, each one of them filled with nine tons of a concentrated insecticide, broke loose from a Danish ship off Ushant. Fortunately again the French navy were able to recover them intact from a depth of 110 meters[360]. The SOLAS Convention provides for regulations on the stowing of dangerous cargoes[361]. But apparently this was not enough. The damage that could be done to the ocean fauna and flora in cases such as this one staggers the imagination.

There is, therefore, a case for strengthening regulations concerning the transport of particularly toxic substances and such regulations should extend to such matters as the sea worthiness of the ships concerned, the qualification of crews and navigation rules and procedures, including the use of pilots and navigational aids in certain areas. But if certain particularly important ecosystems of the ocean are to be preserved, there should be a ban or at least very strict controls on the passage of ships carrying particularly dangerous cargoes in certain areas, especially protected areas and areas adjacent to protected areas where, if the ship should sink, currents could carry the toxic substances in a sufficiently concentrated form to cause serious damage to the area under protection. Among areas which should benefit from this special protection are deep ocean trenches where unique species have evolved separately from the rest of the world[362], other areas of genetic diversity such as coral reefs areas or areas protected because they are the habitat of rare, endangered or endemic species.

With regard to oil, although the economic damage caused by a major accident is often enormous, the ecological damage, provided no toxic dispersants are used subsequently, is generally not as serious as was thought previously. Sea birds, however, may die in large numbers and other animals as well as coastal habitats may also be affected. At certain places, e.g. in the vicinity of large colonies of sea

birds or seal rookeries, and at certain times of the year, i.e. when the animals are present, a major oil spill could therefore have incalculable effects. This could be particularly true in the Arctic and Antarctic environments. The new Convention on the Law of the Sea provides for the possibility to impose navigation restrictions in ice-covered areas. It would seem that such restrictions could be extended to certain other ecologically sensitive areas without undue hardship to shipowners. An inventory of such areas should first be made and the possibility of imposing restrictions to the navigation of ships carrying either dangerous substances or oil in these areas investigated. Appropriate rules and a ship notification system could be developed by IMCO once the most important areas have been determined.

6. Introduction of Alien Species

According to Article 196 of the Convention, states shall prohibit, reduce or control introduction of alien species that can cause significant and harmful damage to the marine environment. Procedures should be developed to prevent accidental introductions and assistance given to states to enable them to implement such procedures effectively. This will not be an easy task but strict controls may nonetheless minimize the risk. As to willful introductions, they should obviously be avoided as much as possible since it is often impossible to determine in advance if they may have significant and harmful effects. In any event, an international notification procedure could be developed when introductions are planned that could affect other states, and possible objections heard. If possible, environmental assessments of proposed introductions should be carried out.

A register of all introductions that have been made or have occurred inadvertently in the marine environment could be a useful reference document. Effects of the introductions in question on the environment in general and on the local fauna and flora in particular should be stressed. Proposed eradication methods should also be listed. Centralization of information on introductions could be carried out by UNEP. Regional Centres could carry out or promote regional research and advise countries in their regions.

7. Compensation for Damages

The problem of the compensation for the damages caused to species and ecosystems has, so far, generally only arisen in cases of pollution, mostly by oil spills. There is no reason, however, why it could not also be raised in respect of the effects of other destructive factors, including even overexploitation.

Damages to the natural environment should be understood as comprising two essential components: the ecological damages which affect ecosystems and species and the economic damages which are the consequence of the fact that certain persons have become unable to utilize certain species or ecosystems for profit.

Economic damages include such items as the cost of clean-up operations, loss of earnings by fishermen or other persons or even losses of amenities. Although by no means an easy task, economic damages can usually be assessed as a basis for compensation.

Ecological damages, although they may result later in direct or indirect economic damages cannot, however, usually be evaluated in monetary terms. Non-

commercial species and unspoiled areas of considerable ecological and scientific value have indeed little or no market value. Similarly the effect of the depletion of certain species, whether commercial or non-commercial, on the ecosystem of which they form a part is difficult to quantify.

As the law stands, in all legal systems, courts may generally go no further than grant compensation for direct economic damage. Furthermore, damages are usually only awarded to these persons or legal entities who can prove that they have suffered an actual economic loss resulting from the damaging action concerned. Other parties with less direct interests have, therefore, as a general rule, no legal standing to sue for damages. This is particularly true in the case of actions that have resulted in the destruction of wild animals since in many legal systems these animals have no legally recognized owner.

The cost to society of the extinction of a species or of the destruction of a unique a highly diversified ecosystem may, however, in the long run, turn out to be normous even though it may prove totally impossible to evaluate it in monetary terms. A purely financial compensation in such a case would anyway be completely meaningless except perhaps a s a deterrent. What is required is to proceed, as diligently as possible to the replacing or restoring of the resources that have been lost. For instance a species can perhaps be saved if propagated in captivity and subsequently released into suitable habitats. In ohter words, one should try to revert ot the situation which existed before the dmage took place. With this objective in mind, damages could be awarded on the basis of the replacement cost of these parts of the natural environment which have been distroyed or degraded.

This concept is extremely new in law and it is inevitable that time will be needed before it becomes more generally accepted by legislators and courts. There are, however, some encouraging signs. The United States Clean Water Act of 1977 and the Outer Continental Shelf Lands Act, Amendments of 1978, both provede that the President of the United States or, within the limits of its jurisdiction, any state, acting on behalf of the public as trustee of the natural resources, may make claims to recover the costs of replacing or restoring natural resources, damaged or destroyed by pollution. These two acts provide, therefore, a solution to the two major problems of legal standing and of compensation on the basis of replacement costs. This new approach, although the polluting incident occurred as early as 1973, has now been tested in the US Federal courts in the S.S. Zoe Colocotroni case[363].

This is, however, only a beginning. A review should, therefore, be made of legal problems relating tot standing and compensation for ecological damages in other legal systems with a view to promoting the necessary changes in legislation and court practices without which the recovery of damags commensurate with replacement costs will be impossible.

Notes

1 On the subject of migration of commercial species of fish, see FAO Fisheries Circular No 148, Revision 1, Rome 1973

2 At the time of the United Nations Technical Conference on the conservation of the living resources of the sea (Rome 1955). Document A/CONF, 10/6 July 1955 para 69 and 73

3 Carroz, J. — Les problèmes de la peche à la Conference du droit de la mer et dans le pratique des Etats, Revue Générale de Droit International Public, 1980, no 3, p. 7

4 Gulland, J.A. — Review of the State of World Fishery Resources — FAO Fisheries circular no 710. 1978, p. 3; Robinson — Prospects of World Fisheries up to the Year 2000, FAO Fisheries circular no 722, 1979, p. 1-2; Carroz — *op. cit.,* p. 7

5 Halstead — Pharmaceuticals from the sea, 7 New Scientist, December 1972, pp. 565 and ff.

6 Perez, R. — Les Algues source de nourrlture, de substances chimlques et d'énergie, 9th Symposium on Ocean Exploitation, Paris, January 1980

7 Greenway, James C. Jr. — Extinct and Vanishing Birds of the World, New York, 1958, pp. 271 and ff. The great auk became extinct in 1844. Other extinct sea birds are the diablotin *(Pterodroma hasitata caribbea),* the Guadalupe storm petrel *(Oceanodroma macrodactyla)* and the spectacled cormorant (*Phalacrocorax perspicillatus)*

8 Red Data Books, Vol. 1: Mammalia, Vol. 2: Aves, Vol. 3: Amphibia - Reptilia, Vol. 4: Pisces (IUCN, Morges)

9 The blue whale *(Balaenoptera musculus),* the fin whale *(Balaenoptera physalus),* the humpback whale *(Megaptera novaeangliae),* the bowhead whale *(Balaena mysticetus)* and the right whale *(Eubalaena glacialis)*

10 The northern bottlenose whale *(Hyperodon ampullatus)* and the cochito *(Phocoena sinus)* which is a porpoise endemic to the Gulf of California

11 The dugong *(Dugong dugon)* and the three species of manatees (*Trichechus inunguis, Trichechus manatus* and *Trichechus senegalensis),* the South American manatee, the North American manatee and the West African manatee respectively

12 The Galapagos fur seal (*Artocephalus galopagoensis),* the Juan Fernandez fur seal *(Artocephalus philippii),* the Guadelupe fur seal *(Artocephalus townsendi),* the Japanese sea lion *(Zalophus californianus japonicus),* the Laptev walrus *(Odobenus rosmarus laptevi),* the Kurile harbour seal *(Phoca kurilensis),* the Saimaa seal *(Phoca hispida saimensis),* the Mediterranean monk seal *(Monachus monachus)* and the Hawaiian monk seal *(Monachus schauinslandi)*

13 Endangered sea birds include 16 forms of petrels and shearwaters, two cormorants, two frigate birds, one booby, one albatross and several gulls and terns

14 The loggershead turtle *(Caretta caretta),* the flatback turtle *(Chelonia depressa),* the green turtle *(Chelonia mydas),* the hawksbill turtle *(Eretmochelys imbricata),* the Atlantic ridley *(Lepidochelys kempi),* the Pacific ridley *(Lepidochelys olivacea)* and the leathery turtle *(Dermochelys coriacea)*

15 Endangered fishes include the totoaba *(Cynoscion macdonaldi),* an endemic species of the Gulf of California, the Key silverside *(Menidia conchorum)* of Southern Florida, a Hawaiian catadromous species of goby (*Lentipes concolor),* the famous coelacanth *(Latimeria chalumnae),* and several species of anadromous sturgeons

16 Quoted by Carroz, J. — Les Problèmes de la pêche à la Conference sur le Droit de la Mer et dans la pratique des Etats, Revue Générale de Droit International public, no 3, p. 8

17 Outlaw whalers, Greenpeace, San Francisco, 1980, p. 30

18 See, however, a recent report of the International Council for the Exploration of the Sea (ICES): Report of the ICES Working Group on Effects on Fisheries of Marine Sand and Gravel Extraction, C.M., 1979/E.3

19 The damming of the Zuyderzee in Holland prevented the North Sea anchovies to reach their spawning grounds. Most of them, in the first year of the damming, crashed against the dyke and died. They now seem to have found alternative spawning grounds in the estuary of the Elbe. If these had not been available the fishery would have collapsed. See Besançon, J. — Géographic de la pêche, Paris 1965, p. 399

20 More than thirty of the United States most Important commercial species of fish, molluscs and crustaceans spend a part or all of their lives in an estuary .... Two small estuary lagoons in Rhode Island — less than four square miles in size — contribute twenty-five per cent of the annual catch to the region's flounder fishery, Marx, W. — The Frail Ocean, New York 1967, p. 101; see also US Ocean policy in the 1970s: Status and issues, US Department of Commerce publication, 1978, p. IV-1 to IV-4

21 For the effects of mercury on marine life, see, for example, Mercury and the Environment, OECD publication, Paris 1974, pp. 158-163

22 *See:* Goldberg, Edward D. The health of the oceans — UNESCO, Paris 1979. This work deals with the major pollutants of the seas, their production, pathways and effects on the environment and human health; *see* also Impact of Oil on the Marine Environment, GESAMP, Reports and studies n° 6, 1977

23 Goldberg, Edward D., *op. cit.,* p. 62, discusses the effects of DDT on the thickness of pelican egg shells. *See* also Bijleveld, Goeldlin and Mayol — Persistent pollutants in Audouin's gull *(Larus audouinii)* in the

Western Mediterranean: A case study with wide implications? 6 Environmental Conservation, 2, 1979, pp. 139-143

24 As an example, a very large proportion of the North West Europe population of the common shelduck (*Tadorna tadorna)* congregate in July and August on a few sandbanks off the Frisian coast and estuaries in the North Sea. An oil spill at this time and place could have a devastating effect on that population

25 The blow-out of an oil well (IXTOC-1) in the Gulf of Mexico in the summer of 1979 could have had disastrous consequences on North American waterfowl wintering along the coast of Texas, and on shrimps growing in coastal lagoons. The disaster was avoided by the placing of booms to prevent the oil from penetrating into the most sensitive areas. The nesting beach of the endangered Atlantic ridley turtle was reached by the oil spill at the time when the young turtles were about to hatch. Hatchlings had to be air-lifted to a safe area. ORYX, Vol. XV, no 3, p. 252

26 "Exotic species .... can have adverse effects on native species in one or more of the following ways: competition for space or food, predation, habitat destruction or degradation and transmission of diseases and parasites"; World Conservation Strategy, IUCN, UNEP and WWF, Gland and Nairobi 1980, para 3.12

27 IUCN — World Conservation Strategy — IUCN, UNEP and WWF, Gland and Nairobi, 1980

28 World Conservation Strategy, *op. cit.,* para. 3.1

29 World Conservation Strategy, *op. cit.,* para. 6-9

30 The Conservation of Wild Living Resources, Report of Workshops held at Airlie House, Virginia, in February and April 1975 by the World Wildlife Fund, the Ecological Society of America, IUCN, the President's Council on Environmental Quality and the Smithsonian Institution, edited by Sydney Holt and Lee Talbot, p. 30

31 Prolific spawners can usually reproduce themselves long before they have reached their maximum size. They continue then to grow during their adult life for a long period of time, although more slowly at the end. Fishing them at an early stage does not therefore affect the reproductive potential of the stock but it obviously affects the economic profitability of the fishery

32 Eggs may be protected by a thick crust as those of sharks and rays, they can be carried on the back of the animal as in certain crabs, or under its belly as in lobsters and shrimps. The animals may be vivigenous as is the case for marine mammals or ovovivigenous as in certain sharks. The young may be kept under the control of the parents and protected by them, or even kept in an incubating pouch as in sea-horses

33 Holt, S.J. — A contribution to discussion of a new economic order with reference to the living resources of the ocean, FAO, Rome, November 1977, p. 20

34 Holt, S.J., *op. cit.,* p. 22

35 IUCN — Resolutions of the 12th General Assembly — Special Supplement of IUCN Bulletin, Vol. 6, no 11, November 1975, Resolution no 8, p. 3

36 Holt, S.J., *op. cit.,* p. 37

37 In the United States, for instance, individual States have jurisdiction over a three miles wide territorial sea and the Federal Government over a breadth of 197 miles extending to the outer limit of the exclusive fishing zone (Fishery Conservation and Management Act of 1976, Pub. L no 94-265). Similar rules exist in Australia (Fisheries Act 1952-1975, Section 4)

38 As an example, in the United States sea turtles are under the jurisdiction of the Department of Commerce while in the water and under the jurisdiction of the Department of the Interior when they come ashore to lay their eggs (CFR 227-4, Note 1); adequate conservation and management require, therefore, the full co-operation of the two Departments concerned

39 Cushing, D.H. — Marine Ecology and Fisheries, Cambridge, 1975, pp. 242-243

40 Christy, F. and Scott, A. — The Common Wealth in Ocean Fisheries, Baltimore, 1965, p. 232

41 Gulland, J.A. — Review of the State of World Fishery Resources, FAO Fisheries Circular, no 710, Rome, 1978 p. 1

42 *See* as an example of this type of regulatory measures, Article 7 of the new Northeast Fisheries Convention signed at London on 18 November 1980

43 Pub. L no. 94-265

44 On the general subject of enforcement of fisheries regulations on the high seas, *see* Carroz, J. and Roche, A., The International Policing of High Sea Fisheries, Canadian Yearbook of International Law, 1968, pp. 61-90

45 Article 218

46 Other important treaties falling in this category are those dealing with the protection of human rights, or the treaties relating to working conditions concluded under the aegis of ILO

47 There is a form of retaliation which exists, however, in the legislation of the United States. Under the Marine Mammals Act of 1972 (16 USC 1361-63, 1371-84, 1401-07) the government may prohibit the importation of certain fish products from countries using commercial fishing methods which result in the incidental kill or serious injury of marine mammals in excess of United States standards, or which are proscribed for United States fishermen; under the Tuna Conventions Act of 1950 (16 USC 955), restrictions can be established on the importation of certain tuna fish from any country which shall fail to take action to prevent the occurrence of certain proscribed activities

48 Leonard, L.L. — International Regulation of Fisheries, Washington, Carnegie Endowment for International Peace, 1944, pp. 175-185

49 Leonard proposed the establishment of a Central Office, a Scientific Council, an Arbitration Panel and a Secretariat. The world was to be divided into regions, each region having its own Regional Fishery Board and Regional Scientific Board. Specialized committees were also to be appointed to serve the same functions in respect of highly migratory pelagic species such as whales. Regional Fishery Boards were to be empowered to adopt conservation measures for their region. Member States were to be entitled to object to those measures. If agreement proved impossible between the Regional Board and the objecting State the matter was to be brought forward to a central Fishery Commission whose decisions were to be binding

50 Carroz, J. — Les problèmes de la pêche à la Conference du droit de la mer et dans la pratique des Etats, Revue Générale du Droit International Public, Paris, 1980, no 3, p. 20

51 Johnston, D.M. — The International Law of Fisheries, New Haven and London, 1965, p. 173

52 There were some exceptions see Johnston, D.J., *op. cit.,* p. 175

53 Slovka, Z.J. — International custom and the Continental Shelf, The Hague, 1968, p. 40

54 The nature of that right differed according to authors. For some it was based merely on occupation, for others on prescriptive acquisition. There exists a considerable amount of literature on the subject which seems, at a certain time, to have attracted the attention of many international lawyers. *See* in particular: Papandreou — Les Pêcheries sédentaires en haute mer, Revue hellénique de Droit international, 1958; Oda, S.— International Control of Sea Resources, Leyden 1963, pp. 181-186. Johnston, D.M., op. *cit.,* p. 226-231

55 Quoted by Leonard, L, *op. cit.,* p. 81

56 Steller's sea cow (*Hydrodamalis stelleri)* was discovered in 1741. It lived in the Bering Sea. Immediately after its discovery, it began to be slaughtered for its meat and became extinct 27 years later; Street, P., Vanishing Animals, London, 1961, p. 85

57 The sea otter *(Enhydra lutris)* was also discovered in 1741 in the Bering Sea. It was hunted for its pelt for more than a century until it was thought to be extinct. It has been fully protected since the beginning of the century and is now recovering, Street, Philip, *op. cit.,* pp. 76-80

58 The North Pacific fur seal *(Callorhinus ursinus)* is also restricted to the Bering Sea. It was hunted for its pelt. It has now fully recovered.

59 Mackintosh, N.A., — The Stocks of Whales, London, 1965, pp. 163-64. Red Data book. Mammals — IUCN

60 On the "abstention principle" see Johnston, D.M. *op. cit.,* p. 275-282; Oda, S. *op. cit.,* p. 67-72; Christy and Scott, *op. cit.,* p. 187.
The problem is well summarized by Johnston, D.M., *op. cit,* p. 281 In the following way: "It has been argued that abstention was a fair counterpart for the investments made by the other countries in the fishery under abstention. Fairness would have entailed a guarantee of a certain return on the investments in question. But did investments result in exclusive property of a resource? Moreover, Japan could have been associated to the investments. The Japanese position, when Japan tried to bring about a revision of the treaty was that it had been actually designed for the protection of the fisheries industries rather than the conservation of the resources".

61 "The report prepared by the rapporteur Suarez stressed that if the present confusion should persist a few years longer, the extinction of the principal species of fish would be the inevitable consequence of unrestricted exploitation", quoted by Oda, S., *op. cit.,* p. 82

62 At its first session in 1949, the ILC decided to choose the high seas as a priority subject for study. Also in 1949, the General Assembly, at Its fourth session, decided to choose the regime of the territorial waters as a priority subject. At its fifth session in 1953, the ILC adopted draft articles relating to the continental shelf, sedentary fisheries and the living resources of the high seas and at its sixth session in 1954 it adopted provisional draft articles on the regime of the territorial sea. On the whole subject of the work of the ILC in the field of the law of the sea, see Visser't Hooft, Les Nations Unies et la Conservation des Ressources de la Mer, The Hague, 1958.

63 UN General Assembly Resolution, 900 (IX)

64 Report of the International Technical Conference on the Conservation of the Living Resources of the Sea, Doc. A/CONF. 10/6. § 81

65 *Ibid.,* § 18

66 *Ibid.,* § 19-22

67 *Ibid.,*§ 23-24

68 *Ibid.,* § 43

69 *Ibid.,* § 67

70 *Ibid.,* § 54

71 *Ibid.,* § 79

72 *Ibid.,* § 80

73 450 UNTS 82; IEL-MT, 958-33

74 516 UNTS 205; IEL-MT, 958-32

75 499 UNTS 311; IEL-MT, 958-30

76 559 UNTS 285; IEL-MT, 958-31
77 Convention on the Continental Shelf, Art. 2
78 See supra, p. 91
79 Convention on Fishing and Conservation of the Living Resources on the High Seas. Art. 13. This right was given to the coastal State provided such fisheries had long been maintained and conducted by its nationals and provided that non-nationals were permitted to participate in such activities on an equal footing with nationals except in areas where such fisheries had by long usage been exclusively enjoyed by such nationals
80 Convention on the Continental Shelf, Art. 2.4
81 Disputes arose in particular in respect of the fishing of the king crab *(Paralithodes camtschaticus)* in the Bering Sea. The United States held the view that the king crab was a natural resource of the continental shelf over which the coastal State (in this case the United States) had exclusive jurisdiction, control and rights of exploitation. Japan, on the other hand, argued that king crabs were a high seas fishery resource and that, as a result, nationals and vessels of Japan were entitled to continue fishing for these crabs in the eastern Bering Sea. The two governments, however, while maintaining their position on the legal status of the resource, agreed through an exchange of notes on 25 November 1964 on a temporary arrangement whereby Japan could continue to exploit the resource in those waters in which migrate the king crab stocks exploited historically by Japan (533 UNTS 31). A similar agreement was concluded between Japan and the USSR by an exchange of notes on 11 May 1973
82 For the preparatory work in the ILC see Visser T'Hooft, *op.cit.,* pp. 182-184, 192-196, 203-212, 220-231
83 Convention on Fishing and Conservation of the Living Resources of the High Seas, Art. 6.1
84 *Ibid.* Art. 6.2
85 *Ibid.* Art. 8
86 *Ibid.* Art. 1.1 and Convention on the High Seas, Art. 2
87 Netboy, A. — The Atlantic Salmon, a Vanishing Species, London, 1968, pp. 40-41
88 For a full account of the respective positions of the United Kingdom and Denmark, see Council of Europe Consultative Assembly — Report on problems concerning salmon fishing in the Atlantic. Doc. 3038, 13 October 1971
89 ICNAF, Annual Proceedings, Vol. 22, 1971-1972, pp. 43-44
90 Convention on Fishing and Conservation of the Living Resources of the High Seas, Art. 2
91 Report of the International Technical Conference on the Conservation of the Living Resources of the Sea, Rome, 1965, Doc. A/CONF. 10/6. § 18
92 The Conference embodied this concept into the Convention without defining it, perhaps in the belief that it was a clear scientific concept that did not require a legal definition. This is, however, clearly not the case. It could be argued that the optimum sustainable yield concept is more flexible than the MSY and takes more factors into consideration, including economic factors. This may be true, but, in the absence of a definition, is difficult to substantiate. Furthermore, the fact that the objective, as stated by the Convention, remains the maximization of food supply seems to be sufficent evidence that the two concepts of optimum sustainable yield and MSY are, to say the least, very similar
93 Convention on Fishing and Conservation of the Living Resources of the High Seas, Art. 1.2
94 *Ibid.,* Art. 5. It must, however, be realized that if it is possible for new entrants to apply to their nationals such restrictive measures as regulation of gear, minimum sizes, protection of females or of immatures, closed areas or closed seasons, the Convention becomes inoperative when conservation measures involve the establishment of a catch ceiling. In such an event a full renegotiation of the fisheries agreement concerned appears inevitable
95 Article 3 of the draft adopted in 1953 by the ILC reads as follows: "States shall be under the duty to accept, as binding upon their nationals, any system of regulation of fisheries in any area of the high seas which an international authority, to be created within the framework of the United Nations, shall prescribe as being essential for the purpose of protecting the fishing resources of that area against waste or extermination. Such authority shall act at the request of any interested State". ILC Report, 1953, p. 17
96 Convention on Fishing and Conservation of the Living Resources of the Sea, Art. 9 to 12
97 Gulland, J.A. — Review of the State of World Fishery Resources, FAO Fisheries Circular no 710, Rome, April 1978, p. 42. These figures do not include catches of whales which have been steadily declining for the past twenty years due to overexploitation
98 Between 1945 and 1970 world catches increased at about seven percent per year. After 1970 the rate of increase went down to one to two percent per year. *Ibid.,* p. 3
This is due to the stabilization of the catches of some fish stocks and the collapse of others *(e.g.* anchoveta and North Atlantic herring) which intensive fishing on newly exploited stocks with very large yields (*e.g.* capelin, Alaskan pollack) is unable to compensate. Holt, S.J., *op. cit.,* pp. 9-10
99 *Ibid.,* p. 10
100 Co-operation among International Agencies in relation to Fisheries, FAO document COFI/71/9(b), 5 March 1971, Annex III
101 World Fisheries and the Law of the Sea, The EEZ programme of FAO, Rome, 1979, p. 7

102 486 UNTS 157; IEL-MT, 959-07
103 377 UNTS 203; IEL-MT, 959-51
104 IEL-MT, 969-79
105 IEL-MT, 973-68
106 673 UNTS 63; IEL-MT, 966-38
107 590 UNTS 292; IEL-MT, 963-52
108 IEL-MT, 973-85; Polar bears range on the ice-pack often at considerable distance from the coast and are therefore considered as marine mammals by scientists
109 IEL-MT, 972-41
110 A Focus for Ocean Research, IOC document, technical series no 20, p. 14
111 This seems to have been the view of ICES from the very beginning. It was also the opinion of the Technical Conference on the Living Resources of the Sea convened by the United Nations in Rome in 1955. UN Document: A/CONF.10/6, para. 74-76
112 Convention for the International Council for the Exploration of the Sea, signed at Copenhagen on 12 September 1964; 652 UNTS 237; IEL-MT, 964-68
113 Convention for the International Council for the Exploration of the Sea
114 Since the adoption of the revised Convention, Canada and the United States have acceded to ICES which has now a membership of 18 states. All are coastal states of the North Atlantic or its adjacent seas
115 ICNAF has now been replaced by a new organization, the Northwest Atlantic Fisheries Organization (NAFO). Article VI.2 of the Convention establishing NAFO provides that the NAFO scientific council may carry out its functions, in co-operation with other organizations having related objectives. Thus, co-operation with ICES will certainly continue
116 Chayé, F. — Le statut international de la chasse à la baleine et l'industrle balelnière mondiale, these, Paris, 1950, pp. 76-77. There existed before 1939 a whaling committee of ICES. In 1938 this committee recommended a limit on the total amount of whale oil produced. This recommendation, however, was not accepted by the States concerned. See, Mackintosh, M.A., The Stocks of Whales, London, 1965, p. 152
117 This Commission was established under the Convention for the Protection, Preservation, and Extension of the Sockeye Salmon Fishery of the Fraser River System signed at Washington on 26 May 1930, 184 LNTS 305, as amended by a Protocol to include pink salmon In the Fraser River System, signed at Ottawa on 28 December 1956, 290 UNTS 103. Article III of the Convention gives the necessary authority to the Commission to carry out its own research
118 This commission was established under the Convention for the Establishment of an Inter-American Tropical Tuna Commission, signed at Washington on 31 May 1949, 80 UNTS 3. Authority for the Commission to carry out research work appears in Article II
119 Report of the ICES Working Group on Effects on Fisheries of Marine Sand and Gravel Extraction; CM. 1979/E.3
120 *See* supra p. 89
121 *See* supra p. 81
122 Review of the State of Exploitation of the World Fish Resources. FAO Document COFI/77/5, Rome, November 1976, p. 1
123 *Ibid.,* p. 1
124 *Ibid.,* p. 2
125 *Ibid.,* p. 3
126 155 LNTS 349; only right whales were protected under this convention
127 190 LNTS 79, in addition to right whales, this agreement gave protection to the grey whales
128 161 UNTS 72
129 Proposals concerning the Cetacea. Document submitted by the United Kingdom to the second meeting of the Conference of the Parties to the Convention on International Trade of Endangered Species of Wild Fauna and Flora, San Jose (Costa Rica), 1979. London, 1979, p. 32
130 *Ibid.,* p. 53
131 *Ibid.,* p. 45
132 Holt, S.J., *op. cit.,* p. 26
133 Inter-American Tropical Tuna Convention, signed at Washington on 31 May 1949; 80 UNTS 3; IEL-MT, 959-41, Article II
134 International Convention for the Conservation of Atlantic Tunas, signed at Rio de Janeiro on 14 May 1966; 673 UNTS 63; IEL-MT, 966-38
135 Carroz, J. and Roche, A.— The proposed International Commission for the Conservation of Atlantic Tuna, 61 AJIL 673
136 African Convention on the Conservation of Nature and Natural Resources, signed at Algiers on 15 September 1968; IEL-MT, 968-68
137 European Treaty Series, no 104. IEL-MT, 979-70. Full protection under this convention is given to the polar bear, the walrus, the Mediterranean monk seal, the common dolphin, porpoises and the great whales

138 The Antarctic Treaty was concluded in Washington on 1 December 1959; 402 UNTS 71; IEL-MT, 959-91
139 The "Agreed Measures" were adopted in Brussels in 1964 at the third meeting of the Antarctic Treaty Powers; TIAS 6058; IEL-MT, 964-41. They were supplemented on several occasions subsequently, mainly for the purpose of designating protected species and protected areas. They had not, however, come into force yet by the end of 1980 but the Parties to the Antarctic Treaty have accepted to apply them nonetheless as interim guidelines (Recommendation III-9-1 of the third meeting of the Antarctic Treaty Powers, Brussels, 1964)
140 Antarctic Treaty, Article VII
141 IEL-MT, 973-18. This Convention establishes a system of export permits for species listed in its Appendix 2 and a more stringent system of import and export permits for particularly endangered species listed in its Appendix 1
142 Convention for the Protection, Preservation and Extension of the Sockeye Salmon Fishery of the Fraser River System, signed at Washington on 26 May 1930; 184 LNTS 305 as amended by a Protocol signed on 28 December 1956, 290 UNTS 103
143 Rooney, Talbot and Clark — Marine Reserves, Vol. 1, Macquaire University (Australia), 1978, p. 187 ff. quoted in UNEP/IG20/INF3 of 2 July 1980, document prepared by FAO on the basis of the work of Ch. du Saussay and M. Prieur, for the Intergovernmental Meeting on Specially Protected Areas in the Mediterranean, Athens, 13-17 October 1980
144 Convention on Nature Protection and Wild Life Preservation in the Western Hemisphere, signed at Washington, 12 October 1940, 161 UNTS 193
145 *See* supra, p. 104
146 Convention concerning the Protection of the World Cultural and Natural Heritage, adopted in Paris on 23 November 1972 by the General Conference of UNESCO; IEL-MT, 972-86
147 Convention on Wetlands of International Importance, Especially as Waterfowl Habitat, signed at Ramsar, Iran, on 2 February 1971; IEL-MT, 971-09
148 For a criticism of the Ramsar Convention, see World Conservation Strategy, IUCN-UNEP-WWF, Gland, 1980, p. 15. For a Legal Review of the Convention, see de Klemm, C, The Ramsar Convention, a Legal Review, Doc. CONF/5 presented at the Conference on the Convention on Wetlands of International Importance, Especially as Waterfowl Habitat, Cagliari, Italy, November 1980.
149 *See* Cagliari Conference Documents CONF/3, Summary of national reports submitted by contracting parties and CONF/4 Scott, D.A. and Smart, M., the Ramsar Convention, a Technical Review
150 Resolution no 2, 3 and 4 adopted by the 12th General Assembly of IUCN, Kinshasa, Zaire, 18 September 1975, Special supplement to IUCN Bulletin, Vol. 6, no 11, November 1975
151 On the general subject of marine pollution and living resources, see Report of the FAO Technical Conference on Marine Pollution and its Effects on Living Resources and Fishing, Rome, 9-18 December 1970. FAO Fisheries Report no 99, FAO, Rome, 1971
152 IEL-MT, 972-96
153 Convention for the Prevention of Marine Pollution by Dumping from Ships and Aircraft, signed at Oslo on 15 February, 1972, IEL-MT, 972-2
154 IEL-MT, 974-23. This Convention was not yet in force in 1980. An Interim Committee and a secretariat have, however, already been set up and have been actively engaged in achieving the objectives of the Convention
155 The Introduction to Annex I emphasizes that the protection of the Baltic Sea area from pollution by the substances listed in the annex can involve the use of appropriate technical means, prohibitions and regulations of the transport, trade, handling, application and final disposition of products containing such substances. The use of DDT has been banned by signatory States even before the Convention came into force
156 They appear, however, at the top of the list, with the mention "for urgent consideration"
157 IEL-MT, 974-43
158 *See* infra, p. 156
159 This convention was signed at Barcelona on 16 February 1976, IEL-MT, 976-13
160 Protocol for the Prevention of Pollution of the Mediterranean Sea by Dumping from Ships and Aircraft, Barcelona, 16 February 1976, IEL-MT, 976-14
161 Protocol for the Protection of the Mediterranean Sea from Land-based Pollution, Athens, 17 May 1980
162 OECD and the Environment — OECD publication, Paris 1979, p. 77
163 Decisions of the OECD Council are binding upon member countries
164 OECD and the Environment — OECD publication, Paris 1979, p. 84
165 The Convention was signed in London on 12 May 1954, 327 UNTS 3; IEL-MT 954-36
166 IEL-MT, 969-77
167 The term "from the nearest land" is deemed to mean "from the baseline from which the territorial sea of the territory in question is established in accordance with the Geneva Convention on the Territorial Sea and Contiguous Zone of 1958 (Article 1 of the 1954 London Convention as amended in 1969)
168 The Convention was signed in London on 2 November 1973, IEL-MT, 973:84, it was amended by a Protocol

signed in London on 17 February 1978, IEL-MT, 978:13

169 Amendments to the International Convention for the Prevention of Pollution of the Sea by Oil, 1954, Concerning the Protection of the Great Barrier Reef, signed in London on 12 October 1971, IEL-MT, 971:71. These amendments provide for a new definition of the "nearest land" along the northeastern coast of Australia thereby including the Great Barrier Reef within the baseline. These amendments have now been incorporated Into the new 1973 convention.

170 Signed in London on 30 December 1972, IEL-MT 972:77

171 *See* supra, note 139

172 With the exception of a very small number of domestic species

173 The establishment of a new United Nations specialized Agency "devoted solely to ocean problems as they affect man's welfare", had been suggested as early as 1963. This would have entailed that the Fisheries Division of FAO and the Office of Oceanography of UNESCO be separated from these organizations to form this new specialized Agency. See Report of the Technical Committee on Fisheries to the Twelfth Session of the FAO Conference, 16 November 1963, Document C/63/FI/7, pp. 17-20

174 UNEP was established by Resolution 2997 (XXVII) of the United Nations General Assembly of 15 December 1972. This resolution provides that UNEP is established "as a focal point for environmental action and coordination within the United Nations system"

175 Examples are the 1972 World Heritage Convention (UNESCO), the Convention for the Prevention of Pollution from Ships (IMCO) and the conventions for the protection of certain regional seas (e.g. the Mediterranean) against pollution (UNEP)

176 Statutes of the IOC, Article 1.2. The Statutes of the IOC were approved by the UNESCO General Conference at its sixteenth session, Resolution 2.343 of 13 November 1970. By January 1979 there were 103 member States of IOC

177 Roll, H.U. — A Focus for Ocean Research, Intergovernmental Oceanographic Commission - History, Functions, Achievements, IOC technical series, no 20, UNESCO, Paris, 1979, p. 5

178 For further details on the contents and results of these programmes see H.U. Roll, *op. cit.,* pp. 15-24

179 IOC technical series no 7, UNESCO, Paris, 1970

180 Roll, H.U., *op. cit.,* p. 24

181 *Ibid.,* p. 27

182 IOCARIBE was established by IOC Resolution IX-13 adopted at the ninth session of the IOC Assembly in 1975

183 The working group of IOC for the Western Pacific (WESTPAC) would seem to be a likely candidate. Institutionalization of that working group has already begun with the appointment of a full time Secretary (IOC Assembly Resolution XI-10, 1979)

184 Roll, H.U., *op. cit.,* p. 13

185 UNESCO General Conference Resolution 2.313, para. 1

186 IEL-MT, 972:86

187 Report on FAO, the FAO Committee on Fisheries and Regional Fishery Bodies — FAO Fisheries Circular no 331, FAO, Rome, 1975

188 *Ibid., p.* 6

189 *Ibid.,* pp. 15-16

190 Agreement for the Establishment of the Indo-Pacific Fisheries Council, signed at Baguio on 26 February 1948, 120 UNTS 59, as amended in 1952, 1955, 1958 and 1961; 190 UNTS 383, 227 UNTS 322, 343 UNTS 343 and 418 UNTS 348. The IPFC decided at its 17th session in 1977 to change its title into Indo Pacific Fisheries Commission

191 Agreement for the Establishment of a General Fisheries Council for the Mediterranean, signed at Rome on 24 September 1944, 126 UNTS 239 as amended on 22 May 1963, 490 UNTS 444

192 Ruivo, M. and Gonçalves, M.E. — Trends in Ocean Uses and Related Institutional Aspects, PACEM in MARIBUS proceedings of VIIth Conference, Algiers, 1976, pp. 59-60

193 *See* supra, p. 112

194 For the terms of reference of ICSPRO, *see* Regional Cooperation in Marine Science, IOC document IOC/INF 407, Paris, 1 September 1979, p. A-3

195 *Ibid.,* p. A-6

196 *Ibid., p.* A-10

197 Ruivo, M. and Gonçalves, M.E., *op. cit.,* p. 56

198 For a critical appraisal of these mechanisms, see Ruivo, M. and Gonçalves, M.E., *op. cit.*, p. 62-63

199 Report of the UN Conference on the Human Environment, UN Document A/CONF.48/14

200 Other principles of some relevance to the conservation of the living resources of the sea are principles 3, 4, 6 and 22

201 Emphasis added

202 Emphasis added

203 Emphasis added

204 Emphasis added
205 IEL-MT, 979-55
206 In particular in the Preamble and in Article 1.2
207 International Court of Justice, Reports of Judgments, Advisory Opinions and Orders, Fisheries Jurisdiction Case (United Kingdom v. Iceland), Merits, Judgment of 25 July 1974, p. 31
208 Resolution no III adopted by the Geneva Conference recommends "that States concerned should co-operate in establishing the necessary conservation regime through the medium of such organizations covering particular areas of the high seas or species of living marine resource"..., Document A/CONF.13/L56
209 40 AJIL (1946), supplement, p. 45
210 On this subject see Visser T'Hooft, *op. cit.,* pp. 101-109; see also Johnston, D.M., *op. cit.,* pp. 333-344
211 See ICJ pleadings, Fisheries Jurisdiction, Vol. I and II and Fisheries Jurisdiction (United Kingdom v. Iceland) Merits, Judgment, ICJ Reports 1974, p. 3. The court did not decide on the legality of unilateral extensions of jurisdiction. It ruled, however, that the extension made by Iceland was not opposable to the United Kingdom in view of earlier agreements between the two countries, p. 34
212 For a complete table of national jurisdiction limits at the beginning of 1980, *see* Carroz, J. — Les Problèmes de la Pêche à la Conference sur le Droit de la Mer et dans la Pratique des Etats, Revue Générale de Droit International Public, 1980, no 3. For a review of the fishery limits situation in 1971, see Limits and Status of the Territorial Sea, Exclusive Fishing Zones, Fishery Conservation Zones and the Continental Shelf, FAO Fisheries Circular no 127, FAO, Rome, 1971
213 The General Assembly had first created in 1967 an *ad hoc* Committee to study peaceful uses of the sea-bed and ocean floor beyond the limits of national jurisdiction. This committee recognized that the sea-bed should Indeed only be used for peaceful purposes. Several delegations, however, felt that the zone to which this principle applied should be clearly delimitated and that this involved, as a consequence, the requirement to lay down new rules relating to national jurisdlctional boundaries.
214 Resolution 2749 (XXV) of 17 December 1970
215 Resolution 2750 C (XXV)
216 This list was approved by the UN Seabed Committee on 18 August 1972. It was agreed that the list was not necessarily complete nor did it establish the order of priority for consideration of the various subjects or issues; Oda, S., The International Law of the Ocean Development, Vol. II, 1975, p. 239
217 Documents A/CONF.62/W.P.8, Parts I, II and III and A/CONF.62/W.P.9
218 Documents A/CONF.62/W.P.8, Parts I, II and III, Rev. 1
219 Document A/CONF.62/W.P.10
220 Document A/CONF.62/W.P.10, Rev. 1
221 Document A/CONF.62/W.P.10, Rev. 1
222 Document A/CONF.62/W.P.10, Rev. 3
223 Article 7, 8, 9 and 10. See also Art. 3, 4, 5, 6 and 7 of the Convention on the Territorial Sea and the Contiguous Zone
224 Article 3
225 Article 2
226 Article 17
227 Article 47
228 Article 48
229 Article 52
230 Article 57
231 Article 56
232 Article 62
233 Article 76
234 Article 77
235 Article 86
236 Article 87
237 They will presumably be more numerous when certain non-conventional resources such as krill, lanternfishes, or squids are exploited unless these species are added to the list of highly migratory species in Annex I
238 Holt, S.J., *op. cit.*
239 Paragraph 2 of Article 64 clearly states that all the other provisions of part V of the Convention, namely the provisions relating to the EEZ, are applicable to highly migratory species
240 Articles 62.3 and 119.1
241 Italy, New Zealand, the Seychelles, Sweden and Switzerland are all non-whaling nations who joined the IWC after 1976
242 The control of pollution in the Area, as it may affect species and ecosystems is dealt with in the next section, see infra, p. 142

243 Article 165.2(d)
244 Article 165.2(c). This supervision, however, can only be exercised upon the request of the Council
245 Article 165.2(e)
246 Article 165.2(f)
247 Annex III, Article 2.1(b)
248 Annex III, Article 3.3
249 Annex III, Article 4.6 (a)
250 Annex III, Article 4.6(b)
251 In earlier drafts of the Convention, e.g. the ICNT, Revision 1, there was a provision in Annex II, (later to become Annex III), Article 2.1(d), which empowered the Authority to close a particular area to prospecting where the available data indicated the risk of irreparable harm to a unique environment. This provision has, however, not been included in the new draft Convention
252 The Arctic Waters Pollution Prevention Act of 1970, R.S.C. 1970 (250 supp.) C.2 (1970)
253 *See* supra, p. 108
254 The recent treaties for the protection of migratory birds which have been concluded between the United States and Japan (Tokyo, 4 March 1972), Japan and the USSR (Moscow, 10 October 1973), Japan and Australia (Tokyo, 6 February 1974) and the United States and the USSR (Moscow, 19 November 1976) all contain provisions relating to the control of introductions of animal and plant species that may cause damage to the bird species covered by the agreements and to their environment. Most of them also contain clauses dealing with the control of introductions into unique island environments. Other recent conventions containing provisions on the Introduction of alien species are the Convention on Conservation of Nature in the South Pacific (Apia, 12 June 1976); IEL-MT, 976-45, the Convention on the Conservation of Migratory Species of Wild Animals (Bonn, 23 June 1979, IEL-MT, 979-55) and the Convention on the Conservation of European Wildlife and Natural Habitats, which was concluded under the auspices of the Council of Europe (Bern, 19 September 1979, IEL-MT 979-70)
255 Article 61.2, for example, requires coastal States to take into account the best scientific evidence to ensure the maintenance of the living resources in their EEZ. A similar provision in Article 119.1(a) relates to the high seas
256 Article 246.2
257 Article 246.3
258 Article 246.5(a). This provision, however, does not apply to that part of the continental shelf that extends beyond the outer limits of the EEZ (Article 246.6)
259 Article 257
260 Article 256
261 Article 143
262 *See*, in particular, Article 243
263 Articles 238, 239, 244, 251, 61.5, 119.2
264 Article 249
265 Article 240(d)
266 Article 266.1
267 Article 266.2
268 Article 272
269 Article 275.1
270 Article 276
271 Article 277
272 Article 62.4(k) and Article 73
273 Articles 63, 64 and 118
274 *See* supra, p. 145
275 Article 277
276 Article 276.2
277 Article 297.3(a)
278 Article 297.3(b)(ii)
279 Article 297.3(b)(iii)
280 Article 297.3(b)(i)
281 Article 297.3(c)
282 Article 297.3(d)
283 Article 297.2(a)
284 It has been calculated that the extension of coastal state jurisdiction to a breadth of 200 miles will cover a total area equal to 25% of the surface of the earth and to 36% of the total surface of the oceans
285 On the other hand, in the Baltic Sea, although until 1977 coastal state Jurisdiction did not extend beyond the 12 miles territorial sea, Sweden followed by all the other riparian states of that sea decided in 1978 to establish a 200 miles exclusive fishing zone. As a result no part of the Baltic is any longer under the regime

of the high seas

286 On this subject, *see* Carroz, J. — Les Problèmes de la pêche à la Conférence sur le Droit de la Mer et dans la Pratique des Etats, Revue Générale de Droit International Public, Paris, juillet-septembre 1980, no 3, pp. 24-28

287 A noteworthy exception is, for instance, the new Fisheries Act of the Gambia (Act no 17 of 1977) which provides, as a fishery policy statement, for the preparation and keeping under continual review of plans for the management and development of fisheries to be based on the best scientific information available and to ensure the optimum utilization of the fishery resources consistent with sound management principles and with the avoidance of overfishing (Section 4 of the Act). It also provides for consultations with governments of nations sharing the same or interrelated stocks with a view to ensuring the closest practicable harmonization or co-ordination of their respective fisheries management and development plans, the collection of statistics, the formulation of regional fisheries management and development plans and the establishment and operation of joint or regional fisheries management bodies (Section 5.1 of the Act).

288 Another example is the Marine Spaces Act, 1977 of Fiji (Act no 18 of 1977 as amended by Act no 22 of 1977). This Act provides that the minister shall from time to time determine on the basis of the best available information the total allowable catch in respect of every fishery within the EEZ and the portion of that catch which Fiji's vessels have the capacity to harvest. It further provides that the minister may apportion among other countries the resulting surplus (Section 11 of the Act). The total allowable catch is defined, in Section 2 of the Act, as the amount of fish that will produce from the fishery the maximum substainable yield, as qualified by any relevant economic or environmental factors and taking into account fishing patterns, the interdependence of stocks of fish and any generally recommended sub-regional, regional, or global standards. It thus reproduces almost verbatim the text of Article 61.3 of the new Law of the Sea treaty. Regulations adopted in 1979 under the 1977 Act (Marine Spaces - Foreign Fishing Vessels) Regulations 1979, provide for the adoption of fishing plans in respect of fishery operations carried out by foreign vessels. See also the law of the Bahamas on Living Resources (jurisdiction and conservation) of 1977

289 The Fishery Conservation and Management Act of 1976 provides also for the setting of fish catch quotas by species on the basis of an ecologically sound optimal yield, within the 200 miles zones (Pub. L 94-265 — 16 and 22 USC) (1976). Under this law, a number of Regional Fishery Management Councils have been established for the purpose of developing fishery management plans, to review assessments of optimum yield and total allowable level of foreign fishing. For details on regional Councils, guidelines for development of management plans regulation of foreign fishing and regulation of certain fisheries in the EEZ of the United States see Code of Federal Regulations, Vol. 50, chapter VI, parts 601 to 674

290 *See*, Carroz, J. — Les problèmes de la pêche à la conférence sur le droit de la mer et dans la pratique des Etats, *op. cit.,* pp. 28, 32

291 Convention on Future Multilateral Co-operation in the Northwest Atlantic Fisheries, Ottawa, 24 October 1978; IEL-MT, 978:79

292 North-East Atlantic Fishery Convention, London, 18 November 1980

293 Protocol amending the International Convention for the High Seas Fisheries of the North Pacific Ocean, Tokyo, 25 April 1978

294 Article XI.2 of the new Northwest Atlantic Fisheries Convention; Article 5.1 of the new North-East Atlantic Fisheries Convention

295 Article XI.3 of the new Northwest Atlantic Fisheries Convention; Article 5.2 of the new North-East Atlantic Fisheries Convention

296 Article 6

297 *See*, Carroz, J. — Les problèmes de la pêche à la conférence sur le droit de la mer et dans la pratique des Etats, *op. cit.,* p. 34. President Reagan has now, however, decided that, in view of strong opposition from New England fishing interests, the treaty would not be submitted to the Senate for ratification (Herald Tribune, 11 March 1981)

298 On the Baltic Sea, see Goralczyk, W. — La Mer Baltique et les problèmes de cooperation des Etats riverains, Revue Générale de Droit International Public, Tome 84-1980/1, pp. 269-283; See also Carroz, J.E. — The Management of Living Resources in the Baltic Sea and the Belts, Ocean Development and International Law Journal, Volume 4, Number 3, pp. 213-232

299 No marine mammals are covered by the new Northeast Atlantic Fisheries Convention, Article 1.2

300 Cetacean stocks managed by the International Whaling Commission are the only sea mammals excluded

301 IEL-MT, 980-39; Article 2.1 and 2.2

302 The Antarctic convergence is the line along which the cold Antarctic surface water sinks below the warmer sub-Antarctic water (Mackintosh, N.A., The stocks of whales, *op. cit.,* p. 51). It more or less coincides with the 60th degree of south latitude, which forms the boundary of the area to which the Antarctic treaty applies, but extends further north sometimes to considerable distances in the South Atlantic and part of the Indian Ocean

303 Article II.3(b)

304 Article II.3(c)

305 Article IX.2(l)

306 Article IX.2(g)

307 Article IX.5

308 Article XI
309 On the problems of the Southern Oceans and the Convention on the Conservation of Antarctic Marine Living Resources, *see* IIED: The Management of the Southern Oceans, 1979.
310 *Ibid.,* p. 52
311 Council Regulation (EEC) no 101/76 of 19 January 1976 laying down a common structural policy for the fishing industry. Official Journal of the European Communities no L 20/19 of 28 January 1976. Regulations adopted by the Council of the Communities are binding and directly applicable in all Member States
312 *Ibid.,* Article 4
313 *Ibid.,* Article 11
314 Council Regulation (EEC) no 2527/80 of 30 September 1980 laying down technical measures for the conservation of fishery resources; Official Journal of the European Communities no L 258/1 of 1 October 1980
315 *Ibid.,* Articles 8 to 10
316 *Ibid.,* Article 13
317 Council Regulation (EEC) no 754/80 of 26 March 1980 concerning, for certain fish stocks occurring in the Community fishing zone, the fixing for 1980 of the total allowable catches, the share available for the Community and the means of making the catches, Official Journal of the European Communities, no L 84/36 of 28 March 1980
318 On the new FAO programme for the Development and Management of Fisheries in Economic Zones *see* ECOSOC document E/1980/68 of 23 May 1980, Report of the Secretary General on the Uses of the Sea, pp. 19-20. *See* also FAO document COFI/79/4; see also Loftas, T., FAO's EEZ Programme, Marine Policy, vol. 5, n°3, July 1981, p. 229
319 Report of the Third Session of CECAF. FAO, Fisheries Report no 132, para. 27
320 Report of the Third Session of the Indian Ocean Fishery Commission, FAO Fisheries Report no 130, para. 40, Report of the Sixth Session of the Indian Ocean Fishery Commission, FAO Fisheries Report no 234, para. 61
321 For a more detailed analysis of the UNEP Regional Seas Programme, see UNEP document: Information on UNEP-Sponsored Regional Seas Programme (RS/PAC gen. 1, 18 February 1980) and work plan of the UNEP Regional Seas Programme Activity Centre (RS/PAC gen. 2, 1 April 1980)
322 Convention for the Protection of the Mediterranean Sea against Pollution, Barcelona, 16 February 1976, IEL-MT, 976-13
323 Regional Convention for Co-operation on the Protection of the Marine Environment from Pollution, Kuwait, 24 April 1978, IEL-MT, 978-31; Convention for Co-operation in the Protection and Development of the Marine and Coastal Environment of the West and Central African Region; *see* final act of the conference of plenipotentiaries, Abidjan, 23 March 1981 — UNEP document, UNEP/IG, 22/7
324 Protocol for the Prevention of Pollution of the Mediterranean Sea by Dumping from Ships and Aircraft (Barcelona, 16 February 1976, IEL-MT, 976-14), Protocol Concerning Co-operation on Combating Pollution of the Mediterranean Sea by Oil and other Harmful Substances in Cases of Emergency (Barcelona, 16 February 1976, IEL-MT, 976-15); Protocol for the Protection of the Mediterranean Sea against Pollution from Land-Based Sources (Athens, 17 May 1980, IEL-MT 980-37)
325 Protocol Concerning Regional Co-operation in Combating Pollution by Oil and other Harmful Substances in Cases of Emergency (Kuwait, 24 April 1978, IEL-MT, 978-32); Protocol Concerning Co-operation in Combating Pollution in Cases of Emergency, adopted in Abidjan on 23 March 1981; UNEP document, UNEP/IG, 22/7
326 Information on UNEP-sponsored Regional Seas Programme; UNEP document RS/PAC gen. 1, 18 February 1980, p. 8
327 *Ibid,* pp. 2-3
328 Details of present and planned activities of UNEP in these fields can be found in the two documents referred to supra (note 321)
329 Conservation of Nature Act, Consolidated Act no 435 of 1 September 1978, Article 46
330 The "Conservatoire du Littoral" was established by an Act dated 10 July 1975
331 16 USC 1451 *et seq.*
332 Section 305 of the Act
333 *See*, Ray, Carleton — Inshore Marine Conservation, Proceedings of the First World Conference on National Parks, Seattle, 1962, p. 84, for a list of the very small number of marine protected areas existing at that time
334 Examples are the International Conference on Marine Parks and Reserves (Tokyo, May 1975), the Regional meeting on Marine Parks and Reserves in the Northern Indian Ocean (Teheran, March 1975) and Resolution adopted by the 11th and 12th General Assemblies of IUCN
335 Marine Reserves Act of 20 September 1971, Act no 15 of 1971
336 16 USC 1431-34
337 Act no 85 of 1975
338 *See* supra, p. 184
339 *See* progress report on MAB project no 5 in the report of the International Coordinating Council of the MAB

Programme, sixth session, Paris, November 1979, MAB report, series no 48, p. 100. For a list of biosphere reserves in November 1979, *Ibid,* p. 89

340 *See* supra, p. 184

341 UNESCO document IOC-XI-6 Annex

342 UNESCO draft programme and budget for 1981-1983, UNESCO document 21C/5, para. 2458-2466

343 UNESCO document MARINF/26, Paris, 19 September 1979, p. 4

344 Committee of Ministers Resolution no 29 of 26 October 1973

345 On the influence of these principles on new French legislation, *see* Despax, M. — Droit de l'Environnement, Paris 1980, p. 215

346 Formerly, the Ocean Economics and Technology Office (OETO). This branch forms part of the Department of International Economic and Social Affairs of the United Nations. For an overview of the activities of OETO, *see*, Uses of the Sea, Report of the Secretary General, ECOSOC document E/1980/68 of 23 May 1980, pp. 16-18

347 Reports of this research work have been published in the form of two manuals in 1980

348 Information on UNEP-sponsored Regional Seas Programme, UNEP document RS/PAC gen. 1, 18 February 1980, p. 10

349 UNESCO document MARINF/26, Paris, 19 September 1979, p. 2

350 IEL-MT, 979-55

351 The definition of Range States is provided in Article l.1(h). It Includes States that exercise jurisdiction over any part of the range of a migratory species as well as States that harvest that species on the high seas

352 Article ll.3(c)

353 Article V

354 World Conservation Strategy, *op. cit*., p. 5

355 Proceedings Second World Conference on National Parks, IUCN, Morges 1974: An ecosystem approach to marine parks and reserves, p. 261

356 It would seem particularly important to include sea-turtles under this definition. The situation is very similar to that of fishes such as salmon. Coastal states may incur considerable costs to protect beaches where turtles lay their eggs, set up hatcheries and carry out breeding programmes. Yet, if the turtles are caught on the high seas, these efforts will have been made in vain

357 *See*, Proceedings IUCN Workshop an Cetacean sanctuaries, Tijuana and Guerero Negro, Mexico, February 1979, pp. 33-34

358 Cushing, D.H., in Marine Ecology and Fisheries, Cambridge, 1975, p. 93 notes that for the North-east Atlantic herring, spawning grounds are often of very limited extent, on gravel and in relatively shallow waters. They are, it would seem, used by the fish every year. These areas are apparently very vulnerable to destruction by dredging, trawling with heavy gear, filling, silting or pollution

359 *See* also further down under pollution

360 "Le Figaro", 2 February 1981

361 International Convention for the Safety of Life at Sea, signed at London on 1 November 1974, IEL-MT, 974-81

362 The French daily "Le Monde" reports on 14 January 1981 that species of small crustaceans (amphipods) new to science have been taken alive from the deepest ocean trench in the world, the Challenger Deep, near the Marianas in the Pacific. Bacteria found on these animals are adapted to extremely high pressures and it is expected to use them in industrial processes to produce enzymes speeding up chemical reactions under high pressure and at relatively low temperature. There may be many other forms of life of equal value in deep ocean trenches that could be destroyed forever if a ship carrying an ultra hazardous cargo sinks to these depths and breaks up

363 The S.S. Zoe Colocotroni was an oil tanker which ran aground on a reef off the south coast of Puerto Rico in 1973. The resulting oils slick heavily polluted a coastal mangrove and caused the death of a large number of marine organisms. A US District Court awarded 5,5 million dollars to the Commonwealth of Puerto Rico as the replacement cost of the organisms destroyed. The case was subsequently brought to a US Court of Appeals (First Circuit, n° 78-1543 and 79-1486). In its judgment of 12 August 1980, the court, whilst recognizing that it had, in that case, "ventured Into unchartered waters" and rejecting to amount of damages granted by the lower court, recognized, however, that "some lesser steps" could be taken that would have a beneficial effect on the mangrove ecosystem without excessive destruction of existing natural resources or disproportionate cost" and that "the costs projected for the carrying out of such reasonable lesser steps would be an appropriate award of damages". It is likely that a final decision on the case will have eventually to be taken by the US Supreme Court.

Abbreviations

AJIL American Journal of International Law

IEL-MT International Environmental Law, Multilateral Treaties, edited by W.E. Burhenne, compiled by R. Muecke, Revised by Daniel B. Navid and Odile Seidel, E. Schmidt Verlag, Berlin, 1972-1980, 4 vols.

LNTS League of Nations Treaty Series

TIAS Treaties and other International Acts Series

UNTS United Nations Treaty Series

Chapter Three
Prevention and Control of Marine Pollution

I Introduction

Martine Rémond-Gouilloud

A. Pollution of the Marine Environment: The Range of Harms and Hazards

There was a time when marine pollution control was considered a "luxury" problem for wealthy nations. The health of the oceans was the concern of poets and bird-watchers. Reasons for the perceptual change that has taken place in recent decades have been suggested at length above[1], and the modern arguments for a comprehensive conservation strategy for the ocean have also been reviewed in some detail[2].

The previous chapter was organized around the theme of ocean conservation: the conservation of marine species, their habitats and related ecosystems[3]. In this chapter the central theme is marine pollution in all its diverse manifestations. There is, of course, a cognate relationship between the two central themes, but the distinction is a real one. On the one hand, the conservation of marine species and ecosystems obviously depends on the preservation of life support systems in the sea. In that respect, the control of marine pollution is indeed a prerequisite to species and ecosystem conservation. By the same token, marine pollution constitutes a threat to human interests since it may affect resources of significant importance to man.

On the other hand, marine pollution threatens a considerable number of other human interests, ranging from public health to the loss of amenities. (No one, after all, actually enjoys the sight of a polluted beach or harbour.) The range of interests and threats is indeed so vast that the field of marine pollution control might be said to extend from relatively minor, minimally threatening, harms at one extreme to potentially catastrophic, genuinely frightening, hazards at the other.

After 30 years of growing awareness — assisted by the Torry Canyon, Minamata, Santa Barbara, Amoco Cadiz, IXTOC I, and other famous incidents — marine pollution control is now a conspicuous part of our civilization. Decision-makers are bound to take account of the need for such control in any intelligent approach to ocean resource development and mangement or to the development and management of coastal areas. The need for such regulation is now widely accepted, especially in heavily industrialized and highly congested coastal communities.

B. Problems of Definition

Yet, because of the extremely wide range of harms and hazards evisaged, it has not been easy to define the concept of marine pollution. Indeed it might be described as a term rather than a concept, precisely because of the impossibility of a scientifically rigorous definition.

Of the many early efforts made[4], perhaps the most successful was that of the International Oceanographic Commission (IOC)[5]. This version served as the basis for a subsequent, quasi-official definition formulated by the Group of Experts on the Scientific Aspects of Marine Pollution (GESAMP). In recent years the GESAMP definition, slightly modified, has been incorporated in the Draft Convention on the Law of the Sea[6] negotiated at UNCLOS III. This latest version reads as follows:

> "Pollution of the marine environment" means the introduction by man, directly or indirectly, of substances or energy into the marine environment (including estuaries) which results or is likely to result in such deleterious effects as harm to living and marine life, hazards to human health, hinderance to marine activities, including fishing and other legitimate uses of the sea, impairment of quality for use of sea water and reduction of amenities".

Despite its extremely broad scope, this definition cannot be accepted as entirely satisfactory for environmentalists. By the late 20th century, the degradation of the oceans can no longer be attributed to the "introduction of substances or energy" only. It is now generally agreed that the dredging of the sea floor for extracting sand and gravel can have long-term sterilizing effects[7] for the affected marine environment. Deep-sea mining technology, currently being developed for harvesting ferromanganese nodules on the ocean floor, is another potential cause of impairment of the seas, which implies no introduction of any kind[8]. Finally, it might be argued that a comprehensive definition of pollution should encompass the disturbances caused by such activities as the construction of a sea-level canal between the Pacific and Atlantic Oceans[9], or the drilling of a hole through the crust of the earth[10]. One important consequence of the breadth of marine pollution concerns is that the problems of control are extremely diverse, requiring the development of rapidly expanding areas of expertise, each in itself representing a particular kind of specialization and involving a particular set of cross-disciplinary interactions and insights. For this reason, the present chapter draws on a number of different kinds of legal specialists, reflecting a variety of expert perspectives as well as a variety of personal styles[11].

The oceans are now subject to such various forms of disturbance that one single word cannot possibly embrace them all. Each time mankind marvels at a new success in mastering another kind of technology at sea, a new threat arises as its counterpart. Any attempt at presenting a definitive definition of marine pollution would indeed be short-sighted, since it would be based on the fallacious assumption that technological progress at sea has come to an end. For that reason it seems wise to seek the widest possible definition of marine pollution to ensureadequate protection of the marine environment.

C. Classification of Pollutants

It must also be emphasized that marine pollution control addresses itself to an extraordinary diversity of pollutants. While the exact nature of a pollutant may be of little importance to a lawyer, diplomat or legislator, the scope, mode and effectiveness of a regulatory approach to any kind of marine pollution may depend in part on a properly scientific appreciation of the characteristics of the polluting substances and of their effects on the affected sectors of the food chain, and at

least an approximate understanding of the assimilative capacity of ocean areas under particular stress.

Listings of the classes of ocean pollutants are somewhat arbitrary at best, but the following is not unrepresentative of such compilations[12]:

(i) chlorinated hydrocarbons;
(ii) municipal wastes;
(iii) solid materials;
(iv) heavy metals;
(v) petroleum;
(vi) heat (thermal pollution);
(vii) radioactivity; and
(viii) carbon dioxide.

As noted above, it is not for non-scientists to quarrel with such listings. For lawyers and other non-scientists, it is usually enough to know — or to be assured — that a particular substance or class of substances is noxious and can adversely affect legitimate interests. Yet, this usually casual attitude of the layman to what is perceived to be a scientific undertaking is likely to be replaced shortly, as it becomes necessary for purposes of negotiation, administration and litigation to understand the rationale of a proposed or existing regulatory regime or management system which is based on such listings. Existing classifications are usually organized around the concept of *potential threat or risk,* graded in accordance with the degree of rigour required for purposes of effective regulation. For example, the "list system" which has been developed for land-based pollution and dumping control agreements in recent years — and for implementing legislation — is based on the degree of toxicity, persistence and bio-accumulation: priority attention is, therefore, given to organo-halogen compounds, mercury, cadmium and other heavy metals and their compounds, persistent synthetic materials which may float, substances possessing carcinogenic properties and high-level radioactive wastes[13]. These substances are "black-listed", and thus subject to total prohibition. As explained in the 1974 Paris Convention[14], these pollutants are unacceptable in any quantity, because they are not readily degradable or otherwise rendered harmless by natural processes; or because they may give rise, individually or in combination, to dangerous accumulations in the food chain or otherwise endanger living organisms causing undesirable changes in marine ecosystems; or because they seriously interfere with the harvesting of sea-food or with other legitimate uses of the sea[15].

Another method of distinguishing pollutants is by reference to the factor of visibility—a matter of little scientific import, but of great legal and political significance because of its "psychological" impact. The visual nature of oil pollution has no doubt played a major role in the promotion of legal and institutional developments, related to oil pollution of the sea. The visibility of a noxious discharge often accelerates the process of detecting and apprehending the polluter responsible. The existence of non-visible pollutants usually makes proof difficult or even impossible, creating the need for more innovative, non-liability approaches to regulation, which are often more direct and more effective techniques of environmental management.

D. Sources of Marine Pollution

Nothing, however, is more fundamental in the general design of marine pollution control policy than the classification of sources of marine pollution. It is now invariable to accept, almost uncritically, that there are six *forms* of marine pollution by virtue of the *source* of the harm or hazard. The "source" in each case is funtionally conceived, referring to a type of activity that causes the harm or hazard, namely:

(i) navigation (ship-generated or vessel-source pollution);
(ii) the disposal of wastes at sea (pollution by dumping);
(iii) the discharge of a wide range of shore-generated effluents (land-based pollution);
(iv) the dissemination of emissions on land (pollution from and through the atmosphere);
(v) offshore petroleum exploration and exploitation (pollution from seabed activities); and
(vi) deep-ocean mining (pollution from activities in the Area).

Five of these six categories require some special comment. (Atmospheric pollution is referred to briefly as a variant of land-based marine pollution.)

1. Vessel-Source Pollution

This was the first form of marine pollution to arouse the attention of the world community, and therefore the first to receive concerted political and legal treatment at the international level. Control techniques are not easily applied to things of many different "nationalities" that move almost continuously around the world, in and out of every kind of jurisdictional regime. Even in the presence of strong evidence that an unlawful discharge has occurred, it is often easy for the suspected offender to avoid or slip out of the harbour of the investigating state and thus evade proceedings against it. The difficulties inherent in normal judicial procedures against vessels arise from the economic structure of the world shipping industry: two-thirds of the world commercial fleet is registered in a small number of states, many of which are frankly more interested in protecting the commercial interests of the ships flying their flag than in protecting the marine environment. Many of these "flag states" are "flags of convenience states": that is, states with an "open registry" system, which permits the registration of any vessel, regardless of nationality which is willing to pay the registration fee[16]. Such states are often reluctant to sign and ratify vessel-source pollution control agreements, or at least reluctant to give effect to such treaty-based restraints on shipping.

Most of these difficulties of treaty enforcement are related to the traditional exclusivity of flag state jurisdiction outside the territorial limits of a coastal state. Although important jurisdiction innovations have recently been introduced at UNCLOS III — the extension of coastal state jurisdiction and the addition of universal port state jurisdiction in carefully defined circumstances applied to certain matters pertaining to vessel-source pollution control[17] — the concurrence of traditional flag state jurisdiction is likely to continue to inhibit the enforcement of this area of international law. Moreover, the influence of flag state interests will continue to limit the possibilities of unilateral environmental initiatives. But the world-wide scope of shipping will always force the treatment of these pollution control issues

to be carried out primarily at the global level, where all interests affected have a chance to be represented.

2. Pollution by Dumping

In contrast, the business of dumping wastes at sea is an unconcealed, easily verified practice: a permitted and more or less legitimate activity, which should, however, be subject to careful regulation. Indeed the harzardous — sometimes ultra-hazardous — nature of some of these wastes certainly necessitates the strictest controls, or even outright prohibition. Since much of the dumping takes place far beyond territorial limits, an international basis for such controls has always been necessary, and this continues to be essential even in the present age of extended coastal jurisdiction out to the 200-mile seaward limits of the extensive economic zone. (Incidentally, much of the most serious dumping is done by military vessels, but these practices are virtually impossible to assess without access to classified information.)

3. Land-Based Pollution

This source is by far the most common form of marine pollution, accounting for 75% or more of the pollutants in the ocean. This figure is certainly not too high, if one includes under this heading the deposition of land-generated emissions *carried from and through the atmosphere*. Land-based marine pollution, by any definition, is derived from an extraordinary diversity of human activities, and therefore poses truly formidable problems of control, not least because of the national sovereignty sentiment which tends to frustrate the imposition of international controls at the source. This form of marine pollution is particularly critical in coastal waters, where the every-day wastes of a country or a region usually end up at a high level of concentration, before dispersal through the ocean areas beyond. These concentrations in coastal areas are of critical concern, but the effects of disposal likely to extend to even the most remote areas of the oceans. Despite the world-wide circulation of shore-generated wastes, global conventions on land-based marine pollution are more likely to serve as an inducement to control rather than as a method of regulation. On the other hand, there is obviously an important role for regional co-operation in the control of land-based marine pollution, especially in semi-enclosed seas and other compact areas that constitute a potential "unit" of management for ocean waste disposal.

4. Pollution from Seabed Activities

Pollution arising from the exploration and exploitation of the continental shelf, as legally defined[18], calls for quite specific treatment, since it can be traced easily and precisely to one of two kinds of operations: platform-based or rig-generated activities associated directly with *exploration* for or *production* of offshore natural gas or oil; and related activities associated with the *transmission* of offshore gas or oil by *pipeline* to on-shore or off-shore reception facilities. (Pollution arising from the transport of offshore oil by *tanker* is classified as ship-generated pollution.) Normally seabed-related pollution takes the form of oil slicks, as in the majority of vessel-source pollution incidents. The most serious spillages are those caused by blow-outs, which can assume "catastrophic" proportions, as in the case of the 1979 IXTOC I blow-out in the Gulf of Mexico. This form of marine pollution comes under the primary responsibility of the coastal state, since it has sovereign rights to the

continental shelf and its resources under customary international law[19], as well as the exclusive right to regulate the construction, operation and use of "artificial islands, installations and structures" under the emerging regime of the continental shelf[20]. Marine pollution control related to offshore operations is, in fact, virtually the unilateral task of the coastal (licensing) state. One looks not to international controls but to national regulations governing the environmental aspects of such operations in all phases, from preliminary exploration to final, full-scale production and distribution. The international aspect of this sector of marine pollution control may be limited to questions of liability and compensation with respect to neighbouring states which may suffer environmental damage and incur clean-up costs as a direct result of a major spillage in the offshore areas of the same region[21]. Such issues are likely to be resolved by bilateral or regional, rather than global, arrangements.

5. Pollution from Deep-Ocean Mining

Deep-ocean mining has not yet begun. When it does begin, it seems likely to create a risk of degradation to the marine environment which may or may not fall within the current concept of marine pollution. Most mining activities, scheduled to begin later in the 1980s, are expected to be concentrated in deep-ocean areas beyond limits of national jurisdiction in the so-called "Area"[22]. In this event the environmental regulation of such operations will come under the jurisdiction of the International Seabed Authority[23], but only after the United Nations Law of the Sea Convention comes into force. Questions regarding interim arrangements before that date are politically controversial and still unresolved, and the possibility of unilateral national initiatives before then, beyond limits of national jurisdiction, cannot be ruled out[24]. Moreover, it is entirely probable that deep-ocean mining might be begun within the next few years *within* limits of national jurisdiction[25]. Added to these political uncertainties is the problem that very little is known about the extent of probable disruptions to the affected areas of the ocean environment as a result of deep-ocean mining. Accordingly, it is extremely difficult to anticipate the most probable course of events, to envisage clearly where to address one's concerns, and to evaluate the harms that might occur.

E. Legal Approaches to Marine Pollution Control

When the *Torrey Canyon* went aground in 1967, two conventions had already been adopted concerning marine pollution; one, the 1962 Convention on Nuclear Powered Ships was inapplicable[26], whilst the other, the 1954 Convention for the Prevention of Pollution of the Seas by Oil, was ineffective[27]. Observers unanimously agreed that bringing evidence of a prohibited oil discharge was impossible in most cases. All efforts before 1967 can, therefore, be regarded as belonging to a "prehistoric" era of marine pollution control. Harbours, oyster beds and some vulnerable fishing grounds were the only marine areas subject to adequate protection under old-fashioned national legislation[28].

As to international control, the real work started in 1967, when the British and French governments, alarmed by the considerable damage they had suffered from the *Torrey Canyon* oil spill, convened an international conference in order to address this issue specifically. Since then much of the legal development that has

taken place in the field of marine pollution control has, very conspicuously, been accelerated, facilitated or even initiated as a direct result of other comparably serious pollution incidents that have threatened to pose a serious hazard[29] or cause major damage.

Most of the law which has developed in response to marine pollution problems can be characterized as either *preventive* or *remedial* in its thrust. In most cases, either the primary purpose is to impose reasonable precautions to reduce the frequency of incidents, and to minimize the damage before they occur; or it is to provide for the availability of compensatory remedies and other forms of relief to the victims of such injuries after they occur. The remedial branch of marine pollution law is based on traditional concepts of fault, negligence, or criminal culpability, or on strict liability or no-fault systems, sometimes accompanied by provisions for victim entitlement to compensation through the establishment of a fund for that purpose. The preventive branch of marine pollution law is more directly concerned with the imposition of regulatory controls, through research or review requirements, higher technical standards, operational conditions, professional criteria, licensing techniques, guarantees of accountability, and other modes of management. As ocean management in general and marine pollution control in particular become more sophisticated, new law may become increasingly "regulatory" in orientation.

Each of these two areas of marine pollution law has certain limitations inherent in it and raises difficult problems in legal or diplomatic practice. The traditional liability approach to marine pollution remedies is, of course, complicated by evidentiary difficulties, as noted above, especially where the pollutant in question is non-visible or the sources are too numerous or too widely scattered to permit the attribution of an injury to the action of any one single polluter. Moreover, the scale of catastrophic spillages of oil, whether by tanker or offshore rig, may be so enormous that the traditional system of individual liability is simply unable to cope with the remedial problems, even with the assistance of third party liability insurance. Literally, hundreds of victims might be able to show in a given situation that they had suffered a loss as a result of a catastrophic spillage.

Theoretically, it is possible to create the concept of *collective liability* for an entire class of polluters and to devise a system of shared financial responsibility for injuries attributable to the class as a whole. But politically, and sometimes legally, such a concept or system is extremely difficult to apply in practice. Moreover, under a doctrine of collective liability it would be more difficult than under that of individual liability for a plaintiff to show that his injury is the result of wrongful acts or omissions, since a system of collective responsibility would be unconscionable unless it could be shown that all members of the class were in some degree culpable.

For these and similar reasons it has been found easier, and perhaps more equitable, in certain marine pollution situations, to develop no-fault compensation schemes, whereby the victim merely has to prove his injury, in the relevant circumstances, in order to become entitled to a compensatory award from a fund established for that purpose. But even this kind of system, whether voluntary or mandatory, is subject to severe limitations. The chief deficiency of a compensation fund approach is that it is simply unfeasible and unacceptable in practice unless certain maximum (ceiling)

limits are prescribed (as in the case of certain areas of non-indemnity insurance), and the existence of such limits means, of course, that in many situations, some or most of the victims cannot be fully compensated or reimbursed.

Quite different kinds of problems tend to arise in the preventive branch of marine pollution law. One familiar difficulty is that of *extraterritoriality:* how to enforce national or international precautionary measures outside national territorial limits. The traditional response has been by reference to flag state jurisdiction, but, as we have seen, the dependence on flag states has tended to frustrate rather than facilitate the application of measures beyond territorial limits. Three innovations have emerged to reduce this problem, but each is limited in its scope and effectiveness. First, the 1969 Intervention Convention[30] provides for unilateral intervention by a contracting state in the case of an emergency, such as a potentially catastrophic spillage, involving the vessel of another contracting state, if it takes place outside its territorial limits in the high seas, and it permits drastic precautionary measures to be taken by the intervening state if there is sufficient risk ("grave and imminent danger") of serious environmental damage to the latter's coastal interests. But the terms of the Convention are rather narrowly restricted[31], and, of course, they are binding only on those states that have chosen to ratify the instrument[32]. Supportive language is contained in the Draft Convention on the Law of the Sea, and fortunately it is less restrictively worded[33].

Second, the dependency on flag state initiatives has been reduced by the extension of coastal state jurisdiction out to the seaward (200-mile) limits of the exclusive economic zone[34], but the entire question of the coastal state's authority over marine pollution control within these extended limits is complicated and clouded by uncertainties due to the nuanced trade-off diplomacy engaged In by the delegations at UNCLOS III[35].

Third, UNCLOS III further diminished the traditional exclusivity of flag state jurisdiction, as noted above, by introducing the additional (concurrent) resort to port state jurisdiction for purposes of vessel-source pollution control. But the effectiveness of this innovation depends, of course, on a particular set of fortuitous circumstances: namely, that the culprit vessel after the incident chooses, or is obliged, to enter the port of a state which is able and willing to initiate investigations.

Even where preventive and remedial elements are combined in a single approach to marine pollution control, as in the case of joint contingency plans[36], a variety of political and technical difficulties often prevent a successful outcome of negotiations or result in quarrels over interpretation at a later stage. Negotiations of a joint contingency plan concerning oil spills — which might apply to maritime casualties, blow-outs, pipeline ruptures, and certain kinds of land-based pollution incidents — sometime break down over the issue of how to share the costs of implementation, because of the unforeseeability of all possible consequences, or result in quarrels over the allocation of expenses involved in the implementation of a (generally worded) provision[37].

But the choice of a legal strategy for marine pollution control at the global level, and sometimes at the regional level also, is sometimes less a matter of choosing between the preventive and the remedial aspects than a matter of determining the ap-

propriate level of norm-making. General, hortatory or aspirational, language — for example, urging states to co-operate or requiring them to enter into negotiations, rather than obligating them to take a concrete form of action or creating a basis of liability — has become very common in the forum of law-making diplomacy. The traditionally trained lawyer, particularly one specializing in private law, is inclined to minimize the legal significance of such language, even when it is contained in what is admitted to be a legally binding treaty. Examples of this "soft law" approach to legal development abound in the context of marine pollution. The truth is that even if such language seems vague by the standards of interpretation applied to contracts or statutes in national tribunals, "soft law" language might be highly influential on future legal developments at the governmental or intergovernmental level — and it is to this level, and not that of national tribunals, that such language is chiefly directed. Normally, "soft law" is written by lawyer-diplomats for lawyer-diplomats, and has to be interpreted from that perspective.

At the global level, the most famous "soft law" of marine pollution control, the 1972 Stockholm Declaration on the Human Environment and the accompanying Action Plan, has been extremely influential on the development of international environmental law since 1972, despite the fact (or because of the fact?) that it did not originate in a legally binding treaty instrument.

Although the legal status of the Stockholm Declaration is controversial, most commentators tend to the view that its Principle 21 reflects the existing state of customary international law on the general obligation of states to take preventive measures for the protection of the environment, marine and otherwise:

> "States have, in accordance with the Charter of the United Nations and the principles of international law, the sovereign right to exploit their own resources pursuant to their own environmental policies, and *the responsibility to ensure that activities within their jurisdiction or control do not cause damage to the environment of other States or of areas beyond the limits of national jurisdiction"*[emphasis added].

Previously, a preparatory meeting of the Inter-Governmental Working Group on Marine Pollution, held at Ottawa in 1971, adopted a Statement of Objectives and 23 General Principles for the Assessment and Control of Marine Pollution. These objectives and principles were endorsed by the Stockholm Conference itself. In addition, Principle 7 of the Declaration and several detailed recommendations of the Stockholm Action Plan dealt specifically with the need to prevent and control marine pollution[38].

In the early 1970s the UN Seabed Committee and its "successor", the Third UN Conference on the Law of the Sea (UNCLOS III), proceeded among other things to "develop" some, indeed most, of these "soft law" principles or recommendations emanating from the UN Stockholm Conference on the Human Environment. First, the 1980 Draft Convention on the Law of the Sea, in Article 192, gives expression to the general environmental responsibility of states in the ocean:

> *Article 192*
>
> States have the obligation to protect and preserve the marine environment.

Part XII of the UNCLOS III Convention contains a number of other, equally general, obligations to this end: for example,

Article 194

1. States shall take all necessary measures consistent with this Convention to prevent, reduce and control pollution of the marine environment from any source using for this purpose the best practicable means at their disposal and in accordance with their capabilities, individually or jointly as appropriate, and they shall endeavour to harmonize their policies in this connexion.

2. States shall take all necessary measures to ensure that activities under their jurisdiction or control are so conducted, that they do not cause damage by pollution to other States and their environment, and that pollution arising from incidents or activities under their jurisdiction or control does not spread beyond the areas where they exercise sovereign rights in accordance with this Convention.

Article 195

In taking measures to prevent, reduce and control pollution of the marine environment, States shall act so as not to transfer, directly or indirectly, damage or hazards from one area to another or transform one type of pollution into another.

Many of the more specific provisions on marine pollution in the Draft Convention of UNCLOS III are even better examples of "soft law" provisions[39], and almost all represent preventive approaches to marine pollution control.

F. The Need for Legal Development

The challenge of the 1980s lies in finding ways of converting general marine pollution control provisions into more concrete, more significant legal and institutional developments at global, regional and national levels. In the following five sections of this chapter the recent history of marine pollution control arrangements is reviewed at some length with a view to laying the foundation for an assessment of future requirements following the conclusion of UNCLOS III.

II Pollution from Vessels*

Jan Schneider

A. Scope of the General Problem

Codification and progressive development of international law for the prevention and control of pollution from vessels is most often seen as a battle between environmentalists and coastal interests on the one hand, and shipping concerns and oil companies on the other. Thus construing the situation in combative terms loses sight of the real problem, which is to ensure that international maritime law makes adequate provision for the protection and preservation of the marine environment, and that the procedures and costs of international shipping realistically reflect the environmental threats inherent therein. In an age when international commerce has become critically dependent on transport of goods by sea, it would be naive for environmentalists to try to promote legal regimes which would cripple international shipping or render it prohibitively expensive. There are, however, reasonable solutions capable of effecting a workable balance between both sets of competing interests. An attempt will be made in this section to show the means and provisions by which the world community is groping toward such effective compromise and the areas in which the resulting legal regime remains incomplete or deficient.

In recent years more and much larger ships have been transporting increasingly dangerous cargoes over greater distances. More accidents are threatened, and more occur. Just twenty years ago, there were about 36,000 ships of at least 100 gross registered tons (grt), totaling about 125 million grt; since then the number of ships has increased to about 70,000 representing a total of more than 400 million grt[40]. In 1954 the world tanker fleet consisted of almost 3,500 ships totalling 37 million deadweight tons (dwt), while by 1977 there were nearly 7,000 of them in operation, aggregating approximately 340 million dwt[41]. As to size of tankers, during the same period the largest tanker in service went from around 30,000 dwt, to over 500,0000 dwt[42]. In little more than twenty years, the amount of oil transported by sea increased by 700 percent: from 250 million tons to more than 1,700 million[43]. Moreover, the problem is not only hydrocarbons. Two decades ago, not one container ship, chemical carrier, or gas carrier was in operation, whereas today there are many ships carrying such potentially dangerous cargoes in the world's sea lanes[44].

As more and larger ships carry greater amounts of hazardous cargo, the potential consequences of accidents have escalated proportionately. It was a major disaster in 1967 when the *Torrey Canyon,* whose dwt of 118,285 ranked her as the third largest in the world at the time, struck a reef and split while carrying 880,000 barrels of crude oil[45]. In 1974, the dramatic loss of the *Metula* in the Straits of Magellan and the grounding of the *Showa Maru* highlighted the dangers of transport of oil in en-

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vironmentally sensitive areas[46]. Shortly, thereafter, a whole series of tanker accidents made headlines: the *Argo Merchant, Sansinena, Oswego Peace, Olympic Games, Daphne, Grand Zenith, Barcola, Mary Ann, Universal Leader* and *Irenes Challenger* all met disaster during the single two-month period of December 1976 and January 1977[47]. The next year produced probably the most dramatic of all tanker casualties when the four-year-old, 230,000 dwt *Amoco Cadiz* broke in half off the beaches of Brittany on March 16, 1978, spilling its full cargo of crude oil[48].

Nor do accidents, although dramatic, constitute the only threat to the marine environment from vessels: operational discharges have also accumulated to cause significant pollution. The tanks in which oil is transported are usually washed while a tanker is returning to its loading port, and some of the tanks are normally filled with water as ballast to make the ship low enough in the water to achieve maneuverability. In the past, the resulting mixtures of oil and water were pumped directly into the sea before a new cargo was taken on[49].

Several available strategies have long been recognized as practicable in controlling these and other pollution problems. Most notable are the "load on-top" (LOT) technique (whereby tank washings and ballast are not discharged, but rather allowed to settle out)[50], segregated ballast tanks (SBT), and double bottoms or hulls[51]. States, however, have been somewhat slow about requiring such measures for new ships, and they have been even more reluctant to mandate retrofitting of second-hand tankers.

Pollution problems and remedies, in general, are viewed by states from differing perspectives. Coastal states, of course, have the most direct interest in protecting their coastal resources and related environment. Many coastal states, however, are also important port states, heavily dependent upon shipping benefits, and therefore influenced by conflicting motivations. Developing countries are customarily the purchasers of second-hand ships, and they are thus usually somewhat opposed to increased stringency in vessel construction and design standards or required retrofitting; nevertheless, the consequences of a major spillage can be extremely serious in developing areas, if only because there are fewer resources available for dealing with them. As a result of such tensions, what has occurred, and is prevalent at every level of international decision-making, is a complex and multivariate process of reconciliation and balancing of interests. The outcomes have been reflected in a whole series of multilateral and regional agreements and in a panorama of state practice, or unilateral actions, and they continue to be the subject of multiple and interpenetrating ongoing processes of codification and progressive development of international law.

B. Responses to Particular Problems

1. Introduction

Over time, states have been groping, collectively and individually, toward an effective accommodation of interests and the crystallization of an agreed regime for the prevention and control of vessel-source pollution. They have proceeded by means of negotiation of a variety of international agreements and measures directed toward particular marine pollution problems, while simultaneously attempting to develop

an overall framework in the context of the "umbrella treaty" of the Third United Nations Conference on the Law of the Sea.

Even when fewer and smaller ships carried less dangerous cargo, they posed a serious threat to the marine environment, and states felt the need to establish at least minimum international standards for environmental protection and preservation. As the risk escalated, the perception of hazard has been clarified, and the will to take effective action substantially reinforced. The focus of attention has been largely on crude oil, which although not necessarily the most dangerous substance carried by sea, is the most significant in tonnage terms. In recent years, both the scope of international comprehension and the extent of measures to deal with marine pollution problems have increased markedly. Today there are a large number of global and regional agreements on the subject, in addition to the relevant national laws.

2. Global Agreements

The current legal thrust for preventing vessel-source pollution has as its initial landmark the 1954 Convention for the Prevention of Pollution of the Seas by Oil[52]. The basic rule of this convention is that discharges containing more than one hundred parts per million of "persistent oils" (a phrase that includes crude and heavy fuel oil but excludes the highly toxic refined products known as "light oils") must occur outside "prohibited zones" — generally speaking, the area lying within fifty miles of the nearest coast. In 1969, amendments were passed to the 1954 Convention[53], which for the first time applied discharge standards to vessels even when outside the narrow prohibited zones. In 1971 there was another set of amendments, which imposed certain standards on tank subdivision and stability[54].

This initial measure received early bolstering by some provisions of the 1958 Convention on the High Seas[55], one of the four instruments emerging from the First United Nations Conference on the Law of the Sea[56]. Article 24 of the High Seas Convention committed states to draw up regulations to prevent vessel-source and other pollution of the seas by oil[51]. Around the same time, states also dealt with certain related liability measures in the 1957 International Convention Relating to the Limitation of the Liability of Owners of Sea-going Ships[58] and the 1962 Convention on the Liability of Operators of Nuclear Ships[59].

The next landmark in the development of international law to prevent and control vessel-source pollution was the International Legal Conference on Marine Pollution Damage, convened by the Inter-governmental Maritime Consultative Organization (IMCO) in Brussels from November 10 to 19, 1969. Two important agreements were prepared and opened for signature by that conference: the International Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties (IMCO "Public Law" or "Intervention" Convention)[60] and the International Convention on Civil Liability for Oil Pollution Damage ("Private Law" or "Civil Liability Convention)[61]. The central thrust of the former is to allow states to take necessary measures on the high seas to prevent, mitigate, or eliminate "grave and imminent danger to their coastline or related interests"[62] from oil pollution following a maritime casualty or acts related to such a casualty. The Private Law Convention established a limit and provided for compensation (up to a maximum of approx-

imately $ 17 million per incident) for damage suffered from pollution resulting from the escape or discharge of oil from ships[63].

In order to provide an effective means to secure compensation for such damage, states in 1971 followed up their earlier work in IMCO and drafted the International Convention on the Establishment of an International Fund for Compensation for Oil Pollution Damage (IMCO "Fund" Convention)[64]. This subsequent convention more than doubles the maximum amount available under the 1969 Civil Liability Convention, up to $ 36 million per incident (which seems likely to be doubled by a decision of the Governing Body of the International Fund). In the manner of international agreements, however, the Fund Convention took most of a decade to come into force, and in the meantime the international community limped along with two ad hoc voluntary arrangements on the part of oil companies: the Tanker Owners Voluntary Agreement Concerning Liability for Oil Pollution (TOVALOP)[65] and the Contract Regarding an Interim Supplement to Tanker Liability for Oil Pollution (CRISTAL)[66]. Even now that the Fund Convention is in force, these agreements will continue to provide certain additional amounts of financial resources and coverage supplementary to the conventional regime[67].

At the global or broadly multilateral level, the next landmark development in the control of vessel-source pollution was the International Convention for the Prevention of Pollution from Ships ("MARPOL" Convention)[68], which was produced in London at an international conference sponsored by IMCO. After nine years however, this important treaty was not yet in force. Basically, the MARPOL Convention extends the 1954 IMCO Convention, discussed above, with the objective of achieving "the complete elimination of intentional pollution of the marine environment by oil and other harmful substances and the minimization of accidental discharge of such substances"[69]. For this purpose, it both established generally applicable norms and expressly allows for more stringent pollution control standards in certain internationally designated "special areas"[70]. And one year later, the MARPOL Convention was complemented by the new International Convention for the Safety of Life at Sea ("SOLAS" Convention)[71], which was adopted at a conference sponsored by IMCO in 1974. Dealing with such matters as partitioning and stability of cargoes, machinery and electrical installations, fire prevention, detection and extinction, life-saving appliances, radio and navigational equipment, etc., the new SOLAS Convention bolsters the aim of the MARPOL Convention to combat accidental as well as operational pollution.

Despite these emerging conventions, the series of tanker accidents in the winter of 1976-1977 led concerned parties to wonder whether indeed enough had been done. Most of these accidents had taken place off the coast of the United States, and that government took the initiative in proposing to IMCO a series of measures designed to improve tanker safety. The result was that at the International Conference on Tanker Safety and Pollution Prevention convened by IMCO in February 1978, two more instruments were adopted, commonly referred to as the MARPOL Protocol[72] and the SOLAS Protocol[73].

While the MARPOL Convention required segregated ballast tanks (SBT) only on new tankers of more than 70,000 dwt, the new MARPOL Protocol extends the requirement to include all new crude oil tankers of more than 20,000 dwt and all new pro-

duct carriers of more than 30,000 dwt; all existing crude oil carriers of more than 40,000 dwt will have to be operated with either SBT, clean ballast tanks (CBT), or approved crude oil washing (COW); all existing product tankers of more than 40,000 dwt will have to have either SBT or CBT from entry into force of the MARPOL Protocol[74]. Under the 1974 SOLAS Convention, requirements for inert gas systems were limited to new oil tankers of more than 100,000 dwt and new combination carriers over 50,000 dwt. Under the new SOLAS Protocol these systems will basically be required on all new and existing oil tankers of more than 20,000 dwt and all existing product carriers larger than 40,000 dwt; the SOLAS Protocol also contains certain other more stringent requirements with regard to stearing gear and to radio and collision avoidance aids[75]. The question is, however, as with the parent conventions themselves, when will all these new and improved measures actually come into force[76]?

3. Regional Agreements

In the absence of global action, or often as a supplement thereto, states have also proceeded to address themselves to problems of ship-generated pollution at the regional level[77]. Not surprisingly, these efforts have tended to be concentrated in areas where regional institutions and co-operation on other matters are already the strongest. Conventions have been drawn up for the Northeast Atlantic, Baltic, Mediterranean, Gulf, and West African regions.

In the Northeast Atlantic region, the first important step was the Agreement concerning Pollution of the North Sea by Oil, which was signed at Bonn on June 9, 1969[78]. The Bonn Agreement — between the governments of Belgium, Denmark, France, the Federal Republic of Germany, the Netherlands, Norway, Sweden, and the United Kingdom — among other things divided the North Sea into zones within which the contracting party or parties responsible were to make assessments to oil pollution casualties and take certain ameliorative actions[79].

In the Baltic area, states have chosen to establish a comprehensive blueprint and fill in the details as appropriate. The Convention on the Protection of the Marine Environment of the Baltic Sea Area (Helsinki Convention)[80] of 1974 seeks to provide a framework for controlling all source of marine pollution — ship accidents and dumping — and also land-based, seabed-source and airborne pollution. In addition, two subregional agreements should be mentioned in connection with the Helsinki Convention: the Agreement Concerning Co-operation in Measures to Deal with Pollution of the Sea by Oil (Copenhagen Agreement)[81] and the Convention on the Protection of the Environment[82], both of which were concluded among the four Nordic countries. The former deals explicity with vessel-source pollution, and the latter established a broad framework of environmental co-operation encompassing, among other things, pollution of the sea.

For the Mediterranean, under the auspices of the Regional Seas Programme of the United Nations Environment Programme (UNEP), the 18 concerned countries in 1976 adopted a Convention for the Protection of the Mediterranean Sea Against Pollution (Barcelona Convention)[83]. They concurrently concluded a Protocol on Prevention of Pollution of the Mediterranean Sea by Dumping from Ships and Aircraft[84] and a Protocol concerning Co-operation in Combating Pollution of the Mediterranean Sea by Oil and Other Harmful Substances in Cases of Emergency[85]. These instruments all

entered into force on February 12, 1978 — a notable political as well as environmental achievement.

Following the model of the Barcelona Convention and also under the auspices of the Regional Seas Programme, the countries bordering the "Gulf area" (*i.e.,* that body of water variously known as the Persian or Arabic Gulf) in 1978 adopted the Kuwait Regional Convention for Co-operation in the Protection of the Marine Environment from Pollution[86] and a Protocol concerning Co-operation in Combating Pollution by Oil and Harmful Substances in Cases of Emergency[87]. Their provisions are more or less modelled on those previously emerging from Barcelona, and as with the Mediterranean effort, the Gulf countries are now also working on additional protocols on dumping and on seabed-source and land-based pollution.

In March 1981, at a conference convened by UNEP at Abidjan, another region followed a similar route by adopting the Convention for Co-operation in the Protection and Development of the Marine and Coastal Environment of the West and Central African Region and the Protocol concerning Co-operation in Combating Pollution in Cases of Emergency. Having benefited from the Barcelona and Kuwait experience, the Abidjan Convention may be the most highly developed of all the conventions adopted under UNEP's Regional Seas Programme.

4 National Laws

Discontent with the pace and content of global and regional efforts to prevent and control ship-generated pollution has motivated some states to "go it alone" on certain issues. This may be heralded as state practice promoting the progressive development of international law, or alternatively condemned as unilateral action in derrogation of international norms, depending on one's perspective and views of the outcome. The point is, however, that recent years have seen the passage of certain pieces of national legislation and accompanying regulations that have profound international legal significance. Dialogue and activity in this area have been most intense in North America, although similar measures have also been taken elsewhere.

A major fury was generated in North America over the passage of the Canadian Arctic Waters Pollution Prevention Act in 1970[88]. While undoubtedly influenced by political motivations, the Arctic Waters Act is clearly of great importance as an environmental precedent in its language and provisions. When Canada passed this legislation inaugurating a 100-mile antipollution zone off its northern coast[89], the United States protested strongly that "[international law provides no basis for these proposed unilateral extensions of jurisdictions on the high seas, and the United States can neither accept nor acquiesce in the assertion of such jurisdiction"[90]. Canada replied equally tersely that "[t]he Canadian Government is unable to accept the views of the United States Government concerning the Arctic Waters Pollution Prevention Bill and the amendments"[91]. And several months later, when Canada created new fishing zones within lines not enclosed by previous territorial sea baselines, its neighbour to the south again vehemently protested that "]t]he United States regards this unilateral act as totally without foundation in international law"[92].

This sharp exchange notwithstanding, the United States has itself taken far-reaching measures to protect its marine environment from vessel-generated pollution. Although the Ports and Waterways Safety Act of 1972[93] did not prescribe a specific coastal zone, it did have a radical and far-reaching impact on international maritime commerce. The Act, *inter alia,* allows the Secretary of the Department in which the Coast Guard is operating to control vessel traffic by promulgating special standards for all ships entering US ports, harbours, and other navigable waters. This includes the setting of standards with regard to matters of design, construction, manning and equipment (DCME standards), and the operation and movement of vessels.

More recently, having extended its own fisheries jurisdiction to 200 miles in 1976[94], the United States expanded certain antipollution competence with respect to vessels in the same zone[95]. The Clean Water Act of 1977[96] extended various provisions of § 311 of the earlier Federal Water Pollution Control Act that dealt with oil and hazardous substances liability. It declared the policy of the United States that there should be "no discharges" of oil or hazardous substances by vessels not only in the territorial sea and the contiguous zone, but also

> in connection with activities under the Outer Continental Shelf Lands Act or the Deepwater Port Act of 1974, or which may affect natural resources belonging to, appertaining to, or under the exclusive management authority of the United States (including resources under the Fishery Conservation and Management Act of 1976)[97].

Many other examples of vessel-source marine pollution control legislation have, of course, been enacted in other countries[98], showing the extend of international concern with this problem around the world.

C. Negotiation of "Umbrella Treaty" Provisions At UNCLOS III

While addressing themselves to particular marine pollution problems at various levels of decision-making, states have also attempted to establish a broad framework in this as in other areas within the context of the Third United Nations Conference on the Law of the Sea, which is responsible for development of a new "umbrella treaty" for the oceans. Vessel-source pollution was a major concern of the Third Committee of UNCLOS III, whose mandate included preparation of draft articles on protection and preservation of the marine environment. When these negotiations began a decade ago with the creation of the UN Committee on the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction, there was a dearth of conventional or customary international law concerning the prevention and control of pollution of the marine environment. The provisions drafted by UNCLOS III have gone a long way toward filling this void, both in and of themselves and also in reflecting and giving impetus to the particular agreements and other measures discussed earlier.

In order to understand the specific provisions prepared by UNCLOS III on vessel-source pollution, it is necessary to have a general idea of the overall structure of the marine pollution text[99]. As a foundation, states have agreed on certain general pro-

visions: they include a general obligation to refrain from polluting the marine environment; a commitment to take necessary measures to prevent, reduce and control marine pollution; a duty not to transform one type of damage of pollution into another; and a provision against introduction of dangerous technologies or alien species into the marine environment[100]. They have also agreed on the need for certain measures of global and regional co-operation and technical assistance[101], and they have for the first time in general treaty form undertaken to assume positive legal responsibility to co-operate in international ocean monitoring programmes and to assess the impacts of their activities on the marine environment[102]. Then follow provisions for international rules and national legislation, as well as enforcement measures, to prevent, reduce and control marine pollution from all the various sources, including land-based sources, national and international seabed activities, dumping, pollution from vessels, and pollution from or through the atmosphere[103].

The standard-setting articles dealing with dumping and other forms of pollution from ships were especially controversial and are extremely complicated. In skeletal form, the regime looks as follows: first, in the territorial sea, coastal states may exercise their sovereignty to establish anti-pollution laws and regulations, provided that they "shall not apply to the design, construction, manning or equipment of foreign ships unless they are giving effect to generally accepted international rules and standards"[104]; second, in the exclusive economic zone, they can prevent dumping and may establish certain other laws and regulations giving effect to generally accepted international rules and standards for ship-generated pollution[105]; and third, there are certain supplementary provisions for "special areas" within the economic zone with particularly sensitive oceanographical and ecological conditions[106].

In the area of enforcement competence, UNCLOS III has, as noted earlier[107], developed several important new concepts, in particular, universal port state enforcement. While traditionally the province almost exclusively of flag states (states of registry), new enforcement powers are now recognized to lie with port and coastal states also. Port states will be empowered to undertake certain enforcement procedures in respect of discharge violations — even those occuring outside their internal waters, territorial seas, or economic zones. Also, under certain carefully delineated circumstances, coastal states will be able to cause proceedings and other measures to be taken in respect of violations of national laws and regulations or applicable international rules and standards.

Enforcement powers are, however, to be subject to certain highly detailed safeguards to make sure that freedom of navigation is maintained along with efforts to ensure environmental protection and preservation[108]. These safeguards include a basic provision for non-discrimination against foreign vessels[109], flag state preemption rights (requiring suspension of proceedings by other states[110]), monetary penalties only for violations beyond internal waters except in cases of "wilful and serious" acts of pollution in the territorial sea[111], and a special article on safeguards with respect to straits used for international navigation[112]. There is even an overall sovereign immunity exception to the anti-pollution provisions that exempts any warship, naval auxiliary, or other vessel or aircraft owned or operated by a state and used on government noncommercial service; states undertake, however, to ensure

that such craft, "so far as is reasonable and practicable", act in a manner consistent with the new umbrella treaty[113].

As noted above[114], for situations where an incident has taken place that results in "pollution or threat of pollution", the Committee III text contains a provision that allows the threatened coastal state to take certain forms of preventive action unilaterally beyond its territorial sea[115]. Basically, this provision may be interpreted as more liberal than the 1969 IMCO Public Law Convention discussed earlier, since it keeps the requirement of proportionality of the response to the actual or threatened damage but speaks only of pollution or threat thereof in place of the prior "grave and imminent danger"[116]. As to liability and compensation for damage that nevertheless results, states are to be liable in accordance with international law, and they shall "co-operate in the implementation of existing international law and the further development of international law relating to responsibility and liability for the assessment of and compensation for damage and the settlement of related disputes, as well as, where appropriate, development of criteria and procedures for payment of adequate compensation such as compulsory insurance or compensation funds"[117]. In this last respect obviously the proposed "umbrella treaty" will leave an important area uncovered for future development.

Despite the positive contributions of UNCLOS III in the progressive development of international environmental law, it must also be recognized that there have been some steps backwards. On the negative side, probably the most objectionable provision on vessel-source pollution, from the environmental perspective, is that forbidding a coastal state from making laws and regulations in its own territorial sea relating to "design, construction, manning or equipment" of foreign ships, as noted above[118]. Certain states maintain that this is a weakening of existing sovereign prerogatives and would render illegal certain provisions of existing national legislation (including certain provisions of the US Ports and Waterways Safety Programme[119]). Similarly distasteful, because of its implications with respect to the nature and extent of sovereignty in the territorial sea, is the provision on innocent passage which states that passage is prejudicial only if the foreign ship engages in, *inter alia,* "[a]ny act of wilful and serious pollution, contrary to this Convention"[120]. This "wilful and serious" qualification is also repeated in a later article that restricts penalties in the territorial sea to monetary ones, except in cases of such violations (again contrary to several countries' existing national legislation)[121].

Given the historical role of the notion of a territorial sea, the issue at stake in these seemingly retrogressive steps could be argued to be whether or not environmental integrity is a basic tenet of national self-protection. If it is recognized, however, that the new "umbrella treaty" will in many respects entirely revise old notions and precedents — extending resource jurisdiction throughout the full extent of the exclusive economic zone while, at the same time, preserving and modernizing the legal protection of navigational freedoms — then the implications for the development of international environmental law are not nearly so serious. On such an understanding it might be concluded that states have, on the one hand, disavowed certain powers previously exercised in a largely anachronistic, arbitrary, narrow belt of immediately proximate waters, but have, on the other hand, gained for the first time in treaty form both a set of more expansive and comprehensive general prin-

ciples on the protection and preservation of the marine environment[122] and the many new practical and effective port and coastal state rights of prescription and enforcement discussed above.

Finally, a word should be added about ice-covered areas, which are extra-special and are set out in a separate and distinct section of the proposed UNCLOS text[123]. In the article in question, states have in effect acknowledged the validity and uniqueness of the Canadian Arctic Waters Pollution Prevention Act mentioned earlier. The article says that the coastal state in question has the right to establish and enforce nondiscriminatory laws and regulations for the prevention, reduction, and control of pollution from vessels in ice-covered areas in the exclusive economic zone "where particularly severe climatic conditions and the presence of ice-covering such areas for most of the year create obstructions or exceptional hazards to navigation, and pollution of the marine environment could cause major harm to or irreversible disturbance of the ecological balance"[124].

What conclusion can be drawn from the trends in development of vessel anti-pollution law? When the present law of the sea negotiations began a decade ago, there was very little conventional or customary international law concerning the protection and preservation of the marine environment. During this period, the framework developed at UNCLOS III, stimulating and stimulated by an interlocking and inter-penetrating network of global and regional agreements and of national measures, has gone a long way toward filling this void in international law. The legal dynamics have been responsive both to newly perceived environmental imperatives and to the age-old demands for freedom of navigation, seeking to satisfy the former while providing clear and livable norms for the latter. Many of the progressive developments set forth on paper have yet to be achieved in actual practice, through elaboration of the basic framework and entry into force of various legal instruments, but at least the process is well begun.

D. Processes and Mechanisms for the Protection and Preservation of the Marine Environment

It should now be obvious that there has been, at least on paper, in recent years an impressive elaboration of international rules, standards, and recommended practices and procedures for the prevention, reduction, and control of pollution from vessels. Questions arise about the processes and mechanisms by which the international community has managed to balance competing interests in arriving at these arrangements, and through which their goals, standards, and hortatory recommendations may be further developed and actually effected in practical outcomes. What is immediately apparent is that the oft-asserted, restrictive, two-tier approach to global decision-making — which attempts to draw a sharp distinction between international law and national legal processes — is inadequate to describe the complexity of the current process of legal development. Rather, states and other actors on the international plane have been working at a variety of levels of decision-making and action to create and implement a viable and effective international legal regime for ship-generated pollution. The broad overview function being performed by general diplomatic negotiations at the Third United Nations Conference on the Law of the Sea has already been discussed at some length, and so

the present focus will be on other mechanisms and institutions. Broadly multilateral work on vessel-source pollution has been accomplished primarily in the forum of the Inter-governmental Maritime Consultative Organization (IMCO), while the United Nations Environment Programme (UNEP) has taken the lead in promoting regional measures; but other international and national mechanisms have been active as well.

Mention has already been made of the 1969 Public and Private Law Conventions, of the 1971 Fund Convention, and of the MARPOL and SOLAS Conventions and their protocols[125] — all of which were concluded at conferences sponsored by IMCO. In fact, the Organization, which came into existence in 1958 and now has 110 member states, in constitutionally charged with two main responsibilities: the promotion of safety at sea and the elimination of marine pollution from ships. Toward the latter of these ends, a special Marine Environment Protection Committee (MEPC), has been created within IMCO, and its Legal Committee has also been very active in these areas. Most recently, for example, in the wake of the *Amoco Cadiz* disaster, France brought a whole collection of considerations before the Marine Safety and Legal Committees, that ranged from a set of proposed amendments to the 1910 Salvage Convention to suggestions for new coastal state intervention provisions[126].

Similarly, there has already been some discussion of the activities of the United Nations Environment Programme (UNEP), which is constitutionally charged with a broad overview mandate as regards international environmental concerns (including the ocean environment), in connection with the conclusion of the Barcelona, Kuwait and Abidjan Conventions[127]. Actually, the launchings of these agreements were only crucial initial steps in UNEP's Regional Seas Programme, and extensive anti-pollution activities are now underway or scheduled for the Mediterranean, Gulf, and West African Regions. For the Mediterranean, for example, 86 marine laboratories in 16 countries of the region are now participating in a co-ordinated pollution-monitoring and research programme[128]. Moreover, these are only three of ten marine regions designated for special attention by UNEP[129]. All of these regional efforts require the co-operation of a number of neighbouring governments, marine facilities, and international organizations.

Differing institutional arrangements have been made in different contexts. Some have established a special regional commission to exercise overall supervision of implementation. Special reference might be made to the Helsinki Convention, which established an intergovernmental Baltic Marine Environment Protection Commission[130].

In a general overview of the world decision-making or constitutive process for the prevention and control of pollution from vessels, it is necessary to identify the basic functions this process must be expected to perform. First, there must be an adequate information or intelligence input to supply and store data from marine scientific research and descriptive information on the characteristics and operations of vessels engaged in maritime transit. Marine environmental monitoring, such as that sought through UNEP's Global Environmental Monitoring System (GEMS)[131], is an essential component of such an information system. Secondly, there must be promotion of the environmental perspective, since international commercial interests are already well represented. Several concerned states (including Canada, Spain,

the United States, and some others) have tried to play this role at UNCLOS III and in IMCO, many of the results of which have already been described; and, in addition, intergovernmental organizations such as UNEP, nongovernmental organizations (e.g., the Sierra Club, Center for Law and Social Policy, etc.), and individual activists have tried to assert suasive influence in many fora. Obviously, of course, there must also be systematic processes for the formulation and projection of policies and for their implementation in concrete circumstances. Both the results of prescription of international law concerning vessel-source pollution by articulated agreements and habitual behaviour and also the available means for application or enforcement of this law have been discussed at length earlier, and it is not the present intention to attempt a long grocery list of agreements or a catalogue of comparative regulations. The point here is merely to emphasize that these functions — as well as dispute-settlement, adjustment, and appraisal functions — are vital to a comprehensive decision-making process; and they must be provided for if the overall goal of a viable legal regime for the prevention, reduction, and control of pollution from vessels is to be achieved.

E. Problem Areas for Clarification or Progressive Development of International Law Related to Vessel-Source Pollution

Somewhat slowly, fitfully, and at many levels of interaction, the international community has been groping toward an effective legal regime for the protection and preservation of the marine environment, which will accommodate the legitimate demands for freedom of navigation. The essential concern is to ensure that the laws and regulations governing the transport of goods by sea adequately take into account the environmental risks and threats involved. A considerable amount of progress toward this end has been recorded to date, but many problem areas remain, where further clarification or progressive development of international law is vitally needed.

On the broadest multilateral level, certain defects in the marine pollution provisions that have emerged from the Law of the Sea Conference have already been pointed out. The problem of coastal state sovereignty in the territorial sea has been discussed and needs no further elaboration here.

Another critical general area where the sought-after UNCLOS "umbrella treaty" will clearly be deficient is that of responsibility and liability for environmental damage, particularly vessel-source pollution. Many of the recent agreements dealing with discharges and accidental pollution from vessels (including the Baltic, Mediterranean, Gulf and Abidjan Conventions)[132] exhort and pledge states to co-operate with a view to the development of a regime for liability and compensation. States have not done much better at UNCLOS III — merely committing themselves to "implementation of" existing law, "further development of" new international law on responsibility and liability, and "development of" criteria and procedures for payment of compensation[133]. There are, however, many gaping holes in the existing legal arrangements concerning substantive standards of fault and of no-fault or objective liability, limitation of liability, evidence requirements, dispute-settlement machinery and other matters which require comprehensive reassessment and supplementary treaty-making[134].

There are certain other glaring omissions or peculiarities in the UNCLOS III provisions on marine pollution. The draft provisions on pollution from vessels recognize the rights of states to establish laws and regulations having the same effect as, conforming and giving effect to, and/or implementing *"generally accepted* international rules and standards established through the competent international organization or general diplomatic conference"[135], and the provisions on enforcement by flag, port, and coastal states are replete with references to compliance with or violation of *"applicable* international rules and standards established through the competent international organization or general diplomatic conference" [emphasis added][136]. Moreover, the draft article on measures relating to maritime casuaties refers to the intervention rights of states under "international law, both customary and conventional"[137]. The trouble is, however, that it has not yet been established just what these terms mean and to what specific norms states are to be bound. If a convention, such as MARPOL, has been signed by many states assembled in conference and is being applied by major maritime nations, even before entry into force, can it not be presumed to be "generally accepted" — and/or "applicable"? And what is the role of customary international law, which certain states refuse to acknowledge at all, in this regard? The problem here is not one of mere words, but rather the very nature of the international legal system and the rights and responsibilities of states herein.

Another area of confusion or ambiguity in the UNCLOS III marine pollution articles relates to a long-standing problem from many points of view: flags of convenience. It is a well-known fact of international maritime law that ships have been registered in accommodating states (notably Panama, Liberia, Honduras, and Greece) to avoid certain national legislation, usually labor and tax laws, in the countries of their owners. The ships have very little genuine linkage to the states whose flags they fly, and the states have correspondingly little incentive or ability to apply various international rules and standards to them. And yet, there is an article in the UNCLOS text providing for suspension of proceedings by other states for pollution violations outside their territorial seas upon the taking of proceedings by the flag state "unless those proceedings relate to a case of major damage to the coastal State or the flag State in question has repeatedly disregarded its obligations to enforce effectively the applicable international rules and standards in respect of violations committed by its vessels"[138]. Clearly this leaves abundant room for discretion, and are the usual minimal fines an "effective" enforcement measure?

And to further complicate matters, while most of the provisions of the text refer simply to the "flag State"[139], in certain articles there are ambiguous references concerning information obligations and/or enforcement duties of a state with respect to "a vessel flying its flag *or* of its registry"[140] [emphasis added]. To most international lawyers, the notion that there could be a difference between the two would seem incredible, or at the very least unjustified by existing international law[141]. Even the suggestion that jurisdiction and responsibility can be shifted around and divided in the manner the quoted provisions would suggest — possibly adding a second level of convenience registries to flags of convenience — could be used to contravene any number of existing anti-pollution conventions, not to mention creating a gaping hole in the protective provisions of the new "umbrella treaty" itself.

An examination of the specific international agreements negotiated by states to deal with vessel-source pollution discloses that the main defect of many of these conventions is that they are not yet in force. As has been noted, at the time of this writing in 1981, the 1971 Amendments to the 1954 Pollution of the Sea by Oil Convention have failed to enter into force[142]; the landmark 1973 Pollution from Ships Convention has not yet come into force[143]; nor has the 1974 Safety of Life at Sea Convention[144] nor, of course, the 1978 Protocols to the MARPOL and SOLAS Conventions[145]. The 1969 IMCO "Public" and "Private Law" Convention took nearly six years to come into force, their respective dates being 6 May and 19 June 1975[146]; the 1971 Fund Convention took even longer, coming into force on 16 October 1978[147]; and the 1972 London Dumping Convention did somewhat better, entering into force on 30 August 1975[148]. To a large extent this delay is due to the weighted international decision-making system governing acceptance of these agreements. The Pollution from Ships Convention, for example, will not enter into force until "not less than 15 States, the combined merchant fleets of which constitute not less than fifty per cent of the gross tonnage of the world's merchant shipping" have become parties[149]. Moreover, certain marine pollution conventions have met outright rejection by important, environmentally minded governments, for a variety of reasons. The United States, for example, has not ratified either the 1969 IMCO Civil Liability Convention or the 1971 Fund Convention, hoping to "go it alone" with its more comprehensive national "Superfund" legislation[150]. Canada, for another example, has failed to ratify both these two conventions and also the 1969 IMCO Intervention Convention, deeming the limitation in the latter to "grave and imminent danger" to coasts more restrictive than customary norms.

Beyond such basic faults, this is not the place to attempt to catalogue all the deficiencies of and the compromises in maritime law provisions that render them undesirable from environmental or other points of view. Several design and operation measures of oil tankers that would significantly reduce both operational and accidental pollution could be implemented in advance of the entry into force of the MARPOL Convention, but this has not occurred. Certain port states are trying to accelerate adoption of these measures by establishing port entry requirements more rigorous than international norms[151], but because of their balancing interests in maritime transit, most of these states have preferred to await international solutions. Other possibly effective and desirable measures, such as an extensive system of computerized ship manifests (comparable perhaps to INTERPOL facilities), have scarcely even been discussed. Then too, states have gone back on already existing compromises. Annex II of the MARPOL Convention, for example, contained Regulations for the Control of Pollution by Noxious Liquid Substances in Bulk (i.e., hazardous liquid cargoes of all types, not just oil), but the 1978 Protocol to MARPOL provided that states need not be bound by this annex for a period of three years after entry into force of the protocol or for such longer period as may be decided by a two-thirds majority of the parties to the protocol in IMCO's Marine Environment Protection Committee[152].

Also, the organizational processes for effecting improvements are often unbalanced and almost invariably painfully slow. IMCO, for example, is well known to be basically a "ship-owners' club". It is slow to respond to environmental imperatives, and the Organization usually acts only in the face of major casualties, such as the

series of spills off United States coasts in the winter of 1976-1977 that prompted the 1978 MARPOL and SOLAS Protocols or the French proposals in the wake of the *Amoco Cadiz* disaster. UNEP too, while constitutionally charged with an overview function of environmental affairs, is painfully slow to action. UNEP has made major strides in the Regional Seas Programme, has efforts underway in the fields of environmental law and of liability and compensation for damage, and has made progress in some related areas, but this organization obviously has a long way to go toward fully and effectively living up to its mandate in the protection and preservation of the marine environment[153].

F. Conclusion

In conclusion, there has been some impressive overall development of international maritime law toward a viable regime, taking better account of environmental variables and considerations, while at the same time accommodating the basic needs of international navigation and transport. A workable balance has not yet been achieved, primarily because the great forces of power and vested interest all tend one way. Environmentalists, for their part, have to recognize that, in an age of critical energy shortages and trade needs, transportation of a great deal of oil and other harzardous substances by sea is virtually inevitable. The aim, therefore, is to develop an effective new international environmental law for the prevention, reduction, and control of marine pollution from vessels, in full awareness of the need for patience and understanding in matters contingent on the complexities of conference diplomacy.

III Pollution from Dumping

Norman G. Letalik

A. Scope of the Problem

Ocean dumping is the seaward transport of land-generated wastes by ships, barges, platforms or aircraft and their disposal in the marine environment. Such wastes may be 'dumped' in bulk in containers, or incinerated, but regardless of the method of disposal used, the ultimate aim is the disposal of undesirable wastes in the sea. Dumping forms part of the overall problem of how to dispose of wastes in an environmentally safe way[154].

While ocean dumping has existed since man first ventured to sea, only in more recent times has it been recognized as an activity which can have adverse impact on the marine environment. This stems partially from the fact that no national or international mechanisms existed for monitoring wastes being dumped at sea. For example, on the New York Bight's Inner Shelf, the practice of transporting wastes by barge to officially designated dumpsites has been taking place since 1888[155]. Unofficial dumping had, of course, existed even prior to that time. In 1968, the United States alone disposed of an estimated 48 million tons of waste by ocean dumping.

Amongst other effects, this dumping created a "dead sea" at the mouth of New York City's harbour[156]. Yet it appears that the effects of the disposed wastes from dumping at sea can be "as varied as the effects of wastes discharged from coasts"[157].

In addition to the tremendous amount of bulk waste which is dumped annually in the oceans, there is also a large amount of containerized waste which is also dumped. Although smaller in quantity than dumped bulk wastes such as dredging spoils, garbage and trash, industrial wastes and sewage sludge[158], dumped containerized wastes can often be more dangerous to the marine environment. This is because containerized wastes usually contain substances whose dispersion into the marine ecosystem is considered dangerous, including such substances as radioactive wastes, toxic chemicals, and chemical weapons[159]. In the late sixties, for example, it was discovered that the Baltic Sea contained high levels of arsenic. Investigations revealed that forty years earlier 7,000 tons of arsenic had been dumped into the Baltic in concrete containers. This quantity of arsenic purportedly would be sufficient to kill the world's population three times over, if widely administered[160].

Unfortunately, the discovery of the dumping of arsenic wastes was not an isolated incident. Danish fishermen in the Baltic have been burned by fish contaminated with German mustard gas dumped by the Allies following World War II. Similar incidents have occurred elsewhere[161]. In August of 1970 the US Navy in 'Operation Chase' placed 418 vaults containing 67 tons of nerve gas on board a rusty vessel and deliberately sank the vessel 282 miles east of the Florida coast in 16,000 feet of water[162]. When word of 'Operation Chase' was made known, it raised a storm of public protest internationally, and especially in the United States. In the word of one writer, the event:

> clearly demonstrated ... the frightening inadequacy of both municipal and international law to provide even minimum safeguards for the protection of the marine environment and to cope effectively with the growing number of reported dumping incidents[163].

Shortly following the furor created by 'Operation Chase', the first major sytematic study of ocean dumping was carried out by the US Council on Environmental Quality[164]. The Council's work was summarized in its report to the President entitled: *Ocean Dumping, A National Policy.* It dealt with the main types of waste disposed by the US through dumping at sea. Specifically, dredge spoils, sewage sludge, solid waste, industrial wastes, construction and demolition debris and radioactive wastes were discussed.

The recommendations produced in the report included a call to action at both the national and international levels to address the problems presented by existing dumping practices. The US's national response came quickly, as on 8 February 1971 the President recommended domestic legislation which would ban unregulated ocean dumping and establish an administrative structure to allow dumping to take place only after the issuance of a permit by the US Environmental Protection Agency. On the same date, the President also instructed the Secretary of State together with the Chairman of the Council to "develop and pursue international initiatives directed toward this same objective on a global basis"[165].

B. Global Responses to Ocean Dumping

1. Introduction

In June 1971 at the first meeting of the Intergovernmental Working Group on Marine Pollution (IWGMP), which was preparing for the UN Conference on the Human Environment to be held in Stockholm in 1972, the US government tabled the first draft of a convention on ocean dumping. That draft was criticized as it did "no more than to require ratifying states to pass certain laws prohibiting the transportation from its territory of any material destined to be dumped in the ocean unless a permit is first obtained"[166]. The Canadian delegation even went so far as to label the draft a "licence to pollute"[167]. The main criticism of the American draft stemmed from the broad discretion which it gave to states to control dumping.

The IWGMP met a second time in Ottawa in November 1971. One of two working groups formed in Ottawa was assigned the task of considering and making amendments to the draft convention. Importantly, the Norwegian delegation, one of 42 delegations present, presented to the Working Group the draft of its amendments based on the final result of the Oslo Conference on Dumping which had been held in Oslo just a few weeks earlier in October.

Just as there had been an impetus in the United States to deal with ocean dumping, so had there been one in Norway. Norway with its strong fishing interests historically had a considerable stake in the preservation of the marine environment. Two events, in particular, helped to galvananize public opinion on the matter of ocean dumping: first, a ship designed to dump industrial wastes from British factories into the North Sea was being built in the UK; second, a Dutch vessel was preparing to dump 600 tons of noxious wastes either into the North Sea or the Atlantic Ocean[168]. The public outcry motivated the Norwegian government in the spring of 1971 to call a diplomatic conference on ocean dumping to be held in Oslo that fall. The invitations to the Oslo Conference were extended to all parties to the North East Atlantic Fisheries Convention[169] and Finland.

Between the extension of invitations to the Oslo Conference and the Conference itself, the first IWGMP meeting on ocean dumping took place. Although the US draft for a convention presented at that meeting was not to serve as a substitute for the efforts of the participants in the upcoming Oslo Conference, the preparatory work for Oslo was heavily influenced by the US draft[170]. The influence was not, however, only to be uni-directional. The Convention produced by the Oslo Conference itself influenced the work of the IWGMP at its second session in Ottawa, especially the black list/grey list system of prohibiting dumping in the Oslo Convention, which was incorporated into the IWGMP's draft.

There had not been sufficient time in Ottawa for the IWGMP to produce a complete convention; consequently, Iceland, the United States, and Sweden, with the assistance of the UN Secretariat, organized an Intergovernmental Meeting on Ocean Dumping in Reykjavik in April 1972. Both the Ottawa draft and the Oslo Convention were used as models at the meeting, which ultimately produced a draft convention. One final meeting regarding several disputed clauses in the Reykjavik draft took place in London in May 1972. It was decided in London that the proposed convention should be concluded only after the Stockholm Conference had ended.

2. The Dumping Convention and its Impact on the Stockholm Conference and UNCLOS III

The timing of the emergence of the draft on ocean dumping raised several questions concerning which forum would be the most appropriate in which to adopt the Convention. Three fora came immediately to mind. First, IMCO's Assembly had decided in October 1969 to convene an international conference on marine pollution. Secondly, the UN Conference on the Human Environment was scheduled to meet in the summer of 1972. Thirdly, the UN General Assembly had already decided to convene the Third UN Conference on the Law of the Sea in 1973. That conference was to deal with matters that included, *inter alia,* the preservation of the marine environment.

At the time when the US advanced its first draft of a dumping convention, it was forwarded with the intention of being discussed within the ambit of the Stockholm Conference. That Conference, however, was only to agree on principles and provide guidelines for states in the form of an Action Plan. It was not convened to draft treaties, especially a treaty which would govern an area which purportedly only accounts for 10% of all marine disposal of waste[171]. It may be speculated that there were three possible motives for the US forwarding its draft for inclusion at the Stockholm Conference: i) that quick action on the matter of dumping at Stockholm would provide a high profile designed to appease the environmentalists in the US; ii) that the relatively uncontroversial area of dumping appeared to be an area in which concrete action could be taken at Stockholm in case the Conference produced little else; and iii) that the US was not pleased with some of the provisions in the Norwegian draft for the Oslo Conference on dumping, and that, as a result, the US tabling its own preferred draft at a global forum like Stockholm might: a) influence the Oslo Conference, or b) prevent the Oslo negotiations from developing from a regional into a global treaty[172]. Regarding this third point, it is interesting to note that the US draft did in fact strongly influence the Oslo negotiations, as the Norwegian draft, which the US found objectionable, was replaced by the North Sea countries' draft as the official negotiating draft at the Oslo Conference. The North Sea countries' draft showed that it had been heavily influenced by the US draft tabled for Stockholm.

The suitability of the draft convention on ocean dumping for inclusion in the Conference's agenda was discussed by one of the Committees preparing the Stockholm Conference. Specifically, Committee III had discussed the Reykjavik draft on ocean dumping within the framework of the Action Plan. The Committee decided that there was not sufficient time available to complete the Convention in Stockholm; consequently, it adopted the following recommendation:

> It is recommended that Governments ... (d) Refer the draft articles and annexes contained in the report of the intergovernmental meeting in Reykjavik, Iceland, in April, 1972, and in London in May, 1972, to the United Nations Sea-Bed Committee at its session in July/August, 1972, for information and comments and to a Conference of Governments to be convened by the Government of the United Kingdom in consultation with the Secretary General of the United Nations before November, 1972, for further consideration with a view to opening the proposed Convention for signature at a place to be decided by that Conference, preferably before the end of 1972[173].

The recommendation, therefore, advised the states which had been attending the IWGMP's meetings to continue with their work and hold a convention conference before November 1972. Hence, Committee III felt that the Stockholm Conference was an inappropriate forum for the completion of the Ocean Dumping Convention.

The use of UNCLOS III as a forum to conclude the Dumping Convention was also considered; as a result, a copy of the Reykjavik draft was sent to Sub-Committee III of the UN Committee on the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiciton for comment. The Sub-Committee pointed out that all jurisdictional questions should be left to the Law of the Sea Conference for decision and that coastal states should not only enforce the provisions of the convention against their own ships but also against foreign vessels in waters under coastal state jurisdiction. Moreover, the representatives from developing states pointed out that the convention was unfair to developing states as it imposed the same burden upon them as it did upon developed states. Finally, representatives from environmentalist states indicated that the basic principle of the convention should be the prohibition of dumping and that the exceptions to that principle should not be so broad as to negate its purpose[174].

That forum was inappropriate, however, for a number of reasons. First, it was anticipated that UNCLOS III negotiations would carry on for a considerable length of time, thus postponing the completion of the Ocean Dumping Convention unnecessarily and risking, of course, that the momentum gained from strong public opinion might dwindle. Secondly, a new law of the sea convention would be constitutive in nature, making the inclusion of anything other than broad principles regarding ocean dumping into the convention unlikely. Both the Reykjavik and Oslo drafts contained far more detail than that which could have been included in a convention on the law of the sea.

The IMCO Conference on Marine Pollution was not adopted as an appropriate forum for the conclusion of an ocean dumping convention primarily for two reasons. First, IMCO was still viewed by environmentalists as being too shipowner-oriented. Secondly, ocean dumping was seen by many as being a land-based source of pollution, even though the pollutants in question were transported in and dumped from vessels[175]. IMCO's Sub-Committee on Marine Pollution and the IMCO Legal Committee, in the course of making preparations for the 1973 Conference on Marine Pollution, did examine the issue of dumping. In view of the forthcoming Stockholm Conference, however, IMCO's Legal Committee decided in early 1972 that it would refrain from drafting an ocean dumping convention[176].

With these three forums rejected, it was decided that the advice of the Stockholm Conference's Committee III should be followed and that a special conference on dumping should meet shortly after the Stockholm Conference as a "follow-up action"[177]. As a result, the United Kingdom's invitation to host the Ocean Dumping Conference in London was accepted and the Conference was convened for October 1972.

3. The London Dumping Convention

The Intergovernmental Conference on the Convention on the Dumping of Wastes at Sea convened in London on 30 October 1972. Representatives from 82 states at-

tended, as did observers from 12 states and seven international organizations. The Conference produced the Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter. The Convention was opened for signature on 29 December 1972 and it entered into force on 30 August 1975. As of late 1980, 45 states had ratified or acceded to the London Dumping Convention (LDC), as it is commonly called[178].

The Convention itself is comprised of a preamble, twenty-two articles and three important annexes. Dumping is defined in the convention to include any deliberate disposal at sea of wastes or other matter from vessels, aircraft, or other man-made structure at sea, or any deliberate disposal at sea of vessels or other man-made structures at sea. Dumping, specifically, is not meant to include the disposal at sea of wastes, etc., derived from the normal operation of vessels, etc., at sea, nor was dumping to include the disposal of wastes related to the exploration or exploitation of sea-bed mineral resources. Hence, normal operational ship pollution and pollution associated with the exploitation of sea-bed minerals are not within the ambit of the LDC.

Operational discharges were to be the subject of the 1973 International Convention for the Prevention of Pollution from Ships (MARPOL). Thus it was not advisable to include these components in the London Dumping Convention. In the same way, the Convention on Civil Liability for Oil Pollution Damage resulting from Exploration for and Exploitation of Seabed Mineral Resources, 1976[179], was scheduled to deal with the problem of pollution resulting from offshore dumping activities from offshore mineral exploitation installations. Hence it was agreed that the latter, rather than the LDC, should regulate this type of activity. Yet it is important to point out that Article XII of the LDC pledged the contracting parties to promote measures to protect the marine environment from pollution caused by:

(a) hydrocarbons, including oil, and their wastes;

(b) other noxious or hazardous matter transported by vessels for purposes other than dumping;

(c) wastes generated in the course of operation of vessels, aircraft platforms and other man-made structures at sea;

(d) radio-active pollutants from all sources, including vessels;

(e) agents of chemical and biological warfare;

(f) wastes or other matter directly arising from, or related to the exploration, exploitation and associated off-shore processing of sea-bed mineral resources.

All of these sources of marine pollution were recognized as not being adequately covered by the convention.

Except for the two areas of operational vessel pollution and pollution from offshore mineral exploitation installations, the LDC's coverage is "desirably broad"[180]. Geographically, the scope of the convention is equally broad. Article III (3) defines "sea"as including "all marine waters other than the internal waters of States", making the LDC truly global in its application, unlike the Oslo Convention which had preceded it.

The essence of the LDC is found in its Article IV, which sets out a so-called black list/grey list method of dealing with different types of pollutants. The dumping of all substances listed in Annex I of the Convention (black list) is prohibited. Those substances listed in Annex II (grey list) may only be dumped if a prior *special* permit has been obtained. All other wastes or matter being dumped requires only the issuance of a prior *general* permit. Consequently, it is only the flag state or the loading state which may issue the general and special permits for dumping. This method of regulating dumping has been criticized by some as legalizing ocean dumping. The Convention's supporters, however, simply point out that in its present form, the LDC recognizes the sea's ability to serve as a waste treatment facility for limited quantities of waste. For this view to receive any credibility, however, one has first to assume that all contracting parties to the convention will act responsibly in not only issuing dumping permits, but that they will observe and enforce dumping practices from their vessels.

Enforcement of the LDC, like many other conventions dealing with the enforcement of ship-generated marine pollution, can present considerable difficulties. The LDC does not create guidelines for the issuance of special or general permits, rather Article VI leaves this wholly to each contracting party to determine. To its credit, the LDC requires contracting states to: i) keep records of the nature and quantities of all matter permitted to be dumped and the location, time and method of dumping; and ii) monitor individually, or with other contracting parties and competent international organizations, the conditions of the seas for the purpose of the Convention[181]. As a result, a conscientious state would only issue as many permits to allow dumping as the sea in the designated dump site is capable of withstanding. Unfortunately, experience shows that past oceanographic monitoring studies have been less than satisfactory. For example, the LDC permits the controversial practice of so-called low-level radioactive waste dumping (grey list) through special permit, and yet in the only known active radioactive waste dump site, located in the North Atlantic, no monitoring studies had taken place as of late 1979[182].

The LDC creates, in effect, a minimum international standard by prohibiting outright those substances listed in Annex I. Other wastes may still be dumped subject to the flag or the leading state issuing a special or a general permit. The permit will be issued by authorities of the flag or loading state subject to discretion of the permit — issuing state, which is first obliged to "carefully consider" factors set forth in Annex III of the Convention[183]. Article III (3) of the Convention also states that authorities issuing permits "shall comply" with Annex III. In this way, the LDC creates guidelines which should be followed, but which, in practice, are difficult for the international community to enforce.

A problem with the LDC that was not contemplated in 1972, but which received the attention of the contracting parties in 1978 at their third consultation meeting, was that of *incineration at sea.* The parties at their third meeting noted that incineration at sea created risks to both the marine and atmospheric environments and that amendments should accordingly be made to the LDC. The resolution adopted at the meeting added a paragraph to Annexes I and II and an Addendum was added to Annex I. The addition to Annex I essentially created an exception to the 'black list', i.e., that some of the substances listed in Annex I could be disposed of at sea by means of incineration. Such incineration could only take place with special permit,

however, and any such incineration had to comply with the Regulations for the Control of Incineration of Wastes and Other Matter at Sea, which are now included in an Addendum to Annex I of the LDC[184]. In essence, the incineration amendments serve as an important exception to the general rule that substances on the 'black list' may not be disposed of at sea.

Other important exceptions to the LDC include: i) its inapplicability to ships and aircraft entitled to sovereign immunity (warship, etc.)[185]; ii) dumping necessary for the safety of human life or vessels[186]; iii) the dumping of substances on the 'black list' by special permit in cases where emergencies pose an unacceptable risk to human health and no other solutions are feasible[187]; iv) the substances on the 'black list' may be dumped if they are 'rapidly rendered harmless' by physical, chemical or biological processes in the sea, provided that they do not: (a) make edible marine organisms unpalatable; or (b) endanger human health or that of domestic animals[188]; and v) that substances on the 'black list' may nevertheless be dumped, if they appear as trace contaminants in wastes (e.g. sewage sludges and dredge spoils)[189].

The enforcement provisions of the LDC contain a significant innovation with regard to the enforcement of environmental treaties; namely, its provisions may be enforced not only by a contracting flag state and a contracting loading state (port state), but also by a contracting coastal state[190]. Although the question of determining the extent of offshore areas over which contracting coastal states would have jurisdiction was left to UNCLOS III, the LDC made it clear that contracting coastal states would have such jurisdiction at least within its territorial sea and left it up to UNCLOS III to determine the extent of areas beyond the territorial sea over which coastal states were to exercise pollution control jurisdiction. An examination of the Draft Convention on the Law of the Sea reveals that coastal states now have the jurisdiction to enforce dumping restrictions within their territorial seas, exclusive economic zones, or onto their continental shelves[191].

4. Other Global Measures dealing with Ocean Dumping

a) The Stockholm Conference

Recommendation 86(c) from the Action Plan for the Human Environment asked all governments, *inter alia,* to ensure that ocean dumping by their nationals or by persons in areas under their jurisdiction is controlled. Furthermore, it was recommended that a global as well as needed regional agreements should be negotiated, especially for enclosed and semi-enclosed seas. The Action Plan, therefore, recommended that action should be taken at three levels:
1) the global level through the London Dumping Convention; 2) the regional level, especially in enclosed and semi-enclosed seas; and 3) national level, with states at the very least implementing the global and regional standards.

b) UNCLOS III

The entry into force of the LDC in 1975 realized one kind of global action envisaged in the Action Plan, while the Draft Convention on the Law of the Sea entrenched the main principles of the LDC into the law of the sea. Article 210 of the Draft Convention on the Law of the Sea requires that states adopt laws and regulations to prevent, reduce and control pollution of the marine environment by dum-

ping. It also requires that such national laws and regulations be no less effective than global rules and standards. With more than 45 adherents, including most of the major maritime states, the LDC clearly articulates the applicable global rules and standards. Moreover, Article 210 contains exhortations that states work through competent international organizations to establish and re-examine global and regional practices and standards.

At the global level, IMCO, now the depository of the LDC, would seem to be the international organization entrusted with reviewing the LDC, although GESAMP is the most active scientific organization determining what impact the LDC is having on the marine environment[192]. At the regional level, UNEP with its Regional Seas Programme seems to be the most likely international organization to establish regional dumping standards. In fact, various UNEP Action Plans contain provisions relating to ocean dumping. At the national level, the Draft Convention on the Law of the Sea requires that coastal states regulate dumping within their territorial seas, economic zones and continental shelves[193].

c) Other forums

Among the major threats to the marine environment resulting from deliberate dumping activities, the disposal of radioactive wastes deserves special attention. Given the highly political implications of this problem, the information available on its exact scope is scarce, especially since a significant part of nuclear waste is being disposed of as a by-product of military activities. Little has been achieved by formal law-making assemblies. Following the 1958 Geneva Convention on the High Seas, the International Atomic Energy Agency (IAEA) promoted the preparation of a draft Convention. A report was prepared on the "legal implications of disposal of radioactive wastes into the sea", but due to major discrepancy of views, the draft Convention was never finalized. (Efforts since then to deal with this problem have been at the regional level, especially within OECD, partly as a follow-up from the approaches adopted to nuclear waste disposal in the Helsinki and Oslo Conventions.)

C. Regional Responses to Ocean Dumping

1. The Oslo Dumping Convention

The Convention for the Prevention of Marine Pollution by Dumping from Ships and Aircraft adopted at Oslo in 1972 (Oslo Dumping Convention or ODC)[194] was the first international convention to deal with the problem of ocean dumping. As has been discussed, the ODC was being formulated at the same time when the IWGMP was preparing what ultimately became the LDC. Because the same governments which were interested in the ODC were also participating in the development of the LDC, earlier drafts of both conventions had considerable impact on the outcomes of the final versions of both conventions. The ODC was heavily influenced by earlier drafts of the LDC and the LDC borrowed its black list/grey list format from the ODC.

Being only regional in scope, the ODC applies geographically to the Northeast Atlantic Ocean, a portion of the Arctic Ocean, and the North Sea. The Baltic and the Mediterranean Seas are specifically outside of the Convention's geographic area of application[195]. The area in question is identical to the area under the control of the former North-East Atlantic Fisheries Commission.

The black and grey lists of the ODC and the LDC are virtually the same, with only a few differences. The ODC's black list includes organo-silican compounds, while the LDC does not, and the LDC's list includes crude oil and other petrochemicals, high-level radioactive wastes, and materials produced for biological and chemical warfare. Both black lists prohibit organo-halogen compounds, mercury and cadmium and compounds of these metals, persistent plastics, and other persistent synthetic materials which may float or remain in suspension so as to interfere with other uses of the sea.

The grey lists for both conventions include arsenic, lead, copper, zinc and their compounds, cyanides, fluorides, organo-silicon compounds (LDC only), pesticides and their by-products not covered on the black list, containers, scrap metal, and other bulky wastes which may present obstacles to fishing or navigation[196].

The ODC was signed by Germany, Belgium, Denmark, Spain, Finland, France, the United Kingdom, Iceland, Norway, Netherlands, Portugal and Sweden, and entered into force in April 1974. Since that time, a commission established under the Convention[197] decided that environmental monitoring was necessary for the effectiveness of the Convention. In co-operation with a commission from the Convention for the Prevention of Marine Pollution from Land-based Sources, a Joint Monitoring Group was formed. As a result of two meetings in 1978, the principles and procedures for establishing a joint monitoring programme under the conventions were adopted. The Group decided that the monitoring programme should be based upon existing national programmes, but that the joint programme would be harmonized by intercalibration of sampling, treatment of samples, methods of analysis and standard reporting procedures[198]. It is clearly most important that such monitoring activities take place. Without them it will be impossible to determine what amounts of wastes the seas can safely absorb. In the past, such monitoring has been especially difficult, because the data obtained by different states has not been comparable. Calibration and monitoring design, therefore, must be made uniform in the same way that the Joint Monitoring Group has done.

2. The Helsinki Convention

In 1974 a Conference on the Protection of the Marine Environment of the Baltic Sea Area was held in Helsinki. The Conference was attended by representatives from Denmark, Finland, German Democratic Republic, Federal Republic of Germany, Poland, Sweden and the Soviet Union. The Conference produced the Convention on the Protection of the Marine Environment of the Baltic Sea Area[199], whose geographic area of application included not only the Baltic but also the Gulf of Bothnia, the Gulf of Finland and the entrance to the Baltic[200]. Article 9 of the Helsinki Convention, as it is called, deals specifically with dumping. The LDC served as the model for this part of the Helsinki Convention. The definition of dumping for both Conventions is identical, except that the Helsinki Convention covers dumping related to the exploration and exploitation of the seabed, whereas the LDC does not. The Helsinki Convention is also far stricter in its prohibition of dumping, as the Convention prohibits the dumping of all substances except dredge spoils, which require the prior acquisition of a special permit. In effect, the Helsinki Convention's 'black list' includes all substances but dredge spoils, which fall under the 'grey list'. In evaluating the role of the Helsinki Convention in the broader context of Dumping, one might conclude that:

> The Helsinki Convention, so far as dumping is concerned, may be said to be a regional agreement within the context of the London Convention on Dumping[201].

Thus, the Helsinki Convention, which finally entered into force in May 1980[202], represents a regional convention which creates higher levels of protection from dumping for a semi-enclosed sea, which in the past has proven to be very vulnerable to dumping.

3. The Barcelona Protocol on Dumping

The 1976 Barcelona Convention for the Protection of the Mediterranean Sea against Pollution[203] is accompanied by a protocol called the Barcelona Protocol for the Prevention of Pollution of the Mediterranean Sea by Dumping from Ships and Aircraft[204]. The Protocol is binding on all parties to the Barcelona Convention. Like the Helsinki Convention, the Barcelona Protocol uses the dumping definition provided in the LDC. The use of a 'black list' and a 'grey list' in the Barcelona Convention also mirrors the practice of the LDC and the ODC. Even the substances found on the respective lists are very similar to those found on the LDC's lists[205]. The influence of the LDC on the Barcelona Protocol is obvious and shows that the LDC can serve as a appropriate model.

Being a component of UNEP's Mediterranean Action Plan, the Barcelona Convention and Protocol form part of a general plan to improve the environmental quality of the Mediterranean. Included in this plan is the creation of a central bank for all pollution information and statistics[206]. Such long-term research and monitoring activity is essential, if the grey and black lists of the Convention are to be amended according to the procedure outlined in Article 17 of the Convention. The Barcelona Convention and Protocol entered into force in 1978, less than two years after its date of adoption, showing that the states of the region had accepted the need to combat this serious pollution problem[207].

4. Other Regional Developments

At the 1978 Kuwait Regional Conference of Plenipotentiaries on the Protection and Development of the Marine Environment and the Coastal Areas, convened under the aegis of the UNEP Regional Seas Programme, a framework convention (which may be referred to as the Kuwait Convention) was adopted[208] along the lines of the 1976 Barcelona Convention, discussed above. Article 5 of the Kuwait Convention dealt with pollution caused by dumping. It required contracting states to take all appropriate measures to prevent marine pollution from dumping in the area and to comply with the international rules controlling this type of pollution. The Convention entered into force on 30 June 1979, or just slightly one year after the Kuwait Conference[209].

While the provisions within the Kuwait Convention on dumping do not compare in detail with the above-mentioned regional agreements, the inclusion of dumping provisions within the Convention shows that consideration of dumping is perceived widely as a requisite to the formation of a regional action plan on marine pollution. Importantly, more detailed provisions on dumping can be added to the Convention in the future, once an assessment of environmental quality dictates the necessity of such additions. An essential component of the Kuwait Regional Action Plan is an assessment of the state of the marine environment in the region[210].

In March 1981 a similar regional initiative was taken at Abidjan at the Conference of Plenipotentiaries on Co-operation in the Protection and Development of the Marine and Coastal Environment of the West and Central African Region. Here too a framework regional treaty (the Abidjan Convention) was adopted, which contained a provision on dumping (Article 6). Again, it may be hoped that this regional framework will serve to stimulate more specific action by the parties to regulate dumping in the region, not least with a view to their commitment in Article 11 to "take all appropriate measures to protect and preserve rare of fragile ecosystems as well as the habitat of depleted, threatened or endangered species and other marine life".

D. National Responses to Ocean Dumping

National responses to ocean dumping were generally considered to be grossly inadequate before 1970. In the United States this prompted the President to direct the Council on Environmental Quality to issue a report to him. By 1972, the US enacted legislation in the area[211]. With the increased public concern over dumping in the late 1960s and early 1970s, Norway also passed laws preventing dumping from Norwegian ships[212]. With the conclusion in 1972 of the Oslo and the London Dumping Conventions, frameworks were created for future national legislation over ocean dumping. Both conventions envisaged that administrative structures be created to carry out a state's obligations under such international agreements. For example, both Conventions required the establishment of a national governmental organization which would issue special and general dumping permits. As a result, all states which are parties to these and the other dumping conventions which follow a similar model require implementing legislation to establish such procedures. To this end, the existing US legislation on dumping had to be amended to take into account the US's ratification of the LDC[213].

The United Kingdom and Canada also passed Dumping Acts as a result of their ratifications of the LDC[214]. As a result the global and regional conventions on dumping have served as international antecedents to national legislation. To a degree it can even be said that the international agreements served as catalysts to national legislation. In the future, as the global and regional standards tighten one would expect commensurate amendments to take place in national legislation. Moreover, as more regional dumping conventions are created, one would expect a greater amount of national legislative activity to take place in the region. As a result, one might focus on the regions subject to UNEP's Regional Seas Programme to find the states which are most likely to promulgate legislation controlling ocean dumping practices.

Such legislation is also likely to generate a significant amount of litigation in national tribunals. Indeed some important judicial precedents related to dumping issues have already been established in recent years, even in the absence of controlling legislation. In France, for example, judicial developments of this kind have been in evidence at least as early as 1968, when fishermen brought suit against a bauxite mining firm which used to dump its red coloured wastes into the bay of Cassis. On that occasion the Conseil d'Etat refused to compensate the plaintiffs on the grounds that "there was no evidence that the so-called red muds deposits" were of such a nature as to prejudice the economical, ecological and touristical interests of the area[215]. Apart from their red-coloured appearance the deposits had no

physical effect on the sea-bed. On another occasion, however, other red-muds of a much more toxic nature, raised French public attention, and more specifically that of the inhabitants of the island of Corsica. The polluting caused by the Italian firm Montedison in 1972 resulted from the dumping of titanium dioxyde residues in the open sea, forty kilometers off the coasts of Corsica. The waste was proven to contain significant quantities of heavy metals, including iron, vanadium, manganese and chromium. After the discovery of six agonising sperm-whales in the dumping area, and repeated catches of fish affected by recrosis, the case led to severe social agitation in Corsica. Various judgements were rendered in that case[216], one of which constitutes an outstanding precedent in the history of environmental law in France by acknowledging the existence of ecological damage as such, for the first time in the French law of civil liability[217].

E. Conclusion

While great progress has been made in the protection of the marine environment from ocean dumping since the Stockholm Conference, work in the area is still far from complete. Although one global international and four regional conventions have been drafted and entered into force since the summer of 1972, one cannot, unfortunately, report that the oceans are no longer used to dump hazardous wastes. Radioactive wastes, for example, may still be *legally* dumped into the seas, though admittedly the dumping of high-level radioactive wastes from non-governmental vessels has been prohibited.

The effectiveness of the conventions now in force will, to a large degree, depend upon the determination of the contracting parties to enforce them. There is considerable room within the Convention for diverse interpretation, and the manner in which the. provisions of a convention are interpreted will dictate the amount of dumping which a contracting state will permit. Moreover, the level of dumping which a contracting state permits to a large degree depends upon the scientific information available to that state regarding the assimilative capacity of a particular dump site. In fact, not nearly enough study has been undertaken to determine what effect dumping practices have had. The regional conventions, especially those developed in conjunction with UNEP's Regional Seas Programme, can provide the vehicles for the establishment of study programmes on a regional basis and provide for the intercalibration of data and the design of monitoring methods. This is not to say that such study will only take place within the ambit of the Regional Seas Programme, as the Commission under the Helsinki Convention has shown that it can prepare monitoring and environmental assessment studies. Nevertheless, in areas where such regional organizations have not yet been established, the Regional Seas Programme, if applicable, seems to be the most capable of establishing monitoring groups.

Importantly, the UNEP Regional Action Plans also consider many other forms of marine pollution, including land-sourced marine pollution. As ocean dumping is really a land-sourced problem tranferred to sea, it can really only be effectively dealt with under a general plan to deal with the disposal of wastes and hazardous substances. In the past it was easy to transfer a waste disposal problem by dumping the wastes at sea, because states were not being held accountable for their ac-

tions. The international conventions now in force and the Draft Convention on the Law of the Sea no longer leave this option open to states, as they now must account for their dumping activities and at least record what is being dumped and designate a site which is best able to assimilate the wastes in question.

Once the environmental impact of disposing wastes on land as compared with the sea is determined, then states will better be able to choose which method of disposal is the least environmentally damaging for given types of waste. Unfortunately, the science of land waste disposal is not as advanced as it might be, especially when considering toxic and noxious substances. If inexpensive and environmentally safe methods for waste disposal are developed on land, then there will be no need to dump substances at sea. Unfortunately, there seems to be little likelihood of such land waste disposal technology being developed in the near future. As a result, the seas will remain vulnerable and dependent to a large degree on the will of the states which have not yet ratified the Dumping Conventions and on the integrity of the contracting states' agencies authorized to issue dumping permits.

In the interim, it is recommended that the effectiveness of the current dumping conventions be assessed to determine whether they permit the dumping of more wastes than the assimilative capacities of the sea permit. Moreover, the science of finding ecologically suitable ocean dumping sites should be developed as well. Finally, states should be encouraged to keep strict records of all dumping activities, as they are required to do under the various dumping conventions. Without such records, it will be impossible to determine the assimilative capacities of the seas.

IV Land-Based Pollution

Martine Rémond-Gouilloud

A. Scope and Nature of the Problem

1. Introduction

By most scientific accounts 75%-85% of the detectable pollutants in the ocean are traceable to land-based sources. Some originate in sewers which discharge directly into the sea; some, in the form of industrial or domestic effluents, are carried down by rivers to estuaries and coastal waters; and much of the rest is associated with atmospheric pollution of the land-mass in general and of congested coastal zones in particular. In sheer magnitude, the problem of land-based pollution of the ocean is the most serious of all. In addition to the magnitude of the problem ecological from an viewpoint, it may also be the most directly related to human health concerns, through the intake of contaminated sea-food[218], and there is growing evidence of the health hazards associated with swimming in polluted coastal waters.

Yet, the international approach to land-based marine pollution is of recent origin. The first multilateral agreements on this topic emerged in the 1970s, and they are still relatively few if compared with other sources of harm to the marine environment. This is, however, easy to understand: first, land-based pollution is "largely in-

visible"; and, secondly, it raises extremely difficult political and economic questions. The fact that most effluents discharged into coastal waters are not visible is certainly one of the basic reasons which explain the lack of attention by decision-makers. Whereas oil-spills have usually helped to arouse public emotion, and thus encourage international action, non-visible discharges do not provoke the same amount of public awareness. The same vacationers who will refuse to bathe on an oil-polluted beach are sometimes seen swimming at the mouth of municipal sewers. Moreover, the fact that a polluting substance is not visually apparent makes the identification of the polluter an extremely difficult task for legal purposes, the establishment of the causation link between damage suffered and a given discharge often proves impossible. Legislators are not inclined to tackle questions which are especially difficult to solve and appear politically non-urgent.

Moreover, the problem of land-based marine pollution presents extremely difficult political problems at the international level, especially for states confronted with fundamental difficulties in economic development. Such states are likely to resist increased financial burdens which might result in a slowing down of their development. The need to reconcile developmental and environmental priorities emphasized by a 1971 Resolution of the United Nations General Assembly concerning "Development and Environment[219]" is reflected clearly in the context of land-based pollution. However, although many states still look on sophisticated sewage equipment and rational coastal management as luxury items they cannot afford, an increasing number are now aware of the vital importance of their coastal waters not only for the conservation of their living resources capital but also as a resource base for the development of the tourist industry. With their sea-coast accepted as a vital economic and employment factor, many coastal states today are eager to keep their air pure, their waters clean and their beaches white[220]. A recent example of this awareness of the need to keep a balance between development and environment is contained in the 1978 Kuwait Convention, which refers to:

> "The need to develop an integrated management approach to the use of the marine environment and the coastal areas which allows the achievement of environmental and development goals in a harmonious manner"[221].

The various action plans adopted under the aegis of UNEP's Regional Seas Programme are intended to provide a framework for an environmentally sound and comprehensive approach to coastal development[222]. But these efforts have just been started; on the whole one must have some doubts about the immediate prospects of adequate preservation of coastal waters in those parts of the world faced with the most serious developmental challenges in the coming years.

2. Special Characteristics of the Problem

Land-based marine pollution has a number of special characteristics, but most of these can be discussed under three general headings: the extreme diversity of causes associated with this particular source of marine pollution, the crucial role of national regulatory action in any effective approach to a long-term solution, and the inadequacy of traditional liability provisions.

a) Diversity of Causes

In contrast with other forms of pollution, land-based marine pollution arises from an extremely wide variety of human activities. The problem varies considerably both

with the pollutant discharged and the source from which it originates. Since nobody can predict the final consequence of a given discharge in a sink or river, it is impossible to make a comprehensive listing of all land-based activities which may contribute, directly or indirectly, to the pollution of the ocean. Even innocuous substances can prove hazardous to the environment, if they are discharged in large quantities. Apart from any chemical effect, their deposits can sterilize the seabed, disrupt the foodchain, increase turbidity, and in other ways impair the assimilative capacity of the marine environment. Faced with these uncertainties, lawyer-diplomats focused initially on substances known to be poisonous: the 1974 Helsinki Convention, one of the first international instruments taking care of this issue, attempted to define guidelines for the categorization of noxious liquid substances[223]. As a result, its Annex II lists substances which call for urgent consideration such as mercury, cadmium and their compounds, followed by another category of substances such as other heavy metals, phenols, halogenated hydrocarbons, pesticides, and so forth[224].

The discharge of such toxic and other harmful substances is now prohibited by all legal instruments concerning land-based pollution. The 1974 Paris Convention and 1980 Athens Protocol, which concern the protection of the Northeast Atlantic and the Mediterranean Sea against pollution from land-based sources, further define the criteria of determining wastes, which are highly hazardous to the marine environment. They select families and groups of substances on the basis of their toxicity, persistence, or bioaccumulation[225].

However, the definition of a highly hazardous substance needs qualification. Banning all discharges containing harmful substances, even if the toxic component of the waste is infinitely small, would unduly affect agricultural and industrial development of a coastal area. For that reason, a second group of wastes is designated not for total elimination, but for strict regulation. This group comprises discharges which contain highly noxious substances in quantities below certain limits[226]. In this group are also listed the substances which, although not intrinsically toxic, may form a dangerous compound in the marine environment. It should be further noted that the list of toxic wastes is deemed to vary according to the progress of scientific knowledge and to increased experience in the industrial and agricultural development of the coastal areas concerned. This constant evolution as well as the highly technical nature of the criteria and methodology involved call for flexible legal instruments subject to frequent modifications. For that reason the "annex technique", enabling the contracting parties to update their lists of pollutants in light of new knowledge and changed conditions, now seems essential to effective control.

The complex nature of land-based pollution is worsened by the fact that the sources of noxious discharges are both numerous and different in nature. These can be indirect (through watercourses) or direct (coastal disposal "including introduction through underwater or other pipelines[227]"). Reflecting this diversity, a US text concerning pollution control adopted in 1980 defines release as follows:

> "release means any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the en-

vironment, but excludes (A) any release which results in exposure to persons solely within a workplace, with respect to a claim which such persons may assert against the employer of such persons, (B) emissions from the engine exhaust of a motor vehicle, rolling stock, aircraft, vessel, or pipeline pumping station engine, (C) release of source, by-product, or special nuclear material from a nuclear incident. Such release is subject to requirements with respect to financial protection established by the Nuclear Regulatory Commission under section 170 of such Act, or, for the purposes of section 104 of this title or any other response action, any release of source by-product, or special nuclear material from any processing site designated under section 102(a)(1) or 302(a) of the Uranium Mill Tailings Radiation Control Act of 1978, and (D) the normal application of fertilizer"[228].

Artificial offshore structures are sometimes treated as significant potential sources of threat to the coastal environment, and are specifically mentioned by international agreements. The question arises whether discharges from offshore structures engaged in exploration of seabed or exploitation of its resources should be covered by land-based pollution agreements. In view of the factory-like appearance of an offshore drilling rig and the fact that it might be the source of threats identical to these associated with ordinary land-based pollution, the first international agreements included them among land-based sources: the 1974 Paris Convention, for instance, specifically mentions such structures as one origin of "pollution from land-based source" falling under its scope of application, and the Helsinki Convention, after defining "vessels and aircraft" as including "fixed or floating platforms" for general purposes, does not exclude them when it specifically deals with land-based pollution (Art. 6). More recent texts do not share this view. The Athens Protocol on the Mediterranean includes "discharges from fixed man-made offshore structures", only in so far as "they serve purposes other than exploration and exploitation of mineral resources of the continental shelf and the seabed and its subsoil" (Art. 4(2)). This consideration deserves approval; oil drilling platforms are now going farther and farther away from the coasts, even from the territorial waters of the coastal states, and the potential threat they represent for the marine environment can clearly not be treated nowadays in the same way as the threat from a sewer. They deserve *ad hoc* treatment[229]. Moreover, the industrial nature of such offshore activity calls for specific environmental treatment by lawyers as well as ecologists. If pollution arising from offshore "rig-based" activity is to be analogized with other sources of marine pollution, it should be with ship-generated pollution, and indeed in many states the former is often dealt with in the legislation governing vessel-source pollution.

Similarly, marine pollution from or through the atmosphere is sometimes regarded as "land-based". As such it is treated both in the 1974 Helsinki Convention and the 1980 Athens Protocol, but neither in the 1974 Paris Convention nor the 1981 Abidjan Convention[230]. Most authors[231] prefer maintaining the distinction. Atmospheric pollution has its own physical characteristics. Dealing with both under a single legal instrument gives rise to unnecessary complexity. Normally separate texts are likely to deal more adequately with each issue. However, the multiplication of texts constitutes in itself another kind of complexity, and at the initial stage of policy-making and legal development there might be a net benefit in dealing with the two

problems together. Nothing prevents a legislature from dealing with them separately at a later stage, when specific regulations have to be introduced. It should be noted, however, that the UN Draft Convention on the Law of the Sea clearly establishes the distinction[232].

This diversity of sources complicates the problem of controlling land-based pollution not only because of its technical complexity, but also because of the various legal methodologies involved. Whereas pollution arising from a factory can be treated along "classical" lines, because the person responsible for the noxious discharge can be identified, isolated, and held liable according to traditional fault or risk principles, the same technique usually proves inappropriate when the pollution results from a river or a sewer.

As noted below[233], land-based marine pollution — like "acid rain" and other "systemic" sources of air pollution — is unlikely to be treated effectively except on the basis of "collective" or "generic" approach to regulation.

As a consequence of the variable harmfulness of the substances discharged into the marine environment, the agreements aimed at controlling land-based pollution utilize a wide range of techniques. Depending on whether the legislator aims at eliminating totally the release of a substance or only at reducing its discharge, the substance will be categorized in a different list. The general formulation pursuant to which states must take measures "to prevent, abate and combat pollution"[234] or control, minimize and "counteract"[235] the introduction of harmful substances therefore encompasses much more specific provisions. These will vary not only according to the goal pursued by the legislator; elimination or control, but also according to the time-table decided upon. In that respect the 1974 Paris Convention established an interesting difference between present and future sources of pollution: whereas "Parties should endeavour" to reduce existing pollution from land-based sources, they must forestall any new pollution from those sources, including that which derives from new substances. The 1980 Athens Protocol has even established separate procedures for the treatment of discharges from existing sources and new installations, but this innovation has been criticized as introducing an unnecessary element of complexity in the interpretation of the text.

Sometimes, of course, the choice of regulatory technique depends on the degree of rigour brought to the task of management. Although under most agreements only the releases are subject to regulation, the Helsinki Convention states that the protection of the Baltic Sea "can involve the use of appropriate technical means, prohibitions and regulations of the transport, trade, handling, application and final disposal of products containing such substances"[236]. This recommendation, inspired by previous agreements concerning carriage and storage of hazardous substances, seems to be extremely ambitious. Given the high technical nature of the regulations and means necessary to implement them, the innumerable loopholes they may permit, and the economic interests involved, it may be very difficult to apply such an ambitious policy. The same may be said, of course, of all proposed systems for the regulation of land-based pollution. While most of the existing formulations suffer from vagueness, others create the risk of inadequate implementation or enforcement because of their high technicality. A judge is by no means a chemical expert, and, when faced with an allegedly unlawful release, he can do

nothing unless there is sufficient evidence that the substance released contained the harmful component in the legally prohibited quantity.

The variable location of sources of land-based pollution raises difficult questions about the desirable physical extent of an international pollution control system. For example, should a regional agreement that purports to establish an effective control system be extended to all, or at least most, of the upstream states that contribute in some degree to the pollution of a river which discharges the pollutants into the ocean? Not surprisingly, different conventions deal differently with questions of geographical coverage. Concerns about "surrendering sovereignty" frequently inhibit states from making explicit commitments that might be subject to this kind of domestic political criticism. For this reason some land-based marine pollution agreements are silent or vague on the question of the scope of their internal applicability.

The Helsinki Convention, for example, does not expressly provide for the applicability of its provisions to the internal waters of the contracting parties[237], whereas both the Paris Convention and the 1980 Athens Protocol are expressly made applicable to the coastal waters of the parties. The Paris Convention does not mention it specifically, but its definition of the "maritime areas" covered includes "the high seas, the territorial seas and the waters on the landward side of the baselines from which the breadth of the territorial sea is measured", and, in the case of watercourses, extends up to the fresh water limit (Art. 3(a)). The 1980 Athens Protocol is even wider in scope: it applies expressly to pollution of the Mediterranean Sea caused by discharges from rivers, coastal establishments or outfalls, or emanating from any other land-based sources within their territories. The fact that a polluting plant is not located on the shore does not, therefore, prevent that plant from falling under the scope of the Protocol, if it actually contributes to the pollution of the Mediterranean environment. As to the limits of that marine environment subject to legal protecton, Article 3 of the Protocol defines it as:

a) the Mediterranean Sea area;

b) waters on the landward side of the baselines from which the breadth of the territorial sea is measured and extending, in the case of watercourses, up to the freshwater limit; and

c) saltwater marshes communicating with the sea.

This specific inclusion of saltwater marshes represents an interesting innovation of the Athens Protocol[238]. The intimate ecological interrelations between these saltwater marshes and the open sea evidently called for such an extension. The acknowledgement of these interrelations in an international agreement is nonetheless indicative of the growing awareness by coastal states of the need for a comprehensive approach, taking into account geographical and ecological realities. This attempt to reflect geographical reality is further demonstrated by the Paris Convention. Its Commission is empowered to fix an inner limit different from that of the freshwater limit in the case of watercourses (Arts. 3(a) and 16(c)).

In respect of watercourses which flow through the territories of two or more parties, or form a boundary between them, no special difficulty is deemed to arise: Article 11

of the Athens Protocol recommends that the parties "co-operate with a view to ensuring its full application". Unfortunately, most watercourses do not flow parallel, but perpendicular, to the coastline. They usually flow down from mountainous areas, often areas under the jurisdiction of a non-party state. This is part of the more general problem of neighbourhood. A country otherwise willing to ratify a land-based pollution control agreement may hesitate to do so, if a neighbouring non-party coastal state is not willing to eliminate or reduce its own noxious releases into the sea. In such cases, the economic burden imposed upon a contracting state may well benefit its neighbour through cleaner water, while the non-party neighbouring state continues to pollute in the same way. Many modern legal instruments attempt to tackle this problem, notwithstanding the well-known principle whereby a convention has binding force only upon those states which are party to it[239]. The 1974 Paris Convention is open for signature to upstream as well as coastal states, whose watercourses flow into the protected maritime areas (Arts. 24-30). The Athens Protocol specifically provides for this. According to Article 11(2), "[a] Party shall not be responsible for any pollution originating on the territory of a non-contracting State. However, the said Party shall endeavour to co-operate with the said State, so as to make possible full application of the Protocol".

These efforts clearly show that the duty of a state bordering a watercourse are of a double nature. It bears obligations not only towards its own neighbours, according to freshwater pollution agreements, but also towards the states bordering the sea into which the watercourse flows, and more indirectly towards all users of that sea.

b) Importance of National Regulation

Invariably, land-based marine pollution originates in areas where national sovereignty is undisputed: land territory or territorial waters. The consequences of this are two-fold. On the one hand, the fact that domestic waters are close to the shore and that fisheries, tourism and other coastal activities are primarily affected by noxious releases provides an incentive for national action. National controls are likely to be carried out, and national regulations are likely to be enforced. On the other hand, international regulation and control will generally prove more difficult. States are inclined to resist any technical limitation which might result in an increasd financial burden or a less competitive economy. In consequence, the most important part of land-based pollution control regulation is usually to be found in national laws. International efforts aimed at promoting uniformity of norms, standardized assessment of noxious wastes, and above all international control, are not likely to be efficacious in most cases. To date international agreements have been limited to very general formulations. However, an interesting exception to this is the set of agreements which have been adopted in Northwest Europe in the 1970s, as noted below.

c) Inadequacy of Traditional Liability Provisions

The traditional approach to social control rests very largely on the possibility of determining that a person — an individual or a corporate body — is culpable, under the concept of civil or criminal liability. But the effectiveness of criminal penalties or civil remedies assumes that the culpable person can be detected and brought to account. In order to cope with situations where such a person cannot be identified, modern laws, have established Funds, aimed at compensating a victim where the

author of the damage remains unknown or proves unable to pay. But this again implies the identification of a given group of persons, who will be asked to contribute financially to the Fund. For that reason, one of the major difficulties posed by land-based pollution arises from the fact that it is basically a collective or generic phenomenon. Although some categories of land-based polluters such as chemical plants can be identified, in most cases, it is virtually impossible to identify the person, or even the category of persons, responsible for a harmful release. Land-based pollution control is, therefore, more dependent on tax-incentive or tax-deterrent systems than on traditional legal instruments. Such collective systems, although indiscriminate and therefore not wholly equitable, prove the only possible means of applying the "polluter pays principle" with respect to land-based pollution[240]. (This feature also serves to differentiate land-based pollution from dumping activities.)

B. Legal and Institutional Responses

1. National Responses

Although international concern with land-based marine pollution is very recent, national regulations governing the discharge of wastes into coastal waters can be traced back to an early period of industrial and agricultural development. Centuries before oil slicks called for attention, noxious wastes were subject to control, in marine areas where human activity was particularly intense, such as harbours, or in areas where water quality is of special importance, such as fishing grounds or oyster-beds. An illustration of this may be found in the French "Grande ordonnance sur la Marine" promulgated in 1681, under the reign of Louis XIV[241].

Today the national regulation of waste disposal in coastal waters is likely to be effected in one of three ways. First, and perhaps most common, is the direct approach to the regulation of a specific activity, such as fishing or fish-farming. Second, and less directly, land-based marine pollution may be controlled in some degree by regulations over industrial development. For instance, the French law concerning the "installations classées", installations subject to control with respect to environment, applies to industrial discharges in marine as well as fresh waters[242]. Third, and more frequently, the problem may be dealt with under coastal management (coastal land use planning) regulations[243]. Difficulties may arise from the fact that different authorities are empowered for activities on land and at sea. The stringency of pollution control is often affected by the lack of motivation on the part of those who should issue the regulations. For that reason, it has often been felt preferable in recent years to tackle land-based pollution as a component of a more general "water-improvement" type of legislation. An example of this trend is found in the legislation enacted by the US Congress in 1970, the Water Quality Improvement Act[244], and in the 1964 French law on the regime of waters and pollution control[245]. Although the applicability of the latter law to marine waters has been challenged on various occasions, there is no doubt today about its applicability to the marine environment[246]. In addition, land-based pollution is sometimes dealt within domestic legislation by more comprehensive environmental texts. A recent example of such laws is the Superfund Act enacted by the US Congress in 1980, which provides for meeting the costs of remedial action and damages for injury through destruction or loss of natural resources, whichever the cause of environmental damage[247].

Both the potentially harmful effects of waste disposal and the extreme difficulty of establishing a causal link between polluter and victim with respect to land-based pollution are illustrated by two major land-based pollution accidents which have taken place in Japan in the past twenty years. Those two cases did much to arouse international attention to the problem of land-based pollution. The Itai-Itai case arose in 1955, when a report was presented at a meeting of clinical surgeons about a strange disease of unknown origin in the Jintsu river basin. In 1959-1960, the theory was announced that Itai-Itai disease was a form of chronic cadmium poisoning. After the establishment of *ad hoc* research committees both at the local and national level, an announcement of the Ministry of Health and Welfare in 1968 revealed that the Itai-Itai disease led to kidney impairment through chronic poisoning, abnormalities in pregnancy, problems in breast feeding, premature aging, calcium and nutritional deficiencies. The only evident cause was the effluent from a mine of the Mitsui Mining and Smelting Co. More than 100 deaths and 600 severe illnesses were officially reported to have been caused by the Itai-Itai disease.

In the Minamata cases a similar number of persons were killed or seriously affected, and similarly protracted efforts were made to establish the cause of the disease. After the first accident, which took place in the Minamata bay area, it took more than twelve years before the view was officially expressed that the disease was caused by the absorption of fish and shell-fish contaminated with methyl mercury compounds discharged from the Minamata plant of the Chiso Corporation. The first report regarding the occurrence of the disease was made public in May 1956, and the announcement of the government's views on the cause of the disease took place in September 1968. More than 4.000 persons are reported to have been severely affected, and 150 killed, by the Minamata mercury poisoning. These cases led the Japanese government to improve national measures for relief of pollution related to health damage, and among others to establish an automatic compensation system based on the designation of certain areas where there is a marked degree of air or water pollution caused by industrial or other activities, with an accompanying high incidence of ailments due to such pollution[248].

Apart from these Japanese disasters, most accidents resulting from land-based pollution have not been proven directly hazardous to human health. Yet an increasing number of incidents are recorded where a discharge, generally of industrial origin, is proven to have severely affected the marine environment. Out of the now numerous law suits brought against industry and municipalities by fishermen, coastal property-owners and vacationers, a new set of precedents is now emerging.

The Corsican "Red Muds Case"[249] although more relevant to dumping than to land-based pollution, contributed significantly to the raising of public awareness on that issue, and consequently provided later litigation with a firm precedent.

In the United States, public awareness of the problem of land-based pollution has probably been raised most by the famous "Wyandotte Case", which involved the release of mercury into the Lake Erie area by Dow Chemical of Canada[250].

2. Regional Responses

a) Introduction

Theoretically, the problem of marine pollution control should be treated as a whole,

since pollution knows no boundaries, but the regional approach has proved to be preferable in many cases. Countries bordering the same sea share the same interests in its preservation. Moreover, apart from the fact that each sea is characterized by special geographical and physical features calling for a specific approach, agreement is reached more easily between states which are used to dealing with one another, especially if they have reached a similar stage of development or share the same cultural background. In the case of ship-generated pollution, the regional approach can be criticized on the ground that it is not conducive to uniformity of norms, but this objection is more difficult to sustain when the pollution in question is associated with a permanently localized source, as it is usually in the case of land-based pollution. It has been further noted that "states which for one reason or another hesitate to ratify a global convention may ratify a regional convention more readily", since the outcome of their efforts is more directly perceivable, and their votes are of a greater relative weight[251]. Special problems frequently arise in regions shared by rich and poor states, where the latter suffer more than they gain from ocean pollution and naturally wish the former to bear the financial burden of pollution control. In such regions pollution control agreements tend to be based on the "polluter pays" principle, though the industrial countries evidently resist that evolution. A good example of this kind of evolution is the Mediterranean, where a vigorous North-South dialogue is clearly in evidence.

b) Northern und Western Europe

The first regional efforts to control land-based marine pollution took place in Western Europe, an area where the threat arising from land-based pollution is probably the most serious, due to high industrialisation and an advanced level of development. The Scandinavians first paid attention to the matter. In March 1974, they adopted in Helsinki a comprehensive regional instrument entitled the Convention on the Protection of the Marine Environment of the Baltic Sea (the Helsinki Convention). Article 6 of this Convention establishes principles and obligations concerning land-based pollution:

"1. The Contracting Parties shall take all appropriate measures to control and minimize land-based pollution of the marine environment of the Baltic Sea Area.

2. In particular, the Contracting Parties shall take all appropriate measures to control and strictly limit pollution by noxious substances and materials in accordance with Annex II of the present Convention. To this end they shall, *inter alia,* as appropriate, co-operate in the development and adoption of specific programmes, guidelines, standards or regulations concerning discharges, environmental quality and products containing such substances and materials and their use.

6. To control and minimize pollution of the Baltic Sea Area by harmful substances the Contracting Parties shall, in addition to the provisions of Article 5 of the present Convention, aim at attaining the goals and applying the criteria enumerated in Annex III of the present Convention.

7. If the discharge from a watercourse, flowing through the territories of two or more Contracting Parties or forming a boundary between them, is liable to

cause pollution of the marine environment of the Baltic Sea Area, the Contracting Parties concerned shall in common take appropriate measures in order to prevent and abate such pollution.

8. The Contracting Parties shall endeavour to use the best practicable means in order to minimize the airborne pollution of the Baltic Sea Area by noxious substances."

During the same year (1974) another Convention was adopted to tackle land-based pollution in the Northeast Atlantic area: the Paris Convention. This instrument applies to the marine area bordered by Western Europe, but the Baltic and Mediterranean Seas are left out of its scope of application. Its aims and means of action are based on a system of lists. The Contracting Parties undertake "to eliminate pollution by certain substances, and to strictly limit pollution by other substances" (Art. 4(1)). This dual regime is affected by the categorization of noxious substances in two lists, usually referred to as the "black list" and "grey list". For the substances included in the black list, listed in Annex I/A; deliberate discharge is prohibited; whereas the substances included in t h e y list (Annex II/A) can be discharged only after approval of the appropriate national authority. In practice they are subject to the issue of an authorization, for which a number of fixed criteria should be taken into account. Part II of Annex I explains the reason for restricting discharges of those substances without providing for their total elimination.

A third European agreement partly related to land-based pollution was also adopted in 1974: the Nordic Environmental Protection Convention. It may be enough here to note that this extremely broad Convention is unusual in the extent to which it focuses on liability and compensation issues, which are seldom dealt with in land-based marine pollution agreements.

Reference should also be made to EEC efforts to control land-based marine pollution. On May 4, 1976, the Council of the Economic European Community adopted a directive on pollution caused by certain dangerous substances discharged in the aquatic environment of the Community. Operational discharges from ships and dumping being expressly excluded from its scope of application, this directive is focussed on land-based pollution[252].

c) The Mediterranean

Elsewhere in this study[253], the evolution and purpose of UNEP's Regional Seas Programme are outlined in some detail. It is sufficient here to note that land-based marine pollution is the most conspicuous component both of the action plans and legal instruments adopted under that Programme. The first of such instruments dealing wholly with land-based marine pollution is the Protocol for the Protection of the Mediterranean Sea against Pollution from Land-Based Sources (the Athens Protocol), adopted in May 1980 pursuant to Article 8 of the Barcelona Convention of 1976. The inediterranean deserves special attention as a victim of land-based pollution. Its coastal areas suffer from a general lack of adequate treatment of domestic sewage and industrial discharges and extensive use of pesticides in agriculture. In addition, the Mediterranean beaches represent the favourite summer vacation playground for the whole of Europe. More than 100 million tourists flock to its coasts every year, thereby contributing to a general pollution load of toxic chemicals in

biota and of nutrients and pathogenic organisms in waters. The geography of the Mediterranean makes this threat worse, since it is an almost enclosed sea with access only through narrow straits leading to the Atlantic and the Black Sea.

After several unsuccessful attempts at environmental co-operation[254], the 1976 Barcelona Convention led the way[255] in providing that "Contracting Parties have an obligation to take all appropriate measures to prevent, abate and combat pollution of the Mediterranean Sea caused by discharges from rivers, coastal establishments or outfalls or emanating from any other land-based source within their territories" (Art. 8). The 1980 Athens Protocol implements this principle in a detailed way. Its definition of land-based pollution is broader than that of similar agreements[256] and it applies expressly to internal waters, unlike the Barcelona Convention itself. Article 3 of the Protocol specifically provides for this. Under this instrument contracting states are responsible for land-based pollution "originating within their territory". Co-operation is recommended between party states in the case of transfrontier pollution sources, such as watercourses, and each party "must endeavour to co-operate with non-contracting States in order to make possible full application of the Protocol".

The "list system" is the basic feature of the mechanism. Parties undertake to eliminate pollution from substances listed in Annex I of the Protocol. This "black list" includes substances which are proven to be particularly hazardous to the marine environment, because of their toxicity or because they are persistent or are subject to bio-accumulation[257]. However, if these substances are emitted in quantities below certain limits, the prohibition does not apply. Programmes are to be established for setting deadlines for the elimination of designated substances. Another list, the "grey list", specifies substances subject to "strict limitation", such as zinc, copper, nickel, lead, arsenic, chromium, tin, uranium, biocides, organosilicon compounds, hydrocarbons, cyanides, fluorides, thermal discharges, and other pollutants.

This list embodied in Annex II of the Protocol is supplemented by a categorization of criteria on the basis of which discharge permits may be granted. The criteria are the following:

a) characteristics and composition of the waste
b) characteristics of waste constituents with respect to their harmfulness
c) characteristics of discharge site and receiving marine environment
d) availability of waste technologies
e) potential impairment of marine ecosystems and sea water uses.

The Protocol further includes provisions for technical assistance to developing countries, technical co-operation, continuing surveillance and environmental assessment, and exchange of information. Despite the fact that the Protocol includes no provisions concerning its enforcement (which of course was a thorny issue, since land-based sources of pollution are subject to strong national sovereignty feelings) and despite the political difficulties which lessen its chances of being implemented rapidly, the Athens Protocol should be regarded as a model protocol ready to be adopted by the states of other regions.

d) Red Sea and Gulf of Aden

Among the components of the Action Plan approved in Jeddah is a regional environmental assessment programme addressed to the following activities:

> "Assessment of the magnitude of pollutants affecting human health and marine ecosystems of the region consisting of:
>
> a) survey of land-based sources of industrial and municipal wastes discharged directly or indirectly into the sea or reaching it through the atmosphere;
> b) studies on the impact of industrial and municipal wastes on human health including micro-organisms;
> c) research on effect of pollutants and other human activities, such as dredging and land-reclamation, on important marine species, communities and ecosystems;
> d) base-line studies and monitoring of the levels of selected pollutants, in particular heavy metals in marine organisms" (Article 12(4) of the Action Plan).

The environmental management programme lists amongst the projects that should be strengthened "assistance programmes on the treatment of industrial wastes" (Art. 17(3)). The "strengthening of the national public health service" and "co-ordination of national water management policies" (Art. 18(4)) should also prove useful as approaches to the land-based pollution. As to legal instruments, a draft regional convention, approved in Jeddah in January 1981, deals with pollution from land-based sources. Its Article VI states that: "The Contracting States shall take all appropriate measures to prevent, abate and combat pollution caused by discharges from land reaching the Red Sea and Gulf of Aden, whether water-borne, air-borne, or directly from the coast including outfalls and pipelines".

e) Other Regions

Similar approaches are evident in the regional action plans adopted in four other regions designated under UNEP's Regional Seas Programme: the Kuwait Action Plan Region[258], the West and Central African Region[259], the Wider Caribbean Region[260], and the East Asian Seas Region[261]. In the first two of these regions legal development has proceeded further. In the Kuwait Action Plan Region land-based pollution is dealt with in Article 8 of the 1978 Kuwait Convention[262]; in the West and Central African Region it is addressed in Article 7 of the 1981 Abidjan Convention[263].

3. Global Responses

The Draft Convention on the Law of the Sea prepared by UNCLOS III confirms that the problem of land-based marine pollution is now of international concern, and no longer to be left to the action of individual coastal states. Following the prescriptions of the 1972 Stockholm Declaration — and going beyond the ambit of UNCLOS I in 1958 — UNCLOS III deals specifically with this problem as a matter of state responsibility under the international law of the sea. Article 207 of the Draft Convention provides that all states have a duty to:

> "Establish national laws and regulations to prevent, reduce and control pollution of the marine environment from land-based sources including rivers, estuaries, pipelines and outfall structures, taking into account internationally agreed rules, standards and recommended practices and procedures".

The wide scope of this provision seems at first encouraging, since the sources of pollution envisaged are not only those situated right on the shore line but include "rivers, estuaries, pipelines and outfall structures". However, the extremely vague formulation of the obligations imposed does not give rise to much hope for effective modes of implementation. States "shall also take other measures" (but what are those measures?) "as may be necessary" (but who decides what is necessary?). They "shall endeavour to harmonize their national policies". Assuming the conflicting interests of most neighbouring states and given the cost of sewage plants and other pollution control devices, one may be pessimistic about the future fate of this provision. In principle, however, state parties to the proposed Convention are obliged to take action with respect to enforcement. According to Article 213, they shall:

> "enforce their laws and regulations established in accordance with article 207 of the present Convention, and
>
> 2. adopt the necessary legislative, administrative and other measures to implement applicable international rules and standards established through competent international organizations or diplomatic conference for the protection and preservation of the marine environment from land-based sources of marine pollution."

But why should a state be required to enforce its own laws? It has been suggested that the provision is motivated by practical considerations. The national authorities entrusted with the application of the law would be more careful when given precise directions and concrete suggestions for the development of practical methods. It has been further noted that the incapacity of flag states to apply anti-pollution controls was largely due to the lack of enforcement provisions[264]. One may find that the reason for the failure of those Conventions lies simply in the lack of political motivation on the part of the states concerned. As regards land-based pollution, one might even consider that UNCLOS III treaty drafters mentioned the obligation of states to enforce their own laws precisely because even in this matter they were aware of a probable lack of political will.

Yet the UNCLOS III provisions concerning land-based pollution do represent the appropriate three-level approach to this problem: priority is rightly given to national domestic laws, but the regional and global approaches are also taken into account. Article 207(4) reads as follows:

> "States, acting especially through competent international organizations or diplomatic conference, shall endeavour to establish global and regional rules, standards and recommended practices and procedures to prevent, reduce and control pollution of the marine environment from land-based sources, taking into account characteristic regional features, the economic capacity of developing States and their need for economic development".

Secondly, Article 207(4) recommends that the "rules, standards, and recommended practices and procedures shall be re-examined from time to time as necessary". The fact that periodical review of international agreements is specifically provided for, which is new[265], seems to imply that institutional structures are to be established for that purpose. This corresponds to a real need.

Finally, the Draft Convention adequately Identifies the various forms of action required for land-based pollution control: the preventing, reducing and controlling of noxious discharges. Apart from prohibiting the release of some toxic, harmful, and noxious substances, minimizing the releases may prove more realistic, and therefore more efficient, in most cases.

C. Conclusions

Whatever the final outcome of UNCLOS III as a treaty-making conference, it should already be evident that there is a duty on all states — certainly all coastal states — to prevent, reduce and control pollution of the marine environment from land-based sources. Despite the very general level of formulation, this norm is of fundamental importance in the evolving international environmental law. As a general norm, it serves at least three important functions at this stage of legal development: first, it provides a stimulus for national legislatures to develop or improve their laws in this area of pollution control; second, it serves to encourage co-operation to this end on the part of neighbouring states; and third, it provides a legal basis for the integration of pollution control policy, and relevant institutional arrangements, especially in coastal areas.

1. National Legislation

It must be hoped that the inclusion of land-based pollution provisions in international agreements and action plans will result in a general raising of awareness about this, the most serious, source of marine pollution. The most significant response would take the form of national legislation, especially in developing states that have not had a long history of industrial abuse to warn them of the hazards associated with waste disposal in their coastal waters. The need to put land-based pollution control among the priorities of sound eco-development should now be more evident than ever before, because of the emphasis placed on this problem in recent legal instruments.

2. Regional Co-operation

The doctrine of the common heritage of mankind appears relevant to the principle of regional co-operation in this particular context of marine pollution control. By altering the shared marine environment, or permitting detrimental uses to continue unchecked, a coastal state prejudices the interests of all other coastal states in the region. In most areas of the world's oceans, effective land-based pollution control is hopeless without a serious and concerted effort at consultation with a view to the common development or harmonization of national laws and practices. But it seldom happens that all coastal states in a region have developed similar laws and institutions providing for efficient co-operation in this field of action. A basic feature of land-based pollution control agreements, therefore, is the establishment of international commissions at a bilateral or regional level[266]. Only through those institutional bodies can discharge standards be harmonized, appropriate research developed along the same lines, and surveillance and control actions adequately co-ordinated. In addition, since elimination or reduction of noxious discharges can be achieved only on a long term basis, institutional structures are needed to develop the programmes provided for by international agreements. Joint commis-

sions are therefore being established by most relevant agreements as soon as they go beyond vague formulations and envisage actual action. In fact, this was the only purpose of the RAMOGE Agreement, adopted in 1976, by Italy, France and Monaco[267], concerning the coastal waters of the Riviera. Other examples of such regional commissions are provided by the agreements developed under UNEP's Regional Seas Programme referred to above.

In most of those emerging commissions, however, there is still no apparent intention to install a permanent secretariat with day-to-day administrative and research capabilities. It must be hoped in these cases that the present arrangements, limited essentially to occasional or regular consultations among the member states, represent the initial step toward the development of more effective modes of co-operation in this area of ocean management.

3. Integrated Planning

Both at national and regional levels, states now have an opportunity to begin planning the development of integrated land-based pollution control system, whether in the form of coastal zone management, sea use planning, or otherwise. Proper planning in an integrated manner can only result in economies, reducing the various forms of wastes in the use of the coastal environment and its resources. Benefits can be expected in the co-ordination of research effort, the harmonization of norms and criteria for discharge controls, the standardization of data and procedures, and in other ways. A fully integrated system of land-based pollution control would, of course, include effective enforcement provisions and practices. Indeed the key to successful pollution control lies in the choice and application of appropriate arrangements for *implementation,* not in the formulation of obligations and objectives, however well intentioned. Not least, provision should be made to afford meaningful remedies to the victims of pollution damage in the state where the damage is caused and in the state where the harmful discharge has taken place. For the time being, however, it may be premature to introduce model legislation. It seems more fundamental at this stage to invoke the spirit of regional co-operation and the sense of a common interest in the environmental welfare of coastal communities around the world.

V Pollution from Seabed Activities

Martine Rémond-Gouilloud

A. The Nature and Scope of the Problem

1. The Threat to the Offshore Environment

Between one-quarter and one-third of the world's proven reserves of oil and natural gas lie in the offshore. In the last decade the tapping of these reserves has become a high priority in all coastal states fortunate enough to have commercial quantities of such resources beyond their shores[268].

As early as 1973 almost a hundred countries were engaged in offshore exploration for oil or natural gas, and 40 were engaged in, or planning for, production. In 1977,

436 offshore rigs were in operation, and it is estimated that more than 1000 rigs will be drilling by the year 2000.

Offshore petroleum activity consists invariably of a series of steps that must be carried out before production is possible. After the determination of an area of commercial interest, but before any licence (permit) is granted, preliminary investigations are made in the form of seismic and geological surveys. After the granting of an exploration permit, the search phase begins. The exploration programme consists essentially of exploratory drillings. For this purpose mobile platforms are used, and these may range from jack-up platforms with metallic supports resting on the sea floor to semi-submersible rigs and drilling vessels, which are simply anchored to the seabed. As oil operations gradually move out to deeper waters, semi-submersible rigs and drilling vessels are preferred to jack-up platforms.

Then comes the exploitation phase, if deposits are found in sufficient quantity to justify further investment. By this time the size of the reserve and its geological and chemical characteristics will have been ascertained in greater detail. In accordance with the scale of the development which is possible, platforms and other forms of equipment are ordered and installed on site. Production wells are drilled and equipped in various ways, with tubing and casing, safety devices known as "blow-out preventers", and other facilities. During the drilling operations, mud mixtures are extensively used. They are composed ofabout 85 per cent of fine, inert material, which can be suspended in the water column without altering it, at least chemically, but can degrade the optical properties of sea water and impede the penetration of solar energy. In addition, soluble caustic sodium hydroxide, iron, chromium and sulfur are added to the mud, which will eventually be discharged directly into the marine environment.

The production phase lasts as long as the field is productive: ten years or more. This explains why production platforms are usually fixed to the seabed, or built upon it if they are heavy enough: they cannot be moved. They will have to be destroyed, when the exploitation is over. This is the final stage of an offshore operation. Most important here is the disposal of the equipment: processing platforms (i.e. platforms with equipment to process petroleum before it is transported on shore), storage platforms, and auxiliary installations and pipelines. These latter "accommodation installations" include supply vessels and lodging units.

As far as pollution is concerned, these platforms and ships should be regarded as ordinary ships and subject to the same constraints, since both can be the source of the same kinds of nuisances. Pipelines, to the extent they may be a source of oil spills, require separate treatment. In principle, a pipeline which has been correctly laid on the seabed and completely covered over will not be exposed to the effects of waves or currents, and should not, therefore, be a source of nuisance. However, fishing and anchoring activities can cause serious damage to a pipeline. After the Ekofisk accident in 1976, extensive surveys of the underwater installations revealed that the Frigg pipelines had all suffered damage due to human activities[269].

2 The History of Environmental Injury in the Offshore

Inevitably, accidents occur in the course of any undertaking as complex, arduous and risky as offshore exploration and exploitation[270]. Above all, the risks of accident

are extremely high for those involved in these activities — higher than in almost any other occupation — and every precaution should be taken to reduce these risks to human life to the absolute minimum. But the risks to the national environment may be less obvious, and less likely to be reduced by the application of reasonable precautions[271].

The adverse environmental impacts of offshore activity were first brought to the attention of the public in 1969, when an oil well ran out of control off the coast of Santa Barbara, California. In 1970, a series of accidents occurred on the Atlantic side of the United States leading to substantial oil pollution of adjacent waters. As offshore activities are often close to densely populated coasts, these effects were bound to be felt by more and more coastal communities. But world attention to these risks has been aroused chiefly by the publicity given to major accidents. Three in particular deserve special comment.

a) Santa Barbara

The Santa Barbara accident, caused by a crack in the seabed outside the territorial waters of California, involved a crude oil release estimated at about 80,000 barrels over the first 11 days. The total oil impinging on coastal areas was 4,500 tons (33,000 barrels). Oil was found on recreational beaches covering ninety kilometers of the California coast, in an area where the fishing industry employs 75 boats and more than 500 people. Many law suits were brought against the oil companies and their insurers. Whilst commercial fishermen obtained compensation for their loss of fishing time, owners of private pleasure boats sustaining physical damage from contact with the oil slick were awarded compensation for physical damage only. They were denied any recovery for their loss of "navigation rights", on the ground that they had been deprived of no more than their "occasional Sunday piscatorial pleasure"[272].

Including legal fees, the Santa Barbara oil spill cost the Union Oil Company consortium and its insurers approximately 60 million US dollars. Additional (uncompensated) environmental damage has been estimated at between 10 to 50 million dollars (US).

b) Ekofisk

On the evening of April 22, 1977, a platform named Bravo 14, operating in the Norwegian sector of the North Sea on the Ekofisk field, suddenly blew out. Eight days later the 180 feet high geyser was controlled and the well sealed. Approximately 21,300 tons of oil were released, and after several days an oil slick, one to two millimeters thick, had spread to cover approximately 1,000 square kilometers. Some 50% of the oil was estimated to have evaporated. The oil slick never reached the coast of any country, due partly to the evaporation of the oil, partly to active cooperation between the industry responsible, the Norwegian authorities, and the North Sea coastal states[273].

c) IXTOC I

Competition works in strange ways within the oil industry! Just as the Santa Barbara blow-out was followed soon after by the *Torrey Canyon* stranding, the *Amoco Cadiz* (the largest ship-generated oil spill) was followed shortly afterwards by the IXTOC I blow-out in the Gulf of Mexico, the largest oil spillage from any single source.

On June the 3rd, 1979, an exploration well operating in the Gulf of Campeche, eighty kilometers from the *city* of Ciudad *del* Carmen, blew out and caught fire. The safety devices proved incapable of containing the pressure, and after fifteen hours the rig and its hoisting equipment collapsed. Estimates of the daily amount of escaped oil vary from 20,000 to 40,000 barrels. Despite a series of efforts to contain the flow (two relief wells were achieved without success), the well was not brought under control until March the 24th, 1980, nine months after the spill had begun. The total amount of oil discharged by the well is generally estimated at about 3 million barrels, or 500,000 tons. (230,000 tons is the amount of oil discharged by the tanker *Amoco Cadiz).* Hearings conducted by a US Senate panel revealed that the drilling crew lacked experience and qualifications in several respects: for instance, due to insufficient records, the exact location of the bottom of the hole was unknown. Clean-up operations and attempts to recover the oil began shortly after the accident. By mid-June, 500 persons and 23 vessels were attempting to combat the oil slick, which moved constantly on the surface. Extensive aerial spraying of dispersants was carried out at the same time. Mexican and US scientists co-operated in removing the hatchlings of Atlantic Ridley turtles to cleaner waters, since this species, which migrates during the winter in the oil slick area, is highly endangered, and since oiled hatchlings had been found dead on the Texas coast. The loss of one year's hatchlings for the Ridley turtle might have led to its total extinction. As to the shrimp fisheries, an essential component of the regional economy, as well as to other affected business activities, the extent of the injury from the oil spill is not known with certainty. An eighty million US dollar fund was established by the US Congress in order to compensate Texas private interests damaged by the spill[274].

B. The Global Response

The extent to which a coastal state has the right to grant licences and issue regulations with respect to offshore operations is a question which must be resolved in accordance with the provisions of international law. The original basis for coastal state jurisdiction over affected shore areas consisted of a combination of customary international law and early treaty law. State practice (beginning in the 1940s or earlier) and juridical opinion (continental shelf doctrine) provided the main ingredients of the former, and the Geneva Convention on the Continental Shelf (adopted at the First UN Conference on the Law of the Sea in 1958) represented the latter. Derived, of course, from a geological term, the *legal concept* of the continental shelf was defined at UNCLOS I by reference to three different, and unrelated, criteria: a fixed depth (200 metres), exploitability, and adjacency to the territorial sea (and therefore the proximate land-mass)[275].

The second of these criteria provided most of the difficulty created by the 1958 definition. It soon became evident that many of the oil and gas reserves lay, or extended, beyond the 200-metre isobath. Since both customary and conventional international law had vested "sovereign rights" to the shelf resources in the adjacent coastal state, national "appetites", whetted by the energy crisis, expanded outwards and soon put the 1958 definition under intolerable stress. The outcome of this expansion of national jurisdiction over the shelf is reflected in Article 76 of the Draft Convention on the Law of the Sea:

1. The continental shelf of a coastal State comprises the sea-bed and subsoil of the submarine areas that extent beyond its territorial sea throughout the natural prolongation of its land territory to the outer edge of the continental margin, or to a distance of 200 nautical miles from the baselines from which the breadth of the territorial sea is measured where the outer edge of the continental margin does not extend up to that distance.

2. The continental shelf of a coastal State shall not extent beyond the limits provided for in paragraphs 4 to 6.

3. The continental margin comprises the submerged prolongation of the land mass of the coastal State, and consists of the sea-bed and subsoil of the shelf, the slope and the rise. It does not include the deep ocean floor with its oceanic ridges or the subsoil thereof.

4. (a) For the purposes of this Convention, the coastal States shall establish the outer edge of the continental margin wherever the margin extends beyond 200 nautical miles from the baselines from which the breadth of the territorial sea is measured, by either:

   (i) A line delineated in accordance with paragraph 7 by reference to the outermost fixed points at each of which the thickness of sedimentary rocks is at least 1 per cent of the shortest distance from such point to the foot of the continental slope; or

   (ii) A line delineated in accordance with paragraph 7 by reference to fixed points not more than 60 nautical miles from the fool of the continental slope.

   (b) In the absence of evidence to the contrary, the foot of the continental slope shall be determined as the point of maximum change in the gradient at its base.

5. The fixed points comprising the line of the outer limits of the continental shelf on the sea-bed, drawn in accordance with paragraph 4(a)(i) and (ii), either shall not exceed 350 nautical miles from the baselines from which the breadth of the territorial sea is measured or shall not exceed 100 nautical miles from the 2,500 metre sobath, which is a line connecting the depth of 2,500 metres.

6. Notwithstanding the provisions of paragraph 5, on submarine ridges, the outer limit of the continental shelf shall not exceed 350 nautical miles from the baselines from which the breadth of the territorial sea is measured. This paragraph does not apply to submarine elevations that are natural components of the continental margin, such as its plateaux, rises, caps, banks and spurs.

7. The coastal States shall delineate the seaward boundary of its continental shelf where that shelf extends beyond 200 nautical miles from the baselines from which the breadth of the territorial sea is measured by straight lines not exceeding 60 nautical miles in length, connecting fixed points, such points to be defined by co-ordinates of latitude and longitude.

The mind-boggling complexity of this formula gives some indication of the diplomatic energy expended on it and of the difficulties of interpretation that might arise with respect to claims by "broad shelf states", whose shelf extends beyond the 200-mile limit. For present purposes in this study, the most important point to be made is that most offshore petroleum activities likely to take place in the near future will come under the exclusive jurisdiction of the "adjacent" coastal state and that the latter will be expected to discharge the primary, if not exclusive, responsibilities for the application of environmental controls related to such activities.

Article 208 may be regarded as the most important provision of the Draft Convention on marine pollution deriving from offshore activities. It reads as follows:

1. Coastal States shall establish national laws and regulations to prevent, reduce and control pollution of the marine environment arising from or in connexion with seabed activities subject to their jurisdiction and from artificial islands, installations and structures under their jurisdiction, pursuant to articles 60 and 80.

2. States shall also take other measures as may be necessary to prevent, reduce and control such pollution.

3. Such laws, regulations and measures shall be no less effective than international rules, standards and recommended practices and procedures.

4. States shall endeavour to harmonize their national policies at the appropriate regional level.

5. States, acting in particular through competent international organizations or diplomatic conference, shall establish global and regional rules, standards and recommended practices and procedures to prevent, reduce and control pollution of the marine environment arising from or in connexion with seabed activities subject to their jurisdiction and from artificial islands, installations and structures under their jurisdiction referred to in paragraph 1. Such rules, standards and recommended practices and procedures shall be re-examined from time to time as necessary.

Enforcement of these regulations is provided for in Article 214[276].

Given the growing demand for energy, it is quite possible, however, that petroleum exploration and exploitation may have been extended by the 1990s beyond the regime of the continental shelf, as defined in Article 76 of the UNCLOS III Convention. Some projections suggest that "offshore" operations may be under way in the middle of the oceans before the end of the century. If so, they will take place in the "Area", that part of the deep-ocean floor regarded as belonging to the "common heritage of mankind"[277]; and intended, therefore, to be governed by the proposed International Seabed Authority[278]. In this event, marine pollution control would be governed by Article 209 of the UNCLOS III Convention[279].

Another recent development has been the work of the UNEP Group of Experts on Environmental Law on the problem of marine pollution resulting from offshore mining and drilling within the limits of national jurisdiction. The conclusions of the Group (released in February 1981), were framed in light of developments at UNCLOS

and advocated the adoption of national legislative and regulatory measures to prevent, reduce and limit pollution and other adverse effects on the environment from offshore operations. Specific recommendations are geared to the various stages of offshore mining and drilling operations, including the need for permit authorization systems for the construction, operation and alteration of offshore sites, the use of environmental assessment in conjunction with the permit process, the use of appropriate monitoring systems, safety measures including contingency planning and normal operational measures, and finally liability and compensation issues.

C. The Regional Response

Regionalism is regarded by many as the key to marine pollution control[280]. Pollution deriving from offshore activities, unlike tanker pollution, necessarily originates under the same jurisdiction as the potential victim, or at least the first to be exposed to environmental harm. But if the operations are conducted under a neighbouring state's jurisdiction, agreement will at any rate be easier to reach amongst countries sharing the same geographical environment and often a similar economic and cultural background.

This may explain why regional pollution agreements have been reached more easily and more effectively than universal ones. Apart from the fact that many rules are aimed at combating pollution generally, whatever its origin, numerous agreements include specific provisions concerning pollution from offshore activities. A good example of the first set of rules is the (Bonn) Agreement for Co-operation in Dealing with Pollution of the North Sea by Oil, signed in June 1969[281]. The first *specific* provisions regarding seabed pollution embodied in a regional agreement are, however, to be found in the 1974 Baltic Convention, referred to above[282]. Article 10 of this Convention states that:

> Each Contracting Party shall take all appropriate measures in order to prevent pollution of the marine environment of the Baltic Sea Area resulting from exploration or exploitation of its part of the seabed and its subsoil or from any associated activities thereon. It shall also ensure that adequate equipment is at hand to start an immediate abatement of pollution in that area.

Under the UNEP Regional Seas Programme regional conventions have begun to emerge with provisions dealing specifically with pollution arising from seabed activities[283], and similar regional developments are expected to take place elsewhere[284].

D. The National Response

The history of national responses to this kind of marine pollution is much too detailed to be recited here. It may be enough to note some trends and special features.

States have been urged by all relevant conventions to take all necessary measures to prevent pollution. Although such a principle seems quite simple to establish, its implementation raises a number of difficult questions. An offshore drilling platform is not a ship, and the shipping conventions concerning the prevention of marine pollution require significant modifications in order to be applicable to the different

conditions of offshore operations. In particular, the scope of application of many a legal instrument needs remodelling. For instance, the 1954 London Convention[285], however modified, could not possibly be adapted to offshore rigs which look like factories. Its technical features are designed for sea-going ships. The dumping conventions[286] carefully draw the line between dumping and discharges of oil and other pollutants due to the operation of a rig, but dumping in the form of intentional sewage disposal from a platform, whether fixed or floating, is covered in the dumping conventions[287].

In order to meet the challenges of hazardous working conditions and the threat to the environment, most oil and gas companies have instituted corporate safety and environmental policies which are equal to, sometimes even stricter than, minimum requirements under national and international regulations. Problems arise from the fact that due to the fast evolution of technology, unduly detailed regulations create the risk of inhibiting offshore development. For that reason national regulations now usually set forth basic environmental goals and principles only[288], and are supplemented by more supple rules which can be revised in the light of new knowledge. UK "guidance notes" issued as an aid to the interpretation of the offshore regulations 1974 are an example of such a procedure[289]. These regulations, although numerous and wide-ranging, are often criticized for not being comprehensive enough. Operators therefore often draw up for themselves "codes of conduct" for safe practice in drilling, production and pipeline operations specially designed for marine areas. In such codes, detailed procedures on both avoidance of pollution incidents arising from oil-spills and emergency procedures are to be found[290].

Also noteworthy is the fact that offshore pollution prevention regulations are not limited to pollution by oil. Other sorts of refuse, such as domestic sewage and in particular drilling mud mixtures which are extensively used during offshore operations, are covered in those regulations[291].

One important legal issue is whether criminal law is an adequate deterrent for pollution prevention from offshore activities. Are civil penalties and third party liability not preferable in a field where the operator himself is not on board the rig? It is to be feared that the *manager* of the offshore installation, like the captain of ship, may be procecuted more as a scapegoat than as the person actually responsible. For that reason offshore legislation usually makes the (corporate) *owner* liable or the *operator* of the rig criminally liable, rather than hold the manager of the installation[292] responsible.

This issue, the adequacy of criminal provisions, is well illustrated by the difficulties faced by the French legislature. Under article 28 of the 31 December 1968 Act on exploration of the continental shelf and exploitation of its mineral resources, the operator's manager, representing him on the platform, used to be treated like the captain of a tanker. He was obligated to keep track of any oil discharges in a special register. Apart from the fact that penalties were insufficient, article 28 was also inappropriate. It is quite unrealistic to prohibit all oil discharges from a production platform. A rig is not a ship, and limiting, not prohibiting, the discharge is the only possible aim. Accordingly, French law has undergone a complete remodelling in 1977. Penalties are now higher[293]. The delict is constituted by:

1) Any discharge of oil during the exploration;
2) During exploitation phase, by any discharge of oil exceeding a fixed criterion;
3) By the lack of biologic and ecologic assessment in the area covered by the licence.

The licence holder is held liable as an accomplice of the platform manager, if he has not ordered him in writing to comply with environmental regulations.

E. Special Problems

1. Preparedness

The Ekofisk and IXTOC I blow-ups, and the numerous minor spills that occur more often, are evidence that in spite of efforts to prevent them, oil spills can take place at any time. Whatever the stringency of the regulations and the perfection of industrial practices, oil eventually finds its way into the oceans. The oil spill can even occur long after the exploitation is over, if the anchor of a ship gets caught in the remaining equipment. Oil-spill preparedness, therefore, appears as necessary as oil-spill prevention, and strengthened efforts have been devoted to that issue for the last ten years[294]. Clean-up equipment is more sophisticated; dispersants are now less harmful to the environment; and continuous monitoring and control systems for oil discharges have been set up[295]. Industry mutual aid organizations are formed to pool the resources of their member companies for increasing the joint capability to clean up oil spills. This is the case, for instance, in the North Sea and in the Gulf Area. Such co-operation appears to be all the more necessary as clean-up costs prove to be more costly each year. Economists show that these costs have become 5.5 times higher in the last ten years[296].

The Draft Convention on the Law of the Sea urges states "to co-operate on a global basis and, as appropriate, on a regional basis, directly or through competent organizations, in formulating and elaborating international rules ... for the protection and preservation of the marine environment, taking into account characteristic of regional features" (Art. 197). They must provide for notification of imminent and actual damage (Art. 198), promote and develop contingency plans for responding to pollution incidents (Art. 199), and promote studies and undertake programmes of scientific research and encourage the exchange of data (Art. 200). More generally they have the duty of taking all necessary measures to prevent accidents and deal with emergencies (Art. 194(3)(d)). Although these obligations concern all sorts of pollution accidents and are not restricted to offshore oil spills, they are being developed by regional agreements with special reference to offshore exploitation, wherever this activity is taking place. A good example of such co-operation in cases of disaster preparedness is the 1969 Bonn Agreement relating to co-operation between coastal states of the North Sea in dealing with pollution by oil. Under this agreement state parties are obliged to inform other parties of oil slicks which are likely to constitute a serious threat to the coastal or related interests of other states. The North Sea is divided into different zones and each party is required to make the necessary assessment of the nature and extent of any casualty within its own zone. A good illustration of the efficiency of this agreement is provided by the 1977 Ekofisk case. Some hours after the blow-out, all other North Sea states were given information about the accident. As the oil slick developed in size and changed

in direction, the other North Sea states were given all relevant information. In addition to the Norwegian air surveillance over the Ekofisk area, the United Kingdom, Denmark and Germany also sent planes, and information from these flights was continuously given to the Norwegian authorities. Shortly after the accident the ambassadors of all North Sea states constituted an *ad hoc* committee at the request of the Norwegian Foreign Minister, in order to ensure that their governments would be given updated information on what was happening. In addition the Committee proved to be a useful forum for receiving offers of assistance and discussing possible help[297].

Co-operation of this kind is not confined to the governments involved. The oil companies are also engaged in a similar process. Special corporate agreements have been designed to allow ready integration and exchange of equipment and materials. They are indicative of the oil industry's ability to work together and to respond in a coherent, integrated way to the risk of massive oil spills. A typical example of such agreements is provided by the North Sea Operations Clean Seas Committee[298]. Eight members of the Committee with offshore interests in the North Sea and adjacent waters agreed on co-operative action to counter any major oil spill from their drilling and production operations. The result is a plan based on mutual understanding between the national operator who suffers an oil spill incident and associate organizations which are members of the Committee.

Small spills which occur through the lapse of good house-keeping practices are not covered by the plan. In case of a significant spill, if a call for assistance is received from an affected operating company, the others through their national associate organizations respond by the provision of materials and equipment. Clean-up operations are to be organized by and under the control of the affected operating company, whereas the transport of equipment is subject to mutual arrangements between this company and those providing assistance. The equipment consists of oil spill dispersants, approved by government authorities and compatible with standard spraying equipment, and mechanized oil retrieval devices. (The agreement includes a list of available equipment and materials as well as a list of persons to be contacted.) In order to keep up with continually improving technology, equipment revisions and additions are provided for in the agreement. As to the costs of assistance and responsibilites incurred, only the affected company is responsible for the replacement of the material used and its return in good order. No responsibility or liability is attached to an organization or company providing assistance in relation to damages arising from the clean-up operations, deficiencies of the materials, fault of their personnel, or damage caused by oil pollution as a result of the spill[299].

2. Liability and Compensation Issues

First, in pollution law a problem of proof often results from the dilution of the effluent in the sea. Oil slicks can, in a matter of hours, spread over large areas of water. The effluent is also carried hither and thither by wind, current and tide. In addition, it is frequently invisible, and the most visible effluents discharged into the sea are not necessarily the most toxic. Yet, in order to obtain compensation, a victim must prove not only his loss but also the origin of the loss: in other words, the source of the pollution and more particularly the chain of causation between the

source of pollution and the damage. If he is unable to establish that the loss was caused by a specific agent, he will obtain no compensation. Besides, most legal systems require him to prove that the discharge complained of is a result of a wrongful act or omission ("fault"), in the absence of which the party causing the damage cannot be required to make reparation.

The fact that the oil is coming out of a hole in the seabed and not out of a ship should be a factor of simplification. The rig does not move while in operation, and the potential risk is therefore located with precision. The first victim to be affected by pollution will be the coastal state which issued the exploitation licence, and it is therefore in a good position to obtain compensation from the operator. But, on the other hand, that state might be held liable for the way in which the drilling operations have been conducted[300]. When the Brussels Civil Liability Convention[301] was adopted in 1969, it was considered to be one of its major deficiencies that the question of liability for damage caused by offshore oil-spills was left out. Then, as offshore exploration and exploitation developed at an accelerated pace in the North Sea, the United Kingdom government, mindful of the resulting potential risk of pollution in that area, undertook to devise a legal regime of liability similar to that of the tanker owners. The intergovernmental conference convened for that purpose concluded a convention in December 1976 (hereafter referred to as the London Convention)[302]. The main purpose of the Convention was to set up a uniform regime of third party liability which would take care of the transnational character of the risk. As to substantive rules, several participating states favoured a mere adaptation of the 1969 Brussels Convention concerning tankers owners to offshore operations. However, the regime finally set up in 1976 appears to be stricter than that applicable to tanker owners, especially regarding the liability ceilings. This explains why the London Convention is not yet in force in 1981, many a state being reluctant to impose too severe a burden upon its offshore operators.

The 1978 London Convention enables compensation in the event of pollution damage resulting from offshore activities in the waters where a state party has sovereign rights over natural resources in accordance with international law. This includes not only the territorial sea and internal waters of such states, but also their continental shelf, as defined under international law, as well as their exclusive economic zone over a width of 200 miles. The North Sea is not the only area concerned. Since any sea area falling within the jurisdiction of the contracting parties is subject to the Convention, waters surrounding their overseas territories would also appear to be included.

The following installations are covered by the Convention:

a) any well or other facility which is used for the purpose of exploring for, producing, treating, storing, transmitting or regaining control of the flow of crude oil;

b) any well which has been used for the purpose of exploring for, producing or regaining control of the flow of crude oil from the seabed or its subsoil and which has been abandoned after the entry into force of this Convention for the Controlling State concerned;

c) any well which is used for the purpose or exploring for, producing or regaining control of the flow of gas or natural gas liquids from the seabed or its subsoil

during the period that any such well is being drilled (including completion) or worked upon (except for normal maintenance operations);

d) any well which is used for the purpose of exploring for any mineral resources other than crude oil, gas or natural gas liquids, where such exploration involves the deep penetration of the subsoil of the seabed;

e) any facility which is normally used for storing crude oil from the seabed or its subsoil.

This definition calls for the following remarks. It has been noted that a platform is often directly connected to several wells; in that case the wells together with such platform or facility, constitute one installation only. This may prove important, since the limits of liability apply to each "installation". Ships, of course, are not subject to the provisions of the Convention, since the 1969 Brussels Convention takes care of them. Therefore barges are not dealt with here. As they are towed and not self-powered, they cannot be regarded as ships. Nevertheless they are as a rule covered by maritime law, as the towing vessel and barge are regarded as a unit.

Another question is solved by Article 1(2)(b) of the Convention. If an installation is abandoned or lost, adrift or sunk, it becomes a *wreck.* Under most legal regimes, a wreck becomes a "res nullius", does not belong to anyone any more. In such cases the former owner, having abandoned his right, will therefore refuse to accept any responsibility. For that reason the 1976 London Convention applies to oil installations abandoned for up to five years, provided that they have been abandoned in accordance with the requirements of the coastal state. In other regimes, wrecks are subject to a special system derogating from the usual maritime regulations. Until 1978, this was the case under French law: the owner of the ship that had become a wreck had the right to abandon it to his creditors in full settlement. The owner of a wreck confiscated by the government is even more fortunate when he ceases to be the owner, since along with ownership the responsibility of compensating for any damage caused is also passed on to the state! Generally speaking, an owner relinquishes all rights to property which he abandons at sea, while entirely escaping liability for it by the same token.

Among the other issues raised by the London Convention, it is perhaps enough to discuss one more: the question of limits of liability. This topic, the most controversial of all during the preparatory conference, prevented any progress for nearly three years. The participants had split into two opposite camps. Some, such as the UK and France, favoured a maximum 25 million dollar limit of liability per incident, whereas Norway and Denmark supported liability unlimited in extent, or at least a 60 million dollar limit. The problem was to ensure that, as far as possible, liability was entirely covered by insurance and that the victims of pollution were properly compensated. It was advocated that it would be too severe a burden upon the industry to impose strict liability without any limitation, and that this risk might induce certain companies to refrain from taking part in activities on the continental shelf. The Norwegian experience appeared to be different, since despite domestic rules based on strict and unlimited liability, there had been no tendency for companies to withdraw from offshore operations for that reason. The compromise eventually agreed upon in 1976 provides that, notwithstanding the conventional limit, set

at 30 million S.D.R. (Special Drawing Rights) per incident, each state party remains at liberty to provide in its national legislation for a higher limit or unlimited liability[303]. This compromise has received criticism from part of the industry, not only because of the possible unlimited liability it provides for, but also because some fear that the absence of uniformity amongst muncipal laws might be a source of difficulties.

The Offshore Pollution Liability Plan (referred to as OPOL) was signed on September the 4th, 1974 by operating companies agreeing to accept strict liability for pollution damage resulting from their activity, up to the maximum of 16 million dollars per accident. Mirroring closely previous voluntary plans adopted by the industry concerning oil pollution by tankers, OPOL is intended to achieve the following:

1) to provide an orderly means for the expeditious settlement of claims arising from offshore exploitation and production operations;
2) to encourage immediate remedial action by the participants (they may themselves recover the cost of the remedial measures they take);
3) to ensure the financial responsibility of participants to meet their obligations;
4) to guarantee that claims would be met;
5) to avoid complicated jurisdictional problems.

Parties to OPOL must establish and maintain their financial responsibilities to meet their obligations for an amount of not less than 16 million US dollars for each occurrence and the double for each year.

OPOL came into effect in 1975. After 1981 it will continue to be in effect on a year-to-year basis.

At its Rio de Janeiro meeting in September 1977, the Comite Maritime International (CMI) discussed and adopted a draft convention on drilling platforms[304]. Article 7 of this draft convention deals with liability for pollution damages. Explicit reference is made to the 1969 CLC Brussels Convention, the provision of which should be extended to offshore platforms. The difficulty here lies in the limitation of liability problem. Since rigs are not measured for tonnage in the normal sense, the Brussels Convention provisions in respect of limitation amounts, calculated according to the ship's tonnage, are unsuitable here. It had been suggested that a fixed total limitation amount based on the size or the construction valve should be established. A mere reference to the 1969 Convention also appears to be largely insufficient, because, since this Convention is not applicable to bunker oil or storage tanks, important potential sources of pollution could be neglected. At any rate, should the Brussels Convention be extended to offshore drilling platforms, significant modifications would prove necessary in its provisions. On the whole, it seems doubtful that such adaptation is desirable. The shipping industry and the offshore industry are based on totally different market structures, and their response to a risk, pollution or otherwise, cannot be dealt with along the same lines.

After the Bay of Campeche oil-spill occurred in 1978, it was noted that the State of Mexico, in contrast to a company, private or public, could not be sued in the Courts of the United States, because the very cause of action against Mexico would be

defeated under the Foreign Sovereign Immunities Act. A question therefore arises about the basis and scope of state liability with *respect to offshore exploration* and exploitation, since these operations pose a significant extra-territorial environmental risk. This problem is directly related to the question of whether a state can be held responsible under international law for damages caused outside its jurisdiction by private activities falling under its control[305]. In light of Principle 21 of the Stockholm Declaration, it might be argued today that there is strict liability on the coastal state responsible for issuing the drilling licence. In the absence of a clear precedent, however, it seems difficult, at least for the time being, to admit that this major step has been taken by international law. Negligence continues to be a necessary element of state liability towards other nations, and the mere fact that a transnational environmental damage has originated in one state does not appear to be sufficient basis for its being held liable[306]. This, however, needs qualifying: the nature of the negligence required may vary considerably, since the controlling state is not only in charge of issuing licences, but also of issuing safety regulations related to the drilling operations, and providing for their sound supervision and control. One will recall that in the Santa Barbara case, the US federal government, which had jurisdiction over the waters where the blow-out took place, was sued by the Attorney General of California on the grounds that having authorized an increase in the drilling operations, it had therefore contributed through its decision to the crack of the seabed which provoked the oil-spill. The negligence necessary to establish a violation of a state's international obligations may also consist in the "inadequacy of an organizational system to assure a safe drilling operation", as was stated by the Norwegian Royal Commission which investigated on the 1977 Ekofisk blow-out. As to the inadequacy of an administrative structure to cope with a major oil-spill, or to control and mitigate the consequences of a blow-out, the question whether such shortcoming can be considered negligence on the part of the coastal state vis-à-vis neighbour countries, and therefore a violation of its International duties, is still uncertain. It should, however, be noted that the situation of the potential victim coastal state differs according to whether or not it was engaged on the offshore operations which caused the oil-spill. If not, the situation is similar to that of an oil-spill caused by a tanker accident, and it seems highly doubtful that the negligence of the victim state and its lack of organization can be held against it. Only a reduced indemnity is conceivable in such a case.

F. Conclusion

Clearly, legal development for the control of oil pollution in the offshore sector is still at an early stage. Most governmental and corporate arrangements worked out, to date, have been in the North Sea region. As offshore oil production proceeds rapidly around the world, it must be hoped that these North Sea arrangements will be taken into account, and improved upon, by governments and companies elsewhere.

VI Pollution from Deep Ocean Mining

James N. Barnes

A. Scope of the Problem

1. Introduction

Of the five kinds of marine pollution problems reviewed in this study, that of pollution from deep ocean mining is, in several respects, the most difficult to deal with. First, unlike the others discussed above, it is a problem that exists in prospect only, not yet in actuality. Deep ocean mining issues have been negotiated at UNCLOS III in advance of deep ocean mining. At the time of writing — in mid-1981 — it is still impossible to predict when and how such operations are likely to begin. Second, there is very little scientific information available to permit a reliable forecast of the most probable environmental effects of such activities. Relatively little is known technically about potential impacts, either short-term or long-term. Third, it is probable that the disruptions on the deep ocean floor and in superjacent waters as a result of mining operations may fall outside the definition of "marine pollution" adopted at UNCLOS III. Moreover, the whole question of deep ocean mining has been the most politically controversial area of the UNCLOS III agenda from the beginning. At an early stage of the Conference it was agreed that the deep ocean floor beyond limits of national jurisdiction (designated the "Area") shall be governed by the principle of the common heritage of mankind (Article 136) and that activities in the Area shall be regulated by a global intergovernmental organization, the International Seabed Authority (Articles 156-185). But as the Conference approached its 1981 summer session in Geneva several major issues remained unresolved and some older issues, thought to be resolved, seemed to have been reopened.

The relevant provisions negotiated at UNCLOS III have aroused heated debate among the nascent deep-sea mining industry, existing land-based producers, developing countries and environmental groups, all of whom are competing to have their concerns addressed in a certain way by the Convention. In order to provide a viable basis upon which mining can take place, the Convention and its implementing framework of rules, regulations and practices must recognize and reconcile these varied concerns.

Mining companies require a stable investment climate within which to continue development of the new industry. If investments are to continue during the years until the Convention is in force, industry requires guaranteed access to chosen sites, security of tenure for the time period necessary to recoup its investments, and fair conditions governing such items as promulgation of rules and regulations and mandatory transfers of technology. Only if these conditions are satisfactorily met will companies be in a position to finance adequately their operations during the interim period before the Convention comes into force.

Developing countries want assurance that they can share in the benefits of exploited seabed resources in accordance with the principles of the "common heritage of mankind", and within a time-frame similar to that of industry. To that

end, the Enterprise (the new international entity designated to conduct common heritage mining operations on behalf of developing countries) requires commitments regarding adequate capital, technology, and training.

Neither the mining companies nor the developing countries have given much attention to the regulatory framework necessary to deal with the adverse impacts deep ocean mining would have on the marine environment. The matter is complicated — or made more sensitive, politically — by the mining industry's desire to continue developing its deep ocean capabilities through the interim period before the UNCLOS III Convention can come into force. Accordingly, it seems important to develop interim as well as permanent arrangements for a system of environmental regulations consistent with the concept of the common heritage of mankind as well as with the economic, technical, and scientific realities of deep ocean mining.

2. Potential Environmental Impacts Associated with Nodule Recovery

Adverse environmental impact will occur in or near the sea floor, in the water column and at the ocean surface, as a result of the process of nodule recovery. As nodules are scraped from the ocean floor, sediment will be suspended and benthic flora and fauna disrupted and possibly destroyed. Nodules raised to the surface will be accompanied by deep ocean water and sediment, and possibly by dormant spores. The dumping of the sediment on the surface will create a "plume" over the ocean that may take years to settle downward. The effect of the plume is unknown; it could be detrimental to important elements in the food chain, like phytoplankton or fish.

Mining inevitably will substantially affect the ocean bottom. The scraping or sucking techniques employed will kill bottom-dwelling organisms. We know, for example, that there is an entire community of life-forms attached to the surface of the nodules, many of which are not identified at this time. One recent discovery, giant worms on the floor of the Pacific ocean, have been found using colonies of bacteria to obtain food and energy in a manner never before observed; for this and other reasons, these worms have been recognized as a new family in the animal kingdom. Furthermore, do not know how any of the benthic organisms are linked to the overall oceanic food chain. A 1000 ton per day nodule mining machine probably will move more than 4000 tons of sediments. Much of this sediment will be quickly redistributed on the ocean floor, killing whatever organisms are there. A cloud of suspended sediment will be formed (the benthic plume) as a result of these operations. Depending on particle size, portions of the sediments may take years to settle. Another unknown is the recolonization rate for disturbed areas. Benthic community metabolism in the deep ocean may be three orders of magnitude lower than in shallow waters. If this is correct, recovery time may be measured in the hundreds of years.

Whatever mining method is used, some water and sediments will be released along the upward path into the water column. More sediments are likely to come to the surface with the suction method. Although it may be technologically possible to pipe most sediments back down for discharge at intermediate depths, it is unknown whether that would be environmentally preferable. If all materials are discharged at or near the surface, a plume will be formed from the particles. It remains unclear

how long it will take these particles to settle, although they will be dispersed laterally quite quickly.

The environmental impact of the suspension and resedimentation of stirred up and discharged sediment has not been measured or analyzed at this time. The disposal of deep ocean sediment and bottom water in the surface areas could result in several types of environmental impacts. The upper layers of the ocean are where phytoplankton exist, where photosynthesis takes place, and where part of the earth's oxygen is produced. By depositing sediment on top of this zone some basic processes of nature would be disturbed. With light penetration limited, phytoplankton could be reduced substantially and photosynthesis and early stages of the food chain impaired. If certain ocean layers are rendered less hospitable, the vertical distribution of phytoplankton may be changed. As sediment sinks, bacteria attached to it may use oxygen in oxygen-scarce zones and adversely affect organisms there.

Locations where mining is going to take place are also, in part, areas of intensive fishing for tuna and other species. Whether these two uses of the ocean — fishing and deep-sea mining — are compatible, is unclear. Whether the sediment discharged from the surface will be detrimental to commercial fish stocks either directly, or indirectly through their effect on the food chain, is not known.

Over the last five years a number of studies of these matters have been carried out by the US National Oceanic and Atmospheric Administration: the project known as DOMES (Deep Ocean Mining Environmental Studies). The most recent DOMES monitoring of industry prototype mining indicates that much of the concern regarding surface plumes and possible problems with spores may be without foundation. On the other hand, it is important to keep in mind how little still has been studied. We do not even really know how the nodules themselves are formed, and what man's activities may do to that production. There is some evidence that the process of nodule formation is, in part, bacteriological. There is the possibility that spores will be brought to the surface, and virtually no research has been undertaken on bacteria which occur in the proposed mining region. NOAA's technical memoranda on these subjects repeatedly stress that, because of our lack of knowledge, true consequences cannot be predicted reliably.

3. Potential Environmental Impacts Associated with Processing

While initial processing operations likely will be land-based, some experts have suggested that at-sea processing ultimately offers the best solution to the problems of transport and the disposal of tailings. Processing at sea would probably involve the use of large ocean thermal energy conversion (OTEC) plants to produce the necessary energy. These plants could have substantial environmental implications in and of themselves. Even if a conventional fossil or nuclear energy plant were used, there could be substantial air and water pollution risks, with resultant impacts on various organisms. A processing plant on land would require considerable energy and water, both of which will have environmental impacts. There would be the same types of air and water pollution problems as with at-sea processing, except that they would have more direct impact on humans.

The largest potential impact, however, comes from the tailings, which would have to be dumped into the oceans or disposed of on land. The toxic reagents used in the processing could have serious impacts on organisms. DOMES has indicated that acids and other toxic materials could kill organisms in the vicinity of the effluent, including tuna or other economic species. Tuna, for example, are known to be particularly sensitive to copper. The extent of this impact would be magnified if such species were attracted to mining vessels or associated plumes. Flocculants might also produce mortalities of pelagic or benthic organisms.

The tailings would contain possibly dangerous metals. Heavy metals are known to cause adverse effects on organisms. DOMES concluded that, upon reaching critical concentration, these metals could produce mortalities at any level and ultimately contaminate products destined for use by man. At present, we do not have any real idea of the seriousness of these problems. The necessary research has not yet been undertaken.

B. Environmental Responses at UNCLOS III

1. Introduction

Does the Convention, as it is presently written, contain the necessary provisions to preserve and protect the marine environment? There is cause for serious concern. Recently changed decision-making procedures threaten to make unworkable Convention mechanisms for the protection of the marine environment. Many crucial policy decisions have yet to be made regarding such issues as research, monitoring and enforcement requirements, all of which will have a substantial impact on the soundness of the regulatory regime from an environmental perspective.

2. Environmental Obligations

From the perspective of environmental concerns, the Convention sets the groundwork for a comprehensive system of protection. The Convention devotes an entire section (Part XII) to provisions detailing the broad duty of all participating nations to protect and preserve the marine environment. All states, under Article 192, "have the obligation to protect and preserve the marine environment." Further, states must take "all necessary measures ... to prevent, reduce and control pollution of the marine environment from any source" (Art. 194). States have an obligation to act "so as not to transfer, directly or indirectly, damage or hazards from one area to another or transform one type of pollution into another" (Article 195).

With respect specifically to activities in the Area (the seabed and ocean floor and subsoil thereof beyond the limits of national jursidiction), Article 145 requires that all necessary measures be taken to ensure effective protection of the marine environment, and requires the Authority to adopt rules and procedures for

(a) the prevention of pollution and contamination and other hazards to the marine environment, including the coastline, and of interference with the ecological balance of the marine environment, particular attention being paid to the need for protection from the consequences of such activities as drilling, dredging, excavation and operation or maintenance of installations, pipelines, and other devices related to such activities;

(b) the protection and conservation of the natural resources of the Area and the prevention of damage to the flora and fauna of the marine environment.

In addition, it is relevant to note some further general pollution control obligations that would fall on individual states that may engage in deep ocean mining. Each state is required to take steps to "observe, measure, evaluate and analyze, by recognized methods, the risks or effects of pollution of the marine environment" (Article 204(1)). Article 204(2) requires in particular that states must analyze "the effects of any activities which they permit or in which they engage to determine whether these activities are likely to pollute the marine environment." Each state must assess the potential effects of planned activities which "may cause substantial pollution of, or significant and harmful changes to, the marine environment" and must report the results of such assessments (Article 206). Each state is bound to co-operate actively in "regional and global programmes to acquire knowledge for the assessment of the nature and extent of pollution, the exposure to it, its pathways, risks and remedies" (Article 200). Consistent with the spirit of environmental interdependence, Convention provisions make substantial requirements for interstate disclosure and dissemination of information. States must notify each other of imminent or actual damages:

> when a State becomes aware of cases in which the marine environment is in imminent danger of being damaged ... it shall immediately notify other States ... as well as the competent international organizations" (Article 198).

Further, as discussed above, states must "publish reports" of findings relating to the risks or effects of marine environmental damage. Again, reports of this nature submitted to international oversight bodies must be made "available to all States" (Article 205). States and international bodies must "promote and facilitate the development and conduct of marine scientific research" (Article 239), and must ensure that such research fully complies with Convention regulations for "the protection and preservation of the marine environment" (Article 240(d)).

These provisions for individual state environmental analysis are reinforoed and expanded on the international level primarily through the mandate of the Legal and Technical Commission. This Commission would be required to "prepare assessments of the environmental implications" of activities in the Area, to "make recommendations to the Council on the protection of the marine environment, taking into account the views of recognized experts in that field", and to recommend "to the Council to disapprove areas for exploitation by contractors or the Enterprise ... [when] evidence indicates the risk of serious harm to the marine environment" (Article 165(2)(d), (e), and (l)).

Wide public interests are represented in decision-making positions within proposed law of the sea governing bodies. With respect to environmental research and advisory issues, it is significant that persons with diverse backgrounds must be involved. Article 165(1) especially stresses the importance of selecting members with backgrounds relevant to mineral resource exploration, oceanography, protection of the marine environment, economics and law.

The public disclosure of information gathered both by the Commission and by other state and international bodies is also provided for in the draft Convention. Article

143 requires the Authority to "co-ordinate and disseminate the results" of marine scientific research and analysis, and similarly requires states "effectively to ensure development or programmes for the benefit of developing states with a view to disseminating the results of research and analysis" which states parties have undertaken in the area. Article 244 broadly requires states and international organizations to "make available by publication and dissemination through appropriate channels information of proposed major programmes and their objectives as well as knowledge resulting from marine scientific research." Finally, environmental disclosure is specifically emphasized in Article 205, which provides that states must publish reports of the result obtained relating to risks or effects of pollution on the marine environment. Provision is also made for reports to relevant international organizations "at appropriate intervals" which must in turn make the reports "available to all states".

3. Amendments and Revisions

Amendments and modifications are provided for at all levels of the proposed UNCLOS III framework, both as a result of periodic reappraisals and as new information compels revision. Article 154 requires that every five years, the Assembly undertake a "general and systematic review" of the overall operation of the international regime in the Area. In light of data gathered as a result of such a survey, and necessarily including environmental information, the Assembly is empowered to "adopt or recommend that other organs adopt" substantive measures to correct the inadequacies uncovered and to improve the operation of the regime.

The Legal and Technical Commission, charged with assessing the environmental implications of activities in the Area and with formulating rules and recommendations based on such assessments, is required to "keep such rules, regulations and procedures under review and recommend to the Council from time to time such amendments thereto as it may deem necessary or desirable" (Article 165(2)(g)). The Commission is also required to recommend to the Council issuance of "emergency orders" for the "adjustment of operations to prevent serious harm to the marine environment arising out of activities in the Area." Such recommendations for revision must be "taken up by the Council on a priority basis ..." (Article 165(2)(k)).

Provision is also made in the Convention for revision of contracts and for amendments in individual exploitation plans. Specifically, Article 17(2)(b)(iii) of Annex III requires that "the total duration of exploitation ... should ... be short enough to give the Authority an opportunity to amend the terms and conditions of the plan at work" in accordance with rules and regulations under the Convention and necessarily including environmental mandates cited above. Article 19(1) of Annex III further requires the "revision of [a] contract" when "circumstances have arisen, or are likely to arise" which make it unlikely or impossible to achieve the contract objectives". This mandate also necessarily comprehends any and all environmental objectives set forth, as required under the Convention.

4. Decision-Making Procedures

The environmental implications of decision-making procedures under the Draft Convention have not yet been thoroughly considered, and there are certain ambiguities in this respect in the present text. In particular, Article 162(2)(n)(ii) gives the

Council authority to adopt rules, regulations and procedures that "relate to prospecting, exploration, and exploitation ..." If that phrase were read to include environmental rules, then Article 161(7)(d) would require consensus decision-making. Moreover, all other possible actions of the Legal and Technical Commission apparently would be subject to whatever voting rules are recommended by the Preparatory Commission for the Convention. These functions cannot be carried out responsibly if subjected to consensus decision-making. Governments should be urged to ensure that the final Convention not require consensus procedures for environmental decisions.

Specifically, there is concern about the proposal procedure relating to the issuance of emergency orders for the suspension or adjustment of mining operations to prevent serious harm to the marine environment. Under Article 161(7)(c), a three-fourths majority is required to issue an emergency order, but only on the proviso that such orders may be binding for no more than 30 days unless confirmed by a consensus vote. Such a requirement makes the issuance of such an order for more than 30 days extremely difficult to accomplish, since consensus decision-making would apply after the 30-day period. It also fails to distinguish between orders for suspension and orders for adjustment. In addition to the recommendation that the general consensus requirement be modified, it should be recommended that the 30-day limitation on emergency orders apply only to suspension orders, and that orders issued for the adjustment of operations be based on non-discriminating performance regulations applicable to all mine sites without imposing any predetermined cut-off date.

One potential way to deal with this problem would be to require by way of regulation that the terms of contracts under which deep ocean mining activities take place must provide for appropriate modification or other changes necessary to accommodate new factors and information.

C. Priority Requirements

1. General requirements for deep ocean mining

As outlined above, the UNCLOS III Draft Convention contains many principles and obligations related to the protection of the marine environment from deep-sea mining activities. They include (1) reviewing detailed written plans of work, (2) supervising activities in the area, (3) preparing assessments of environmental implications of activities in the area, (4) making recommendations to the Council regarding protection of the environment, (5) formulating rules, regulations and procedures, (6) keeping such rules, regulations and procedures "under review" and recommending from time to time such amendments as are "necessary and desirable", (7) establishing regular monitoring and research programme, including that needed to "ensure that existing regulations are adequate", (8) recommending that emergency orders be issued, including those for "suspension or adjustment of operations" that are needed to prevent serious harm to the marine environment. The Preparatory Commission will draft the detailed rules, regulations and procedures to implement these requirements.

It seems that the newly revised text would require that all rules be set by consensus, and that later changes also be made by consensus. This has been agreed to, ap-

parently, in order to protect the companies against requirements that could prevent mining, but it is inappropriate for sound environmental regulation. What is required is establishment of a separate body that is given the specific obligation to generate environmental requirements. That "Environmental Committee" should be staffed only by qualified professionals and should be insulated from politics insofar as is possible. Such a body would be the sole original source of environmental rules or changes thereto, so that the companies would not need to fear politically motivated manipulation of the regulatory structure.

2. Need for More Research and Information

In view of the fact that little is known about the character of the marine environment of the deep ocean and its susceptibility to harm from deep seabed mining, general research should be directed to establishing this character, in addition to research on the particular impacts of such mining. This general research should be embarked upon as soon as possible, preferably on a co-ordinated international basis. Comprehensive environmental assessments should be carried out prior to any mining activities taking place. Governments should consider inviting an appropriate international body to prepare an environmental assessment of deep seabed mining under the Draft Convention. Monitoring and review of the impact of deep seabed mining activities on the marine environment should be undertaken in a routine and regular manner by recognized scientific methods.

Mechanisms for co-operative international and inter-governmental research, inspection and enforcement should be established and should draw if possible on existing practices and precedents. Governments should consider working through such organizations as the International Oceanographic Commission, the Scientific Committee on Oceanographic Research, UNEP, GESAMP, ICES and, as appropriate, OECD, to develop a comprehensive ocean research programme related to development of the deep seabed mining industry. This could be focused initially on developing a greater understanding of benthic ecosystems and their relationship to other ocean systems.

With a view to strengthening the environmental safeguards to be developed by the International Seabed Authority under the UNCLOS III Convention, it is recommended above that an Environmental Committee should be established. Such a committee should have the power to require collection and dissemination of data on the causes, effects and possible means of controlling marine pollution from all aspects of mineral exploitation of the ocean beyond the limits of national jurisdiction, and the power to compel disclosure of relevant information from states, private parties and the Enterprise about potentially harmful activities. Monitoring of all exploitation activities should be required, in order to expand the scientific data base and to ensure that prevailing standards yield the desired level of protection.

The Environmental Committee proposed above should also have the power both to authorize research on its own and to design research plans for the Enterprise and other interested operators. It should have the responsibility to analyze the potential environmental impacts of proposed activities, no matter who is conducting the activities.

3. Need for a Comprehensive System of Environmental Management

Because of the possible dangers to the marine environment, and in light of the pervasive lack of information and analysis of actual consequences, environmental organizations have stressed the need for a comprehensive regulatory structure. With a view to future mining activities in the Area, governments, international organizations and private parties should join to create a management structure along the following lines:

1. Baseline studies, monitoring and assessment should be a principal component of the system.

2. The resulting information should be used to develop standards and measures for environmental protection which reflect the genuine needs of the oceans. In this connection, the definition of pollution falling under the aegis of the regulatory agency should include both damage caused by the introduction of any substance associated with mining and processing into the marine environment, and damage from any activity associated with mining and processing, whether or not a new substance is introduced. The preparation of site-specific environmental impact statements is most desirable.

3. Measures should be enforced without regard to political exigencies, and conflicts of interest between development and regulation must be avoided whenever possible. This requires adequate representation of the non-economic interests in the decision-making process and a bureaucratic separation of the development and protection functions.

4. Environmental rules, regulations and measures based on new data and research should be applied to on-going mining and processing operations.

5. National and international conservation programmes for the critical minerals in question should be established, so that excess mining can be avoided.

It seems clear that efforts to assess the environmental implications and to control the adverse effects of proposed actions by nations, private companies or the Enterprise are unlikely to be effective unless performed according to a broad mandate of protection and under a framework of scientific integrity. The ultimate objective should be to create a regulatory entity with sufficient responsibility, authority, staff, funds, and expertise to exercise the functions needed for maximum protection of the marine environment.

4. Need for Establishment of Protected Areas

While quite a lot of attention has been paid to the possible environmental effects of nodule recovery and processing in the Area in general, and especially in waters adjacent to sites selected for commercial deep-ocean activities, it is striking that so little attention has been devoted to the concept of protected areas.

Even since UNCLOS III was convened, late in 1973, some important discoveries have been made:

— the presence of hot metalliferous brines in the Red Sea and elsewhere;
— the existence of entire communities of animal forms new to science whose

primary source of nourishment is derived not from photosynthesis but from sulphur metabolising bacteria;
— the venting of gases and suspended materials from the sea floor, with compositions extraordinarily high in a variety of chemical elements, including metals such as manganese, copper and silver;
— the fact that forms of life in the deep oceans, many similar to some of those in shallow water and others apparently quite different, have extremely long lifespans; and
— the realization that diversity of life in the deep sea, previously thought to be low, may in many instances be comparable to the high diversity and abundance characteristic of tropical shallow water communities.

These and other considerations are all relevant to human evaluation of the resources of the deep seas. Other properties, such as thermal layering in the open sea and volcanism of the seabed, are being considered as sources of recoverable energy and materials. This is largely because the development of underwater technology, which makes new resource use feasible, has been as rapid as scientific discovery. These natural deepsea systems, which nodule recovery could change in ways we cannot now imagine, have existed for millions of years before the appearance of humans on the land. Our intervention by exploitation and even by vigorous exploration is likely to disrupt these systems, destroying some of their potential value to us and impeding further discovery and understanding.

On the land, the growth of economies and the explosion of technology have led us to hasten to protect and set aside areas for fear of irrevocable loss. It is imperative that we take similar steps regarding the deep ocean. The vast area of the ocean and the difficulties of operating at great depths will not offer incidental protection for long. Heretofore, protection has been achieved by default, because the deep sea was too remote to be affected greatly by human activity. Now such protection must begin to involve conscious action.

There is a provision in the US deep ocean mining law for the establishment of "stable reference areas", in which no mining shall take place. Similar precautions might be expected under the legislation of other nations preparing to engage in deep ocean mining. National legislation will not be enough, even in the interim period, but however deep ocean mining gets under way, the same questions have to be addressed:

— How many such areas are needed, how big, and what shapes should they be?
— Should they be contiguous or scattered?
— How important is the need for some protected areas to be adjacent to mined areas (e.g., for recolonization by organisms)?
— How should different special features be treated, insofar as they are already known (e.g., deep trenches, areas of high volcanism, locations of discontinuities of temperatures and chemical composition, and of concentrations of deepsea organisms)?
— How can it be ensured, as far as possible on the basis of existing knowledge, that some of the protected areas are comparable to those for which licences are issued with respect to nodule distribution and composition, and to ecological, topological and oceanographic features?

Since the protected areas can only be established by international agreement, there is a strong *prima facie* case for the involvement of the international scientific organizations, particularly the Scientific Committee on Oceanographic Research (SCOR), at an early stage in formulating selection criteria, recommending suitable areas, and preparing research and monitoring proposals.

The actual design and conduct of investigations in designated areas would require intergovernmental co-ordination. Co-operative arrangements must, of course, involve the potential deep ocean mining states. It is desirable, however, that such co-operation remain the ultimate responsibility of the community of nations as a whole, and UNESCO/IOC is probably the appropriate instrument for that.

Although the details of preferred locations and the choice of the protected areas are primarily scientific questions, the decision regarding the *scale* of protected areas as a whole is in part a political question. In terms of the prime need to provide a hedge against the unknown, these areas should be as large as possible.

A reasonable policy would be that for every area for which a permit or licence is issued, an area of equal size be designated for protection. The ultimate result would be that when the international regime is eventually in operation, the areas designated for exploration and exploitation by both the Enterprise and private industry would be equal in size to the areas designated for protection.

5. Need for Avoidance of Conflicts of Interest

All nations have had experience with conflicts between economic or development interests on the one hand, and environmental, conservation and health interests on the other. Almost inevitably the development interests win out, unless (i) adequate representation of the non-economic interests is provided in the decison-making process, and (ii) there is a separation bureaucratically between the development and protection functions. In designing a mechanism to develop the common heritage of mankind, those conflicts should be avoided. Efforts to assess the environmental implications and to control the adverse effects of proposed actions by states, private companies or the Enterprise are unlikely to be effective unless performed according to a broad mandate of protection and under a framework of basic independence and scientific integrity. The Draft Convention does not now provide such a framework.

The Environmental Committee proposed above should be assigned the obligation, power and resources to:

(a) analyze the environmental implications of proposed mining and processing activities, including those to be undertaken by the Enterprise, (b) plan and provide for the carrying out of necessary scientific research, (c) promulgate standards, general regulations, and specific terms and conditions for licences obtained both by private companies, states, and the Enterprise, (d) monitor all mining and processing activities of licencees and the Enterprise, (e) enforce rules and regulations against private licencees, states parties, and the Enterprise, (f) issue suspension orders and revoke licences on sensible environmental grounds, pursuant to stated criteria, (g) apply environmental regulations based on new data and research to on-going mining and processing operations, and (h) institute appropriate actions before the settlement of disputes tribunal, or act in other appropriate ways to enforce its man-

date. While proposed rules and regulations should be ratified by the Council, such ratification should be the norm and not the exception. Once approved, rules and regulations should be binding on states.

6. Need for Development of Standards and Measures

Independent experts in the relevant fields, working for the Legal and Technical Commission, should have the obligation, given the current state of knowledge, to establish appropriate international standards and measures for preventing harm to the marine environment from any activities associated with exploitation of the common heritage of mankind. With respect to deep ocean mining, environmental regulations must be required before any mining under the Convention takes place. The definition of pollution falling under the aegis of the Commission should include damage caused by the introduction of any substance associated with mining and processing into the marine environment, and damage from any activity associated with mining and processing. All aspects of processing at sea must be covered, whether or not it takes place immediately over a mine site. The Commission should have the obligation and power to modify its standards and measures in light of later research results, as appropriate.

7. Need for Public Participation and Access to Information

Private parties should be accorded adequate opportunities to participate in the various decision-making processes to be used by the Authority and the proposed Environmental Committee, in order to prevent or redress harm to their particular environmental objectives or on behalf of the international community. When acting on their own behalf, private parties may legitimately be required to demonstrate that the action involved affects their particular or special environmental and/or natural resources interests. When acting on behalf of the international community as a whole, non-governmental individuals or groups should be required to demonstrate an established concern and knowledge regarding the environmental matters at issue.

In particular, non-governmental individuals and groups should be permitted to participate in the process of preparing international pollution control standards and measures; to challenge the Legal and Technical Commission's actions as being arbitrary, unreasonable or not in accord with environmental needs; to support or challenge selection of areas for mining and/or preservation; to participate in the process of licensing; and to initiate proceedings that can prevent or redress harm to their environmental interests or harm generally to the marine environment.

Whatever interim arrangements are made for deep ocean mining in the period prior to the entry into force of the final UNCLOS III Convention, licencees and permittees should be required to submit all data that reasonably is necessary or relevant to determinations with respect to the issuance, revocation, modification or suspension of any licence or permit. An effective public participation procedure would make all such data routinely available to other governments and to non-governmental groups as well.

In addition to public disclosure requirements, interim arrangements should provide for public participation and commentary in the process of formulating and carrying out a deep seabed mining programme.

8. Need for Interim Arrangements

Reference has been made to the need for special environmental safeguards for deep ocean mining in the interim period before the Convention on the Law of the Sea comes into force.

It is enough for the purposes of this study to point out that virtually all the environmental precautions discussed above should be initiated as soon as possible in whatever institutional form seems appropiate for the interim period. The more protracted this period, the more essential it is that sensible environmental (and other) restraints should be accepted by those governments and corporations that take the first steps into this new area of ocean resource production.

Notes

1 Douglas M. Johnston, "The Environmental Law of the Sea: Historical Development", *supra*, pp. 17 *et seq.*
2 Cyrille de Klemm, "Conservation of the Living Resources of the Ocean", *supra,* pp. 163 *et seq.*
3 *Ibid.,* p. 79
4 *See*, for example, Allen L Springer, "Towards a Meaningful Concept of Pollution in International Law" (1977), 26 *Int'l and Comp. L.Q.* 531; James Barros and Douglas M. Johnston, *The International Law of Pollution* (1974), pp. 6-7; and Michael Hardy, "Definitions and Forms of Marine Pollution" in Churchill, Simmonds and Welch, *eds., New Directions in the Law of the Sea,* vol. III (1973), pp. 73-77. For more recent discussions, *see* Gregorias J. Timagenis, *International Control of Marine Pollution* (in 2 vols., 1980), pp. 22-25, and Jose A. Yturriaga, "Regional Conventions on the Protection of the Marine Environment", Academie de Droit International, *Recueil des Cours* 1979, vol. I, pp. 331-332. See also J. McLoughlin, *The Law Relating to Pollution: An Introduction* (1972), pp. 1-7.
5 UN Doc. A/77.50, Part 1-2 (10 November 1963).
6 UN Doc. A/CONF. 62/WP. 10/Rev. 3 (27 August 1980).
7 *Ibid.,* Article 1.
8 For this reason, it has been suggested that the concept of marine pollution has two elements: "a) the introduction by man of certain substances or energy into the marine environment; and b) the harm that is caused to other legitimate activities that depend in some way on the use of the marine environment". Yturriaga, *supra* note 4, at p. 332.
9 During the debate on the future status of the Panama Canal, in the late 1970s, proposals were made for the construction of an alternative canal through less politically sensitive territory
10 The best known project of this kind was the Mohole project conducted in the United States in the 1960s
11 Many marine pollution law specialists — including those who have contributed to this chapter — specialize at present in more than one (source-defined) area or aspect of marine pollution, but the current trend toward further specialization seems likely to continue, as the field of inquiry becomes more voluminous and more technical.
12 James E. Mielke, "Effects of Man's Activities on the Marine Environment" (May 1975), study prepared for the US Federal Committee on Commerce and National Ocean Policy Study, 94th Congress, 1st Session.
13 Not all of these substances are invariably included in all black lists. *See infra,* pp. 278-280.
14 *Infra,* p. 253.
15 For an early compilation of technical studies of these threats to the marine environment, *see* Mario Ruiuo, *ed., Marine Pollution and Sea Life* (FAO, 1972).
16 For a general study, see Boleslaw A. Boczek, *Flags of Convenience: An International Legal Study* (1962). For more recent developments, see Emmanuel du Pontavice, "Les pavilions de complaisance" (1977), 29 *Droit Maritime Français* 503, and Rene Rodiere and Martine Rémond-Gouilloud, *La mer, droits des hommes ou des Etats* (1980), p. 176.
17 The principle of universal port state jurisdiction represents an enormous potential advance in the enforcement of international rules for the control of vessel-source pollution. In accordance with Article 218 of the Draft Convention on the Law of the Sea:
"When a vessel is voluntarily within a port or at an off-shore terminal of a State, that State may *undertake investigations* and, where warranted by the evidence of the case, *institute proceedings* in respect of any discharge from that vessel in violation of applicable international rules and standards established through the competent international organization or general diplomatic conference, *outside* the internal waters, territorial sea, our exclusive economic zone of that State" [emphasis added].
Smilarly, recent 200-mile extensions of coastal state jurisdiction give heightened significance to Article 220:
"When a vessel is voluntarily within a port or at an off-shore terminal of a State, that State may ... *institute proceedings* in respect of any violation of its laws and regulations adopted in accordance with this Convention or applicable international rules and standards for the prevention, reduction and control from vessels, when the violation has occurred *within* the territorial sea or the exclusive economic zone of that State" (emphasis added].
See Timagenis, *supra* note 4, pp. 486-515.
18 The latest legal definition of the continental shelf is contained in Article 76 of the Draft Convention on the Law of the Sea. *See infra,* p. 249.
19 On the early trends toward exclusive coastal state control over the resource of the shelf, *see* Myres S. McDougal and William T. Burke, *The Public Order of the Oceans: A Contemporary International Law of the Sea* (1962), pp. 631-663.
20 Draft Convention on the Law of the Sea, Article 80.
21 *See infra,* p. 245.
22 Draft Convention on the Law of the Sea, Article 1.
23 *Ibid.,* Articles 156-185.

24 *Infra,* p. 271.
25 Commercially significant quantities of manganese modules are believed to lie within the national jurisdictional limits of the United States and Mexico and perhaps other coastal states.
26 In 1967, the United States was the only country which actually had nuclear-powered vessels in operation, but it has never ratified the 1962 Convention.
27 *See infra,* p. 205. The unwillingness of flag states to prosecute their vessels for violations of the 1954 Convention limited the latter's effectiveness very considerably.
28 Until 1964 the French fishery law prohibiting noxious discharges was based on Colbert's "Ordonnance sur la marine" passed in 1681.
29 On the role of fear in the development of international environmental law, see Johnston, *supra* note 1, pp. 25 *et seq.*
30 International Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties (1969). Text reproduced in (1975) 1 *UST* 765, *TIAS* No. 8068.
31 *See infra,* p. 205.
32 For a recent list of states that have ratified the 1969 Intervention Convention, see *110.*
33 *Draft Convention on the Law of the Sea, Article 221.*
34 *Ibid.,* Article 56.
35 *See infra,* p. 200.
36 *See infra,* p. 253.
37 *See infra,* p. 254.
38 Principle 7 provides: "States shall take all possible steps to prevent pollution of the seas by substances that are liable to create hazards to human health, to harm living resources and marine life, to damage amenities or to interfere with other legitimate uses of the sea".
39 *See* list of environmentally significant provisions of UNCLOS III, *infra*, p. 387.
40 IMCO, The International Conference on Tanker Safety and Pollution Prevention 2 (1978) [hereinafter TSPP]
41 *Ibid.,* p. 4
42 *Ibid.*
43 *Ibid.*
44 *Ibid.,* p. 2
45 On the *Torrey Canyon* disaster, *see, e.g.* Brown, *The Lessons of the Torrey Canyon* (1968), 21 Current Legal Problems 223; McGurren, *The Externalities of a Torrey Situation: An Impetus for Change in Legislation* (1971), 11 Natural Resources J. 349; Utton, *Protective Measures and the Torrey Canyon* (1968), 9 B.C. Ind. & Com, L Rev. 613; (Comment), *Post Torrey Canyon: Toward a New Solution to the Problem of Traumatic Oil Spill* (1970), 2 Conn. L. Rev. 632
46 *See* McManus & Schneider, *Shipwrecks, Pollution & the Law of the Sea*, National Parks & Conservation Magazine 10 (June 1977). For a history of prior spills, see Senate Comm. on Commerce, Science, and Transportation, An Analysis of Oil Tanker Casualties: 1969-74, 95th Cong., 2d Sess. (1978)
47 *Ibid.*
48 The *Amoco Cadiz* is a textbook case illustrating the character of modern tanker operations. It was owned by a United States company, built in Spain, chartered to a multinational corporation, manned by an Italian crew, and flew a Liberian flag. The tanker was wrecked in French territorial waters, while bound for the United Kingdom and the Netherlands, with a cargo of Arab oil. Legal proceedings in respect of the casualty are now underway in Chicago. On the *Amoco Cadiz* and related problems and international legal responses thereto, *see* NOAA/EPA Special Report, The AMOCO CADIZ Oil Spill (W. Hess ed., USGPO, 1978); *Tanker Pollution: An Ocean World* Special, Ocean Reporter (April-May 1978)
49 For discussion of the normal cycle of tanker operations, see TSPP, *supra* note 40, at pp. 6-8
50 With LOT, after the mixture has settled the water is drawn off the bottom, so that oil and a small amount of water remain in the tanks. These consolidated slops are then transferred to a shore-based reception facility or combined with the next cargo (and hence the term "Load-on-top")
51 The SBT design concept requires that tankers have sufficient tanks to be able to reserve some for ballast water alone. Segregated ballast, whether or not it is placed in double bottoms, in addition to providing some defensive capability to vessels involved in incidents, also obviously helps reduce operating discharges. A similar concept is that of clean ballast tanks (CBT), whereby certain cargo tanks are reserved solely for ballast water. Since CBT, unlike SBT, does not require separate pipes and pumps, it is usually considered an operational, rather than a design or equipment, measure. Another operational measure which reduces the quantity of oil remaining on board after discharge (and therefore the potential for pollution) is crude oil washing (COW), a technique involving the cleaning or washing of cargo tanks with high pressure jets of crude oil (which has the property to dissolve its own residues). For discussion, *see* TSPP, *supra* note 1, at pp. 10-12
52 UST 2989, TIAS No 4900, 327 UNTS 3: with amendments *adopted* 11 April 1962, [1966] 2 UST 1523, TIAS No 6109, 600 UNTS 332; 21 Oct. 1969 (1977) 1 UST 1207, TIAS No 8505; 15 Oct. 1971, *reprinted* in 11 *Int'l Leg. Mats.* 267(1972)

53 These amendments came into force on 20 Jan. 1978
54 These amendments became a foundation for the subsequent 1973 International Convention for the Prevention of Pollution from Ships
55 2 UST 233, TIAS No 5200, 327 UNTS 3
56 The other three conventions to emerge in 1958 from that Conference were: Convention on the Continental Shelf, (1964) UST 471, TIAS No 5578, 499 UNTS 311; Convention on Fishing and Conservation of the Living Resources of the High Seas (1966) 1 UST 138, TIAS No 5969, 559 UNTS 285; and Convention on the Territorial Sea and the Contiguous Zone, (1964) 2 UST 1606, TIAS No 5639, 516 UNTS 205
57 Article 24 of the High Seas Convention reads in full as follows:
"Every State shall draw up regulations to prevent pollution of the seas by the discharge of oil from ships or pipelines or resulting from the exploitation and exploration of the seabed and its subsoil, taking account of existing treaty provisions on the subject"
Article 25 on radioactive wastes reads in full as follows:
"1) Every State shall take measures to prevent pollution of the seas from the dumping of radioactive waste, taking into account any standards and regulations which may be formulated by the competent international organizations
2) All States shall co-operate with the competent international organizations in taking measure for the prevention of pollution of the seas or air space above, resulting from any activities with radioactive materials or other harmful agents"
58 3 US Dept. State Bull. 759 (1957)
59 Text reproduced in 57 *Am. J. Int'l* L 268 (1963)
60 (1975) 1 UST 765, TIAS No 8068
61 Text in 9 *Int'l Leg.Mats.* 45 (1970). *See also* Protocol to the International Convention on Civil Liability for Oil Pollution Damage, 29 Nov. 1976, in *ibid.,* vol. 16, at p. 617 (1977)
62 Article 1
63 Since the protocol, the amounts have been denominated in terms of special drawing rights (SDRs)
64 11 *Int'l Leg. Mats.* 284 (1972). *See also* Protocol to the International Convention on the Establishment of an International Fund for Compensation for Oil Pollution Damage, 19 Nov. 1976, *in ibid.,* vol. 16, at p. 621 (1977)
65 8 *Int'l Leg. Mats.* 497 (1969)
66 10 *Int'l Leg. Mats.* 137 (1971)
67 For the workings of all the various agreements in this area, including TOVALOP and CRISTAL, *see* Carven, Becker, *Oil Spill Liability and Compensation* (1978) 23 Exxon Marine 2, at p. 24 (Summer 1978)
68 12 *Int'l Leg. Mats.* 1319 (1973)
69 *Ibid.,* preamble
70 *Ibid.,* Annex I. In this annex, a "special area" is defined as "a sea area where for recognized technical reasons in relation to its oceanographical and ecological condition and to the particular character of its traffic the adoption of special mandatory methods for the prevention of sea pollution by oil is required". Reg. 1(10). For its purposes, there are five internationally agreed-upon special areas: the Mediterranean Sea area, the Baltic Sea area, the Black Sea area, the Red Sea area, and the "Gulfs area". *Ibid.* Reg. 10
71 14 *Int'l Leg. Mats.* 959 (1975)
72 Protocol of 1978 relating to the International Convention for the Prevention of Pollution from Ships, 1973, in 17 Int'l Leg. Mats. 546 (1978)
73 Protocol of 1978 relating to the International Convention for the Safety of Life a Sea, 1974, 17 Int'l Leg. Mats. 579(1978)
74 For discussion, *see generally* TSPP, *supra* note 40, at pp. 10-12
75 *Ibid.* at pp. 12-13
76 On the procedural arrangements for entry into force and implementation of MARPOL and its protocol and SOLAS '74 and its protocol, *see ibid.* at pp. 21-23
77 On this whole question, see the lectures delivered at the Hague Academie de Droit International by Jose Antonio Yturrlaga Barberan, in Yturriaga, *supra* note 4, p. 323
78 9 *Int'l Leg. Mats.* 359 (1970)
79 The zones are described by latitude and longitude in the Annex to the Agreement.
80 13 *Int'l Leg. Mats.* 544 (1974)
81 822 UNTS 311
82 13 *Int'l Leg. Mats.* 591 (1974)
83 15 *Int'l Leg. Mats.* 290 (1976)
84 *Ibid.,* at p. 300
85 *Ibid.,* at p. 306
86 17 *Int'l Leg. Mats.* 511 (1978)
87 *Ibid.,* at p. 526
88 RSC 1970 (1st Supp.) c.2 (1970)
89 Among other things, the Arctic Waters Act prohibits and prescribes penalties for the deposit of "waste" in

Arctic waters, provides for civil liability for such deposits, enables the promulgation of regulations in relation to navigation in shipping safety control zones, provides for prohibiting of ships from such zones unless they meet the regulations, and makes arrangements for Pollution Officers to take enforcement measures. At the same time as passage of the Arctic Waters Act, Canada passed an accompanying Bill to Amend the Territorial Sea and Zones Act, RSC 1970 (1st Supp.) c.2 & 45 (1970)

90 US Statement on Canada's Proposed Legislation, *reprinted in* 9 Int'l Leg. Mats. 605 (1970)

91 Canadian Reply to the US Government, *ibid.* at p. 607

92 US Regrets Canada's Extension of High Seas Jurisdiction (1971), 64 Dept. State Bull. 139

93 33 USC & 1221 *et seq.* (Supp. V 1975). The purpose of this act is stated as the prevention of damage to various structures and "to protect the navigable waters and the resources therein from environmental harm -." *Ibid.* § 1221. *See also* Deepwater Port Act of 1974, 33 USC § 1501 *et seq.* (Supp. V 1975)

94 Fishery Conservation and Management Act of 1976 (FCMA), 16 USCA § 1801 *et seq.* (West 1979). The prohibition of foreign fishing without permits in a 200-mile zone off US coasts came into effect on 1 March 1977

95 Pub. L. No 95-217, 91 Stat. 1566 (1977) (codified at 33 USCA § 1251 and scattered subsequent sections (West 1978))

96 Clean Water Act, § 58, 33 USCA § 1321(b)(1) (West 1978)

97 *Ibid.* § 1321(c)(1)

98 For discussion of some early marine pollution efforts, *see* Hayashi, *Comparative National Legislation of Offshore Pollution* (1975), 1 Syr. J. Int'l & Com. 250

99 At the time of this writing the current set of draft treaty articles before UNCLOS III is Draft Convention on the Law of the Sea (Informal text) UN Doc. A/Conf. 62/W.P.10/Rev. 3 (27 Aug. 1980)

100 *Ibid.* Part XII, Articles 192-96

101 *Ibid.* Articles 197-203

102 *ibid.* Articles 204-6

103 *Ibid.* Articles 207-22. The standard-setting provisions are found in the first six of these Articles followed by those on enforcement

104 The general provision on pollution from vessels is found in Article 211. The limitation as to DEME standards appears in an earlier part on the territorial sea and contiguous zone in Article 21(2)

105 Dumping standards are dealt with in Article 210. *See also* Article 211

106 *Ibid.* Article 211(6)

107 *See supra,* p. 196

108 *Ibid.* Articles 223-33. At the time of this writing, there is still outstanding an informal proposal by the Union of Soviet Socialist Republics, which would make a separate part of the umbrella treaty on "general safeguards", rather than including such protections in Part XII on environmental provisions. *See Report by the Chairman of the Third Committee, Ambassador A. Yankov (Bulgaria),* 10 Third United Nations Conference on the Law of the Sea: Official Records 173, 186 (1978); *See also Report by the Chairman of the Third Committee, Ambassador A. Yankov (Bulgaria),* UN Doc. A/Conf.62/L34, at 3 (26 April 1979). The basic effect of the USSR proposal would seem to be to extend the safeguards to fishing vessels.

109 *Ibid.* Article 227

110 *Ibid.* Article 228

111 *Ibid.* Article 230

112 *Ibid.* Article 233

113 *Ibid.* Article 236

114 *See supra*, p. 210

115 *Ibid.* Article 221

116 *See supra*, p. 205

117 *Ibid.,* Article 235

118 Article 21(2)

119 *See* Ports and Waterways Safety Act, *supra* note 65. Regulatory jurisdiction for the prevention of pollution under the provisions of this act has been extended to the territorial sea. 33 USC § 1362(7)

120 Article 19(2)(h)

121 Article 230(2). In the wake of the *Amoco Cadiz* disaster, and in light of its own national legislation providing for imprisonment, the delegation of France has been particularly persistent in opposing this provision and advocating its amendment. *See, e.g.* Report by the Chairman of the Third Committee, Ambassador A. Yankov (Bulgaria), UN Doc. A/Conf.62/L34, at 2 (26 April 1979). The US Ports and Waterways Safety Act, *supra* note 65, also envisions Imprisonment among its penalties. 33 USC § 1227

122 Articles 192-96

123 Article 234

124 *Ibid.*

125 *See supra*, pp. 205-207

126 See Government of France, Note on Steps to be Taken in Order to Avoid Any Recurrence of Disasters such as (hat of the "Amoco Cadiz". IMCO Doc. MSC XXXVIII/21/Add. 1 (12 April 1976); see also Report of the

Legal Committee on the Work of its Thirty-Fifth Session, IMCO Doc. LEG XXXV/4, at 9-16 (7 June 1978). For the most recent outcome of these efforts at the time of this writing, see Consideration of Legal Questions Arising from the "Amoco Cadiz" Disaster: Working Paper Presented by the Delegations of France, Mexico and Uruguay — Draft Articles resulting from Informal Consultation with Several Delegations which Contain Elements for the Development of International Law relating to Intervention of Coastal States, IMCO Doc. LEG XI/2/1 (6 March 1979)

127 *See* de Klemm *supra,* p. 71 *et seq.;* and Johnston and Enomoto, *infra,* p. 285 *et seq.*

128 For discussion, *see* Mediterranean Scientists Join Forces. UNEP, The Siren No. 2, at 8 (Sept. 1978). For summary discussion of the Mediterranean, *ibid.* No 1 (June 1978). The Siren is a news bulletin from UNEP's Regional Seas Programme, which is issued four times a year

129 Johnston and Enomoto, *op. cit.*

130 *See* Rémond-Gouilloud, *infra,* p. 239, and Johnston and Enomoto, *infra,* p. 348

131 On GEMS and its objectives and principles, see the description presented by UNEP to the Law of the Sea Conference, UNEP, *The Global Environmental Monitoring System of the United Nations Environment Programme* (UN Doc. A/Conf.62/C.3/L23,17 March 1975), *reprinted in* 4 Third United Nations Conference *on the Law of the Sea: Official Records 207 (1975)*

132 Johnston and Enomoto, *op. cit.,* p. 324 *et seq.*

133 On the role of "soft law" provision, *see* Rémond-Gouilloud, *supra.*

134 The Finland delegation introduced in the Third Committee of UNCLOS III a resolution calling for a general diplomatic conference to conclude an international convention on criteria and procedure for the determination of liability, the assessment of damage, the payment of compensation, and the settlement of related disputes. Some delegations favour this approach, while others prefer to work on these subjects regionally, nationally, or otherwise. In short, while there seems to be general agreement that these are areas where there is need for further development of international law, states have not yet agreed on how or where

135 Article 211(2) and (5). For other uses of "generally accepted" in the ICNT/Rev. 3 *see also* Articles 21(2), 21(4), 39(2), 41(3), 53(8), 60(5), 60(6) 94(2)(a) and 94(5). *Compare* Articles 207 on pollution from land-based sources and 212 on pollution from or through the atmosphere, which say that national laws and regulations shall take into account *"internationally agreed* rules, standards and recommended practices and procedures" (emphasis added); and Article 208 on pollution from sea-bed activities, which says that they shall be no less effective than simply "international rules, standards and recommended practices and procedures"

136 *See* Articles 217, 218, 220; see *also* Articles 216 on enforcement with respect to dumping, 219 on measures relating to seaworthines of vessels to avoid pollution, 228 on suspension and restrictions on institutions of proceedings, and 230 on monetary penalties and the observance of recognized rights of the accused. Similar language appears in Article 213 on enforcement with respect to land-based sources of pollution, 214 on enforcement with respect to pollution from sea-bed activities, and 222 on enforcement with respect to pollution from or through the atmosphere. For other references to "applicable" provisions *see also* Articles 42(1)(b), 60(5) and 94(3)(b)

137 Article 221

138 Article 228(1)

139 *See*, *e.g.,* Articles 216(1)(b), 218(2), 220(4), 228(1) & (3), and 231

140 Article 211(3). Similar references to vessels flying their flags *or* of their registry are found in Articles 217(1), (2) and (3)

141 Apparently, however, there has been considerable behind-the-scenes debate at UNCLOS III among a few countries on the issue. Certain Liberian representatives claim that the Federal Republic of Germany has a law which would permit a German flag to be flown on a charter vessel of another registry, and Panama is contemplating legislation that would enable such split jurisdiction. The benign explanation given by other prominent UNCLOS figures is that these references are to cover the case where a ship of the new Enterprise (to be the operating arm of the International Sea-Bed Authority) may charter a ship of some country, which would then fly a United Nations flag. This may be a valid concern, but surely it can be dealt with in a better and more efficacious manner than introducing such a huge ambiguity in the text, potentially destructive not only to the UNCLOS marine pollution provisions but also to many existing conventions which use one term or the other

142 *Supra,* p. 206

143 *Supra,* pp. 205-207

144 *Ibid.*

145 *Ibid.*

146 *Ibid.*

147 *Ibid.*

148 *Ibid.*

149 *Supra* note 38, Article 15(1). The SOLAS Convention, Article 10(a), has a requirement that not less than 25 states, representing 50 percent of the world's merchant shipping tonnage, become parties for its entry into force. Both the MARPOL Protocol, Article 5(1), and the SOLAS Protocol, Article 5(1), contain the 15-state, 50

percent-shipping tonnage requirements of the MARPOL Convention model

150 For a comparison between, and on the interrelationships of, the IMCO regime and the United States "Superfund" proposals, *see* Carven and Becker, *supra* note 29. At the 95th Congress in 1978, the Superfund Bill passed the House and was prevented from reaching the Senate floor by a procedural move by a single member — Senator James Abourezk, who retired after that session of Congress. For discussion, see 20 Ocean Science News 41, at 1 (16 Oct. 1978). As finally passed by the 96th Congress, The Superfund Act of 1980, Pub. L. No 96-510, 94 Stat. 2767, did not include liability for oil pollution

151 Especially noteworthy in this regard is the United States Ports and Waterway Safety Programme

152 The provisions on implementation of Annex II of the convention are contained in Article 2 of the protocol

153 For some of the recent reports of UNEP working groups in these areas, see Report of the Working Group of Experts on Environmental Law on Its Second Session, UN Doc. UNEP/WG.14/4 (12 April 1978); Report of the Group of Experts on Liability for Pollution and Other Environmental Damage and Compensation for such Damage, UN Doc. UNEP/WG.8/3 (6 April 1977). See *also* Report of the Intergovernmental Working Group of Experts on Natural Resources Shared by Two or More States on the Work of Its Fifth Session, UN Doc. UNEP/IG. 12/2 (8 February 1978).

154 IMCO Report on the Questionnaire on Pollution of the Marine Environment, IMCO Doc. OPS/Circ. 15 (May 13, 1969) or Annex to GESAMP/22 (Feb. 10, 1970) referred to in Schachter and Serwer, "Marine Pollution Problems and Remedies" (1971), 65 *Am. J. Int'l L.* 84 at p. 105

155 Williams, "Geologic Effects of Ocean Dumping on the New York Bight Inner Shelf" in Palmer, H.D. and Gross, M.G. (eds.), *Ocean Dumping and Marine Pollution,* (1979), at p. 52

156 Schachter and Serwer, *supra* note 154, at p. 106

157 *Ibid.*

158 In the United States in 1975 about 87, 826, 362 cubic yards of dredge spoils, 3,446,000 tons of industrial wastes and 5,039,000 tons of sewage sludge were dumped in coastal waters

159 Schachter and Serwer, *op. cit.,* at 106

160 "Spectrum", 11 *Environment* S-2 (July-August, 1969) as cited in *ibid.*

161 Schachter and Serwer, *op. cit.,* at p. 107

162 Cundick, "Army Nerve Gas Dumping: International Atropine" (1972), 56 *Military L. Rev.* 165 at pp. 167-68

163 Timagenis, *supra* note 4, at p. 110

164 The study had been carried out in response to a request made by the President to study the problems and alternatives to ocean dumping and to recommend further actions. Leitzell, "The Ocean Dumping Convention — A Hopeful Beginning" (1973), 10 *San Diego L. Rev.* 502 at p. 503

165 *Ibid.*

166 Mendelsohn, "Ocean Pollution and the 1972 United Nations Conference on the Environment" (1972), 3 *J. Mar. L. & Comm.,* 385 at p. 395. The Draft Treaty is reprinted in 3 *J.Mar. L. & Comm.* 423 (1972)

167 Timagenis, *supra* note 4, at p. 173

168 *Ibid.,* at p. 124

169 Belgium, Denmark, France, Ireland, Spain, Sweden and the United Kingdom all ratified, while Italy, Poland and Portugal acceded to the Convention.
See Oda, S., *The International Law of the Ocean Development* (1975), at p. 131

170 Timagenis, *op. cit.,* at pp. 125, 126

171 Kullenberg, G. "Ocean Dumping Sites" (1974), 2 *Ocean Mgmt.* 189 at p. 193

172 Timagenis, *op cit.,* at p. 182

173 Recommendation 86(d), A/CONF. 48/14, reprinted in 11 *Int'l Leg. Mat.* 1416 (1972) at p. 1454

174 Timagenis, op. *cit.,* at p. 178

175 *Ibid.* at pp. 182-83

176 *Ibid.* at p. 183

177 *Ibid.* at p. 184

178 26 UST 2403; TIAS, No 8165. On recent accessions, *see* IMCO, *Status of the IMCO Convention and Amendments thereto and of Multilateral Conventions and Instruments in respect of which the Inter-Governmental Maritime Consultative Organization or its Secretary General Acts as Depositary*, 1980

179 16 *Int'l Leg. Mat.* 1450-55 (1977)

180 Schenker, M. "Saving a Dying Sea? The London Convention on Ocean Dumping" (1973), 7 *Cornell Int'l L.J. 32* at p. 42

181 Article VI(1)(c) and (d)

182 Batey, G. "An International Perspective on an international Problem: Regulation and Assessment *of Ocean* Dumping" (1979), 20 *Sea Technology* 17 at p. 19

183 Article Ill(2). The factors in Annex III are:

A . Characteristics and Composition of the Matter

1. Total amount and average composition of matter dumped (e.g. per year)
2. Form, e.g. solid, sludge, liquid, or gaseous
3. Properties: physical (e.g. solubility and density), chemical and. biochemical (e.g. oxygen demand,

nutrients) and biological (e.g. presence of viruses, bacteria, yeasts, parasites)

4. Toxicity
5. Persistence: physical, chemical and biological
6. Accumulation and biotransformation in biological materials or sediments
7. Susceptibility to physical, chemical and biochemical changes and interaction In the aquatic environment with other dissolved organic and inorganic materials
8. Probability of production of taints or other changes reducing marketability of resources (fish, shellfish, etc.)

B. Characteristics of Dumping Site and Method of Deposit

1. Location (e.g. co-ordinates of the dumping area, depth and distance from the coast), location in relation to other areas (e.g. amenity areas, spawning, nursery and fishing areas and exploitable resources)
2. Rate of disposal per specific period (e.g. quantity per day, per week, per month)
3. Methods of packaging and containment, if any
4. Initial dilution achieved by proposed method of release
5. Dispersal characteristics (e.g. effects of currents, tides and wind on horizontal transport and vertical mixing)

a Water characteristics (e.g. temperature, pH, salinity, stratification, oxygen indices of pollution-dissolved oxygen (DO), chemical oxygen demand (COD), bio-chemical oxygen demand (BOD)-nitrogen present in organic and mineral form including ammonia, suspended matter, other nutrients and productivity)

7. Bottom characteristics (e.g. topography, geochemical and geological characteristics and biological productivity)
8. Existence and effects of other dumpings which have been made in the dumping area (e.g. heavy metal background reading and organic carbon content)
9. In issuing a permit for dumping, Contract-Parties should consider whether an adequate scientific basis exists for assessing the consequences of such dumping, as outlined in this Annex, taking into account seasonal variations

C . General Considerations and Conditions

1. Possible effects amenities (e.g. presence of floating or stranded material, turbidity, objectionable odour, discolouration and foaming)
2 Possible effects on marine life, fish and shellfish culture, fish stocks and fisheries, seaweed harvesting and culture
3. Possible effects on other uses of the sea (e.g. impairment of water quality for industrial use, underwater corrosion of structures, interference with ship operations from floating materials, interference with fishing or navigation through deposit of waste or solid objects on the sea floor and protection of areas of special importance for scientific or conservation purposes)
4. The practical availability of alternative land-based methods of treatment, disposal or elimination, or of treatment to render the matter less harmful for dumping at sea

184 The regulations require: that prior to issuing an incineration permit, a contracting party must first consider the practical availability of a land-based method of disposing the waste (Regulation 2); that the incineration facilities meet the requirements outlined in the Regulations (Regulation 3); that the requirements set for the operation of the incineration system are met (Regulation 5); and that the incineration process is monitored and certain information on the incineration is recorded (Regulations 6 and 8)

185 Article Vll(4)

186 Article V(1)

187 Article V(2)

188 Annex I, para. 8

189 Annex I, para. 9

190 *See* Rémond-Gouilloud, *supra,* p. 196

191 Article 210(5)

192 Kullenberg, *supra,* note 171, on GESAMP's recent research

193 *See* note 191, *supra*

194 *Int'l Leg. Mats.* (1092) II 262

195 Article 2 of the ODC states the exact geographic co-ordinates of the ODC's geographic area of application

196 The LDC's Black list includes:

ANNEX I

1. Organohalogen compounds
2 Mercury and mercury compounds
3 Cadmium and cadmium compounds
4. Persistent plastics and other persistent synthetic materials, for example, netting and ropes, which may float or may remain in suspension in the sea in such a manner as to interfere materially with fishing, navigation or other legitimate uses of the sea

5. Crude Oil, fuel oil, heavy diesel oil, and lubricating oils, hydraulic fluids, and any mixtures containing any of these, taken on board for the purpose of dumping.
6. High-level radioactive wastes or other high-level radioactive matter, defined on public health, biological or other grounds, by the competent international body in this field, at present the International Atomic Energy Agency, as unsuitable for dumping at sea
7. Materials in whatever form (e.g. solids, liquids, semi-liquids, gases or in a living state) produced for biological and chemical warfare
8. The preceding paragraphs of this annex do not apply to substances which are rapidly rendered harmless by physical, chemical or biological processes in the sea provided they do not:
   (i) make edible marine organisms unpalatable, or
   (ii) endanger human health or that of domestic animals.
   The consultative procedure provided for under Article XIV should be followed by a Party if there is doubt about the harmlessness of the substance
9. This Annex does not apply to wastes or other materials (e.g. sewage sludges and dredged spoils) containing the matters referred to in paragraphs 1-5 above as trace contaminants,. Such wastes shall be subject to the provisions of Annexes II and III as appropriate.

While the ODC's includes:

ANNEX I

The following substances are listed for the purposes of Article 5 of the Convention:

1. Organohalogen compounds and compounds which may form such substances in the marine environment, excluding those which are nontoxic, or which are rapidly converted in the sea into substances which are biologically harmless;
2. Organosilicon compounds and compounds which form such substances in the marine environment, excluding those which are nontoxic, or which are rapidly converted in the sea into substances which are biologically harmless;
3. Substances which have been agreed between the Contracting Parties as likely to be carcinogenic under the conditions of disposal;
4 Mercury and mercury compounds;
5 Cadmium and cadmium compounds;
6 Persistent plastics and other persistent synthetic materials which may float or remain in supension in the sea, and which may seriously interfere with fishing or navigation, reduce amenities, or interfere with other legitimate uses of the sea.

The LDC's Grey List includes:

ANNEX II

The following substances and materials requiring special care are listed for the purposes of Article Vl(1)(a).

A. Wastes containing significant amounts of the matters listed below:
   arsenic)
   lead)
   copper) and their compounds
   zinc)
   organosilicon compounds
   cyanides
   fluorides
   pesticides and their by-products not covered in Annex I.
B. In the issue of permits for the dumping of large quantities of acids and alkalis, consideration shall be given to the possible presence in such wastes of the substances listed in paragraph A and to the following additional substances:
   beryllium)
   chromium)
   nickel) and their compounds
   vanadium)
C. Containers, scrap metal and other bulky wastes liable to sink to the sea bottom which may present a serious obstacle to fishing or navigation.
D. Radio-active wastes or other radio-active matter not included in Annex I. In the issue of permits for the dumping of this matter, the Contracting Parties should take full account of the recommendations of the competent international body in this field, at present the International Atomic Energy Agency.

While the ODC's includes:

ANNEX II

1. The following substances and materials requiring special care are listed for the purposes of Article 6:
   (a) Arsenic, lead, copper, zinc and their compounds, cyanides and fluorides, and pesticides and their by-products not covered by the provisions of Annex I;
   (b) Containers, scrap metal, tar-like substances liable to sink to the sea bottom and other bulky wastes

which may present a serious obstacle to fishing or navigation;

(c) Substances which, though of a nontoxic nature, may become harmful due to the quantities in which they are dumped or which are liable to seriously reduce amenities.

2 The substances and materials listed under paragraph 1(b) above should always be deposited in deep water.

3. In the Issuance of permits or approvals for the dumping of large quantities of acids and alkalis, consideration should be given to the possible presence in such wastes of the substances listed in paragraph 1) above.

4. When, in the application of the provisions of Annexes II and III, it Is considered necessary to deposit waste in deep water, this should be done only when the following two conditions are both fulfilled:

(a) that the depth is not less than 2000 metres,

(b) that the distance from the nearest land is not less than 150 nautical miles.

197 *See* Articles 16-18

198 Batey, supra note 182, at p. 18

199 13 *tnt'l Leg. Mat.* 546 (1974)

200 Article 1

201 Timagenis, *supra* note 4, at p. 293

202 All of the States attending the Helsinki Conference have ratified the Convention

203 15 *Int'l Leg. Mat.* 290 (1976)

204 *Ibid.*

205 The "black list" of the Barcelona Convention includes:

1. Organohalogen compounds and compounds which may form substances in the marine environment, excluding those which are nontoxic or which are rapidly converted in the sea Into substances which are biologically harmless, provided that they do not make edible marine organisms unpalatable.
2. Organosilicon compounds and compounds which may form such substances In the marine environment excluding those which are nontoxic or which are rapidly converted in the sea into substances which are biologically harmless, provided that they do not make edible marine organisms unpalatable.
3. Mercury and mercury compounds.
4. Cadmium and cadmium compounds.
5. Persistent plastic and other persistent synthetic materials which may materially interfere with fishing or navigation, reduce amenities, or interfere with other legitimate uses of the sea.
6 Crude oil and hydrocarbons which may be derived from petroleum, and any mixtures containing any of these, taken on board for the purpose of dumping.
7. High and medium and low-level radioactive wastes or other high and medium and low-level radioactive matter to be defined by the International Atomic Energy Agency.
8 Acid and alkaline compounds of such composition and in such quantity that they may seriously impair the quality of sea water. The composition and quantity to be taken into consideration shall be determined by the Parties in accordance with the procedure laid down in Article 14, paragraph 3, of this Protocol.
9. Materials in whatever form (e.g. solids, liquids, semi-liquids, gases, or in a living state) produced for biological and chemical warfare, other than those rapidly rendered harmless by physical, chemical or biological processes in the sea provided that they do not:

(i) Make edible marine organisms unpalatable; or

(ii) Endanger human or animal health.

This Annex does not apply to wastes or other materials, such as sewage sludge and dredge spoils, containing the substance referred to in paragraphs 1-6 above as trace contaminants. The dumping of such wastes shall be subject to the provisions of Annexes II and III as appropriate.

The "grey-list" includes:

1. (i) Arsenic, lead, copper, zinc, beryllium, chromium, nickel, vanadium, selenium, antimony and their compounds;

(ii) Cyanides and fluorides;

(iii) Pesticides and their by-products not covered in Annex I;

(iv) Synthetic organic chemicals, other than those referred to in Annex I, likely to produce harmful effects on marine organisms or to make edible marine organisms unpalatable.

2. (i) Acid and alkaline compounds the composition and quantity of which have not yet been determined in accordance with the procedure referred to in Annex I, paragraph A.8.

(ii) Acid and alkaline compounds not covered by Annex I, excluding compounds to be dumped in quantities below thresholds which shall be determined by the Parties in accordance with the procedure laid down in Article 14, paragraph 3 of this Protocol.

3. Containers, scrap metal and other bulky wastes liable to sink to the sea bottom which may present a serious obstacle to fishing or navigation.
4. Substances which, though of a nontoxic nature may become harmful owing to the quantities in which they are dumped, or which are liable to reduce amenities seriously or to endanger human life or

marine organisms or to interfere with navigation.

5. Radioactive waste or other radioactive matter which will not be included in Annex I. In the issue of permits for the dumping of this matter, the Parties should take full account of the recommendations of the competent international body in this field, at present the International Atomic Energy Agency.

206 Thacher, P. and Meith, N. "Approaches to Regional Marine Problems: A Progress Report on UNEPs Regional Seas Programme" (1980), 2 *Ocean Y.B.* 153 at pp. 161-65

207 As of 1 June 1979, Egypt, EEC, France, Greece, Israel, Italy, Lebanon, Libya, Malta, Morocco, Spain, Syria, Tunisia and Yugoslavia had all become parties to the Barcelona Convention and Protocol

208 Reprinted in 2 *Ocean Y.B.* 516 (1980) at pp. 527-37

209 The contracting parties were Bahrain, Iraq, Kuwait, Oman and Qatar

210 Thacher and Meith, *supra* note 206, at p. 170

211 Marine Protection, Research and Sanctuaries Act of 1972, 86 Stat. 1052 (1972), reprinted in 11 *Int'l Leg. Mat.* 1394 (1972)

212 Norwegian Law of 11.6.1971 on the prevention of dumping at sea of noxious substances

213 See 88 Stat. 50 (1974), reprinted in 13 *Int'l Leg. Mat.* 611 (1974)

214 UK: Dumping at Sea Act, 1974; Canada: Ocean Dumping Control Act, 1975

215 Commune de Cassis, Conseil d'Etat 15th March 1968, Recueil Lebon, p. 189

216 Charles Alexandres Kiss, "Un cas de pollution international: l'affaire des boues rouges", Clunet 1975, p. 207

217 Tribunal de grande instance de Bastia, 8th November 1976

218 Goodbody, Hodges, and Martinez, "Tourism and Natural Resources: Caribbean study project", Working Papers for Pacem in Maribus IV (1973), p. 87

219 "No environmental policy should adversely affect the present or future development possibilities of the developing countries". See UN GA Resolution 20 December 1971, No. 2849, para 416. The 1972 Stockholm Declaration on the Human Environment, taking this constraint into account, proclaimed that "the protection of the human environment is a major issue which affects .... economic development throughout the world" (Principle 13) and that "States should adopt an integrated and co-ordinated approach to their development planning so as to ensure that development is compatible with the need to protect and improve the environment for the benefit of their population" (Principle 14)

220 Goodbody, Hodges and Martinez, *op. cit.*

221 Preamble to the Kuwait Regional Convention on the Protection of the Marine Environment from Pollution, para. 4

222 S. Keckes, "The Regional Seas Programme of UNEP", *UNEP Industry and Environment,* vol. 3, no. 3 (July-Sept, 1980), p. 5-7

223 The class of noxious liquid substances is divided into several categories in Annex II:

Category A

Substances which are bioaccumulated and liable to produce a hazard to acquatic life or human health; or which are highly toxic to acquatic life; and additionally certain substances which are moderately toxic to acquatic life when particular weight is given to additional factors in the hazard profile or to special characteristics of the substance.

Category B

Substances which are bioaccumulated with a short retention of the order of one week or less; or which are liable to produce tainting of the sea food; or which are moderately toxic to acquatic life; and additionally certain substances which are slightly toxic to acquatic life when particular weight is given to additional factors in the hazard profile or to special characteristics of the substance

Category C

Substances which are slightly toxic to acquatic life and additionally certain substances which are practically nontoxic to acquatic life when particular weight is given to additional factors in the hazard profile or to special characteristics of the substance

Category D

Substances which are particularly non-toxic to acquatic life, or causing deposits blanketing the seafloor with a high biochemical oxygen demand (BOD); or highly hazardous to human health, with an LD 50 of 5 mg/kg or more, but less than 50 mg/kg and produce slight reduction of amenities

Other Liquid Substances

Substances other than those categorized in Categories A, B, C, and D above

224 The B. List of Annex II reads as follows:

1. Mercury, cadmium, and their compounds
2. Antimony, arsenic, beryllium, chromium, copper, lead, molybdenum, nickel, selenium, tin, vanadium, zinc, and their compounds, as well as elemental phosphorus
3. Phenols and their derivatives
4. Phthalic acid and its derivatives
5. Cyanides

6. Persistent halogenated hydrocarbons
7. Polycyclic aromatic hydrocarbons and their derivatives
8. Persistent toxic organosilic compounds
9 Persistent pesticides, including organophosphoric and organostannic pesticides, herbicides, slimicides and chemicals used for the preservation of wood, timber, wood pulp, cellulose, paper, hides and textiles, not covered by the provisions of Annex I of the present Convention
10. Radioactive materials
11. Acids, alkalis and surface active agents in high concentrations or big quantities
12. Oil and wastes of petrochemical and other industries containing lipid-soluble substances
13. Substances having adverse effects on the taste and/or smell, colour, transparency or other characteristics of the water seriously reducing its amenity values
14. Materlals and substances which may float, remain in suspension or sink, and which may seriously interfere with any legitimate use of the sea
15. Lignin substances contained In industrial waste waters
16. The chelators EDTA (ethylenedinitriloteraacetic acid or ehtylenediamlnetetraacetic acid) and DTPA (diethylenetriaminopentaacetic acid)

225 *See* Annexes of 1974 Paris Convention and 1980 Athens Protocol
226 *See*, for example, Part II of Annex A of the Paris Convention and Annex I B of the Athens Protocol
227 Paris Convention, Article 3(c)
228 H.R. 7020:4-se
229 As offshore platform technology advances in sophistication, it may become increasingly artificial to draw analogies between shore-based and platform-based activities
230 In the Abidjan Convention atmospheric pollution is dealt with in Article 9, and land-based pollution in Article 8
231 *See*, for example, Yturriaga, *supra* note 4, p. 387
232 Article 207 and 212
233 *See Infra,* p. 245
234 *See* Article 1(2) of the Paris Convention, Article 8 of the Barcelona Convention, and Article 1 of the Athens Protocol
235 *See* Article 5 and 6 of the Helsinki Convention
236 Annex I of the Helsinki Convention
237 Helsinki Convention, Article 1
238 Gregorias J. Timagenis, "Protocol for the Protection of the Mediterranean Sea against Pollution from Land-Based Sources" (1980), 1 *Hellenic Review of International Relations* 123 at p. 132
239 Lucchini, "A propos de I'Amoco-Cadiz: Evolution on revolution du droit international", 1979 *Annuaire français de droit international* 24
240 OECD, *Problems of Transfrontier Pollution* (1974) and *Legal Aspects of Transfrontler Pollution* (1977)
241 Article 1, title 1, line IV, of the Marine Ordinance of August 1681. This text was apparently still considered operational on May 16th, 1947, when the Conseil d'Etat was last recorded to have applied it. Recuell Lebon, 16th May 1947, p. 202. But with the introduction of a general water pollution control statute, the 1681 Ordinance is no longer in force
242 *See* Act of 19th July 1976, which codifies the 1917 law on "établissements classes insalubres, Incommodes ou dangereux"
243 Renout and Headley, "Coastal Zone Management: Regional Governance as a Function of National Interest" (paper prepared for Madrid Conference on the Law of the World, September 1979)
244 33 USC 1151-1175 (1970). This law was re-enacted two years later as the Federal Water Quality Control Act, 33 USC 1251-1372 (1972), then amended again in 1977 to provide for damages In the case of destruction or loss of natural resources
245 *See* Act of 16 December 1964 relative au régime et à la repartition des eaux et à la lutte contre leur pollution
246 The Conseil d'Etat confirmed on 21st November 1973 that industrial waste released into marine waters is subject to payment of the tax established under the 1964 Act. See Recueil Lebon 1973, p. 654
247 Rashkow, "An Analysis of Alternatives to the Traditional Approach to Damage Assessment" (paper presented to OECD Seminar on the Economic Consequences of OH Spills, June 1981)
248 It is worth noting that the Japanese system of environmental law is now regarded as one of the most advanced in the world, and the Japanese environmental movement one of the most alert
249 *See* Letalik, supra, p. 217 *et seq.*
250 Ohio v. Wyandotte Chemical Corp. 401 US 493 (1971). *See* McCaffrey, *Private Remedies for Transfrontier Environmental Disturbances* (IUCN Paper No. 8, 1975)
251 Timagenis, *supra* note 238, p. 129
252 76/464/EEC. See Journal officiel des Communautés, no. 129 (18th May 1976), p. 23
253 Johnston and Enomoto, *infra,* pp. 324 *et seq.*
254 On the Neuilly 1972 Conference and other earlier efforts, *see* Timagenis, *supra* note 238, at p. 126

255 In 1976 two other protocols had already been adopted under the Barcelona Convention: one concerning co-operation in combatting oil pollution in cases of emergency, and the other concerning dumping from ships and aircraft
256 *See* discussion at page 329, *supra*
257 *See* Annex I of the Protocol
258 Johnston and Enomoto, *infra,* p. 330
259 *Ibid.,* p. 331
260 *Ibid.,* p. 331
261 *Ibid.,* p. 332
262 Land-based pollution in this coastal area — bordered by Bahrain, Iran, Iraq, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates — deserves special attention, since it is undergoing the most rapid industrialization of any region in the world
263 The implementation of Article 7 cannot be expected to proceed rapidly without substantial assistance from outside the region, where most of the countries suffer from serious developmental problems
264 Timagenis, *supra* note 4, p. 616
265 A provision for *automatic* review, which was first put forward, was later rejected in negotiations
266 Such commissions should be regarded as a mechanism designed to facilitate the harmonization of national pollution control laws and policies in the area rather than another level of bureaucracy encouraged to impose new sets of controls on the member governments
267 Accord de Monaco (10th May 1976) relatif à la protection des eaux du littoral méditerranéens. For French law, *see Journal Officiel* of 9th June 1977.
268 Comite Professionel du Pétrole, *Pétrole 77: Elements Statistiques* (1978)
269 Rein, Sogn, et al., *Norwegian Petroleum Law* (1978). pp. 1-4
270 Until the 1970s the accident record of the offshore oil industry was relatively good, considering the nature and Intensity of its activities around the world. In 1972 a study by the US National Academy of Engineering showed that in the year 1971, when more than 10,000 offshore wells were in production and another 5,000 were in the process of development, eleven "blow-outs" (involving loss of control over the well) had occurred, and 729 pollution incidents. But these statistics are distortive of the situation today. In 1971 most of those operations were being conducted on the US continental shelf in relatively safe waters. North Sea offshore operations, which where then just starting, have proved infinitely more hazardous. Moreover, the accidents reported are usually rig-related and do not include pollution incidents associated with support activities
271 It is believed that pollution from "seabed activities" — that is, from offshore oil production and related activities — contributes only 3-4% of the total pollution of the ocean: 80,000 out of 6, 113,000 tons annually according to a 1973 estimate of the US National Academy of Sciences. But none of the estimates available take account of the IXTOC I blow-out of 1979
272 Union Oil v. Oppen, 501 Fed 2d 558 (1974); Oppen v. Aetna Insurance Co., 26 Amer. Law Rep. Fed. 340(1973)
273 Royal Norwegian Government, *Uncontrolled Blow-out on Bravo Report of Commission of Injury* (1977)
274 Advisory Committee on Oil Pollution of the Sea, *Annual Report* (1979), p. 11
275 1958 (Geneva) Convention on the Continental Shelf, Article 1
276 Article 214: "States shall enforce their laws and regulations adopted in accordance with article 208 and shall adopt laws and regulations and take measures necessary to implement applicable international rules and standards established through competent international organizations or diplomatic conference for the protection and preservation of the marine environment from pollution arising from sea-bed activities subject to their jurisdiction and from artificial islands, installations and structures under their jurisdiction, pursuant to articles 60 and 80"
277 Draft Convention on the Law of the Sea, Article 136
278 *Ibid.,* Articles 156-185
279 Under Article 209 it is intended that environmental obligations arise directly for coastal and flag states as well as for the International Seabed Authority (and, therefore, indirectly the member states)
280 McGonigle and Zacher, *Pollution, Politics and International Law* (1979), p. 70
281 For an evaluation, *see* Balleneger, *La Pollution en Droit International* (1975), p. 148
282 *See* also Kiss, "La convention nordique sur l'environnement" (1974), 1974 *Annuaire Français de Droit International* 808
283 *See*, for example, Article 7 of the Barcelona Convention, Article VII of the Kuwait Convention, and Article 8 of the Abidjan Convention. For a general review of the UNEP Regional Sea Programme, see Johnston and Enomoto,*infra,* pp. 324 *et seq.*
284 *See*, for example, Article VII of the Draft Regional Convention for the Conservation of the Coastal Area of the Red Sea and Gulf of Aden
285 The Convention for the Prevention of Pollution of the Seas by Oil (1954). *See* Schneider, *supra,* p. 205
286 The Oslo and London Conventions of 1972, *see* Letalik, *supra,* pp. 221, 225
287 Compare Article 19(2) of the Oslo Convention and Article III(1)(c) of the London Convention

288 The normal wording is that "operations must be undertaken in a proper and workmanlike manner in accordance with the methods and practices customarily used in good oil field practice"

289 H.R. George, "Safety in Offshore Hydrocarbon Exploration and Exploitation" (paper presented at Petromar 80, 1980), p. 77

290 *See*, for example, UK Institute of Petroleum, *Code of Safe Practice lor Drilling in Marine Areas*

291 It is estimated that a 10,000 foot well requires about 253 tons of dry mud components

292 *See*, for example, UK Prevention of Oil Pollution Act (1971), ss. 2 and 3

293 They consist of a fine of 10,000 to 100,000 francs or imprisonment for a period from three months to two years, or both. Act no. 77-485 (11th May 1977). These penalties may be double in the case of a recurring offence

294 Cormack, "The Relative Effectiveness of Current Techniques for Oil Spill Clearance" (paper presented at Petromar 80, 1980), p. 135

295 *See* papers by Guiges, West, Ross, and others, also prepared for Petromar 80

296 Smets, "Some Economic Aspects of Accidental Oil-spills" (paper presented at the Third Assises pour l'environnement, Paris, December 1980)

297 *Uncontrolled Blow-out on Bravo, supra* note 273

298 Dubais, "The 1976 London Convention on Civil Liability for Oil Pollution Damage from Offshore Operations" (1977), 9 *J. Maritime Law and Commerce* 61

299 OECD, Environment Directorate, Document no. 3 (16) (September 27, 1979)

300 After the Santa Barbara blow-out, the State of California brought suit against the US federal Department of the Interior, alleging that the latter had proceeded with offshore development in the area without proper consultation with the State and without sufficient care

301 International Convention on Civil Liability for Oil Pollution Damage. *See* Schneider, *supra,* p. 205

302 The Convention on Civil Liability for Oil Pollution Damage Resulting from the Exploration and Exploitation of Sea-bed Mineral Resources was concluded in December 1976 and opened for signature in May 1977. It is known variously as the 1976 London Convention and the 1977 London Convention! *See* Kwandowsky, "Civil Liability for Oil Pollution Damage on the Norwegian Continental Shelf" (1978), 5 *Ocean Development and International Law* 4

303 Dubais, *supra* note 298

304 Pierron, "Projet de Convention Internationale sur les engins mobiles off-shore", 1978 *Droit Maritime Français* 131

305 Handl, "State Liability for Accidental Transnational Environmental Damage Due to Activities of Private Persons or Entities" (paper prepared for OECD, December 1979)

306 *See* Rémond-Gouilloud, *supra*, p. 199

Chapter Four

Regional Approaches to the Protection and Conservation of the Marine Environment

Douglas M. Johnston and Lawrence M.G. Enomoto

A. Introduction

1. Regions and Regionalism

In its popular meaning[1], the term "region" is generally accepted as a convenient way of focusing on a part of the planet. Because of the frequency of its use in diverse contexts, it is a term that simply cannot be dispensed with.

Yet to political geographers, no concept is more elusive. All efforts to define it with any degree of precision are faced with the overwhelming arbitrariness of the task. Regions, like boundaries, may be characterized in many ways, according to the kind of significance attributed to them. Although possessing some kind of geographical character, a region may be perceived as political, cultural, economic, or institutional. Indeed, natural features are not normally the predominant consideration in referring to a region; and it rarely, if ever, makes sense to describe a region as "natural"[2]. It is equally idle, therefore, to try to bring any precision to the task of delimiting regions in general[3]. Even when purely physical features are used for the purpose of demarcation, incongruities are the rule rather than the exception. More often than not, only the "core" of a designated region can be defined with absolute precision, and the delineation of the periphery must often be left vague or approximate[4].

For the political scientist these imprecisions complicate the task of formulating hypotheses regarding political behaviour at the regional level. Whether the focus is on regional co-operation, regional organizations, regional systems and subsystems, regionalism, or regional integration[5], the diverse usage of the underlying concept of a region limits the potential utility of such theorizing. The imprecision of the regional focus presents particularly acute problems in the theory of international relations where serious efforts are made to acquire "insights into the process of community formation" at the international level[6].

Yet one should not exaggerate the importance of the problems inherent in the definition and delimitation of geographical regions. In looking at regionalism and its variants, the political scientist is normally studying "regional approaches" to problems and issues: that is, he is focusing on a level of treatment that is intermediate between the national and the global. To that extent, his immediate interest lies less in the geography of the region than in the structure and dynamics of a geographically defined "regional arrangement". Less immediately, he is concerned with the phenomenon of "regionalization": the trend toward a more or less localized pattern of co-operative action within the international political system, explainable in terms of an apparent perception by two or more states of links resulting from geographical proximity[7]. Since the perceived links are often described in specific functional terms, the areas of co-operative action are usually referred to as

"institutional regions". An "international resource region"[8] is a familiar example of this type. In such a region, the importance of a precise definition of the outer limits may vary considerably with the administrative functions assigned to the institution.

2. Marine Regions and Marine Regionalism

The problems of definition and delimitation are scarcely less complicated when applied to the ocean. Indeed they are aggravated, in some degree, by a number of considerations: for example, concepts and practices of regional affiliation and alignment are essentially land-oriented; there is little cultural tradition to support broad-based regional co-operation at sea; the surface waters of the ocean present relatively few "natural features" to assist efforts at regional definition and delimitation; ecologically sensible, and administratively convenient, divisions of ocean space are frequently incompatible with political preferences; and recent nationalistic and acquisitive tendencies in the law of the sea seem less than conducive to even the concept of regional sharing. But although the goal of global ocean management is politically unpopular, few argue for eliminating the regional level of regulation and administration. In several sectors of ocean management, the national approach seems too narrow, and the global approach too broad. For reasons of efficiency — even in the absence of supportive traditions of regional co-operation — it seems necessary to invent regional mechanisms as an intermediate component in what may be termed a "split-level" system of authority on the oceans[9].

In the context of "ocean affairs" one sees two distinct approaches to the concept of regionalism: that of the political geographer, on the one hand, and that of the political scientist, on the other. The former, it might be suggested, is primarily interested in the conceptual significance of efforts to divide the marine environment into regions, as reflected in various kinds of regional arrangements; whereas the political scientist is chiefly concerned with the strategic, political, operational, or conceptual significance of regional arrangements based as they are on various concepts of marine region.

This difference of emphasis has led both disciplines into the art of rather subtle distinctions. One prominent political geographer, for example, has distinguished three principal forms of marine regions: physical regions ("differentiated from other areas on the basis of coastal configuration"), management regions ("representing situations where there is a well-defined management problem, capable of being handled as a discrete issue"), and operational regions ("sites of one or more regional arrangements")[10]. Physical marine regions, he suggests, can be divided in two ways: into nine *ocean basins and, somewhat more arbitrarily, 24 semi-enclosed seas*[11]. In addition, there are various kinds of physical marine subregions, such as the Sea of Azov, the Gulfs of Bothnia, Thailand and Aqaba, and the northwestern approaches to the Strait of Malacca. One could also include archipelagic regions, including those of Indonesia and the Philippines which are large enough to encompass a number of partially enclosed seas (e.g. Java, Flores, Banda, and Molucca), and the waters adjacent to island areas such as the Leeward and Windward Islands of the Caribbean[12]. Each of these types of regions or subregions could be regarded as a potential site or "unit" for certain ocean management purposes. Accordingly, some but not all of these physical regions might also qualify as a management or operational region[13].

An equally prominent political scientist, on the other hand, has drawn a distinction between two approaches to the definition of region. "The first is to use location/contiguity as the determining criterion and infer that there is a direct causal relationship between this characteristic and the pattern of activities and policy problems dealt with. The second alternative is to treat location as being secondary and focus instead on the pattern of activities and perceived policy problems that should be at least analytically separable from the rest of the world"[14].

This allows him to go further and distinguish between "fully regional" arrangements ("[i]f the countries involved are all situated around the locus of the perceived problems") and "quasi-regional" arrangements ("[l]f countries from other parts of the world are also involved")[15].

However one treats these terms in the abstract, the normal mode of reference in contemporary writings on marine regionalism is to two kinds of regional arrangements: *regional agreements* and *regional organizations.* Both kinds of arrangements are apparently intended to have an important role to play, both in the normative development of the law of the sea and in the institutionalization of ocean management.

3. Regional Arrangements and UNCLOS III

At the commencement of the Third UN Conference on the Law of the Sea (UNCLOS III) it was hoped, no doubt by most delegations, that as much as possible would be provided as a global foundation for the new law of the sea. Indeed it may have been widely assumed that one of the major purposes of such a large-scale, virtually comprehensive, law-making convention was to produce a nearly complete system of global principles and procedures for ocean management, and thus to minimize the need for resort to regional arrangements. If so, many delegates must have been dismayed to learn just how many political and institutional limitations are inherent in the task of negotiating global arrangements for managing the world's oceans, and how much would have to be left to regional diplomacy and organization.

The significance of this is particularly striking at a law-making conference that has devoted so much effort to the allocation of extensive areas of *national* authority to coastal states. Despite the dramatic seaward extensions of limits of national jurisdiction around the world, the Conference has created the universal need for an unprecedented range of regional arrangements, not only to implement but also to supplement these global provisions. Two principal kinds of regional arrangements will be required after the conclusion of UNCLOS III: regional organizations to assist in the implementation of the final provisions, and regional arrangements to supplement the new global principles and procedures. Even if the Conference fails to produce a generally ratifiable final text, so that the new law of the sea lacks a universal treaty foundation, many of the UNCLOS III formulations which survived unchanged throughout the second half of the 1970s are likely to remain uncontested by most nations in the 1980s, and thus to acquire some degree of juridical significance in customary international law. In such a situation, there would still be a need for implementation of these formulations by regional organizations, and for explanatory or supplementary rule-making by regional agreements.

Even those commentators who are sceptical that there will be any pronounced and significant trend towards regionalism in the immediate future[16] concede that there are certain problems that coastal states will probably have to deal with through regional arrangements: such as, "the management of regional pollution problems, particularly those of enclosed or semi-enclosed seas; the management of certain regional pelagic fisheries; certain regional efforts at scientific inquiry and information gathering; possibly some attempts to establish denuclearized 'zones of peace' in certain areas; and perhaps certain regional attempts to deal with areas where jurisdiction is uncertain, such as Antarctica"[17].

In the view of other commentators[18], there will be increasing need in the 1980s to seek regional or subregional solutions to a large number of problems left in the wake of decolonization. Some of these problems are likely to be perceived as special to a particular locality, and not amenable to effective treatment through global rule-making[19]. In particular, it has been suggested, regionalism may have to play an important role in the management of two kinds of special conflicts: those between developed and developing nations over the development and transfer of skills and technology, and those between coastal and landlocked (or otherwise geographically disadvantaged) states over special access or sharing arrangements[20].

The latest text of UNCLOS III — the Draft Convention on the Law of the Sea (Informal Text) (ICNT/Rev. 3)[21] — contains numerous references, both explicit and implicit, to the need for regional arrangements. Of the explicit references, there are six principal kinds. First, there are numerous omnibus references to the need for states to co-operate in a particular way, "either directly or through appropriate organizations global or regional"[22]. Such a phrase is intended to encompass *inter alia* the prospect both of regional (and presumably subregional) *agreements* and of regional (or subregional) *organizations.* Second, there are equally many express references to the need for co-operation with appropriate organizations, where it has to be inferred whether global or regional organizations would be involved[23]. Third, there are a few cases where the specific need for "bilateral, subregional or regional agreement" is expressly recognized[24]. Fourth, there are several references to subregional and/or regional organizations[25]. Fifth, a number of draft articles envisage some kind of special arrangement, without specifying the level or mode of co-operation[26]. Sixth, and last, a few references are apparently limited to bilateral arrangements[27].

In addition, the Conference breaks new ground in dealing with the problems of development and transfer of marine technology by providing for the establishment of regional (as well as national) "marine scientific and technological centres". Such centres are designed "to stimulate and advance the conduct of marine scientific research by developing States and foster the transfer of technology"[28].

B. Regional Arrangements for the Protection and Conservation of the Marine Environment

1. Regional Agreements

It has always been difficult to give a precise definition of a regional agreement. Part

of the difficulty comes from the diversity of meanings assigned to the concept of "region" or "subregion". Moreover, some agreements are "regional" not only in the sense that they focus for a particular purpose on a physically defined geographical area, but also in the sense that the parties consist of countries which belong to that area. If all the countries of a region have become parties to an agreement, then that agreement may be described as "fully regional", (or "fully subregional"). If, on the other hand, some of the parties do not belong to the region which is the focus of the agreement — that is, if some of the. parties are "extra-regional" — then the arrangement might be described as only "quasi-regional".

The compilation of Table I involves a somewhat arbitrary process of selection. The difficulty of deciding what to include in a list of regional agreements for the conservation of marine species is compounded by the recent proliferation of bilateral fishery agreements, many of which are subregional, if not regional, in focus. Most of these agreements are, however, chiefly concerned with access to designated stocks in newly established fishing zones rather than with conservation proper, and are therefore omitted from Table I. Only a few bilaterals are regarded as sufficiently concerned with conservation to be included.

Two features of Table I are perhaps worthy of comment. First, the four marine regions which have attracted conservation agreements — the Atlantic, the Arctic, the Indo-Pacific, and the Southern Oceans — vary considerably in the number of levels of approach taken to regional arrangements. All four have been the focus of at least-one region-wide ("general") agreement. Of these agreements, three are expressly concerned with a single species (viz. Atlantic tunas, Arctic polar bears, and Antarctic seals), one is primarily (albeit non-explicitly) concerned with a single species (viz. krill), and the fifth is simply the means of establishing a fisheries council for the region (Indo-Pacific). But whereas the Southern Ocean is limited to region-wide agreements, the Atlantic Ocean, at the opposite extreme, has attracted conservation agreements at no less than nine other levels within the region. Moreover, seven of these Atlantic conservation agreements are "multilateral: that is, they have four or more parties. The Atlantic, in short, is seen to be most "developed" in the range and diversity of regional arrangements for the conservation of marine species. The Indo-Pacific shows five levels of approach below the region-wide level, and only three of the Indo-Pacific conservation agreements are multilateral.

TABLE I
CURRENT REGIONAL AGREEMENTS FOR THE CONSERVATION OF MARINE SPECIES

Region	Agreement	Parties	Type	Citation
A. Atlantic				
1. General	International Convention for the Conservation of Atlantic Tunas, 1966	multilateral	quasi-regional	673 *UNTS* 63
	Agreement relating to Conservation of Atlantic Salmon, 1972	Denmark, USA	fully regional	*TIAS* 7402
2. Northeast	Agreement on Measures for Regulating the Catch and Conserving the Stocks of Seals in the Northeastern Part of the Atlantic Ocean, 1957	Norway, USSR	fully regional	309 *UNTS* 280
	Northeast Atlantic Fisheries Convention, 1959	multilateral	fully regional	486 *UNTS* 157
	Agreement Concerning the Regulation of Fishing of the Atlantic Scandinavian Herring, 1973	Iceland, Norway, USSR	(sub)regional	ST/LEG/SER. B/18 p. 563
3. Skagerrak, Kattegat and Sound	Convention Concerning the Preservation of Plaice and Dab, 1937	Denmark, Norway, Sweden	fully (sub) regional	186 *UNTS* 419
4. Baltic	Agreement Concerning Measures for Protection of the Stocks of Deep-Sea Prawns, European Lobsters, Lobsters and Crabs, 1952	Denmark, Norway, Sweden	regional	175 *UNTS* 208

	Agreement Concerning the Protection of the Salmon Population in the Baltic Sea, 1962 (with Protocol, 1972)	multilateral	regional	ST/LEG/SER.B/15, p. 859 ST/LEG/SER.B/18, p. 561
	Agreement on Fishing and Sealing (in the Gulf of Finland), 1969	Finland USSR	fully (sub) regional	739 *UNTS* 78
	Convention on Fishing and Conservation of the Living Resources in the Baltic Sea and Belts, 1973	multilateral	fully regional	12 *ILM* 1292 (1972)
5. Northwest	Agreement on Sealing and Conservation of Seal Stocks in the Northwest Atlantic	Canada, Norway	quasi-regional	870 *UNTS* 85
	Convention on Future Multilateral Co-operation in the Northwest Atlantic Fisheries, 1978	multilateral	quasi-regional	1979 *Can. TS* No. 11
	Agreement on East Coast Fishery Resources, 1979 [unperfected]	Canada, US	(sub)regional	9 Nordquist Simmonds, *eds., New Directions in the Law of the Sea* (1980) p. 178
6. Mediterranean	Agreement for the Establishment of a General Fisheries Council for the Mediterranean, 1949 (as amended 1963)	multilateral	regional	126 *UNTS* 237 490 *UNTS* 444
7. Black Sea	Convention Concerning Fishing in the Black Sea, 1959	Bulgaria, Romania, USSR	fully (sub) regional	377 *UNTS* 203

8. Southeast	Convention on the Conservation of the Living Resources of the South-east Atlantic, 1969	multilateral	quasi-regional	801 *UNTS* 10
9. Caribbean	Agreement concerning Shrimp, 1972	Brazil, Trin-idad and Tobago	(sub)regional	1 Lay, Churchill, and Nordquist *eds., New Directions in the Law of the Sea* (1973), p. 463
10. Southwest	Agreement on Fishing and Conserva-tion of Living Resources, 1968	Brazil, Uruguay	regional	ST/LEG/SER. 16, p. 510
	Agreement Concerning Shrimp, 1972	Brazil, Netherlands	quasi (sub) regional	ST/LEG/SER. 18/Add. 2 p. 348 (1975)
B. Arctic				
1. General	Agreement on Conservation of Polar Bears, 1973	multilateral	fully regional	13 ILM 13 (1974)
2. Northeast	Agreement on the Regulation of the Fishing of Northeast Arctic Cod, 1974	Norway, UK, USSR	quasi-regional	13 ILM 12 (1974)
C. Indo-Pacific				
1. General	Agreement for the Establishment of Indo-Pacific Fisheries Council, 1948	multilateral	quasi-regional	120 *UNTS* 59
2. North Pacific	Convention for the Preservation of the Halibut Fishery of the North-ern Pacific Ocean and Bering Sea, 1953 (as amended 1979)	Canada, US	regional	222 *UNTS* 77 9 Nordquist & Simmonds, *eds., New Direction in the Law of the Sea* (1980) p. 233

	Interim Convention on the Conservation of North Pacific Fur Seals, 1957 (extended, 1969)	multilateral	fully regional	314 *UNTS* 105 719 *UNTS* 313
	Agreement Relating to Fishing for King and Tanner Crab, 1974	Japan, US	regional	*TIAS* 798
3. Northwest Pacific	Agreement Regarding the King and Tanner Crab Fisheries in the Eastern Bering Sea	US, USSR	(sub)regional	(re-negotiated annually)
	Agreement relating to the Crab Fisheries in the Northwest Pacific Ocean	Japan, USSR	regional	5 Churchill, Nordquist and Lay, *eds. New Direction in the Law of the Sea* (1979) p. 762
	Convention in Concerning High Seas Fisheries of the Northwest Pacific Ocean, 1956	Japan, USSR	regional	53 *AJIL* 763
	Agreement on Fisheries, 1965	Japan, South Korea	(sub)regional	4 ILM 1128 (1966)
	Agreement on Fisheries, 1975	China, Japan	(sub)regional	6 Churchill, Nordquist, and Lay, *eds,. New Direction in the Law of the Sea* 1977), p. 766

4. Northeast Pacific	Convention for the Protection, Preservation and Extension of the Sockeye Salmon Fishery of the Fraser River, 1939 (Protocol, 1956)	Canada, US	(sub)regional	184 *LNTS* 305 290 *UNTS 104*
	International Convention for the High Seas Fisheries of the North Pacific, 1952 (as amended 1978)	Canada, Japan US	quasi-regional	205 *UNTS* 67; *TIAS* 4493; *TIAS* 4992; *TIAS* 5385
	Agreement on Certain Fisheries Problems in Northeastern Part of the Pacific Ocean off the US Coast	US, USSR	quasi-regional	(re-negotiated annually)
	Agreement Concerning Fishing off the West Coast of Canada, 1979	Canada, US	fully (sub) regional	9 Nordquist Simmonds, *eds., New Direction in the Law of the Sea* (1980) p. 213
5. Eastern Pacific	Convention of the Establishment of an Inter-American Tropical Tuna Commission, 1949	multilateral	quasi-regional	80 *UNTS* 3 (196 *UNTS* 4)
6. Southeast Pacific	Agreements on the Exploitation and Conservation of Marine Resources of the South Pacific, 1952 (as amended 1954)	Chile, Ecuador Peru	fully regional	ST/LEG/SERV. B/6, p. 723 (ST/LEG/SER. B/6, p. 729
D. Southern				
1. General	Convention on the Conservation of Antarctic Seals, 1972	multilateral	quasi-regional	11 ILM 251 (1972)
	Convention on the Conservation of Antarctic Marine Living Resources, 1980	multilateral	quasi-regional	19 *ILM* 841 (1980)

Second, eleven of the 38 agreements listed in Table I are characterized as "fully regional" (or "fully subregional"), and 12 as "quasi-regional" (or "quasi-sub-regional"). For physical and legal reasons, there is no fully regional agreement for the Southern Ocean[29]. Both the Atlantic and Indo-Pacific regions are fairly evenly balanced between fully regional and quasi-regional arrangements; seven and five, respectively, for the former, and three and four, respectively, for the latter.

Table II shows that only two regions — the Atlantic and the Indo-Pacific — have attracted regional agreements for the prevention and control of marine pollution. Of the 15 instruments listed, 12 belong to the Atlantic, at six levels below the region-wide level. The other three agreements belong to the Indian Ocean, at two subregional levels. Twelve agreements are multilateral, one is trilateral, two bilateral. Eight are fully (or almost fully) regional (or sub-regional). But the striking feature of Table II is the absence of any quasi-regional agreements for marine pollution prevention and control. In the case of regional conservation agreements, no less than 12 of those listed in Table I were quasi-regional.

In addition to the 38 conservation agreements listed in Table I and the 15 pollution prevention agreements in Table II, there are also a number of instruments which reflect "evolving regional commitments" for either the conservation of marine species or the prevention or control of marine pollution. (Taken together, these two areas may be described as the "protection and conservation of the marine environment").

TABLE II
CURRENT REGIONAL AGREEMENTS FOR MARINE POLLUTION PREVENTION AND CONTROL

Region	Agreement	Parties	Type	Citation
A. Atlantic				
1. Northeast	Convention for the Prevention of Marine Pollution by Dumping from Ships and Aircraft (Oslo) 1972	multilateral	regional	ST/LEG/SER.B/16 p. 457
	Convention for the Prevention of Marine Pollution from Land-Based Sources (Paris), 1974	multinational	regional	13 *ILM* 352 (1974)
2. North Sea	Agreement for Co-operation in Dealing with Pollution by Oil (Bonn), 1969 (Amended, 1972)	multilateral	fully regional	704 *UNTS* 3
	Convention on Civil Liability for Oil Pollution Damage Resulting from Exploration and Exploitation of Seabed Mineral Resources (London), 1977	multilateral	regional	UK Misc. No. 8 (1977) (Mnd. 6791, p. 6)
3. Sound	Agreement Concerning the Protection of the Sound (Oresund) from Pollution, 1974	Denmark, Sweden	fully (sub) regional	ST/LEG/SER.B/18/ Add. 2 p. 297
4. Baltic	Agreement Concerning Co-operation in Measures to Deal with Pollution of the Sea by Oil (Nordic) 1971	multilateral	regional	822 *UNTS* 324

	Convention on the Protection of the Marine Environment of the Baltic Sea Area (Helsinki), 1974	multilateral	fully regional	4 Churchill & Nordquist, *eds., New Directions in the Law of the Sea* (1975), p. 455
5. Mediterranean	Convention for the Protection of the Mediterranean Sea Against Pollution (Barcelona) 1976	multilateral	regional	15 *ILM* 290 (1976)
	Protocol Concerning Co-operation in Combating Pollution by Oil and Other Harmful Substances in Cases of Emergency, 1976	multilateral	regional	15 *ILM* 306 (1976)
	Protocol for the Prevention of Pollution by Dumping from Ships and Aircraft, 1976	multilateral	regional	15 *ILM* 300 (1976)
	Protocol for the Protection of the Mediterranean against Pollution from Land-Based Sources, 1980	multilateral	(almost) fully regional	19 *ILM* 869 (1980)
6.Adriatic	Agreement on Co-operation for the Protection of the Waters of the Adriatic Sea and Coastal Zones from Pollution	Italy, Yugoslavia	fully (sub) regional	6 Churchill, Nordquist, & Lay, *eds.,New Directions in the Law of the Sea* (1977), p. 456
B. *Arctic*				
C. Indo-Pacific 1. Indian	---	---	---	---

2. Persian Arabian Gulf	Kuwait Regional Convention for Co-operation and Protection of the Marine Environment from Pollution, 1978	multilateral	fully (sub) regional	17 *ILM* 511
	Protocol Concerning Regional Co-operation in Combating Pollution by Oil and Other Harmful Substances in Cases of Emergency, 1978	multilateral	fully (sub) regional	17 *ILM* 526
3. Malacca & Singapore Straits	Tripartite Agreement on the Safety of Navigation in the Straits of Malacca and Singapore, 1977	Indonesia, Malaysia & Signapore	fully (sub) regional	IMCO Res. A375 (1977)
D. Southern	---	---	---	---

Of the 14 instruments listed in Table III, all but four were (at the end of 1980) still in draft form and therefore not yet the embodiment of official and formal commitments. Moreover, only four[30] of these 10 draft instruments seem intended to result in legally binding "treaty commitments". Furthermore, none of the four completed instruments is a formal agreement of "treaty character"[31]. Yet, despite the tentative, informal and incomplete nature of these instruments, they reflect the range and diversity of significant "policy commitments" contemplated and developing at regional levels for the protection and conservation of the marine environment. The most conspicuous feature of Table III is the number of Action Plans and other instruments in the making as a result of UNEP initiatives under its Regional Seas Programme, which will receive more detailed treatment later[32]. In the meantime, it should be noted especially that all 14 of these emerging arrangements are "fully regional" (or subregional) in scope.

Finally, Table IV lists seven instruments which have been concluded in a form which seems to take them outside the normal definitions of "regional agreements", and yet may be regarded as having significant regional implications for the protection and conservation of the marine environment. The Whaling Convention, for example, is a global instrument covering whaling in all the waters of the world, but has been chiefly concerned with whaling in the Southern Ocean. No doubt other instruments of this kind could be added to the list, but the main point to be made is that regional treaty commitments may be accepted outside the orthodox forums of regional treaty-making.

TABLE III
SOME EVOLVING REGIONAL COMMITMENTS FOR THE PROTECTION AND CONSERVATION OF THE MARINE ENVIRONMENT

Region	Instrument	Parties	Type	Status
A. Marine Pollution Prevention and Control				
1. Atlantic				
(a) Mediterranean	UNEP Action Plan	multilateral	fully regional	Approved in January 1975
(b) West Africa (Gulf of Guinea) and adjacent waters)	Draft UNEP Action Plan	multilateral	fully regional	To be submitted for approval in 1981
	Draft UNEP Regional Convention for the Protection of the Marine Environment of the West African Region	multilateral	fully regional	To be submitted for approval in 1981
(c) Caribbean (Caribbean Sea, Gulf of Mexico and adjacent waters)	Draft UNEP Action Plan	multilateral	fully regional	To be submitted for approval in March 1981
2. Arctic	---	---	---	---
3. Indo-Pacific				
(a) Red Sea/Gulf of Aden	UNEP Action Plan	multilateral	fully sub-regional	Approved in January 1976

	Draft UNEP Regional Convention for the Environmental Protection of the Red Sea and Gulf of Aden	multilateral	fully sub-regional	Subject to continuing consultations
(b) Persian/ Arabian Gulf ("Kuwait Action Plan Region")	UNEP Action Plan	multilateral	fully sub-regional	Approved in April 1978
(c) East Asian Seas	Draft Action Plan for ASEAN sub-region (South China Sea and adjacent waters)	multilateral	fully sub-regional	Subject to continuing consultations
(d) Southeast Pacific	Action Plan (short term)	multilateral	fully regional	Adopted in November 1978
	Draft UNEP Convention on the Protection of the Marine Environment against Pollution in the Southeast Pacific	multilateral	fully regional	Subject to continuing consultations
	Draft UNEP Protocol on Regional Co-operation in cases of Pollution Emergencies	multilateral	fully regional	ditto
	Draft UNEP Action Plan	multilateral	fully regional	ditto
(e) Southwest Pacific	Draft Declaration of Principles on the Management and Improvement of the Environment in the Southwest Pacific	multilateral	fully regional	ditto
	Draft for a UNEP Regional Plan on Development and Environmental Protection	multilateral	fully regional	ditto

4. Southern	---	---	---	---
B. Conservation of Marine Species				
1. Atlantic	---	---	---	---
2. Arctic	---	---	---	---
3. Indo-Pacific	---	---	---	---
4. Southern	---	---	---	---

TABLE IV
MISCELLANEOUS INSTRUMENTS WITH REGIONAL IMPLICATIONS FOR THE PROTECTION AND CONSERVATION OF THE MARINE ENVIRONMENT

Region	Instrument	Parties	Type	Citation
1. Atlantic				
a. Northeast	Council of Europe Resolution (73) 29 on the Protection of Coastal Areas, 1973	multilateral	regional	21 *Europe Y.B.* 371 (1973)
	Convention for the Protection of the Environment (Nordic Convention), 1974	Denmark, Finland, Norway, Sweden	regional	13 *ILM* 591 (1974)
	Council of Europe Recommendation (847) on European Action to Prevent Oil Pollution of Waters and Coasts, 1978	multilateral	regional	19 Rüster-Simma, *eds., International Protection of the Environment* (1979), p. 9940
b. Southwest	Treaty of the La Plata River and its Maritime Limits, 1973	Argentina, Uruguay	fully sub-regional	Shigeru Oda, *International Law Development* (1977) Vol. II, p. 72
2. Arctic	---	---	---	---
3. Indo-Pacific				
a. Indian	Declaration of Indian Ocean as Zone of Peace, 1971	multilateral	non-regional	(GAOR Res. 2992 (XXVIII)
b. South Pacific	Treaty on Sovereignty and Maritime Boundaries in the Area Between the Countries, 1978	Australia & Papua New Guinea	fully sub-regional (bilateral)	?
4. Southern				
	International Convention for the Regulation of Whaling, 1946	multilateral	non-regional	161 *UNTS* 74

2. Regional Organizations

Unlike "regional agreements", the term "regional organization" is given a relatively narrow definition for the purpose of this study.

In view of the proliferation of regional organizational initiatives in marine affairs in recent years — including a host of research programs or projects[33] — it seems more useful to list only those organizational frameworks or mechanisms which are chiefly designed for making decisions or recommendations concerning the conservation of marine species or the prevention or control of marine pollution. Accordingly, organizations responsible mainly for the conduct or co-ordination of research, or the dissemination of data, are omitted — even though this means the exclusion of long-established and respected institutions such as the International Council for the Exploration of the Sea (ICES)[34].

Subject to these exclusions, a comparison of Tables V and VI shows a much larger number of regional organizations in the field of marine species conservation than in that of marine pollution prevention and control: 29 against 8. It is worth noting here that this ratio reflects an even sharper disparity between regionalization trends in the two fields than the 38:15 ratio between conservation and pollution control agreements listed in Tables I and II, respectively. Regionalism in marine pollution prevention and control seems likely, however, to be significantly increased as the UNEP Regional Seas Programme results in a growing number of regional arrangements in the 1980s. Whether that expected increase will be matched by a significant increase in the number of regional conservation arrangements will depend on a number of imponderables, such as the probability that nation states in the early 1980s will encourage FAO to lead them into regional fishery conservation arrangements in the wake of world-wide acceptance of extended national jurisdiction over living marine resources.

TABLE V
SOME CURRENT REGIONAL ORGANIZATIONS FOR THE CONSERVATION OF MARINE SPECIES

Region	Organization	Members (No.)	Type	Established by
A. Atlantic				
1. General	International Commission for the Conservation of Atlantic Tunas (ICCAT)	(12)	quasi-regional	ICCAT Convention 1966
2. Northeast Atlantic	Sealing Commission for the Northeast Atlantic (SCNEA)	(2) (Norway, USSR)	fully regional	SCNEA Convention 1957
	North-East Atlantic Fisheries Commission (NEAFC)	(14)	fully regional	NEAFC Convention 1959
3. Baltic	Shellfish Commission for Skagerrak and Kattegat	(3) (Denmark, Norway, Sweden)	fully regional	Agreement of 1952
	International Baltic Sea Fishery Commission (IBSFC)	(7)	fully regional	IBSFC Convention, 1973
4. Northwest Atlantic	Sealing Commission for the Northwest Atlantic (SCNWA)	?	quasi-regional	SCNWA Convention
	Northwest Atlantic Fisheries Organization (NAFO)	?	quasi-regional	NAFO Convention 1978
	US-Canada East Coast Fisheries Commission (apparently abortive)	(2) (Canada, US)	fully sub-regional	Fisheries Convention, 1979

5. East Central Atlantic	Fishery Committee for the Eastern Central Atlantic (CECAF)	(27)	quasi-regional	FAO Council Resolution, 1967
6. West Central Atlantic	Western Central Atlantic Fishery Commission (WECAFC)	(23)	quasi-regional	FAO Conference
7. Mediterranean	General Fisheries Council for the Mediterranean (CFCM)	(18)	fully regional	GFCM Convention, 1949
8. Black Sea	Mixed Commission for Black Sea Fisheries (MCBSF)	(3) (Bulgaria, Rumania, USSR)	fully regional	MCBSF Convention, 1959
9. Southeast Atlantic	International Commission for Southeast Atlantic Fisheries (ICSEAF)	---	quasi-regional	---
10. Southwest Atlantic	Regional Fisheries Advisory Commission for the Southwest Atlantic (CARPAS)	(3)(Argentina, Brazil, Uruguay)	fully regional	FAO Conference Resolution, 1962
11. Caribbean	---	---	---	---
B. *Arctic*	---	---	---	---
C. Indo-Pacific				
1. General	Indo-Pacific Fisheries Council (IPFC)	(18)	quasi-regional	IPFC Convention, 1948
2. Indian	Indian Ocean Fishery Commission (IOFC)	(29)	quasi-regional	FAO Council Resolution, 1948
3. North Pacific	International Pacific Halibut Commission (IPHC)	(2) (Canada, US)	regional	IPHC Convention 1953 (replacing 1924 and other Conventions

	North Pacific Fur Seal Commission (NPFSC)	(4) (Canada, Japan, US, USSR)	fully regional	NPFSC Convention, 1957
4. Northeast Pacific	International Pacific Salmon Fisheries Commission (IPSFC)	(2) (Canada, US)	sub-regional	IPSFC Convention, 1930
	International North Pacific Fisheries Commission (INPFC)	(3) (Canada, Japan, US)	quasi-regional	INPFC Convention, 1952 (amended 1978)
5. Northwest Pacific	Japanese-Soviet Northwest Pacific Fisheries Commission (JSFC)	(2) (Japan, USSR	regional	JSFC Convention, 1956
	Commission for Fisheries Research in the Western Pacific	(4) (China, North Korea, USSR, Vietnam)	quasi-regional	Convention of 1956
	Japan-Republic of Korea Joint Fisheries Commission (JKFC)	(2) (Japan, South Korea)	sub-regional	JKFC Convention, 1965
	Japan-China Joint Fisheries Commission (JCFC)	(2) (China, Japan)	sub-regional	JCFC Convention, 1977
6. Eastern Pacific	Inter-American Tropical Tuna Commission (IATTC)	(6)	quasi-regional	LATTC Convention, 1949
7. South Pacific	---	---	---	---
8. Southeast Pacific	Permanent Commission of the Conference on the Use and Conservation of the Marine Resources of the South Pacific (PCSP)	(3) (Chile, Ecuador, Peru)	fully regional	Agreement of 1952
	Fisheries Council of South Pacific Forum	?	regional	?

9. Southwest Pacific	South East Asian Fisheries Development Centre (SEAFDEC)	(4) (Japan, the Philippines, Singapore, Thailand)	quasi-regional	Agreement of 1967
D. Southern				
	Commission for the Conservation of Antarctic Marine Living Resources	(14)	quasi-regional	Convention of 1981

TABLE VI
CURRENT REGIONAL ORGANIZATIONS FOR MARINE POLLUTION PREVENTION AND CONTROL

Region	Agreement	Members (No.)	Type	Established by
A. Atlantic				
1. Northeast	(Mixed) Commission (for the Prevention of Pollution by Dumping from Ships and Aircraft), 1972	(9)	regional	Oslo Convention, 1972
	(Mixed) Commission (for the Prevention of Marine Pollution from Land-Based Sources), 1974	(8)	regional	Paris Convention, 1974
2. North Sea	Committee (on Civil Liability for Oil Pollution Damage Resulting from Exploration and Exploitation of Seabed Mineral Resources), 1976	(6)	regional	London Convention, 1976
3. Sound	Danish-Swedish Commission, 1974	Denmark-Sweden	fully sub-regional	Oresund Agreement, 1974
4. Baltic	Baltic Marine Environmental Protection Commission, 1974	(7)	fully regional	Helsinki Convention, 1974
5. Mediterranean	Regional Center (for Combating Pollution by Oil and Other Harmful Substances in Cases of Emergency, 1976	(17)	regional	Barcelona Convention, 1976
B. Arctic	---	---	---	---
C. Indo-Pacific	---	---	---	---
1. Indian				

2. Persian Gulf	Marine Emergency Mutual Aid Centre, 1978	(8)	fully sub-regional	Kuwait Convention, 1978
3. Malacca & Singapore Straits	Council for the Safety of Navigation and Control of Marine Pollution, 1977	Indonesia, Malaysia, Singapore	fully sub-regional	Tripartite Agreement
D.Southern	---	---	---	---

Another striking contrast brought out in Tables V and VI is in the number of quasi-regional arrangements: 13 of the 29 listed in the former, none of the 8 in the latter. The ratios are to be compared with those of 12:38 and 0:15 shown in Tables I and II, respectively. Some change is to be expected in the field of conservation, as regional blocs attempt to reduce or even eliminate extra-regional control over the management of their marine resources. In marine pollution prevention and control, on the other hand, the exclusion of extra-regional influence from regional arrangements is likely to continue under the sympathetic aegis of the UNEP Regional Seas Programme.

C. Regional Developments (1945-1973)

1. The Pre-Stockholm Years (1945-1972)

a) Introduction

For the protection and conservation of the ocean environment, little was attempted at the regional level between 1945 and 1971, except in the field of fishery conservation. In the first place, other priorities took precedence over environmental concerns: the reduction of East-West tensions; decolonialization and political consolidation in the Third World; foreign aid programming; arms control diplomacy; and peace-keeping missions. Moreover, most of these issues were dealt with at the global level, normally within the framework of the United Nations. Where the global level of diplomacy was found inappropriate, remedies were generally sought in bilateral or *ad hoc* arrangements. Regionalism was of political interest chiefly in the form of operationally successful regional communities, such as the EEC, and of continental organizations, such as OAS and OAU, which served as convenient forums for consultation and discussion[34]. Mainly perhaps because these were the years of early political development for so many nations, regional arrangements in environmental co-operation were still essentially an untried experiment[35].

b) Conservation of Marine Species

Apparently, 23 international fishery organizations were established and became operative between 1945 and 1972, and of these no less than 20 were regional or subregional in scope[36]. Perhaps no other category of "functional" organizations reflects the regional trend as clearly as fishery organizations. By 1972, however, there was fairly general agreement among informed observers that very few, if any, of these intergovernmental fishery organizations were coping effectively with conservation problems[37]. However, diverse explanations for these deficiencies were offered, and diverse remedies advanced.

Most of the solutions suggested were derived from one of eleven principal criticisms. Three of these criticisms were "systemic" in character, in the sense that they were based on the belief that the level of approach to fishery conservation was of primary importance. The remaining eight criticisms were "operational" in character, focusing on alleged sources of managerial or administrative ineffectiveness.

Of the three systemic criticisms, the most popular was the political criticism that regional fishery commissions deprived coastal states — especially developing coastal states — of their entitlement to special or exclusive fishing rights and

fishery conservation authority over stocks adjacent to their shores. Most of the existing regional fishery organizations were quasi-regional arrangements, whereby the coastal states had to share control over conservation and related policies with extra-regional participants. Since the latter tended to be heavily capitalized industrial fishing powers committed to the intensive utilization of stocks, such arrangements of shared control tended to work to the considerable disadvantage of the less capitalized, more vulnerable coastal states, particularly those which were struggling with basic problems of economic development. Solutions advanced from this line of criticism did not necessarily preclude the continuance of regional fishery commissions, but they did imply a reduction in the jurisdictional scope of such bodies[38].

Others argued, much less fashionably, for the establishment of a global fishery agency, or at least for a global system within which a universal approach to the problems of fishery conservation could be adopted. Such an approach was held out as the best chance of developing a rational and equitable treatment of conservation problems, applied consistently if not uniformly to comparable conditions in different regions of the world[39].

A third type of systemic criticism was more eclectic, warning against the dangers of "over-investing" in any one level of approach to the problem of fishery conservation. Each level — global, regional, subregional, national, and subnational — was characterized as having both advantages and disadvantages for particular purposes, and a "split-level" approach to the structure of fishery management was recommended as the most rational and the most likely to be effective in the longer run[40].

The "operational" criticisms, on the other hand, attached less importance to level or structure, and more to the characteristics of a management system. Some of these critics felt that the primary requirement was to grant direct regulatory authority, instead of merely recommendatory powers, to the existing regional commissions, with a view to eliminating the political veto powers vested in the member governments and reducing the prospects of political interference[41]. Others believed that some of the existing commissions were ineffective because their geographical scope was too restricted[42], or that the focus on species or area was inappropriate[43]. Still others felt that the objectives of existing commissions were too scientific, or otherwise too restrictive, in formulation, and that conservation objectives would only be accepted by certain fishing states if they were more explicitly integrated with fishery development and other economic objectives[44]. Similarly but more technically, many experts attacked the theory of conservation[45] explicitly or implicitly underlying the conventions on which the commissions were based, and argued for a better understanding of the interface between fishery biology and fishery economics[46]. Some critics believed that the critical deficiency was, more specifically, the lack of entrepreneurial competence, and that this would be corrected only by replacing intergovernmental modes or systems of operation by public enterprises, or a blend of public and private enterprises[47]. Last but not least, the failures of regional fishery commissions have been ascribed to inherent difficulties in enforcement[48].

In short, then, the history of regional fishery conservation arrangements between 1945 and 1972 is one of dramatic growth and mounting dismay. By 1972 frustration over the relative failure of most of these organizations was widespread, but there was very little agreement on how to proceed in order to deal more effectively with the conservation of the world's fisheries. The only thing that was clear to all was that UNCLOS III would be convened in a period of confusion and disarray.

c) Prevention and Control of Marine Pollution

In sharp contrast, the 1945-1972 period was one of very little institutional growth for marine pollution control purposes at the regional level[49], and yet despite the lack of regional innovation there was a slowly maturing understanding that regional, as well as global and national, arrangements would have to be developed in order to deal effectively with these complex environmental problems. ,

Looking back, we can see, from the general environmental perspective, that the 1945-1972 period was one of learning rather than institution-building. As far as marine pollution was concerned, most of the emphasis was placed on the visible and newsworthy threats associated with shipping, especially on the problem of catastrophic spillages, and then on the more pervasive practices of deliberate discharge and commercial dumping[50]. To deal with these matters an existing international agency (IMCO) had to renovate itself[51], but as long as the problem of ship-generated pollution received most of the attention, under the auspices of a global agency such as IMCO, there was little incentive to develop appropriate regional arrangements.

2. Marine Regionalism and the Stockholm Conference (1972)

a) Introduction

The Stockholm Conference on the Human Environment, held in June 1972, was of course a global undertaking: global not only in the sense that it was convened under the universalist auspices of the United Nations, at the call of the UN General Assembly, but also in the sense that it was designed with much care to represent the first, all-encompassing, planetary perception of the environmental disorders affecting the human race. At such an eventful inauguration of globalism in environmental planning it might have been expected that the case for regionalism would be understated. Yet a perusal of the main documents of the Stockholm Conference shows frequent references to the need for environmental action at the regional level and an occasional emphasis on the central role of regional arrangements.

Of the three main documents emanating from Stockholm[52], the Declaration on the Human Environment is the most global in tone, the most expressive of the need to perceive environmental problems in biospherical terms. Indeed, the word "regional" is used only once in the document: paragraph 7 of the preamble to the Declaration acknowledges, *inter alia*, that "[a] growing class of environmental problems, because they are regional or global in extent or because they affect the common international realm, will require extensive co-operation among nations and action by international organizations in the common interest"[53]. The 26 Principles enumerated in the Declaration contain no references at all to the need for regional action[54]. Moreover, the third document, the Resolution on Institutional and Financial Ar-

rangements, in prescribing the mandate for the Governing Council of UNEP[55], makes no express reference to the need for regional programming by this new global agency. In the absence of almost any such reference[56], let alone emphasis, the emergence of UNEP's Regional Seas Programme is to be seen as a development rather than a reflection of Stockholm intentions.

The Stockholm references to regionalism are confined, then, mostly to the second and most detailed of the documents approved at Stockholm, the Action Plan for the Human Environment. This document consists of 109 Recommendations, organized under five main headings: I, Planning and management of human settlements for environmental quality (Recommendations 1-18); II, Environmental aspects of natural resources management (19-69); III, Identification and control of pollutants of broad international significance (70-94); IV, Education, informational, social and cultural aspects of environmental issues (95-101); and V, Development and environment (102-109). Of these five sections, section II makes by far the most references (14), explicitly or implicitly, to the need for regional measures. Section I makes six such references, Sections III and IV two each, and Section V only one. It is normally difficult to be definite in quantifying *implicit* references, but the amount of implicit references is less arbitrary than might be supposed to the extent that the Stockholm Action Plan seems fairly consistent in using the word "international" restrictively to mean "global". This is evident in the frequency of the distinctions drawn between "international", "regional", and "national" measures, standards, and organizations[57], and between "international" and "regional" agreements and consultations[58]. On several occasions, however, the term "international" is used ambiguously in the Action Plan, in contexts where it might mean global or merely multilateral[59].

When one focuses more sharply on recommendations regarding the role of regional arrangements in the environmental management of the oceans, fourteen in particular stand out as especially noteworthy. Of these fourteen, eleven deal with the conservation of marine species, two with marine pollution prevention and control, and one is more general.

b) Conservation of Marine Species

Five of these eleven regionally oriented Stockholm recommendations deal with fishery conservation, three with "genetic resources", and one each with more particular problems: the protection of "species inhabiting international waters or those which migrate from one country to another", "protected areas", and "ecosystems of international significance".

i) Fishery conservation

All five of the Action Plan's recommendations dealing with fishery conservation contain explicit or implicit references to the role of regional arrangements. Recommendation 46 calls for general support of the "action programmes initiated by FAO and its Intergovernmental Committee of Fisheries and approximately 24 other bilateral and multilateral international commissions, councils and committees"[60]. Almost all of these 24 bodies are, of course, regional organizations[61].

Recommendation 47 advocates the close participation of "fishery agencies and interests" — "international [i.e., global] and regional fishery bodies" — in prepara-

tions for UNCLOS III[62]. Although FAO has been able to maintain a continuous presence as official observer at the Conference, this recommendation has apparently had little effect as far as the direct input of regional fishery organizations is concerned. Governments have spoken in fishery conservation matters at UNCLOS III almost exclusively in their national capacity, and hardly at all in their capacity as members of regional bodies. The regional perspective on these matters has not had any direct representation at the Conference.

Recommendation 48 calls for co-operation in the "research, control and regulation of the side effects of national activities in resource utilization where these affect the aquatic resources of other nations"[63]. The reference here is to "international co-operation", but the call seems to be addressed to global rather than regional bodies. In view of the nature of the problem addressed, the absence of any explicit reference to regional fishery conservation organizations is rather curious — especially, it might be added, since it follows the call for support of such organizations in Recommendation 46. It was at least one year too early for the Stockholm Conference to be inhibited by the trend in state practice and conference diplomacy toward the 200-mile extension of national fishery jurisdiction, and toward a resultant diminution in the jurisdictional scope of regional fishery commissions. If the Conference had foreseen that trend, this would have been evident presumably in the phrasing of Recommendation 46 rather than in that of Recommendation 48.

Recommendation 49 urges measures "to develop further and strengthen facilities for collecting, analysing and disseminating data on living aquatic resources and the environment in which they live"[64]. Here again, the call is apparently addressed to global agencies and national governments, but there is an express reference to the need for "local and regional data networks"[65].

Finally, however, in Recommendation 50, the Stockholm Conference urges that steps be taken, in various designated ways, "to ensure full co-operation among governments by strengthening the existing international and regional machinery for development and management of fisheries and their related environmental aspects and, in those regions where these do not exist, to encourage the establishment of fishery councils and commissions as appropriate"[66]. Few commentators, perhaps, would argue that UNCLOS III has devoted itself seriously to that end.

ii) Genetic resources

Recommendations 39 to 45 deal with genetic resources in general, and not just aquatic or marine organisms. None of these recommendations has an explicitly regional marine significance, but three in particular are of marine as well as terrestrial significance, referring to salt as well as fresh water habitats, and have implicit regional implications. Recommendation 39 advocates the establishment of "an international programme to preserve the world's genetic resources"[67], and gives special priority to conservation, "which represents the crucial element to which all other programmes relate"[68]. Recommendation 40 urges governments to make inventories of "the genetic resources most endangered by depletion or extinction"[69]. Presumably this task would be difficult, in shared or neighbouring coastal areas, without a degree of regional co-operation. Last, Recommendation 41 includes "aquatic organisms" in the list of genetic materials which should be included in the registers of existing collections, which it is proposed to compile or ex-

tend[70]. Again, regional co-operation would presumably be useful in this process: for example, between the universities and other research stations in neighbouring countries.

iii) Shared or migratory species

Anticipating UNCLOS III, the Stockholm Action Plan urges, in Recommendation 32, that governments give their attention to "the need to enact international conventions and treaties to protect species inhabiting international waters or those which migrate from one country to another"[71]. Two proposals are advanced: first, a "broadly-based convention should be considered which would provide a framework by which criteria for game regulations could be agreed upon and the over-exploitation of "resources curtailed by signatory countries"; and second, a "working group should be set up as soon as possible to consider these problems and to advise on the need for, and possible scope of, such conventions or treaties"[72]. The most significant development along these lines is the global Bonn Convention on the Conservation of Migratory Species of Wild Animals (1979). It is envisaged that the provisions of this convention shall be implemented in part through regional arrangements.

iv) Protected areas

Recommendation 37 addresses itself to the need for intergovernmental co-ordination and co-operation in the "management of neighbouring or contiguous protected areas"[73]. The regional nature of such measures is clear, though only implicit, and there is an express inclusion of "fishery regulations" in the list of matters proposed for agreements between neighbouring or contiguous states.

v) Ecosystems of international significance

Recommendation 38 advocates that governments "take steps to set aside areas representing ecosystems of international significance for protection under international agreement"[74]. Both uses of the word "international" are ambiguous, but the first should perhaps be interpreted as meaning "global or regional" and the second as "multilateral". In practice, it is easier to envisage regional arrangements rather than wider arrangements to accomplish this end, since ecosystems are relatively localized, but it can be argued that extra-regional states should be encouraged to participate in "quasi-regional" arrangements for the maintenance of ecosystems upon which their nationals have an adverse impact.

c) Prevention and Control of Marine Pollution

Of the 25 recommendations in the Stockholm Action Plan which deal with the "identification and control of pollutants of broad international significance", 16 deal with pollution in general and nine with marine pollution in particular. Only two seem to have explicit regional implications. Without specific reference to marine biology, Recommendation 80 urges that "regional and global networks of existing and, where necessary, new research stations, research centres, and biological reserves be designated or established within the framework of the Man and the Biosphere Programme (MAB) in all major ecological regions, to facilitate intensive analysis of the structure and functioning of ecosystems under natural or managed conditions"[75]. Regional marine parks or reserves are the sort of co-operative measure that one would wish to see adopted within the MAB for the protection of the marine environment[76]. In the section dealing specifically with marine pollution,

Recommendation 86 urges support for all ongoing international (including regional) efforts to curb the various forms of marine pollution. Special reference is made to enclosed and semi-enclosed seas, "which are more at risk from pollution", and to the need for "an overall instrument for the control of ocean dumping as well as needed regional agreements within the framework of this instrument"[77]. More than any other part of the Stockholm Action Plan, this recommendation seems to provide the foundation and rationale of the UNEP Regional Seas Programme[78].

d) General

In the final section of the Stockholm Action Plan Recommendation 102 identifies numerous steps that should be taken by " appropriate regional organizations" to help reduce incompatibilities between developmental and environmental planning. Of particular importance in the protection and conservation of the ocean environment are these three roles:

(a) Preparing short-term and long-term plans at regional, subregional and sectoral levels for the study and identification of the major environmental problems faced by the countries of the region concerned as well as the special problems of the least developed countries of the region and of countries with coastlines and inland lakes and rivers exposed to the risk of marine and other forms of pollution;

(b) Evaluating the administrative, technical and legal solutions to various environmental problems in terms of both preventive and remedial measures, taking into account possible alternative and/or multidisciplinary approaches to development;

(c) Preparation, within the framework of international agreements, of legislative measures designed to protect marine (and fresh-water) fisheries resources within the limits of their national jurisdiction ..."[79].

As we shall see later[80], in the absence of appropriate regional organizations in most regions of the world, the Regional Seas Programme of UNEP would undertake these roles in assisting the coastal states of ten designated regions to co-operate for the protection and conservation of their shared regional marine environment.

3. Marine Regionalism and the UN Sea-Bed Committee (1968-1973)

a) Introduction

Although the 1972 Stockholm Conference was primarily a global and cross-sectoral effort to preserve the human environment, its impact on marine regionalism was by no means negligible. Another influential event was the series of discussions initiated by the UN *Ad Hoc* Committee[81] and its successor Committee on the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction[82] (generally called the "Sea-Bed Committee"). Since its life-span (1968-1973) both precedes and follows the Stockholm Conference, the Sea-Bed Committee documents reflect the changing and diverse perceptions of marine regionalism even more clearly.

An early perception was that of the need for practical co-operation in economic and other matters at regional and local levels, especially between coastal states border-

ing on marginal and internal seas. The *Ad Hoc* Committee's Economic and Technical Working Group's[83] report pointed out in 1968 how "(g)roups of nations can join together to explore and develop the resources of a limited area of the world ocean ..."[84]. Moreover, this Working Group cited the Committee of Co-ordination of Joint Prospecting for Mineral Resources in Asian Off-shore Areas (CCOP), under ECAFE (now ESCAP) sponsorship, as a useful model for such regional co-operation affecting coastal states[85].

At the same time, the *Ad Hoc* Committee's Legal Working Group, having surveyed existing international agreements concerning the seabed and ocean floor, could find only two multilateral treaties of a regional character — both of which were agreements by Chile, Ecuador and Peru regarding the establishment of a 200-mile maritime zone[86]. Thus, from its "survey of past and present activities of the United Nations and other intergovernmental bodies with regard to the seabed and ocean floor, and of existing international agreements concerning those areas ..."[87], the *Ad Hoc* Committee found some regional arrangements for the protection and conservation of the marine environment.

A growing awareness of the importance of systematic regional investigations of selected ocean regions was reflected in a report published by UNESCO's Intergovernmental Oceanographic Commission (IOC): the "Comprehensive Outline of the Scope of the Long-Term and Expanded Programme of Oceanic Exploration and Research"[88]. In this 1969 report to the Sea-Bed Committee, IOC pointed out: (a) how it had acquired extensive experience in organizing and conducting large-scale international expeditions in the Indian Ocean, the equatorial part of the Atlantic Ocean, and the Kuroshio region of the Pacific; and (b) how it had planned or projected further investigations in the Northern Atlantic, the Mediterranean, the Carribean, and Antarctic waters. IOC then urged that "(p)lans for these regional investigations should be developed in close collaboration with the existing regional organizations such as ICES (International Council for the Exploration of the Sea), ICNAF (International Commission for the Northwest Atlantic Fisheries, now succeeded by the Northwest Atlantic Fisheries Organization (NAFO)), etc., which have accumulated valuable experience in organizing co-operative investigations in their respective regions with great benefit for fisheries development there"[89].

Conflicting perceptions of the role of regional organizations, either in conjunction with or as part of the international machinery for a seabed regime, were reflected in a 1971 analytical summary of proposals and suggestions of delegations to the First Committee of the General Assembly at its 25th session and to the Sea-Bed Committee[90]. In urging that regional differences and particularities of a geographical, social and economic nature should be recognized, and citing several regional declarations and agreements (Santiago, 1952; Montevideo and Lima, 1970; Baltic States, 1967), this summary concluded that "[i]t would be clearly unrealistic ... not to recognize that existing trend which permeated the whole field of the law of the sea"[91]. The arguments of those who expressed a preference "for an international regime comprising regional or subregional systems co-ordinated on a world scale, in accordance with principles of common interest ..." were these:

(1) Such systems would help ensure that the exploration and exploitation of the area were undertaken for the benefit of mankind.

(2) Regionalization would make it easier to produce the most satisfactory solutions to problems affecting the interests of each individual region.

(3) If regionalized, the system would be more flexible and would provide a better guarantee of effective international co-operation at the regional and inter-regional levels[92].

Others, however, felt that such variations accommodating customary regional arrangements "should not infringe on the principles already agreed upon by the world community"[93]. Exactly which principles they meant was not clear, but probably those in the 1970 UN Declaration of Principles[94] which clearly emphasized the global approach:

> 11. With respect to activities in the area and acting in conforming with the international regime to be established, States shall take appropriate measures for and shall co-operate in the adoption and implementation of *international* [emphasis added] rules, standards and procedures for, *inter alia:*
>
> (a) The prevention of pollution and contamination, and other hazards to the marine environment, including the coastline, and of interference with the ecological balance of the marine environment;
>
> (b) The protection and conservation of the natural resources of the area and the prevention of damage to the flora and fauna of the marine environment[95].

Opponents of the regional approach also feared that "exploitation of the seabed and ocean floor on a regional basis, which could place countries in different parts of the world on a different footing, was not likely to ensure equitable sharing by all states in the benefits to be derived"[96].

One example of the uncertainty how to accommodate such regional arrangements, institutionally and procedurally, was the draft ocean space treaty proposed by Malta in 1971[97]. This draft treaty contained only one article on regional arrangements[98]. That article was relegated to the chapter on "Miscellaneous Provisions" and the footnote to that article clearly indicated the supplementary nature of such regional arrangements[99]. This proposal, not to allow regional arrangements until *four years* after the treaty came into force, reflected the globalism of its author[100] and his perception that such regional arrangements were auxiliary.

Further evidence of uncertainty about the role of regionalism was found in the varying treatment of the terms "regional criteria" and "regional arrangements" in the various comprehensive lists of subjects and issues relating to the law of the sea submitted by different groups of countries[101]. For example, two of these lists — one submitted primarily by Latin American countries[102], the other submitted primarily by Afro-Asian countries[103] — treated these terms quite differently. The former list included "regional criteria" under three items (breadth of territorial sea; rights of coastal states; enclosed and semi-enclosed seas and archipelagos) and "regional arrangements" under one item with four sub-items (management of ocean space, including living and non-living resources; scientific research; preservation of marine environment; and zones of peace and security)[104]. The latter list, however, included the term "regional arrangements" only twice — once under the heading of the ex-

clusive economic zone beyond the territorial sea, and once as a separate item without any sub-items[105].

Although these are only two of the 21 lists submitted to the Sea-Bed Committee, they indicated a divergence of viewpoints on how to discuss, let alone accommodate, regional issues related to the protection and conservation of the marine environment. On the one hand, the Latin American approach, associating "regional criteria" with several subjects and functions, suggested that the role of regional arrangements should be addressed in debating every issue affecting regional interest. On the other hand, the Afro-Asian approach, invoking "regional arrangements" for general rather than subjective or functional considerations, suggested that the case for regionalism should be subordinated when issues affecting regional interests were discussed. In the final comprehensive list recommended by the Sea-Bed Committee to "serve as a framework for discussion and drafting of necessary articles"[106], a compromise approach was evident because the phrase "global or regional criteria" was included under only one rubric on the breadth of the territorial sea, and the topic of "regional arrangements" was retained as a separate item for discussion[107].

The actual draft articles proposed by various delegations, and later consolidated in the final Sea-Bed Committee records[108], also reflected this divergence of treatment. Under Item 15 ("Regional Arrangements") can be found draft articles submitted by Columbia, Mexico and Venezuela[109], Ecuador, Panama and Peru[110], Iran[111], and Yugoslavia[112]. However, four of the draft articles submitted by Jamaica[113] with both global and regional criteria were duplicated under the sub-item "Breadth of the Territorial Sea" and under the item "Exclusive Economic Zone beyond the Territorial Sea"[114]. One of the draft articles submitted by the Netherlands relating to an intermediate zone was included under the sub-item "Coastal State Preferential Rights or Other Non-Exclusive Jurisdiction over Resources beyond the Territorial Sea"[115].

Before the end of its mission, the Sea-Bed Committee proved unable to reconcile the diverse aspects of marine regionalism in the official design for UNCLOS III. In 1971 the Committee was reorganized from a system of two named Sub-Committees (Economic and Scientific, and Legal) to one of three numbered Sub-Committees (I, II, & III)[116], with more or less discrete subjects and functions allocated to each. Henceforward, regional concerns would have to be dealt with by each Sub-Committee[117], and the Committee as a whole would be structurally incapable of presenting a systematic view of the role of regional arrangements to UNCLOS III.

Finally, although individual countries' perceptions of "ideal" regional arrangements might have varied considerably, one group of countries — the landlocked states — considered that their interests would be "better and more appropriately safeguarded" by global agreements[118]. Thus, in the face of some strong opponents, the proponents of marine regionalism in the Sea-Bed Committee had to try to accommodate the landlocked with global as well as regional concessions.

b) Conservation of Marine Species

Sea-Bed Committee documents also reveal that the problem of finding the most effective institutional mechanism for balancing antagonistic fishery interests remained an unsettled issue. For example, at its 1971 session the Sub-Committee on

the Law of the Sea of the Afro-Asian Legal Consultative Committee (AALCC), a non-UN regional organization, reported that "problems of fisheries and conservation of living marine resources be approached on a regional or ocean basis, the States in the region or ocean being encouraged to enter into agreements among themselves regulating the rights and obligations of each other in relation to fishing, *free of outside interference"* [emphasis added][119]. This position in favour of excluding extra-regional influences from regional fishery arrangements is interesting in two respects. First, 35 members of AALCC also belong to the 39-member Indian Ocean Fisheries Commission (IOFC); and 11 members of IOFC are clearly from outside the Indian Ocean region — *i.e.* extra-regional states. Second, this call by AALCC coincided with increasing concern for coastal states' rights. This latter concern was clearly reflected in the 1973 Organization of African Unity (OAU) Declaration on the Issues of the Law of the Sea[120], and in the 1973 final report of the Sea-Bed Committee II. Reference was made to "the special interest of a coastal state in the management and conservation of living resources in the area subject to its jurisdiction". The role of "international *or* [emphasis added] regional fishery organizations in connection with the regulation of fisheries in general, and in particular with regard to highly migratory resources", was also stressed[121].

Finally, the Sea-Bed Sub-Committee II consolidated proposals of 15 countries into 19 articles under the sub-item: "Management and Conservation of Living Resources"[122]. Although only five of the 19 articles mentioned either "regional arrangement" or "regional organization", many of the other articles alluded to the obligation for co-operation in conserving marine stocks.

c) Prevention and Control of Marine Pollution

The 1972 Stockholm Conference had its most apparent impact on the Sea-Bed Committee in the deliberations of Sub-Committee III, whose mandate was "(t)o deal with the preservation of the marine environment (including, *inter alia,* the prevention of pollution) and scientific research and to prepare draft treaty articles thereon"[123]. Before Stockholm, the Sub-Committee III view was that marine regional arrangements should be supplementary and preliminary to a general treaty on prevention and control of marine pollution[124]. However, after Stockholm the Sub-Committee III view had shifted to the position that arrangements to prevent and control marine pollution should be integrated in a holistic scheme of rules and standards involving *all* levels concurrently[125].

Thus, the emphasis of the Sea-Bed Committee's approach to marine regionalism for the control of marine pollution appeared to have been significantly affected by the 1972 Stockholm Conference.

4. Conclusions

The period from 1945 to 1973 seems, in retrospect, to have alternated between phases of predominantly regional accomplishments and phases of predominantly global accomplishments in the protection and conservation of the ocean environment. The first phase, from 1945 to 1958, was featured by the emergence of the first generation of post-war regional fishery commissions[126]. Much of this period also witnessed the preparation by the International Law Commission of draft articles for the First UN Conference on the Law of the Sea (UNCLOS I) held at Geneva in 1958. But this globally-oriented work took place in a period when the regime of the high

seas was conceived in the most spatially extensive way, so that there was little interference with the notion of granting maximum jurisdictional scope to regional fishery commissions[127]. Significantly, this was also a period of international idealism, or at least a period when regional commissions were not seen essentially as a threat to the sovereign prerogatives of nation states[128].

The second phase, from the convening of UNCLOS I in 1958 to that of its successor UNCLOS II in 1960, was predominantly a period of global adjustments in the law of the sea. In this phase, as in the first, conservation concerns were limited to fisheries, and the approach was almost exclusively scientific, reflected for example in the biological concept of maximum sustainable yield[129], and in the emphasis on the need for scientific evidence[130]. UNCLOS I did take cognizance of the role of regional arrangements[131], but until the outcome of UNCLOS II was known[132] there was little incentive to proceed with the further development of regional fishery commissions[133].

The third phase, from 1960 to 1967, was a period of extremely little institutional development, at any level, in the protection and conservation of the ocean environment. The few accomplishments of this phase were all at the regional level[134]; and it was also in those years that certain countries, especially those of Western Europe, first became seriously concerned with a widening range of international environmental problems, including that of ship-generated marine pollution[135].

The fourth phase, from 1967 to the end of 1973, began with Arvid Pardo's universally oriented call for a reformulation of the law of the sea, which would be rational and comprehensive and thus give due weight to all environmental and technological considerations of the late 20th century. While the Sea-Bed Committee and its terms of reference expanded, preparations were concurrently under way for the Stockholm Conference, and for a sophisticated (and largely Western) formulation of a biospherical approach to international environmental problems. In this final (pre-UNCLOS III) phase of predominantly global accomplishments, the paramount factor was perhaps the mutual influence of Stockholm and Sea-Bed Committee.

The Sea-Bed Committee — as its name reminds us — was first established in 1968 to deal only with the issues of deep ocean mining. In 1969 and 1970 an increasing number of delegates to the Sea-bed Committee began to refer to the importance of other law of the sea issues, including issues related to marine pollution prevention and control and fishery conservation[136]. Some of the delegates were, at this time, beginning to become involved also in their government's preparations for Stockholm, and in the minds of a few there began to evolve the idea of developing a linkage between the Stockholm Conference, the Sea-bed Committee (and eventually UNCLOS III), and other major forums for law-making in the fields of law of the sea and international environmental law[137]. Particularly for purposes of marine pollution control, it was believed that "systematic" legal development might be nurtured through a succession of distinct but thematically related conferences. According to such a strategy of legal development, early conferences (such as Stockholm and the Sea-bed Committee) provide general formulations in technically nonbinding documents[138]; later conferences (such as the 1972 London Conference on Dumping and the 1973 IMCO Conference on Marine Pollution by Ships) would be devoted to the elaboration of treaty commitments in designated areas, which would be derived

from those formulations; and, finally, UNCLOS III itself would provide the most authoritative and comprehensive framework of universal principles and procedures. Such a strategy, in short, presented a challenge to the new generation of lawyer-diplomats: the orchestration of conference diplomacy in the interests of the "progressive development" of international law[139].

This strategy seemed to be working successfully when, in the fall of 1970, the General Assembly adopted the Declaration of Principles Governing the Sea-Bed and the Ocean Floor, and the Subsoil Thereof, Beyond the Limits of National Jurisdiction[140]. Although not chiefly concerned with the protection of the ocean environment[141], this document could be regarded as a contribution to the general purpose of legal development within the law of the sea context.

At the same time[142], the General Assembly decided: that the Sea-bed Committee should become the official body entrusted with the preparations of UNCLOS III[143]; that, as such, the Committee should henceforward devote itself to a greatly expanded agenda of issues, which would include a range of environmental concerns[144]; and that the Committee should be reorganized in order to carry out its work more efficiently[145]. On the face of things, these decisions seemed conducive to the stategy of systematic legal development, taken as they were at a time when a holistic, comprehensive, and systematic approach to the human environment was being taken in preparations for the UN Stockholm Conference. Moreover, steps were being taken to draft a Stockholm Declaration of Principles, which might be expected to have some degree of juridical significance in a "soft law" form comparable to the 1970 Declaration of Principles on deep ocean mining. But the decision to reorganize the structure of the Sea-bed Committee, involving a diffusion of diplomatic effort among the three new Sub-Committees, can be seen in retrospect as defeating the prospect of a holistic, comprehensive and systematic approach to the environmental issues in the law of the sea[146]. To that extent, the reorganization of the Sea-bed Committee — albeit one justified by the advantages of greater efficiency at the technical level — was a setback for the cause of systematic legal development. Accordingly, despite frequent invocations in the Sea-bed Committee in 1973 to the holistic Stockholm view of the ocean environment[147], a partial acceptance of the Stockholm framework[148], the compartmentalization of the Sea-bed Committee resulted in a loss of vision[149].

Prominent among the casualties was the question of the role of regional arrangements, for conservation and other purposes, in the new law of the sea. By 1973 it was already inevitable that the potentialities of regionalism would not be fully grasped and systematically developed at UNCLOS III.

D. Regional Developments (1973-1980)

1. Regional Reactions to Extended Jurisdiction

Claims to extended coastal state jurisdiction have taken various forms both before and during UNCLOS III: territorial sea claims beyond the traditional three-mile limit; exclusive fishing or fishery conservation claims beyond three miles; and continental shelf claims beyond traditional territorial limits[150]. It was not until 1974, however, *that the first official claim to a 200-mile exclusive economic zone (EEZ) was ad-*

vanced. By the middle of 1979 the first such claimant (Bangladesh) had been joined by 46 other EEZ claimants, as well as by 42 other states claiming exclusive fishing and fishery conservation rights out to 200-mile limits, either as part of a 200-mile territorial sea claim (14) or as a separate claim solely in terms of fishery jurisdiction (28)[151].

Of the 47 EEZ claimants up to mid-1979 Africa had produced 15, Asia 10, mainland Latin America 7, the Caribbean island states 5, Western Europe 4, and Oceania/Australasia 6. Of the 14 200-mile territorialists, 7 were from mainland Latin America and 7 from Africa. Of the 28 200-mile fishing zone claimants, 3 were African, 4 Asian/Pacific, 2 mainland Latin American, 1 Caribbean, 10 Western European, 4 Eastern European, 2 North American, and 2 Oceanic/Australasian. The 200-mile phenomenon in extended coastal state jurisdiction had, then, rapidly become global.

In this process of expansion, only two regional patterns are discernible. First, the process originated in the 1950s in mainland Latin America, with 200-mile claims mostly in territorial or quasi-territorial form[152]. Second, the African nations were the most conspicuously responsive to the Latin American example, albeit in a mixture of territorial and non-territorial forms of claims[153]. It is well-known that the desire for exclusive fishing rights played a major part in this process of jurisdictional expansion, and that it owed much less to an explicit willingness by coastal claimant states to accept responsibility for protection and conservation of the ocean environment. Species conservation issues received relatively little attention during Committee Two debates on the EEZ provisions at UNCLOS III[154], and marine pollution issues were discussed and negotiated exiusiveiy within the jurisdictionally less confining context of Committee Three[155].

2. Regional Seas Programme

a) Introduction

The United Nations Environment Programme (UNEP) has been a member of the UN family since 1973. The idea of an international environmental agency had been widely discussed in official[156] and academic[157] circles in the years of preparation leading up to the UN Conference on the Human Environment held at Stockholm in 1972[158]. By that time a consensus had emerged that the primary need was for a mechanism to promote environmental initiatives within the United Nations and coordinate relevant activities involving the UN agencies and the member states. The Stockholm Conference recommended the establishment of a "permanent institutional arrangement"[159], and acting upon this the UN General Assembly gave birth to UNEP on 15th December 1972[160].

Since it was clearly impracticable to secure a total involvement by UNEP in all environmental issues, within a single comprehensive framework, it was decided at an early stage to build up programmes in certain designated areas of priority. By 1976 these priorities were: human settlements and human health; the management and control of terrestrial ecosystems; environment and development; oceans; energy; and natural disasters[161]. The decision to include oceans in its list of priorities was consistent with the oceanic emphasis provided at Stockholm[162], and apparently reflected a continuing concern with the problems of marine pollution[163]. Even as ear-

ly as the summer of 1974, as UNCLOS III began its substantive sessions in Caracas, it was evident that important changes in the law of the sea concerning ship-generated pollution would require UNEP to play an important co-ordinative role in that area. Equally clearly, the critical problem of land-based pollution of the sea, which could not be dealt with in any detail by UNCLOS III, would be of special concern to UNEP, since land-based activities were the most important source of marine pollution and created a problem that could be treated effectively only through intergovernmental co-operation among inland as well as coastal states.

The Stockholm approach to the problems of marine pollution had emphasized the tasks of assessment and control. In developing a general strategy for its oceans programme, UNEP's Governing Council at its third session in 1975 adopted the following objectives: (i) the promotion of international and regional conventions, guidelines, and action for the control of marine pollution and for the protection and management of aquatic resources; (ii) the assessment of the state of pollution and of living resources; and (iii) the monitoring of marine pollution and aquatic resources[164].

The identification of UNEP's objectives, in any of the designated areas, was not easily accomplished. In the first place, the arrival of a new agency within the UN bureaucratic system was seen by many as an additional competitive strain on scarce budgetary and other resources, and UNEP had to contend intially with defensive, even resentful, attitudes on the part of the older agencies. Second, although UNEP's mandate was described chiefly in *co-ordinative* terms[165], its field of concern was extremely broad and it proved difficult for the agency's enthusiastic leadership to avoid giving the impression of seeking a *directive* role in UN affairs. Third, in the wake of the Stockholm Conference UNEP inevitably inherited a climate of suspicion or apathy on the part of some developing countries, which still maintained that environmental concerns should have a lower priority than their developmental needs[166]. Fourth, UNEP's interest in promoting international environmental conventions — and, to that extent, in the development of international environmental law — was regarded by some national governments as improper, especially at a time when forums such as UNCLOS III were engaged in difficult and controversial issues of that kind[167].

Despite some resistance along these lines UNEP was able to make considerable progress in the early stages of its oceans programme. This may be attributed chiefly to its decision to adopt a regional strategy[168]. Such a strategy was designed to deflect unnecessary criticism in a number of ways: by concentrating on selected marine regions that seemed to be significantly vulnerable to environmental clangers; by taking into account the physical, political and economic characteristics of each of these areas of the ocean environment; by following a schedule that permitted realistic programme objectives to be met within a time-frame appropriate to each region; and by recognizing the need for close and continuing participation by the countries affected at all stages of preparation and implementation of each regional plan. Given sufficient support from the participating countries it was hoped that it would prove possible to implement in each region a "comprehensive" action plan. Each regional action plan would be followed, it was hoped, by a general framework convention, supplemented by appropriate protocols.

Subject to regional variations, each regional action plan would be designed around the Stockholm model, which consisted of three components: environmental assessment, environmental management, and supporting measures[169]. Of these three components the first was the most clearly envisaged. Nothing, it was generally agreed, could be accomplished, or even understood, without amassing the relevant data for assessment in the service of environmental management. Data would be needed on such matters as "the sources, amounts, behaviour, and effects of pollutants in sea water, sediments, and biota; effects of these pollutants on human health and coastal ecosystems; status and trends of exploitation of living and non-living resources; ongoing socioeconomic development practices which have direct or indirect effect on the environment; and the status and proficiency of local institutions and experts available for participation in the action plan"[170]. Unfortunately, effective environmental assessment is an expensive undertaking, and this area of UNEP activities, dependent as it is on the work of other agencies, has been slow in developing[171].

The concept of "environmental management", on the other hand, is more difficult to articulate. Most theorists would argue, for example, that it consists of several functions, including "assessment"[172]. Even if the latter is treated separately as a premanagerial activity, the need for assessment does not cease after the introduction of a management system: on the contrary, it might be asserted, assessment becomes more specific, once administrative opportunities and requirements have been clarified. Management continues, moreover, after the need for "supporting measures" has arisen.

Yet, whatever one's view on the theory of environmental management it is in this area — the *planning* of environmental management systems — that UNEP's practical success has been most conspicuous. Arguably, the most significant test of the success of UNEP's regional seas strategy is its acceptability by the governments with which UNEP has had to consult. With this consideration in mind, as well as the "framework for environmental action" inherited from Stockholm, UNEP no doubt had to adopt a "practical" approach even to the "theory" of its strategy[173].

In any event, it should be noted that UNEP's present concern with "environmental management" in its oceans programme is a narrow one, focusing only on two goals: "to assist Governments in making environmentally wise decisions about development, and to improve their ability to make such decisions on their own"[174]. The first, a special kind of "assessment" function, and the second, a "supporting measure", are both important aspects of environmental management, but aspects also that were no doubt found to be politically popular among the governments consulted.

Under the title of the Regional Seas Programme in recent years, UNEP's oceans programmes understood to have the best chance of acceptance if it remains limited to a selected number of regional areas, whose environmental well-being appears especially vulnerable to actual or potential harms. Initially, it was decided to concentrate on four regions: the Mediterranean, the Persian-Arabian Gulf (the Kuwait Action Plan Region), the (Wider) Caribbean, and West Africa (the Gulf of Guinea). By the late 1970s however, it had been agreed to add four more regions to the Programme: the East Asian Seas (including the Malacca Straits), the Red Sea and Gulf of Aden, the Southeast Pacific, and the Southwest Pacific. In 1980 two more regions were added, East Africa and the Southwest Atlantic, bringing the total number of

designated "regional seas" to ten. A typical Action Plan includes the following components:

Legal component: In most cases a legal framework for co-operative action is adopted, consisting of a broadly conceived but legally binding regional convention (a "parent" or "framework" treaty, a "loi cadre"), elaborated by a number of specific and more technical protocols.

Assessment component: All Plans list a large number and variety of current and projected acitivites aimed at assessing and evaluating the causes, magnitude and consequences of pollution and related problems in the marine region.

Management component: Each Plan also includes a wide range of management activities to be undertaken, such as: co-operative regional projects on rational exploitation of marine living resources; utilization of renewable sources of energy; management of fresh-water resources; protection of soil from erosion and desertification; development of tourism without ecological harm, mitigation of environmental damage associated with human settlements; and so forth[175].

At first impression, the ten designated ocean regions are strikingly dissimilar in size, and in the diversity as well as scale of environmental problems that they seem to present. The Red Sea and the Persian-Arabian Gulf are only a fraction of the size of the Southwest Pacific and Southeast Pacific. The Mediterranean is one of the more researched marine regions in the world[176], and the South Pacific regions are certainly among the least researched. Moreover, great disparities seem to exist in the potentiality of the ten regions for co-operative action, although none, it must be hoped, is so conflict-prone as to place the objectives of co-operative environmental management beyond reach[177]. Any attempt to make a fair evaluation of the choice of regions must, however, recognize that preconceptions of what constitutes a suitable "unit of management" are bound to yield to practical considerations of administrative feasibility and political acceptability; and that, if they did not, the best efforts at co-operative management would be destined to fail.

b) The Mediterranean

Almost from the time the Regional Seas Programme was first conceived, it seemed obvious that UNEP should turn its attention initially to the Mediterranean Sea. None of the other regions under consideration had such a well documented history of abuse. Along with the Baltic and Black Seas, the Mediterranean is proven to be one of the world's most seriously polluted semi-enclosed seas[178]. It has no tradition of en *bloc* co-operative action[179]; and yet, since all coastal states contribute to the land-based sources of marine pollution in the region[180], nothing less than a total collective effort by all the Mediterranean states would be sufficient to deal effectively with the problem. Moreover, all, without exception, had much to gain from the introduction of higher environmental standards in the Mediterranean basin: in improved water and sanitation for their coastal communities; in healthier and more productive fishery stocks[181]; and in cleaner, more attractive beaches and scenic areas important for tourist development. Despite disparities in living standards among the Mediterranean states, it seemed politically feasible to launch a pan-regional environmental initiative, provided the UNEP proposals were sound and the coastal countries of the region properly consulted. In short, of all the designated regional seas, the Mediterranean seemed to offer by far the best prospect of success as a regional "laboratory" in the development of marine and environmental management.

The Mediterranean initiative was apparently set in motion as early as December 1969, when the General Fisheries Council for the Mediterranean (GFCM), a regional fishery commission of FAO, initiated a study on the state of pollution in the Mediterranean. This study was submitted to the FAO Technical Conference on Marine Pollution and its Effects on Living Resources and Fishing, which advocated a regional approach to marine pollution control in enclosed and semi-enclosed seas. A revised version of this study was examined by the GFCM in 1972, and two years later a proposal for a convention for the control of pollution affecting the living resources of the Mediterranean was submitted to a formal consultative meeting of the GFCM member states at FAO headquarters in Rome. After discussion of a suggested set of principles, these states agreed to call for the adoption of a comprovensive framework convention, which would be supplemented by several protocols on dumping, pollution emergencies, pollution from ships, land-based sources of pollution, and pollution from sea-bed exploration and exploitation. To these ends it was agreed to prepare a set of guidelines[182].

At this stage, the primary promotional effort fell to UNEP, which had taken over the co-ordinative work assigned to it after the Stockholm Conference on the Human Environment. The next phase in the process of regional organization consisted of two integovernmental conferences: the Intergovernmental Meeting on the Protection of the Mediterranean held at Barcelona in February 1975; and the Conference of Plenipotentiaries of the Coastal States of the Mediterranean Region, also held at Barcelona in February 1976[183]. The 1975 conference adopted the proposed Action Plan[184], and discussed a draft framework convention, and two draft protocols[185]. The 1976 Plenipotentiary Conference adopted and opened for signature the final versions of these three agreements: the (Barcelona) Convention for the Protection of the Mediterranean Sea against Pollution[186]; the Protocol for the Prevention of Pollution of the Mediterranean Sea by Dumping from Ships and Aircrafts[187], and the Protocol Concerning Co-operation in Combating Pollution of the Mediterranean Sea by Oil and Other Harmful Substances in Cases of Emergency[188]. A number of Resolutions were also approved[189], including one which accepted the proposal for the establishment in Malta of a "regional oil-combating centre" for the Mediterranean staffed by IMCO[190]. In May 1980 another important instrument, the Protocol for the Protection of the Mediterranean Sea against Pollution from Land-Based Sources, was signed by 12 Mediterranean states and the EEC[191].

A fourth protocol on marine parks and protected areas is now at an advanced stage of preparation and is expected to be submitted for intergovernmental approval in 1982. Some progress has also been made in drafting a fifth protocol on pollution associated with sea-bed exploration and exploitation. In effect, then, the Mediterranean states will soon have completed most of thelegal work envisaged by the GFCM ten years earlier.

Despite the successful negotiation of these legal instruments it may be too early to assess UNEP's initiative in the Mediterranean. As a prototype it has to be evaluated not only with a view to its *acceptability*[192] and *efficacy*[193] in the region, but also with a view to its value as a *model*[194] to be followed, subject to appropriate adjustments, in other regions. It may be very nearly impossible to satisfy everyone under any of these widely different criteria. Success under one criterion by no means guarantees success under either of the others[195].

The *acceptability* of the Mediterranean initiative is the least in question. The first Intergovernmental Review Meeting of Mediterranean Coastal States, convened in Monaco in January 1978 to discuss the various components of the 1975 Action Plan, was attended by 17 of the 18 Mediterranean countries. Within two years of the adoption of the 1976 Barcelona Convention and the two related protocols, all three agreements had entered into force with the deposit of instruments of ratification[196]. By mid-1980 a total of 15 states and the EEC had become parties. The record of signatures and ratifications is especially impressive in a region where political antagonisms continue to block most efforts at co-operation[197].

The *efficacy* of the UNEP initiative in the Mediterranean is not, of course, so easily assessed in the short term. Perhaps the only reasonable test in the early years of the Action Plan is the number and diversity of studies, assessments, and monitoring exercises initiated by the states under the Plan; or, more exactly, the number and diversity of such projects that would not have been initiated in any event, in the absence of an Action Plan for the region[198]. Since the later can never be quantified with certainty, it may be necessary to judge the Plan's efficacy solely by reference to the total number and nature of research projects generated, until such time as the findings of these projects can be evaluated and later actions, based on the findings, can be assessed. The best hope so far springs from the acceptance of the Land-Based Protocol. This agreement commits the contracting states to undertake massive measures, which are likely to cost something like $ 15 billion[199]. UNEP believes that a commitment on this scale marks a critical breakthrough in government attitudes, and the measures agreed to seem bound to have a significant remedial effect in the near future[200].

It is also premature to judge the success of UNEP's Mediterranean initiative by the third criterion: as a *model* for initiatives in other regions. The other nine designated regions under the Regional Seas Programme are still at an earlier stage in the formulation and implementation of their Action Plans. In most of these regions questions have been raised about the comparability of the problems encountered and, thus, about the utility of a single model[201]. On the other hand, it is generally conceded to be scientifically unnecessary and economically wasteful to insist on approaching each region *de novo,* as if it were totally unique in every respect. In practice, considerations of institutional convenience and good sense are perhaps likely to -prevail over reservations based on scientific exactitude and political pride. Although there will be many regional variations of approach, and occasional important deviations from the Mediterranean Action Plan elsewhere[202], the original model framework is unlikely to be abandoned.

At the time of writing, the most important test of the Mediterranean initiative lies in *the financial arrangements for implementation. It has always been intended by* UNEP that the financial burden assumed by it in the early stages or formulation should be passed over, soon after adoption of the Action Plan, to the countries of the region[203]. At the Second Intergovernmental Review Meeting, held at Geneva in February 1979, the Mediterranean states agreed on the establishment of a $3.28 million Trust Fund, by which they will provide the major financial support for the regional programme[204]. But this represents only half of the total operational budget for the programme in 1979 and 1980, and since the Mediterranean is one of the more affluent of the ten designated regions, reluctance there to accept full financial

responsibility for its environmental programme may bode ill for UNEP in its efforts elsewhere[205].

c) The Red Sea and Gulf of Aden

This region was the second of the designated ten regions to adopt an Action Plan[206], at Jeddah in January 1976, one year after the Mediterranean Action Plan received approval. Co-ordination of the programme has been entrusted to the Arab League Educational, Cultural and Scientific Organization (ALECSO), and the co-operation of teaching and research institutions at Amman, Aden, Port Sudan, and Ghardaga has been secured[207].

The relatively high degree of cultural and scientific homogeneity within this region and its special vulnerability to pollution[208] may explain the fairly rapid acceptance of the Action Plan. A Draft Regional Convention for the Protection of the Red Sea and Gulf of Aden Environment has been under consideration for some time. A revised draft was presented for governmental approval at an ALECSO-sponsored conference, early in 1981[209].

The potential effectiveness of the Action Plan will be clarified in the years ahead, after the findings of current environmental research have become available as the basis for future management in the region[210]. Special attention has been given to the problems of oil pollution and the need for training courses in Red Sea ecology. Because of its small size and the relatively high degree of ecological and hydrographic homogeneity, the Red Sea and Gulf of Aden region is an obvious candidate for a transnational approach to *coastal zone management.* In this sense, the Red Sea and Gulf of Aden might be regarded as a special opportunity — a model for imitation — outside as well as inside the framework of UNEP's Regional Seas Programme[211].

d) Kuwait Action Plan Region (The Arabian-Persian Gulf)

This region, consisting of eight of the world's richest oil-producing countries[212], adopted its Action Plan in April 1978, at a regional conference of plenipotentiaries held at Kuwait[213]. Since then the region has been referred to officially, within the framework of UNEP's Regional Seas Programme, as the "Kuwait Action Plan Region".

The Action Plan provides for the initiation of various programmes, such as:

(i) studies on the origin and magnitude of oil pollution;
(ii) evaluation of all industrial wastes and municipal sewage reaching the sea;
(iii) stock assessment of commercially important species of fish and shellfish;
(iv) ecological effects of coastal engineering and mining; and
(v) contingency planning for accidents related to oil exploration, exploitation and transport.

At the same conference the eight countries of the region also adopted two anti-pollution agreements[214], set up a regional trust fund[215], established a marine emergency mutual aid center to co-ordinate action by the coastal states against oil spills in the region[216], and approved the proposal for their own Regional Organization for the Protection of the Marine Environment to manage the Action Plan[217]. By the end of June 1979 both the Kuwait Convention and the accompanying Protocol

had entered into force[218], but current hostilities and other political upheavals in the region make it difficult to expect rapid progress in implementing the Plan.

As the wealthiest of the ten designated regions, the Kuwait Action Plan region has perhaps the best opportunity to finance the implementation of the Action Plan on its own, independently of UNEP, and thus to secure a high degree of regional autonomy in the environmental management of the region. By the same token, it may be the least likely to serve as a model for other, less affluent regions to follow. Presumably spin-off benefits are most likely to accrue in the neighbouring region of the Red Sea and Gulf of Aden. As a region undergoing extraordinarily rapid industrialization[219], the Kuwait Action Plan Region may also have some influence on, and be influenced by, similar Action Plan programming in the Mediterranean and East Asian Sea.

e) West Africa

After a number of preparatory activities between 1976 and 1978[220] a draft Action Plan for the West African Region[221] was sent to the appropriate governments in October 1978, and a UNEP mission visited most of these governments[222] early in 1979. These preparations culminated in the convening of a conference of plenipotentiaries at Abidjan in the Republic of the Ivory Coast in May 1981. Invitations were sent to the Governments of all coastal and inland states regarded as belonging to the region, which was described officially as West and Central Africa. The 16 states attending adopted three documents: the Action Plan for the Protection and Development of the Marine Environment and Coastal Areas of the West and Central African Region; the Convention for Co-operation in the Protection and Development of the Marine and Coastal Environment of the West and Central African Region; and the Protocol concerning Co-operation in Combating Pollution in Cases of Emergency. In many respects the Abidjan Convention is the most highly developed of these regional conventions for the protection of the marine environment, having benefited from the experience gained in drafting the earlier Barcelona and Kuwait conventions[223].

Despite a moderately high degree of ecological and hydrographic homogeneity off the northern two-thirds of the west coast of Africa, regional co-operation in the implementation of an Action Plan may prove to be difficult until the ferociously difficult boundary delimitation issues in the Gulf of Guinea have been dealt with[224]. This may be accomplished either through their resolution by means of a regional network of interrelated bilateral boundary agreements, or through a regional administrative arrangement to set these legal and political issues aside, at least temporarily, and to pool resources for more effective environmental management. In a tightly congested marine region such as the Gulf of Guinea, where the various zones of national jurisdiction have much in common with one another, ecologically and hydrographically, the expedient of setting aside their boundary problems may prove tempting to the coastal states[225].

f) Caribbean

Like the Red Sea and Gulf of Aden, the Caribbean is a tropical environment inhabited by small countries with serious problems of resource deficiency[226]. Like the Gulf of Guinea, it is confronted by special difficulties in regional co-operation because of sensitive and difficult boundary delimitation issues in a congested area.

Like the Mediterranean, it is a semi-enclosed sea apparently tailor-made for a holistic, single-unit approach to *environmental management*[227]. *But more than any* of these other regions, the Caribbean is an intricate network of fragile and distinctive, but interlocking, ecosystems. Moreover, the future management of the marine environment of the Caribbean has a particularly intimate relationship with the planning of national economies, and the islands of the region "dare not risk following patterns of development created for continental land masses"[228].

The scientific and institutional complexity of environmental management in the Caribbean has attracted a considerable amount of governmental and non-governmental attention since 1976[229]. Extensive consultations between UNEP, the Economic Commission for Latin America (ECLA), other agencies, and the governments of the region, led to the formulation in 1979 of a Draft Action Plan[230]. At the same time a large number of supporting studies[231] were undertaken and completed in time for submission to the Meeting of Government Nominated Experts to Review the Draft Action Plan for the Wider Caribbean Region held at Caracas, Venezuela, in January/February 1980. As a result of this successful conference a revised Action Plan has been produced[232]. At the samte time a large number of supporting studies[231] were undertaken and comleted in time for submission to the Meeting of Government Nominated Experts to Review the Draft Action Plan for the Wider Caribbean Region held at Caracas, Venezuela, early in 1980. A revised Action Plan received intergovernment approval in April 1981[232]. A regional conventin and protocol are expected to be adopted by the governments in 1982.

With so many countries belonging to the Caribbean region — close to the total number of nations on the African continent (39) — it is a matter of urgency to utilize all existing infrastructures and to follow initiatives proved to be successful elsewhere. For example, the Regional Seas Programme Activity Centre has been given responsibility for co-ordinating marine pollution prevention activities under the proposed Caribbean Action Plan and the Secretariat of the IOC Association for the Caribbean and Adjacent Regions (IOCARIBE)[233] has already assisted in the preparation of a directory of marine research centres, on the model of the Mediterranean prototype[234]. Like the Red Sea and Gulf of Aden, the Caribbean is a region threatened by oil spills, and this problem has received particular attention since 1978[235]. The Action Plan, now adopted, is one of the most thoroughly prepared of all Plans under the Programme, and much is expected of it, since the area covered is a truly regional, semi-enclosed sea.

g) East Asian Seas

No marine region in the world presents more diverse problems of environmental management than East Asia[236]. Gulfs and peninsulas, sluggish estuaries and narrow straits, indented land masses and clusters of islands large and small, all contribute to a physical configuration of unequalled complexity. Peopled by teeming populations of widely varying cultures, ideologies, and socio-economic systems, this region offers a formidable challenge to resource diplomacy and the making of environmental policy. Few would deny that East Asia is the most audacious of UNEP's designations under its Regional Seas Programme.

To reduce the problem to manageable size, it was recommended at an early stage that the region should be divided into a number of "subregions": the Bay of Bengal,

the Strait of Malacca, the Gulf of Thailand, the South China Sea, the Sea of Japan, the Yellow and East China Seas, and the seas of the Eastern Archipelago[237]. Since then developments have been swiftest in the Southeast Asian sector (the South China Sea and adjacent waters)[238], where the Association of South East Asian Nations (ASEAN) has joined UNEP in a number of meetings designed to result in the approval by the five member states[239] of a Draft Action Plan for the Protection and Development of the Marine Environment and Coastal Areas of the Southeast Asian Region[240]. The rate of preparations in the South China Sea (ASEAN) subregion accelerated throughout 1980. In June 1980 UNEP organized at Baguio in the Philippines a Meeting of Experts to Review the (sub-regional) Draft Action Plan for the East Asian Seas. As in the Caribbean, this review meeting, with the benefit of numerous supporting studies[241], prepared a revised draft of the Action Plan[242]. It also recommended a second, combined meeting of government-designated scientific/technical and legal experts to develop further the components of the Action Plan. These preparations resulted in an intergovernmental meeting at Manila in April 1981, where the five ASEAN states offically adopted the Action Plan for the "East Asian Seas"[243].

At the same time, voices are heard for a continuation of the effort to involve a larger number of East Asian countries in UNEP's Regional Seas Programme[244]. For example, a Draft Action Plan for the Bay of Bengal subregion has been initiated[245]. Moreover, outside the UNEP framework the energies of the Economic and Social Commission for Asia and the Pacific (ESCAP)[246] and other agencies[247] are also directed to that end. Significantly, the ESCAP Regional Meeting on the Protection of the Marine Environment and Related Ecosystems in Asia and the Pacific, held at Bangkok in August 1980, was attended not only by the five ASEAN nations but also by ten non-ASEAN states[248].

As in the Caribbean and the Red Sea and Gulf of Aden, a special interest has been taken in combating the dangers of oil pollution in general as well as the special threat to the fragile mangrove ecosystems of the region[249]. As early as April 1976, at the Penang International Workshop on Marine Pollution in East Asian Waters held under the joint auspices of IOC, FAO and UNEP, oil pollution was identified as a first priority concern in the ship-congested area of the Straits of Malacca and Singapore. Since then the three straits states of Indonesia, Malaysia and Singapore have received the approval of IMCO for a vessel traffic control system in the area[250].

h) Southwest Pacific

As recently as 1977[251] it was UNEP's intention to treat the South Pacific as a whole, as a single designated region under the Regional Seas Programme: an area even more extensive than the East Asian Seas. But taking advantage of the intergovernmental infrastructure of the South Pacific Commission (SPC)[252] in the southwestern sector, stretching from Australia as far east as the Tuamotu archipelago, UNEP has separated this sector from its southeastern counterpart and treated them as two distinct regional seas[253].

Discussions on the need for an environmental policy for the Southwest Pacific were begun as early as 1974 by the SPC. Despite extreme difficulties in cultivating intergovernmental co-opeation in such a widely diffused region, a steady effort has been made to prepare working papers[254] for a high governmental level Regional Con-

ference on the Human Environment in the Southwest Pacific Region planned for 1981. With UNEP's support, ESCAP, SPC, and the South Pacific Bureau for economic *Co-operation (SPEC)*[255] *have played leading roles in the preparation of* this conference. It is hoped the participating governments will be prepared to adopt the Action Plan for the region in the near future.

i) Southeast Pacific

This region encompasses the waters adjacent to the western coast of South America, stretching from Panama in the North to the southern tip of Chile. Since this area covers the entire spectrum of hydrographic conditions — from tropical, through subtropical and temperate, to sub-antarctic — it may be questioned whether it lends itself to a single systematic approach to environmental management[256]. Nevertheless, the prospect of a regional action plan has proved attractive to the five governments of the Southeast Pacific[257], and an existing organization in the region, the Permanent Commission for the South Pacific (CPPS), has been ready and willing to play a central role in developing a programme of research and regional agreements[258]. At a workshop held in Santiago in November 1978 government representatives recommended that a number of measures should be adopted. Since then intergovernmental consultations have been held, but present evidence suggests that the Southeast Pacific Action Plan may be less influenced by UNEP than most of the others envisaged under the Regional Seas Programme.

j) Southwest Atlantic

Since this region was added to the Regional Seas Programme as recently as 1980, it is too early to trace any legal or institutional developments. A scientific workshop on the regional ocean environment, organized by IOC, will be held at Montevideo in 1981.

k) East Africa

This region also is too new a designation to have produced any concerted response to the problems of conservation of the ocean environment under UNEP auspices. The fact that UNEP's head office is located in Nairobi may give a special prominence to Action Plan developments in East Africa and adjacent areas of the Indian Ocean.

l) Conclusions

It seems to be five or even ten years too early to evaluate UNEP's Regional Seas Programme from an operational standpoint. It may be appropriate, however, to comment on the programme as a concept and as a framework for action.

Like so many other undertakings of the United Nations bureaucracy, the Programme purports to be a "comprehensive" working out of a "systematic" approach to a multi-faceted (meta) problem. Such an effort is a familiar phenomenon in the recent history of public administration, both at national and international levels. Conceptually, the Programme could be viewed as an audacious combination of four ambitious institutional innovations: a novel maritime approach to *regional integration;* a selective effort to decentralize what is essentially a global undertaking to protect and conserve the *human environment;* an experiment in the promotion of *coastal zone management* through international co-operation; and a brave attempt to com-

bat the potential dangers of excessive nationalism inherent in *economic zone management.*

Each of these four modes of thinking about the Regional Seas Programme is capable of raising excited hopes, at least on the part of the conceptually inclined. But this may be the place to note the reservations of those who refuse to surrender uncritically to the prevailing fashion of mounting comprehensive, systematic, integrated responses to the immensely complex problems of policy-making, government, and administration, which characterize the modern world. Some theorists in public administration, such as Charles Lindblom[259], are prepared to make the case for *incremantalism,* even for "muddling through", as a generally more effective mode of approach to institutional problems of arduous complexity[260]. Without making an *a priori* commitment either way, one might still wish to preserve a degree of scepticism about the "human cost effectiveness" of elaborate undertakings such as the Regional Seas Programme. Even so, putting this reservation aside, how should we assess the Programme's significance in the 1980s?

From the viewpoint of the *regional integrationist,* the excitement of the Programme lies in the hope that, focusing as it does on ocean areas of "common interest", it represents a new and potentially successful approach to the building of "regional communities". After all, it might not be unduly fanciful to suppose that the future of ocean-oriented regionalism has more to show, in terms of human co-operation, than the history of land-bound regionalism. If so, an environmental framework such as UNEP's may be appropriate to the prosecution of such an effort.

As a matter of intention, of course, the Regional Seas Programme attempts to implement the Stockholm Action Plan for the protection and conservation of the *human environment.* Yet, although the Plan made frequent references to the role of regional arrangements in international environment affairs, none of the Stockholm documents attempted to prescribe a regional strategy for UNEP. Indeed, UNEP — or, more properly, its Governing Council — seems to have been given a free hand in determining the need for decentralization in its programming. In this perspective, the Regional Seas Programme should be interesting to compare with the projects of other "co-operating" bodies within the UN family, and even of those bodies entrusted with a more "directive" mandate. If this effort to regionalize the Stockholm movement is generally successful, it may have a significant impact on the development of international environmental law and on the formation of international environmental organizations.

Around the time of the Stockholm experiment in global environmentalism, some nations were attempting a comparably integrative approach at the national level to complex problems of *coastal zone management*[261]. But for some coastal problems, the only feasible national approach may be something less than holistic, or "comprehensive" or "systematic". An integrative approach to some of these problems may be possible or desirable only at the international level, especially in the case of small countries located in a tightly congested region whose marine environment is characterized by intricate ecological interdependencies, such as the Mediterranean, the Caribbean, the Gulf of Guinea, and other partially enclosed seas. Even though the Regional Seas Programme is not officially presented as a response to this kind of problem, it overlaps sufficiently with the environmental problems

special to the land-sea interface to be judged, at least partially, from this vantage point. In most regions it is now recognized that the treatment of marine pollution *rests above all on achieving effective control over the land-based sources of the problem,* and that progress in this sector at home may depend on the existence of a well-conceived, regionally appropriate, "framework" of ideas and intentions.

Furthermore, it should be kept in mind that the years of evolution for UNEP's Regional Seas Programme will also be the years of national efforts to develop *ocean management* programmes within extended limits of coastal jurisdiction in the wake of UNCLOS III[262]. There is a legitimate concern that many coastal states will be unable to mount effective national programmes on their own initiative, that their best efforts to discharge their environmental responsibilities within their exclusive economic zone will be defeated by the very extensiveness of these new jurisdictional limits. These concerns are matched by fears that some coastal states may be moved by nationalist sentiment to use boundary and other resource issues to obstruct efforts to produce environmental co-operation in the region[263]. But the Regional Seas Programme itself may be the best evidence in support of the more optimistic view that the possession of a 200-mile economic zone will, sooner or later, force most coastal states to develop environmental management measures, and that the best way of doing so will be found to lie in regional consultation and co-operation, especially for states sharing a semi-enclosed sea or a similar geographical configuration.

Finally, how should we interpret the significance of the designations applied within the Programme? Of the ten designated regional seas, none is a natural ocean basin, but two (the Southwest and Southeast Pacific Regions) together are approximately co-extensive with an ocean basin (the South Pacific)[264]. Five of the remaining eight are either semi-enclosed seas (the Mediterranean and the Arabian-Persian Gulf) or consist mostly of a semi-enclosed sea (West African Region, including the Gulf of Guinea), or consist of two neighbouring semi-enclosed seas (the Caribbean and the Red Sea and Gulf of Aden)[265]. What, one might ask, is a sensible "unit" for managing the ocean environment? The tentative answer might be put in negative terms:

(i) an area that is not too extensive;
(ii) an area that is not too ecologically heterogenous; and
(iii) an area that is not too politically inactive, competitive, or conflict-prone to discharge its environmental responsibilities effectively on a collective and co-operative basis.

Judged by these three criteria (given equal weighting), the Mediterranean seems the "safest" designation, and despite continuing difficulties and frustrations-commensurate with the scale of the undertaking, the Mediterranean Action Plan must already be regarded as an impressive experiment in regional co-opeation. Each of the other nine designated regions seems to present a more formidable set of obstacles to overcome before even a qualified judgment can be passed on the relevant legal and institutional developments. "Success" is not, of course, achieved by the mere approval of an action plan, or even by the adoption of a legally binding, but generally worded, convention. Management of the ocean environment, despite its appearance as a matter of common interest or mutual convenience, may prove to be merely another forum for symbolic diplomacy. Cosmetics and substance will have

to be distinguished with care in the 1980s as regional responses to UNEP's Regional Seas Progamme unfold.

3. Other UN Regional Programmes Concerned with the Protection and Conservation of the Marine Environment

a) FAO

The Food and Agriculture Organization was the first specialized agency to be established under the United Nations[266]. Since the beginning the FAO's Fisheries Division has had to contend with a lower status, and a much smaller budgetary allocation, than the other FAO Divisions[267]. Throughout its history the Fisheries Division (later Fisheries Department) has been frustrated, and at times demoralized, by its inability under these conditions to discharge effectively its broad responsibilities around the world. Faced with the need to operate within a system of strict priorities, the Division has been forced to spend most of its inadequate resources on helping national governments with their immediate problems of fishery development. As a result, too little has been available to spend on conservation, management, and research — matters which might be regarded as its basic and long-range responsibilities.

In August 1947, at its third sesson, the FAO Conference recommended that FAO establish regional bodies for scientific exploration of fishery resources. Seven marine regions were put forward as candidates for such bodies: (i) the Northwest Atlantic; (ii) the Southwest Pacific and Indian Ocan; (iii) the Mediterranean Sea and contiguous waters; (iv) the Northeast Pacific; (v) the Southeast Pacific; (vi) the Western South Atlantic; and (vii) the Eastern South Atlantic and Indian Ocean. The boundaries of these regions were to be defined not by FAO but by the interested governments which chose to respond to this recommendation[268]. Interested governments, it was hoped, would wish to accept much of the responsibility for conservation, management and research in their region, and for the financing of these regional bodies.

Of the first three of the designated regions which had been given initial priority, the Northwest-Atlantic, the most affluent of them , chose to establish the International Commission for the Northwest Atlantic Fisheries — as an independent regulatory commission outside the framework of the FAO[269]; the Southwest Pacific and Indian Ocean responded by supporting the establishment in 1948 of the Indo-Pacific Fisheries Council[270]; and the Mediterranean region followed suit in the following year by supporting the creation of the General Fisheries Council for the Mediterranean[271]. But efforts by FAO to establish similar bodies in the other regions ran into greater difficulties in the 1950s. An attempt to establish a Latin-American Fisheries Council in 1951 was unsuccessful, because the Agreement failed to obtain the required ratifications[272]. The regionalization programme was suspected of several deficiencies: a tendency to duplicate the work of existing organizations in some regions[273]; an overambitious experiment in other regions still lacking the technical capability to support such a body[274]; a trend that would weaken the overall effort of the headquarters in Rome[275]; and most basic of all, a luxury that the Fisheries Division simply could not afford[276]. All four criticisms had some weight, and FAO's regional fishery activities — and indeed its fishery activities in general — continued to be plagued with difficulties into the 1960s[277].

The overall weakness of the Fisheries Division was finally recognized by the FAO Conference in 1963, when at its 12th Session it urged the Director-General to "ensure that FAO, through its Fisheries Division, has in future years the status of being the leading intergovernmental body in encouraging rational harvesting of food from the oceans and inland waters, bearing in mind the dynamic relationship between the living aquatic resources and the environment, and also bearing in mind the importance of fisheries in providing needed animal protein"[278]. The Conference also authorized the Director-General to strengthen the Fisheries Division in 1964-1965, "within the limits of the funds available", in designated ways; and it endorsed the Director-General's proposals for the engagement of consultants to assist regional fishery commissions and for the outposting of more technical staff to the regions[279].

This important policy decision resulted in the creation in 1966 of the permanent Committee on Fisheries (COFI)[280] and in the reorganization and upgrading of the Fisheries Division to the status of Department[2*1]. This was intended to be merely the first step in the expansion of FAO's fishery activities, but insufficiency of funds once again intervened, slowing down the expansion effort[282]. By the late 1960s, however, it was generally recognized that a higher priority must be given to FAO's regional fishery activities[2113], and by 1967 three more regional fishery commissions had been added to the family: the Regional Fisheries Advisory Commission for the South West Atlantic (1962)[284]; the Indian Ocean Fishery Commission (1967)[285]; and the Fishery Committee for the Eastern Central Atlantic (1967)[286]. Moreover, FAO was attempting to establish international consultative and coordinating arrangements for other regions (e.g. Caribbean, Gulf of Mexico, and the Persian-Arabian Gulf)[287]; and COFI played an important role in promoting the establishment of independent commissions outside the FAO framework (e.g. the International Commission for the Conservation of Atlantic Tunas in 1966[288] and the International Commission for the Southeast Atlantic Fisheries in 1969)[289].

In the 1970s the FAO Fisheries Department gradually began to enjoy some benefits accruing from its higher priority and status within the Organization. With important jurisdictional changes in prospect as a result of Seabed Committee discussions and UNCLOS III negotiations, the member states attending the biennial Conference encouraged the Department into greater efforts. Most of these efforts were directed at fishery development rather than stock conservation, but the two objectives came together in the Comprehensive Programme of Assistance in the Development and Management of Fisheries in Exclusive Economic Zones, which was approved at the 20th Session of the FAO Conference in November 1979[290]. Four years earlier the Conference had called for a strengthening of FAO's regional fishery commissions, and agreed that they should have "increasing responsibility for stock assessment and management and also be the forum for working out development strategies suited to each region"[291]. It is significant, then, that the Programme of Assistance is to be executed in a decentralized manner, through the existing network of regional fishery bodies and associated field programmes, which will be "adjusted in their structure and reoriented in their activities to reflect the needs of groups of states sharing fish stocks or fisheries, common problems or opportunities, and other natural affinities"[292]. Under this new approach to the structuring of international fishery conservation and management, existing regional commissions would be encouraged to establish subregional committees, whose membership would corres-

pond to all the littoral states belonging to a designated "natural management area"[293]. Already the South China Sea, the Bay of Bengal, and the Southwest Indian Ocean have been so designated, and corresponding committees have been established within the Indo-Pacific Fisheries Council (for the South China Sea) and the Indian Ocean Fisheries Commission (for the other two)[294].

To all appearances, then, FAO has embarked upon a new, and perhaps more hopeful, venture in the promotion and co-ordination of regional fishery commissions.

b) ECOSOC

The 54-member Economic and Social Council, one of the principal organs under the General Assembly, "co-ordinates the economic and social work of the United Nations and the specialized agencies and institutions — known as the 'United Nations family' of organizations"[295]. Because ECOSOC makes recommendations and initiates activities covering a broad spectrum of economic, social, cultural and humanitarian questions, its subsidiary bodies include six functional commissions[296], five regional commissions[297] and seven standing bodies[298]. ECOSOC not only oversees 14 other related UN bodies[299], but also co-ordinates the economic and social activities of the entire UN family, including 17 related intergovernmental agencies[300].

i) Restructuring of the economic and social sectors of the United Nations system[301]

For more than a decade, senior UN officials have recognized the problems of coherence and efficiency arising from the proliferation and variety of its institutional mechanisms. Acknowledging the relevancy of these problems to the issues of economic and social development, as perceived by the new member states and as reflected in the historical imbalances in the pattern of the world economy, ECOSOC directed in 1973 that a review of the terms of reference of its subsidiary bodies should be undertaken[302]. The next year, after the UN General Assembly's "Declaration and Programme of Action on the Establishment of a New International Economic Order"[303], ECOSOC requested its regional commissions to adjust their programme of work and activities in response to this new direction[304].

In 1975, the UN General Assembly established an *Ad Hoc* Committee on the Restructuring of the Economic and Social Sector of the United Nations System[305]. After two years of general debate open to all states, in which almost all of the executive heads of UN organizations and organs as well as senior officials of the UN Secretariat participated, the UN General Assembly endorsed the *Ad Hoc* Committee's conclusions and recommendations by adopting resolution 32/197 on 20 December 1977[306], which outlined the major transfers of functional elements and redeployment of staff resources, particularly to the regional commissions. The next year, the UN General Assembly decided "that the regional commissions shall have the status of executing agencies, in their own right"[307], for intersectoral, subregional, regional and interregional projects beyond the sectoral responsibilities of specialized agencies and other UN bodies. Thus, the UN General Assembly has launched not only the restructuring of the economic and social sectors of the UN system but also the decentralization of economic and social activities to, and strengthening of, the regional commissions[308].

The restructuring, decentralization and strengthening of regional programmes concerned with the marine environment have had consequences within the UN Secretariat itself. The Ocean Economics and Technology Office (OETO)[309], whose activities "relate mainly to research and analysis in regard to the integrated development of coastal areas and marine resouces ..."[310], was retained in the new Department of International Economic and Social Affairs (DIESA), but its services were made available to the other new Department of Technical Co-operation for Development (DTCD) as required.

In 1979, ECOSOC invited the Secretary-General "to pursue actively the process of decentralization to the regional commissions ..." and "to expedite the decentralization of activities, including the redeployment of the necessary and appropriate resource ..."[311]. Based on agreements of the Executive Secretaries of the Regional Commissions on the necessity and priorities for decentralization, the Secretary-General proposed a redeployment of five positions from DIESA, one to each of the regional commissons[312], and the transfer of resources from DTCD for the employment of five additional regional advisers to be used in highly specialized areas[313]. Although the effect of these proposals on decentralization does not directly concern the marine environment, the impact of one of the Secretary-General's proposals relating to the strengthening of regional commissions does involve OETO: namely, in the matter of technical assistance to the Economic Commission for West Asia (ECWA)[314]. Thus, these proposals "constitute the first phase of decentralization and strengthening"[315] of regional programmes according to regionally-determined priorities.

ii) Regional commissions

The five regional commissions — serving Europe, Asia and the Pacific, Latin America, Africa, and Western Asia — were established by the Economic and Social Council "in recognition of the fact that many economic problems are best dealt with at the regional level"[316].

(a) ECE

The 34-member[317] Economic Commission for Europe, established in 1947, operates from its headquarters in Geneva. One of ECE's four priority objectives is improvement of the environment. "While all ECE principal subsidiary bodies are studying environmental problems arising from industrial or other activities in their sectors, the Senior Advisers on Environmental Problems consider the broad policy aspects of environmental questions and their interrelationships"[318]. Although this group examined the problems of transboundary water pollution and the protection of flora and fauna and their habitats[319] in preparation for a high level meeting[320], these two topics were later excluded from the agenda because preparations were not sufficiently advanced. However, work on these two areas is continuing. A revision of the ECE Declaration of policy on the prevention and control of water pollution was prepared jointly with the Committee on Water Problems[321]. Also, an ECE Group of Experts on Aspects of Water Quality and Quantity, as a follow-up to the seminar on the protection of coastal waters against pollution from land-based sources, held in Lisbon in 1975, was charged with implementation of the project on protection of the marine environment from pollutants emanating from land-based sources[322]. This Group of Experts was also charged with compiling a synoptic list of existing conventions and international agreements between ECE countries concerning water

resources[323]. Thus, ECE, whose socio-economic region spans three continents[324], has made a commitment to the protection and conservation of not only the marine environment but other aquatic environments as well.

(b) ESCAP

The 34-member[325] Economic and Social Commission for Asia and the Pacific, also established in 1947, serves a region that contains more than half the world's population from its headquarters in Bangkok. Of the important initiatives undertaken by ESCAP to stimulate co-operative approaches to regional marine-related problems, three activities must be noted: (i) the Interim Committee for Co-ordination of Investigations of the Lower Mekong Basin (Interim Mekong Committee)[326]; (ii) the Committee for Co-ordination of Joint Prospecting for Mineral Resources in Asian Offshore Areas (CCOP)[327]; and (iii) the Committee for Co-ordination of Joint Prospecting for Mineral Resources in South Pacific Offshore Areas (CCOP-SOPAC)[328].

Supported by 25 countries, 18 UN agencies and related organizations, four foundations and many private institutions and businesses[329], the original Mekong Committee's[330] work programme for 1977 included a three-part fisheries programme[331] involving eight projects[332]. However, because of the political uncertainties in that region, some of these projects have been held in abeyance. For the same reason, a Committee proposal to UNDP in 1974 for assistance in a two-year environmental health planning project has also been held in abeyance[333].

CCOP has recognized that many of its member countries[334] have attached increasing importance to marine environmental problems related to offshore petroleum and mineral development. CCOP has earmarked UNEP funds to engage consultants who will advise and assist the ASEAN Council on Petroleum (ASCOPE) in its project on "Mitigation of Marine Pollution and Environmental Protection", by initially developing interim environmental protection regulations with respect to offshore petroleum exploration and production in several of the member countries[335]. CCOP has also endorsed other recommendations of its Technical Advisory Group (TAG) on environmental problems encountered in the CCOP region[336].

A link between fundamental scientific research and exploration of mineral deposits was forged in 1973 by a programme, under the joint sponsorship of CCOP and the Intergovernmental Oceanographic Commission (IOC), "to undertake investigation on tectonic structure and history of East Asia and western Pacific with metalliferous ore and hydrocarbon genesis" as part of the International Decade of Ocean Exploration programme (IDOE). Subsequently, a joint CCOP-IOC Working Group[337] was established to facilitate implementation of the programme and to promote co-operation among CCOP Member Countries as well as with other states of IOC"[338]. Although this CCOP/SEATAR Programme was scheduled to terminate at the end of IDOE (1971-80), CCOP has proposed a post-IDOE CCOP-IOC Joint Working Group with a new name and terms of reference to come into force in 1981.[339].

CCOP-SOPAC, on the other hand, apparently has not yet initiated programmes which focus on protection of the marine environment of its region. Given (a) the fact that offshore exploration for petroleum and mineral resources has not progressed as fast or as far as it has in the CCOP region; (b) the perennial problems of obtaining funds; and (c) the ESCAP-familial relation between CCOP-SOPAC and CCOP, it may

be that CCOP-SOPAC has either temporarily deferred such expensive programmes until funds are available, or wisely avoided needless duplication of programmes already conducted by CCOP.

As evidence of ESCAP's commitment to seek environmentally sound strategies, it established an Environmental Co-ordinating Unit (ECU) in its secretariat[340]. This unit has followed up on the "Task Force on the Human Environment" project[341] and conducted the 1978 ESCAP/UNEP Intergovernmental Meeting on Environmental Protection Legislation[342].

In outlining the role of the environment in development planning in the 1980s, ESCAP argues that "there are limits to what can be achieved by national action alone ... Ultimately, what is required is some form of global authority to measure the stock and changes in world natural resources, review the basic ecological cycles whereby the flow of renewable resources is maintained and establish broad management rules which would ensure an environmentally sound global use of resources. Such global measures will take time to develop and win acceptance, however, and there is much which can be done at the regional or sub-regional level to promote environmentally sound practices"[343].

For the 1980-81 biennium, ESCAP noted four priority areas: "(a) creation of environmental awareness within the region; (b) development and/or strengthening of the institutional and legal framework for environmental protection and management at the national level; (c) environmental management of land ecosystems, ...; and (d) *protection of the marine environment and related ecosystems"* [emphasis added][344].

Finally, "[i] n the ESCAP region, moreover, with the main exception of ASEAN, there is a lower propensity for forming subregional economic groupings than in other regions, in part because of a lack of homogeneity among countries ... Nevertheless, the 1970s have witnessed a growing sense of willingness to co-operate among developing countries of the ESCAP region, as manifested by the many new bodies that have been created to foster specific aspects of technical co-operation..."[345].

(c) ECLA (in Spanish CEPAL)

The 32-member[346] Economic Commission for Latin American established in 1948, assists Latin American governments to promote the economic development of their countries from its headquarters in Santiago. Its Natural Resources and Environment Division has surveyed the main environmental problems confronting the region and the proposed solutions.

The focal area of marine environmental concern is the Caribbean region. In conjunction with UNEP's Regional Seas Programme, ECLA has substantially supported the Action Plan for the management of the environment in that region. To avoid institutional duplication in that area and insure greater harmonization of UN system activities at the subregional level, the Caribbean Development and Co-operation Committee was established in 1975 and has as its Secretariat the ECLA Office for the Caribbean at Port of Spain. Moreover, in 1977, ECLA convened the Second Seminar on the Economy of the Oceans at Buenos Aires.

These and other recent activities seem to reflect ECLA's commitment toward the preservation of the marine environment primarily in the Caribbean region.

(d) ECA

The 51-member[350] Economic Commission for Africa, established in 1958, promotes the modernization of Africa from its headquarters in Addis Ababa. Since 1977, ECA has launched within its five subregions Multinational Programming and Operational Centres (MULPOCs) for "indentifying and formulating multidisciplinary and multinational projects"[347]. Although ECA's conference of Ministers recommended that operational activities and resources of the commission should be decentralized "in line with the spirit behind the establishment of (MULPOCs)"[348], they reaffirmed "the need for member states, individually and collectively, to move expeditiously, in the development, management and conservation of their resources of the sea; ... (and requested) adequate provisions for the staffing of the Resources of the Sea Unit at (ECA) headquarters..."[349].

In conjunction with UNESCO, the ECA secretariat "has prepared a project on the development of marine science and technology in Africa. The project has three principal objectives, namely, to enable the maritime states in East and West Africa better to explore and exploit their marine resources, to develop indigenous manpower in these countries in marine science and technology research, and to promote co-operation among the countries concerned in the development and utilization of their marine resources"[350].

As far as protection of the marine environment is concerned, other than the three UNEP regional seas programmes surrounding the ECA region — Mediterranean, Red Sea and Gulf of Aden, and West Africa (Gulf of Guinea) — ECA has not yet initiated any other regional or subregional programme. However, in 1978, the ECA Executive Secretary established the Environmental Co-ordination Office in his cabinet and abolished the Environment Unit in the Natural Resources Division[351]. Although the impact of this move on regional concerns for the marine environment is unclear, with adequate staffing, this office and the Resources of the Sea Unit would provide a nucleus for further efforts in conserving and protecting the marine environment of the ECA region.

(e) ECWA

The 14-member[352] Economic Commission for Western Asia, established in 1973 by integrating the United Nations Economic and Social Office in Beirut (UNESOB), covers most of the Middle East from its new permanent headquarters in Baghdad[353]. ECWA has concluded that "technical and legal frameworks for the development of marine resources integrating the marine and coastal area ecosytems are highly significant aspects of ocean economics at the national, regional, and international levels. Most of ECWA member States are bordering small but internationally important seas: The Gulf, the Red Sea, and the Mediterranean. In view of this it is important that legal, environmental and economic safeguards be established against exploitation and, at the same time, to assist member States of ECWA to achieve maximum benefits in utilizing the resources of these seas"[354]. To achieve these objectives, ECWA has proposed a two-part activity[355] to be implemented in the biennium 1980-81 in collaboration with UN headquarters (DIESA/OETO), WHO, FAO, UNESCO, and UNEP[356]. Guidelines will be prepared ... tailored to regional requirements on management of the resources of the sea taking into consideration social, economic, and environmental impacts of development and outlining a regional

strategy in this connection"[357]. This ECWA project would satisfy UNEP's concept of eco-development because it clearly stresses a "holistic" rather than "sectoral" planning approach.

c) UNESCO

The 145-member[358] United Nations Educational, Scientific and *Cultural Organization,* one of the UN specialized agencies, was formed in 1946 "to contribute to peace and security in the world by promoting collaboration among nations through education, science, culture and communication ..."[359].

i) Division of Marine Sciences (SC/OCE)

Within the UNESCO Secretariat in Paris, this Division primarily concentrates on development of: (a) the scientific basis for the understanding of marine (especially coastal) systems[360]; and (b) national and regional infrastructures in the marine sciences[361]. This division: (a) disseminates knowledge in the marine sciences through marine information services[362], publications[363], and the International Oceanographic Data Exchange (IODE)[364]; supports the training and education of specialists in marine sciences through SC/OCE fellowships, training courses, course material and technical advice; and through IOC Training, Education and Mutual Assistance in Marine Sciences (TEMA) activities[365].

ii) IOC

Through the concerted action of its member States[366], the Intergovernmental Oceanographic Commission (IOC) primarily promotes scientific investigations of the nature and resources of the ocean by means of global[367] and regional[368] ocean science and ocean service activities. In the field of marine pollution monitoring, areas in which activities are being carried out or have been proposed within the framework of the Global Investigation of Pollution in the Marine Environment (GIPME), are the Mediterranean Sea[369], Caribbean and adjacent regions[370], East Asian waters[371], South-east Pacific[372], South-east Atlantic/Gulf of Guinea Region[373], Western Pacific[374], and North and Central Western Indian Ocean[375]. The Working Committee for GIPME also planned two regional marine pollution workshops during 1979[376].

IOC's transition to regionalization began in 1976 when it terminated the Co-operative Investigations of the Caribbean and Adjacent Regions (CICAR)[377] and concurrently established IOCARIBE "for the purpose of continuing and developing regional co-operation in marine sciences ..."[378]. This first IOC regional association has its own secretariat and newsletter, and conducts its own regional workshops[379].

The second regional association (although only designated as a Working Group) is Western Pacific (WESTPAC) which emerged from the Co-operative Study of the Kuroshio and adjacent regions (CSK) which IOC terminated in 1979[380]. A full-time WESTPAC secretariat was also planned for its region.

The latest regional association (again not yet so designated) is Co-operative Investigation of Central and Western Indian Ocean (CINCWIO) also established in 1979 and scheduled to get its own regional secretariat in 1980 to help member countries in setting up national research laboratories[381].

Thus, the trend toward regionalization in IOC is expected to continue not only under the impetus of restructuring of the economic and social sectors of the UN system, but also at the insistence of member states participating in other IOC regional co-operative investigations in the Mediterranean, South-east Atlantic and South-east Pacific. These IOC regional associations, even if formed in all regions presently being investigated co-operatively, are not yet seen as rivals of UNEP's Regional Seas Programmes. In fact, at the 11th IOC Assembly, a resolution[382] was passed reaffirming IOC/UNEP co-operation under the Regional Seas Programme. However, it is suggested that these IOC regional associations, because they are based on good intentions of the member states to investigate the marine environment of their regions, rather than on formal treaty agreements to preserve and protect them, may become alternative regional programmes to UNEP's Regional Seas Programme.

d) IMCO

The 111-member Intergovernmental Maritime Consultative Organization, another of the UN specialized agencies, was formed in 1958 to facilitate "intergovernmental co-operation and exchange of information on technical matters affecting international shipping, and [to achieve] the highest practicable standards of maritime safety and efficient navigation, with special emphasis on safety of life at sea and on the control and prevention of marine pollution"[383].

IMCO's Marine Environment Protection Committee has a global responsibility under the 1973 International Convention for the Prevention of Pollution by Ships[384] to develop "guidelines on port reception facilities for oil waste, information on oil discharge monitoring and control equipment, revision of recommended performance specifications for oily-water separating equipment, and a manual on methods of dealing with oil pollution"[385].

The first of its regional activities is the regional centre in Malta, established in 1976 with UNEP, to facilitate co-operation among Mediterranean states in cases of massive oil spillages, and to assist those states in developing anti-pollution capabilities[386]. This may be the forerunner of similar regional centres in other enclosed or semi-enclosed seas, which may or may not be part of UNEP's Regional Seas Programme. Possible locations for such IMCO regional centres may coincide with some of the more than 70 congested shipping areas such as the Straits of Malacca and Singapore where IMCO has developed traffic separation schemes (TSS) for safer navigation. These areas also pose a greater potential for collision or grounding and resultant oil spillage.

In 1976 to strengthen its technical co-operation role with Asia and the Pacific, IMCO based one regional adviser at Bangkok, and another at Jakarta. It also opened a regional office for Latin America at Rio de Janeiro[387]. Technical assistance provided by IMCO included "aid in the prevention and control of marine pollution, the training of maritime personnel and the drafting of maritime legislation"[388].

IMCO has been designated as the secretariat for the 1972 London Ocean Dumping Convention[389]. As such it is responsible for assuring that regional agreements on prevention of pollution from ships and aircraft contain criteria and standards for ocean dumping of hazardous and other toxic materials which are not less stringent than those in the annexes of the London Dumping Convention[390].

Finally, through two inter-agency co-ordinating groups, IMCO is indirectly involved in activities affecting the investigation and preservation of regional marine environments conducted by the other agencies. One is the Intersecretariat Committee on Scientific Programmes Relating to Oceanography (ICSPRO)[391]; the other is the Group of Experts on the Scientific Aspects of Marine Pollution (GESAMP)[392]. Therefore, through linkage with both groups, IMCO has contributed more to conservation and protection of regional marine environments than would be expected from its original mission to facilitate safety of navigation.

e) UNDP

The United Nations Development Programme was created in 1965 through the merger of two funds[393] to assist developing countries in building long-term, self-sustained development capacities[394]. UNDP, governed by a 48-member council elected by ECOSOC from member states, specialized agencies or IAEA"[395], works closely with governments and over 24 international agencies in support of some 8000 development assistance projects in Africa, Asia, Latin America and parts of Europe[396].

During the period of March 1974 through August 1978, the following data extracted from a UNDP project summary"[397] for that period reveales some interesting statistics concerning projects approved for conservation, protection and development related to the marine environment.

TYPE/LEVEL	NATIONAL	REGIONAL	INTERREGIONAL	GLOBAL	TOTAL
Total Projects	1593	213	27	4	1847
Marine-related	45(2.8%)	13(6.1%)	7(25.9%)	0	65(3.5%)
Conservation	(0)	(0)	(0)	(0)	(0)
Protection	(1)	(0)	(0)	(0)	(1)
Development	(44)	(13)	(7)	(0)	(64)

Marine-related projects comprised 3.5 percent of all projects approved for funding. When these projects are compared to each other, the ratio is 45 national to 20 regional and interregional combined, but when compared proportionately, more marine-related projects were approved at the latter levels (6.1 percent regional, and 25.9 percent interregional) than were approved at the former level (2.8 percent national). Without knowing the total number of marine-related projects submitted for approval, it would be hazardous to reach any conclusions about these data. However, it may be significant to note that of the marine-related regional projects seven out of 13 were approved for the ESCAP region, four out of 13 were approved for the ECLA region, and one out of 13 for each of the ECE and ECA regions. This is not inconsistent with earlier discussions of the extent of marine-related activities in those regions.

It should be pointed out that, "|a|s an executing agency of UNDP and as and operational arm of the United Nations, executing projects financed from extra-budgeting sources and the regular programme of technical co-operation, TCD (in UN headquarters) provides technical expertise in the formulation, implementation and evaluation of country and inter-country programmes and of specific projects[398]. In keeping with General Assembly resolutions on restructuring (32/197) and on decentralization of activities (32/202), meetings were held between TCD and representatives of the regional commissions, with a view to establishing criteria and a

timetable for further decentralization to the latter of management responsibility for regional and subregional projects"[399]. Although UNDP primarily functions as a conduit for project financing between contributors and recipients, the executing agency responsibility for regional marine-related projects is gradually changing from UN (TCD) to the regional economic commissions for UNDP-funded projects[400].

f) UN Regional Training Programmes

i) UNU

The United Nations University, with its headquarters in Tokyo, has neither a campus nor any faculty. Instead it has to rely on co-operating institutions to furnish both to students who are trained and perform research on UNU-directed programmes[401]. One UNU programme is on the Use and Management of Natural Resources (NRP) which began in 1977. It focuses on problems including "environmental deterioration and mismanagement of natural resources in the humid tropics"[402]. One of the more specific areas of interest to UNU is water-land interactive systems. Using the facilities of Bogor (Indonesia) Agricultural University, UNU students are concentrating their research on brackish water fish ponds *(tambaks)* as a resource system. Closely related to this is an emerging project on coastal zones[403].

Because almost 90 percent of the world's demand for marine protein is derived from coastal areas, and more than half of the world's potentially recoverable hydrocarbon resources lie beneath coastal waters, in 1978 UNU launched a coastal studies project to be done at the Indonesian Institute of Oceanology[404]. This took on a marine regional implication because the students are young scientists from Southeast Asia who spent a year with Indonesian and other experts in preparing individual theses and contributing to a group project. The results of their project was scheduled for presentation to government decision-makers at a regional conference[405].

ii) ILO

The International Labor Organization, one of the oldest members of the UN[406], has been assisting states in "establishing, developing or improving their maritime industries in practical implementation of international maritime labor standards through such means as ... regional and interregional training courses and seminars"[407]. Although these activities are not directly related to marine regionalism, they contribute to the development of more and better qualified mariners who would be directly involved in conservation or protection of their national or regional marine environments.

iii) WHO

The World Health Organization, established in 1948, provides assistance to member states in all aspects of water supply, water pollution control and waste disposal, including some aspects of marine pollution"[408]. During 1970-73, WHO, in collaboration with the government of Denmark, conducted Inter-Regional Training Courses on Coastal Water Pollution Control for personnel from developing countries[409].

iv) IAEA

The International Atomic Energy Agency, from its headquarters in Vienna, has offered regional and inter-regional training courses and study group meetings; granted research contracts on marine science and scientific aspects of pollution;

and conducted special missions on siting of nuclear facilities on rivers and sea coasts and in the offshore[410]. Thus IAEA, like WHO and ILO, has contributed to marine regionalism by educating those involved with the marine environment.

4. Other Regional Arrangements for the Protection and Conservation of the Marine Environment

a) Introduction

The years since 1973 have also seen the growth of many regional agreements and organizations outside the framework of the United Nations. Even those regional developments concerned directly or indirectly with the protection and conservation of the marine environment are quite numerous. The major agreements and organizations of this kind have already been identified in this study, but in the final part of this section it may be useful to look at some of these non-UN regional arrangements in order to group the diversity of recent and current developments at the regional level.

For this purpose it is not necessary to canvass all regions of the world. To keep this study within monographic proportions, it seems best to look only at three distinct, and widely scattered, regions: the Baltic Sea, Asia and the Pacific, and the Antarctic.

b) Baltic

This semi-enclosed sea is one of the largest bodies of brackish water in the world[411]. Because of the Baltic's peculiar hydrography through the Belts and Sound[412], which "has an adverse effect upon the living resources which have not entirely adapted themselves to their brackish environment"[413], and also because of the "roughly threefold increase in phosphate content in the deeper layers of the Baltic in the post-World War II period ..., the delicate ecological balance of the Baltic, otherwise strained by the unique features of that sea, has been further disturbed by man"[414]. This disturbance, primarily from land-based sources — phosphate nutrients, nitrates, other harmful and toxic substances, wastes deliberately dumped from vessels, and oil spills[415] — "has contributed heavily to the deterioration of the environmental quality of the Baltic Sea area"[416].

i) Conservation of marine species

Since 1925, there have been a series of co-operative efforts[417] to promote fishery conservation and management in the Baltic and the Belts. The regional organizations to accomplish similar purposes, however, were not necessarily created by these regional agreements. For example, the International Council for the Exploration of the Sea (ICES)[418], through its Working Groups for Baltic Bottom Fish and Baltic Pelagic Fish, have accomplished much of the research and conservation work of interest to all Baltic coastal states[419].

However, the current regional organization responsible for conserving this subregional marine environment is the International Baltic Sea Fishery Commission (IBSFC) established by the 1973 Baltic Convention on Fishing and Conservation of Living Resources[420]. As is the case for most regional fishery bodies, IBSFC has no mandate, nor is it equipped to carry out direct scientific investigations with its own staff. Instead, ICES will serve as the advisory body on scientific matters[421].

ii) Prevention and control of marine pollution

Unlike the Baltic Convention on Fishing and Conservation of Living Resources which came into force within two years of signing, the Convention on the Protection of the Marine Environment of the Baltic Sea Area[422] came into force after a seven-year delay[423]. "Even before it officially came into force, it had been hailed as the main reason the Baltic had become cleaner over the past six years. Since 1974, the seven Baltic nations have banned almost totally the use of DDT and prohibited the dumping of waste into the sea"[424].

The Baltic Marine Environment Protection Commission (BMEPC), at its first meeting immediately following the coming into force of the so-called Baltic Protection Convention, recommended that beginning in 1981, "all ships over 20,000 tons, primarily oil tankers, and ships over 1600 tons carrying chemicals will be required to report as they enter the Baltic Sea and to notify authorities at designated ground stations of their position at regular intervals"[425].

At the same meeting, BMEPC "also agreed to start work on banning mercury and polychlorinated biphenyls (PCB).... (and) were further reported ready to establish exact norms limiting the discharge of land-based toxic substances"[426].

Moreover, BMEPC assigned the task of monitoring a particular hazardous substance to each country (e.g. Denmark, PCBs; USSR, mercury; Poland, DDT; and West Germany, lead). However, because the Eastern countries, particularly the USSR, agreed only to the use of "general definitions, which preserved national control over coastal waters", each country will monitor its territorial sea[427]. Although BMEPC's enforcement powers are limited, it "is intended to provide a permanent center of information and co-ordination for the parties to the treaty" — something it has done as an interim commission since 1974[428].

In one of the shallow outlets between Denmark and Sweden to the North Sea — the Sound (Oresund) — bilateral co-operation began as early as 1955. In 1960 these nations established the joint Sound Water Committee[429]. The 1974 Oresund Agreement[430] on the protection of the Sound from land-based pollution launched a five-year programme of cleaning up waste reaching the Sound under a Danish-Swedish Commission[431].

In other parts of the Baltic, bilateral arrangements continued to be formed. For example, "in 1968, co-operation to combat pollution in the Gulf of Finland had started between Finland and the USSR[432]. Four years later, a joint Finnish-Swedish Gulf of Bothnia Committee was set up" to co-ordinate water quality research programmes"[433].

Finally, although other Baltic regional and subregional arrangements have been concluded for protection of the marine environment, this discussion of non-UN regional organizations in the Baltic area should indicate that efforts to establish similar intergovernmental organizations for marine regions elsewhere do *not* need an organizational structure within UNEP's Regional Seas Programme.

c) Asia and the Pacific

In this immense and highly heterogeneous region — surely the most difficult of all regions to "organize" for purposes of regional co-operation in any context — a

variety of regional initiatives have been taken in recent years for the conservation of marine species and the prevention and control of marine pollution. In additon, some thought is now being given to adopting regional approaches to what are conceived as problems of "coastal zone management". Most of these initiatives have been taken by UN agencies such as UNEP, ESCAP, and the FAO-sponsored Indo-Pacific Fishery Commission, but the contributions of non-UN regional organizations such as the Association of South East Asian Nations (ASEAN) and the South East Asian Fisheries Development Center (SEAFDEC) should not be overlooked.

As noted above[434], UNEP originally designated the "East Asian Seas" as one of the ten "concentration areas" under its Regional Seas Programme. Owing to unique difficulties in promoting co-operative action for such a vast and heterogeneous region, UNEP decided in 1978 to limit its early initiatives to a Draft Action Plan for the Protection and Development of the Marine Environment and Coastal Areas of the South East Asian Region. Even Southeast Asia proved too large and complex a "subregion" to deal with, and by the early months of 1979 the scope of the Draft Action Plan had been further reduced, at least for "the purposes of the initial stages of the Action Plan", to "the marine environment and coastal area of the following States: Indonesia, Malaysia, Philippines, Singapore, Thailand"[435]. The purpose was, of course, to take advantage of the infrastructure of co-operative arrangements being developed by these five Southeast Asian countries as the members of ASEAN[436].

Even so restricted, the venture was diplomatically perilous in a region characterized by sweeping claims to extended national jurisdiction which left none of the South China Sea region uncontested under the regime of the high seas. In particular, the jurisdictional claims of Indonesia and the Philippines both to archipelagic waters and 200-mile exclusive economic zones purported to bring fishery conservation authority in much of the region under their national sovereignty or sovereign rights[437]. Recognizing the sensitivity of these jurisdictional matters, the drafters of the First Draft of the Action Plan were careful to add: "The geographic limitation of the marine environment and coastal area to be considered as part of the region will be identified by the Governments concerned on an *ad hoc* basis depending on the type of activities to be carried out as part of the Action Plan"[438]. More optimistically, the Draft concluded: "Future extension of the region to comprise the marine environment and the coastal areas of all the States bordering the entire South China Sea is considered desirable"[439]. But these ASEAN member states are only five of 12 territories bordering the South China Sea, and it will not be politically easy, for conservation or other environmental purposes, to extend the Action Plan to the other seven: the People's Republic of China, Taiwan (the Republic of China), Vietnam, Kampuchea, and the colonies of Hong-Kong, Macao, and Brunei.

Even restricted to the five ASEAN member states, the Action Plan is unlikely to deal in any detail with the conservation of species: first, because of the national jurisdictional sensitivities referred to; and second, because of the recognition that FAO, not UNEP, has "primary jurisdiction" over fishery matters within the system of UN agencies. Yet it should be noted that the First Draft of the Action Plan included among its objectives the "(m)anagement of ... marine and coastal development activities that may have an impact on ... the protection and use of renewable marine resources on a sustainable basis"[440].

The ASEAN countries have also attempted to establish co-operative measures for the prevention and control of marine pollution, especially to combat the threat of ship-generated oil pollution in the Malacca Straits and adjacent waters[441].

The first effort to respond internationally to the marine pollution problem in the region was the International Workshop on Marine Pollution in East Asian Waters, organized jointly in 1976 by UNEP, FAO and UNESCO. Among the various major projects discussed, three were singled out by the ASEAN countries as having special priority: (a) assessment of oil pollution and its impact on living resources; (b) effects of agro-industrial wastes on the coastal ecosystem; and (c) impact of pollution on the mangrove ecosystem and its productivity[442]. Since then the ASEAN countries have been particularly active, both with and without UNEP collaboration in attempting to deal regionally with the first of these problems.

For example, the ASEAN Expert Group on Marine Pollution[443] has produced an ASEAN Contingency Plan for the Control and Mitigation of Marine Pollution[444]. Indonesia, Malaysia and Singapore issued in 1977 a Joint Statement on Navigation Safety which provides *inter alia* that all tankers and other large vessels navigating the Straits of Malacca and Singapore must have adequate insurance and compensation cover[445]. At the Fourth Meeting of the ASEAN Expert Group on Marine Pollution in Manila, in April 1977, it was recommended that the assistance of UN agencies be sought in the training of personnel to combat oil pollution, and also in the drawing up of baseline study guidelines for the ASEAN region[446]. In June 1979 the First ASEAN Working Group Meeting on Marine Sciences identified marine pollution problem areas requiring strong scientific inputs through regional co-operation[447].

In recent years much hope has been invested in the prospect of collaboration between UNEP and ASEAN as a mode of regional co-operation for the prevention and control of marine pollution. The First Meeting of the ASEAN Experts on the Environment, held in December 1978, recommended the creation of a Sub-Committee on the Environment under the ASEAN Committee in Science and Technology. This recommendation failed to win approval at the Fifth Meeting of the ASEAN Standing Committee held at Bali in June 1979, but the Standing Committee did approve in principle of ASEAN-UNEP co-operation in the areas identified by the ASEAN experts on the Environment. At its Second Meeting, held at Penang in September 1979, the ASEAN Experts on the Environment reviewed UNEP activities in the region and included the UNEP Regional Seas Programme among its four "urgent priority areas"[448]. In September 1980, at a workshop on nature conservation held at Denpasar, Indonesia, the ASEAN Experts on the Environment agreed to develop simultaneously both an Action Plan and a convention for the conservation of nature, which would include measures, *inter alia,* for the conservation of marine resources and the establishment of marine parks[449].

At the Third Ministerial Conference for the Economic Development of Southeast Asia (MEDSEA), held at Singapore in April 1968, ten nations agreed on the establishment of the Southeast Asian Fisheries Development Center (SEAFDEC)[450]. Shortly afterwards a Secretariat, headed by a Director-General, was based in Bangkok to co-ordinate the functions and programmes of the Center. Today the Center has three Departments: the Marine Fisheries Training Department in Bangkok, the

Marine Biology Department in Singapore, and the Aquaculture Department at Tigbanan, lloilo, on the Island of Panay in central Philippines. But at present only four nations are active in SEAFDEC: Japan, the Philippines, Thailand, and Singapore. Moreover, SEAFDEC is interested in fishery development rather than conservation. Indeed its approach to fishery management is almost anti-conservationist[451].

Because of their limited memberships, both ASEAN and SEAFDEC have inherent limitations as forums of regional co-operation for the protection and conservation of the South China Sea. SEAFDEC, moreover, is a "quasi-regional" organization, since one of its members (Japan) does not belong to the region. Outside the South China Sea area, other developing countries of Asia and the Pacific, and most other regions of the world, face the same dilemma: whether to persevere, often in adversity, with their own inadequate but "fully regional" arrangements, or to exploit the advantages of greater potential capability within the framework of "quasi-regional" organizations.

d) Antarctic

Conservation in the Antarctic has had an unusually interesting history. Despite its inhospitable nature for human habitation, the land mass has attracted a very considerable volume of scientific research geared *inter alia* toward the protection of the flora and fauna. Much of this research has been conducted internationally under the promotional or co-ordinating auspices of the Scientific Committee on Antarctic Research (SCAR) of the International Council on Scientific Unions (ICSU)[452]. When the Antarctic Treaty was signed in 1959, the region was designated in effect as a reserve for scientific research. SCAR was appointed as the body responsible for maintaining Antarctic research on a co-operative basis. Since then the extent and importance of the international research effort in the Antarctic has been extraordinary and the conservation record impressive[453].

The research and conservation effort in the Antarctic (or Southern) Ocean has been less successful and less impressive than the work done on the land mass of Antarctica. The conservation of whales in that region has been a sad failure for decades, although encouraging improvements have been achieved in recent years under the International Whaling Commission[454]. In 1978 the Convention for the Conservation of Antarctic Seals, signed in 1972, came into force, but serious environmental threats have risen from the prospect of offshore petroleum production and the rapidly increasing scale of harvesting the planktonic crustaceans known as krill *(Euphausiasuperba)*[455].

The krill threat is the more immediate of the two. Excluded from 200-mile national fishing zones around the world, distant fishing vessels have begun to take large and increasing catches of these protein-rich crustaceans, to the point that some commentators fear that actual overfishing in certain areas may soon occur. This trend is particularly disconcerting, since krill form the bottom layer of the marine food chain in the region[456]. Other species, such as Antarctic cod, Patagonian hake, and blue whiting are also beginning to attract commercial attention. Although much more remains to be learned about the marine productivity of the Southern Ocean[457], there is, on present evidence, a serious danger of ecological disturbance.

On the face of things, the danger has been recognized and steps taken to avert the worst kind of disaster. Recently the Convention on the Conservation of Antarctic Marine Living Resources[458] has been signed with a view to establishing a system of conservation controls based on scientific research[459], but this instrument is subject to a number of serious criticisms. Most of these are political objections that may be raised against the exclusive regime of the Antarctic Treaty states under whose exclusive auspices the Convention was drafted[460]. Some have argued in favour of opening up a system of krill (and general marine species) management in the Southern Ocean for all nations, preferably in a manner that would directly benefit those countries with the most serious problems of protein deficiency[461]. Yet the Convention is one of the first efforts in the world to establish a regulatory system for a jurisdictional area that has been defined in such a way as to correspond to the entire ecosystem which the system is designed to maintain[462]. Jurisdictionally, the main flaw is that the text, to be acceptable to the seven claimant states[463], has been drafted with "clever ambiguity" in order to permit the interpretation that these states may exclude from the applicability of the Convention marine areas which they may wish to claim as exclusive economic zones extending beyond the land areas and territorial waters which they already claim[464]. These earlier claims were "frozen" under the Antarctic Treaty, and no progress has been made since 1959 on the resolution of the issue of national jurisdiction in Antarctica.

The political sensitivity of these jurisdictional issues in the region has already had a negative effect on "outside" efforts to place ocean management there on a fully international basis[465]. The future of "regional" arrangements in the Southern Ocean is almost impossible to predict.

E. Conclusion

1. The Regional Level of Treatment

Regionalism is, above all, a level of treatment, intermediate between the national and the global, at which appropriate co-operative action can be taken by national and international agencies to deal effectively with such problems as the protection and conservation of a shared ocean environment. The feasibility of such co-operative action depends chiefly upon the perception of the nations in question that they do in fact have a "shared" or *common interest* in the ocean areas of their neighbourhood, whatever the legal status of the various areas. In the age of extended national jurisdiction in the ocean, the concept of common interest is losing much of its traditional association with that of inclusiveness and with the principle of reciprocity. Rather, it is becoming a more complex combination of elements such as interdependence, mutual benefit, complementarity, and joint responsibility as well as inclusiveness and reciprocity. At a time when the principle of the common heritage of mankind is beginning to assume an institutional significance at the global level, it would be perverse if the parent concept of common interest were not also to acquire institutional significance at more modest regional levels.

Regional arrangements, through which regional co-operative action can be taken in the common interest, have been referred to in this paper as taking the form of regional agreements or regional organizations. But for purposes of analysis and advocacy alike, it should be recognized that regional arrangements take more than

these two forms. As a matter of principle as well as convenience, they may be classified according to the nature, degree, and duration of *commitment* involved, for co-operative action lacking any commitment by the nations in questions lacks any real significance. Accordingly, four kinds of regional arrangements may be distinguished: minimal commitments, special commitments, formal (or transactional) commitments, and organizational commitments.

Minimal regional commitments are those accepted as routine matters of courtesy and low-cost investments in neighbourly goodwill. They are reflected in informal and occasional exchanges of information, and in official but occasional notification and consultation, outside any agreed upon framework of co-operative behaviour. Regional arrangements of this sort are everyday occurrences around the world, and generally pass unnoticed outside the world of staff-level communications in diplomatic and other sectors of bureaucracy.

Special regional commitments are those accepted as significant but finite, *ad hoc,* medium-cost investments in collaboration with neighbouring nations in order to secure a narrowly defined objective of joint interest. They may take the form of limited joint projects; more ambitious, medium-term, programmes; or joint development or planning exercises that may have more intimate, longer lasting implications, albeit outside any framework of indefinitely continuing commitments.

Formal (transactional) regional commitments are those accepted in a publicized, legally significant form, in the name of the state, often with long-term or indefinite consequences for the nation which has accepted them. Generally, these are treaty arrangements of one kind or another: bilateral or plurilateral neighbourhood agreements, or regional agreements. Sometimes, however, they may be issued as joint communiques or joint declarations which lack full treaty character but are, nonetheless, public and formal expressions of policy commitment with some juridical significance. Normally, questions arising out of formal commitments are dealt with in accordance with the principles, if not always the procedures, of international law.

Organizational regional commitments are those accepted by the creation of, and participation in, joint institutional mechanisms. Most international institutions, whether global, regional or subregional, belong to a *circuit* of organizations. Accordingly, many new organizational regional commitments are additions to the appropriate circuit, designed to serve a supplementary or developmental function of a fairly specific kind. Such initiatives might be described as "linkage arrangements", since their purpose can only be understood and evaluated by reference to the appropriate circuit. Some "linkage arrangements" are created primarily to enable existing organizations to focus more sharply and work more efficiently within a more limited regional setting, as when as regional commission creates a subregional committee within its membership. A higher level or regional organization is represented by regulatory commissions with some degree of autonomy or discretionary authority, independently of the governments of the member states. The most highly developed mode of organizational regional commitment is the permanent, collective or quasi-communal, organization. This category ranges from communal, multi-functional entities such as the Organization of American States, the Organization of African Unity, and the League of Arab States, to more loosely con-

ceived mechanisms for consultation, co-ordination, and the occasional joint initiative, such as the Nordic Council[466].

2. The Role of Regional Co-operation in the Protection and Conservation of the Marine Environment

It is important, in any given context, not to exaggerate the virtue or importance of regional co-operation. International lawyers are not alone in being susceptible to this kind of ideal, particularly in an age when so many nations seem to be turning away from the ideal of global co-operation. But a degree of caution or scepticism about the desirable scale of regional co-operation should be entertained, even by the advocate for such initiatives.

Like it or not, we live in a world of nation states. In varying degrees, all national governments pay lip service to the principle of international co-operation, both global and regional, but most recognize such utterances as obligatory rhetoric in international politics and diplomacy. Few, if any, regard international co-operation as an end in itself[467]. Co-operative initiatives are seen rather as means to other ends, and these ends are almost always conceived and defined in national terms. Regional co-operation for the protection and conservation of the marine environment will be construed almost everywhere, by national governments, as a possible level of approach to a variety of national, ocean and coast related purposes. This attitude may be mistaken or shortsighted, but it is the predominant official attitude with which advocates for marine regionalism will have to contend, especially in the age of extended national jurisdiction. The expectations of national enrichment engendered by this trend may be largely illusory in many cases, but proposals for marine regional arrangements in the 1980s are likely to be evaluated, in the main, by the extent to which they seem conducive to, or at least consistent with, these somewhat illusory ends.

But scepticism also can be carried too far. It is anticipated that most coastal states will suffer frustrations in the 1980s, as they try to take advantage of extended zones of national jurisdiction, not least in the difficult process of integrating conservation and other environmental considerations with development planning. Sooner or later, regional conservation arrangements of one kind or another are likely to be accepted by most coastal states as an important contribution to developmental objectives conceived in the national interest. Regional co-operation will be perceived as an opportunity to strengthen national capabilities, rather than as a constraint upon national will. *As national caution dissipates in the longer run, many coastal states might be expected to advance from minimal and special regional commitments to a wider range of regional agreements and regional organizations.*

3. The Role of Regional Arrangements in the Protection and Conservation of the Marine Environment

a) General

Most nations are somewhat reluctant to enter into regional arrangements — that is, formal or organizational commitments — except in unusually auspicious or desperate circumstances. Regional arrangements of both kinds are generally avoided unless there is a clear prospect of national benefits to be derived or a common clanger to be confronted. Such commitments tend to be long-term — indeed, they

are often of indefinite duration; in the long-term unexpected difficulties may arise; and such commitments are usually conspicuous undertakings, for which statesmen and officials may be accountable. Given a real choice, governments may prefer minimal and special commitments as modes of regional co-operation.

Government reluctance to resort to regional agreements in particular is likely to be evident in the context of environmental protection, which has to fight for recognition and support against other, allegedly higher, priorities. In such a context, the reluctant states may be won over only after watering down of proposed treaty commitments to general, vague, or non-mandatory language. Reluctance to participate in environmental co-operation through regional organizations may continue until assurance is given that membership in a proposed organization will not result in a imposition of onerous costs. Any recommendations for environmental co-operation among the governments of the region must take cognizance of these realities[468].

b) Regional Agreements

International lawyers, bureaucrats, and diplomats do perhaps tend to overrate the value of international treaty arrangements as a mode of regional co-operation. In normal, non-regional relationships between two or more states there are, of course, many important purposes to be served by treaty-making, in its broadest sense; and in regional settings treaty-making may be necessary to create certain kinds of organizational commitments between neighbouring states. But in the specific context of *regional co-operation,* treaty-making should be viewed, typically, as a means to an end, rather than an end in itself, and sometimes the end can be served better without a formally binding treaty instrument.

Almost every kind of international agreement belongs — if not exclusively — to one of four categories: administrative, distributive, resolutive, or demonstrative. The ends of regional co-operation in ocean affairs might be served, in theory, by all four categories or agreements. An *administrative* agreement — designed primarily to initiate and maintain inter-state services — might be negotiated regionally to produce a joint shipping service, a co-operative enforcement arrangement between the coast guards of neighbouring states, or a joint fishery management system. A *distributive* agreement — designed primarily to distribute, redistribute or exchange resources (people, money, knowledge, skills, data, commodities or wealth-creating entitlements) — might be negotiated regionally to allocate fishing rights, authorize a joint venture, sell marine products, exchange scientific information, inculcate management skills, or transfer technology. A *resolutive* agreement — designed primarily to resolve an outstanding issue — might be negotiated regionally to settle a maritime boundary dispute or some other kind of jurisdictional quarrel. And a *demonstrative* agreement — designed primarily to dramatize good intentions — might be negotiated regionally to pledge good faith and co-operative action in the search for solutions to designated problems.

Applied to the specific context of the protection and conservation of the ocean environment, all four types of regional agreements might conceivably be entered into, but the prospects seem to vary appreciably from category to category. Most states appear reluctant at present to enter into formal treaty commitments of any *administrative* character, and the ideal of environmental co-operation is unlikely to overcome this aversion. Again, most nations engaged in serious ocean-related

disputes will be slow to resort to treaty-making solely for *resolutive* purposes, especially in an age when most coastal states are apt to make a liberal, discretionary interpretation of their resource-related rights within their exclusive economic zone. Certain kinds of *distributive* agreements are more conceivable, but matters such as research, training, and exchange of data can usually be carried out regionally without a treaty. *In the short term, the primary purpose of regional treaty-making for the protection and conservation of the marine environment may often be symbolic or demonstrative — and yet this may be crucial to more substantive progress.*

If a case is to be made, then, for a regional or subregional treaty of such a kind, what should be advocated initially, especially in a developing region, is a "parent convention": a framework of good intentions, couched in general language that looks to the development of more specific commitments in the future. A framework convention, being essentially demonstrative in purpose, should not be expected to resolve outstanding issues or to make a significant contribution to resource production or distribution, much less to forge an organic linkage between existing state services. It should be enough that such a convention between neighbouring states serves to promote the *idea* of regional co-operation in the protection and conservation of their shared ocean environment. Less intangible, short-term benefits may be more likely to accrue from the resort to *special* and *organizational* commitments.

c) Regional Organizations

The emerging pattern of regional arrangements for the protection and conservation of the marine environment is likely to be affected profoundly by the restructuring of the social and economic sectors of the UN system. This restructuring is accompanied by a number of related trends: a general disposition to support the continuing decentralization of existing UN agencies in all social and economic sectors; a gradual increase in the autonomy of the regional offices of these agencies; an increasing emphasis on the UN regional economic commissions in particular; a growing pressure by the commissions on UN agencies, non-UN regional organizations, and interested extraregional states to work in collaboration with these commissions; and increasingly successful efforts by the commissions to induce these agencies, organizations and states to reconcile their regional objectives and integrate their regional activities with those of the commission in question.

If these trends continue and are likely to become a dominant influence in the pattern of regional arrangements generally, we may see significant divergences emerging in the level of regional treatment accorded to such matters as the protection and conservation of the marine environment. The macro-regional level may be more successful, or at least more popular, in relatively homogeneous macro-regions such as Latin America and Western Asia; *structural* subregionalization may become prevalent in Africa; and *programmatic* subregionalization may prevail in the most heterogeneous of the regions, Asia and the Pacific. In Europe, North America and Soviet Asia, other regional initiatives outside the ECE framework are likely to continue to play a dominant role in regional co-operation.

We may have to learn all over again, in a new way, that what is a "region" for one purpose may not be a "region" for another — for political as well as cultural or technical reasons.

Another major influence, closely related to the restructuring of the UN system, is the strong desire of most developing countries to avoid or reduce the influence of extra-regional powers, without at the same time depriving themselves of valuable developmental assistance. A conscious attempt may be made in some regions to replace "quasi-regional" arrangements with fully regional (or sub-regional) arrangements and to seek assistance in the strengthening of national and regional capabilities though a variety of *ad hoc* bilateral arrangements with extra-regional states outside the framework of regional organizations. Certain non-UN regional groupings of nations may, however, prove to be more tolerant of association with extra-regional states than the macro-regional UN commissions; and this, of course, raises the possibility that these non-UN regional organizations may prove to be more effective in the short term in dealing with the complex problems associated with the protection and conservation of the marine environment.

Finally, it might be anticipated that ocean industries (transnational corporations) and non-governmental organizations will also be able to play an important supportive role in co-operative efforts to protect and conserve the marine environment: the shipping and energy industries, for example, in the context of ship-generated and offshore platform sources of marine pollution, and NGOs In coastal zone management studies and the study of problems associated with land-based sources of marine pollution.

4. Final Considerations

a) General

Ideally, of course, many marine regions and sub-regions should be brought under a single, comprehensive, regional or subregional system of environmental protection and conservation. Although there is no such thing as a "natural" region for all purposes, for each of the specific (sectoral) purposes of the management of the marine environment identified in this study it may be possible to move toward a closer understanding of what would constitute a "natural (*i.e.* most suitable) management area" for all regions of the world.

b) Conservation of Marine Species

i) Fishery conservation

In order to classify the probable or desirable future pattern of regional co-operation, it is important to distinguish at least six fishery situations:

(a) the situation where the species in question (both mobile and sedentary) are concentrated mainly in coastal stocks that fall within legally permissible limits of national jurisdiction;
(b) the situation where the species in question are concentrated mainly In stocks that are shared across national jurisdictional boundary lines by two or more neighbouring states;
(c) the situation where the species in question are concentrated mainly in stocks that straddle the line dividing the area of national jurisdiction of an adjacent coastal state and high seas area beyond;
(d) the situation of highly migratory species that are found to be widely distributed around the world;

(e) the situation of anadromous species that are spawned in the internal waters of a costal state and return eventually to spawn in the same waters of that state or region, but spend most of the intervening years feeding and growing in the coastal or offshore waters of another state; and
(f) the situation of catadromous species that are spawned in the offshore waters of another state and eventually return there to spawn, but spend most of the intervening years feeding and growing in the internal and inshore waters of another coastal state.

The case for regional co-operative arrangements vary with the situation. The first situation (a) is one where *special* commitments might be entered into from time to time (e.g., projects and programmes for research and training purposes), but where *formal* and *organizational* co-operative arrangements may be unnecessary. Even in circumstances where outside assistance is sought in the establishment of an effective national conservation and management system, it may not be *necessary* to resort to regional agreements or regional organizations.

The second situation (b), by contrast is one in which the neighbouring states have an interest — a short-term, rather than a long-term, interest — in negotiating a *formal* agreement for co-operative management of the shared stocks, or at least for regular consultations and joint research to ensure that the neighbouring management systems are compatible and conducive to effective conservation of the species. Ideally the neighbouring states should enter into an *organizational* commitment, perferably in the form of a joint regulatory commission or at least a joint advisory mechanism that is authorized to make appropriate conservation or management recommendations to the governments.

The third situation (c) is one in which the coastal state should make an *organizational* arrangement for research and regulatory purposes within the framework of the neighbouring regional commission. This may result in the occasional *special* project or programme in response to particular problems, as they arise.

The fourth situation (d) — typically that of tuna — is *not* one in which a limited (subregional) co-operative approach is useful. Both in theory and practice, conservation and management problems of this kind are best treated within a broad-based organization, whose membership includes most of the interested states.

The fifth situation (e) presents the need for *bilateral* (not necessarily *regional)* co-operation between the state of origin and the "feeding state" for purposes of research, conservation, and (if both wish to exploit the resource) the allocation of fishing rights. While the first of these three purposes might be served through *special* commitments (e.g. projects or programmes), the second seems to require an *organizational* arrangement and the third some kind of public *formal* commitment, presumably in treaty form.

The sixth situation (f) seems to create the same requirements as situation (e).

ii) Protection of marine wildlife and related ecosystems
It should first be advocated that all governments give appropriate attention to existing international conventions for the protection of wildlife. Very few states, if

any, have an interest in *not* signing, ratifying or adhering to such conventions. Formal support for these unexceptionable treaties rarely involves a state in serious controversy, and generally represents a low-cost investment in acquiring an international reputation as an environmentally responsible state.

More specifically, the littoral states of each marine region should accept a special responsibility for the preservation of those marine species inhabiting their marine environment which have been identified by scientific experts as "endangered". It may be unnecessary to urge them to accept *formal* or *organizational* commitments for this purpose: it may be more urgent for them to participate in *special* projects or programmes designed to identify and monitor the species in question. But in the worst cases, where the species are threatened with extinction, and where the causes are known and correctable, then the littoral states have a legal and moral obligation to co-operate, either *formally* or *organizationally* or both, with other concerned nations in efforts to restore the species to a sustainable population level.

In the case of special, ecologically valuable areas which are threatened with degradation (e.g., coral reefs, mangrove swamps), the neighbouring coastal states have an interest as well as a responsibility to co-operate in *special* protective research projects and research and training workshops. The same suggestion applies also to neighbouring coastal states with similar or shared interests in the establishment and maintenance of marine parks.

c) Prevention and Control of Marine Pollution

i) General

The case for regional co-operation has been outlined at UNCLOS III. Some emphasis has been placed on the need to take "characteristic regional features" into account in "formulating and elaborating international rules, standards and recommended practices and procedures consistent with [the final UNCLOS III] Convention, for the protection and preservation of the marine environment"[469]. Particular kinds of co-operation are identified: the obligation to give notification of imminent or actual damage[470]; the obligation to co-operate in the joint promotion and development of contingency plans against pollution[471]; the obligation to co-operate in the promotion of studies, research programmes and exchange of information and date[472]; and the obligation to co-operate in establishing scientific criteria for the formulation and elaboration of rules, standards and recommended practices and procedures[473]. Of these four purposes, the first two in particular seem to lend themselves to *regional* arrangements. The first refers to minimal commitments, as defined in this paper, the second to *organizational* commitments. The fourth purpose, developing scientific criteria, is a global rather than a regional undertaking; but the third purpose — research and exchanges — is both regional and global, and seems to envisage a mixture of special and organizational commitments.

ii) Vessel-source pollution

Within this section, UNCLOS III envisages (implicitly regional) co-operative arrangements, whereby two or more coastal states may seek to harmonize policy (and resulting national legislation) by establishing on a joint basis "particular requirements for the prevention, reduction and control of pollution of the marine environment as a condition for the entry of foreign vessels into their ports or internal waters or a call at their offshore terminals"[474]. Notification of such co-operative ar-

rangements will have to be given to the "competent international organization": namely, IMCO. It is also envisaged that some coastal states will wish to designate "special areas" of their waters where particularly stringent environmental precautions are justified, and that the approval of IMCO will be sought for such a designation. This provision does not explicity foresee the possibility that two or more neighbouring states may wish to designate jointly a special area which cuts across their boundary. Finally, the *general* obligation to co-operate in the joint development of contingency plans has special application to the danger of ship-generated pollution.

It should be added that two or more neighbouring states may wish to co-operate, through *organizational* or *formal* arrangements, in the establishment and maintenance of certain types of vessel traffic control systems, such as traffic separation schemes for congested areas shared by such states, in areas such as the Straits of Malacca and Singapore.

iii) Pollution from seabed activities

In this sector, UNCLOS III recognizes an obligation on states to "harmonize their national policies at the appropriate regional level"[415], and to establish regional (as well as global) "rules, standards and recommended practices and procedures to prevent, reduce and control pollution of the marine environment arising from or in connection with seabed activities[476]. *Formal* agreement on such regional standards would probably have to be preceded by *special* consultation and studies.

iv) Pollution from dumping

Regional rules and standards for combating this source of pollution are also envisaged, but it is implied that regional laws, regulations and measures, like national ones, shall be no less effective than global rules and standards, such as those set out in the 1972 London Ocean Dumping Convention[477].

v) Pollution from land-based sources

Again, UNCLOS III requires that coastal states "endeavour to harmonize their national policies at the appropriate regional level"[478], and to establish regional (as well as global) "rules, standards and recommended practices and procedures to prevent, reduce and control pollution of the marine environment from land-based sources, *taking into account characteristic regional features, the economic capacity of developing States and their need for econmic development* [emphasis added]"[479]. The importance of the considerations underlined is likely to mean that some neighbouring coastal states will wish to explore the possibility of approaching these shared problems within the integrative, eco-developmental context of *coastal zone management,* perhaps first through a series of *special* investigative *projects and programmes.*

vi) Pollution from or through the atmosphere

the possibility of regional arrangements within this section is also recognized at UNCLOS III[480]. Presumably the establishment of nuclear free zones (or maritime zones of peace), promulgated through *formal* commitments of a *demonstrative* character, could be regarded as an appropriate example of such an initiative.

Notes

1 *The Shorter Oxford Dictionary,* vol. II (1944-1967), defines "region" as "a more or less defined portion of the earth's surface ... as distinguished by certain natural features, climatic conditions, special fauna or flora, or the like"

2 Lewis M. Alexander, *Regional Arrangements in Ocean Affairs* (Office of Naval Research, 1977), p. 11. Alexander quotes Derwent Whittlesey's definition of region as "an Intellectual concept, an entity for the purposes of thought, created by the selection of certain features that are relevant to an areal interest or problem and by the disregard of all features that are considered to be Irrelevant". It is worth noting that most modern treatises make little or no mention of regions as an "element" of political geography. *See,* for example, W. Gordon East and J.R.V. Prescott, *Our Fragmented World: An Introduction to Political Geography* (1975). Although regions occasionally receive special attention in geopolitical writings (e.g. Saul N. Cohen, "Geostrategic and Geopolitical Regions", In Roger E. Kasperson and Julian V. Minghi, *eds., The Structure of Political Geography* (1969), pp. 178-186), and as "core areas" at the subnatlonal level (e.g. Harm J. de Blij, *Systematic Political Geography* (2nd *ed.,* (1973), pp. 83-115), the contemporary emphasis in regional analysis tends to be on "functions of politically organized areas". See Richard Hartshorns, "the Functional Approach in Political Geography", in Kaperson and Minghi, *op. cit.,* pp. 34-69

3 "It is quite absurd and illogical to seek to establish regional boundaries in detail. They must remain vague, for they are boundaries of a generalization. If, for practical purposes, one finds that he must draw boundaries, then he should do so in a frankly arbitrary manner". Attributed to Preston E. James, in Bruce M. Russett, *International Regions and the International System: A Study In Political Ecology* (1967), at pp. 6-7

4 Oran Young, *Resource Management at the International System: The Case of the North Pacific* (1977), pp. 25-26. *See* also Maria Eduardo Goncalves, "Concepts of Marine Region and the New Law of the Sea" (1979), 3 Marine *Policy* 255 at pp. 257-258

5 These areas of Inquiry are distinguished in Ernest B. Haas, "The Study of Regional Integration: Reflections on the Joy and Anguish of Pretheorizing", in Leon N. Lindberg and Stuart A. Scheingold, *eds., Regional Integration: Theory and Research* (1971), pp. 7-8

6 Ernest B. Haas, "The Challenge of Regionalism" (1958), 12 *Int'l Organ.* 440 at p. 442. According to Haas, one of the leading exponents, the pioneering work in the theory of regional integration is Karl W. Deutsch, *Political Community at the International Level: Problems of Definition and Measurement* (1954). *See* also Joseph S. Nye, *International Regionalism: Readings* (1968), and Peace *In Parts* (1971)

7 Sometimes this trend is advocated as a buffer against power politics; sometimes It Is viewed as screening less visible trends toward the establishment of "spheres of Influence". For an even-handed account, see Hedley Bull, *The Anarchical Society: A Study of Order in World Politics* (1977), pp. 305-311

8 An "International resource region" has been defined as a "geographical area demarcated in terms of the functional issues associated with the management of natural resources and the maintenance of environmental quality". Young, *supra* note 4, at pp. 22-23

9 Douglas M. Johnston, "New Uses of International Law in the North Pacific" (1968), 43 *Wash. Law Rev.* 77

10 Lewis M. Alexander, "Regionalism at Sea: Concept and Reality", in Douglas M. Johnston, *ed., Regionalizatlon of the Lew of the Sea* (1978), p. 5

11 The nine ocean basins are:
(i) North Atlantic; (ii) South Atlantic; (iii) Indian; (iv) Arctic; (v) Antarctic; (vi) North Pacific; (vii) South Pacific; (viii) West Central Pacific; and (ix) East Central Pacific.
The 24 semi-enclosed seas are:
(i) the Gulf of Aden; (ii) the Arabian Sea; (iii) the Andaman Sea; (iv) Baffin Bay — Davis Strait; (v) the Baltic Sea; (vi) the Bay of Bengal; (vii) the Bering Sea; (viii) the Black Sea; (ix) the Caribbean Sea; (x) the Celebes Sea; (xi) the East China - Yellow Sea; (xii) the Gulf of Guinea; (xiii) the Sea of Japan; (xiv) the Mediterranean Sea; (xv) the Gulf of Mexico; (xvi) the North Sea; (xvii) the Sea of Okhotsk; (xviii) the Gulf of Oman; (xix) the Persian (Arabian) Gulf; (xx) the Red Sea; (xxi) the Solomon Sea; (xxii) the South China Sea; (xxiii) the Sulu Sea; and (xxiv) the Timor-Arafura Seas. Alexander, *supra* note 2, pp. 17-18

12 *Ibid.,* pp. 19-20

13 Alexander, *supra* note 10, pp. 5-6; and Alexander, *supra* note 2, pp. 22

14 Edward L. Miles, "On the Utility of Regional Arrangements In the New Ocean Regime", In Johnston, *supra* note 10, p. 250

15 *Ibid.,* pp. 259-260. For Miles' comments on the "regional integration" writings of Haas and Nye, *see ibid.,* pp. 262-264

16 Richard B. Bilder, "The Consequences of Regionalization in the Treaty and Customary Law of the Sea" in Johnston, *supra* note 10, pp. 31-44

17 *Ibid.,* p. 32

18 For example, L.D.M. Nelson, "The Functions of Regionalism In the Emerging Law of the Sea as Reflected in the Informal Composite Negotiating Text" in Johnston, *supra* note 10, pp. 17-30

19 *Ibid.,* p. 18

20 *Ibid.,* p. 25

21 Doc. A/CONF. 62/WP. 10/Rev. 2 (11 April 1980)

22 Many of these references are contained in Part XII of the ICNT/Rev. 3 ("Protection and Preservation of the Marine Environment"). For a listing and analysis of such references, see James D. Kingham and D.M. McRae, "Competent International Organization and the Law of the Sea" (1970), 3 *Marine Policy* 106. Note, however, that this useful study deals only with the global organizations which are apparently Intended In these references

23 *See,* for example, ICNT/Rev. 3, Article 22 (sea lanes and traffic separation schemes In the territorial sea), where the reference to "competent International organizations" is apparently directed only at global agencies such as IMCO, UNEP, FAO, and IAEA; Article 65 (marine mammals), where the reference to "International organization" Is both global (e.g. International Whaling Commission) and regional (e.g. North Pacific Fur Seal Commission); and Article 64 (highly migratory species), where the reference to "appropriate International organizations" Is to be Interpreted principally as a reference to the regional fishery commissions that deal with tunas

24 For example, Articles 69 (right of land-locked states) and 70 (right of states with special geographical characteristics)

25 For example, Articles 63 (stocks straddling economic zone boundary), 118 (conservation of high seas fishery resources), and 123 (enclosed or semi-enclosed seas)

26 For example, Articles 62 (access to surplus stocks in economic zones), 67 (catadromous species), 74 (delimitation of economic zones boundaries), and 83 (delimitation of continental shelf boundaries)

27 For example, Articles 15 (delimitation of territorial sea boundaries), 47 (archipelaglc baselines), 51 (fishing rights in archipelagic waters), and 66 (anadromous stocks). The first three of these examples are of bilateral neighbourhood-type regional arrangements, but the fourth example (agreements between the state of origin and the "feeding" state with respect to the anadromous studies) need not necessarily be a regional arrangement, e.g. Canada and Denmark

28 Article 276 (1). For an enumeration of the functions of such regional centres, see Article 277

29 A number of countries, both regional and extrareglonal maintain territorial claims to portions of the Antarctic land-mass, but all such claims have been "frozen" under the provisions of the Antarctic Treaty of 1959. The legal result Is that no state has an acknowledged territorial existence on the southern perimeter of the Southern Ocean and no "fully regional" arrangement in the region is therefore possible

30 The four are: the Draft Protocol for Protection of the Mediterranean against Pollution from Land-Based Sources; the Draft Convention for the Environmental Protection of the Red Sea and Gulf of Aden; the Draft Convention of the Protection of the Marine Environment Against Pollution in the Southeast Pacific; and the Draft Protocol on Regional Co-operation in Cases of Pollution Emergencies (In the Southeast Pacific)

31 On the terminology of the law of treaties, *see* Arnold D. McNalr, *The Law of Treaties* (1961), pp. 3-34

32 Alexander, *supra* note 2, Includes 42 scientific research programmes, projects, or organizations in his listing of "regional organizations" In ocean affairs (pp. 220-230)

33 *Ibid.,* p. 229. *See* also Albert W. Koers, *International Regulation of Marine Fisheries: A Study of Regional Fisheries Organizations* (1973), who Includes the ICES (pp. 77-79) in his listing of international fishery organizations

34 The proliferation of regional fishery commissions between 1945 and 1972 is totally neglected in the mainstream Anglo-American literature on regionalism and International organizations. See, for example, Evan Luard, *International Agencies: The Emerging Framework of Interdependence* (1977); David M. Lelve, *International Regulatory Regimes: Case Studies In Health, Meteorology and Food* (in 2 vols., 1976); D.W. Bowett, *The Law of International Institutions* (3rd *ed.,* 1975); Paul A. Tharp, Jr., *ed., Regional International Organizations: Structure and* Functions(1971); Louis J. Cantori and Steven L Spiegel, *eds., The International Politics of Regions: A Comparative Approach* (1970); Nye, *supra* note 6; and Russett *supra* note 3

35 Some writers ascribe the rise of "incompatible regionalism" after World War Two to Cold War tensions. *See* Minerva M. Etzionl, *The Majority of One: Toward a Theory of Regional Compatibility* (1970), pp. 90-124

36 The figure of 23 is taken from Koers, *supra* note 33. The 20 regional newcomers were (l) the International Commission for the Northwest Atlantic Fisheries (ICNAF), 1949; (ii) the Inter-American Tropical Tuna Commission (IATTC), 1949; (III) the International North Pacific Fisheries Commission (INPFC), 1952; (iv) the Permanent Commission of the Marine Resources of the South Pacific, 1952; (v) the Japanese-Soviet Northwest Pacific Fisheries Commission, 1956; (vi) Commission for Fisheries research in the Wester Pacific, 1956; (vii) the North-East Atlantic Fisheries Commission (NEAFC), 1956; (viii) the Joint Commission for the Black Sea, 1959; (ix) the Mixed Commission, 1962; (x) the Japan-Republic of Korea Joint Fisheries Commission, 1965; (xi) the International Commission for the Conservation of Atlantic Tunas (ICCAT), 1966; (xii) the International Commission for the Southeast Atlantic Fisheries (ICSEAF), 1969; five FAO regional fishery organizations (xiii) the Indo-Paclflc Fisheries Council (IPFC), 1948; (xiv) the General Fisheries Council for the Mediterranean (GFCM), 1949; (xv) the Regional Fisheries Advisory Commission for the South West Atlantic (CARPAS), 1962; (xvi) the Indian Ocean Fishery Commission (IOFC), 1967; (xvii) the Fishery Committee for the Eastern Central Atlantic (CECAF), 1967; and three lesser known commissions (xviii) Shellfish Commis-

sion for Skagerrak and Kattegat, 1952; (xix) Baltic Sea Salmon Standing Committee, 1962; (xx) Sealing Commission for the Northeast Atlantic (SCNEA), 1957

37 For the most detailed analysis, *see* Koers, *supra* note 33. *See* also William T. Burke, "Aspects of Decision-Making Processes in Inter-governmental Fishery Commissions" (1967), *43 Wash. Law Rev.* 115; and more generally, Douglas M. Johnston, *The International Law of Fisheries: A Framework for Policy-Oriented Inquiries* (1965), and Francis T. Christy, Jr., and Anthony Scott, *The Common Wealth In Ocean Fisheries: Some Problems of Growth and Economic Allocation* (1965). See *also FAO Dept. of Fisheries, International Fishery Bodies* (UN Doc/COFI/66/6, 20 May 1966)

38 On proposals to enhance the fishery authority of coastal states, which were being discussed in the early 1970s, *see* Koers, *op. cit*., pp. 231-251

39 For a model draft Convention for the Establishment of a World Marine Fisheries Organization, *see Ibid.,* Appendix I (pp. 331-339). On the functional possibilities available for a global fisheries agency such as FAO's Committee on Fisheries (COFI) — namely, research, regulation, conflict resolution, and enforcement — *see* Edward Miles, *Organizational Arrangements to Facilitate Global Management of Fisheries* (Resources for the Future, Inc., The Programme of International Studies of Fishery Arrangements, Paper No. 4, 1974)

40 Johnston, *supra* note 9, pp. 80-81. By the early 1970s it was evident that a great deal of the responsibility for fishery conservation would pass down to the national level, and within a few years It was foreseeable that in several countries some of this responsibility would pass further down to subnational levels of organization. This seemed to mean that most of the world's "coastal" species would eventually be withdrawn from the jurisdiction of existing regional commissions. The future of regional fishery conservation organizations would be restricted, it seemed, to highly migratory oceanic species (e.g. tunas) and to stocks shared between two or more neighbouring coastal States. Miles, *supra* note 39, and William T. Burke, *Some Thoughts on Fisheries and a New Conference on the Law of the Sea* (Law of the Sea Institute, Occasional Paper Series No. 9, 1971), pp. 14-17

41 Koers, *op. cit.,* pp. 178-196

42 It was argued that the establishment of ICCAT in 1966 to cover all tunas In the Atlantic was a precedent which should be followed elsewhere. Others supported the idea of establishing a single multi-species conservation commission for the North Pacific, or at least the Northeast Pacific. *Ibid.,* pp. 265-268

43 *Ibid.,* pp. 126-130. *See* also Christy and Scott, *supra* note 37, pp. 235-236

44 On the responsibility of existing regional fishery conservation commissions to ensure "full utilization", *see* Koers, *op. cit.,* pp. 171-178

45 Some fishery scientists are inclined to attribute failures in conservation to failures in communication between science and management. See, for example, David Cushing, *Fisheries Resources of the Sea and their Management* (1975), pp. 68-82. Others put more emphasis on the changes that have taken place in the theory of fishery management, and thus in the thinking that goes Into the formulation of conservation objectives and the kind of technical advice given to management. Several earlier texts (e.g. International Convention on the Northwest Atlantic Fisheries, 1949; the UN Convention on Fishing and Conservation of the Living Resources of the High Seas, 1958; and the International Convention for the Conservation of Atlantic Tunas, 1966) enshrined the concept of "maximum sustainable yield" as the official objective of conservation. Yet by the late 1960s most fishery experts were prepared to concede that some weight should be given to the concept of "maximum net economic yield". See, for example, J.A. Gulland, *The Management of Marine Fisheries"* (1974), pp. 105-126; and, more recently, Sidney Holt, "Marine Fisheries in *Ocean Yearbook I* (*eds.* Borgese and Ginsberg) 1979, pp. 38-83 at 80-82

46 Christy and Scott, *supra* note 37, *passim.*

47 On the prospect of multinational public fishing enterprises, *see* Koers, *op. cit.,* pp. 263-265. On the problem of investment, *see* Robert H. Neuman, "Multinational Investment in Fisheries: Implications for International Agreements on Living Marine Resources", in H. Gary Knight, *ed., The Future of International Fisheries Management* (1975), pp. 93-102

48 H. Gary Knight, *Managing the Sea's Living Resources: Legal and Political Aspects of High Seas Fisheries* (1977), pp. 47-48. For an analysis of these difficulties, *see* Robert M. Goldberg, "Ends and Means: The Role of Enforcement Analysis in International Fisheries Regulation", In Knight, *supra* note 47, pp. 183-211

49 Table VI, *supra,* shows that the first regional organization designed specifically for marine pollution control purposes was not established until 1972

50 R.M. M'Gonigle & M.W. Zacher, *Pollution, Politics and International Law: Tankers at Sea* (1979)

51 Douglas M. Johnston, "The Environmental Law of the Sea: Historical Development", *supra,* p. 17 *et seq.*

52 Declaration of the United Nations Conference on the Human Environment, the Action Plan for the Human Environment, and the Resolution on Institutional and Financial Arrangements. These form sections I, II, and III, respectively, of the *Report of the United Nations Conference on the Human Environment,* UN Doc. A/CONF. 48/14 and Corr. 1, and are reproduced at 11 *Int'l Leg. Mats.* 1416, *et. seq.* (1972)

53 11 *Int'l Leg. Mats.* 1417 (1972)

54 Some of these principles do, of course, have important regional implications. For example, Principles 22, which upholds the *Trail Smelter* principle of liability for polluting another state, has particular significance

for neighbouring countries

55 11 *Int'l Leg. Mats.* 1467 (1972)

56 The sole reference of a regional kind in the Resolution is made in paragraph 17, which "(i)nvites the regional economic commissions and the Economic and Social Office in Beirut, in co-operation where necessary with other appropriate regional bodies, to identify further their efforts directed towards contributing to the implementation of environmental programmes in view of the particular need for rapid development of regional co-operation in this field". *Ibid.,* p. 1469

57 *See*, for example, Recommendations 1, 4, 23, 24, 39, 43, 45, 46, 47, 51, 52 and 80

58 *See*, for example, Recommendations 3, 26, 32, 50, 86 and 96

59 *See*, for example, Recommendations 14, 19, 20, 21, 38, 48, 60, 61, 62, 67, 68, 72, 74, 75 and 106

60 11 *Int'l Leg. Mats.* 1441 (1972)

61 For a listing of current regional fishery conservation organizations, *see* Table V, *supra*

62 11 *Int'l Leg. Mats.* 1442 (1972)

63 *Ibid.*

64 *Ibid.*

65 *Ibid.,* p. 1443

66 *Ibid.*

67 *Ibid.,* p. 1435

68 *Ibid.,* p. 1436

69 *Ibid.*

70 *Ibid.,* p. 1437

71 *Ibid.,* p. 1434

72 *Ibid.*

73 *Ibid.,* p. 1435

74 *Ibid.*

75 *Ibid.,* p. 1452

76 IUCN, *Proceedings of an International Conference on Marine Parks and Reserves* (1976)

77 11 *Int'l Leg. Mats.* 1454 (1972)

78 Five of the eight designated regions under the UNEP Regional Seas Programme are co-extensive with a semi-enclosed sea or with a combination of two or more contiguous single enclosed seas. Moreover, most of the action plans drafted for these designated regions include, among their objectives, the negotiation of a general regional treaty for the environmental preservation of the region and two or more supplementary protocols

79 11 *Int'l Leg. Mats.* 1461 (1972)

80 *See infra,* pp. 324 *et seq.*

81 Composed of 35 member States, the *Ad Hoc* Committee to Study the Peaceful Uses of the Sea-Bed and the Ocean Floor beyond the Limits of National Jurisdiction was established in Dec. 1967 by UN General Assembly Resolution 2340 (XXII), 22 UN GAOR, Supp. 16 (Doc. A/6716) at p. 14 (1968)

82 Composed of 42 (later enlarged to 91) member states, the Committee on the Peaceful Uses of the Sea-bed and Ocean Floor beyond the Limits of National Jurisdiction was established in Dec. 1968 by UN General Assembly Resolution 2467A (XXIII). 23 UN GAOR, Supp. 18 (Doc. A/7218) at p. 15 (1969)

83 *See* 23 UN GAOR (Doc. A/7230) at p. 39 (1968)

84 *Ibid.*

85 *Ibid.,* p. 41

86 These were the 1952 Santiago and 1954 Lima agreements reprinted in the UN Doc. A/AC.135/10/Rev.1, at pp. 11-13(1968)

87 28 UN GAOR, 1 Supp. (Doc. A/9021) at p. 1 (1973)

88 UN Doc. A/AC.138/14, at p. 27 (1969)

89 *Ibid.*

90 UN Doc. A/AC.138/41 (1978)

91 *Ibid.,* p. 28

92 *Ibid.*

93 *Ibid.*

94 Adopted In UN General Assembly Resolution 2749 (XXV), reprinted in 25 UN GAOR, Supp. (Doc. A/8020) at pp. 24-25 (1970)

95 *Ibid.*

96 UN Doc. A/AC.138/41, at p. 28 (1971)

97 Originally issued as UN Doc. A/AC.138/53 (23 Aug. 1971), reprinted in 26 UN GAOR, Supp. 21 (Doc. A/8421) at p. 105 (1971)

98 Article 198

"1. Regional Arrangements for any or all of the purposes of this Convention may be established with the consent of the Council at any time not less than four years after the entry into force of this Convention[127].

2. The Council shall determine the requirements for the establishment of regional arrangements and their scope, taking Into account the wishes of the States In the region and the need for economic and efficient operation of the Institutions." *Ibid.* at p. 190

99 "It Is desirable that regional arrangements should have been established only after the Institutions have had the opportunity of becoming solidly established". *Ibid.* note 127

100 Malta's representative, Arvid Pardo, who In 1967 had advanced the "common heritage of mankind" principle

101 *See* working papers submitted by:
Belgium (A/AC.138/35); Bulgaria and Poland (A/AC.138/45 and Add. 1); Canada and Norway (A/AC.138/52 and Add. 1); Greece (A/AC.138/54); Argentina *et al.* (A/AC.138/56); United Kingdom (A/AC.138/57); Afghanistan *et al.* (A/AC.138/58); Algeria *et al.* (A/AC.138/66 and Corr. 2); Malta (A/AC.138/67); USA (A/AC.138/68); Greece and Italy (A/AC.138/69); Japan (A/AC.138/70 and A/AC.138/78); USSR (A/AC.138/71); Afghanistan *et al.* (A/Ac.138/72 and Corr. 1); Turkey (A/AC.138/74 and Corr. 1); Malta (A/AC.138/75 and Corr. 1); France *et al.* (A/AC.138/76) and Poland (A/AC.138/77)

102 Working paper submitted by Argentina, Brazil, Columbia, Chile, Ecuador, El Salvador, Guatemala, Guyana, Jamaica, Panama, Peru, Spain, Trinidad and Tobago, Uruguay and Venezuela. UN Doc. A/AC.138/56 (1971) reprinted in 26 UN GAOR, Supp. 21 (Doc. A/8421) at p. 198 (1971) (hereinafter cited as *Latin American List)*

103 Working paper submittd by Afghanistan, Algeria, Cameroon, Ceylon, Democratic Republic of the Congo, Ethiopia, Gabon, Ghana, India, Indonesia, Iran, Iraq, Ivory Coast, Kenya, Kuwait, Liberia, Libya, Madagascar, Malaysia, Mauritania, Mauritius, Morocco, Nigeria, Philippines, Singapore, Somalia, Sudan, Tunisia, United Arab Republic, United Republic of Tanzania, Yemen and Yugoslavia. UN Doc. A/AC.138/58 (1971) reprinted in 26 UN GAOR, Supp. 21 (Doc. A.8421) at p. 202 (1971) (hereinafter cited as *Afro-Asian List)*

104 *See Latin American List,* p. 200

105 *See Afro-Asian List,* p. 204

106 27 UN GAOR, Supp. 21 (Doc. A/8721) at p. 8 (1972)

107 *Ibid.*

108 *See* 28 UN GAOR, Supp. 21 (Doc. A/9021) vols. V and VI (1973), (hereinafter cited as *Final Report* V or *Final Report VI)*

109 "Article 18: No provision of this Treaty shall be Interpreted as preventing or restricting the right of any State to conclude regional or subregional agreements to regulate exploitation or distribution of the living resources of the sea, preservation or the marine environment or scientific research, or as affecting the legal validity of existing agreements." UN Doc. A/AC.138/3C.II/C.21 (1973) reprinted in *Final Report V,* Item 15, p. 1

110 "Articles 13.1 and 14:
Article 13:
1. "In regions or subregions in which certain coastal States, owing to geographical or ecological factors, are unable, before all their coastlines, to extend the limits of their sovereignty and jurisdiction up to distances equal to those adopted by other coastal States In the same region or subregion, the former States shall enjoy, In the seas of the latter States, a preferential regime vis-a-vis third States In matters relating to the exploitation of renewable resources, the said regime to be determined by regional, subregional or bilateral agreements taking Into account the Interests of the respective States.
The coastal States of a single region or subregion shall promote such forms of co-operation and consultation as they consider most appropriate In the legal, economic, scientific and technical spheres relating to maritime questions." UN Doc. A/AC.138/SCII/L.27 and Corr. and 2 (1973) reprinted in *Final Report V,* Item 15 at 1

111 " 1 . States of a region or subregion may, in order to co-ordinate matters relating to the legal, economic and technical aspects of the Law of the Sea in their region or subregion, conclude appropriate arrangements between themselves.
"2. These arrangements will take into account:
(a) The legitimate interests of the States concerned;
(b) The orderly development of the renewable resources of the sea under their jurisdiction." UN Doc. A/AC.138/SCII/L.62 (1973) reprinted in *Final Report V, Item 15 at 2*

112 " 1 . The States or a region of subregion may, subject to the general provisions of this Convention, on the basis of equality and mutual respect, promote consultations and negotiate such forms of co-operation as they consider most appropriate with respect to all maritime matters, Including those relating to legal, geographical, economic and ecological aspects, as well as to scientific research and the transfer of technology.
"2. These arrangements should take Into account (a) the legitimate interests of all States concerned, as well as (b) the orderly development and rational management of resources of the ... Zone." Un Doc. A/AC.138/SCII/L.63 (1973) reprinted in *Final Report V,* Item 15 at 2

113 "Article 1
" 1 . In any region where there are geographically disadvantaged coastal States, the nationals of such States shall have the right to exploit, on a regional and preferential basis, the renewable resources within maritime

zones beyond 12 miles from the coasts of the States of the region for the purpose of fostering the economic development of their fishing industry and statisfying the nutritional need of the population.

"2. The procedures regulating the preferential regime referred to In paragraph 1 above shall be determined by regional, subregional and bilateral agreements." UN Doc. A/AC.138/SCII/L.55 (1973)

114 *Ibid.,* reprinted in *Final Report V,* Item 2.3.2, pp. 22-23 and Item 6, pp. 1-2

115 *"Article 2: Issuing of licenses*

All exploration for, and exploitation of, the living or non-living resources of the Intermediate zone shall be Icensed by the coastal States, subject to the rules and regulations established by the competent International authorities (global, regional and/or subregional authorities)" (sic). UN Doc. A/AC.138/SCII/L.59 (1973) reprinted in *Final Report V,* Item 7.1, p. 7

116 *See* "Agreement Reached on Organization of Work" UN Doc. A/AC.138/SR/45 (1971)

117 *See* Item 5, UN GAOR, 1 Supp. 21 (Doc. A/9021) at p. 48 (1973)

118 UN Doc. A/AC.138/83, at p. 15 (1972)

119 Report of the AALCC Sub-Committee on the Law of the Sea (1971) reprinted in UN Doc. A/AC.138/34 at p. 7 (1971)

120 "Regional Arrangements

"That the African States in order to develop and manage the resources of the region take all possible measures including co-operation In the conservation and management of the living resources and the prevention and control of pollution to conserve the marine environment, establish such regional institutions as may be necessary and settle disputes between them In accordance with regional arrangements." UN A/AC.138/89 (1973) reprinted in 28 UN GAOR, 4 Supp. 21 (Doc. A/9021) at p. 4 (1973)

121 28 UN GAOR, 1 Supp. 21 (DOC. A/9021) at p. 55 (1973)

122 *See Final Report VI,* at pp. 97-105

123 UN Doc. A/AC.138/SR/45 (1971)

124 "Many delegations felt that a single comprehensive treaty of general application could be supplemented by treaties of regional application or of a technical nature, and by national legislation. It was also stated that the preparation of a general treaty would be facilitated if preceded by the drafting of multilateral treaties of regional application. Some delegations expressed the view that the elaboration of a treaty or treaties for the protection of the marine environment should be undertaken in conformity with the international regime and machinery for the exploration and exploitation of the seabed and ocean floor beyond the limits of national Jurisdiction ... Several delegations were of the view that the Sub-Committee, in draft-Ing a comprehensive treaty on marine pollution, should await the results of the UN Conference on the Human Environment. They felt that the items not adequately dealt with by the Stockholm and IMCO conferences should be dealt with by this Sub-Committee." UN Doc. A/AC.138/62, at p. 5 (1971)

125 "It was suggested that the basic materials for the work of the Sub-Committee should be the Declaration of the Human Environment, the 23 principles on marine pollution, and the statement of objectives, adopted at Stockholm, and referred to this Committee as well as the three principles drafted at Ottawa ...

"It was stressed that marine pollution could effectively be dealt with bey a combination of global, regional and national rules and standards, with the global ones fixing the minimum provisions to be made for the preservation of the marine environment, and the regional and national ones laying down particular and stricter provisions as may be required to deal with special situations prevailing in a region or a country. It was observed that broad guidelines would Improve regional efforts and could also prevent the emergence of a series of piecemeal conventions. Proliferation of Independent regional agreements could lead to difficulties in subsequent co-ordination.

"It was expressed that the task of the Sub-Committee Included examining the feasibility of drafting, for the 1973 Conference on the Law of the Sea, treaty articles of a general nature concerning pollution from all sources in ocean space as a whole so as to replace Articles 24 and 25 of the 1958 Geneva Convention on the High Seas. It was further pointed out that existing technical conventions, already concluded or under consideration, on various aspects of marine pollution or on pollution in specific regions of the world, could find their proper place within the framework of such general treaty articles". UN Doc. A/AC.138/84, at p. 7 (1972)

126 Of the 37 regional fishery or related conservation agreements listed in Table I, eleven were concluded between 1945 and 1958. Several of these agreements were general in scope, establishing the framework for all fishery co-operation In the region

127 For an argument in favour of the trend to "a system of voluntary negotiated schemes on a regional basis", *see* Johnston, *supra* note 37, pp. 456-458

128 O.J. Lissitzyn, "International Law in a Divided World" (1963), *Int'l Conciliation,* No. 542, p. 69

129 H. Gary Knight, "International Fisheries Management: A Background Paper" in Knight, *supra,* note 47, pp. 1-49, esp. 23-25; and Christy and Scott, *supra* note 37, pp. 217-221

130 Johnston, *supra* note 37, pp. 460-462

131 UNCLOS I references to regional arrangements are somewhat uncertain, because of its use of the ambiguous phrase "international conservation organizations". *See*, for example, Resolutions III ("International Fishery Conservation Conventions") and IV ("Co-operation in Conservation Measures") in the Final

Act of the Conference, reproduced in *Ibid.,* pp. 496-497

132 The Second UN Conference on the Law of the Sea was concluded on April 27th, 1960. For the Final Act of UNCLOS II, *see Ibid.,* pp. 499-500

133 Note, however, that two regional fishery conventions were concluded between UNCLOS I and UNCLOS II: the Northeast Atlantic Fisheries Convention and the Convention Concerning Fishing in the Black Sea, both in1959

134 Of the 37 regional fishery (or related) conservation agreements listed in Table I, only one was concluded between 1960 and 1967: the International Convention for the Conservation of Atlantic Tunas in 1966

135 Concern was chiefly aroused by the famous *Torrey Canyon* grounding at the mouth of the English Channel on 18th March, 1967

136 Johnston, *supra* note 51, p. 48

137 The most articulate advocate for this kind of linkage strategy for the development of international environmental law was Ambassador J. Alan Beesley of Canada. *See*, for example, his paper "The Canadian Approach to International Environmental Law" (1973), 11 *Can. Y.B. Int'l L.* 3

138 The Stockholm Conference was originally intended to be, *Inter alia,* a treaty-making exercise, but this objective was dropped in 1971 in favour of a Declaration, which would serve as a framework of guidelines. Douglas M. Johnston, "International Environmental Law: Recent Developments and Canadian Contributions" in R.St.J. Macdonald, Gerald L. Morris, and Douglas M. Johnston, *eds., Canadian Perspectives on International Law and Organization,* pp. 555-611 at p. 563. For an analysis of the Declaration, *see* Louis B. Sohn, "The Stockholm Declaration on the Human Environment" (1973), 14 *Harvard Int'l L.J.* 423

139 On the general principles and concepts emerging within these forums, *see* Jan Schneider, *World Public Order of the Environment: Towards an International Ecological Law and Organization* (1979), pp. 107-137

140 UN General Assembly Resolution 2749 (XXV), adopted 17th December 1970, reproduced in *United Nations Juridical Yearbook 1970,* pp. 55-57

141 Note, however, that paragraph II is exclusively devoted to environmental matters:

11. "With respect to activities in the area and acting in conformity with the international regime to be established, States shall take appropriate measures for and shall co-operate in the adoption and implementation of international rules, standards and procedures for *inter alia:*

(a) The prevention of pollution and contamination, and other hazards to the marine environment, including the coastline, and of interference with the ecological balance of the marine environment;

(b) The protection and conservation of the natural resources of the are and the prevention of damage to the flora and fauna of the marine environment".

(b) The protection and conservation of the natural resources of the area and the prevention of damage to the flora and fauna of the marine environment"

142 UN General Assembly Resolution 2750 C (XXV), adopted 17th December 1970, reproduced in *Yearbook of the United Nations 1970,* pp. 82-83

143 *Ibid.,* para. 6

144 *Ibid.,* para. 2

145 *Ibid.,* para. 7

146 Johnston, *supra* note 51, p. 49

147 *See*, for example the marine pollution proposals of Australia, Canada, the Soviet Union, and Malta, summarized in Shigeru Oda, *The Law of the Sea in Our Time: The United Nations Sea-bed Committee, 1968-1973* (1977), p. 252-256

148 UN General Assembly Resolution 2750, *supra* note 142, recognized (in the final paragraph of its preamble) that the Stockholm Conference might serve as a kind of "model" or analogue for UNCLOS III

149 Johnston, *supra* note 51, pp. 48-51

150 Douglas M. Johnston and Edgar Gold, "Extended Jurisdiction: The Impact of UNCLOS III on Coastal State Practice" (presented at the 13th Annual Conference of the Law of the Sea Institute at Mexico City, October 20th-24th, 1979)

151 *Ibid.,* Table 3, pp. (56-57)

152 *Ibid.,* pp. (5-17)

153 *Ibid.,* pp. (22-23)

154 Cyrille de Klemm, "Living Resources of the Ocean", *supra,* pp. 120 *et seq.*

155 Jan Schneider, "Pollution from Vessels", *supra,* p. 209

156 Most of the "official" debate merged with the unofficial. For a listing of the "hybrid" conferences held on this subject in 1971, *see Brian Johnson, "The United Nations' Institutional Response to Stockholm: A Case Study in the International Politics of Institutional Change", in David A Kay and Eugene B. Skolnikoff, eds., World Eco-Crisis: International Organizations in Response* (1972), pp. 87-134 at p. 88. On general trends in official thinking, *see* Richard N. Gardner, "The Role of the UN in Environmental Problems", *Ibid.,* pp. 69-86

157 Kay and Skolnikoff, *op. cit.,* and John Lawrence Hargrove, *ed., Law, Institutions, and the Global Environment* (1972). *See*, especially, Abram Chayes, "International Institutions for Environment" in Hargrove, *op. cit.,* pp. 1-26. *See* also Thomas W. Wilson, Jr., "Creating Mechanisms for International Environment Action:

Requirements and Response" (1975), 7 *Stanford J. International Studies* 113; and Minda R. Shecter, "Organizational Instruments for Effectiveness: A Comparative Study with Recommendations to a United Nations Environmental Agency", *ibid.,* p. 123

158 The Stockholm Conference adopted three principal instruments: the UN Declaration on Human Environment; an Action Plan consisting of 109 recommendations; and a detailed resolution on institutional and financial arrangements. UN Doc. A/CONF.48/14., reprinted in 11 *Int'l Leg. Mats.* 1416 (1972)

159 *Ibid.,* at p. 1466. This "arrangement" was to consist of a Governing Council for Environmental Programmes, an Environment Secretariat headed by an Executive Director, and Environment Fund, and an Environmental Co-ordinating Board. For a listing of the functions assigned to the Governing Council, *see ibid.* at p. 1467. In recent years, UNEP has been described, by two of its officials, as a "mobilizer and coordinator of efforts at all levels to protect and preserve the biosphere from unnatural stresses due to human activities". Peter S. Thacher and Nikki Meith-Avcin, "The Oceans: Health and Prognosis", in *Ocean Yearbook I* (1978) (Borgese and Ginsburg, *eds.),* pp. 293-339 at p. 317

160 G.A. Res. 2997 (XXVII), 26 UN GAOR, Supp. 30, pp. 43-45 (1972). This decision was taken after receiving the Report of the Secretary-General on the United Nations Conference at the Human Environment, UN Doc. A/8783 (26 September 1972). A comparison of these two documents shows that some time between September 26th and December 15th the reference to a "Governing Council for Environment Programmes" — the language used at Stockholm — was replaced by the reference to a "Governing Council of the United Nations Environment Programme". Similarly, the "Executive Director" became the "Executive Director of the United Nations Environment Programme". It must be inferred that views on the format and status of UNEP, as a single agency, had crystallized behind the scenes in the fall of 1972, but the process of determination is not reflected in the official UN records

161 The first listing of UNEP programme priorities, contained in UNEP/Report of the Governing Council on the work of its first session (217 UN GAOR, Supp. 25 (Doc. A/9025) at pp. 39-43 (1973)), was somewhat different: human settlements, human health, habitat and well-being; land, water and desertification; education, training, assistance and information; trade, economics, technology and transfer of technology; oceans; conservation of nature, wildlife and genetic resources; and engergy. This listing in the following year's Report remained unchanged, except that the category "education, training, assstance and information" dropped out. 28 UN GAOR, Supp. 25 (Doc. A /9625) at pp. 59-64 (1974). In the 1975 Report the listing of "priority subject areas" was: human settlements and habits; health of people and of the environment; terrestrial ecosystems, their management and control; environment and development; oceans; energy; and natural disasters. 29 UN GAOR, Supp. 25 (Doc. A/10025) at pp. 26-40 (1975). Since 1976 the listings have remained unchanged, with "oceans" occupying the fourth priority spot in a seven-point programme

162 Johnston, *supra* note 51, at p. 46 and note 228

163 *See,* for example, UNEP, *Report of the Governing Council on the work of its fifth session* (9-25 May 1977), UN GAOR, Supp. 25 (Doc. A/32/25) at p. 42-43 (1977)

164 Thacher and Meith-Avcin, *supra* note 159, p. 318

165 It is normal now to refer to UNEP's "co-ordinating and catalytic role" and its responsibility for "promoting and uniting the effort of all countries and organizations to solve major environmental problems". *See*, for example, UNEP, Report of the Governing Council on the work of its seventh session (18 April - 4 May 1979) (GAOR 34th Session, Supp. No. 25) (hereafter referred to as "UNEP, *Seventh Governing Council Report")*

166 This concern affects much of the work undertaken by UNEP, directly or indirectly. Yet most developing countries are coming round to the view that wise preventive measures, both environmental and non-environmental, can keep down the costs of development. "Prevention Costs Less than Cure: Caribbean", 274 *Economist* 87 (Jan. 26, 1980)

167 On UNEP's "legal development" policy, *see* "Developing Environmental Law" (1977), 3 *Environ. Policy and Law* 57

168 In addition to the advantages inherent in a regional strategy, UNEP's oceans programme has also derived considerable benefit from its location in Geneva. This has kept the staff of the Regional Seas Programme conveniently close to the nerve centre of the system of UN specialized agencies, and detached, in large part, from the internal politics with which its division (the Environmental Management Division) has had to contend in Nairobi. Relatively free from head office constraints, reasonably well funded, and yet able to preserve a low visibility, the Geneva staff has been impressively successful in conducting its diplomatic campaign with the missions and agencies

169 "Environmental assessment" was envisaged as including evaluation and review, research, monitoring, and information exchange. "Environmental management" encompassed goal-setting and planning, and international consultations and agreements. "Supporting measures" consisted of education and training, organization, financing, technical co-operation and public information. Action Plan for the Human Environment, 11 *Int'l Leg. Mats.* 1421 (1972)

170 Peter S. Thacher and Nikki Meith, "Approaches to Regional Marine Problems: A Progress Report on UNEP's Regional Seas Programme", in *Ocean Yearbook II* (1980) (Borgese and Ginsburg, *eds.)* pp. (3-4)

171 UNEP, *Seventh Governing Council Report,* p. 36

172 S. Keckes, "Theory and Practice of the United Nations Environment Programme in Dealing with Regional Marine Problems (unpublished, 1977), cited in Thacher and Meith, *supra* note 164, n. 3. For a challenge to the orthodox goal-setting approach to environmental planning, *see* Gerhard Kade, "Introduction: The Economics of Pollution and the Inter-disciplinary Approach to Environmental Planning. 21 *Int'l Soc. Scl. J.,* (no. 4) (1970) reprinted in Morton and Marsha Gofden, *eds., Environmental Management: Science and Politics* (1972), pp. 7-13; and W.R. Derrick Sewell and Harold D. Foster, "Environmental Risk: Management Strategies in the Developing World (1976), 1 *Environ. Man.* 49

173 (Comment) "The Challenge of International Environmental Management: A Critique of the United Nations Environment Programme" (1978), 18 *Virg. J. Int'l L.* 269 at pp. 274-276

174 Thacher and Meith, *op. cit.*, p. (4)

175 For a recent review, *see* S. Keckes, "The Regional Seas Programme of UNEP", *UNEP Industry and Environment,* vol. 3, no 3 (July-Sept. 1980), pp. 5-7. *See* also Stjepan Keckes, "Regional Seas: An Emerging Marine Policy Approach", in *Comparative Marine Policy: Perspectives from Europe, Scandinavia, Canada and the United States* (Center for Ocean Management Studies, University of Rhode Island 1981), pp. 17-20

176 Yet even in the Mediterranean not nearly enough Is known about the population of fish stocks. Sidney Holt, "Mediterranean Fisheries" in Norton Ginsburg, Sidney Holt, and William Murdoch, *eds., The Mediterranean Environment and the Development of the Region* (Pacem In Marlbus III, 1974), pp. 154-172

177 The ranking of marine regions according to the degree of risk of conflict is bound to be a somewhat impressionistic undertaking. *See*, for example, Douglas M. Johnston, "Development, Environment and Marine Resources In the North pacific", in Hedley Bull, *ed., Asia and the Western Pacific: Towards* a New *International Order* (1975), pp. 232-261

178 Every year the Mediterranean attracts about 100 million tourists, and its coastal communities provide a permanent residential population of another 100 million. As the sea remains the most popular recreational centre in the world, it is worth recalling that a yacht produces as much *organic* waste as an oil tanker. For an early warning about the problem, *see* Lord Ritchie-Calder, *The Pollution of the Mediterranean* (1971). *See* also W.W. Murdoch and C.P. Onuf, "The Mediterranean: An Ecological Overview", in Ginsburg, Holt and Murdoch, *supra* note 176, pp. 12-47

179 Although some have emphasized the fundamental unity of the "Mediterranean civilization" (e.g. Andre Siegfried, *The Mediterranean* (tr. Hemming, 1948), most scholars have taken a less romantic view of the history of the region (e.g. Ernie Bradford, *Mediterranean: Portrait of a Sea* (1971). Even in recent years It has been common to divide the region terrestrially Into four regions (the Europe Western Basin; the African Western Basin, the Balkan peninsula, and the Near East). D.S. Walker, *The Mediterranean Lands* (2nd *ed.,* 1962). It is now more fashionable, however, to look at the Western Mediterranean as a single arena of regional interaction. Alvin J. Cottress and James D. Theberge, *eds., The Western Mediterranean: Its Political, Economic and Strategic Importance* (1972); and Robert M. Burrel and Alvin J. Cottrell, *Politics, Oil and the Western Mediterranean* (1973)

180 Most of the Mediterranean states contribute to the problem of dumping untreated sewage directly into the sea. Indeed it has been estimated that 90% of all the raw sewage in the region is pumped Into the waters off-shore. "Mediterranean Pollution: The Tie that Binds", 3 *The Interdependent* (April 1, 1976), at p. 2, cited in Debora de Hoyo's "The United Nations Environment Programme: the Mediterranean Conferences" (1976), 17 *Harvard Int'l L.J.* 639, note 2. In many countries the rapid growth in resident or non-resident population — or, in a few cases, both — aggravates this form of pollution. In addition, Industrialization in most of these countries has led to a serious deterioration in the problem of controlllog the disposal of industrial wastes in river estuaries and other coastal areas. Moreover, the development of oil fields in Libya and Algeria has increased the danger of oil pollution

181 Traditionally the Mediterranean has been regarded as a biologically poor sea. Today It produces only about 830,000 tonnes of fish annually, but this total catch is worth $ 1.3 billion in value (at 1980 prices). With UNDP assistance, several Mediterranean states are involved in a major mariculture project in their coastal zone

182 Dominique Alheritière, "Regional Approaches to Marine Pollution Control" (unpublished paper, 1981)

183 Cyprus attended the second, but not the first, of these conferences. Albania attended neither. All other (15) Mediterranean coastal states attended both

184 Text in 14 *Int'l Leg. Mats.* 475 (1975)

185 *Ibid.,* pp. 481, 490 and 495

186 Text in 15 *Int'l Leg. Mats.* 290 (1976)

187 *Ibid.,* p. 300

188 *Ibid.,* p. 306

189 *Ibid.,* p. 311

190 *Ibid.,* p. 314. The Centre was established in December 1976 with the primary objectives of providing a mechanism for exchange of data relevant to oil pollution training in oil pollution control techniques and assistance in designing contingency plans. All of these texts — the Plan, the Final Act, the Resolutions, the Convention, and the two Protocols — are also published in UNEP, *Mediterranean Action Plan and the Final Act of the Conference of Plenipotentaries of the Coastal States of the Mediterranean Region for the*

Protection of the Mediterranean Sea (1978)

191 Text In 19 *Int'l Leg. Mats.* 869 (1980)

192 "Acceptablity"is normally first Indicated by the grant of government "approval" to the Action Plan itself. The act of approval seems to vary in degree of formality from state to state, but nowhere Is It so formal as to be legally equivalent to an act of consent, such as Is required to accept an International agreement. It might be assumed, however, that the government approval of a UNEP Action Plan normally requires the support of a Cabinet-level decision. To the legally minded, a state (as distinguished from a government) has not accepted obligations until Its consent is given through adoption (by signature and ratification, or accession) of a legally binding treaty Instrument, such as a convention, agreement, or protocol

193 The "efficacy", or beneficial effect, of an environmental arrangement may be difficult or Impossible to establish with any degree of certainty without the investment of years of scientific research. There is a tendency, however, to make a preliminary assessment of an environmental Initiative as efficacious if it generates an Impressive volume or kind of activitity that Is believed to be desirable

194 The value of an initiative as a "model" has declining internal relevance as the programme develops from region to region, but later, more carefully developed initiatives within the programme may have a higher value than earlier Initiatives as a model for other programmes or other agencies

195 For example, because of costs involved in mounting an efficacious programme, a government may be more likely to regard a programme as "acceptable" if it does not commit the government to a high standard of "efficacy". By the same token, an initiative that is seen to be efficacious In one region may not be followed too readily elsewhere If the costs are substantial

196 By the end of December 1978 ten states (Egypt, France, Israel, Lebanon, Malta, Monaco, Spain, Syrian Arab Republic, Tunisia, and Yugoslavia) and the EEC had become parties to the Barcelona Convention; the same ten states had become parties to the Oil Pollution Protocol; and all ten except Israel had become parties, along with the EEC, to Dumping Protocol. Moreover, all three instruments had been signed, but not yet ratified, by all other Mediterranean states except Albania: viz. Algeria, Cyprus, Greece, Italy, Libya, Morocco and Turkey. Since the seven non-ratifying states were all "(a) in the process of undergoing the necessary national formalities toward eventual ratification, and (b) actively participating in the Mediterranean Action Plan as a whole", they were Invited, along with the ten ratifying states and the EEC, to participate in the First Meeting of the Contracting Parties held at Geneva in February 1979. Doc. UNEP/1G.14/9, 20 April 1979 United Nations Environment Programme, *Report of the Inter-governmental Review Meeting of Mediterranean Coastal States and First Meeting of the Contracting Parties to the Convention for the Protection of the Mediterranean Sea against Pollution and its Related Protocols,* at p. 1

197 The Barcelona Convention was the first agreement to be signed by Israel and the Arab countries. In speaking to the International Ocean Institute at Vienna in October 1980, Dr. Keckes, Director of the Regional Seas Programme, emphasized that none of the Programme's meetings had ever been seriously disrupted by political arguments

198 As one of their first decisions under the Action Plan in 1975, even before adoption of the Barcelona Convention and accompanying Protocols, the governments of the Mediterranean littoral states approved a Coordinated Mediterranean Monitoring and Research Programme (MEDPOL). This programme encompassed seven pilot projects to be carried out by approved national institutions and supervised by UNEP in cooperation with FAO, UNESCO, WHO, WMO, and IAEA. These studies confirmed the suspicion that most pollution in the Mediterranean comes from land-based sources (municipal sewage, industrial wastes, agricultural fertilizers, and pesticides). For a survey of the evidence, see UN Doc. UNEPnG.11/INF.5 (10 November, 1977) ("Pollutants from Land-Based Sources in the Mediterranean"). For a brief description of MEDPOL, *see* Lawrence Juda, "The Regional Effort to Control Pollution in the Mediterranean Sea" (1979), 5 *Ocean Management* 125 at pp. 141-143.

The other area of research to be undertaken by the Mediterranean governments is outlined in the Integrated Planning Component of the Action Plan. A prominent feature of that component is the "Blue Plan", a coordinated programme of regional co-operation, which was adopted at an intergovernmental meeting in Split early in 1977. See UN Doc. UNEP/1G.5/7 (21 February, 1977) ("Report of the Intergovernmental Meeting of Mediterranean Coastal States on the "Blue Plan"). This Plan, whose primary objective Is the economic and social development of the region on a sound ecological basis, is of special interest to the less developed countries of the region. To them, the success of the "Blue Plan" would be the most significant test of the Action Plan as a whole. But it is precisely in this context, where environmental planning is integrated with economic development, that the greatest difficulties arise from differences in national priorities. At the Split meeting in 1977 six subjects were selected for integrated planning on a priority basis under the "Blue Plan": soil protection; water resource management; fisheries and aquaculture; human settlements; tourism; and technologies for energy. Thacher and Meith, *op. cit.,* pp. (21-26). Despite this apparent consensus on priorities, the "Blue Plan" has been slow In overcoming the problems caused by Inadequate funding and the lack of technical expertise. Several of the governments are particulary reluctant to commit resources to the study of the critical problems associated with land-based pollution of the Mediterranean. Juda, *op. cit.,* pp. 143-145. Another apparent accomplishment under the Action Plan was the

establishment in Malta of the Regional Oil Combating Centre directed by UNEP in co-operation with IMCO and the host government. *Ibid.,* p. 140. But this Centre also suffers in operational effectiveness from a scarcity of funds and expertise

199 This estimate was given by Dr. Keckes at the IOI conference referred to above, In footnote 197

200 Under Articles 5 and 6 the Protocol will bind the parties to "elaborate and implement ... programmes and measures" designed to fulfill two general objectives: the "elimination" of pollution of the Protocol area from land-based sources by substances listed in Annex I, and the "strict limitation" of pollution by substances or sources listed in Annex II. The Protocol goes so far as to require the adoption of guidelines, standards and criteria for such purposes as "the control and progressive replacement of products, installations and industrial and other processes causing significant pollution of the marine environment" (Article 7(1)(d))

201 Behind this kind of debate lies a philosophical issue concerning the relationship between form and substance. See Ernest B. Haas, "Is There A Hole in the Whole? Knowledge, Technology, Interdependence, and the Construction of international Regimes" (1975), 29 *Int'l Organ* 827

202 For example, other regions may find it difficult or unnecessary, in the 1980s, to follow the precedent set by the 1980 Land-Based Protocol, which commits the contracting states to undertake an expensive environmental modification of their Industrial economy. *See* note 200, *supra*

203 The Governing Council of UNEP operates an Environment Fund, from which it may draw to help finance designated undertakings on a temporary basis

204 UN Doc. UNEP/1G.14/9. This was approved by the UNEP Governing Council in May 1979 (text of resolution reproduced in 5 *Environ. Pol. & Law* 151 (1979))

205 If this reluctance is due chiefly to a perceived clash between environmental management and development priorities in the developing world, future difficulties may be particularly severe in some regions, such as West Africa, the Caribbean, and the Southwest Pacific. If it is due rather to developed-versus-developing tensions, then East Asia may expect to face the stiffest test

206 Text reproduced in 18 *Int'l Leg. Mats.,* (1979)

207 The eight participating countries of the region are Jordan, Saudi Arabia, Yemen Arab Republic, People's Democratic Republic of Yemen, Somalia, Ethiopia, Sudan and Egypt

208 Like the Arabian-Persian Gulf, "the Red Sea Is an area of rapid increase in oil exploration, production, and shipping and the environmental impact is expected to be considerable. Pollution by oil from spills and intentional discharge as well as the physical disturbance of construction and dredging operations are potentially stressful to local benthic communities". Thacher and Meith-Avcin, *supra* note 159, p. 335

209 A Draft Final Act, including a regional convention and protocol, was prepared by a meeting of legal and environmental experts at Jeddah in January 1981

210 The Action Plan calls for intensive development of the region's capabilities In basic and applied marine sciences, through seminars, workshops, training courses, and the establishment of laboratories. It may, therefore, be several years before local research can be carried out to provide the basis of environmental management. But the Red Sea, which seems to be a "new" sea, is intriguing to marine scientists around the world, and has aroused the special interest of the International Union for Conservation of Nature and Natural Resources (IUCN)

211 UN Doc. E/5971, 18 May 1977 ("Coastal Area Development and Management and Marine and Coastal Technology"), reproduced in *Ocean Yearbook I, op. cit.,* pp. 350-375

212 Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, United Arab Emirates, and Oman

213 UNEP, Final Act of the Kuwait Regional Conference of Plenipotentiaries on the Protection and Development of the Marine Environment and the Coastal Areas (1978), reproduced in 17 *Int'l Leg. Mats.* 501 (1978). The Conference was also attended by representatives of a dozen interested organizations. For a listing, *see* Lawrence D. Neuman, "The Protection and Development of the Marine Environment and Coastal Areas of the Kuwait Conference Region: The Programme of the United Nations Systems" in *Proceedings of the 1979 Oil Spill Conference, Los Angeles* (in press), pp. 287-291

214 The Kuwait Regional Convention for Co-operation on the Protection of the Marine Environment from Pollution, and the accompanying Protocol Concerning Regional Co-operation in Combating Pollution by Oil arid Other Harmful Substances in Cases of Emergency. 17 *Int'l Leg. Mats.* 511 and 526 (1978)

215 This fund amounted to almost $ 6 million for an initial period of 2-1/2 years. At present the governments are paying into the fund more money than they can spend

216 About 60% of all oil carried in ships around the world — around a billion tonnes per year — is exported from this region. Thacher and Meith, *supra* note 170, p. (29)

217 For a description of the co-operative projects to be managed by the Organization, *see Ibid.,* pp. (30-37). In the meantime, UNEP continues to provide an interim secretariat

218 These agreements entered into force on 30th June 1979, after the deposit of five instruments of ratification (Bahrein, Iraq, Kuwait, Oman, and Qatar)

219 There are no less than 20 existing or planned major industrial centres along the Gulf coast, representing something between 20 and 40 million dollars per kilometer of coastline, *Ibid.,* p. (29)

220 *Ibid.,* p. (38)

221 This region has been defined as including the marine environment and coastal area of 19 states from Senegal in the north to Namibia in the south, encompassing the Gulf of Guinea and adjacent waters to the north and south

222 Benin, Cameroon, Gabon, Gambia, Ghana, Ivory Coast, Liberia, Nigeria, Senegal, Sierra Leone, and Togo

223 Final Act of Conference of Plenipotentiaries on Co-operation in the Protection and Development of the Marine and Coastal Environment of the West and Central African Region (1981)

224 No less than 12 of these 19 West African states are "shelf-locked" by virtue of the concavity of the Gulf of Guinea. The remaining seven have short, congested coastlines along an approximately 200-mile facade facing the East Central Atlantic

225 Because of the length of time it often takes to resolve maritime boundary disputes, through negotiation or adjudication or both, it may become fairly common for neighbouring states (e.g. Canada and the United States in the Gulf of Maine) to seek some kind of tentative or provisional management arrangement pending such a resolution. The West African situation is special only in the number of neighbouring states bordering on an area that appears to be a "natural" management unit

226 Moreover, like the Southwest Pacific, the Caribbean consists in large part of extremely new micro-states, with the infrastructure of a county rather than of a nation, as well as several colonial territories still evolving toward national independence. Altogether, the region includes 19 island nations, a dozen of which might be regarded as having these special problems

227 Strictly speaking, the Caribbean region consists of two semi-enclosed seas: the Caribbean Sea proper and the Gulf of Mexico. For a recent discussion of the future of marine regionalism in the region as a whole, see Andres Rosental, "Some Brief Considerations on a Caribbean Condominium" (1978), and the resulting discussions, in Johnston, *supra* note 10, pp. 47-62. In recent years the region to be covered by the Action Ran has been described as the "Wider Caribbean Region". This region is defined as comprising "the States and South America from Colombia to French Guiana, Panama, the States of Central America, Mexico, the Gulf States of the United States, as well as the coastal and open waters of the Gulf of Mexico, and the waters of the Atlantic Ocean adjacent to the States and Territories mentioned above"

228 Thacher and Meith, *op. cit.,* p. (32)

229 For example, a detailed mapping inventory of the Caribbean marine environment, under the direction of Dr. Carleton Ray for IUCN, was completed in 1979

230 UN Doc. UNEP/RS/PAC, p. 11

231 For example, "Options for Financial and Institutional Arrangements" (Doc. E/CEPAUPROY.3/L4); "Principles and guidelines which may be used for a regional legal agreement" (Doc. E/CEPAL/PROY. 3/L5); "Development and Environment in the Wider Caribbean Region: A Synthesis" (Doc. E/CEPAL/PROY. 3/LINF.3); "The State of Marine Pollution in the Wider Caribbean Region" (Doc. BCEPAL/PROY. 3/L.INF.4); "Overview on Energy and Environment in the Caribbean Area" (Doc. E/CEPAL/PROY. 3/L.INF.9); "Overview on Natural Resources for Food and Agriculture in the Wider Caribbean Region" (Doc. E/CEPAL(PROY. 3/L.INF.10); "Overview on Environmental Health in the Wider Caribbean Region" (Doc. E/CEPAL/PROY. 3/I.INF.11); "Marine and Coastal Area Development in the Wider Caribbean Area: Overview Study" (Doc. E/CEPAL/PROY. 3/L.INF.13); and "Review of International Conventions relevant to the Environmental Protection of the Wider Caribbean Area" (Doc. E/CEPAL/PROY. 3/L.INF.15)

232 IUCN Bulletin, vol. 12, no. 5-6 (May/June 1981), p. 27

233 Donald K. Atwood, "The International Oceanographic Commission (IOC) and its Regional Association for the Caribbean and Adjacent Regions (IOCARIBE)" (1979) (in press)

234 A second, updated version of the Directory of Mediterranean Marine Research Centres, listing more than 140 institutions, was issued in November 1977

235 Thacher and Meith, *op. cit.,* pp. (35-39)

236 Even the South China Sea "subregion" is an area of immense complexity. Mark J. Valencia, "South China Sea: Present and Potential Coastal Area Resource Use Conflicts" (1979), 5 *Ocean Management* 1

237 This division of the region was recommended at the Penang International Workshop on Marine Pollution in East Asian Waters

238 For an excellent analysis of the prospects of regionalism in the South China Sea, *see* Mark J. Valencia, "Southeast Asia: National Marine Interests and Marine Regionalism" (1978), 5 *Ocean Development and Int'l Law* 421

239 The member nations are Indonesia, Malaysia, the Philippines, Singapore, and Thailand

240 The preparation of this Draft Plan, originating in the Penang Workshop of April 1976, has been accelerated by the prospect of an emerging framework of subregional co-operation in the marine sciences in accordance with an Indonesian prosposal approved by the ASEAN Permanent Committee on Science and Technology at its sixth meeting at Kuala Lumpur in December 1976. The ASEAN Working Group on Marine Sciences, which held its first meeting at Jakarta in June 1977, has agreed on two projects: the ASEAN Co-operative Studies on Tides and Tidal Phenomena and the ASEAN Co-operative Programme on Oceanography. Other ASEAN groups have been involved in the preparations, such as the ASEAN Experts

Group on Marine Pollution, which had met five times by February 1979, and the ASEAN Experts on the Environment, which had met twice by September 1979. Much of the Institutional research resulted from a series of UNEP visits to all five nations in 1977 and follow-up discussions with ASEAN officials concerned with environmental matters

241 For example "Report of the FAO/UNEP (in collaboration with IOC) Expert Consultation Meeting on Assessment of OH Pollution and its Impact on Living Aquatic Resources in Southeast Asia" (Doc. UNEP/WG 41/INF.11); "Research into Toxicity of Oil Dispersal Chemicals on Tropical and Sub-Tropical Marine Species" (Doc. UNEP(WG 41/INF.12); "Proposal for Monitoring Studies on the Contamination of the East Asian Seas by Trace Metals and Organochlorines" by D.J.H. Phillips (Doc. UNEP/WG. 41/INF.13); "Status Report on Research and Degradation Problems of the Coral Reefs of the East Asian Seas" by E.D. Gomez (Doc. UNEP/WG. 41/INFS.15); "Activities and Achievements of the Indo-pacific Fishery Commission (IPFC) and the South China Sea Development and Co-ordinating Programme (SCSP)" (Doc. UNEP/WG. 41/INF.16); "Review of Environmental Management Needs Related to Coastal Area Developments and Coastal Water Pollution Problems in the East Asian Region" (Doc. UNEP/WG. 41/INF.20); "Guidelines for a Convention on the Protection of the Marine Environment in the East Asian Region" (Doc. UNEP/WG. 41/INF.24); "Draft Protocol Concerning Co-operation In Combating Pollution In the East Asian Region by Oil and Other Harmful Substances in Cases of Emergency" (Doc. UNEP/WG. 41/INF.25); and "Options for Institutional and Financial Arrangements Required for the Implementation of the Action Plan for East Asian Seas" (Doc. UNEP/WG. 41/INF.26)

242 UNEP, *Report of the (First) Meeting of Experts to Review the Draft Action for the East Asian Seas* (Doc. UNEP/WG. 41/4), Annex IV

243 Doc. UNEP/IG. 26/6 (Report of Intergovernmental Meeting on the Protection and Development of the Marine Environment and Coastal Areas of the East Asian Region)

244 UNEP, *Report of the Governing Council on the work of its seventh session* (18 April - 4 May 1979), 34 UN GAOR, Supp. 25, pp. 22-23 (1979)

245 For the special problems of this subregion, *see Report of the IOC/FAO(IPFC)/UNEP International Workshop on Marine Pollution in East Asian Waters,* held at Penang in April 1976 (IOC Workshop Report No. 8, 1976), p. 25

246 *See* UN Doc. E/ESCAP/L. 46 (No. 13). Since 1978 much of ESCAP's work on the protection of the marine environment and related eco-systems has been undertaken under a project implemented by its Environmental Co-ordinating UNIT (ECU) in Bangkok, with extra budgetary assistance from Sweden. A series of national seminars conducted in ESCAP countries with severe marine pollution problems have procured significant data. *See*, for example, *Proceedings of the Thailand National Seminar on Protection of the Marine Environment and Related Ecosystems* held at Bangkok in June 1979 (Office of the National Environment Board, 1979). The same ECU project organized a regional seminar In the summer of 1980 to examine technological implications and legal developments. ECU has been particularly active in the collection and analysis of national legislation on marine pollution in selected countries of the ESCAP region. *See* "Legislative Aspects of Protection of the Marine Environment in the ESCAP Region", Doc. ECU/PMERE/2 (4 July 1980)

247 For example, FAO and UNEP collaborated with UNESCO and IUCN In organizing the Expert Consultation Meeting on the Impact of Pollution on the Mangrove Ecosystem and its Productivity in Southeast Asia, held at Manila in February 1980. *See* Doc. UNEP/WG. 41/INF. 14

248 Namely, Bangladesh, China, Hong Kong, Japan, Pakistan, Papua New Guinea, the Republic of Korea, Sri Lanka, the USSR, and Viet Nam. For recommendations, *see* ESCAP, *Report of the Regional Meeting on the Protection of the Marine Environment and Related Ecosystems in Asia and the pacific* (Doc. E/ESCAP/IHT. 4/21)

249 At Kuala Lumpur in August 1980 the "Asian Symposium on Mangrove Environment: Research and Management" focused sharply on these problems. The IUCN Commission on Ecology has recently formed an International Working Group on Mangrove Ecosystems to broaden the research effort

250 Munadjat Danusaputro, "Elements of an Environmental Policy and Navigational Scheme for Southeast Asia, with Special Reference to the Straits of Malacca (1978), in Johnston, *supra* note 10, pp. 171-178

251 UNEP *An Introduction to the United Nations Environment Programme* (1977), p. 5

252 South Pacific Commission, *Regional Co-operation In the South Pacific: History, Aims and Activities* (1975). The Commission has developed an interest in conservation problems over the years. In 1971, for example, it organized the "Regional symposium on the Conservation of Nature: Reefs and Lagoons"

253 In the process UNEP split up the only designated region that coincided with an ocean basin. *See* Alexander's listing, *supra* note 11

254 Report on the State of the Environment in the South-West Pacific Region; Draft Declaration of Principles on the Management and Improvement of the Environment in the South-West Pacific Region; Draft for a Regional Plan in Development and Environmental Protection; and Proposals for the Administration and Financing of the Action Plan

255 UNEP, *The Siren,* no. 8 (Spring 1980), pp. 3, 11

256 UN Doc. UNEP/RS/PAC, p. 13

257 Chile, Colombia, Ecuador, Panama, and Peru
258 UN Doc. UNEP/RS/PAC, pp. 13-15. *See* also UNEP, *The Siren,* no. 8 (Spring 1980), pp. 2-3
259 *See*, for example, Charles E. Lindblom, "The Science of Muddling Through" (1959), 19 *Public Admin. Rev.* 79, and "Still Muddling, Not Yet Through" (1979), 39 *ibid.* 517
260 David Braybrooke and Charles E. Lindblom, *A Strategy of Decision: Policy Evaluation* as a *Social Process* (1963)
261 This trend is most developed in the United States. For an early rationale, see Bostwick H. Ketchum, *ed., The Water's Edge: Critical Problems of the Coastal Zone* (1972)
262 For a recent survey of these trends, *see Johnston and Gold, supra* note 150
263 On the need to combat the "mystique of the national frontier", *see* Richard Falk, *This Endangered Planet: Prospects and Proposals lor Human Survival* (1979)
264 *See* note 11, *supra*
265 It is difficult to envisage the remaining designated regions — the East Asian Seas, the Southwest Atlantic, and East Africa — as scientifically "natural" units of management
266 For an account of its origins, *see* Gove Hambridge, *The Story of FAO* (1955). The International Labor Organization was earlier, but it had its antecedent under the League of Nations
267 On the frustrations of the Fisheries Division, *see* Joseph M. Jones, *The United Nations at Work: Developing Lands, Forests, Oceans — and People* (1965), pp. 209-215
268 FAO, *Report of the Third Session of the Conference* (1947), p. 11
269 Koers, *op. cit.,* pp. 92-95
270 *Ibid.,* pp. 108-109
271 *Ibid.,* pp. 109-110
272 *Ibid.,* p. 110
273 As early as 1949 the FAO Conference acknowledged that FAO should give "full consideration to work being carried on by international fisheries commissions or similar bodies already functioning in international waters". FAO, *Report of the Fifth Session of the Conference* (1950), p. 29
274 At the same meeting, the Conference warned that "care should be taken, in the formation of further regional fisheries councils, that there is sufficient technical fisheries personnel available in the potential member countries to ensure the councils' success". *Ibid.*
275 The question of "balance" between Headquarters and the regional offices has been one of continuing tension inside FAO, as in most other major UN agencies. *See* FAO, *Report of the Ninth Session of the Conference* (1958), pp. 129-135, where a number of recommendations concerning the strenthening of FAO's regional structure was set forth. *See* also FAO, *Report of the Fourteenth Session of the Conference* (1968), p. 61
276 The problem arose partly from the overall budgetary inadequacies of the FAO Fisheries Division, and partly from the unwillingness or inability of the members of the regional commissions to contribute to the administrative costs involved. See, for example, FAO, *Report of the Seventh Session of the Conference* (1954), p. 79; and *Report of the Ninth Session of the Conference* (1958), p. 102. Even in 1980 the entire cost of administering the Indo-Pacific Fisheries Council has to be met by FAO
277 Significantly, only one new FAO regional fishery commission came into operation between 1949 and 1967: namely the Regional Fisheries Advisory Commission for the South West Atlantic (CARPAS), in 1967
278 FAO, *Report of the Twelfth Session of the Conference* (1964), at p. 30
279 *Ibid.*
280 Koers, *op. cit.,* pp. 106-107. *See* also FAO, *Report of the Thirteenth Session of the Conference* (1966), pp. 78-81. For its terms of reference, Rule XXX (revised), *Ibid.,* pp. 80-81
281 *Ibid.,* p. 48
282 FAO, *Report of the Fourteenth Session of the Conference* (1968), p. 60
283 "The Conference was aware that the Organization's support for its regional fishery bodies had been weak. It seemed still not strong enough. The Conference, therefore, called for more vigorous technical and organizational support of the regional fishery bodies". *Ibid.,* at p. 61. The Conference went on to deplore the time it takes to produce conservation action at the original level, from the point of conception through the process of analysis to adoption and execution of an agreement. In referring to the Atlantic tuna, hake, and the Indian Ocean tuna, it commented: "During these periods the fishery situations for the regions and the stocks concerned had worsened: lack of adequate secretariat resources contributed to these delays". *Ibid.*
284 Koers, *op. cit.,* pp. 110-111
285 Ibid., pp. 111-112
286 *Ibid.,* pp. 112-113
287 FAO, *Report of the Fourteenth Session of the Conference* (1968), p. 60
288 Koers, *op. cit.,* pp. 113-114
289 Ibid., pp. 114-115
290 FAO, *Report of the Twentieth Session of the Conference* (1980), pp. 30-31
291 FAO, *Report of the Eighteenth Session of the Conference* (1976), at p. 39

292 FAO, *Report of the Twentieth Session of the Conference* (1980), at p. 31

293 "Within the areas served by Commissions like IPFC we intend to establish fisheries management areas which will in fact be *natural* management areas. This means areas whose borders are drawn, not against the grain of biological and other realities, but *with* the grain. This means, for Instance, that we would group together nations with common fisheries problems and opportunities — for Instance, countries which share fish stocks and fisheries and which have the co-operative advantage of cultural and other affinities. Each of these natural management areas will be served by an organization reporting to the Commission. Each will be equipped with its own multi-disciplinary technical support team: biologists, technologists, economists, legal experts and others." Opening statement by Kenneth C. Lucas, Assistant Director-General (Fisheries), FAO, to the 19th session of the Indo-Pacific Fishery Commission, Kyoto, Japan, 21-30 May, 1980 at p. 2

294 FAO, *Report of the Twentieth Session of the Conference* (1980), p. 29

295 United Nations, Department of Public Information, *Everyone's United Nations (9th ed.)* (1979), p. 15. *See* also (1976) 30 *YBUN* (1979) pp. 1076-88

296 The functional commissions consist of the Statistical Commission; the Population Commission; the Commission for Social Development; the Commission on Human Rights; the Commission on the Status of Women; and the Commission on Narcotic Drugs. (*Note:* None of these conduct regional activities related to the marine environment.)

297 These are the Economic Commission for Africa (ECA), the Economic Commission for Europe (ECE), the Economic Commission for Latin America (ECLA), the Economic Commission for West Africa (ECWA) and the Economic and Social Commission for Asia and the Pacific (ESCAP). *(Note:* Their regional activities related to the marine environment, if any, are discussed later in this section.)

298 There are six standing Committees on Non-Governmental Organizations — Negotiations with Intergovernmental Agencies, Programme and Co-ordination, Natural Resources, Review and Appraisal, and Science and Technology for Development, and one Commission on Transnational Corporations. *(Note:* None of these conduct regional activities related to the marine environment.)

299 These are the Committee on Food Aid Policies and Programmes; the Council of the United Nations University (UNU); the Trade and Development Board of the United Nations Conference on Trade and Development (UNCTAD); the Industrial Development Board of the United Nations Industrial Development Organization (UNIDO); the International Narcotics Control Board; Office of the High Commissioner for Refugees (UNHCR); Executive Board of the United Nations Children's Fund (UNICEF); Governing Council of the United Nations Development Programme (UNDP) and United Nations Environment Programme (UNEP); United Nations Institute for Training and Research (UNITAR), Board of Directors of the United Nations Research Institute for Social Development (UNRISD); Boards of Governors of the United Nations Special Fund and the United Nations Special Fund for Land-locked Developing Countries; and the World Food Council. *(Note:* The regional activities related to the marine environment conducted by UNEP were discussed in the preceding section; those conducted by UNU and UNDP are discussed later in this section.)

300 "The intergovernmental agencies are separate, autonomous organizations related to the United Nations by special agreements. They have their own membership, legislative and executive bodies, secretariats and budgets, but work with the United Nations and with each other through the co-ordinating machinery of the Economic and Social Council".

Fifteen agencies are known as 'specialized agencies', a term used in the Charter. They report annually to the Economic and Social Council. The specialized agencies are:

International Labor Organization (ILO)
Food and Agriculture Organization of the United Nations (FAO)
United Nations Educational, Scientific and Cultural Organization (UNESCO)
World Health Organization (WHO)
International Bank for Reconstruction and Development (World Bank, IBRD)
International Finance Corporation (IFC)
International Development Association (IDA)
International Monetary Fund (Fund, IMF)
International Civil Aviation Organization (ICAO)
Universal Postal Union (UPU)
International Telecommunication Union (ITU)
World Meteorological Organization (WMO)
Inter-Governmental Maritime Consultative Organization (IMCO)
World Intellectual Property Organization (WIPO)
International Fund for Agricultural Development (IFAD)

The International Atomic Energy Agency (IAEA) and the General Agreement on Tariffs and Trade (GATT) are also listed among the related agencies. *Everyone's United Nations supra* n. 295 at pp. 16-17. *(Note:* The regional activities related to the marine environment conducted by FAO were discussed earlier in this section; those conducted by UNESCO, IMCO, ILO, WHO, WMO, and IAEA are discussed later in this section.)

301 For a concise summary, *see* UN Doc. ST/OPI/624 (Nov. 1978) (hereinafter cited as *UN Restructuring Summary)*

302 ECOSOC Res. 1768 (LIV), 18 May 1973
303 GA Res. 3201 and 3202 (S-VI), 1 May 1974
304 ECOSOC Res. 1896 (LVII), 1 Aug. 1974
305 GA Res. 3362 (S-VII), 16 Sept. 1975
306 32 UN GAOR, Supp. No. 45 (Doc. A/32/45), pp. 121-127 (1978). The following is extracted from the UN *Restructuring Summary, supra* n. 301 at pp. 5-14

"THE GENERAL ASSEMBLY, its role as supreme organ in the economic and social fields strongly reaffirmed, will:

— Set overall policy for the system;
— Review and evaluate developments within and outside the UN system;
— Rationalize allocation of work among its committees dealing with economic and social questions;
— Ensure documentation is concise, action-oriented and relevant.

"THE ECONOMIC AND SOCIAL COUNCIL will:

— Be the central forum for discussion of international economic and social issues;
— Formulate relevant policy for the system and member States;
— Monitor and evaluate budgets, policies, programmes and operational activities of the system;
— Assume, where appropriate, direct responsibility for work of subsidiary bodies, which will be dissolved, regrouped, streamlined or redefined as necessary;
— Hold more frequent, shorter, subject-oriented sessions;
— Hold special or ministerial-level sessions when necessary;
— Find ways to make itself fully representative of the United Nations;
— Have more active participation and assistance by the Secretary-General and Executive Director of the system.

"OTHER FORUMS will:

— Promptly and fully give effect to Assembly and ECOSOC policy recommendations;
— Take into account needs of developing countries.

Also:

— The major role of UNCTAD in negotiating international trade and economic co-operation areas is reaffirmed.

"THE REGIONAL COMMISSIONS will:

— Exercise team leadership and responsibility for regional co-operation and co-ordination, taking into account activities of other family members, particularly UNDP;
— Take part in global policy-making and programme implementation, *operating where appropriate as executing agencies* [emphasis added];
— Together with other family members, reach a common definition of regions and subregions, locating offices accordingly;
— Expand co-operation among countries and at the subregional, regional and interregional levels;
— Streamline and rationalize their subsidiary machinery;
— Be given the necessary authority and financial resources to carry out their work programmes.

"OPERATIONAL ACTIVITIES:

— Will have four objectives:
  (a) to increase real resource flows for development assistance;
  (b) to conform with recipient country objectives;
  (c) to reflect overall Assembly and ECOSOC policies; and
  (d) to improve efficiency and lower administrative overheads to increase resources available to recipient countries.
— Will be subject to integration measures gradually undertaken with due regard to current levels of voluntary contributions, a major premise being that integration will stimulate higher levels of contributions.
— Without prejudice to other fund-raising arrangements, or the earmarking of contributions, to particular programmes, will include the holding of a single annual UN Pledging Conference
— Will be carried out by a system which has maximum uniformity of financial, administrative, personnel, planning and procurement procedures.

Also:

— The General Assembly may consider setting up a single, widely representative governing body to oversee operational activities. This would replace existing bodies (UNEP, UNICEF and WFP excepted);
— A single officer will be responsible for all system activities in a country, with country offices unified;
— More developlng-country representation will be provided at executive and policy-making levels in secretariats dealing with operational activities.

"INTERAGENCY CO-ORDINATION:

— Has two aims:
  (a) to assist preparations for Intergovernmental decision-making; and
  (b) to translate these decisions into action

— Will focus on three tasks:
(i) preparation of recommendations;
(ii) bringing into concert work of UN family; and
(iii) developing joint planning, programming and execution.

Also:

— Work of ACC (Administrative Committee on Co-ordination) will be streamlined and minimized with more reliance on *ad hoc* arrangements;
— ACC will take over the work of ECB (Environment Co-ordination Board), IACB (Inter-Agency Consultative Board) and UNIDO Advisory Committee;
— Better communication will be achieved between intergovernmental bodies and ACC, including appropriate intergovernmental representation during Its discussions.

"UN SECRETARIAT SUPPORT SERVICES in the economic and social areas will be restructured to perform six main functions:

(a) Interdisciplinary research and analysis;
(b) Cross-sectoral analysis of programmes and activities;
(c) Substantive support for technical co-operation activities;
(d) Management of technical co-operation activities;
(e) Provision of support services for Assembly and other bodies; and
(f) Other research and analysis.

— The reconstituted Secretariat will allocate these functions among three units: (a) and (b) in one (Department of International Economic and Social Affairs (DIESA)); (c) and (d) in the second (Department of Technical Co-operation for Development (DTCD)); (e) in a third (office of Secretariat Services for Economic and Social Matters); with (f) distributed among all three as appropriate.
— Secretary-General to appoint a Director-General for Development and International Economic Co-operation (Kenneth K.S. Dadzie of Ghana was appointed) to assist him in the economic and social fields. The Director-General will:
(a) provide leadership and co-ordination to ensure system-wide co-ordination; and
(b) ensure coherence, co-ordination and efficient management in these fields".

307 GA Res. 33/202 (XXXIII), 29 Jan. 1979. 33 UN GAOR, Supp. No. 45 (Doc. A/33/45), p. 131 (1979)
308 *See* the *Report of the Secretary-General* to the Thirty-fourth Session of the General Assembly on this restructuring and decentralization. UN Doc. A/34/649 (8. Nov. 1979)
309 This office has published *Manganese Nodules: Dimensions and Perspectives* (1979) based on a meeting of experts in Nov.-Dec. 1977. *See* IMS (Int'l Marine Sciences) *Newsletter* (no. 24), p. 2 (winter 1979-80). Other OETO activities have included: (a) Regional background studies in such developing areas as the Caribbean and the Straits of Malacca (*see* Lewis M. Alexander, "Regionalism as an Approach to Marine Science and Technology Transfer" in 13 *Marine Tech. Soc'y J.* (1979), pp. 30-34 at p. 31); and (b) Joint OETO-CCOP Workshop on Coastal Area Development and Management for East Asian Countries (3-12 Dec. 1979) (*see* UN Doc. E/ESCAP/170, p. 107 (1979))
310 UN Doc. E/1978/118 (19 Jul. 1978) at p. 6
311 ECOSOC Res. 1979/64, 3 Aug. 1979. (1979) ESCOR, Supp. No. 1A (Doc. B1979/79/Add.1) at p. 13
312 UN Doc. A/34/649 (8 Nov. 1979) at p. 6
313 The priority sectors identified are: water resources for ESCAP and ECLA; public administration and finance for ECA and ECWA; and transport for ECE. *Ibid.* at p. 14
314 This proposal submitted by ECWA relating to OETO will be discussed later in this section
315 UN Doc. A/34/649 (8 Nov. 1979) at p. 5
316 *Everyone's United Nations supra* note 295 at p. 217
317 Members: Albania, Austria, Belgium, Bulgaria, Byelorussia, Canada, Cyprus, Czechoslovakia, Denmark, Finland, France, German Democratic Republic, Federal Republic of Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Spain, Sweden, Switzerland, Turkey, Ukraine, USSR, United Kingdom, United States, and Yugoslavia. *Ibid.,* at p. 221
318 *Ibid.,* at p. 220
319 *See ECE Annual Report,* (1979) ESCOR, Supp. No. 12 (Doc. B1979/47; E/ECE/983), at 11-14, 61-64
320 This meeting was scheduled to be convened from 13 to 16 November 1979. *See ECE Annual Report,* (1979) ESCOR, Supp. No. 12 (Doc. E/1979/47/Add.1; E/ECE/983/ADD.1), p. 5
321 *Ibid.,* at p. 223
322 *Ibid.,* at p. 224
323 *Ibid.*
324 North America, Europe and part of Asia
325 Members: Afghanistan, Australia, Bangladesh, Bhutan, Burma, China, Democratic Kampuchea, Fiji, France, India, Indonesia, Iran, Japan, Lao People's Democratic Republic, Malaysia, Maldives, Mongolia, Nauru, Nepal, Netherlands, New Zealand, Pakistan, Papua New Guinea, Philippines, Republic of Korea, Samoa, Singapore, Sri Lanka, Thailand, Tonga, USSR, United Kingdom, United States, and Vietnam

326 *See Mekong Committee Annual Report, 1978.* UN Doc. E/CN.11/WRD/MKG/L.417 (1977).
327 *See Report of CCOP sixteenth session.* UN Doc. E/ESCAP/17O (1978)
328 *See Report of CCOP-SOPAC seventh session.* UN Doc. E/ESCAP/L.29 (1978)
329 *See Mekong Committee Annual Report, 1977, supra* note 326 at pp. 146-47
330 The original Committee was comprised of representatives of the four riparian States: Laos, Cambodia (Kampuchea), Thailand and Vietnam. However, an Interim Committee without Kampuchea was formed in 1978
331 "The fishery programme of the Committee has three parts: (a) surveys and studies of fish and other aquatic life, to identify problems, anticipate adverse effects and devise development and management measures; (b) development and management activities, to increase fish production; and (c) rehabilitation measures, to reverse possible adverse trends in the production of commercially Important fin-fish and shell-fish likely to be endangered by dam construction". *Ibid.,* p. 121
332 FISHERY MANAGEMENT PROJECTS:
— Basinwide fishery studies
— Great Lake/Toule Sap fisheries
— Brackish water fish seed prospecting
— Pilot projects for reservoir fisheries
— Pilot fish farms
— Fish feed mills study
— Prawn/shrimp farming
— Catfish propagation
— *Ibid.,* at p. 165. These projects are described in detail, *ibid.,* at pp. 121-132
333 *Ibid.,* at p. 144
334 Members: China, Indonesia, Japan, Republic of Korea, Malaysia, Papua New Guinea, Philippines, Singapore, Thailand, Trust Territory of the Pacific Islands and the Socialist Republic of Vietnam. *See Report of CCOP sixteenth session, supra* note 327 at pp. 34-36
335 *Ibid.,* at p. 20
336 These include:
(a) Review of governmental regulations and administrative machinery for environmental protection with respect to off-shore petroleum exploration and exploitation activities, as requested by individual member countries or groups of member countries;
(b) Co-operation with other international organizations which might be invited to undertake regional syntheses of accumulated oceanographic and environmental data, to prepare descriptive atlas of physical, chemical, and biological oceanographic data and hydrometeorology, which would provide the basis for guiding off-shore exploration and exploitation plans, and to develop environmental protection measures;
(c) Periodic review of environmental base-line studies and field monitoring of specific marine environmental parameters conducted by other organizations in CCOP areas; and
(d) Study of geological and other natural hazards (such as earthquakes, tsunamis, submarine volcanic eruptions, coastal and submarine landslides) which could hinder off-shore petroleum and mineral development, and of technical incidents influencing the marine environment such as damage by dredging or explosives, landslides and spills caused by exploraton work. *Ibid.,* at pp. 99-100
337 The Joint CCOP-IOC Working Group on IDOE Studies of East Asia Tectonics and Resources (SEATAR). *Ibid.,* at p. iv
338 *Ibid.,* at p. 116
339 *Ibid.,* at p. 24
340 *See* UN Doc. E/ESCAP/75 (12 Jan. 1978) at p. 15
341 *Ibid.,* at p. 16. *See* also *ESCAP 1979 Annual Report* (1979) ESCOR, Supp. No. 13 (Doc. E/1979/48; BESCAP/142) at p. 28
342 This meeting, held in Bangkok during July 1978, "endorsed the general guidelines for the formulation of environmental protection legislation in various areas but emphasized the need to pay special attention to three areas: forestry, land use planning and marine pollution". *ESCAP 1979 Annual Report, supra* note 341 at pp. 27-28
343 UN Doc. E/ESCAP/80 (16 Jan. 1978), at p. 48 "In the medium term, a number of proposals commend themselves. First, at the regional level much needs to be done to understand better the interaction of natural and social processes. Better measures are required to record changes in the stock of natural capital and to quantify present patterns of resource use. Research on such aspects should be fostered at the regional and the subregional level and the results of such research disseminated in appropriate forums. Secondly, development planning techniques and required data and information flows need to be generated which would better incorporate environmental dimensions in all aspects of planning and as an inherent part of the planning process. At the regional level, the formulation of such techniques, generation of such data flows and the dissemination of such results could be developed considerably.

"... Thirdly, the concept of 'eco-development', as sponsored by the United Nations Environment Programme (UNEP), could be actively fostered at the subregional level. Eco-development stresses the "holistic' rather than the 'sectoral' nature of the planning process in such a way that it gives rise to the ecologically sound use of the resources of a given eco-system in order to satisfy the basic needs of the resident population. River basin development in the ESCAP region is an exellent case where subregional action could foster significant eco-development systems.

"... A fourth area where regional action could be taken is in sponsoring of alternative energy systems ...

"... Finally, while any comprehensive treatment of the exploitation of the resources of the world's oceans must await global developments, regional co-operation could proceed independently in certain areas. Action must be initiated to deal with problems of pollution, with the management of fish resources and with maritime research. Steps could be taken to assist groups of countries in promoting regional institutes for marine science and technology, to provide for the participation of land-locked or sea-locked countries in specified marine economies and to encourage developing country participation in world ocean trade". *Ibid.,* pp. 48-49

344 *ESCAP 1979 Annual Report, supra* note 341 at p. 77

345 UN Doc. E/1979/71 (17 May 1979)

346 Members: Argentina, Bahamas, Barbados, Bolivia, Brazil, Canada, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, France, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Netherlands, Nicaragua, Panama, Paraguay, Peru, Surinam, Trinidad and Tobago, United Kingdom, United States, Uruguay, Venezuela.

347 UN Doc.A/34/649 (8 Nov. 1979), pp. 23-26

348 ECA Res. 335 (XIV), 27 Mar. 1979. *ECA 1979 Annual Report* (1979) ESCOR, Supp. No. 15 /Doc. E/1979/50; E/CN.14/725), at p. 137

349 ECA Res. 340 (XIV). *Ibid.,* pp. 141-42

350 *Ibid.,* at p. 25

351 *Ibid.,* at p. 20

352 Members: Bahrain, Democratic Yemen, Egypt, Iraq, Jordan, Kuwait, Lebanon, Oman, Qatar, Saudi Arabia, Syria, United Arab Emirates, Yemen, PLO. *Everyone's United Nations, supra* note 295, p. 234

353 *Ibid.,* at pp. 232-33

354 UN Doc. A/34/649 (8 Nov. 1979) at p. 29

355 "The first part involves missions to the region to survey existing and proposed national and regional programmes released to seas such as fishing, mining, desalination, as well as coastal area development projects, particulary industrial complexes. The social and economic impacts of these projects are to be assessed in a quantitative manner. Furthermore, since national legislation on the subject is often ill-defined, inconsistent and poorly enforced, an overall review of such legislation will be conducted. Based on the survey a regional strategy will be developed in order to enable member States to explore and exploit their territorial waters and develop their coastal areas, within their sovereign rights and in harmony with the environment." *Ibid.,* at p. 30

356 *Ibid.*

357 *Ibid.*

358 As of 31 December, 1978. *UNESCO 1977-1978: Report of the Director-General,* UNESCO Doc. 21/C/3 (1979), p. 99

359 *Everyone's United Nations, supra* note 295 at p. 358

360 *Approved Programme and Budget lor 1979-1980,* UNESCO Doc. 20C/5 Approved (1979), at p. 250. This includes "State-of-the-art-studies ... on mangrove and coral reef topics of the coastal region, as well as on other fundamental marine science problems ... in co-operation with other bodies within the United Nations system. The scientific basis of the management of the marine environment and resources will be promoted". *Ibid.,* at p. 251

361 For example research pilot projects on coastal lagoons, mangrove ecosystems and estuaries will be emphasized in Latin America; the mangrove eco-system programme in Asia will culminate in a symposium to review results and recommend relevant research and development for Member States; this Regional Symposium on Mangrove Environment in Asia was scheduled to be held in Kuala Lumpur, Malaysia, 25-29 August, 1980. See IMS Newsletter (No. 23) September 1979 at p. 1. In Africa, the development of marine scientific manpower will be emphasized and pilot research projects will be promoted on coastal lagoons, mangrove ecosystems and estuaries of the West Coast of Africa; for the region of the Mediterranean, Red Sea and adjacent gulfs of the Indian Ocean, research projects will be developed to fill particular needs in the marine science infrastructure concerned, such as on ecosystem and physical modelling, or reference collections, or tidal measurements and for providing scientific data for applied programmes ... *Ibid.* at p. 52

362 The Joint FAO/IOC/United Nations programme on the Aquatic Sciences and Fisheries Information System (ASFIF). *Ibid.,* at p. 250

363 The quarterly periodical "International Marine Science (IMS) Newsletter", UNESCO Technical Papers in Marine Science, UNESCO Reports in Marine Science, and the International Indian Ocean Expedition

364 IODE is "aimed at facilitating the exchange and handling of, and access and referral to oceanographic data and information resulting from or relating to marine programmes". Also, in co-operation with UNEP, IOC has developed the Marine Environmental Data Information Referral Systems (MEDI). *Ibid.*

365 For example, "fellowships will be awarded and training courses supported: marine pollution chemistry, Liverpool (United Kingdom); marine chemistry (Asia); marine science, Duke University (United States); marine science technicians (Red Sea Region)... Three (TEMA) training courses will be supported; two for marine technicians from Latin America and Southeast Asia, the third on coastal area development and management". *Ibid.,* at p. 254

366 103 members as of 1 March, 1979. *See* Hans V. Roll, *A Focus For Ocean Research: Intergovernmental Oceanographic Commission History, Functions, Achievements* (UNESCO, 1979), at p. 53

367 IOC's global ocean science and ocean services activities include *inter alia:*

(a) General Bathymetric Chart of the Oceans (GEBCO);
(b) Global Atmosphere Research Programme (GARP) with WMO/ICSU;
(c) Global Investigation of Pollution in the Marine Environment (GIPME);
(d) Integrated Global Ocean Station System (IGOSS) with WMO;
(e) Ocean Data Acquisition Systems, Aids and Devices (ODAS) with IMCO; and
(f) International Tsumami Information Center (ITIC). UNESCO, *Approved Programme and Budget tor 1979-1980, supra* note 364, at pp. 246-248

368 IOC's regional associations and ocean science activities include *inter alia:*

(a) Co-operative Investigations in the Mediterranean (CIM) with FAO (GFCM) and International Commission for the Scientific Exploration of the Mediterranean Sea (ICSEM);
(b) International Co-ordination Group for the Southern Oceans (SOC);
(c) IOC Association for the Caribbean and Adjacent regions (IOCARIBE);
id) Working Group for the Western Pacific (WESTPAC);
(e) Co-operative Investigations in the North and Central Western Indian Ocean (CINCWIO) with IOFC;
(g) Studies on East Asia Technics and Resources (SEATAR) with CCOP and CCOP-SOPAC;
(h) Co-operative Investigations of the Northern Part of the Eastern Central Atlantic (CINECA) with International Council for the Exploration of the Sea (ICES) and FAO (CECAF);
(i) Central Eastern Atlantic Region (proposed) (also called Atlantic-Ibero-African Region (AIA) by IOC resolution XI-16, at its 11th Session, Nov. 1979); and
(j) Atlantic and Pacific Geological-Geophysical Atlases. *Ibid.,* at pp. 247-248

369 Two new Joint IOC/UNEP co-ordinated pilot projects have replaced two earlier projects which ended in 1978. *Ibid.,* at p. 249. *See* also, *IOC Biennial Report tor 1976-1977,* UN Doc. IOC/INF-376 (30 Aug. 1978), p. 9

370 Seven projects were proposed at the International Workshop on Marine Pollution in the Caribbean and Adjacent Regions held in Port-of-Spain (Trinidad), 13-17 Dec. 1976. *Ibid.*

371 Four regional and 20 subregional projects were proposed at the International Workshop on Marine Pollution in East Asian Waters held in Penang (Malaysia), 7-13 Apr. 1976. *Ibid.*

372 An International Workshop on Marine Pollution in Southeast Pacific Region was planned for 1978. *Ibid.*

373 An International Workshop on Marine Pollution in the Gulf of Guinea and Adjacent Regions was planned for 1978. *Ibid.*

374 Projects may be conducted through IOC Working Group for WESTPAC. *Ibid.*

375 Projects may be conducted through CINCWIO. *Ibid.*

376 *UNESCO Approved Programme and Budget for 1979-1980, supra* note 364, at p. 247

377 Member states participating in CICAR (1967-1976): Brazil, Columbia, Cuba, France, Guatemala, Jamaica, Mexico, Netherlands, Panama, Trinidad and Tobago, USSR, United Kingdom, United States and Venezuela. *See* Roll, *supra* note 366, at p. 23

378 "The present membership comprises 16 countries from the Caribbean region and three from outside." *Ibid.*

379 *UNESCO Approved Programme and Budget for 1979-1980, supra* note 364, at p. 247

380 *See* Roll, *op. cit.,* at pp. 21-22

381 *UNESCO Approved Programme and Budget tor 1979-1980, supra* note 364, at p. 247

382 IOC Res. XI-31 (Nov. 1979). *See IMS Newsletter* (no. 24) Winter 1979-1980, at p. 6

383 (1976) 30 *YBUN* (1979), at p. 1024

384 12 *Int'l Leg. Mats.* 1319 (1973)

385 (1976) 30 *YBUN* (1979), at p. 1024

386 *Ibid.*

387 *Ibid.,* at p. 1025

388 *Ibid.*

389 *UKTS* no. 54 (1972), Cmd. 5169

390 *Ibid.*

391 Other agencies: UN/FAO/UNESCO/WMO

332 Other agencies: FAO/UNESCO/WMO/WHO/IAEA/UN/UNEP

393 United Nations Expanded Programme of Technical Assistance, established in 1950, and the United Nations

Special Fund, which began in 1959

394 (1976) 30 *YBUN,* (1979), p. 367

395 *Ibid.,* at p. 1086

396 *Ibid.,* at p. 367

397 Source of this data is UN Docs. DP/PROJECTS/INF.1 and INF.2 (1978)

398 UN Doc. DP/RP/20 (4 Apr. 1979), at p. 4

399 *Ibid.* at p. 5

400 For example, "TCD, in co-operation with (IESA) participated In the formulation and format adoption of the Kuwait Action Plan for the Development and Protection of Coastal Areas and the Marine Environment. The programme was Initiated by the (UN) and (UNEP) in 1975, and was in 1978 supported by a $ 5.8 million trust fund ... The major trust of TCD activities was the rational development and management of resources of the region's coastal areas while minimizing environmental degradation in the common marine environment of the Gulf". *Ibid.* at pp. 12-13

401 *Everyone's United Nations, supra* note 295, p. 129

402 W. Manshard and L MacDonald, "Survival, development, welfare" (1979) *Mazingira* (no. 12), pp. 51-59 at p. 56

403 *Ibid.*

404 *See* "Coastal Zone Studies to Begin to Indonesia", 2 *UNU Newsletter* (no. 4) at p. 2 (June 1978)

405 *Ibid.*

406 Established in 1919, ILO became a UN specialized agency in 1946.

407 *Everyone's United Nations, supra* note 295, p. 12

408 *Ibid.*

409 *Ibid.*

410 *Ibid.,* at p. 13

411 For a description of the Baltic marine environment, *see* B.A. Boczek, "International Protection of the Baltic Sea Environment against Pollution; A Study in Marine Regionalism" (1978), 72 *Amer. J. Int'l L.* 782 at pp. 783-786. See also 3 *Int'l Envt'l Rep.* (1980) pp. 231-232

412 "The fresh waters from the rivers and run off flow on the surface into the North Sea while a deeper counter current of the saltier North Sea flows into the baltic through the Belts (chiefly the Great Belt) and — to a lesser extent — through the Sound." *Ibid.,* at p. 785

413 *Ibid.*

414 *Ibid.,* at p. 786

415 *Ibid.,* at pp. 787-792

416 *Ibid.,* at p. 786

417 These include: Convention for the Protection of Fish and Regulation of Fishing signed by Estonia and Latvia on 28 October 1925 *(LNTS)* vol. LIV, no 1286 (1926) p. 231 ff.); Convention on the Regulation of Fishing for Plaice and Flounder in the Baltic Sea signed by Denmark, Germany, Poland, Sweden, and the Free City of Danzig on 17 Dec. 1929 *(LNTS* vol. CXV, no. 2679, p. 93ff.); Agreement concerning the Protection of the Salmon Population in the Baltic Sea concluded on 20 Dec. 1962 between Denmark, the Federal Republic of Germany, Poland and Sweden (UN Doc. ST/LEG/SER.B/15, pp. 859-862); and the Convention on Fishing and Conservatin of Living Resources in the Baltic Sea and Belts signed by Denmark, the Federal Republic of Germany, Finland, the German Democratic Republic, Poland, Sweden and the USSR on 13 Sept. 1973 (12 *Int'l Leg. Mats.* at pp. 1291-1297). *See* J.E. Carroz, "The Management of Living Resources in the Baltic Sea and the Belts" (1977), 4 *Ocean Dev. Int'l L.* 213

418 One of the world's earliest oceanographic institutions, ICES, founded in 1902, has conducted "concerted biological and hydrographical investigations for the promotion of a planned exploitation of the resources of the sea". Membership: 18 countries. 1 *Europa YB* (1977) at p. 397

419 Carroz, *op. cit.,* at p. 217

420 This Baltic Sea Convention was ratified in July 1974

421 Carroz, *op. cit.,* at p. 223

422 *See* 13 *Int'l Leg. Mats.* 546 (1974). Parties are: Denmark, the Federal Republic of Germany, Finland, the German Democratic Republic, Poland, Sweden and the USSR

423 "The delay in ratifying the convention arose not because of disagreements but because of difficulties in co-ordinating translations of the original English texts, especially between East and West Germany. West Germany, the last to ratify on March 3rd (1980), had the additional hold-up of having to contact its 11 regional governments for their approval — even though many of them have no maritime borders." C. Cork, "The Baltic Sea Convention is ratified after seven years" (1980), 6 *World Em. Rep.* (no. 12), at p. 2

424 3 *Int'l Envt'l Rep,* (1980) at p. 231

425 *Ibid.*

426 *Ibid.*

427 *Ibid.,* at p. 232

428 *Ibid.*

429 *See* Boczek, *op. cit.,* at p. 796
430 UN Doc. ST/LEG/SER.B/18/Add.2 at p. 297
431 *Ibid.*
432 *See* Boczek, *op. cit.,* at p. 797
433 *Ibid.,* at p. 796
434 *See supra,* p. 332
435 Draft Action Plan for the Protection and Development of the Marine Environment and Coastal Areas of the South East Asian Region (Doc. UNEP/EAS.1, 5 April 1979), p. 3, para. 5
436 For a general and recent account of ASEAN activities, *see* Thomas W. Allan, *The ASEAN Report* (ed. Wain, in 2 vols., 1979)
437 *See infra,* pp. 67 and 69
438 Draft Action Plan, *supra* note 441, p. 3, para. 5
439 *Ibid.*
440 *Ibid.,* p. 2, para. 4.2
441 The first serious oil pollution incident in the South China Sea was the spilling of 3,380 tonnes of crude oil into the Malacca Straits by the Japanese tanker *Showa Maru* in January 1975. There have also been two relatively minor spillings. In August 1976 a Philippine tanker *Deigo Silang* collided with another vessel in the Malacca Straits and spilled about 6,000 gallons of Kuwait crude. A larger spillage — 590,000 gallons — resulted from an earlier collision in the Gulf of Thailand in April 1974, when a coastal vessel *Visahakit,* loaded with fuel oil, was hit by a freighter
442 ASEP, *op. cit.,* p. 33
443 The terms of reference of this Group are:
"(a) to lay down the fundamental obligation of the ASEAN member countries through the implementation of appropriate anti-pollution measures, taking Into account international conventions;
(b) to consider further national and regional or international measures for the discharge of the fundamental obligation;
(c) to recommend broad principles for dealing with certain problems arising in connection with such national and regional or international questions relating to jurisdiction, compensation for damage and settlement of disputes, and also technical assistance schemes; and
(d) to conduct fact-finding activities in the ASEAN region concerning marine pollution"
444 The Plan provides for:
"(a) an effective reporting system to alert the member countries in case of a major oil spill;
(b) a programme of identification and exchange of information of existing anti-pollution operational capabilities within member countries; and
(c) A system of providing assistance to a member country in the event of a large oil spill which it alone cannot cope with, and/or which threatens a neighbouring country". *Ibid.*
The Plan seems to co-ordinate and integrate the actions of individual member countries in combating discharged oil under national contingency plans, but it does not envisage large-scale joint operations in areas remote from those provided for under these national contingency plans.
445 Text reproduced in Michael Letter, *International Straits of the World: Malacca, Singapore, and Indonesia* (1978), at p. 205
446 ASEP, *op. cit.,* p. 34
447 These areas were:
"(i) monitoring the quality of the marine environment;
(ii) scientific research pertaining to marine pollution and its effects on the living aquatic environment;
(iii) Intercalibration exercises ...; and
(iv) training of scientific and technical staff engaged in marine pollution investigations".
Report of the First ASEAN Working Group Meeting on Marine Sciences (Jakarta, July 12-14, 1979), pp. 4-5. The meeting also pledged its scientific support in
"(i) the development and establishment in the ASEAN region of a regional oil combating centre; and
(ii) the formulation of marine-environmental laws, regulations, and standards". *Ibid.,* p.5
448 *Report of the Second Meeting of the ASEAN Experts in the Environment* (Penang, 17-20 September, 1979), p. 3. It may be inferred from this Report that UNEP's perceptions of the region's environmental priorities had been questioned outside the context of the ASEAN Experts on the Environment
449 UNEP, *Report of the ASEAN Workshop on Nature Conservation of the Asean Experts on the Environment* (Held at Denpasar, Indonesia, 15-19 September, 1980)
450 These nations were: Burma, Cambodia, Indonesia, Japan, Laos, Malaysia, the Philippines, Singapore, Thailand, and Vietnam. Only four, however, became members of SEAFDEC: Japan, the Philippines, Thailand and Singapore. For text of Agreement, *see* SEAFDEC Aquaculture Department, *First Annual Report* (1974), pp. 36-44. Two weeks later, the Agreement was amended by three states which had signed it in order to keep it open longer for signature by the other countries. *Ibid.,* p. 45
451 *See*, for example, the *Integrated Fisheries Community Development Project* (1976), the first of a series of

plans put forward experimentally by the SEAFDEC Aquaculture Department in conjunction with the Philippines Bureau of Fisheries and Aquaculture Resources and the Southeast Asian Regional Center for Graduate Study and Research in Agriculture

452 The antecedents of ICSU date from the turn of the century. On its origins and purposes, *see Britain and International Scientific Co-operation.* (RFP 5814/68, pp. 4-5, 46-47. Although more than half of its annual budget Is derived from UNESCO, ICSU is a non-governmental organization. SCAR is charged with furthering the co-ordination of scientific activity in Antarctica, with a view to framing a scientific programme of circumpolar scope and significance". M.W. Holdgate, *ed., Antarctica Ecology* (in 2 vols., 1970), in vol. I at p. xi.

453 It is said that to date man may have caused the total extinction of only two species of birds in the region, and perhaps only 10 species are endangered. B.B. Roberts, "Conservation in the Antarctica" (1977), 279 *Phil. Trans. of Royal Soc.* 97. For an overview of ecological research, see Holdgate, *op. cit.,vo*\. 1. For recent findings, *see* the *SCAR Bulletin incorporated* in the *Polar Record. In recent years specially protected areas* and sites of special scientific interest have been designated, and a voluntary system of environmental impact assessment, somewhat along the lines of the US NEPA directives on environmental impact statements, has been established in Antarctica. See Bruce C. Parker, *ed., Environmental Impact in Antarctica* (1978) (Select Papers by Scientists Addressing Impact Assessment, Monitoring and Potential Impact of Man's Activities in the Antarctica)

454 Barbara Mitchell and Lee Kimball, "Conflict over the Cold Continent" (1979), 35 *Foreign Policy* 124 at pp. 129-130

455 *Ibid.,* pp. 127-129

456 Krill is said to be the main food supply for five whale, three seal, 20 fish, three squid, and numerous bird species. *Ibid.,* p. 128. On the importance of krill to whales, see N.A. Mackintosh, "Whales and Krill in the Twentieth Century", in Holdgate, *supra* note 456, vol. I, pp. 195-210

457 *See* J.A. Gulland, "The Development of the Resources of the Antarctica Seas", *ibid.,* pp. 195-212

458 19 *Int'l Leg. Mats.* 841 (1980)

459 The applicable principles of conservation are set down in Article ll(3). These principles will be put into effect by the Commission for the Conservation of Antarctic Marine Living Resources, which is established under Article VII. Under Article IX, the Commission is vested with regulatory authority: member states are obligated to implement conservation measures adopted by the Commission. The Commission, in turn, is required to "take full account" of the rcommendations and advice of the Scientific Committee for the Conservtion of Antarctic Marine Living Resources, which is established under Article XIV. Both the Commission and the Scientific Committee are required to co-operate with the Antarctic Treaty Consultative Parties "on matters falling within the competence of the latter", as provided in Article XXIII. It is also worth noting that Article IX(5) requires the Commission to take "full account of any relevant measures or regulations established or recommended by ... existing fisheries commissions responsible for species which may enter the area to which this Convention applies", so as to prevent any inconsistencies from arising between the Convention and other existing conservation arrangements

460 The Antarctic Treaty is restricted to 14 full parties: the original signatory states (Argentina, Australia, Belgium, Chile, France, Japan, New Zealand, Norway, South Africa, the Soviet Union, the United Kingdom and the United States), and other acceding states which are conducting "substantial scientific research activity (in the area), such as the establishment of a scientific station or the dispatch of a scientific station". So far only three acceding states have been able to satisfy this condition: Poland, the German Democratic Republic, and the Federal Republic of Germany. Since few nations can afford to engage in such research, the Antarctica club is likely to remain fairly exclusive

461 "Krill may be unique in its potential contribution to food supplies. The untouched nature of the resource, its high protein content, and its plentiful supply merit its international management for the benefit of poor countries. But if industrialized countries exploit krill for their own benefit, its potential as a high protein human food may never be realized. Equity considerations aside, its extensive native habitat and key importance to the marine food chain make krill a natural subject of global interest. Moreover, the secondary economic implications will be widespread. Full-scale exploitation is bound to affect sharply the markets for other high protein processed foods such as fishmeal, shrimp, and soybean products. Finally, if the Antarctic Treaty nations seek to exclude other countries, this may incite wildcat operations, if not outright challenges." Mitchell and Kimball, *op. cit.,* at p. 134

462 Although it would have been universally easier politically to have limited the regime of the area south of 60° S (the northern boundary of the Antarctic Treaty) it was decided instead to extend it to a line approximating the Antarctic Convergence, whose waters from the ice shelf meets warmer water flowing south providing an ecological barrier to all except highly migratory species. *Ibid.* at p. 136

463 Argentina, Australia, Chile, France, New Zealand, Norway, and the United Kingdom

464 Only Chile and Argentina made maritime jurisdictional claims in the Southern Ocean before 1959. In the last two years several of the seven claimants to sectors of the land-mass have been reported to be interested in making claims to 200-mile zones offshore.

465 The Antarctic Treaty states have dissuaded UNEP and ECOSOC from "interfering" In the region, and their

hostility was partly responsible for the withdrawal of an FAO-UNDP proposal in 1976 for a $ 45 million, 10-year programme, one of whose goals was the rational utilization of the living resources of the region for the benefit of the world as a whole, and the developing countries in particular. Mitchell and Kimball, *op. cit.,* p. 133

466 Another classification of regional organizations might be noted:

(a) multi-functional and geographically comprehensive organizations (e.g. OAS, OAU, League of Arab Nations);

(b) Co-operation- or integration-oriented and geographically limited (subregional) organizations (e.g. Council of Europe, EEC, ASEAN);

(c) technical or otherwise narrowly functional organizations (e.g. Colombo Plan for Co-operative Economic Development in South and Southeast Asia). Introduction to Berhanykun Andemicoel, *ed., Regionalism and the United Nations* (UNITAR, 1979) at p. 9

467 The phrase "interntional co-operation" is used so variously and so loosely that it must be regarded as "nearly meaningless". Sir Peter Smithers, "Towards Greater Coherence among Intergovernmental Organizations through Governmental Control", *Ibid.,* pp. 13-93 at pp. 23-24

468 The matter has been put bluntly by a senior international civil servant. "Whatever may be said upon polite occasions, Governments will not entrust a project to which they attach importance to a body which seems to be out of control, or to a body the composition of which is unacceptable to them for the purpose in question". *Ibid.,* at p. 31

469 ICNT/Rev. 3, Article 197

470 *Ibid.,* Article 198

471 *Ibid., Article 199*

472 Ibid., Article 200

473 *Ibid.,* Article 201

474 *Ibid.,* Article 211 (3)

475 *Ibid.,* Article 208 (4)

476 *Ibid.,* Article 208 (5)

477 *Ibid.,* Article 210

478 *Ibid.,* Article 207 (3)

479 *Ibid.,* Article 207 (4)

480 *Ibid.,* Article 212

Chapter Five

The Environmental Law of the Sea: Conclusions and Recommendations

Douglas M. Johnston and Cyrille de Klemm
in collaboration with Françoise Burhenne-Guilmin and Daniel B. Navid

A. General Conclusions

1. Introduction

As is shown In the preceding chapters, the first decade after the Stockholm Conference on the Human Environment has seen an unprecedented effort made by the Governments of the world to negotiate arrangements for the regulation of all ocean activities.

By far the most important of these developments is the Third UN Conference on the Law of the Sea (UNCLOS III).

Although many of the environmental responsibilities embodied in the Draft Convention on the Law of the Sea are new, many others are already established in customary international law or derive from existing treaties concluded prior to UNCLOS III. Moreover, of the numerous kinds of global, regional and national measures referred to in the new Draft Convention, many have already been initiated years before the Convention comes into effect. In short, the implementation of some of the environmental provisions of UNCLOS III has already begun.

It is timely, therefore, to identify tasks of implementation facing states and agencies which have responsibilities for the protection and the preservation of the marine environment.

2. The Draft Convention on the Law of the Sea: Implementation of the Obligations of Major Environmental Significance

The Draft Convention on the Law of the Sea (Informal Text) (Doc. A/CONF. 62/WP. 10/Rev. 3, 27 August 1980) contains no less than 320 Articles and eight Annexes, which together constitute a document of 180 pages. Most of these Articles and Annexes might be regarded as directly or indirectly significant for the future protection and conservation of the marine environment. Of the Articles dealing directly with environmental questions, forty-eight (48) in particular should be singled out for special attention on the ground that they contain obligations, or refer to responsibilities, of *major* environmental importance. This listing, it should be noted, omits a great deal: for example, (i) Articles which merely *refer to* environmental (as well as non-environmental) considerations (e.g. Article 165); (ii) Articles which deal with matters which are not strictly environmental but may be related to environmental concerns (e.g. Articles in Part XIII on marine scientific research); and (iii) all Annexes, even though some may be of major environmental significance (e.g. Annex I; Annex III, Article 17(2)(f)).

Part V: Exclusive Economic Zone

Art. 56 implicit obligations inherent in "jurisdiction ... with regard to ... the protection and preservation of the marine environment" vested in the coastal state within its EEZ
Art. 61 obligations of coastal state in the conservation of the living resources of the EEZ
Art. 62 obligations of coastal state in seeking to achieve the optimum utilization of the living resources of the EEZ
Art. 63 obligation of coastal state in respect of "straddling" stocks
Art. 64 obligation of coastal state, other states, and international organization(s) in repect of highly migratory species
Art. 65 obligation of coastal state and international organization(s) in respect of marine mammals
Art. 66 obligations of state of origin and other harvesting states in respect of anadromous stocks
Art. 67 obligations of coastal state and other harvesting states in respect of catadromous species

Part VII: High Seas

Art. 116 implicit obligation to exercise the right to fish in the high seas subject to existing treaty obligations and to duties provided for, *inter alia,* in articles 63-67
Art. 117 general obligation to conserve living resources of the high seas
Art. 118 obligations to co-operate in the management and conservation of living resources of the high seas
Art. 119 obligations in determining the allowable catch and establishing other conservation measures for the living resources of the high seas
Art. 120 obligation of states and international organizations in respect of marine mammals (see Article 65)

Part IX: Enclosed or Semi-enclosed Seas

Art. 123 obligation to co-ordinate the management, conservation, etc., of the living resources

Part XI: The Area

Art. 145 obligation of the International Seabed Authority to adopt rules, etc., for the protection of the marine environment
Art. 147 obligations to ensure that activities in the Area are carried out with reasonable regard for other activities in the marine environment
Art. 150 obligation to ensure the rational management of the resources of the Area and, in accordance with sound principles of conservation, the avoidance of unnecessary waste

Part XII: Protection and Preservation of the Marine Environment

Section 1. General Provisions

Art. 192 general obligation to protect and preserve the marine environment
Art. 194 obligation to prevent, reduce and control pollution of the marine environment from any source
Art. 195 obligation not to transfer damage or hazards from one area to another or transform one type of pollution into another
Art. 196 obligation to prevent, reduce and control pollution in respect of the use of technologies and the introduction of alien or new species

Section 2. Global and Regional Co-operation

Art. 197 general obligation of states to co-operate directly or through international organizations
Art. 198 obligation to give notice of imminent or actual damage
Art. 199 obligation to co-operate in the development of joint contingency plans to deal with pollution incidents
Art. 200 obligation to co-operate in the promotion of studies, research programmes, and exchange of data related to marine pollution problems
Art. 201 obligation to co-operate in the development of scientific criteria and the elaboration of rules, etc., for the prevention, reduction, and control of marine pollution

Section 3. Technical Assistance

Art. 202 obligation to promote and provide scientific and technical assistance to developing states for the protection and preservation of the marine environment
Art. 203 obligation of international organizations to grant preferential treatment to developing states for purposes of the prevention, reduction and control of marine pollution

Section 4. Monitoring and Environmental Assessment

Art. 204 obligation to monitor risks and effects of marine pollution
Art. 205 obligation to publish reports
Art. 206 obligation to assess potential effects of projected activities on the marine environment

Section 5. International Rules and National Legislation to Prevent, Reduce and Control Pollution of the Marine Environment

Art. 207 obligation to adopt laws, etc., to deal with pollution from land-based sources
Art. 208 obligations to adopt laws, etc., to deal with pollution from seabed activities
Art. 209 obigations to adopt laws, etc., to deal with pollution from activities in the Area
Art. 210 obligation to adopt laws, etc., to deal with pollution by dumping
Art. 211 obligations to establish international rules, to adopt laws, etc., in respect of pollution from vessels
Art. 212 obligations to establish international rules, to adopt laws, etc., in respect of pollution from or through the atmosphere

Section 6. Enforcement

Art. 213 obligation to enforce laws adopted in accordance with article 207 and to implement international rules, etc., established in respect of pollution from land-based sources

Art. 214 obligation to enforce laws adopted in accordance with article 208 and to implement international rules, etc., established in respect of pollution from seabed activities

Art. 215 obligation to enforce international rules, etc., established in respect of pollution from activities in the Area

Art. 216 obligations for the enforcement of laws, international rules, etc., for the prevention, reduction, and control of pollution by dumping

Art. 217 obligations related to the enforcement by flag states of laws, international rules, etc., in respect of pollution from vessels

Art. 218 obligations related to the enforcement by port states of laws, international rules, etc., in respect of pollution from vessels (e.g. duty to comply with request for investigation)

Art. 219 obligation of port state to prevent departure of vessel suspected of violating international rule relating to seaworthiness of vessel

Art. 220 obligation of flag state to adopt laws, etc., to ensure that its vessels comply with vessel identification requirements

Art. 222 obligation to enforce laws adopted in accordance with article 212 and to implement international rules, etc., established in respect of pollution from or through the atmosphere

Section 9. Responsiblity and Liability

Art. 235 obligations related to the provision of prompt and adequate compensation

Section 11. Obligations under other Conventions on the Protection and Preservation of the Marine Environment

Art. 237 obligation to respect existing agreements and to carry them out in a manner consistent with the general principles and objectives of the (Caracas) Convention

Several other environmentally important provisions, cast in terms of *rights* instead of obligations (or responsibilities), could of course be added to the list. The provisions on innocent passage (Articles 17-26, 45, 52) and ice-covered waters (Article 234) are conspicuous examples. But these provisions are omitted here on the grounds that they do not give rise directly to questions of implementation or development.

These 48 Articles in the Draft Convention on the Law of the Sea can be classified in several ways. Two of these methods are of particular importance for purposes of implementation and development: legal and functional. Unfortunately neither of these methods results in a clear-cut classification. The *legal* method of classification requires that each Article be brought under one juridical regime. Those Articles induced in Parts V, VII, and XI belong to the regimes of the exclusive economic zone, the high seas, and the Area (of the deep ocean floor beyond limits of national jurisdiction), respectively; but many of the other Articles listed have to be divided between two or more regimes, or can be assigned to a specific regime only through

the process of interpretation. The *functional* method of classification requires that each Article be brought under one of several functionally differentiated headings: for example,

(a) protection and preservation of the marine environment (general);
(b) conservation of marine species
    i) fishery management and conservation;
    ii) nature conservation;
(c) marine pollution control
    i) pollution from land-based sources;
    ii) pollution from seabed activities;
    iii) pollution from activities in the Area;
    iv) dumping
    v) pollution from vessels;
    vi) pollution from or through the atmosphere.

But here too it is difficult to achieve a clear-cut classification. In some Articles it is not clear whether the reference to "protection and preservation of the marine environment" is intended to be wider than "the prevention, reduction and control of marine pollution". Most of the articles dealing with conservation of marine species refer to the "management and conservation of living resources" and yet seem to have been framed around commercial fishery concerns. A few provisions (e.g. Articles 65 and 120 on marine mammals) are clearly addressed to a designated class of species including both those with and those without a commercial value. Some provisions might be interpreted as dealing primarily with nature conservation, or equally with nature conservation and fishery management and conservation (e.g. Articles 61 and 119). Moreover, some of these articles, such as those dealing with land-based marine pollution, will require developments that will necessarily have to take place outside the framework of the Caracas Convention since they belong to the area of *coastal zone management* concerns, which have clearly been left out of the draft convention; even some of the fishery access provisions (e.g. Articles 61(3), 62(3), 69(2), 70(3), and 70(5) contain "eclectic" language of the sort associated with multiple-use problems in the coastal zone).

In common with the non-environmental provisions of the Draft Convention, these 48 environmentally significant Articles reflect a wide variety in the "language of commitment". Some are precisely worded, but many are couched in general, vague, hortatory, discretionary, or otherwise restrictive language. Motivated by diverse and sometimes conflicting objectives, UNCLOS III has produced an immense and complex treaty that presents unprecedented tasks of implementation and further legal and institutional development. Precisely because of this heterogeneity in the language of the text, it should now be a major purpose of states and relevant international organizations to undertake these tasks of implementation and development in a systematic manner. Such an undertaking should be a matter of priority in the second decade after the 1972 United Nations Conference on the Human Environment.

Despite its great scope, UNCLOS III did not provide a comprehensive, much less a holistic, treatment of the problems of the protection and conservation of the marine environment. At the global level, the tasks of implementation and development of

the environmental provisions of the Caracas Convention must be integrated with the principles enunciated in the Stockholm Declaration on the Human Environment and the recommendations contained in the Stockholm Action Plan as well as with recent treaty developments outside of UNCLOS III.

3. World Ocean Conservation Strategy

These tasks of implementation and development will be facilitated, if states and relevant international organizations can agree on a common "system-wide" strategy designed specifically to enhance and co-ordinate global, regional and national efforts directed at more effective protection and conservation of the marine environment. Such a strategy for ocean conservation should be developed out of the more general "World Conservation Strategy: Living Resource Conservation for Sustainable Development", which was prepared in 1980 by the International Union for Conservation of Nature and Natural Resources (IUCN) with the advice and co-operation of UNEP and the World Wildlife Fund and in collaboration with FAO and UNESCO. The major objectives of the "World Conservation Strategy" are:

(i) the maintenance of essential ecological processes and life-support systems;
(ii) the preservation of genetic diversity; and
(iii) the sustainable utilization of species and ecosystems.

The first of these objectives requires the maintenance of the general health of the oceans as well as rational planning and allocation of ocean uses, especially in the most intensively exploited and immediately vulnerable of marine areas which lie close to heavily populated shorelines. The second objective requires the surveillance and preservation of genetic material through *inter alia* the establishment and maintenance of marine parks and other protected areas. The third objective requires better knowledge of the productive capacities of marine living resources and measures to ensure that utilization does not exceed these capacities.

More specially, a World Ocean Conservation Strategy would be addressed to objectives such as these:

(i) the advancement of research in ocean processes, ecosystems, habitats and species;
(ii) increased knowledge of population dynamics and biology of harvested species;
(iii) better understanding of present and potential threats to natural systems and species;
(iv) improved statistics on harvested species, including those that are incidentally taken;
(v) the introduction and strengthening of national laws and regulations concerned with the protection and preservation of the marine environment;
(vi) the enhancement of national capabilities to enforce such laws and regulations;
(vii) better geographical coverage of marine areas and exploited stocks by international fishery agreements and commissions;
(viii) the enhancement of the regulatory capabilities of competent international organizations responsible for the protection and conservation of the marine en-

vironment, and of their capabilities to enforce relevant, internationally agreed, rules;

(ix) the protection of marine habitats — especially the more vulnerable habitats — against all forms of degradation;

(x) the establishment of marine and coastal protected areas to preserve scientifically important ecosystems, representative samples of all types of marine habitats, and endangered, rare or endemic marine species;

(xi) the introduction of controls over the introduction of alien species and the release of substances which adversely affect the general health of the oceans;

(xii) the protection of particularly sensitive marine aras, ecosystems and species from accidental pollution; and

(xiii) the development and implementation of plans for the rational management of coastal zones.

In short, the proposed World Ocean Conservation Strategy would provide guidelines for the whole range of conservation measures that have to be taken to preserve the ocean ecosystem in its entirety.

4. Adoption of Global Treaties

After signature of the Caracas Convention, states should move as swiftly as possible toward formal ratification of that instrument. Few nations are likely to be wholly satisfied with every one of its provisions, as finally formulated, but the Convention will represent the closest approximation to a constitution for the world's ocean that this century is likely to witness. For this reason, unnecessarily protracted delays in bringing the Caracas Convention into effect around the world would not be beneficial and, imperfect as it may be from an environmental point of view, deficiencies in the text are not a reason for rejection of the Convention or procrastination in its implementation, but, rather, are a reason for further development.

The cause of ocean conservation does not, however, only depend on the implementation of the Caracas Convention, but also in large part on widespread acceptance of the growing number of interlocking global treaties of major importance to the management and protection of the marine environment. Conspicuous among such instruments are these global treaties:

(i) International Convention for the Regulation of Whaling (1946);

(ii) International Convention for Prevention of Pollution of the Sea by Oil (1954) (as amended in 1962, 1969, and 1971);

(iii) International Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties (1969) (with 1973 Protocol);

(iv) International Convention on Civil Liability for Oil Pollution Damage (1969) (with 1976 Protocol);

(v) International Convention on the Establishment of an International Fund for Compensation for Oil Pollution Damage (1971) (with 1976 Protocol);

(vi) Convention on Civil Liability in the Field of Maritime Carriage of Nuclear Material (1971);

(vii) Convention on Wetlands of International Importance Especially as Waterfowl Habitat (1971);

(viii) Convention Concerning the Protection of the World Cultural and Natural *Heritage (1972);*

(ix) Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter (1972) (as amended in 1978);
(x) International Convention for the Prevention of Pollution from Ships (1973) (with 1978 (Protocol);
(xi) Convention on International Trade in Endangered Specis of Wild Fauna and Flora (1973);
(xii) Convention on the Conservation of Migratory Species of Wild Animals (1979).

5. Global Co-ordination

Within the United Nations system marine matters are dealt with by several agencies. Though often closely related, these matters are generally treated within separate compartments. All these global agencies — FAO, IMCO, IOC, UNEP, and others — share an interest in co-operation and consultation for particular purposes. There would seem, however, to be a need to establish, at the governmental level, a system that would ensure *continuous and effective co-ordination* of all efforts directed at the protection and conservation of the marine environment.

UNEP does, of course, have a co-ordinative mandate in the general field of environmental affairs, but its scope is extremely wide and only a fraction of its time and resources can be spent on the specific problems of ocean conservation. Most of UNEP's programme in ocean affairs is devoted to the promotion and development of *regional* co-operation with a view to the implementation of the oceanic components of the Stockholm Action Plan. But this effort (the Regional Seas Programme) is not designed to deal with more than a few of the tasks of implementation and development inherent in the environmentally significant provisions of the Draft Convention. Most questions concerning vessel-source pollution come within the terms of reference of IMCO, those on fishery conservation under those of FAO and the development of marine scientific research under those of IOC. The Regional Action Plans emerging from UNEP's Regional Seas Programme give most attention to the problems of pollution from land-based sources and to other aspects of coastal zone management. Yet, with or without an expanded budget, UNEP seems likely to have a central involvement in any new global mechanism that may have to be established to co-ordinate efforts to protect and conserve the marine environment.

Ideally, a *global* system of co-ordination should be designed also to co-ordinate the work of the growing family of *regional* arrangements in the field of ocean management. Many regional fishery commissions established outside the UN system have little opportunity to benefit from the scientific and technical services of the UN agencies, and have to rely mostly on the slender resources of their own staff or on research carried out by member states. FAO regional fishery commissions and councils, IOC regional programmes and the UNEP Regional Seas Programme would also be likely to benefit from a global system of co-ordination. Moreover, the Draft Convention envisages the establishment of regional marine scientific and technological research centres "in order to stimulate and advance the conduct of marine scientific research by developing States and foster the transfer of marine technology" (Article 276). Accordingly, it seems important that a global co-ordinating mechanism should assign as high a priority to the co-ordination of these various regional arrangements as to that of existing global arrangements.

The proposed global co-ordinating body should also be available to provide advisory and clearing-house services to *national* agencies responsible for ocean development and management.

6. Regional Co-operation

The full participation of and implementation by concerned states of existing regional conventions of environmental significance would also further the cause of ocean conservation. However, with regard to the conclusion of new regional arrangements, it seems inevitable that different regions will respond in different ways to the call for regional co-operation in the Caracas Convention. In the area of ocean conservation, we are likely to witness a period of experimentation, a process of trial and error within different and sometimes shifting regional or subregional coalitions, involving coastal states (and occasionally extra-regional states), the regional programmes of UN global agencies (e.g. UNEP, FAO, IOC, and increasingly IMCO), the UN regional economic commissions, and regional organizations outside the UN system.

Any expectation of uniformity in the regional responses to the environmental provisions of the Caracas Convention will be defeated by the reality of economic and other kinds of disparity among nations and regions. Even if a global co-ordinating mechanism is established after UNCLOS III, the main result is likely to be the filling of gaps and the correction of anomalies rather than the imposition of uniform thought and action. *It may be a mistake to urge all nations in all regions to concentrate on the need for formal treaty ("transactional") commitments in order to implement the environmental provisions of UNCLOS III. Similarly, it may not be necessary for all states to enter into the same kinds of "organizational" commitments. In many cases, modest, routine commitments (e.g. to notification, consultation, and the exchange of data), or special ad hoc commitments (e.g. to collaborative research projects) may be more cost effective in the short run and, if successful, may point the way to better designed, long-term regional arrangements of the "transactional" or "organizational" sort.*

In the early 1980s, the success of UNEP's Regional Seas Programme is likely to be the best single test of the feasibility of global co-ordination of regional co-operation in the protection and conservation of the marine environment. If this Programme continues to win co-operation and support an interesting variety of regional coalitions may be expected to emerge, but the success of these coalitions, in terms of effective protection and conservation of the marine environment, may depend finally on their ability to persuade economic and social planners at national levels to accept the role of conservation in "sustainable development".

B. Recommendations

1. Discharge of existing obligations

All states are urged, directly or through appropriate global and regional organizations, to discharge their existing responsibilities for the protection and conservation of the marine environment, as re-stated in the Draft Convention on the Law of the Sea.

One of the purposes of the Third UN Conference on the Law of the Sea (UNCLOS III) was to re-affirm and consolidate existing rights and duties under customary international law. Some of the general environmental provisions contained In the Draft Convention seem to be declaratory of existing principles. The most obvious is Article 192: "States have the obligation to protect and preserve the marine environment." Several other general obligations appear to share the same character: for example, Articles 197 (duty to co-operate on a global or regional basis for the protection and preservation of the marine environment) and 117 (duty to take measures for the conservation of the living resources of the high seas). These obligations exist now, and are in no way contingent upon acceptance of the projected Convention.

2. Adoption of the projected Convention on the Law of the Sea

All states are urged to proceed as expeditiously as possible, after the conclusion of negotiations at UNCLOS III, to become a party to the resulting Convention, and pay particular attention to the implementation of these 48 articles which contain obligations or refer to responsibilities of major environmental significance (see p. 387 above). UNEP, should already, pursuant to its catalytic role initiate and expedite the groundwork required for these obligations to be implemented at the earliest possible time.

UNCLOS III has made an important contribution to the development of the international environmental law of the sea. But many of the actions called for in the text will not be easily taken until the Convention on the Law of the Sea has secured widespread acceptance around the world, not only by signature but also by ratification or accession, as envisaged in Articles 305, 306 and 307. To the extent that the Convention, unlike almost all other existing treaties, has a universal and "constitutional" significance, it is particularly important that it be treated by all states as a crucial instrument for the development of the world community. Within the framework provided by the Convention, it will be possible for nations to address themselves to more specific tasks of ocean development and management in conformity with the need to protect and conserve the marine environment.

3. Discharge of environmental provisions of the Convention on the Law of the Sea

After signature, states are urged, under Article 300, to discharge in good faith the environmental obligations contained in the projected Convention on the Law of the Sea and to exercise the rights, jurisdictions and freedoms conferred therein in a manner which would not constitute an environmental abuse of rights.

Because of the heterogeneous character of the language contained in the Convention — not least in the area of environmental obligations and responsibilities — the principles of good faith and abuse of rights are important norms in support of the intentions and aspirations shared by the delegations at UNCLOS III. There is an obligation under general international law, it may be recalled, not to defeat the object and purpose of a signed but still unratified treaty. In the case of a universal law-making treaty like the projected Convention on the Law of the Sea, it is particularly important that conforming states should be able to proceed to the tasks of im-

plementation on the assumption that all other signatory states are precluded from frustrating the major purposes of that treaty.

4. Review of existing environmental treaties

States should review all existing treaties concerned with the protection and conservation of the marine environment, with a view to becoming a party to those which are open to them, thus providing a broader base for the future development of the international environmental law of the sea.

Global treaty-making for the protection and conservation of the marine environment has been particularly active in the last decade or more *(supra* p. 393). A growing number of regional arrangements for the protection and conservation of the marine environment have also begun to emerge in recent years. The final stage of UNCLOS III marks a convenient time for states to review their record in the signature and ratification or accession of these instruments, since the projected Convention on the Law of the Sea represents, among other things, the latest effort to develop a system of related treaties devoted to the problems of ocean conservation.

5. Global co-ordination

UNCLOS III in its final session should pass a resolution calling on the UN General Assembly to appoint a Task Force to study and report on alternative methods of instituting a global co-ordinating body with designated responsibility for the development and implementation of a World Ocean Conservation Strategy.

In light of the existing allocation of responsibilities for the protection and conservation of the marine environment within the UN system, the need for a co-ordinating body seems clear *(supra,* p. 394). The draft Convention on the Law of the Sea text does not allocate implementation responsibilities to particular organizations, but, rather, consistently refers to the "competent international organization". One of the first tasks of implementation after UNCLOS III is to secure meaningful agreement on how to allocate, re-allocate, and co-ordinate implementation responsibilities within the family of UN agencies. Ultimately the matter may have to be resolved at the political level of the UN General Assembly, since it involves questions of budgetary provisioning, but initially the UN agencies themselves should have the opportunity to think out the difficulties involved at the bureaucratic level. Presumably a resolution of the kind proposed would be addressed as soon as possible after the formal signature of the Convention to a session of the UN General Assembly, which would be asked to instruct the UN Secretary-General to establish the Task Force and ensure that it report back to the General Assembly at its next session. The Task Force might be mandated, among other things, to seek independent reports from intergovernmental bodies outside the UN system (e.g. the International Council for the Exploration of the Sea) as well as non-governmental organizations (e.g. International Union for Conservation of Nature and Natural Resources, International Council of Scientific Unions). The first undertaking of the proposed co-ordinating body would be to develop a World Ocean Conservation Strategy as a framework of principles, purposes, and guidelines designed to assist in the setting of priorities *(supra,* p. 392), as well as to review the adequacy of present relevant treaties and of their implementation.

6. Role of UN agencies

All relevant UN agencies should, pending the establishment of a co-ordinating body, address themselves, as a matter of priority, to the question of the role(s) they should play in the implementation and development of the environmental provisions of the Convention on the Law of the Sea.

The UN agencies most likely to be involved are IMCO, FAO, UNEP, and UNESCO, including IOC but three others may also be concerned: the World Health Organization (WHO), the World Meteorological Organization (WHO), and the International Atomic Energy Agency (IAEA). It might also be useful to request a report on the tasks of implementation from the Group of Experts on Scientific Aspects of Marine Pollution (GESAMP), an inter-agency advisory body. It would be sensible, of course, to orchestrate this effort in such a way that it assists and complements the work of the Task Force on global co-ordination, proposed above.

7. Regional co-ordination

Regional mechanisms should be established to co-ordinate all activities at the regional level concerned with marine resources and the preservation and protection of the marine environment.

At the regional level, especially in those areas that constitute ecological units such as semi-enclosed seas, priority should be given to the development of conservation strategies and the co-ordination of the activities of all organizations dealing with marine research, conservation, the harvesting of marine resources and protection and preservation of the marine environment (see, e.g., 8, 9 and 10 below). Several methods could be used to achieve this result. For instance, a regional co-ordinating body or a co-ordinating agency could be desgnated by the global co-ordinating body referred to in recommendation 5 above. Alternatively, the various organizations active in a particular region could decide to set up their own co-ordinating mechanism. A further possibility would consist in making full use of the regional centres provided for by Articles 276 and 277 of the Law of the Sea Convention. In all cases a linkage with the global mechanism proposed earlier would have to be maintained.

8. Role of UNEP's Regional Seas Programme

The Governing Council of UNEP should give increased support to the Regional Seas Programme in Its efforts to establish Regional Action Plans for the protection and conservation of the marine environment around the world.

For several years the Regional Seas Programme of UNEP has been actively engaged in the effort to promote inter-governmental measures for the protection and conservation of the marine environment in certain designated regional seas. Ten regions have thus far been designated for these purposes. In each case steps have been taken, or are being taken, to produce a Regional Action Plan which, designed in accordance with the characteristics and requirements of the region, will serve as a framework for marine and environmental planning and the development of appropriate legal and institutional arrangements. These Regional Action Plans provide a suitable vehicle for the implementation of many of the environmental provisions of the Caracas Convention. States within each of these ten designated regions are urged to support these efforts as an appropriate mode of regional co-

operation in the protection and conservation of the marine environment. It is hoped also that further Action Plans will be developed to cover regions not presently the subject of UNEP Regional Action Plans or similar arrangements.

9. Role of FAO's Regional Fishery Development and Management Assistance Programme

The coastal States concerned should give support to FAO's Comprehensive Programme of Assistance in the Development and Management of Fisheries in Exclusive Economic Zones.

Many developing coastal states which have recently acquired extended areas of national jurisdiction under the new law of the sea are now facing new opportunities for fishery development and new responsibilities in fishery management and conservation. FAO's Department of Fisheries has responded to these opportunities and responsibilities by offering assistance under a new programme. But the resources of the Department are too modest to make a significant contribution on a nation-by-nation basis. The success of this assistance programme depends upon the willingness of neighbouring and adjacent states to develop a common fishery development and management strategy for their region or sub-region, preferably in liaison with the appropriate regional or sub-regional fishery commission or council. States are urged to co-operate in the establishment of such bodies, where they do not already exist. Where the existing body is too large to provide an effective vehicle for this kind of co-operation, it is suggested that neighbouring and adjacent states continue efforts to form and support sub-regional committees within the existing regional commissions or councils rather than establish new organizations.

10. Role of UNESCO's regional research programmes

Concerned states are urged to support and participate in the regional programmes of the International Oceanographic Commission.

The International Oceanographic Commission (IOC) of UNESCO started to develop regional programmes in 1976. At present three such associations have been organized: the IOC Association for the Caribbean and Adjacent Regions (IOCARIBE), the Working Group for the Western Pacific (WESTPAC), and Co-operative Investigations in the North and Central Western Indian Ocean (CINCWIO) with the Indian Ocean Fisheries Council (IOFC). This trend toward the regionalization of IOC activities is expected to continue, both through the emergence of other regional associations and through regional projects conducted within the framework of the Global Investigation of Pollution in the Marine Environment (GIPME). States are urged to participate in these regional programmes. Because of their relative informality — they are not dependent on formal treaty commitments — these regional IOC arrangements can be executed without delays of the kind that often accompany official diplomatic negotiations.

11. Role of UN Economic Commissions

States should make more use of the five economic commissions of ECOSOC, and related offices of the UN Secretariat and specialized agencies, in the development of regional and sub-regional projects in coastal zone management.

In recent years the economic and social programmes entrusted to ECOSOC have been re-structured, decentralized and strengthened. Some of these programmes, including those of most of the regional economic commissions, deal in part with certain problems of protection and conservation of the marine environment as well as problems of development related to marine resources. In view of the combination of economic and environmental concerns embraced by them, it is believed that these commissions have a significant role to play in the execution of regional and, more particularly, sub-regional projects in coastal resource development and management (i.e. coastal zone management). Member states are urged to support efforts by these commissions to form coalitions with relevant global and regional organizations in the common cause of establishing viable systems of coastal zone management.

12. Development of ocean management

Coastal states should co-ordinate all governmental and non-governmental activities at the national level concerned with marine resources and the preservation and protection of the marine environment.

Coastal states have a special obligation to develop national ocean development and management programmes, based upon a national ocean strategy expressing national concern and commitment, in a manner consistent with their responsibilities for the protection and conservation of the marine environment. This is particularly important for those many coastal states around the world which acquire national sovereignty, sovereign rights, or jurisdiction over vast areas of the marine environment. With these rights commensurate responsibilities are also acquired, and not least a range of duties specifically addressed to the protection and conservation of the environment within these marine areas. With the jurisdictional issues resolved in their favour at UNCLOS III, coastal states now have to accept the role of guardian of these areas in defence of their environmental integrity. To this end, states are urged, with the co-operation and assistance of relevant international agencies and organizations, to develop and effectively co-ordinate appropriate protective policies and measures.

13. Development of coastal zone management

Coastal states should as far as possible adopt an integrated ("coastal zone management") approach to the resolution of ocean use conflicts in coastal areas of the marine environment under their national jurisdiction.

The integrity of the marine environment is most seriously threatened in areas relatively close to shore, and it is there that multiple use issues present the most difficult challenges to effective management. All coastal states are, therefore, urged to explore the opportunities for an integrated and systematic approach to the environmental management of coastal areas, including the adoption of appropriate *legislation. Before coastal zone management legislation can be enacted, there will* be, in many cases, a need to clarify the legal status of the various parts of coastal area, from above the high tide mark to the sea-bed, and to resolve national jurisdictionai conflicts between the different administrations or agencies which are territorially competent over these parts.

The harmonious development of coastal areas will also depend, to a very great extent, on improved knowledge of the nature and functioning of coastal ecosystems. The current work of UNESCO's Division of Marine Sciences may in that respect prove invaluable. Technical and financial assistance will often be required to help developing countries in their coastal zone research and planning, in the development of legislation, and in the implementation of management measures.

14. Protection of habitats and establishment of protected areas

With a view to the need for special protection in ecologically vulnerable areas and the habitat of depleted, threatened or endangered species and of other forms of marine life, coastal states should confer special status on such areas within the limits of their national jurisdiction, and to co-operate with other states and relevant international organizations in the designation of areas located beyond the limits of national jursidiction, which may appropriately be brought under special international protection.

It is important that protected areas should be established, at national, regional, and global levels, to counter some of the most serious threats affecting marine species and ecosystems and thereby the ecological and *genetic diversity* of the marine environment. It should be recalled that Article 192 of the Convention on the Law of the Sea sets down the general obligation of all states to protect and preserve the marine environment, and that Article 194(5) provides more specifically for co-operative measures to "protect and preserve rare or fragile ecosystems as well as the habitat of depleted, threatened or endangered species and other forms of marine life". It might be added that Recommendation 38 of the Stockholm Action Ran requests governments to set aside ecosystems of international significance for protection under international agreement. Hence, in waters under their jurisdiction, coastal states are urged to establish protective areas, either at their own initiative, or pursuant to international treaties and programmes. Within the exclusive economic zone, advantage should be taken of the "special areas" procedure provided for in Article 211(6) of the projected Convention on the Law of the Sea, so that special mandatory measures can be adopted for the prevention of vessel-source pollution in areas of particular vulnerability. Coastal states should implement conservation treaties which foster the establishment of protected areas in marine areas under national jurisdiction, such as protocols developed under the UNEP Regional Seas Programme to that effect. Other relevant treaties or programmes include: the World Heritage Convention, especially indicated in the case of unique ecosystems, the Wetlands Convention, as a tool for the protection of certain coastal habitats as are the biosphere reserve network of UNESCO and that of biogenetic reserves of the Council of Europe. The OAU and OAS Conventions might also serve a useful purpose, especially if strengthened. But it seems even more important to conclude separate, and more specialized, agreements with a view to elaborating special protective rules in light of the specific nature of marine ecosystems.

On the high seas, of course, protected areas can only be established by treaty. Such treaties are more likely to be developed, if UNEP, IMCO, FAO, UNESCO, and other agencies with conservation responsibilities, are urged by the member states to collaborate in studies designed to identify the most vulnerable areas of the marine en-

vironment and to assess appropriate treaty arrangements that might be made for protecting such areas in the high seas.

On the seabed and ocean floor beyond the limits of national jurisdiction, the Authority would undoubtedly be competent to set up protected areas and could therefore adopt special rules for their establishment, preservation and management. Such areas should comprise unique ecosystems, such as deep ocean trenches and representative samples of benthic fauna habitat, some of which could be used as controls when monitoring the effects of deep sea mining in other areas. In addition, large areas, comparable in size to the proposed mining tracts, should be set aside as wilderness areas; these areas should be dispersed so that some would be adjacent to mining sites and therefore constitute a basis for the recolonization of these sites by marine organisms, while others would, on the contrary, be located at a considerable distance away and, as a result, remain undisturbed.

15. Preservation of rare and endangered species

Concerned states should enter into arrangements with a view to preserving the natural genetic diversity of the marine environment and, in particular, to the more effective protection of rare and endangered marine species.

The safeguarding of species is one of the objectives enumerated in Principle Two of the 1972 Stockholm Declaration on the Human Environment. Moreover, Recommendations 39 and 40 of the Stockholm Action Plan call for the launching of an international programme to preserve genetic diversity, suggesting that inventories be made of all endangered species and that special attention be given to the means of checking the disappearance of the rarest species. Since the Stockholm Conference little has been done to implement these recommendations, and the Convention on the Law of the Sea reflects minimal concern with this problem. In addition to global and national action, there are a number of co-operative actions that could be taken at the regional level. First, the protection of endangered and rare species should be set out as one of the objectives of the *Regional Action Plans* being negotiated in various regions under the UNEP Regional Seas Programme, and co-operative measures to this end could be developed under appropriate protocols to Regional Seas Conventions negotiated within the framework of these Plans. Second, the same objective should be one of those accepted under the terms of reference of *regional research centres,* in conformity with the provisions of Article 277 of the projected Convention on the Law of the Sea. Third, the protection of endangered *migratory species* (e.g. sea-turtles, dolphins and certain seals) should be the subject of special regional agreements under the 1979 Convention on the Conservation of Migratory Species of Wild Animals. Last, regional groupings of coastal states should explore the possibility of creating *sanctuaries,* where the taking of designated endangered species would be prohibited and other harmful activities in the area proscribed or controlled. A recent example of a sanctuary for the preservation of certain species is the one established in the Indian Ocean by the International Whaling Commission. Although, for the time being, only the killing of whales is forbidden in that area, other international conservation measures could be adopted in the future, possibly by other organizations competent in the area or under other agreements relating to the area. In certain cases, the conclusion of treaties, either independently or in the form of protocols to existing treaties such as

the Regional Seas Conventions, could provide for the necessary machinery to set up a management authority for the sanctuary area.

Where on-site preservation is not sufficient, an additional measure might be the establishment of gene banks on land. Indeed it should be possible to establish a global network of institutions, both public and private, which are equipped to participate in the collection of genetic materials from the marine environment, with a view to ensuring the perpetuation of rare and endangered species.

16. Introduction of alien species

UNEP and other interested organizations should develop guidelines relating to the measures to be taken to prevent or control accidental introductions of alien species into the marine environment and to restrict intentional introductions of such species, with the ultimate goal of securing global and regional arrangements. They should also establish and maintain a registry of all introductions of alien species into the marine environment. Intentional introductions should be the subject of an international notification procedure.

UNCLOS III gave little attention to problems of conservation associated with non-commercial species. But Article 196 takes cognizance of the danger of environmental harm caused by the intentional or accidental introduction of species, alien or new, to a particular part of the marine environment, and states are urged to take all necessary measures to reduce this danger. It is often impossible in practice to determine in advance whether intentional introductions will have significant and harmful effects on the marine environment and hence they should be restricted to those cases where prior research has shown that harmful effects of such introductions on the marine environment are minimal. It is further suggested that an international notification procedure should be developed, so that other states or interested agencies or organizations might lodge objections before it is too late. UNEP and other bodies, such as IOC and IUCN, might co-operate in order to compile and maintain an updated registry of all introductions, both intentional and inadvertent. A list of methods of control or eradication could eventually be made up for the guidance of state agencies responsible for the protection and conservation of the marine environment within limits of national jurisdiction.

17. Fishery management and conservation: protection of critical habitats

Coastal states should preserve critical habitats of commercially important species.

The problems associated with the protection of habitats in the marine environment vary from area to area, depending on whether the area is used for commercial exploitation. In the case of *commercial* species, the connection between fishery productivity and the safeguarding of critical habitats, such as spawning grounds or nursery areas, is obvious. Yet few protective measures of this kind have been taken in commercial fishing grounds and for many commercial species critical habitats have still to be identified. No population dynamics model in the literature on fishery management takes account of the prospect of reduced recruitment due to the degradation or alteration of the relevant habitat. Indeed little research effort has been devoted to the question whether habitat destruction has contributed in a measurable degree to the collapse of sea fisheries, as demonstrated in the case of

many fresh water fisheries. This kind of research seems appropriate for the *regional research centres,* which states are encouraged to help establish under Article 276 of the Convention on the Law of the Sea. Such research should be carried out in co-operation with the appropriate *fishery commission* or council in the region.

18. Fishery management and conservation: unit management

States and fishery organizations should, as far as practicable, base their fishery management and conservation policy on the principle of unit management

Ideally all fishery conservation efforts should be conducted within a regulatory system that is applicable to the entire stock under management, to associated or dependent species, to the relevant habitat, spawning or nursery areas, and to the relevant range of migration in the case of migratory species. In this sense, management should be effected on the basis of unitary control over an extensive, but environmentally coherent, spatial "unit". Internationally, this principle is not easily applied, but one ecological advantage of extended national fishery jurisdiction under the new law of the sea is that many coastal states now have an easier opportunity to put unit management into effect, especially for non-migratory species, within the 200-mile limits of their exclusive fishing (or economic) zone. However, co-operative efforts by all states and fishery bodies concerned will generally be required if unit management is to be achieved in the case of shared stocks, straddling stocks and stocks limited to the high seas.

19. Fishery management and conservation: shared stocks

Neighbouring and adjacent states sharing the same stocks or stocks of associated species within their exclusive economic zones should conclude arrangements for common management and conservation of such stocks.

Every coastal state should strive to bring its fishery management and conservation policies into conformity with the principle of *unit management.* In the situation where stocks are shared between two or more neighbouring or adjacent exclusive economic zones it is especially important to seek co-ordination and harmonization of national policy and legislation between the managing states, so that these stocks are not subjected to different or incompatible conservation strategies and techniques. Article 63(1) of the Convention on the Law of the Sea recognizes the seriousness of this problem under the conditions of extended coastal states jurisdiction and calls for states in this situation to negotiate such arrangements either directly or through appropriate subregional or regional organizations. It is suggested that FAO might offer to develop unit management guidelines designed to facilitate the co-ordination of fishery conservation measures by neighbouring or adjacent coastal states with shared stocks.

20. Fishery management and conservation: straddling stocks

Coastal states managing a stock or stocks of associated species straddling the boundary line between their exclusive economic zone and the high seas areas beyond should conclude arrangements for common management and conservation measures with the other states interested in such stocks.

Likewise, in light of the principle of unit management, states with management responsibilities for stocks extending beyond the seaward limits of their fishing zones into areas of the high seas beyond must recognize the necessity to seek co-ordinative arrangements with the states fishing the same stocks in these areas, and *vice versa.* Article 63(2) of the Convention on the Law of the Sea calls upon all states in this situation to seek agreement on common or at least compatible conservation measures for straddling stocks either directly or through appropriate subregional or regional organizations. But this provision, it should be noted, calls for such co-operation only with respect to those sectors of straddling stocks that are located in the adjacent areas of high seas. The management of straddling stocks is unlikely to be workable if it is based on a separate dual system of this kind. At the least, the coastal state and the neighbouring regional fishery commission should proceed on the basis of an understanding of the need to ensure *consistency* between national and international measures applied to different sectors of the same stock. Ideally, of course, the coastal state and the commission should try to establish a joint or consultative mechanism for the management of the entire stock as a unit. Special attention should be given to the need to protect spawning, nursery, and other critical habitats affecting straddling stocks, on *both* sides of the line between national and international waters.

21. Fishery management and conservation: total allowable catch

Coastal states should use the best scientific evidence available to them in determining the total allowable catch of stocks within their exclusive economic zone, and to co-operate with neighbouring and other states and relevant international organizations to ensure compliance with their conservation measures.

Under Article 61 of the projected Convention on the Law of the Sea the coastal state is entitled to determine the total allowable catch of the living resources in its exclusive economic zone. The purpose is to maintain or restore populations of harvested species at levels which can produce the *maximum sustainable yield* as qualified by relevant environmental and economic factors, and thus to ensure that the maintenance of these resources is not endangered by over-exploitation. Conservation measures should be based on consideration for *unharvested* as well as harvested species within the same ecosystem, as provided in Article 61(4).

The concept of maximum sustainable yield has received considerable scientific criticism in recent years; but despite the new qualifications to the concept introduced at UNCLOS III many coastal states are likely to have severe problems both in determining the most appropriate catch limits for stocks within their zone and in implementing conservation measures based on such limitations. Such states should work closely with FAO and other relevant international organizations in order to obtain the best possible scientific evidence required for making ecologically appropriate determinations of allowable catches and for the formulation of conservation measures in national legislation. In view of the difficulties in ensuring compliance, special precautions should be taken to monitor fishing activities conducted under *joint venture arrangements.* It might be helpful for FAO to draft conservation guidelines, perhaps in the form of a model joint venture contract, which would be available for *adaptation to local conditions* in light of the appropriate scientific research. In addition, neighbouring coastal states are urged to consult

with one another, directly or in appropriate regional forums, both in the determination of total allowable catches for shared or straddling stocks and in the enforcement of conservation measures based on catch limitations.

22. Fishery management and conservation: early warning system

Regional fishery commissions and coastal states are urged to co-operate with FAO in the development and improvement of early warning systems for stocks which are in danger of becoming overexploited

Article 61(5) of the Convention on the Law of the Sea requires the coastal state and other states whose nationals are permitted to fish in the exclusive economic zone of that coastal state to exchange available scientific information, catch and fishing effort statistics, and other data relevant to the conservation of fish stocks. In this way, it is suggested, efforts should be made to develop or improve "early warning systems" around the world to give warnings to the states concerned when a particular stock seems to be in danger of becoming over-exploited in violation of the Convention and of national legislation based upon it. To this end the FAO should assume the primary role in promoting the co-operation of regional fishery commissions and coastal states. Moreover, use might b e made of the national and regional marine scientific and technological centres envisaged in Articles 275 and 276 of the Caracas Convention. These centres should, of course, be staffed with specialists qualified not only to assess the information received but also to recommend the appropriate remedial action when warnings have to be issued.

23. Fishery management and conservation: assessment of socio-economic aspects of the new law of the sea

FAO in co-operation with regional fishery bodies should develop sub-regional, regional or global minimum standards as well as technical and socio-economic guidelines to assist coastal states in assessing new approaches to fishery management and conservation.

An important factor to be determined by coastal states under the projected Convention is the maximum sustainable yield (MSY) for stocks under their jurisdiction. However, MSY, as qualified by Article 61.3, is no longer a purely scientific factor. Since sub-regional, regional or global minimum standards are to be taken into consideration for its determination, it will be necessary to develop these standards which, for the time being, do not exist. Overall general standards may presumably be established by a world organization, i.e. FAO, and regional standards by regional centres in close co-operation with regional or sub-regional fishery bodies. Socio-economic aspects should likewise be the subject of guidelines developed at the world level in a form available for adaption to local fishing patterns and the needs of coastal communities. This could also be performed by regional centres and regional fishery organizations. Further refinements could be made in later years as experience with the socio-economic approach accumulates and the data base becomes sufficiently firm and extensive to permit comparative study of alternative methods of socio-economic management and conservation of fisheries in different settings.

24. Incidental take: research and conservation requirements

FAO and other interested agencies and organizations, particularly fishery bodies, should develop appropriate guidelines on incidental take, and to that effect should undertake a long-range research programme to monitor and measure incidental catches, as well as the ecological impact of such catches.

A serious problem in fishery management and conservation is that of dealing with the "incidental take": that is, the familiar situation in which a net or trawl brings up a mixture of commercially valuable and commercially worthless species. The latter are normally thrown back, but usually they are already dead or dying. FAO and other interested agencies, in particular fishery bodies, should initiate a long-range, worldwide research programme to measure the size of incidental catches in different regions, to compare the harvesting technologies available, and to evaluate the impacts of this source of fish mortality on the species concerned as well as on the related ecosystem. The purpose of this research would be to develop more appropriate gear or fishing methods designed to facilitate the escape of non-target species and to determine other conservation measures to maintain the population of these species at a satisfactory level. The seriousness of this problem was recognized at UNCLOS III and is reflected in Articles 61, 63 and 119 of the Convention on the Law of the Sea. The research programme proposed would be a practical global response to the responsibilities implicit in these provisions.

25. Incidental take: emergency and preventive measures

States and fishery organizations concerned should take emergency measures to combat the existing problem of species endangered as a result of incidental catch, and take preventive measures within the context of appropriate legal and institutional arrangements.

It is known that a number of marine species — for example, the Ridley turtle and certain species of porpoises — are endangered today as a result of mortalities induced incidentally, when they are caught in nets or trawls along with targeted species. States and organizations are urged to take appropriate remedial and preventive action. Flag states whose vessels are involved in the incidental capture of endangered species should make a special effort to institute an effective catch inspection system and address themselves to the task of formulating and implementing appropriate measures for gear restriction or modification or changes in fishing methods. Coastal states in whose waters such problems recur persistently should effect permanent or seasonal closure of breeding and other critical habitats of species endangered as a result of incidental catches. Regional and national fishery conservation organizations should undertake to collect relevant statistics with a view to providing a reliable basis for appropriate action by the relevant coastal states and regional commissions.

26. Highly migratory species: amendment of Annex

Parties to the Law of the Sea Convention, when in force, should develop a procedure for the review and possible amendment of the listing of highly migratory species contained in Annex I of the Convention.

The highly migratory species dealt with in Article 64 of the projected Convention on the Law of the Sea are defined by means of a listing provided in Annex I. It must be presumed that new scientific evidence will, sooner or later, raise questions about the adequacy of this listing, which was drawn up in the 1970s. Yet there is no provision in the Convention specifically for the review and amendment of this listing. The general procedure for formal amendment by a review conference, established in Article 312, cannot be instituted until 10 years from the date of entry into force and is much too clumsy for a matter of this kind. The simplified procedure of amendment by written notice, set out in Article 315, is much less complicated and less dilatory, but any one single state which is party to the Convention may veto amendment by this informal procedure by lodging an objection. It seems unlikely that a rigorous single-veto system was intended to apply to the amendment of the scientific listing contained in Annex I. It is suggested that a simpler method of amendment be devised, preferably by. means of scientific committee, such as the Advisory Committee on Marine Resources Research.

27. Highly migratory species: regional arrangements

States and fishery organizations concerned should conclude effective regional arrangements for the management and conservation of highly migratory species.

It has always been especially difficult to implement the principle of unit management in the case of highly migratory species such as tunas that roam widely throughout the ocean environment. This problem is aggravated by the further compartmentalization of the sea into different zones, mostly zones of coastal state jurisdiction, as a result of UNCLOS III. Article 64 of the Convention on the Law of the Sea, recognizing this difficulty, calls on coastal states and other states whose nationals fish the same highly migratory species to co-operate in the establishment of "appropriate international organizations" to ensure the effective conservation as well as the optimum utilization of such species throughout the region "both within and beyond the exclusive economic zone". Some experts believe that only a worldwide agreement, such as the International Convention for the Regulation of Whaling, setting out global management policies, would be a workable framework for the conservation of tunas, but it is likely that, with or without such a global strategy, the actual management will have to be effected by regional organizations, as contemplated by Article 64. In some parts of the world such mechanisms should perhaps be extremely extensive, covering for example the entire Indo-Pacific region. Within such very large organizations it might, however, be desirable to establish sub-regional management committees.

28. Marine mammals: regulation of whaling

The International Whaling Commission should ensure the effectiveness of its regulatory system on the basis of the principle of unit management.

UNCLOS III did not attempt to make any new provisions for the international regulation of whaling. But in opening up extensive new areas of coastal state jurisdiction around the world, the Convention on the Law of the Sea has entrusted a number of important whaling grounds to the responsibility of coastal states, some of which have little scientific or regulatory experience with respect to whales and whaling. There is ground for concern that some coastal states may fail to accept the

regulatory controls developed by the International Whaling Commission (IWC). Article 65 of the Convention on the Law of the Sea makes it clear that nothing in Part V, which establishes the regime of the exclusive economic zone, "restricts the rights" of a coastal state to apply such controls to its zone or to apply other controls which may be more stringent than those provided for in Part V with respect to marine living resources in general. Hence the International Whaling Commission may continue asserting its global management responsibilities, in accordance with Article 237 of the Convention, and adopt regulatory measures which apply to the whole range of the species, irrespective of jurisdictional zones.

29. Anadromous stocks and catadromous species: listing

FAO or a review body established under the Law of the Sea Convention should compile an official list of those species falling under the provisions of the Convention relating to anadromous stocks and catadromous species.

The Convention on the Law of the Sea provides a listing of "highly migratory" species (in Annex I), but fails to do so for anadromous stocks and catadromous species. It is suggested that a scientific committee should be established to provide such listings. Presumably this should be done by the same committee called for to review Annex I (see Recommendation 26).

30. Anadromous stocks and catadromous species: conclusion of agreements

Coastal states with responsibility for the management and conservation of anadromous stocks and catadromous species under the Convention on the Law of the Sea should conclude agreements pursuant to Articles 66 and 67 with other interested states to ensure the effective implementation of management measures.

Articles 66 of the Convention on the Law of the Sea vests primary responsibility for the management of *anadromous* stocks (e.g. salmon) in the coastal state in whose rivers such stocks originate. In return for the exclusive right of regulation, the state of origin of such stocks is obligated to consult with other states affected by its management controls. Similarly, Article 67 vests responsibility for the management of *catadromous* species (e.g. certain species of eels) in the coastal state in whose waters such species spend the greater part of their life-cycle. In both situations, the managing state should initiate, with other interested states entitled by the Convention to harvest these resources, proposals for negotiation of an appropriate conservation agreement and effective enforcement procedures.

31. Environmental assessment: assistance

UNEP, in co-operation with other global agencies and regional organizations, should help states to assess the environmental effects of planned activities under their jurisdiction or control on the marine environment and to develop criteria, standards and procedures to be followed in the preparation of such assessments.

Article 204 of the Convention on the Law of the Sea creates an obligation on all states to monitor the risk or effects of marine pollution caused by activities which they permit or in which they engage. Under Article 205 states have a duty to publish the results of such studies. More specifically, Article 206 incorporates in the new environmental law of the sea an obligation to carry out *environmental assessment* studies in advance of planned activities which may cause "substantial pollution of, or significant and harmful changes to, the marine environment". This represents a significant step forward in the development of international environmental law, but requires the development of criteria, standards and procedures to ensure that consistent approaches be followed, that all potential damages have been taken into consideration, and that when a proposed project is likely to cause such damage all possible steps are taken to avoid them. In addition, in many regions of the world this progress may prove to be illusory unless the appropriate kinds of technical assistance are made available to states endeavouring to comply with these obligations. As the environmental co-ordinating body within the UN system, UNEP seems to be the most appropriate agency to develop such criteria, standards and procedures and to undertake the establishment of an inter-agency programme designed to provide assistance of this kind.

32. Deep ocean mining: environmental protection

The states signatory to the Law of the Sea Convention are urged to establish by resolution a provisional Committee on Environmental Matters, to be converted to a Standing Committee under the International Seabed Authority when the Convention comes into force, with a first task of developing environmental guidelines for deep-ocean mining.

Little is known about the probable environmental effects of deep-ocean mining, and most of what is known has not been readily available to the delegations negotiating deep-ocean mining issues at UNCLOS III. It is extremely difficult to overcome this problem of uncertainty, but the Conference showed its awareness of the need for care by agreeing in Article 145 that the International Seabed Authority would have the obligation to adopt appropriate rules, regulations and procedures for the "prevention of pollution and contamination, and other hazards to the marine environment ... and of interference with the ecological balance of the marine environment." The Authority would also be responsible for the "protection and conservation of the natural resources" of the deep ocean floor, and the "prevention of damage to the flora and fauna of the marine environment".

It may be several years before the Authority becomes operational. In the years intervening, however, it is important that preliminary work be undertaken to facilitate the Authority's operations, once it is established. It is proposed, therefore, that a provisional environmental committee be established to develop environmental guidelines for deep-ocean mining as a first step, which would be staffed with scientists of appropriate expertise and funded sufficiently to participate effectively in research projects related to the environmental effects of deep-ocean mining. One future project should attempt to identify deep ocean areas of special vulnerability or of ecological importance and other areas requiring protection, with a view to their designation as "protected areas", in which ocean mining activities would be prohibited.

33. Marine pollution: harmonized methodologies

UNEP and other concerned organizations should promote the monitoring of pollution in the marine environment and harmonize standards, techniques and procedures related thereto.

To assess the consequences of the release of noxious waste into the marine environment and identifying the current state of marine pollution especially in highly vulnerable areas, monitoring programmes have become essential. Moreover, they provide the only means of controlling the efficiency of existing regulatory systems. But in order to be effective, this monitoring must be internationally co-ordinated so as to allow for comparisons; this is only possible if methodologies and procedures are harmonized in regard to standards and measurements (e.g. inter-calibration of instruments).

34. Marine pollution emergencies: duty to notify

All coastal states are urged, pursuant to Articles 197 and 198 of the Law of the Sea Convention, to notify at operational level, other states that might be affected in marine pollution emergency situations, preferably under special treaties and other institutional arrangements providing appropriate mechanisms and procedures.

UNCLOS III has played an important role in "hardening" the "soft law" obligations to co-operate and to give notice of imminent and actual pollution damage, which have been evolving in international environmental law in a number of treaties since the Stockholm Conference on the Human Environment. The latest crystallization of these obligations is found in Articles 197 and 198 of the Convention on the Law of the Sea. In particular, the duty to notify, as re-stated in Article 198, should now become the basis of special treaties and other institutional arrangements designed to institute more specific rules and procedures for marine pollution emergency situations, to enable competent foreign authorities to take immediate action.

35. Marine pollution emergencies: contingency planning

All coastal states should develop contingency plans for responding promptly and efficiently to pollution incidents in and adjacent to the areas of the marine environment within their national jurisdiction.

The only reference in the Convention on the Law of the Sea to the need for such contingency planning is in Article 199, which calls for the promotion and development of *joint* contingency plans by neighbouring states. *A fortiori,* each coastal state has an obligation to develop its own national contingency plan or plans, though preferably it should be done in consultation with its neighbours, so that a basis for joint contingency planning is provided. Hence such plans should provide for mutual assistance and joint action, exchange of information, expertise and equipment, as well as for conditions under which compensation should be due for such assistance.

36. Marine pollution from land-based sources: interregional comparisons

UNEP should (a) assist states from different regions to share experiences in the common effort to control marine pollution from land-based sources; (b) ac-

celerate the development of regional rules, standards,practices and procedures to that effect; and (c) participate with other agencies in the effort to secure global agreement on the elimination of the introduction of persistent and other ultrahazardous substances into the marine environment.

Most pollutants in the ocean come from the land. But this, the most important, source of the problem of marine pollution has proved difficult to deal with at a conference such as UNCLOS III. Article 207 re-affirms the obligations of states to adopt appropriate national measures and to co-operate in the development of global and regional rules, standards, and recommended practices and procedures directed to these ends.

A considerable body of knowledge and experience will gradually develop in most regions of the world with respect to the control of marine pollution from land-based sources. Much of this knowledge and experience will derive from measures adopted within the framework of Regional Action Plans initiated by UNEP's Regional Seas Programme. On the basis of such knowledge, states should agree on global and regional rules, standards and recommended practices and procedures as called for in Article 207(4) of the Convention on the Law of the Sea.

Special attention should be devoted to the problems of prohibiting or controlling at the global or regional level the release of "toxic, harmful or noxious substances, especially persistent substances", which are accorded priority status in Article 207(5).

A review of the work of international river basin commissions and other organizations with regulatory authority over the utilization, management, and conservation of water courses, which contribute to the transportation of pollutants into the marine environment, should also be carried out.

37. Marine pollution from sea-bed activities: environmental standards

UNEP should promote the conclusion of a treaty providing for environmental standards relating to offshore exploration and exploitation of mineral resources.

The world today is witnessing extraordinary technological accomplishments in the offshore, especially in the pursuit of the mineral resources on and under the seabed in areas that fall within the regime of the continental shelf, as re-defined at UNCLOS III. Article 208 of the Convention on the Law of the Sea emphasizes the obligations of coastal states engaging in these activities to take appropriate measures, nationally and internationally, to prevent, reduce, and control marine pollution from this source.

In recent years a great deal of environmental knowledge has accumulated as a result of blow-outs and other less spectacular incidents of pollution from seabed activities. With the extent of coastal state jurisdiction over such activities agreed upon under Article 76 of the Convention on the Law of the Sea, states and corporations should now be prepared to share their experiences with these problems with a view to assessing the "state of the art" of this sector of marine pollution control. Many states active in seabed drilling and related activities have not yet adopted any

regional arrangement such as the 1977 Convention on Civil Liability for Oil Pollution Damage Resulting from Exploration and Exploitation of Seabed Mineral Resources. It is suggested that coastal states with related interests would benefit from concluding a treaty providing, *inter alia,* for updating and permanent adaptation to technological progress. The work of the UNEP Working Group of Environmental Law Experts should be drawn upon in this process.

38. Marine pollution associated with shore-based oil refineries: model contract

The UN Centre on Transnational Corporations should draft a model contract with appropriate environmental safeguards for coastal states intending to establish refineries in coastal zones.

Whether the oil is conveyed by vessel or pipeline, a shore-based refinery is the site of environmental risks for a coastal state. With a view to reducing these risks, it is suggested that appropriate environmental standards with respect to the design, construction, and operation of such refineries should be developed at the international level. Certain coastal states interested in negotiating a contract for such an installation on their shoreline with a foreign oil company should have access to a model contract with standard environmental safeguard clauses. Such a model contract might be drafted by the UN Centre on Transnational Corporations in consultation with appropriate agencies and organizations.

39. Marine pollution by dumping: wastes, toxic substances, and radioactive materials

UNEP, IMCO, WHO, IAEA, and other relevant agencies and inter-agency consultative groups (eg. GESAMP) are urged to monitor the effects of dumping wastes, especially toxic substances and radioactive materials, in the marine environment, with a view to the establishment or improvement of standards for the regulation or prohibition of these practices.

Some of the most dangerous substances on this planet are still being dumped in the ocean, despite several efforts in recent years to reduce this serious hazard to the marine environment. The London Dumping Convention of 1972 represents a significant attempt to deal with this problem through the establishment of minimum global standards. Article 210 of the Convention on the Law of the Sea reaffirms the obligations of states to prevent, reduce and control pollution of the marine environment by dumping. However, it is difficult to evaluate the adequacy of the London Convention in the absence of any comprehensive and systematic effort to monitor the effects of dumping around the world. Such an undertaking seems to require the combined resources of a number of agencies and groups with expertise and responsibility in marine pollution control.

The London Convention seems to have a number of grave deficiencies: several of the regulated ("grey-listed") substances should be prohibited ("black-listed"); all government vessels are exempt (despite the fact that many dangerous toxic substances and most radioactive materials are carried in government vessels); there is no provision for the uniform prescription and enforcement of standards; and some of the language of the text is weak or ambiguous. In the absence of reliable scientific data, it is not possible to measure the environmental shortcom-

ings of the London Convention, but the best evidence available suggests that the minimum global standards created by it are seriously inadequate. In particular, there is an evident need for regional agreements like the 1974 Baltic Protection Convention to establish more rigorous dumping standards than those created by the London Convention. In situations where the minimum global standards appear inadequate in light of the existing scientific evidence, consideration should be given to the promulgation of higher standards within the framework of the emerging Regional Action Plans which are being co-ordinated by UNEP under its Regional Seas Programme.

40. Marine pollution from vessels: standard setting

Interested governments, agencies and organizations should review and update existing standards, and where necessary accelerate the development of new standards concerning ship design, construction, manning and equipment, with special reference to technologies for the carriage of hazardous substances.

UNCLOS III reflected acute concern on the part of some coastal states regarding the adequacy of existing international standards in ship design, construction, manning and equipment, especially in light of the hazardous nature of many bulk cargoes transported along congested sea routes and close to vulnerable shorelines. As a result of negotiations at UNCLOS III, Article 211 of the Convention attempts to strike a balance between coastal concerns for environmental protection on the one hand, and concerns for the traditional freedom of navigation on the other. Read along with the provisions on innocent passage (especially Articles 19, 21, 22, 23, 24 and 25), Article 211 makes it difficult for a coastal state to apply rigorous national standards to foreign vessels carrying hazardous substances, even in its territorial sea, for ship design, construction, manning and equipment, except in compliance with international agreements. This places a special importance on existing treaties such as the 1973 International Convention for the Prevention of Pollution from Ships (MARPOL), the Convention for the Safety of Life at Sea (SOLAS), the 1976 ILO Convention on Merchant Shipping (Minimum Standards), and the 1978 IMCO International Convention on Standards of Training, Certification and Watchkeeping for Seafarers. All states and international organizations concerned, such as IMCO, IAEA, UNEP, ILO and WHO should co-operate to re-evaluate the adequacy of these arrangements as they relate to the carriage of dangerous substances, review the state of the existing technologies, and modify present standards or establish new ones as required.

41. Marine pollution from vessels: traffic management

Interested governments and agencies (e.g. IMCO) should gradually establish effective systems of vessel traffic management.

At UNCLOS III a compromise was reached on the allocation of regulatory authority over navigation for the purpose of controlling vessel-source pollution: *maximum* authority (subject to innocent passage provisions) was vested in the coastal state within its territorial sea; special *discretionary* authority was granted to coastal states in *critical* areas of their exclusive economic zone characterized by ice-covered waters (Article 234); special *conditional* authority was assigned to coastal states in *special* areas of their exclusive economic zone (as defined in Article 211(6);

and a *limited* authority is entrusted to coastal states over all other areas of their exclusive economic zone, in conformity with "generally accepted international rules and standards" (Article 211(5)). This diplomatic compromise should now lead to the establishment of effective systems of vessel traffic management in accordance with the new law of the sea.

42. Marine pollution from vessels: special areas

UNEP in co-operation with other competent agencies and organizations should assist coastal states in making and maintaining an inventory of those sites which, because of their ecological vulnerability, may warrant the establishment of "special areas" in the EEZ to preserve them from vessel source pollution. IMCO should undertake to advise governments on legislative techniques for the establishment of "special areas" with a view to developing guidelines for the preservation of marine ecosystems and habitats in water under coastal state jurisdiction.

Under Article 211(6) of the Convention on the Law of the Sea provisions are made for the adoption of special mandatory measures by coastal states in *special areas* of their exclusive economic zone, where international rules and standards are deemed to be inadequate because of the particular environmental vulnerability of these areas. States with responsibility for the protection of endangered species and habitats, and other environmentally vulnerable species and areas, are urged to take advantage of this provision by adopting measures for the establishment and maintenance of specially protected areas in accordance with Article 211(6).

As scon as several coastal states begin to adopt such measures, IMCO should open a register of relevant legislative techniques for the benefit of other coastal states contemplating similar action. Eventually, after the evolution of sufficient state practice, IMCO should undertake to provide a set of guidelines designed to harmonize existing national policy and legislation and to provide guidance for further legislative and administrative action on the part of coastal states with environmentally vulnerable marine areas under their jurisdiction. Such guidelines might provide criteria for identifying areas where navigation in general, or the carriage of particularly dangerous cargoes, should be prohibited with a view to protecting endangered species and habitats, or limited to certain seasons of the year with a view to protecting migratory species.

43. Marine pollution from vessels: emergency assistance to ships in distress

States should establish a global network of facilities wherein ships carrying oil or other hazardous substances could call and be provided emergency assistance without threatening populated or ecologically vulnerable areas.

For the last few years public attention has been drawn to a severe new threat to the marine environment arising from the carriage of persistent oil or other noxious substances — when a ship carrying such substances is imperiled. It happens more and more often that the harbours, where she would normally call for repair, refuse to let her in due to the risks involved both for the safety of the surrounding populations and the local marine environment (see for example the Andros-Patria case in 1979

when this ship, being prohibited to enter any North Eastern Atlantic harbour had to be towed down to the latitude of Gibraltar straits in order to be lightened). For such ships, the number of which is constantly increasing (due to the growing demand of chemicals, pesticides, fertilizers, LNGs etc.....), relief harbours ready to provide them with appropriate facilities in case of emergency becomes an urgent need. Since the potential profit resulting from the exploitation of such harbours is not big enough an incentive compared to the threat they represent to the marine environment, effective action can only be achieved through international co-operation. This is essential, otherwise ships carrying dangerous substances are doomed, whenever imperiled to be abandoned, as has already been the case on several occasions, then capsize, and degrade entire areas of the oceans.

44. Marine pollution from vessels: wrecks

States and interested organizations should develop rules establishing a duty for the "owners" of wrecks to notify the existence, location and content of these wrecks to IMCO, as well as a duty to recover the wrecks whenever feasible.

More than one million wrecks are considered to be littering the world ocean floor. The vast majority of those, have nothing in common with the fabulous, treasure-loaded gallions of the past. Having been built out of inferior materials, low quality metallic alloys, they are not worth recovering and their owners or insurers abandon them wherever they are sunk. Their increasing number in coastal areas where shipping is dense and the waters shallow poses already a significant threat to navigation, and as a consequence, to the marine environment. It is all the more the case since a number of these wrecks contain oil or extremely hazardous substances which inevitably will leak and poison the environment. An international mechanism for the monitoring of and information about wrecks has become essential.

45. Marine pollution: Ice-covered areas

Coastal states with responsibility for the environmental protection of ice-covered areas of their exclusive economic zone should co-operate with one another and avail themselves of the special authority conferred upon them under the Convention on the Law of the Sea.

At UNCLOS III coastal states acquired special environmental authority under Article 234 over the control of pollution from *vessels* in ice-covered areas of their exclusive economic zone. These states are encouraged to exercise this authority with respect to the routing of traffic, discharge controls, and appropriate aspects of vessel construction and design, manning and equipment standards, with a view to securing effective management of these usually sensitive sectors of the marine environment. Coastal states should also give special attention to the reduction of environmental risks associated with platform and other forms of *non-vessel* technology introduced into ice-covered waters in areas under their jurisdiction and control.

46. Marine pollution: responsibility, liability and compensation

The United Nations should promote the conclusion of an international convention relating to damage caused by marine pollution providing for rules concern-

ing responsibility and liability, uniform determination of costs to be reimbursed for remedial measures, assessment of damages caused to natural resources, especially to those without market value and concerning the establishment of expeditious standard procedures for rapid compensation.

These questions received some attention in the Third Committee at UNCLOS III, but because they were dependent upon the resolution of jurisdictional issues in the Second Committee, it was decided to postpone detailed consideration until after the adoption of the Convention on the Law of the Sea. As matters now stand, Article 235(3) anticipates future action of this kind and calls upon all states to co-operate towards this end. One of the difficulties in this politically sensitive area is that the problems of developing principles of responsibility and liability and related compensation practices vary significantly from sector to sector: the approach for vessel-source pollution may be substantially different from that for land-based pollution and the other sources of marine pollution dealt with at UNCLOS III. In light of existing sensitivities, it might be realistic to begin this undertaking in the contexts of vessel-source pollution and marine pollution from activities on the continental shelf. The former initiative should be undertaken by IMCO, with special reference to the problem of standard-setting for the carriage of hazardous substances (see Recommendation 39). The latter initiative should be undertaken by UNEP and other interested agencies as an extension of their responsibility for standard-setting (see Recommendation 36).

47. Marine pollution: review of shipping law

States with the assistance of both governmental and non-governmental organizations should examine and adapt shipping law to take better into account ecological factors; in particular, collision at sea should cover movable structures as well as ships and rules relating to salvage should not be exclusively concerned with the hull and cargo but should also take into consideration the risk of pollution from ships in distress.

In light of recent developments in both law and technology, there is an evident need to review and update shipping laws. These laws have been designed at a time when the only interests worth protecting at sea were those of crews, ships and cargos. Today attention must be given also to coastal states interests and the protection of the marine environment. The legal treatment of a number of traditional concepts, such as salvage, collision and general average, therefore, require reevaluation. It should also be borne in mind in this review process that ships are no longer the only man-made structures being operated at sea, so that appropriate measures should also cover platforms and other off-shore structures.

48. Marine scientific research: regional arrangements

Regional marine organizations and commissions should strengthen their scientific research capabilities in ocean management and conservation.

Articles 242, 243 and 244 of the Convention on the Law of the Sea call upon all states and "competent international organizations" to promote international co-operation in marine scientific research, to create "favourable conditions" for the conduct of such research in the marine environment, and to disseminate the resulting knowledge and information. Most existing regional organizations and

commissions with responsibilities for ocean management and conservation are dependent on the scientific advice and research of either their member states or a global agency. It is suggested that these regional bodies should attempt to enhance their own research capabilities either by developing an independent research staff or their own, or by making consultative arrangements with the International Council for the Exploration of the Sea (ICES) or bodies performing similar functions, such as the regional centres provided for by Article 276 of the projected Convention.

49. Development and transfer of marine technology: national training and research

States should establish national training and research programmes in ocean management and participate in the establishment of regional training and researchcentres.

In the final analysis, the effectiveness of regulatory and administrative systems for the protection and conservation of the marine environment depends, above all, on the knowledge and efficiency of government officials assigned responsibility for these tasks. In particular, all *coastal* states, as guardians of their sector of the marine environment, have an obligation to prescribe training requirements for their officials and to endeavour to enrol them in appropriately designed, national or international, training programmes. All coastal states in particular also have an obligation to participate, with appropriate assistance from relevant agencies and private organizations, in research programmes designed to enhance the capabilities of their officials to deal effectively with the tasks of ocean resource development and management and of the conservation of the marine environment. To these ends states should endeavour to establish *national marine scientific and technological research centres,* as envisaged in Article 275 of the Convention on the Law of the Sea.

In recent years the United Nations University (UNU), based in Tokyo, has attempted to provide training and research opportunities related to the field of ocean development and management through a loosely co-ordinated system of university training programmes in various countries. Due to the lack of widespread support, these efforts have not yet had a substantial effect on the capabilities of government officials to deal with the problems of protection and conservation of the marine environment. A serious effort should now be made by governments to support training and research programmes of this kind; otherwise, all efforts to raise regulatory standards around the world are likely to be self-defeating.

Article 276 of the Convention on the Law of the Sea calls upon all states, in co-ordination with relevant agencies and organizations, both national and international, to promote the establishment of *regional marine scientific and technological research centres.* The suggested training and research functions of such centres are enumerated in Article 277. In several regions, such centres may be of critical importance in assisting coastal states to implement the Regional Action Plans now being negotiated within UNEP's Regional Seas Programme. In some regions it may be more feasible to establish two or more *subregional* centres rather than one single centre to serve the entire region. Indeed there may be little reason for any two

regions to seek identical institutional arrangements for the establishment and maintenance of regional or subregional centres. In any region the coastal states should be expected to identify the most appropriate coalitions of agencies and private organizations for the servicing of such centres.

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